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# PAKISTAN ARMY

# INFANTRY BATTALION IN BATTLE

Inspector General Training and Evaluation
Infantry Directorate

# INFANTRY BATTALION IN BATTLE

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## INTRODUCTION

1. Infantry being the largest appendage of Pakistan Army; forms the mainstay of its combat potential. Owing to the nature of its versatility; Infantry can be employed in all operations of war under all environments with requisite modification and support of other arms and services. Infantry battalion is the basic combat unit that is self-sufficient within its allotted combat potential, has the ability to operate under all kind of battlefield environments.

2. The tactics at infantry battalion level is predominantly battle drills and procedures for undertaking different operations of war. In order to enhance operational worthiness of this combat potential, these battle drills and procedures are a guideline to commanders at all tiers. Adherence of these battle drills/ procedures, assertiveness on fire and manoeuvre and mastering in tactical concepts will further enhance the efficacy of this outfit.

3. This pamphlet lays down the tactical guidelines for battle drills and procedures for conventional operations of war along with employment of integral combat support components in the overall framework of brigade/division.

## PURPOSE AND SCOPE

4. The publication provides the guidelines for battle drills and procedures for major operations of war i.e. Defence, Attack, Withdrawal and Advance at battalion level within framework of brigade. It also provides necessary guidance on employment of battalion's integral fire and manoeuvre elements and administrative aspects.

5. The publication serves following purposes:-

   a. Tactical employment of infantry battalion in all major operations of war and in all types of terrain within framework of brigade.

   b. Lay down guidelines for employment of integral fire and manoeuvre elements i.e. Combat Support Company.

   c. Provide necessary guidelines for integration/employment of elements of supporting arms i.e. Armour, Artillery, Light/Heavy Anti-Tank Battalion and Engineers.

   d. Lay down battle drills and procedures for all operations of war.

   e. Assist battalion commander in command, control and administration of infantry battalion in battle.

# CHAPTER I – THE INFANTRY BATTALION

## SECTION 1 – ORGANIZATION OF INFANTRY BATTALION

### Characteristics and Role of Infantry

1. An infantry battalion is infantry's basic tactical unit. Its mission is "to defeat the enemy by skilful use of firepower and manoeuvre, by day or by night, in any weather or terrain". Its main capabilities are:-

a. Conduct operations in all types of terrain and climatic conditions by day or night.

b. Move by any method i.e. on foot or byroad, rail, sea and air.

c. Seize and hold ground.

d. Provide limited anti-tank protection.

e. Repulse enemy attack by fire and counter-attack.

f. Close with the enemy by means of fire and movement in order to destroy him.

g. Provide base of fire and manoeuvre elements.

h. Conduct helicopter borne/anti-helicopter borne operations.

i. Participate in air transported operations when provided with sufficient air effort.

j. Participate in amphibious operations.

k. Conduct independent operations on a limited scale when suitably reinforced.

2. Infantry has been described as the cutting edge of battle. Though battles are usually won by the close and effective cooperation of all arms and services, the issue is finally decided by infantry in close combat. In attack, the infantryman must close with and complete the destruction of enemy and in defence, he must hold his position and finally repel enemy's assault with maximum possible destruction. A high degree of training is required to achieve and maintain these capabilities.

3. To carry-out this role the infantry battalion contains two essential elements. Men, whose fighting spirit is the most decisive factor in battle and the weapons. Rest of the paraphernalia in the battalion i.e. all the transport and equipment are solely there to assist the infantry soldier in carrying out these tasks with his weapons. It is, therefore, above all else, on the skilful and determined use of weapons in conjunction with ground and movement, both by the individual infantryman and those who lead him, that the battalion will succeed or fail in battle.

**Guidelines Based on Army's Doctrine**

4. The Pakistan Army's doctrine is based on five cardinal features summed in the catchword FATAH (Victory). It provides guidelines for planning, employment of battalion and conduct of different operations of war. The cardinal points are given in subsequent paragraphs.

5. ***Flexibility***. The envisaged employment of infantry battalion in current and projected operational environment relies heavily on flexibility in both mental and the physical spheres. Development and inculcation of this cardinal feature is essential for rapid deployment, speedy grouping/regrouping and repeated employment of forces:-

a. ***Physical Flexibility***. Physical flexibility is achieved through organisational modularity in the field coupled with subunit elements that are trained to perform a variety of roles within and across all operations of war. Infantry battalion as part of holding forces must have the ability to form part of and contribute to an offensive effort and infantry battalion as part of offensive forces must develop the capability to hold ground when necessary. Flexible subunits allow commanders to group and regroup these to achieve tactical proficiency.

b. ***Mental Flexibility***. Physical flexibility is insufficient unless commanders are able to rapidly identify new challenges or opportunities and take timely decisions to exploit them. Without mental flexibility, physical flexibility serves little purpose. This quality is built overtime and must be embedded in peacetime operational, leadership and administrative training. Rigidity of thought, dogma and inflexibility to adapt to fluid battle conditions must be ruthlessly discouraged.

6. ***Agility***. Agility, is the sum total of flexibility, mission specific training, effective leadership and rapid decision making. Organisational and tactical versatility, self-contained sub sectors, adaptable subunits, and redundancy in command, control and communications are essential ingredients of agility. Commanders must strive to develop agility as an indispensable property for an infantry battalion to operate in the contemporary and projected operational environment. It allows the force to operate across the entire threat and conflict spectrum.

7. ***Timeliness***. Timeliness links every cardinal principle and is the scarlet thread of success that runs through development, planning and employment of infantry. It is intimately related to forward planning, force protection, timely orientation, decision making and speedy execution. Timeliness requires accurate information and

intelligence as well as efficient response capacity of different subunits (flexibility and agility), filtering out unnecessary delays. It demands the audacity to take calculated risks for exploiting transient opportunities. Timeliness warrants exhaustive staff work and informed decision making to disrupt synergy in enemy's decisions and actions to wrest initiative at all tiers of command and at all stages of an operation. Timeliness is not attainable without aggressiveness and adequate freedom of action.

8. ***Aggressiveness***. As a cardinal feature, aggressiveness instils the urge to seize and retain initiative. Aggressiveness is as much an attitude towards planning and thinking as it is in the actual employment of forces. As a state of mind, it translates into the nature and type of decisions commanders arrive at, particularly under stress. In action, it implores a balance between calculated risk-taking, boldness and audacity. Aggressiveness requires information but does not imply fighting for information and helps achieve shock and surprise by exploiting fleeting opportunities. The Army's War Fighting Doctrine places special emphasis on aggressiveness. The character of future war and the environment in which it is likely to ensue precludes long drawn operational cycles and suggests the creation of equitable effects at each stage of the operation by way of tactical advantage or success. The doctrine provides conceptual tools and structures that allow both dimensions of aggressiveness to be optimally manifested through mental and physical agility and flexibility to seize fleeting opportunities and half chances identified on the battlefield. Combined in attitude and capability, aggressiveness is built during peacetime in all elements and forces to achieve its signature effects in a holistic response during war.

9. ***Holistic***. Holistic response infers a very high degree of coordination and cooperation with supporting arms including different weapon system contingent upon an in-depth understating of each other's capabilities. At tactical level, a wholesome evaluation of threat and response; employing all available means at right time, characterise this notion.

## Detailed Organization

10. Organization of infantry battalion is given in Figure 1:-

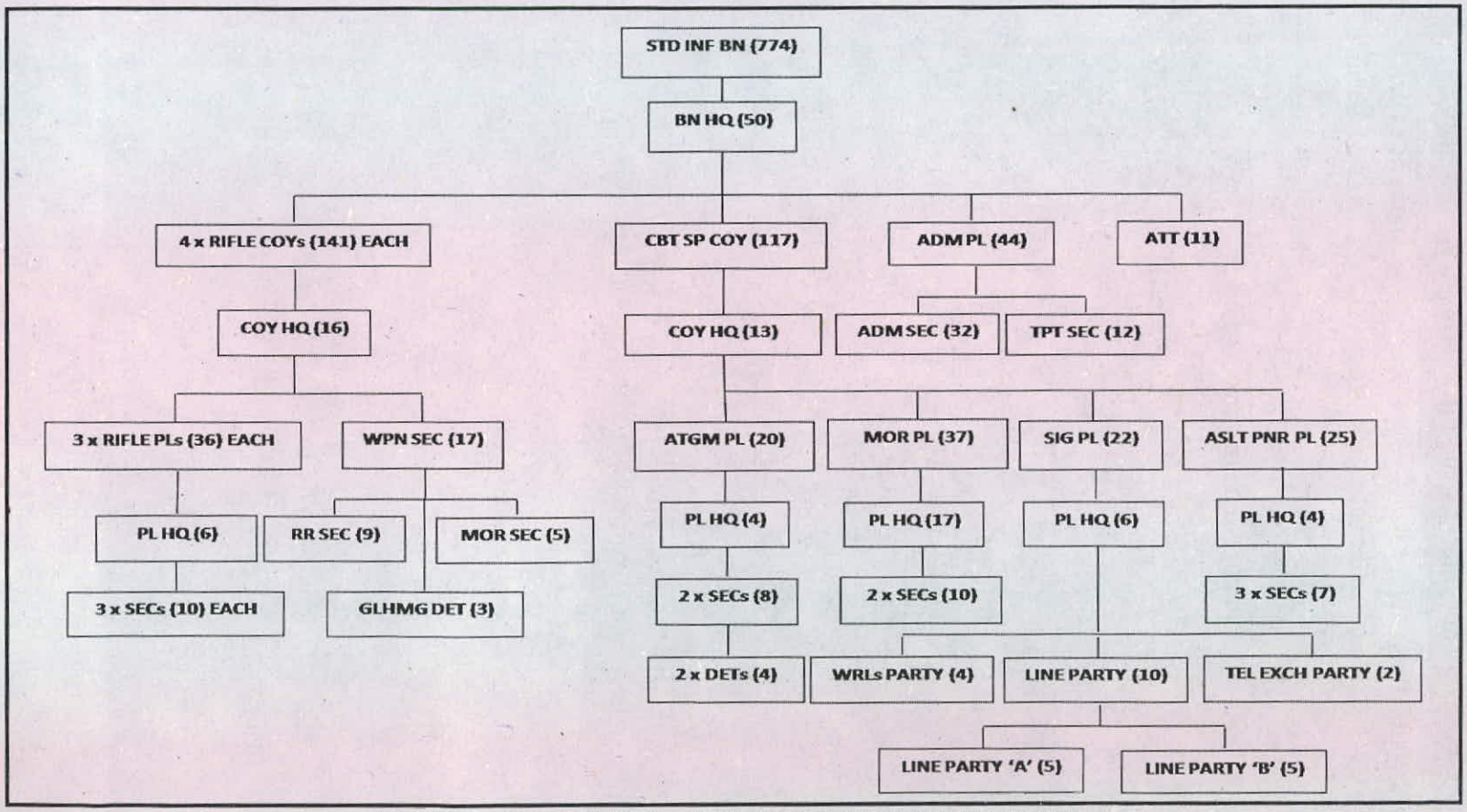


FIGURE 1 – ORGANIZATION OF STANDARD INFANTRY BATTALION.

11. Establishment of standard infantry battalion is as follows:-

| Serial No. | Details | | Present | Filed Increment for War Only |
|---|---|---|---|---|
| a. | ***Personnel*** | | | |
| | (1) | Officers | 24 | – |
| | (2) | Junior Commissioned Officers | 24 | – |
| | (3) | Soldiers | 695 | 28 |
| | (4) | Non-Combatant Enrolled | 16 | – |
| | (5) | Civilians | 4 | – |
| b. | ***Attached*** | | | |
| | (1) | Officers | – | 1 |
| | (2) | Junior Commissioned Officers | 0 | – |
| | (3) | Soldiers | 11 | 4 |
| | (4) | Total | 774 | 33 |
| c. | ***Transport*** | | | |
| | (1) | Truck ¼ ton 4 x 4 General Service | 11 | – |
| | (2) | Truck utility ¼ ton 4 x 4 106 millimetres Recoilless Rifle modified | 8 | – |
| | (3) | Truck ¼ ton Jeep Anti-Tank Guided Missile Launcher | 4 | – |
| | (4) | Truck ¼ ton 60 millimetres Mortar Ammunition/Double Cabin vehicle | 4 | – |
| | (5) | Truck Utility ¼ ton 4 x 4 106 mm Recoilless Rifle Ammunition/Double Cabin Vehicle | 4 | – |
| | (6) | Truck ¼ ton Jeep Anti-Tank Guided Missile Ammunition/ Double Cabin Vehicle | 4 | – |
| | (7) | Truck Ambulance ¼ ton 4 x 4 | 1 | – |
| | (8) | Truck ¼ ton Jeep Radar | 2 | – |

| Serial No. | Details | | Present | Filed Increment for War Only |
|---|---|---|---|---|
| | (9) | Truck ¼ ton 4 x 4 Jeep for Heavy-Machine gun/Grenade Launcher Heavy-Machine gun | 4 | – |
| | (10) | Truck ¾ ton 4 x 4 15 cwt/1.5 ton 4 x 4 Dodge | 13 | – |
| | (11) | Truck 2-½ ton 6 x 6 Lorry 3 ton | 8 | – |
| | (12) | Motorcycle | 2 | – |
| | (13) | Trailer Cargo ¼ ton | 16 | – |
| | (14) | Trailer Cargo ¾ ton | 11 | – |
| | (15) | Trailer Cargo 1½ ton | 6 | – |
| | (16) | Mobile Field Kitchen (MFK) 5 ton | 5 | – |
| | (17) | Water Trailer 1½ ton (400/450 Gallons) | 2 | – |
| | (18) | Bicycles | 3 | – |
| | (19) | Total transport (Less Trailers, Motorcycles and Bicycles) | 63 | – |
| d. | ***Weapon*** | | | |
| | (1) | Pistol/MP5A2 | 58 | 1 |
| | (2) | Rifle 7.62 millimetres G-3/G3A4 | 375 | Including 8 Sniper Rifles G-3/Styre complete with tele-scope and accesso-ries. |
| | (3) | Sub-Machine Gun/MP5A2 | 243 | – |
| | (4) | 7.62 millimetres Machine-gun 1A3/Light-Machine gun 7.62 millimetres | 74 | – |
| | (5) | 40 millimetres Grenade Launcher Heavy-Machine gun | 4 | – |
| | (6) | Rocket Launcher RPG-7 | 12 | – |
| | (7) | 106 millimetres Recoilless Rifle | 8 | – |

| Serial No. | Details | | Present | Filed Increment for War Only |
|---|---|---|---|---|
| | (8) | Baktar Shikan weapon system complete with accessories | 4 | – |
| | (9) | 0.50 Spotting Rifle | 8 | – |
| | (10) | Grenade Arges | 755 | 33 |
| | (11) | Grenade No. 80 WP (White Phosphorus) | 72 | – |
| | (12) | 75 mm Super Grenade HEAT (High Explosive Anti-Tank) | 288 (8 per rifle section ) | – |
| | (13) | Sniper Rifle with telescope and accessories | 4 | Heavy calibre. |
| | (14) | 60 millimetres Mortar (2 x 60 millimetres Mortars per detachment rifle company and 1 x 60 millimetres Mortar with each 106 millimetres Recoilless Rifle) | 16 | – |
| | (15) | 81 millimetres Mortar | 6 | – |
| | (16) | Pencil Shooting/Pistol Flare Signal | 19 | – |
| | (17) | 12.7 millimetres Anti-Aircraft Machine-gun with mount | 2 | Will be issued in lieu of 2 x Machine-gun 1A3 to infantry battalions of offen-sive formations only. |
| | (18) | Anza/Stinger Launchers | 4 | Will be issued in lieu of 4 x Machine-gun 1A3 to infantry battalions of offen-sive formations only. |

12. For details of organization, attention is drawn to the following:-

a. Summary of Personnel – Annex A.

b. Summary of Transport – Annex B.

**Distribution of Transport in Echelons**

13. First line transport consists of three echelons:-

a. ***'F' Echelon***. Vehicles essential to fight the battle.

b. ***'A' Echelon***. Vehicles readily available to support the troops taking part in battle.

c. ***'B' Echelon***. Transport carrying stores and equipment that will not be required until there is a pause in the fighting.

d. Suggested distribution of battalion transport in echelons is given in Appendix 1 to Annex B.

14. The composition of 'F', 'A' and 'B' echelons is normally at the discretion of the battalion commander. The grouping will vary from unit to unit and the commanding officer may decide its composition, according to the requirement and situation. It will vary according to the conditions of each operation. 'F' echelon must be restricted to the absolute minimum. The battalion commander will centrally control 'F' echelon and probably 'A' echelon. 'B' echelon will normally be under the control of brigade or divisional headquarters.

15. A battalion should include standard echelon grouping of transport in its standing orders, covering the organization, allotment, control, march discipline and local protection of transport. Such a standard allotment can also be quickly modified according to the circumstances. Troops carrying capacity with full combat load of various vehicles available in infantry battalion is given below:-

| | | | |
|---|---|---|---|
| a. | ¼ ton Jeep | – | 5 Persons. |
| b. | 1 ton Truck | – | 10 Persons. |
| c. | 1½ ton Dodge | – | 13 Persons. |
| d. | 2½ ton Truck | – | 22 Persons in Isuzu truck.<br>25 Persons in Hino truck. |
| e. | 3 ton Truck | – | 25-27 Persons. |
| f. | 5/6 ton Truck | – | 28-30 Persons. |

**Characteristics of Infantry Battalion Weapons**

16. Characteristics of infantry battalion weapons are given at Annex C.

**Characteristics of Infantry Battalion Night Vision Devices**

17. The chief feature of the battlefield at night is the lack of visibility. In poor visibility the target acquisition becomes difficult and the reconnaissance and observation are strictly limited. Night fighting aids are designed to help overcome

the effects of poor visibility. Following types of systems are used in Night Vision Devices authorized to an infantry battalion:-

a. Infra-red.

b. Image Intensification.

c. Thermal Imaging.

18. ***Infra-red***. These devices illuminate the object to be viewed with infra-red light and receive the infra-red energy reflected back. Infra-red being outside the visible spectrum cannot be seen with naked eye, so the object being illuminated is unaware of the fact. However, its use introduces a number of problems, particularly when used against enemy equipped similarly. In such a case the infra-red beam can be detected by an infra-red detector to actuate an alarm to tell the enemy that he is being watched. Under certain conditions it is possible to use the ambient infra-red light to illuminate a target; that is to say under conditions of moonlit night or at dusk or dawn. There may be sufficient natural infra-red light to activate the viewer. It is also quite possible to use the viewer alone to locate enemy sources of infra-red at considerable distance. Viewing under such conditions is known as passive viewing and it is completely secure, provided that the light source is kept covered. If the light source is not kept covered when not in use, the reflector will reflect and enemy infra-red beam directed towards it and so disclose its position. Some of the limitations of infra-red devices are given below:-

a. Infra-red sources may be detected by an enemy who is himself suitably equipped.

b. Continuous viewing puts a considerable strain on the operator.

c. Infra-red images have unusual shadow grading and perception of depth is difficult.

19. ***Image Intensification Devices***. These devices amplify the ambient light about 50,000 times. Only natural light (moon, stars or sky glow) is necessary to activate the image intensifier tube. The range and resolution of the night vision sights are dependent upon' the brightness of the sky and clearness of the atmosphere. The performance of night vision sights may be improved by using auxiliary visible or infra-red sources. They are un-detectable except at very short-range. Their characteristics are:-

a. Can use ambient light from stars, moon, sky glow or indirect white light.

b. Performance is degraded by rain, fog, mist or smoke.

c. Bright light sources such as flares will cause devices to black out for several seconds.

d. Devices are very expensive and get damaged by rough handling.

e. Devices are comparatively bulky.

f. Can be fitted to individual or crew served weapons.

g. They are of British origin.

20. ***Thermal Imagers***. The principle on which thermal imaging works is that all objects radiate electromagnetic energy. This energy can be detected and used to form an image of the object being viewed. Different parts of an object radiate different amounts of energy, for example the amount of energy radiated by wheels differs from what the iron body of a vehicle radiates. The quality of thermal imaging depends on the heat conditions and particularly on the contrast of heat emitted by the target to that emitted by the surroundings. It is just possible that some form of protective clothing for men could be developed which would reduce the heat they radiate to no more than that radiated by the surroundings. It would, however, be difficult to protect things like vehicles, aircraft and guns. Some of the advantages are given below:-

a. It can be used in complete darkness, whilst image intensifiers require some form of ambient light, be it, only starlight.

b. It is virtually unaffected by smoke, dust or haze and can detect objects even through camouflage nets. So far no effective way of thermal imaging camouflage has been developed.

c. It is not blinded by searchlight, flares and fires, which often completely shut down an image intensifier.

d. It can reveal details which even human eye could not detect. For example, the eye cannot tell whether a tank engine had recently been running, but the thermal imager can, because the engine would be hotter than the rest of the tank.

e. It can be used to spot the fall of the shot since the position is indicated by the heat generated by the explosion.

f. In Built-Up Areas (BUAs) the thermal imagers are of greater value in detecting men because the flaring around the street lamps make it very difficult for the image intensifier to pick-up anything.

g. Rain has little effect on thermal imager.

21. ***White Light***. It is one of the best aids for target acquisition and engagement. It facilitates the use of binocular. It can be used in two ways:-

a. Direct-searchlights, flares and illuminations of varying range and brightness.

b. Indirect-by reflecting searchlights off the cloud base and increasing the ambient light.

22. White light has following characteristics:-

a. Men lose their night vision unless one eye is kept closed.

b. Although individual positions may not be disclosed. The general area in which friendly forces are located may be revealed and some measure of surprise lost.

c. There is a risk of self-illumination when used less than 500 metres from the friendly positions.

d. Scales of illuminating ammunition are small, however, when used correctly it enhances the capability of image intensification equipment.

e. It greatly aids observation and target engagement.

**Limitations of Night Vision Devices**

23. The limitations of Night Vision Devices are:-

a. ***Limited Field of View***. Night Vision Devices have limited field of view like any other viewing device. At ranges where this limitation is overcome, it starts suffering from poor definition of the object.

b. ***Depth Perception Problem***. Some of the Night Vision Devices suffer from inadequate depth perception, which makes the user shy of using them.

c. ***Range in Relation to Ambient Light***. Actual ranges of night vision devices are lesser than the ones claimed and their performance is also directly proportional to the ambient light.

d. ***Thermal Blindness***. Prolonged use of Night Vision Devices and constant scanning through them can cause temporary blindness to a user, which may vary from 2 to 5 minutes.

e. ***Search by Scanning***. It is difficult to search the target, especially when its direction is not known. Similarly constant practice is required by the operator to understand the way the targets appear on the image screen. Human eyes get fatigued by constant scanning through the Night Vision Devices. It causes considerable strain to the eyes of the sentries using Night Vision Devices/Night Observation Devices for a period exceeding 15 minutes at a stretch.

f. ***Flash Effect***. High intensity light produces blind spot on the image screen, which could even be of permanent nature. Likewise flash

produced by artillery or mortar bursts can also damage the image intensifying tube making the device ineffective.

g. ***Effects of Weather***. Infra-red and image intensifiers cannot penetrate haze and mist. Bad weather, poor visibility, humidity and blizzard have pronounced adverse effect on the efficiency of Night Vision Devices.

h. ***Effects of Vegetation***. Dense undergrowth, shrubs, foliage and trees serve as a shield against the Night Vision Devices. Troops merging with the undergrowth or moving with the folds of the ground cannot be observed even at 100 metres.

i. ***Limitation of Power Supply***. Being battery powered, performance remains directly proportional to the input, which is susceptible to decay with constant use and replacement of power source becomes a recurring necessity.

j. ***Suitable for Static Role***. Most of these systems are effective in defensive or static positions. Use of Night Vision Devices in offensive role is comparatively restricted. Therefore, in certain phases of an operation; these would be just a redundant weight. However, these are quite useful for the patrols.

k. ***Delicateness***. Technically advanced Night Vision Device systems are very delicate and require sophisticated, skilful handling and regular maintenance.

24. ***Additional Information***

| | | | |
|---|---|---|---|
| a. | Authorization of Night Vision Devices to an infantry battalion | – | Annex D. |
| b. | Characteristics of Night Vision Devices of infantry battalion | – | Appendix 2 to Annex D. |
| c. | Data/capability of Night Vision Devices of own tanks | – | Appendix 3 to Annex D. |
| d. | Details about Night Vision Devices held by the Indian Army | – | Appendix 4 to Annex D. |

**Training**

25. Night Vision Devices/Night Observation Devices would now be commonly used in a future battlefield. These devices are very expensive and also quite sensitive. Therefore these must be used with lot of care to avoid damage and their maintenance given top priority. However, these devices must be frequently used in individual and collective training to achieve proficiency in their handling. Not only the desired training standards in the use of Night Vision Devices should be achieved

through regular training, but also it must be ensured that these devices are used with the utmost care for their preservation. Following guidelines are given with regard to training and maintenance to enable units/formations to formulate their standard operating procedures.

26. ***Standards to be Achieved by Night Vision Device Users***. The user must:-

a. Know all the characteristics and technical data of the Night Vision Devices that they are operating.

b. Know all checks to be performed before and after use.

c. Be proficient to fix Night Vision Devices on respective weapons during day/night and under all weather conditions.

d. Know complete functioning, stoppages, their rectification and remedial measures under all environments.

e. Be proficient in zeroing Night Vision Devices and altering/adjusting Night Vision Devices with respective weapons.

f. Be proficient in aiming with Night Vision Devices under all weather conditions during day/night.

g. Know how to fire from different positions, i.e., while standing, sitting, kneeling and lying etcetera with the Night Vision Devices.

h. Be proficient in judging distances with the help of Night Vision Devices/Night Observation Devices.

i. Attain at least 1st class firing standard using Night Vision Devices with different weapons on long-ranges under all weather conditions.

j. Be capable of firing uphill/downhill while firing with Night Vision Devices.

k. Know all safety precautions for Night Vision Devices, to be adopted during and after their use.

27. ***How to Train for Use of Night Vision Devices***

a. The operators of Night Vision Devices/Night Observation Devices must be imparted complete theoretical knowledge during various courses at army institutions and also during individual training cycles at unit level.

b. Initial practical training should be conducted during daytime and later during night.

c. At the initial stage, firing practice should be carried out with different Night Vision Devices at short-ranges during day/night.

d. Firing at short-ranges must be followed by firing at the long-ranges.

e. Individual training should culminate in liberal use of Night Vision Devices/Night Observation Devices during summer and winter collective training.

f. Field firing exercises provide yet another opportunity to unit commanders for imparting requisite training in efficient handling and use of Night Vision Devices/Night Observation Devices.

g. Troops need to be made cautious about the use of Night Vision Devices/Night Observation Devices by the enemy. They must understand that the observation environment at night is almost similar to the day and any abnormal move/activity will be picked up by the enemy from a far distance. Therefore, they must take necessary precautions to avoid detection at night.

h. For the purposes of surveillance with Night Observation Devices, the area of responsibility should be divided in three distinct brackets, i.e. near, middle and far distance. The surveillance should be carried out from the left to the right and the observers must rest their eyes to avoid undue stress.

28. ***Maintenance of Night Vision Devices/Night Observation Devices***. They should be operated only by a qualified operator after he has read and understood all operating instructions and has been sufficiently trained. As a general rule, the Night Vision Devices/Night Observation Devices should be handled with maximum care to avoid possible damage to lenses and tubes. Instructions for maintenance of important components and precautions during operation in different conditions and their storage, are given below:-

a. ***Battery***. Ensure the following:-

(1) ***Nickel Cadmium Rechargeable Battery Cells***. Nickel cadmium rechargeable battery cells must be used duly charged. These should be charged regularly as per instructions. Use of discharged batteries could damage Night Vision Devices.

(2) ***Dry Battery Cells***

(a) Use good quality cells, preferably the alkaline battery cells.

(b) Weak cells should not be used in the Night Vision Devices. Replace weak/discharged cells with new cells for proper operation.

(3) ***Miscellaneous***

(a) Make sure that the ON/OFF switch is set to OFF position before installing/removing batteries.

(b) Remove the battery before storage when the Night Vision Device is not in use for a longer period and even during exercises/training periods.

(c) Clean the battery housing and contacts with alcohol/ spirit at least once a month.

b. ***Night Vision Devices/Night Observation Devices Lenses***

(1) Keep the lenses clean and do not touch them with fingers. Clean them carefully with folded lens tissue paper after the dirt has been removed using camel hair brush.

(2) Keep the protective lens covers on, when the Night Vision Devices are not in use.

c. ***Reticle Brightness***. Reticle in the Machine-gun (MG) and Recoilless Rifle sights should not be over-brightened as it damages image intensifier tube of the Night Vision Devices.

d. ***Moving Parts***. Keep all moving parts of the Night Vision Devices/Night Observation Devices clean to ensure their free movement and to avoid damage through friction.

e. ***Exposure to Day/Intense Light***. Night Vision Devices working with image intensifier tubes should never be exposed to natural or artificial intense light e.g. direct sunlight, higher voltage bulb lights and the head lights of vehicles. These Night Vision Devices should be switched ON only during darkness.

f. ***Operation of Night Vision Devices/Night Observation Devices***

(1) ***Dusty or Sandy Area***. Operation in dusty or sandy areas can pit and scratch lenses unless precautions given below are observed:-

(a) Avoid pointing Night Vision Devices/Night Observation Devices into fast wind unless necessary for operation.

(b) Keep the carrying case closed except when removing or replacing the contents.

(c) Ensure that all the dust and sand are removed from the Night Vision Devices/Night Observation Devices (less lenses) and the carrying case, after operation with the help of a camel hair brush and a damp, soft and clean cloth.

(2) ***Rainy or Humid Areas***. Operation in rainy or humid conditions can cause corrosion and deterioration of Night Vision Devices/ Night Observation Devices unless the precautions given below are observed:-

(a) Install the dimmest shields where available.

(b) Keep the carrying case and storage case closed unless removing or replacing the contents.

(c) Dry all parts except lenses with soft clean cloth after exposure to rain or high humidity.

(d) Do not store Night Vision Devices/Night Observation Devices in wet carrying case or wet storage space.

29. ***Storage of Night Vision Devices/Night Observation Devices***

a. Storage of Night Vision Devices/Night Observation Devices for prolonged periods without using them, especially in humid and high temperature areas, enhances their failure rate. Therefore, all Night Vision Devices/Night Observation Devices must be switched ON as following:-

(1) For 20 minutes after 2 weeks, during humid months.

(2) For 20 minutes after every 2-3 months, during dry season.

b. Night Vision Devices/Night Observation Devices should be stored in dry and well ventilated rooms, with measures taken to maintain storage temperature as low as possible, without the use of desert cooler.

c. Sufficient quantity of active Silica Gel must always be used with Night Vision Devices/Night Observation Devices and it should be checked periodically for its effectiveness.

**Marching Speed of Infantry**

30. Marching speed of infantry cannot be quantified as a constant factor. There are a number of variables like combat load, terrain, weather conditions and

distances to be covered. Keeping in view all these aspects, following can be considered as guidelines for marching speeds in different types of terrain:-

| Serial No. | Terrain | Day | Night | Remarks |
|---|---|---|---|---|
| a. | Plains | 4 kmph | 3 kmph | – |
| b. | Desert | 3 kmph | 2.5 kmph | Firm Sandy Surface, fair weather. |
| | | 2.5 kmph | 2 kmph | Loose Sandy Surface, fair weather. |
| c. | Mountains | 3 kmph<br>2 kmph | 2 kmph<br>1 kmph | Descent (upto 10,000 feet).<br>Ascent (upto 10,000 feet). |

31. ***Distances Covered with/without Combat Load in 24 Hours***. The distances covered would vary according to the aspects highlighted above. Keeping in view the average human endurance, nature of employment after the march and the rest time required i.e. approximately 12 hours march and 12 hours rest, the distances likely to be covered are:-

a. ***Distances Covered in 24 Hours with Combat Load***

| Serial No. | Terrain | Distance | Remarks |
|---|---|---|---|
| (1) | Plain | 30-35 kilometres | 15-20 minutes rest after a march of 2 hours. |
| | | | 1 hour rest after a march of 4 hours. 2 hours rest for meals. |
| (2) | Desert | 15-20 kilometres | Depending upon the ground friction. |
| (3) | Mountain | 10-15 kilometres | Depending upon the degree of ascent/descent. |

b. ***Distances Covered in 24 Hours without Combat Load***

(1) Plains – 40-45 kilometres.

(2) Desert – 25-30 kilometres.

(3) Mountains – 20-25 kilometres.

# CHAPTER II – COMMAND AND CONTROL

## SECTION 2 – INFORMATION

### General

1. Successful exercise of command and control in a battalion depends on:-

   a. Good communications.

   b. Efficient exercise of command by the battalion commander.

   c. An efficient, well- organized and well laid out battalion headquarters.

   d. Good and timely dissemination of information.

   e. Good battle procedures, especially deployment drills.

### Sources of Information

2. The chief sources of information are:-

   a. Personal reconnaissance.

   b. The intelligence section.

   c. Patrols, troops in the forward positions and on the flanks and the supporting arms observation posts.

   d. Air photographs and maps.

   e. Prisoners.

   f. Local inhabitants.

   g. Ground surveillance radars.

   h. Higher formations' intelligence summaries, including tactical reconnaissance reports.

   i. Pakistan Air Force and Army Aviation (in-flight reports etcetera).

### Battalion Intelligence Section

3. The intelligence section is a small subunit under the battalion intelligence officer whose duties are:-

   a. Limited collection, collation and dissemination of all information.

   b. Maintenance of up to date battle maps.

   c. Passing all information quickly to brigade headquarters, subunits and flanking units. Submission of situation reports, mortar reports, bomb reports, as and when required.

d. Preliminary interrogation of prisoners and refugees, when situation permits.

e. Up keep of order of battle of the enemy.

f. Preparation of up to date going map.

g. Navigation for the battalion, both by day and night

h. Study and interpretation of air photographs.

4. To carry-out these tasks, members of the intelligence section must be highly trained in intelligence duties, map-reading, making clear and concise reports, manning observation posts and in the voice procedure.

**Use of Global Positioning System (GPS)**

5. The use of Global Positioning System (GPS) is not recommended. It must be remembered that Global Positioning System works on supply of data from satellites. Presently, Pakistan does not own navigational satellites. Like all other supplies, in view of our experience, this is also bound to restrictions. Global Positioning System is being issued to infantry units as training aid only. It should be used during training for attaining perfection in firing and navigation. During peace support operations and when operating with allied armies, Global Positioning System may be used due to obvious surety of data (information) supply.

**Two Way Flow of Information**

6. It is not only essential for the forward troops, including artillery observers, to send all the information back to the battalion headquarters, but also for the battalion headquarters, to send all the useful information to the subunits. Therefore within a battalion a continuous two way flow of information should exist. This will entirely be dependent on good communications.

**Ground Surveillance Radar**

7. The present early warning system at infantry unit/subunit level is usually based on sentries, patrol, listening posts, trip flares and observation posts in the forward areas using binoculars, Night Vision Devices and Night Observation Devices etcetera for observation. However, it has following inadequacies:-

a. Limited observation.

b. It is affected by terrain and weather conditions.

c. Lack of continuity and reliability.

d. Vulnerable to detection by the enemy and counter action.

e. Burden on manpower and greater physical and mental fatigue.

8. Infantry units need to enhance their surveillance capabilities in order to ensure timely response in terms of decision making and timely reaction to prevent enemy achieving surprise. Ground surveillance radar will enhance and strengthen the existing early warning system. It is usually a Doppler radar that can be operated from ground mounted tripod or from a vehicle. A trained operator can identify the targets, from their signals, received by the radar Ground surveillance radars, once intermixed with Platoon Early Warning System (PEWS) will have following advantages:-

a. All weather continuous watch in day and night.

b. Manpower is not required to be employed ahead of defences to enhance the surveillance range.

c. Difficult for enemy to observe and take countermeasures.

d. Greater reliability.

e. Ease of deployment and lesser fatigue.

9. ***Limitations of Radar***. The radar has following limitations:-

a. Being an active system it can be detected, located, jammed or fed deceptive information. Jamming usually takes lesser enemy effort than locating but effectiveness of both can be significantly reduced by properly siting the radars.

b. Although elevated remote aerials can be used, careful siting is however essential as it requires clear line of sight.

c. Each type of radar has a limit of discrimination, below which, two targets closer together, will appear as one.

d. Heavy rain and snow in the target area may reduce efficiency of the radar.

e. A number of radars operating in closer proximity of each other can cause mutual interference.

10. ***Employment of Ground Surveillance Radar in Different Operations of War***

a. ***Defensive Operations***

(1) To cover gaps, flanks or boundaries forward of defensive positions and to determine possible direction of enemy attacks.

(2) To monitor move of enemy troops, soft vehicles and armoured vehicles.

(3) Coordinated employment with observation posts, standing patrols and listening posts to extend range of surveillance and provide early warning.

(4) To augment security system in rear areas.

(5) To improve units' surveillance beyond the ranges of its Night Vision Devices during periods of limited visibility.

(6) To augment security of the headquarters and administrative installations.

(7) To identify the returning patrols in defence.

(8) To cover the artificial/natural obstacles, ahead of defensive position when deployment of standing patrols/listening posts is otherwise difficult.

b. ***Offensive Operations***

(1) Protection of deployment areas (concentration area, assembly area, etcetera).

(2) Provide security against enemy actions by surveillance of the flanks of an attacking unit.

(3) If possible, infiltration forces to carry along the equipment to provide security while in hideouts and deployment areas.

(4) During reorganization, for early warning against enemy counter-attack.

(5) Protection of harbour areas during advance and follow in zone operations.

c. ***Peacetime Use and Internal Security Operations***

(1) Provide early warning against intrusion by insurgents, saboteurs and guerrillas.

(2) Enhance surveillance of sensitive targets and installations.

(3) Protect assembly of troops against guerrilla action.

## SECTION 3 – BATTALION HEADQUARTERS IN THE FIELD

**General**

1. No battalion can be effectively controlled without an efficient and smooth working headquarters, which is moved only when necessary. Before an attack command elements should be well forward. In defence, battalion headquarters

should be placed centrally within the battalion locality, preferably behind the depth localities.

**Organization**

2. A battalion headquarters should be clearly sign posted by day and night and organized as follows:-

a. ***Regimental Police Post***. This should be located at the entrance.

b. ***Vehicle and Motorcycle Parking Area***. This will normally be sited under cover near the entrance. When a headquarters is dug in the open, vehicles (except essential vehicles dug in with the headquarters) should remain under suitable cover. In defence, the bulk of transport will normally be located within the battalion area except B echelon vehicles.

c. ***Command Post Area***. This will normally include the command post, signal office and both forward control and rear link wireless sets. The commander of the affiliated field battery and possibly officer commanding combat support company or some of his platoon commanders will also be here. This area will be laid out so that visitors come first to the signal office then they meet Adjutant, so that the commanding officer remains free from distraction.

d. ***Administrative Area***. This might include rest areas for officers and soldiers, cooking and sanitary areas. It may also include the regimental aid post.

**Forward Command Post**

3. Although efficient control of the battalion and its supporting arms can best be exercised from battalion headquarters, which is the hub of the communication system. The battalion commander must often go forward to see for himself what is going on. He may go to a previously prepared observation post or to a company headquarters. When away from his own headquarters, he will require his rover set. He will usually take with him his intelligence officer along with other supporting arms commanders. All these personnel together will then constitute the forward command post, the tactical headquarters of the battalion.

**Local Protection**

4. Arrangements for local protection must include:-

a. Sentries by day and night.

b. Alarm posts which must be carefully sited and prepared.

c. Practice alarms should be held immediately after arrival and regularly thereafter.

d. Action to be taken in the event of air attack.

5. All ranks at the headquarters must be ready to man their alarm posts at short notice, by day or night, in an emergency. In addition it may, on certain occasions, be necessary to detail personnel from outside the battalion headquarters to provide immediate protection to it.

**Siting and Layout**

6. The detailed siting (within the area specified by the battalion commander) and layout of battalion headquarters is the responsibility of the Reconnaissance and Layout Group. Following must be kept in view while siting the battalion headquarters:-

a. The headquarters should be concealed from air observation. It must be dug in and camouflaged.

b. The site chosen should be suitable for local protection. Since there are 55 all ranks in the battalion headquarters, it must be planned as a defended locality and sited accordingly. Manpower and weapons of headquarters must be put to full use.

c. Buildings, unless particularly prominent, are often more suitable than dug in positions in the open as it is possible to operate in them more efficiently in bad weather and lights can be more easily concealed, at night. In such cases, windows should be sandbagged.

d. The site must allow for good communications (including wireless both within the battalion and with brigade headquarters).

e. Track discipline must be enforced immediately on arrival.

f. If possible, the site should be close to the brigade axis from which it must be sign posted. Mechanical transport should be able to approach reasonably near it.

g. Approaches should, preferably be invisible to enemy from air.

h. An alternative position should be planned, chosen, reconnoitred and where possible, dug for quick occupation.

7. The layout of battalion headquarters as a guideline is given at Figure 2 as explained below:-

| Serial No. | Area Earmarked for | Reference Number |
|---|---|---|
| a. | Officers | 1 |
| b. | Runners | 2 |
| c. | Mess staff | 3 |
| d. | Drivers | 4 |
| e. | Radio/telephone operators | 5 |
| f. | Clerks | 6 |
| g. | Intelligence section | 7 |
| h. | Miscellaneous persons of battalion headquarters/ headquarters company | 8 |
| i. | Persons of battalion headquarters/headquarters company (when additional troops not allotted) | 9 |
| j. | Signal platoon (when additional troops not allotted) | 10 |
| k. | Signal platoon | 11 |
| l. | Regimental Aid Post/Regimental Police | 12 |
| m. | Protection section | 13 |
| n. | Pioneer platoon | 14 |
| o. | ***Spare***. For additional elements placed under command battalion e.g., engineers etcetera. | 15 |

**Note**: The jeep/dodge etcetera of signal platoon, pioneer platoon, regimental aid post commanding officer. Battery commander, adjutant and other supporting arms commanders will be within their respective areas well camouflaged/dispersed.

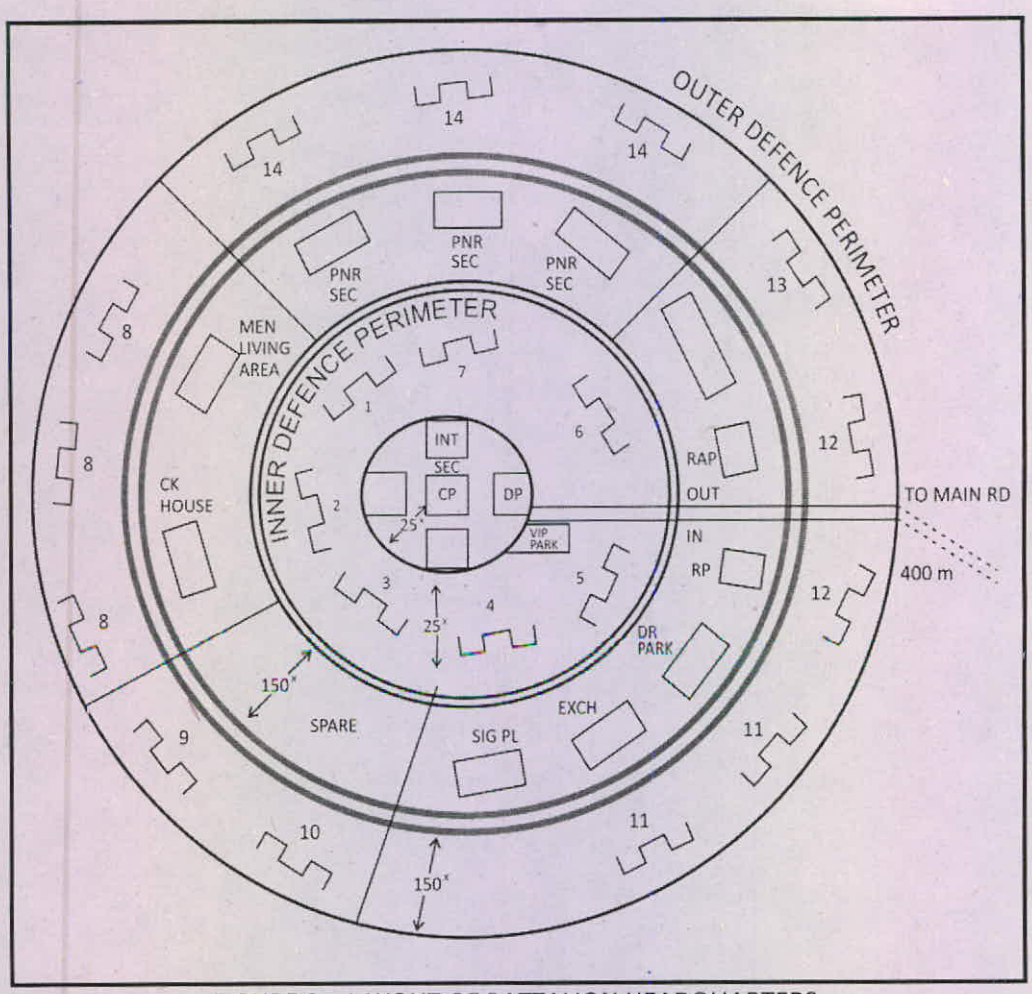


FIGURE 2 – LAYOUT OF BATTALION HEADQUARTERS.

**Routine**

8. A general pattern of routine at the battalion headquarters is as follows:-

   a. Changes in the position of battalion headquarters must be immediately notified to brigade headquarters and subunits.

   b. Reports and returns will be submitted as per Standing Orders for War, unless otherwise specified by the brigade headquarters.

   c. Battalion headquarters is manned at all times by an officer. Generally, adjutant mans the command post but is very often relieved by the signal officer, intelligence officer or any other officer of the battalion headquarters, if available.

   d. Rest must be organized for all officers and men.

e. Visitors should first report to the Adjutant, who should ensure that they have been given the latest information before they see the Commanding Officer.

f. All incoming orderlies must report at the signal centre with the messages. Motorcycles must not be brought beyond the motorcycle park.

g. Before visiting another headquarters all officers should call at their respective signal centres for any messages that they could take along. Likewise, all visitors should also, before leaving, call at the signal centre for collecting the messages.

## SECTION 4 – BATTLE PROCEDURES AND BATTLE DRILLS

**General**

1. Battle procedures and battle drills at battalion and company level are evolved for telescoping timings and stream lining actions required to be taken by various commanders and units/subunits during battle. The procedures aim at quick and timely translation of commanders' orders for successful, steady and continuous actions in battle.

2. In order to launch a successful operation, following actions are necessary:-

a. Reconnaissance.

b. Planning.

c. Issue of orders.

d. Deployment of troops.

e. Organization of fire support.

f. Administrative support.

3. All the above actions require considerable time. But, the time being invariably short and always vital, it becomes imperative that the procedures and drills are evolved, to carry-out simultaneous actions, at various levels. This will reduce the time taken for preparation, to the barest minimum.

4. The principle governing these drills, therefore, is that while reconnaissance is being carried out and the orders issued by various commanders, simultaneous and continuous actions are being taken by all the participating troops, for deployment and organization of fire and administrative support.

5. The aim of battle procedure and battle drill can only be achieved when every soldier knows the part he has to play. It is only through constant practice that the units will be able to attain a high standard in affecting the required deployment with:-

a. The least delay.

b. The minimum orders.

c. The optimum efficiency.

**The Group System**

6. There are five main groups connected with deployment at battalion level:-

a. ***Reconnaissance or 'R' Group***. This is the commander's group for reconnaissance and planning.

b. ***Orders or 'O' Group***. This is the group that receives orders. This group will normally consist of subordinate commanders' 'R' Group.

c. ***Main Body***. Fighting troops including the F echelon transport which comprising the essential vehicles required to accompany the fighting troops in action.

d. ***'A' Echelon***. The personnel and vehicles which must be readily available to support the main body in action.

e. ***'B' Echelon***. Consisting of personnel and transport not required in 'F' and 'A' echelons.

7. This nomenclature saves time when issuing orders. As a guideline, a suggested composition of groups is given at Annex E. The 'R' group of a unit or formation also forms part of the 'O' group of the next higher formation e.g. unit 'R' group attends the brigade 'O' group. Care must be taken to use correct nomenclature. For instance, brigade headquarters will call for the brigade 'O' group and not battalion 'R' group, etcetera.

**Reconnaissance and Layout Group**

8. When a unit is required to move from a concentration area to an assembly area or in case of other similar moves, a group known as the 'Reconnaissance and Layout Group' is added. Its function is to precede the main body, to layout an assembly area or harbour area and to see that the unit moves into these areas in the right order without confusion or delay. In advance to contact the main body will move in bounds from harbour to harbour and the drill described above applies here

with particular importance. The group in such cases is called 'harbour party'. This group might be composed as follows:-

a. Officer Commanding Combat Support Company (Officer-in-Charge Party).

b. Signal officer (or his representative) with a dispatch rider.

c. Battalion Havildar Major.

d. Two Regimental Policemen.

e. A working party and escort.

f. Representatives from rifle companies and specialist platoons.

**Basic Procedures**

9. The basic sequence of battle procedure to be followed depends on the nature of operation and the availability of time. In a deliberate attack, or a planned occupation of defence, when there is ample time available, it is desirable to follow detailed sequences, so that all minor details are worked out and coordinated before the execution. Keeping these factors in view, there can be two distinct sets of battle procedures that can be followed.

10. ***Quick Battle Procedure***. If the time available is less, following sequence will be generally applied in all operations of war:-

a. Issue of warning order.

b. Quick reconnaissance.

c. Outline plan.

d. Coordination and final orders.

e. Final coordination on ground.

11. ***Normal Battle Procedure***. In case enough time is available, following sequence can be applied:-

a. Quick reconnaissance.

b. Warning order including outline plan.

c. Detailed reconnaissance.

d. Detailed verbal orders.

e. Coordination and final orders on ground.

12. It should be noted that due to scarcity of time the detailed reconnaissance and verbal orders are not included in quick battle procedure, the outline plan issued in this case must furnish sufficient details to enable subunit commanders, to formulate their plans. Final orders will be issued after the coordination and would generally consist of changes/readjustments in the outline plan.

13. ***Warning Order***. The warning order may be issued verbally, by message or by wireless. Its contents will normally include such items as:-

a. Brief statement of the situation.

b. A brief statement of the task.

c. Rendezvous and time for the 'O' group.

d. Time before which there will be no move of the main body.

e. Location of assembly area (if known), instructions for move of main body, 'A' and 'B' echelons.

f. Limitations on reconnaissance.

g. Any other instructions or restriction e.g. administration etcetera.

14. Items at sub paragraphs 13 b, c and d above are considered essential and the remainder desirable. The issue of a warning order should never be held up because the remainder is not complete. On occasions, the warning order may also include the outline plan (see paragraph 20 below).

15. ***Quick Reconnaissance***. The methods by which a commander can carry-out quick reconnaissance are as follows:-

a. Off the map.

b. On ground.

c. From the air, if facilities are made available.

16. Depending upon the availability of time, commander may resort to one, two or all of the above modes of carrying out reconnaissance. Whichever method is adopted, the essence is that the reconnaissance should be quick and the commander should not spend more time on it than is required to formulate his plan. He must not, however, hasten to conceive an outline plan, as it may have to be radically changed, as a result of more detailed reconnaissance.

17. Every unit must have its own reconnaissance drills, which must be known to all. The reconnaissance drills should embrace personnel, transport, communication, sign posting arrangements, order of march of reconnaissance party, officer-in-charge party, control and guidance of 'O' group, preparation of sketches for diagrammatic orders, etcetera. The local protection of reconnaissance parties must be an integral part of the reconnaissance drills.

18. In an encounter battle, elaborate arrangements for reconnaissance cannot be made. Therefore commanders must initially conduct a detailed reconnaissance off the map. They should move well forward in the order of march to personally get firsthand information about the enemy as well as terrain during advance. They should also get useful information from troops/commanders in direct contact with the enemy.

19. For deliberate attack reconnaissance must be carefully organized, otherwise surprise may be lost and the plan compromised. Before a coordinated attack, the commander will usually lay down forward extent of reconnaissance in his warning order. Arrangements must also be made for:-

a. Guiding commanders' vehicles to RV, their concealment at the RV and guiding the commanders from rendezvous to the observation point.

b. Improving concealment, camouflage, etcetera on the way upto and at the observation point.

c. Strict control of all movement near the observation point.

d. Local protection of the observation point. Trenches should be prepared for quick shelter in case of enemy's shelling.

20. ***Outline Plan***. This should include only such information as is required to enable subordinate and supporting commanders to carry-out their quick reconnaissance and formulate their outline plans. It can be given out:-

a. To 'O' group at the battalion headquarters, company headquarters or any other place, providing good observation.

b. By message on wireless, telephone or by dispatch rider.

c. By the battalion second-in-command, adjutant or intelligence officer.

d. When convenient the outline plan should be included in the warning order.

21. ***Detailed Reconnaissance***. At this stage battalion commander carries out detailed reconnaissance of the ground and prepares his plan.

22. Based on his detailed reconnaissance of the ground, commander gives out his orders to the 'O' group, to be confirmed later, after coordination with lower and higher commanders. Alternatively, he may coordinate while doing detailed reconnaissance and issue final orders.

23. ***Coordination***. Coordination is carried out after the lower commanders have received orders and carried out their reconnaissance, the battalion commander meets his subordinate commanders on the ground primarily to confirm his plan and affect adjustments on the basis of subordinate commanders' recommendations if required. The coordination can take following forms:-

a. Coordinating conference with lower commander followed by a coordinating conference with the higher commander.

b. Coordination with the lower commanders carried out on the ground, the commander visiting his lower commanders in turn, followed by a coordinating conference with the higher commander.

c. Coordination with the lower commanders on the ground followed by coordination with the higher commander also on the ground. Applicable when the higher commander is visiting the lower commander.

24. Before a commander coordinates with his higher commander, he must have coordinated with his lower commanders those aspects which do not affect the higher commanders' plan. Those aspects which are likely to affect the higher commanders plan e.g., any change in allocation of fire support or defence stores made in the outline plan, deployment areas, timings, guidance, security, objectives and allotment of troops are coordinated with higher commander and final orders are issued accordingly.

25. ***Final Orders***. Decisions made in coordination are followed by formal orders verbally/in writing; however, at times confirmatory notes are also issued if required. The method adopted will depend on the level and on the urgency. Possible examples are:-

| | | | |
|---|---|---|---|
| a. | At divisional/brigade level | – | A coordinating conference followed by a written operational order. |
| b. | At battalion level | – | Coordination on ground and final orders. |

26. The specialist platoons will attend battalion 'O' group and receive orders from battalion headquarters/officer commanding combat support company. The sequence of rest of the battle procedures would remain the same as explained under the platoon column. These platoons will follow the deployment drills of each weapon, which are explained in the respective weapons' pamphlets. Battle procedures and battle drills are given in the following Annexes:-

| | | | |
|---|---|---|---|
| a. | Battle Procedure – Advance | – | Annex F |
| b. | Battle Procedure – Attack | – | Annex G |
| c. | Battle Procedure – Defence | – | Annex H |
| d. | Battle Procedure – Withdrawal | – | Annex I |
| e. | Battle Procedure – Patrols | – | Annex J |
| f. | Battle Drills – Advance | – | Annex K |
| g. | Battle Drills – Attack | – | Annex L |
| h. | Battle Drills – Defence | – | Annex M |
| i. | Battle Drills – Withdrawal | – | Annex N |
| j. | Battle Drills – Patrols | – | Annex O |

# CHAPTER III – ADMINISTRATION IN THE FIELD

## SECTION 5 – WATER SUPPLY

**Responsibility**

1. ***Formation Staff***
   a. Authorizing scale of issue.
   b. Deciding the general policy of distribution on engineers' advice.
   c. Arranging for traffic control and discipline at water points etcetera.
   d. Deciding what provision, if any, shall be made for water for fire-fighting.
   e. Transportation of water byroad, rail etcetera when circumstances require it.
   f. Notifying units of:-
      (1) Location of water points.
      (2) Routes.
      (3) Timing for drawing.
      (4) Closure of water points for any reason.
      (5) Special precautions to be carried out for testing in certain circumstances.
2. ***Engineers***
   a. Reconnaissance and development of source of supply.
   b. Collection, treatment and storage of water.
   c. Provision, erection, operation and maintenance of water supply equipment other than unit equipment of other arms.
   d. Delivery of water at selected water points as required.
3. ***Medical***
   a. Advising as to its suitability.
   b. Testing and continuing to test any source of water to be used by troops for drinking or washing.
   c. Recommending the method of treatment to be adopted and any steps considered necessary to prevent pollution at source.
   d. Recommendation of scales of issue.
   e. Training of water duty men.

4. ***Infantry Battalion***. Units can also be made responsible for operation of their own units' water supply equipment and for duties in connection with water, whenever necessary. This may include special precautions against the contamination of such water by radioactivity, chemical and biological agents. Where contamination by hostile agents is suspected, all water must be tested and monitored shortly before use.

**Requirements**

5. ***Quality***

a. ***Source of Supply***. On active service water from any untested source must be considered unfit for drinking/cooking and must be treated accordingly. Water to be used solely for washing may also require treatment, especially in tropical climates.

b. ***Drinking Water***. The requirements for drinking water are:-

(1) No disease producing organisms.

(2) No radioactivity beyond the limits laid down and no poisons or bone oil.

(3) No colour, taste or smell.

(4) No appreciable amount of dissolved chlorides (e.g. salt) sulphates or bicarbonates (e.g. hardness).

(5) No appreciable amount of suspended matter, i.e. cloudiness.

6. ***Quantity***. The quantity of water required by men and animals varies considerably according to climate. In hot climate men and animals need more water than they do in cold or temperate climates and figures given should be increased by upto 50 percent. Men can be trained to reduce the quantity of water needed for drinking. Following are the water requirements of a military force:-

a. Each man requires daily:-

| | | | |
|---|---|---|---|
| (1) | In barracks, in the plains of Pakistan | – | 36 litres. |
| (2) | In barracks, in the hills of Pakistan | – | 22.5 litres. |
| (3) | In standing camps, when clothing is washed | – | 22.5 litres. |
| (4) | In camps when clothing is not washed | – | 13.5 litres. |
| (5) | In bivouacs, for drinking and cooking only in cold weather | – | 4.5 litres. |
| (6) | For drinking only as a minimum in cold weather | – | 2.24 litres. |
| (7) | For garrison hospitals, per bed strength includes all purposes | – | 135-225 litres. |

7. The quantity laid down has to be increased in hot weather and when troops are doing hard work. Under certain conditions of high temperature and hard muscular work a man may drink as much as 16.2 litres of water a day. Such consumption of water needs an addition of salt to replace salt loss. In addition:-

a. Each Animal requires daily:-

| | | | |
|---|---|---|---|
| (1) | Horse (7.2 litres at a time) | – | 45 litres. |
| (2) | Camel | – | 45 litres. |
| (3) | Mule (7.2 litres at a time) | – | 225 litres. |

b. Mechanical Transport requires daily:-

| | | | |
|---|---|---|---|
| (1) | For washing down at field parks | – | 225 litres. |
| (2) | For radiator, filtered water average | – | 18 litres. |

## SECTION 6 – REINFORCEMENTS

1. This section should be read in conjunction with "Reinforcement Procedure – War, Pakistan Army 1969 issued by General Headquarters".

**Calculation of Reinforcement**

2. For infantry battalion, the war wastage rate, per month, would be as under:-

a. ***Intense***

| | | | |
|---|---|---|---|
| (1) | Officers/Junior Commissioned Officers | – | 20% |
| (2) | Soldiers | – | 20% |

b. ***Normal***

| | | | |
|---|---|---|---|
| (1) | Officers/Junior Commissioned Officers | – | 10% |
| (2) | Soldiers | – | 8% |

**Holding**

3. The reinforcement will be held as under:-

a. Field Formations/Units – 7 days.

(1) 7 days reinforcement requirements are known as first line reinforcement.

(2) Casualties will be replaced from first line reinforcement.

b. Reinforcement camps – 15 days.

c. Centres – 8 days.

**Reinforcement Demands**

4. Casualties for which reinforcements are demanded are:-

   a. Deaths.

   b. Sick and wounded, evacuated beyond Main Dressing Station/ boundaries of formations (division/independent brigades).

   c. Captured by enemy or believed prisoner of war.

   d. Missing due to enemy action.

   e. Posted to extra regimental employment.

   f. Undergoing detention in military prison/civil jail.

   g. Overstay of leave for more than 15 days.

5. Following must be kept in view while forwarding the reinforcement demand:-

   a. A reinforcement demand once placed is not repeated in the next demand, otherwise it results in duplication.

   b. There should normally be one demand per day.

   c. Reinforcement already despatched but not taken into account by units; should be kept in view while submitting consolidated reinforcement demand.

   d. Each demand must be numbered serially.

6. ***Action by the Units***. Infantry battalion submits its demands for reinforcement direct to their formation headquarters daily at a time laid down in the formation standing orders for war (but not later than 2000 hours) in accordance with requirements as at the time. Specimen is given below:-

**SPECIMEN SIGNAL MESSAGE**

(Reinforcement Demand from Units to Formations)

**IMMEDIATE** DTG_____

FROM: _____ (UNITS) **CLASSIFICATION**

TO: _____ (FORMATION HQ) NUMBER

INFO: PARENT FORMATION (IF APPLICABLE)

---

REINFORCEMENT DEMAND NO 3 (.) OFFICERS (.) MAJS TWO CMM CAPTS THREE TOTAL FIVE (.) JCOS (.) SUB ONE CMM N/SUB CLERK ONE TOTAL TWO (.) OR (.) SEPOYS GD TWENTY CMM COOK MESS ONE TOTAL TWENTY ONE (.) ACK

---

RANK: _____

CASE NO _____ DATED _____ TEL MIL: _____

**Reception of Reinforcements by Units**

7. Reinforcements will usually make their first contact with an infantry battalion at B echelon where their reception can do much to give them a good start. The quartermaster may be made responsible for this job. Points that may be included in the reception, when operations permit, are as follow:-

a. Checking of clothing and equipment to scale.

b. Giving an opportunity to write to their next of kin giving their new address.

c. Explaining the present operational position of the unit and the task assigned.

d. Organizing some rest.

8. It is desirable that the reinforcements settle before going into action and they do not arrive in the heat of the battle. On arrival in the units, where possible, they should be incorporated into companies and platoons, while the battalion is having a quiet spell.

## CHAPTER IV – ORGANIZATION AND CAPABILITIES OF WEAPON SECTION

### SECTION 7 – ORGANIZATION AND CHARACTERISTICS

**Organization**

1. Weapon section consists of seventeen soldiers. It has two sections, one of Recoilless Rifles with two 106 millimetres Recoilless Rifles, the other of 60 millimetre Mortars with two 60 millimetres Mortars and a Grenade Launcher Heavy-Machine gun (GLHMG) detachment which holds one 40 millimetres GLHMG. Both the sections and GLHMG detachment are commanded by a havildar each. One weapon section is organic to each rifle company.

**Capabilities**

2. The capabilities are as follows:-

a. Speedy deployment due to availability of vehicles, the weapons can also be transported on man pack basis.

b. Fire support can be coordinated and controlled by the battalion commander.

c. Concentration of fire support.

d. Provide direct and indirect fire support.

e. Sections can be detached if required.

f. Centralized employment in the battalion area, when needed.

3. The transport allotment in the weapon section is as under:-

| | | | |
|---|---|---|---|
| a. | Truck ¼ ton for Recoilless Rifles | – | 2 |
| b. | Truck ¼ ton for Recoilless Rifles ammunition | – | 1 |
| c. | Trailer Cargo ¼ ton for Recoilless Rifles | – | 1 |
| d. | Truck ¼ ton for 60 millimetres Mortars ammunition | – | 1 |
| e. | Trailer Cargo ¼ ton for 60 millimetres Mortars | – | 1 |
| f. | Truck ¼ ton for Grenade Launcher Heavy-Machine gun | – | 1 |

**Characteristics**

4. ***Recoilless Rifle (RR) Section***. It has two Recoilless Rifles and is completely mobile. Its primary task is to destroy enemy armour trying to penetrate into or overrun a company locality. Its characteristics are:-

a. ***Mobility***. It is light weight and can be transported on man pack basis. It is highly mobile when mounted on jeep and can easily be moved from one place to another, even in the thick of the battle, either to engage the target or for self-protection, or when located by the enemy.

b. ***Range and Penetration***. It range is 2000 metres in direct role and 7600 metres in indirect role. Long-range prevents it from being spotted and easily neutralized by tanks. The shell can penetrate 41 to 51 centimetres of homogeneous armour, at extreme range.

c. ***Recoilless Operation***. Being recoilless, the rifle causes no gun shyness to a new anti-tank gunner and little time is required to lay the weapon on the target and to fire the second round. There is not the slightest movement to the jeep, when the rifle is fired from it.

d. ***Accuracy***. Spotting rifle is designed to help in the determination of range and lead. Ranging, etcetera, is done by the spotter tracer ammunition. Once the spotter tracer round hits the target, the 106 millimetres Recoilless Rifle can be fired within a split second interval and the first round hit is obtained. The best range with the spotting rifle is 1189 metres, beyond which it is not a sure guide.

e. ***Flexibility***. Due to its tremendous fire effect, it can be usefully and most effectively employed in all operations of war, not only in anti-tank role, but also in anti-personnel role. It can also be employed in special roles like destroying strong points, pillboxes, concrete emplacements and other pinpoint targets. It fires anti-tank, high explosive and white phosphorus types of ammunition. The rifle has 360 degrees traverse.

5. ***Limitations of the Recoilless Rifle (RR)***. Rifle has following limitations, which must always be kept in view, when operating the weapon:-

a. ***Back Blast***. It causes a very severe and dangerous back blast, with a danger zone of 100 feet to the rear of the breech and 175 feet on either side of it. For further 200 feet troops must not face this weapon. Approximately 100 feet of clearance to the rear is, therefore, generally required, which must be remembered in the siting of this weapon. The dust caused by the back blast obscures observation from the weapon position for a few seconds. It is very difficult to conceal the back blast from the enemy. As a precautionary measure dusty areas should be avoided and ground selected should be hard, rocky or damp. Areas with some vegetation are suitable and help in absorbing the back blast.

b. ***Vulnerability***. The weapon as well as its crew is vulnerable to all types of fire especially small arms fire as there is no protection for the crew. The rifle is more vulnerable when mounted on the jeep, particularly during its move. It is necessary that rifle occupies a carefully selected defiladed and concealed position, so that it is not neutralized from the front.

## SECTION 8 – PRINCIPLES OF SITING – RECOILLESS RIFLE

**Tank Tactics**

1. Knowledge of certain fundamental points common to all armour tactics is essential to correctly apply the principles of siting the anti-tank rifles. Following points are considered essential for all, who will be responsible for deploying the anti-tank rifles:-

a. To develop their full power and achieve decision, tanks are used in mass, in areas which give scope for manoeuvre; although smaller number of tanks, working in cooperation with other arms, can be used effectively in close country and BUAs. Anti-tank measures, however, assume greater importance in operations in open country.

b. Tanks may attack by themselves, but usually have infantry with them. They become very vulnerable when separated from their supporting infantry. The separation of tanks from their supporting infantry should, therefore, always form part of anti-tank plan.

c. Because the turret and frontal armour of a tank is generally thicker than the armour on the side of the hull, the tanks closing with the defences will try to attack head on, with their flanks covered by fire

from artillery, infantry or other tanks. An enfilade shot at this time from a defiladed position is most damaging to the attacking tank.

d. With the use of 'Stabilizers', tanks can fire with reasonable accuracy even on the move, however, it is the hull down tank supporting the other tanks on the move, which will be the chief menace to an anti-tank rifle. Neutralization of the hull down tanks by anti-tank rifles, artillery fire and smoke is, therefore, very important.

e. Mobility of a tank depends on limitations imposed by terrain. The careful appreciation of ground is, therefore, of utmost importance and must be given due consideration when planning anti-tank defence of an area.

f. All tanks lose speed when turning especially on soft ground and when climbing. When moving across a steep slope, tanks tend to overturn. This entails constant use of the steering and further slows down the tanks. Tanks, therefore, usually attack straight up a steep hill.

**Principles of Siting**

2. a. ***All-Round Defence and Depth***. Although all-round defence is essential, the rifles will, primarily be sited to cover the most likely tank approaches. They must be sited in depth to deal with any enemy tanks, which succeed in penetrating the position.

b. ***Defilade***. Every possible effort should be made to site the rifles in defiladed position. This will ensure that flash and the dust raised by back blast are concealed from enemy observation posts or supporting tanks and that the rifle is able to engage enemy tanks from the side, where they are more vulnerable.

c. ***Mutual Support***. The rifle should be sited to cover each other, as far as possible, so that should a rifle be attacked by tanks the other rifle(s) will be able to engage them. Dead ground in the area of fire of one rifle must be covered by another. Thus a hull down tank not visible to one rifle could be engaged by the others. Due to the mobility of these rifles and their long-ranges, the distances between rifles can be considerably increased without losing mutual support. On occasions when, due to large number of tank approaches, it is not possible with the number of available rifles, to achieve this mutual support, rocket launchers must be used to cover the exposed flanks.

d. ***Concealment and Camouflage***. The correct siting of the rifle is of primary importance. However, great attention must be paid to concealment from ground and air observation. For a quick occupation

and when on the move, full use must be made of natural cover and readily available material to break the outline of the rifles and to make it blend with the background. Natural garnish can be bound to the barrel by means of strings etcetera. Rigid track discipline should be maintained in order to avoid detection of the rifle by enemy air. Whenever possible, the rifle should be dug in. This will provide protection to the men and the equipment. Concealment must be good at all times and movement in and around positions should be restricted to the absolute minimum.

e. ***Local Protection***. As compared to other anti-tank weapons, this rifle has got a considerably larger danger zone because of its pronounced flash and back blast. Consequently, it may not always be possible to site the rifle in infantry localities without leaving unacceptable gaps in the defensive system. On the other hand, if left by itself, the rifle will be very vulnerable to enemy infiltrating parties/patrols etcetera. Therefore, it will be necessary to site it during the daytimes as close to infantry localities as safety permits. During hours of darkness, it must be withdrawn within infantry localities. Normally, the defence is built around Recoilless Rifles and machine-guns, which are sited first.

f. ***Field of Fire***. The field of fire should cover flat or slightly sloping ground, while avoiding plunging fire. If obstacles and broken ground are included in the area of fire, tanks will admittedly slow down and thus present an easier target. However, it must be remembered that the broken ground will also offer opportunities to the tanks to take up hull down positions.

g. ***Alternative Positions***. A minimum of one alternative position must be selected for every rifle. When selecting alternative positions, following requirements should be borne in mind:-

(1) The rifle should be able to cover the same area of fire and carry-out the tasks originally allotted to it.

(2) As far as possible, there should be a covered approach between the two positions.

(3) The intervening ground between two positions should permit easy movement. The detachment commander will be at liberty to move to his alternative position on his own initiative, once he has fired and his position has been spotted by the enemy; but he must keep the rifle platoon headquarters/company headquarters fully informed of the move. In case his movement affects his tasks, he will move only on orders from

the platoon/company headquarters. Full digging and preparation of the alternative position must be carried out as soon as the main position is developed.

h. ***Fire-Control***. Fire-control is most essential. A Recoilless Rifle should only open up when it is sure of a kill, otherwise, its position will be given away, which will provide an opportunity to the tanks to destroy it.

## SECTION 9 – TACTICAL HANDLING – RECOILLESS RIFLE

**Appreciation and Plan**

1. The anti-tank appreciation and plan will probably be made, in the first instance, at divisional level. The battalion Commander, however, will decide from which direction the enemy's main tank threat is likely to come and what are likely no of tanks which may approach his locality. This plan for the anti-tank defence of the locality will then have to be fitted into the brigade and divisional anti- tank plan. His eight Recoilless Rifles, together with Baktar Shikan and any additional resources allotted to the battalion from the divisional light anti-tank battalion form the framework; 3.5 inch Rocket Launchers and anti-tank grenades fill in the gaps.

**Coordination**

2. At the brigade level, the anti-tank plan will be largely governed by the need to conform to divisional plan, already laid down in some detail. The brigade commander is responsible for the detailed implementation of the divisional plan and for coordination between units. This coordination, in all probability will be undertaken by the company commander from the divisional light anti-tank battalion, allotted to the brigade or the tank squadron commander. Finally, points requiring the decisions of brigade commander are settled in the brigade coordinating conference.

3. The anti-tank plan of the battalion will be based on the plan of the brigade, to ensure its implementation. Coordination at this level will primarily be done by the battalion commander himself. The final coordination, bearing in mind the other battalions of the brigade and in relation to other things like obstacles and anti-tank mines, will be undertaken by the company commander of the divisional light anti-tank battalion allotted to the brigade or any other officer specially detailed for the purpose.

**Advance**

4. The main purpose for having anti-tank resources integral to the infantry battalion is to ensure that the battalion is not surprised by enemy tanks before anti-tank effort from brigade or divisional resources can be made available.

5. The task of the Recoilless Rifles, during any move is to protect the part of the column with which they are moving. There may be a need for centralized employment of the battalion Recoilless Rifles, the battalion commander concentrating these weapons for employment as the situation demands. The advance guard for a brigade usually consists of a force of all arms built around an infantry battalion. In this case, the protection of the column during the advance to contact is best achieved by allotting some rifles to the vanguard and the remainder are grouped together with the main guard.

6. The Recoilless Rifles with the main guard may:-

a. Move at the head of the column, near the battalion 'O' group, so that they may be quickly deployed. This method will be used when tank approaches on the axis of advance are not clearly defined and cannot be anticipated.

b. Be used for the flank protection of the column by picketing the route by bounds in pairs. Marching troops should be allowed to pass them before they in turn pull out of position and move on to the next bound. This method will be used when there is a tank threat. The tank can inflict casualties on the infantry, initially firing high explosives from long-ranges and then overrunning the infantry, physically, if necessary. The Recoilless Rifles, therefore, should deploy at such a distance from the axis that the tanks are unable to inflict casualties on the infantry by direct aimed fire.

7. In case the battalion is provided transport for move, the Recoilless Rifles will be distributed throughout the column or preferably deployed as in paragraph 'b' above. If the battalion is deployed during the approach march, the Recoilless Rifles should be allotted to the companies and the siting coordinated with tanks if they also form part of the advance guard. In such a case, it will be better to use these tanks for the destruction of enemy tanks, rather than Recoilless Rifles, which have no armour protection. The Recoilless Rifles may form part of the base of fire or be used on security missions.

8. When accompanying un-armoured columns, battalion Recoilless Rifles are permitted to open at long-ranges in order to keep the enemy tanks away from the column. Recoilless Rifles are invariably deployed ready to engage any armour threat, which may develop against the troops in the concentration area or in the harbour.

9. In the event of meeting an opposition during advance which ultimately develops into an encounter battle, the task and role of the Recoilless Rifles would be exactly the same as for attack, once the orders for deployment have been received. It should, however, be noted that owing to the speed required for encounter battle, the orders and the battle procedure at various levels will be cut down to the minimum essential.

**Attack**

10. ***General***

a. Whenever the ground is suitable for armour action, attacking troops are liable to immediate counter-attack by tanks at various stages of the attack. Support by Recoilless Rifles must, therefore, always be provided by deploying rifles forward and to the flanks for this purpose.

b. The objective or position reached may be quite unprotected by any form of anti-tank obstacles. It is, therefore, essential that the Recoilless Rifles should reach the objective soon after its capture to give protection to the troops on the objective.

c. Recoilless Rifles, to be of value, must be handled boldly. Early reconnaissance of the ground over which the anti-tank rifles are to move is essential. Before an attack, besides ground reconnaissance full use should be made of maps and air photographs. During the attack, continuous forward reconnaissance must be carried out by recoilless detachment commanders.

d. The action by Recoilless Rifle detachment commander must be characterized by speed, quick decision and initiative, combined with the skilful use of ground. Recoilless Rifles are most effective when mounted on jeep, permit high speed and flexibility in their employment. The effectiveness of Recoilless Rifles lies in their ability to combine surprise with effective and controlled fire.

11. ***Employment***. The Recoilless Rifles can be employed in attack as follows:-

a. ***Before the Attack***

(1) The protection of assembly area and the release area, where the threat of attack by enemy armour exists.

(2) For the protection of flanks against enemy armour.

(3) To take part in the initial fire plan of the battalion by providing fire support from a base of fire.

b. ***During Attack***

(1) To protect flanks of the attacking troops against the enemy tanks.

(2) To give fire support either on timed program as part of the fire plan, engaging known enemy targets or give fire 'on call'.

(3) Take on opportunity targets e.g. enemy anti-tank weapons, automatics etcetera.

c. ***Reorganization Phase***

(1) As soon as the objective has been captured, the rifles should take part in the reorganization phase. Because of their mobility, firepower and long-range, they will be of great assistance in beating back enemy tank and infantry counter-attacks. They will also be able to move forward of the objective and support rifle companies in exploitation.

(2) Recoilless Rifles will deploy immediately on arrival on the objective, after its capture, to cover the most likely tank approaches. Rifle company commander allocates area and tasks to the Recoilless Rifles bearing in mind the ground, likely approaches for enemy tanks, the final positions of their platoons and pattern and type of existing obstacles.

(3) Immediate deployment of the short-range anti-tank weapons is of equal importance. They should be sited in relation to the positions planned for the Recoilless Rifles. When allotting tasks to the Recoilless Rifles, during the attack, the ammunition requirements for the reorganization phase should be borne in mind. It will be best if rifles fire dumped ammunition prior to reorganization phase.

(4) ***Suitable Targets***. Recoilless Rifles are essentially direct firing weapons, ideal targets for them, therefore, are:-

(a) Tanks, moving or stationary.

(b) Known weapon positions.

(c) Pillboxes, bunkers and concrete replacements.

(d) Troops in the open if high explosive ammunition is available.

(5) The rifle can also be effectively used in a BUAs employment in attack defence provided open space for the back blast is available.

(6) The deployment of the Recoilless Rifles should be pre-planned off the map, considering the tank approaches and the rifles should be immediately deployed on reaching the objective, with any adjustments, which may be necessary.

**Defence**

12. In defence, the Recoilless Rifles should be used as follows:-

a. With reconnaissance elements.

b. With covering troops/advance positions.

c. With screens.

d. In the Main defensive position.

e. With counter-attack force.

f. For ambushes.

g. Covering.

13. Because of their mobility and long-range, the Recoilless Rifles are ideally suited for employment with covering troops, screens and advance positions. They can impose delay, destroy enemy armour and disengage quickly.

14. ***Main Defensive Position***

a. The role of the Recoilless Rifle to destroy tanks approaching battalion locality. Full advantage of the long-range of the weapons must be taken to separate enemy infantry from tanks before they close on to defensive position.

b. On occasions it might not be possible to adequately cover the flanks or gaps between defensive localities by the rifles sited in main defensive position. It might, therefore, often be necessary to send out detachments to cover the flanks or gaps within the battalion locality. These may be sent out at their own or as part of standing patrols.

**Ambushes**

15. Besides the roles mentioned above, the rifles, due to their mobility and the fact that they present a small target can also be employed offensively during defence. If situation permits, the rifles can be used in ambushing and destroying tanks. Following are the main requirements when rifles are used in the offensive role:-

a. The tank must not see the rifles until it has been fired at.

b. The tank following must not see the tank in front being knocked out, otherwise it will be able to locate the rifle and destroy it, before the rifle is able to engage the next tank or change its position.

c. There should be easy and hidden route for withdrawal of the rifles.

d. Where possible the position should be covered by artillery fire to facilitate withdrawal.

**Withdrawal**

16. In withdrawal Recoilless Rifles should be used to prevent the enemy armour from interfering with own with knowing forces and to protect the exposed flanks. Deployment of Recoilless Rifles should be carefully coordinated with the demolition plan.

# CHAPTER V – THE SIGNAL PLATOON

## SECTION 10 – ORGANIZATION

### General

1. The function of the signal platoon is to provide communication within the infantry battalion and to coordinate additional communication form divisional resources that it does not itself provide. This chapter covers the responsibilities of the signal platoon.

### Organization

2. The platoon forms part of Combat Support Company and depends on that company for administration. The platoon is commanded by a captain (Adjutant), who has a Naib Subedar as his second-in-command. It has a headquarters, a wireless party, a telephone exchange party and two line parties. For outline organization see Figure 3. Besides the signallers in the signal platoon there are 32 sepoy operators distributed within the battalion.

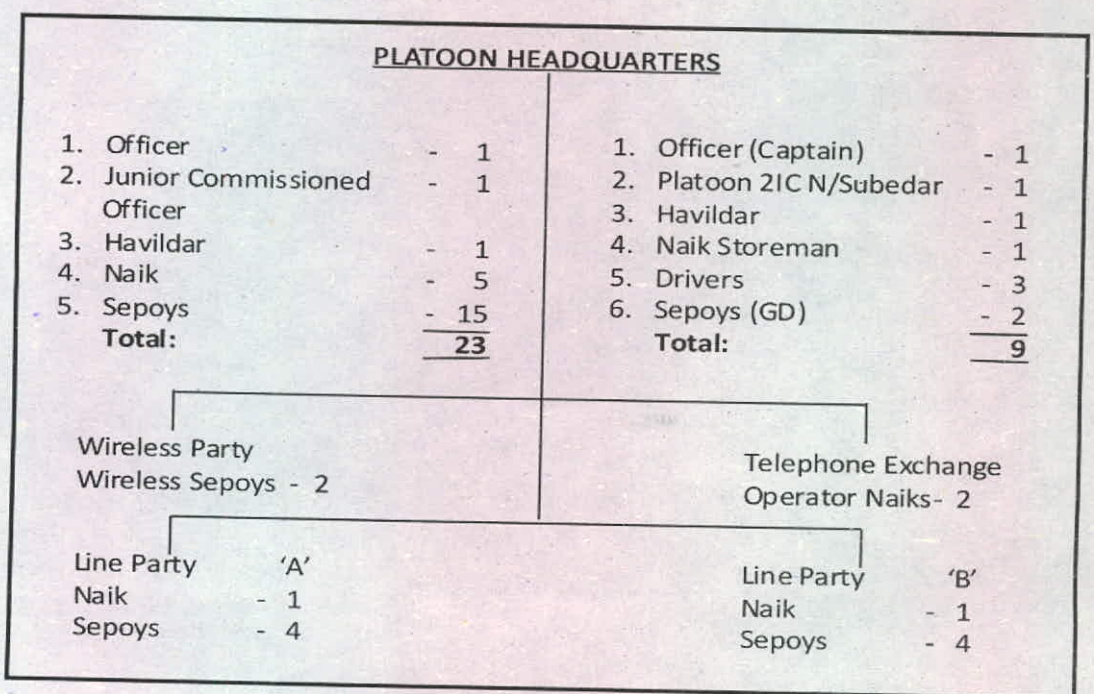


FIGURE 3 – ORGANIZATION OF SIGNAL PLATOON.

### Training

3. Unlike other arms, the selection of regimental signallers is made by the commanding officer of an infantry battalion. No signal training is given at the regimental centre. After about 3 years' service in a rifle company, individuals who have aptitude for signal work and are reasonably educated are selected and trained as regimental signallers by the regimental signal officer. They are then absorbed in

the signal platoon or are distributed among companies as operators. The responsibility of training of 44 operators, distributed in other subunits of the battalion is also that of the signal platoon commander.

4. Signallers are trained to carry-out signal office, wireless and line duties. All ranks must be fully interchangeable. For further details, refer to 'Signal Training (All Arms), Pamphlet Number 9, 1961'.

**Equipment**

5. The means of intercommunication in the battalion consist of wireless, line, despatch riders and runners. The despatch riders are given to the battalion headquarters. The rifle company and mortar platoon have their own signalmen and runners.

6. In mobile operations, the battalion will rely largely on wireless but line should also be laid, if and when the situation permits.

7. The main items of signal equipment held are:-

| | | | |
|---|---|---|---|
| a. | Walkie-Talkie (PK/PRC-2505) | – | 57 |
| b. | PRC-77/PRC-4620 Radio Sets | – | 46 |
| c. | Telephone Exchange | – | 7 |
| d. | Telephones | – | 44 |
| e. | Wire WD-1/TT | – | 90 kilometres |

8. ***Characteristics of Radio Equipment***

| Serial No. | Type of Set | Utility | Frequency Band MHz | Maximum Range (kms) | Channels |
|---|---|---|---|---|---|
| a. | AN/PRC 77/ PRC 4620 | Battalion to company, company to platoon, specialist platoons and echelons as above. Pool sets can be used for communication between battalion and special patrol, detachment as booster or for replacement at battalion commanders discretion. | 30-75.95 | 8 | 920 |
| b. | Walkie-Talkie (PK/ PRC-2505) | Company/platoon to section/weapon section detachments, patrols at company commanders dir. | 47-54 | 1 | 4 |

9. Maintenance of this equipment is the responsibility of battalion signallers, in respect of day to day care, inspection and the keeping of records as required by the working instructions issued with the equipment.

10. Unit maintenance is the responsibility of the brigade signal company. Owing to the complexity of the internal workings of modern wireless sets and due to the fact that most of them have water proofing seals very little unit maintenance is possible and the majority of faulty wireless sets will have to be evacuated. The allotment of wireless sets is completely flexible and can be varied according to the particular operation.

## SECTION 11 – DUTIES AND LIAISON WITH BRIGADE SIGNAL COMPANY

**Battalion Signal Officer**

1. Besides his duties as officer commanding signal platoon, the signal officer is responsible to the battalion commander for all communications, within the battalion. Either the signal officer or his second-in-command must be available permanently at the main centre of communication, normally the battalion headquarters.

2. Duties of the signal officer are:-

   a. From part of the battalion commander's 'R' group, if required, before each action; making a signal plan and giving signal orders.

   b. Advising the battalion commander on all signal matters.

   c. Supervising all communications in the battalion.

   d. Organizing directly the communications at the battalion headquarters.

3. These duties involve:-

   a. Issuing codes, slidex and griddle keys, call signs, frequencies and channels.

   b. Making necessary communication arrangements for supporting arms.

   c. Maintenance of the signal platoon stores.

   d. Supervising company signal stores.

   e. Instituting, carrying out and supervising all signal training in the battalion.

   f. Turning over of signallers in the platoon, in order to maintain laid down reserve to replace casualties.

   g. Close liaison with the brigade signal officer.

h. Deploying the brigade rear link set and any other signal detachment temporarily attached to the battalion. The brigade signal detachment is under command of the battalion signal officer for all purposes.

**The Signal Platoon**

4. The signal platoon is responsible for the provision of:-

a. A signal office, including an exchange at the battalion headquarters and a temporary signal centre elsewhere, if required.

b. A control set, on the battalion command wireless net. The set is normally manned by the adjutant.

c. A rover set, on the battalion command wireless net, used when the battalion commander moves out separately or away from the battalion headquarters.

d. Line communication from battalion headquarters down to the company headquarters level and to the affiliated elements. There are two line parties of five soldiers each, who carry-out this task.

e. Assistance to companies in the field and on training.

f. Communication with neighbouring units and the major unit on the right of the battalion. The signal officer is responsible for arranging wireless communication for this purpose and for providing any line communication required. He will arrange communication to the supporting arms. The signal platoon can expect all possible assistance from supporting arms in particular from artillery.

g. Additional communications which must be met as occasion demands, for example, wireless or line between main and forward command post, to regimental aid post or patrols, or for traffic control.

h. The care and custody of the reserve wireless sets.

5. The battalion signal officer must keep the close liaison with the brigade signal officer. Matters affecting the battalion in which the brigade signal officer is directly concerned are:-

a. Classification of signallers.

b. Assistance in training the signal platoon.

c. Maintenance of signal equipment.

d. Replacement of controlled signal equipment.

e. Rear link set at battalion headquarters is provided and manned by a detachment of the brigade signal company.

f. Allocation of frequencies and call signs.

g. Extra wireless sets or telephones for special tasks.

h. Moves and positions of battalion headquarters.

## SECTION 12 – TACTICAL HANDLING

1. The signal platoon plays very important role in all the operations of war. The success of any operation cannot be guaranteed without an effective line and radio link between the subunits in an infantry battalion.

2. The authorization of signal equipment and its distribution is very flexible. Rigidity in allotment of the signal assets must be avoided. Commanding officer must use his full authority while sanctioning the signal resources as per the dictates of the operation. A tactical layout in the field is suggested for guidance in Figures 4 and 5.

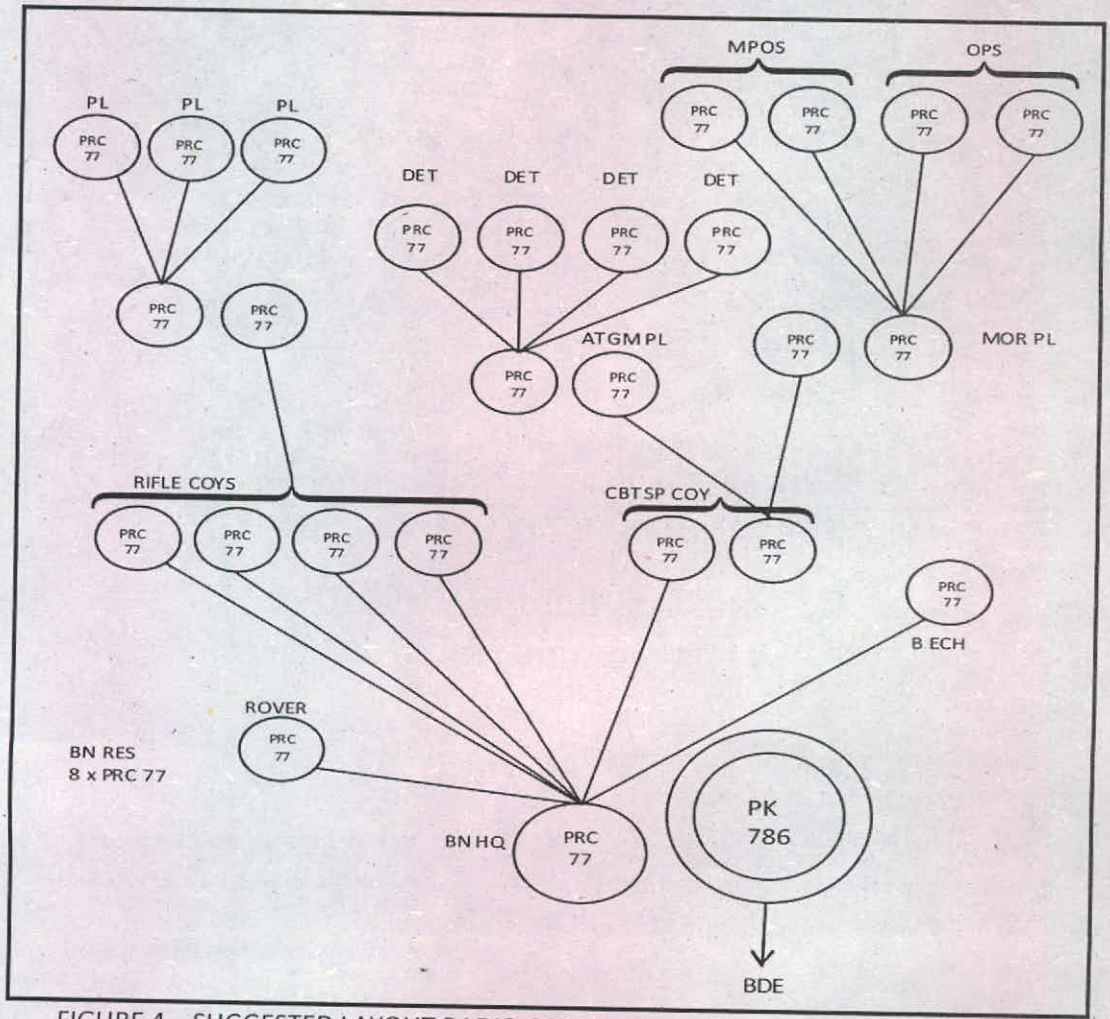


FIGURE 4 – SUGGESTED LAYOUT RADIO COMMUNICATION – INFANTRY BATTALION.

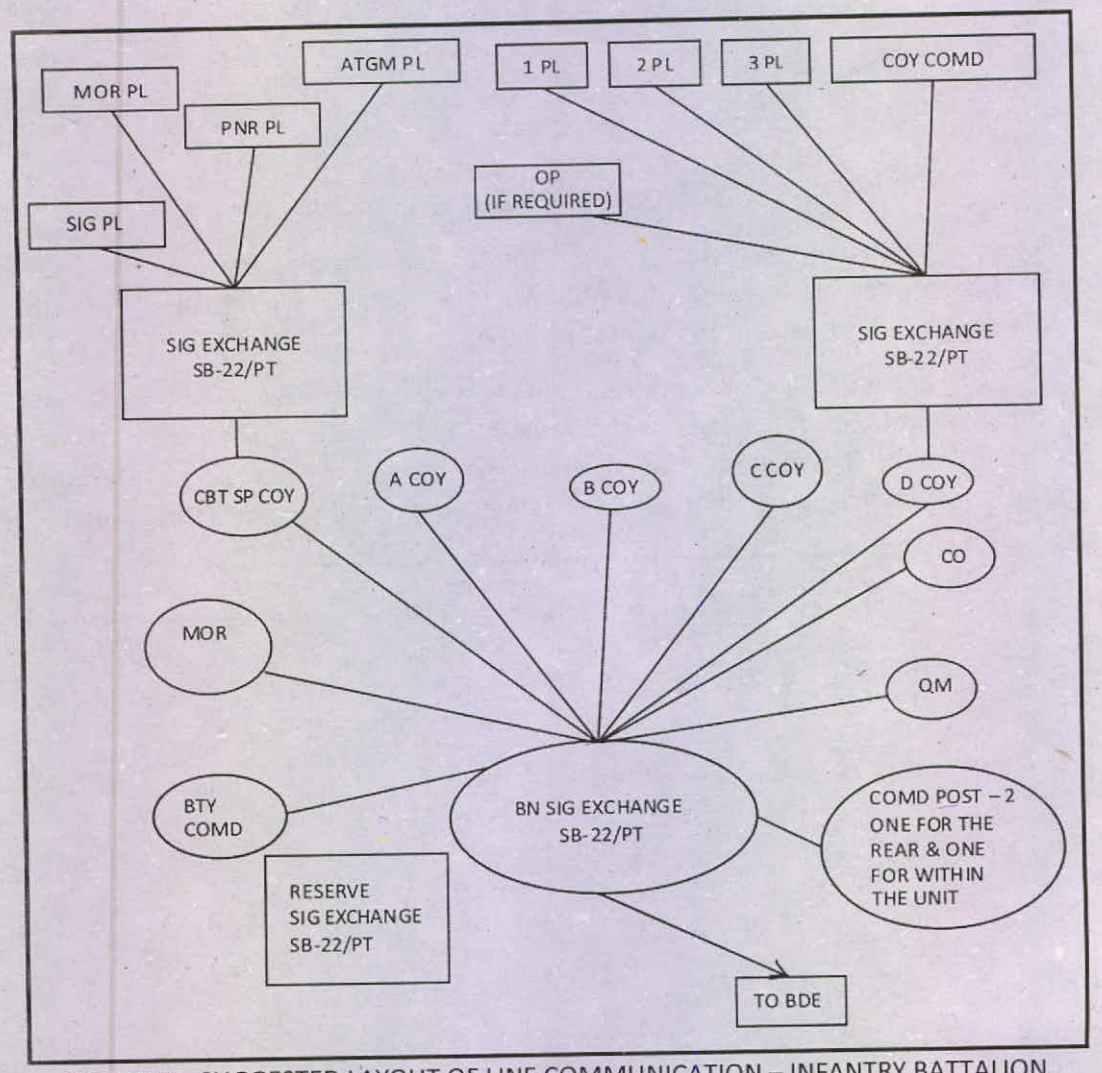


FIGURE 5 – SUGGESTED LAYOUT OF LINE COMMUNICATION – INFANTRY BATTALION.

# CHAPTER VI – THE MORTAR PLATOON

## SECTION 13 – ORGANIZATION AND CHARACTERISTICS

**General**

1. Mortar is one of the oldest and simplest form of a cannon. Its importance, in mobile warfare is comparatively recent. Mortar platoon is considered to be the battalion commander's personal artillery. Their successful employment depends principally upon good communications and fire-control. Detailed organization of 81 millimetres Mortar platoon is given at Figure 6.

**PLATOON HEADQUARTERS (17)**

| | | |
|---|---|---|
| 1. | Platoon Commander (Captain) | 1 |
| 2. | Platoon 2IC (Naib/Subedar) | 1 |
| 3. | Havildar OP (MFC) | 2 |
| 4. | Havildar MPO | 2 |
| 5. | Naik OP | 2 |
| 6. | Naik Ammunition | 1 |
| 7. | Driver MT | 4 |
| 8. | Sep (OP Radio & Wire) | 2 |
| 9. | Sep GD | 2 |
| | **Total:** | **17** |

Section (10)
(3 x 81 mm Mortars)

Section (10)
(3 x 81 mm Mortars)
Section Commander Naik - 1
Mortar Sepoys - 9

**SUMMARY**

| Personnel | | Transport | | Weapons | |
|---|---|---|---|---|---|
| Captain | - 1 | Truck ¼ ton | - 1 | 81 mm Mortars | - 6 |
| Naib/Subedar | - 1 | With Trailor | | Pistol/Revolver | - 8 |
| Havildar | - 4 | Cargo ¾ ton | - 3 | Sten 9 mm | - 15 |
| Naik | - 5 | With Trailor | | Rifles | - 14 |
| Sepoy GD | - 20 | **Total:** | **4** | Pistol Signal | - 1 |
| Driver MT | - 4 | | | **Total:** | **44** |
| Sepoy (OP Radio & Wire) | - 2 | | | | |
| **Total:** | **37** | | | | |

FIGURE 6 – DETAILED ORGANIZATION OF 81 MILLIMETRES MORTAR PLATOON.

2. To obtain effective fire rapidly, at the right place and time, it is essential for all commanders to understand the characteristics, capabilities and limitations of mortars. The principles characteristics have been divided into the following three categories:-

a. ***Principles of Employment***. The principles that govern tactical employment of the weapon.

b. ***Deployment Characteristics***. The characteristics that affect tactical employment of the weapon.

c. ***Firepower Characteristics***. The characteristics that affect firepower of the weapon in an operation.

**Principles of Employment**

3. ***Concentration***. The principle of concentration of force applies to the fire of mortars. The platoon should basically be employed as a whole. It may only be split in two sections if the entire battalion frontage cannot be covered from one position. The object of concentration is that if the target is small or pinpoint the lines of fire of all mortars are focused on the point.

4. ***Flexibility***. It is the ability to switch fire from one target to another depending on the tactical situation. Flexibility will depend on the speed and reliability of command and delegation of authority to call for fire, to the subordinates, in touch with the situation.

5. ***Observation***. If possible observation posts should be sited to cover wide front. Communication should be duplicated. Apart from controlling fire the observation post will also be a useful source of information. Observation is provided by a static observer or mobile fire controller. A mortar platoon is capable of establishing three observation posts which are provided in organization of the platoon. They also act as mobile fire controllers during advance and attack operations. The platoon commander can also act as an observer or a mobile fire controller when necessary. In an infantry battalion all officers/junior commissioned officers should be trained to call for mortar fire.

6. ***Continuity***. Mortars should be able to provide continuous fire support during all operations including attack or advance. This is only possible to achieve when one section is deployed to engage the enemy and other section is on the move. Careful arrangements must be made for replenishment of ammunition.

7. ***Simplicity***. It is one of the characteristics of mortar fire. While making fire support plan, attention must be paid to this principle.

**Deployment Characteristics**

8. ***High Trajectory***. The mortar is a high trajectory weapon. Height of its trajectory varies between 1415 metres at charge II with 80 degrees elevation and 340 metres at charge 1 with 45 degrees elevation. This vertex with Steep angles of ascent and descent give following capabilities to the weapon:-

a. Mortar can be sited and brought into action from behind high cover; where it will be safe from the fire of enemy's low trajectory weapons, including field artillery.

b. Mortar can engage targets which cannot be engaged by own low trajectory weapons i.e. reverse slopes, dead ground, deep nullahs, BUAs etcetera.

c. Mortar can give overhead fire support.

9. ***Mobility***. It is a comparatively light weapon and can be carried by various means in any type of terrain. Whatever is the mode of transport, mortars can be taken across any type of terrain to support the battalion in all types of operations. In action, following means are normally employed to carry the weapon and its ammunition:-

a. ***Mechanical Transport***. Mortars along with their ammunition can easily be carried into action in mechanical transport i.e. jeep or ¾ ton truck, right up to the mortar position, if situation permits.

b. ***Man Pack***. Where terrain does not permit use of vehicles, the mortars can be on man pack basis. The crew should easily be able to carry the mortars over long distances of 24-32 kilometres. Personnel of the rifle companies will be required to carry the ammunition.

c. ***Mule Pack***. In a hilly country and thick jungles, mules can be used to transport the weapon and its ammunition. A rough breakdown of load table and mules required for one mortar and its pouch ammunition based on one mule load of 72.5 kilograms is given as under:-

(1) A mortar along with its accessories requires one mule.

(2) 60 bombs of pouch ammunition in boxes of 6 rounds each require one mule to carry two boxes.

(3) Five mules to carry the pouch ammunition of one mortar.

**Firepower Characteristics**

10. ***High Rate of Fire***. Mortars can fire more than 30 rounds per minute, but owing to limited supply of ammunition, it is not practicable to maintain such a high rate of fire. Following rates of fire have been laid down:-

a. Intense rate of fire is 8 rounds per minute.

b. Rapid rate of fire is 6 rounds per minute.

c. Normal rate of fire is 4 rounds per minute.

d. Slow rate of fire is 2 rounds per minute.

11. With these rates of fire, first line high explosive pouch ammunition will last as under:-

a. With intense rate of fire for 5 minutes 30 seconds.

b. With rapid rate of fire for 7 minutes 30 seconds.

c. With normal rate of fire for 11 minutes 30 seconds.

d. With slow rate of fire 22 minutes 30 seconds.

It is evident from the above that mortars should be used very discreetly in battle and arrangements for its ammunition supply should be made before hand, where extensive use of mortars is planned or expected in an operation.

12. ***Flexibility***. Flexibility means the mortar's ability to switch fire from one target to another without delay. Mortars can engage targets over a wide area and widely separated ranges with little delay. This is due to two factors:-

a. The use of eight charges with standard M57D bomb:-

| | | | |
|---|---|---|---|
| (1) | Charge 0 | – | 100 to 550 metres. |
| (2) | Charge 1 | – | 300 to 1400 metres. |
| (3) | Charge 2 | – | 1000 to 2050 metres. |
| (4) | Charge 3 | – | 1500 to 2650 metres. |
| (5) | Charge 4 | – | 2200 to 3250 metres. |
| (6) | Charge 5 | – | 3100 to 3800 metres. |
| (7) | Charge 6 | – | 3200 to 4250 metres. |
| (8) | Charge 7 | – | 3500 to 4550 metres. |

b. ***The Easy Traverse Provided by the Bipod***. Mortars can be switched through 360 degrees by moving the bipod. Traversing gear can also be used to achieve a wide traverse.

13. ***Fragmentation***. The high explosive bomb has much better fragmentation than shells fired by artillery pieces. It bursts into numerous small fragments, with a lethal area of 91 metres all around the point of burst.

14. ***Ability to Produce Smoke Screen***. Under favourable terrain and weather conditions the mortars can produce a limited, but effective smoke screen. Two types of screening smoke i.e., White Phosphorus (WP) and Base Ejection (BE) are used. Another type of BE smoke bomb is used to produce a sky trail for target indication.

15. ***Ability of Fire at Night, in Fog or Smoke***. Effective fire can be brought down during night, in fog, smoke or in conditions of poor visibility on targets, which have been previously registered. Mortars can also engage unregistered targets by prediction during night, but fire may not be very effective, as it will be difficult to observe and adjust.

16. ***Long Dispersion Zone***. The dispersion zone for four charges is:-

| | | | |
|---|---|---|---|
| a. | Charge 1 | – | 100 metres long 16 metres wide. |
| b. | Charge 3 | – | 200 metres long 32 metres wide. |
| c. | Charge 5 | – | 300 metres long 40 metres wide. |
| d. | Charge 7 | – | 300 metres long 48 metres wide. |

17. ***Range***. The minimum range of 81 millimetres Mortar with M57D bomb is 550 metres and the maximum range is 4550 metres.

**Limitations**

18. From the point of view of accuracy, long dispersion zone is also a disadvantage. Mortars should not be ordered to neutralize or destroy a pinpoint target, as it is an area neutralizing weapon. The other limitations are:-

a. ***Non Calibration***. The mortar sight is not coordinated with the barrel and as such, (apart from many other factors like wind, ammunition and crew fatigue), the bombs may not fall at the exact place, for which range and line was set on the sight. This error can be adjusted only through the process of ranging because of which, lot of ammunition is spent, in order to bring down effective fire on the target.

b. ***Long Time of Flight***. The muzzle velocity of the mortar bomb is low, therefore, effect of wind on its path will be considerable. Using different charges the following are the minimum timings during which the bomb remains in the air:-

| | | | |
|---|---|---|---|
| (1) | Charge 0 | – | 11 to 16 seconds. |
| (2) | Charge 1 | – | 18 to 25 seconds. |
| (3) | Charge 2 | – | 22 to 28 seconds. |
| (4) | Charge 3 | – | 25 to 32 seconds. |
| (5) | Charge 4 | – | 28 to 35 seconds. |
| (6) | Charge 5 | – | 31 to 37 seconds. |
| (7) | Charge 6 | – | 33 to 41 seconds. |
| (8) | Charge 7 | – | 35 to 42 seconds. |

c. ***Night Deployment***. Mortars cannot be effectively deployed during hours of darkness, unless targets are registered during daylight. No correction or adjustment of fire is possible during darkness.

d. ***Vulnerability***. Due to distinctive sound and flash produced while firing mortars can easily be detected by the enemy, if not sited behind cover. They can be located by radar and sound ranging equipment, even if sited behind high cover; this makes them liable to enemy counter mortar fire. The solution to this problem is:-

(1) Good siting and use of reverse slope positions behind high cover.

(2) Frequent use of already selected and prepared alternative positions.

**Safety Precautions**

19. ***Overhead Safety***. Responsibility for ensuring safety of the troops being supported, when moving or stationary, is as follows:-

a. ***Stationary***. In defence, the responsibility is entirely of the observer.

b. ***Moving***. The responsibility is shared between the observer and the commander of the moving troops. Mobile fire controller will act as adviser to the commander of the troops on all safety matters.

**Safety Limits**

20. Safety limit has been defined as a line or point on ground, upto which own troops can advance with complete safety, while the bombs are falling on the target. The limits are as under:-

a. ***Safety Distance***

(1) ***Overhead Safety Distance***

| | | | |
|---|---|---|---|
| (a) | Charge 1 | – | 290 metres. |
| (b) | Charge 3 | – | 467.5 metres. |
| (c) | Charge 5 | – | 630 metres. |
| (d) | Charge 7 | – | 705 metres. |

(2) Flanking safety distance – 351.5 metres.

b. ***Safety Limits***

(1) ***Overhead Safety Limit***

| | | | |
|---|---|---|---|
| (a) | Charge 1 | – | 150 metres. |
| (b) | Charge 3 | – | 200 metres. |
| (c) | Charge 5 | – | 250 metres. |
| (d) | Charge 7 | – | 250 metres. |

(2) Flanking safety limit – 124 metres.

21. The above rules of safety apply to all types of high explosive and smoke ammunition.

**Exception to the Rules of Safety**

22. There are three exceptions to the rules of overhead and flanking safety:-

a. When own troops are dug in and are under cover, then safety limit can be reduced by 91 metres.

b. If own troops are in properly designed and fully closed down armoured personnel carrier, the danger area of the bomb, can be disregarded.

c. When supporting tanks, fully closed down and without any infantry nearby, no safety measures are required; but care and common sense must be exercised in the uses of white phosphorus because of fire hazard and discomfort of crews.

**Communication**

23. The means of communication in the mortar platoon are:-

a. 6 x AN/PRC-77.

b. 5 x telephones TA/1/PT with 10 kilometres wire.

24. One AN/PRC-77 will be permanently on the Combat Support Company Commanders' net and will be the platoon commander's rear link. The remaining AN/PRC-77 can be used as required by the platoon commander. They may be given to Mobile Fire Controllers (MFC) or observation posts. When all three observers are deployed, one additional AN/PRC-77 radio set may be provided to the platoon from the battalion pool.

25. Communication with observation posts must be reliable and, whenever the situation allows, they should be duplicated by laying telephone lines.

**Observation**

26. The mortar platoon is capable of establishing four observation posts which are manned by two havildar and two naik observers, as provided by the organization of the platoon. They also act as mobile fire controllers during advance, attack and withdrawal operations.

27. The platoon commander can also act as an observer at the observation post or act as a mobile fire controller, if necessary.

28. In an infantry battalion, all officers and Junior Commissioned Officers should be trained to take/conduct mortar shoot.

## SECTION 14 – TACTICAL HANDLING

**General**

1. A mortar section is the smallest firing unit, as maximum fire effect can only be obtained by firing two or more mortars on a target. However, single mortar may also be fired, if considered sufficient. Breaking up the platoon in several detachments should be avoided, as it reduces fire effect on the target.

2. To achieve the principle of concentration while planning the mortar fire, command should be centralized at platoon level and whenever possible, the platoon should engage targets as one firing unit. But, if large frontage have to be covered by mortar fire, it may be necessary to site the platoon in two section positions in order to take on all the targets.

**Advance**

3. Mortars are used well up in the advance guard to enable the commanders to overcome light resistance quickly and effectively.

4. In advance, it is normal to allot a section of mortars to the vanguard company. Two Mobile Fire Controllers will accompany the vanguard commander or the point platoon commander and will be in communication with the mortar positions, by wireless.

**Attack**

5. When an attack is planned at battalion level, the roles of mortars will be:-

a. To provide fire support as per fire support plan.

b. To provide immediate close support by engaging opportunity targets.

c. To assist in reorganization by engaging quickly planned defensive fire tasks.

d. To provide smoke screen.

6. ***Initial Fire Plan and Immediate Close Support***. Following must be considered while planning these types of fire support:-

a. Availability of ammunition must always be kept in mind and, if there is sufficient fire support available from other arms, the mortar platoon may be left out of the initial fire plan to conserve ammunition for immediate close support and for the reorganization phase.

b. Initial fire plan can best be controlled from a static observation post, while close support is best obtained through mobile fire controllers, moving with the assaulting company commanders.

c. Mortar fire must be coordinated with that of the other supporting weapons. This coordination will normally be done by the affiliated field battery commander.

7. ***Reorganization***. Employment of the mortars for the reorganization phase should be planned as under:-

a. All defensive fire tasks selected for the reorganization phase must be preselected and included in the overall fire plan of the battalion.

b. In the reorganization phase, mortars must move up, as quickly as possible with the 'F' echelon. This move should be carried out in a manner that there is always one section on the ground. The mobile fire controllers should reach the objective with the assaulting troops.

c. During the attack, the mortars should be so sited that they are able to fire well beyond the objective. This will enable them to engage defensive fire tasks in the reorganization phase without having to move forward.

**Defence**

8. In defence the roles of the mortar platoon will be:-

a. ***To Support Protective Detachments and Patrols***. To achieve this, the mortar platoon or one of its sections may be deployed in a forward position and mobile fire controller allotted to the screens or an important patrol. Smaller patrols and outposts may ask for fire support on wireless or by prearranged visual signals.

b. To assist in the defensive fire plan by engaging areas that cannot be adequately dealt with by small arms and artillery. The number of defensive fire tasks should be limited to 6 to 8 and one defensive fire (SOS) per mortar platoon.

c. To support counter-attack and counter penetration.

d. ***Harassing Fire***. Whenever possible control should be centralized under the platoon commander. In order not to disclose the main mortar position, harassing fire should be carried out from temporary positions.

e. ***Counter Mortar***. Mortars may be called upon for counter mortar fire against hostile mortars.

9. Observation posts should, be sited to cover the whole of the battalion front. It will normally be necessary to deploy more than one observation post in the forward companies. Communication from observation posts to the mortars position should be duplicated.

10. It may not be always possible to site the mortars as a platoon if a battalion is holding a wide frontage. Mortar sections may therefore have to be deployed in company localities to cover the complete frontage.

11. To avoid detection by enemy, mortars should be well dispersed. Once mortars are sited, they should be dug in as quickly as possible and mortar positions concealed from ground and air observation.

12. If it is considered necessary to dump ammunition at the mortar position, it should be placed in pits with overhead protection away from the mortar position.

**Withdrawal**

13. Owing to their great mobility, the mortars can be employed to provide fire support to withdrawing troops till the very last stage of abandoning a position. Roles of mortar platoon in a withdrawal operation are:-

a. To provide fire support to the standing patrols.

b. To delay the enemy by firing at long-ranges, thereby forcing him to deploy.

c. To assist in the overall defensive fire plan. Mortars may have to be deployed well forward and placed under command of companies.

d. To cover final withdrawal of the rear parties.

e. To form part of the rearguard. The mortars can help in any counter-attack or assist in holding intermediate positions.

14. In a withdrawal operation, if artillery support is not available, mortar sections should move back in a manner that there is always one section on the ground to ensure continuous fire support to the rear parties. It is, however, desirable that this leapfrog backwards should be coordinated with the affiliated field battery so that mortar platoon, as far as possible, moves back as a whole.

15. ***Night Employment***. Normal defensive fire and defensive fire (SOS) tasks, planned and registered during the day can be effectively engaged by night. Mortar platoon is also capable of firing at night or in fog on targets that have not been previously registered. This is done as an unobserved shoot planned from the map and requires accurate map-reading. This however, causes a heavy expenditure of ammunition.

## CHAPTER VII – THE ASSAULT PIONEER PLATOON

### SECTION 15 – ORGANIZATION AND CHARACTERISTICS

**Organization**

1. The assault pioneer platoon is organized in a platoon headquarters and three assault pioneer sections, each consisting of one naik and six soldiers. The platoon is commanded by a subedar with an engineer havildar as platoon second-in-command. The platoon havildar and one naik section commander are attached from the engineers.

2. Detailed organization of the assault pioneer platoon is given in Figure 7.

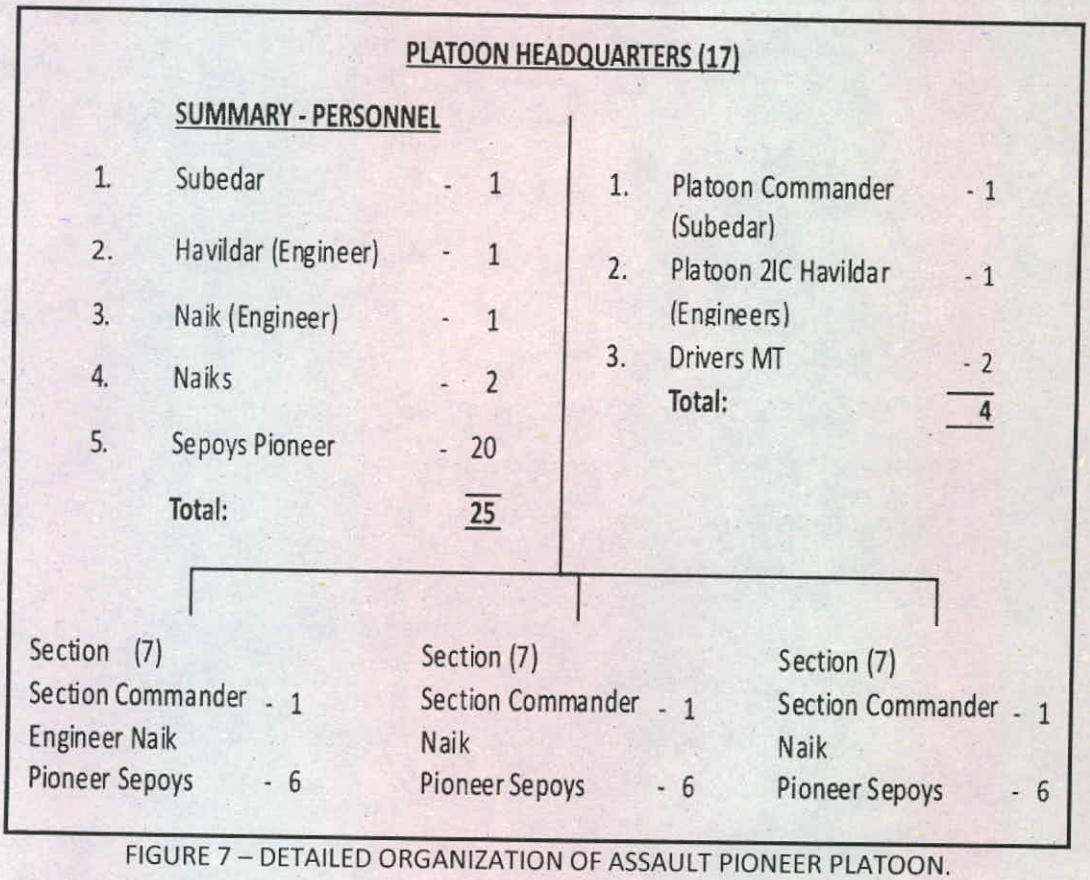


FIGURE 7 – DETAILED ORGANIZATION OF ASSAULT PIONEER PLATOON.

**Characteristics**

3. The platoon is designed and equipped to undertake minor engineer works in the battalion.

4. It can carry only a small amount of equipment in the limited transport available. It must, therefore, be trained to rely to a very large extent on improvisation with the material available locally.

5. In addition to its normal pioneer tasks, the platoon can also provide a small reserve of firepower for the battalion. It should, however, be used to augment battalion's firepower only as a last resort in the defensive battle, otherwise it will not be able to carry-out its main task.

6. All members of the platoon are trained in anti-gas duties and can be usefully employed for this task in the battalion headquarters.

7. The platoon should maintain close liaison with the engineer element, affiliated with the parent formation.

**Tasks of the Assault Pioneer Platoon**

8. Normal tasks of an assault pioneer platoon are:-

a. Laying, arming, neutralizing and lifting of all types of mines and instructions in these procedures, upto the standards laid down, within the battalion.

b. Arming and neutralizing of booby-traps.

c. Reconnaissance of mined areas before breaching.

d. Advice and assistance on improvised bridging rafting and waterman ship.

e. Destruction of enemy weapons and equipment.

f. Clearing of lanes through mined area.

g. Mine clearance of roads, tracks and BUAs.

h. Assault demolition and cratering.

i. Destruction of masonry, wire and other obstacles.

j. Construction of roadblocks.

k. Hasty demolition of bridges and cratering of defiles.

l. Improvement of buildings for defence.

m. Anti-gas duties to cover gas proofing of the command post and regimental aid post, etcetera.

n. Knots and lashings.

9. The tasks mentioned in preceding paragraphs are many and varied. To enable the assault pioneers to perform these tasks efficiently, planning must be done keeping in view the capability and the limited strength of the platoon. In most cases manpower assistance must be provided from the rifle companies.

**Employment**

10. The platoon normally has many tasks. It is, therefore, best employed as a platoon under direct control of the battalion commander. In an attack or other operations, assault pioneer sections may however, be attached to the rifle companies for limited periods. Time must be allowed for the pioneers to rest.

**Administration**

11. The responsibility of providing administrative cover is that of the Combat Support Company. If at any time, an assault pioneer section is attached to a rifle company, it must be administered by that company.

**Communication**

12. The platoon has no wireless sets or telephones. Necessary communication must either be done by runner or through the rifle companies to which the assault pioneer sections are attached.

## SECTION 16 – TASKS OF THE ASSAULT PIONEER PLATOON

**Tasks in Advance**

1. The main tasks will be:-

   a. Construction, repairs and maintenance of minor roads and tracks which are essential to battalion's advance and are not the responsibility of engineers.

   b. Reconnaissance and clearance of small mine strips or scattered enemy mines.

   c. Destruction and removal of the obstacles either with explosives or by the use of the tools and equipment carried by the platoon.

   d. To assist the battalion in the crossing of water obstacles.

2. ***Mines***. The platoon is capable of locating, marking and lifting small scattered mixed mine strips mine strips.

3. ***Explosives***. Sufficient explosives and hand tools are carried in the platoon for the destruction or removal of small obstacles.

4. ***Bridging and Rafting***. The platoon must be trained in the construction and use of class 5 rafts. It must also be capable of providing improvised rafts and small bridges with the local material when standard equipment is not available or is in short supply.

**Tasks in Attack**

5. The likely tasks are:-

   a. Clearance of route for 'F' echelon vehicles.

   b. Reconnaissance and marking of enemy minefields.

   c. Use of minor assault demolitions (e.g. the pole charge and the Bangalore Torpedo).

   d. Destruction and removal of obstacles.

   e. Repair and maintenance of routes.

   f. Use and operation of viper-lane charge.

   g. Assist rifle companies in destruction of enemy pillboxes/bunkers.

6. In an attack at brigade or divisional level, it is probable that the direction of minefield reconnaissance will be the task of engineers. The assault pioneer platoon will, however, either be required to make individual reconnaissance within the overall pattern or will be employed on local detailed reconnaissance on the battalion front.

7. ***Minefield Clearance and Lanes***. The platoon may be required to provide a number of lanes through a minefield for the assaulting infantry. This will probably be done by devices such as the viper-lane charge, if available, which will be either operated or supervised by the platoon. Further lanes will be made by the use of mine detectors. The pioneer platoon is also responsible for the marking of the lanes which should be done as clearly as possible.

8. ***Explosive***. The platoon may also be employed to destroy wire obstacles by the use of explosives carried by the platoon. It will also assist in the assault of strong and defended buildings using the pole charges.

**Tasks in Defence**

9. In defence, the tasks of assault pioneer platoon will be:-

a. To assist rifle companies in the laying and arming of protective minefields.

b. To supervise and carry-out laying of all anti-personnel mines laid in the battalion protective minefields. It will also assist in the laying and arming of booby-traps, trip flares and other warning devices.

c. To advise and assist in the construction of all types of wire obstacles including roadblocks.

d. To assist in the provision of defensive earth work, in particular for battalion headquarters and signals shelters. It will also advise on the reverting and draining of positions and the correct method of providing overhead protection.

e. To assist with use of explosives where these are necessary on hard ground.

f. To strengthen houses for defence and provide loopholes for weapons. It will also carry-out shoring up of cellars and rooms required as headquarters, regimental aid posts etcetera and creating of obstacles within the defence buildings.

g. To carry-out repair and maintenance of the roads and tracks within the battalion locality. Improvised materials are generally used for the purpose to construct small culverts and side and cross drains etcetera.

10. ***Explosives***. Small demolitions to fell trees in defiles, crater roads and tracks and to impede the enemy in built-up areas may be carried out in conjunction with engineers. The platoon can also carry-out hasty demolitions of bridges and in defiles.

**Tasks in Withdrawal**

11. In a withdrawal operation the platoon will be required to impose maximum delay on the enemy while assisting the move of own battalion. To achieve this, it may be required:-

a. To advise and assist in the laying of all types of minefields.

b. To lay booby-traps.

c. To carry-out minor demolitions of small bridges and culverts.

d. To provide adequate maintenance of routes on the axis of withdrawal of the battalion.

## CHAPTER VIII – ANTI-TANK GUIDED MISSILE PLATOON

### SECTION 17 – ORGANIZATION AND CHARACTERISTICS

1. Anti-Tank Guided Missile (ATGM) platoon equipped with 4 x Baktar Shikans wire guided missiles has enchased the anti-tank capability of infantry battalion against the latest enemy tanks. Baktar Shikan also provides limited ability to engage helicopters and slow moving low flying aircraft with accuracy. The organization of Anti-Tank Guided Missile platoon is given in Figure 8.

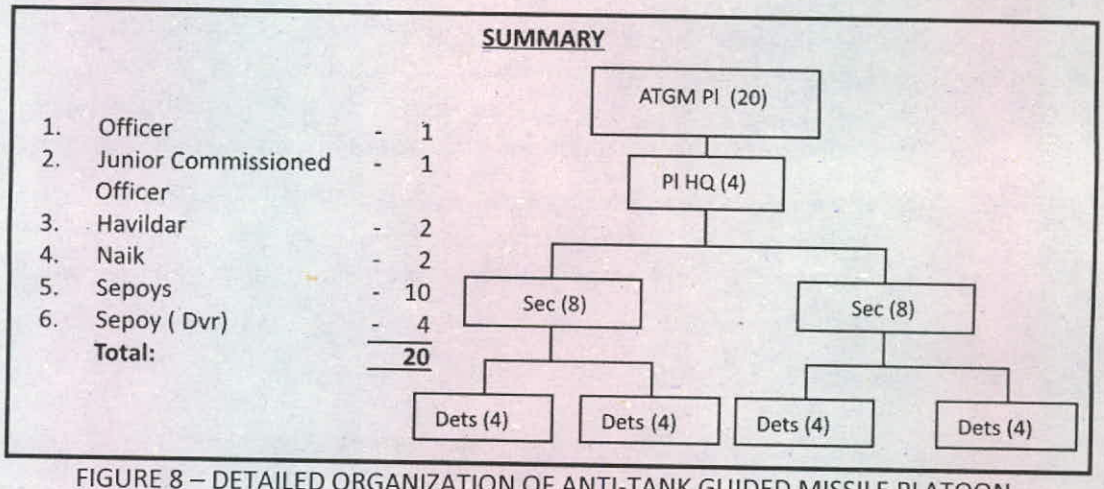


FIGURE 8 – DETAILED ORGANIZATION OF ANTI-TANK GUIDED MISSILE PLATOON.

2. Basic considerations for employment of Baktar Shikan should be kept in mind for employment of Anti-Tank Guided Missile platoon, preferably it should be employed as one entity, however, it can also be employed in sections in conjunction with the Recoilless Rifles. Baktar Shikan should be employed in pairs. When deployed

in independent role, requisite protection and administrative support will have to be provided. The characteristics of Recoilless Rifles have already been explained in Section 7. The characteristics of Baktar Shikan weapon system are explained in the succeeding paragraphs.

**Characteristics of the Baktar Shikan Weapon System**

3. ***Capabilities***

a. ***Range and Accuracy***. This weapon can fire accurately upto 3000 metres. There is no need to apply range or lead on the sight, instead the gunner only has to constantly align the cross-hair on the target and keep it there till the impact. All targets within a range of 100 to 3000 metres can be engaged with near perfect accuracy. Against moving targets, a tracking rate commensurate with the speed of the target, has to be maintained i.e. the weapon is traversed and elevated or depressed by the gunner so as to continuously keep the cross-hair of the sight on the target and the missile automatically flies along the gunners line-of-sight till its impact.

b. ***Mobility***. It is a light weight weapon and can be carried by a crew of four men. Its total weight with one missile is 88 kilograms. It is highly mobile when mounted on a jeep or Armoured Personnel Carrier and can fire and operate in any terrain. It can be easily switched from one place to another even in the thick of the battle, to engage the targets or for self-protection on being located by the enemy.

c. ***Flexibility***. Being highly mobile and accurate at long-ranges, this weapon can be effectively employed in all operations of war. It is used not only to kill tanks but also to destroy strong points, pillboxes, concrete weapon emplacements and other pinpoint targets.

d. ***Penetration***. Its warhead is made on the hollow charge principle and its penetration in homogeneous armour is in excess of 800 millimetres. The warhead functions and maintains full penetration power even at the impact angles of 65 degrees.

e. ***Recoilless Operations***. The weapon is fully recoilless; therefore its position is not disturbed on firing. It can be fired from any platform i.e. ground, jeep or Armoured Personnel Carrier.

f. ***Simplicity***. Despite the fact that the weapon system is composed of highly advanced technical instruments, it is simple and easy to operate. These instruments are provided in the weapon system in a manner that the user handles them with ease. For the user there are only few knobs, latches and power cables with the help of which he

assembles and operates the launcher. Beyond this, the user needs to have no technical knowledge in the handling and use of the weapon system.

g. ***Night Firing Capability***. The weapon can be used at night with the help of a night firing device mounted on its optical sight. In the absence of night firing device it can engage targets using flares/ illuminating rounds.

4. ***Limitations***. It has the following limitations which must be kept in view when operating this weapon:-

a. ***Back Blast***. The missile has a back blast which extends 40 metres to the rear of the launcher. The blast is caused by firing and it is dangerous for the personnel/material directly behind the launcher. The back-blast forms a 90 degree area and is divided in two zones, a dangerous zone of 20 metres and a caution zone of another 20 metres. The reason for extending these zones upto 40 metres is due to the following:-

(1) Launcher container.

(2) Flame.

(3) Heat.

b. There are minimum safety limitations when employing the Baktar Shikan weapon system in tactical situations. Where tactically feasible the 20 metres danger zone must be observed. However, where it may not be possible it must be kept in mind that the troops:-

(1) Are at least 20 metres behind the launcher.

(2) Are in a protected position such as a trench, or behind a solid object/cover.

(3) Know that the Baktar Shikan is being fired.

(4) Take protective measures to cover their eyes and do not look at the firing launcher.

(5) Wear steel helmets and ear plugs.

c. ***Vulnerability***. In ground as well as jeep mounted role the weapon and its crew are vulnerable to all types of fire. During movement the weapon is even more vulnerable. It is, therefore, necessary that the weapon is carefully sited, concealed and is located in a defiladed position.

d. ***Silhouette***. It has a peculiar shape when mounted on the jeep which gives out its locations. Its shape must therefore be broken by proper camouflage and concealment measures.

e. ***Time of Flight***. A peculiar characteristic of the weapon is that after firing the missile the gunner has to constantly keep the sight on the target till the missile's impact. At maximum range (3000 metres) time of flight of the missile is 15 seconds. Due to this the weapon crew must have steady nerves, particularly when the missile is launched and the enemy the counter action by artillery fire etcetera is also in progress. Any shakiness or flinching on the part of the gunner would result in unsteady tracking and a possible loss of the missile.

5. ***Technical Data***. Technical data of the weapon and missile is as follows:-

| | | | |
|---|---|---|---|
| a. | Weight with one missile | – | 88 kilograms. |
| b. | Range | – | Minimum 100 metres.<br>Maximum 3000 metres. |
| c. | Weight of missile with container | – | 25 kilograms. |
| d. | Penetration | – | 800 millimetres in steel. |
| e. | Crew | – | 4 men. |
| f. | Firing platforms | – | Ground and Jeep/Armoured Personnel Carrier. |
| g. | ***Scale of Missiles*** | | |
| | (1) First Line | – | 2 |
| | (2) Second Line | – | 4 |
| h. | Guidance method | – | Wire guided, command to line of sight. |

**Techniques of Employment**

6. A significant feature of the Baktar Shikan weapon system is that it is more accurate than most of the tanks at ranges beyond 1500 metres. With the Baktar Shikans adding tremendously to the anti-tank firepower, the tanks can be relieved for the classic, aggressive action.

7. The disadvantages of the Baktar Shikan system are that the crew are exposed to enemy fire while engaging targets. The principal factor while employing the Baktar Shikan weapon system should be two-fold that is exploit the capabilities of the weapon and protect the crew from countermeasures.

**Principles of Siting**

8. Following principles should be considered while siting the weapon system:-

a. ***Use of Terrain***. Every piece of ground has features which can enhance or reduce usefulness of the Baktar Shikan weapon system. Terrain should be made use of so as to maximize the capabilities of Baktar Shikan and minimize its vulnerability due to detection by enemy.

b. ***Exploit its Accuracy and Range***. Site the weapon system in such a way as to take full advantage of its accuracy and maximum range so as to engage enemy armour and yet be relatively safe from accurate enemy tank fire. This is due to the stand-off achieved when the maximum range of the weapon is made use of.

c. ***Cover and Concealment***. The weapon system should be sited in a manner that it is covered from enemy fire and concealed from his ground and aerial observation. Full use of natural features must be made. Any movement or activity by the crew members which might give away the location of the weapon system must be avoided. Positions which are prominent and obvious to an approaching enemy must be discarded. The position of Baktar Shikan must enhance effectiveness of the weapon system and provide protection to its crew from enemy tank fire.

d. ***Mutual Support***. The detachments must be able to mutually support each other. It must be borne in mind that the Baktar Shikan has a relatively slow rate of fire and the gunner is exposed for a considerable time while engaging target. Therefore, attempt should be made to ensure that while one detachment is engaging a target the other is reloading or tracking. This can be achieved if the detachments are able to give mutual support to each other. Similarly, as the weapon is usually deployed in a defiladed position, it must effectively dominate the area in front of the neighbouring Baktar Shikan and vice versa. This will also ensure complete and continuous coverage of a given sector of the area of responsibility. Baktar Shikan must be employed in pairs to be able to overcome the limitation of slow rate of fire and to achieve mutual support within the platoon. This principle must also be adhered to when employing Baktar Shikan with the tanks and other anti-tank weapons.

e. ***Local Protection***. Baktar Shikan crews are not only vulnerable to enemy artillery and close range tank fires but also to the dismounted infantry attacks in the form of raids and ambushes. Baktar Shikan should, therefore, be provided full protection by the unit/subunit whom they are giving the anti-tank support.

f. ***Dispersion***. Baktar Shikan detachments must be positioned well apart so that one enemy shell/salvo may not knock out more than one weapon system at a time. Good communications and a longer range of the Baktar Shikan allow it to be dispersed on ground. The distance between two detachments will often be dictated by the terrain but as a guideline it should be minimum 400-500 metres.

g. ***Flanking Fire***. Frontal fire, wherever possible, should be avoided. While advancing, the firepower and observation of the tank is generally oriented to the front which makes it difficult for him to detect and counter a missile launched from the flank. The vulnerability of the weapon system is reduced by its ability to take on enemy tanks from the flanks. A defiladed position will achieve flanking fire against enemy armour and at the same time protect itself from the direct return fire.

h. ***Clear Back Blast***. The back blast area must be free of personnel and equipment. Loose debris that could fly and give away position or dry vegetation that could be ignited by the flame should be removed. When firing from ground, a tarpaulin pegged down directly to the rear will help in concealing the flash.

i. ***Resupply Potential***. When sited in a position for a long period, the weapon system requires resupply of charged batteries and missiles. To ensure resupply, concealed routes from rear/flanks must lead to the firing position. The first line reserve, charged batteries and spares should not be located too far away from the firing positions.

j. ***Alternative Positions***. Depending on the terrain and visualized enemy threat, alternative position must be selected and prepared before the defensive battle begins. There should be more than one such position for each weapon system. Whenever possible alternative positions should be at least 800 to 1000 metres apart.

9. ***Siting in Built-Up Area (BUA)***. Care must be exercised in use of Baktar Shikan weapon system in BUAs. It cannot be fired from a completely enclosed building. However, it can fire effectively from the rooftops, street openings or from inside a building under the following conditions:-

a. The building is strong so that it can bear the blast effect when the Baktar Shikan is fired.

b. The ceiling is at least higher than 7 feet.

c. The floor is at least 17 x 24 feet.

d. The room has 6 square feet of ventilation. An open door 7 feet by 3 feet will provide the necessary ventilation.

e. Glass is removed from all openings.

f. All personnel in the room at the time of firing should wear ear plugs.

g. There is at least 2.4 feet clearance for the launch tube.

h. All personnel in the room are forward of the rear end of the weapon system.

***Note***: The Baktar Shikan may be fired from smaller enclosures provided the area immediately to the rear of the launch tube is open and 2.4 feet clearance is available in front.

**Salient of Existing Concept of Anti-Armour Warfare**

10. ***Formulation and Coordination of Anti-Armour Plan***. The highest commander is responsible for making the overall tactical plan for an operation and the anti-armour plan is an important part of it. The anti-armour plan should be made at the earliest stage in planning. It should integrate all the organic and non-organic anti-armour weapons. It should also be planned and coordinated at all the subordinate levels.

11. ***Engagement of Armour at Maximum Ranges***. Enemy armour should be engaged at the longest possible ranges, utilizing the Heavy Anti-Armour Weapons (HAWs). The surviving armour should be engaged with an increased number of Medium Anti-Armour Weapons (MAWs) as it comes within range and finally by Light Anti-Armour Weapons (LAWs) placed within the defensive area.

12. ***Concentration of Anti-Armour Weapons in Vital Areas***. Anti-armour weapons should be so distributed that they create greatest density at the most vital area. No attempt should be made to have an even distribution of weapons all along the front.

13. ***Depth of Anti-Armour Defences***. Anti-armour defence should be planned in depth basing it on defended localities or strong points organically combined with the overall defence plan and deployment of the troops. Depth can be considerably increased by initially deploying them well forward, a substantial proportion of long-range anti-armour weapons (depending upon area and nature of enemy threat) and their subsequent redeployment in the main defences. As enemy tanks are not likely to attack forward defended localities along with infantry, specially in the plains of Punjab, so the deployment of anti-armour weapons in the forward defended localities should be minimum. This may not be applicable in an open terrain like desert, where some weapons will have to be deployed in the forward defended localities.

14. ***Reserves***. Some tanks and anti-armour weapons may be kept as reserve in order to reinforce a weak sector and to deal with an armour attack from an unexpected direction. But this may not always be possible because of paucity of resources.

**Employment Considerations for Baktar Shikan**

15. Deployment of Baktar Shikan should be guided by certain basic considerations which are as followings:-

a. Firing positions.

b. Field of fire.

c. Long-range.

d. Concentrated Baktar Shikan employment.

e. Security.

f. Mutual support and integration.

## SECTION 18 – EMPLOYMENT OF BAKTAR SHIKAN WEAPON SYSTEM/ RECOILLESS RIFLE IN VARIOUS OPERATIONS OF WAR

**Defence**

1. Baktar Shikan is basically a defensive weapon. To make use of its strengths Baktar Shikan should be deployed to cover the likely tank approaches in areas that offer good visibility and clear fields of fire. Its likely employment in defensive operations can be as follow:-

a. ***Protective Detachments***

(1) ***Covering Troops***. Baktar Shikan and Recoilless Rifles can be effectively employed in conjunction with covering troops. Baktar Shikan with its long-range and accuracy is ideal for employment as they can force the enemy to deploy, thus causing delay. On withdrawal of covering troops Baktar Shikan must pull back first to provide cover to Recoilless Rifles and tanks when these have to fall back to the next position. Recoilless Rifles can supplement the fire of Baktar Shikan at medium ranges. The number of Baktar Shikans to be employed in covering troops will depend on terrain, enemy threat and the delay desired by the commander.

(2) ***Advance Positions***. Baktar Shikan weapon system can be effectively employed as part of the advance positions. Baktar Shikan will fight the enemy armour from covered and

concealed battle positions at long-ranges. Together with Recoilless Rifles it will form a base of anti-tank fire to cause delay and attrition on the enemy armour. It can also canalize enemy armour into the desired areas for their ultimate destruction. When advance position is ordered to withdraw, Baktar Shikan weapon system will pull back first and take up firing positions to cover the move back of the Recoilless Rifles and tanks (if allotted). Infantry must be employed to provide protection to Baktar Shikan positions.

b. ***Main Positions***. In the main defences, Anti-Tank Guided Missiles must be positioned closer to infantry and armour. Baktar Shikan can be located in the forward defended localities, depth or counter penetration positions depending upon terrain and armour threat. Recoilless Rifles can also be employed together with the Baktar Shikan. Apart from the main defences, Baktar Shikan can also be sited on the flanks or in the gaps, provided local protection is ensured. The enemy is not likely to attack the main defensive positions with tanks because of the obstacle system. Enemy tanks are likely to manoeuvre and bypass our main defence and hit the depth. A substantial number of Baktar Shikan weapon systems should be earmark to deal with the tank manoeuvre by deploying these in the depth or in counter penetration positions. In the main positions Baktar Shikan will be used to separate enemy tanks from his infantry to delay/stop his advance and break the momentum of his attack. Employment considerations of Baktar Shikan/Recoilless Rifles specific to the plains and desert terrain are:-

(1) ***Plains***. Our defences in the plains are generally based on linear obstacle system. It is visualised that the enemy would first secure a bridgehead with infantry, pump in armour and break-out for deeper objectives. Thus the enemy would present its maximum armour while building up in the bridgehead or in the break-out phase. Enemy may employ some tanks to knock off the bunkers on the water obstacle and trawls for mine breaching. Tanks enjoy stand off against Recoilless Rifles, therefore, some Anti-Tank Guided Missiles can be employed initially on the water obstacle. Most Baktar Shikans should be positioned initially to provide fire to the forward companies. These positions may be within the forward defended localities or behind the forward companies. Commanders should position Baktar Shikan in depth so that they are integrated with other anti-tank weapons.

Baktar Shikan in the main defensive position will normally hold fire and wait till enemy tanks are clear of cover and their kill is certain. They may allow enemy tanks to develop their manoeuvre and when they come within the designated killing areas, surprise fire from flanking/defiladed positions may then be delivered. Baktar Shikans in conjunction with other anti-tank weapons should be employed to:-

(a) Destroy enemy tanks in the bridgehead as induction/ build up takes place.

(b) Contain the bridgehead and assist own forces in eliminating the enemy penetration by launching counter-attack.

(c) Destroy enemy tanks once they break-out from the bridgehead.

(d) Form part of the force going for counter-offensive.

(2) ***Desert***. Mobile defence or a combination of mobile and area defence is visualised in the desert terrain. Baktar Shikan along with other anti-tank weapons could be employed as following:-

(a) Well ahead of the main defences in the form of anti-tank nests. These nests covered with artillery fire can be sited along enemy armour approaches. Disintegration of the enemy force and canalizing effects can be achieved by these anti-tank nests. Baktar Shikan, Recoilless Rifles along with tanks can fight retrograde battle from selected positions till the main defences.

(b) Form part of the force which would contain the canalized enemy in the killing zone.

(c) Form part of the main force which would be employed for destruction of the enemy in the killing zone.

(3) ***Defence Based on Water Obstacles***

(a) In such a situation, enemy will undertake purely an infantry operation in establishing a bridgehead across the water obstacle. Enemy is likely to employ limited number of tanks, purely in fire support role during the bridgehead operation. On the water obstacle some Anti-Tank Guided Missiles should be employed

covering the main tank approaches whereas the bulk of Anti-Tank Guided Missiles should be positioned in depth. They should be well dug in with overhead protection and should cover all possible break-out approaches likely to be used by the enemy armour. These Anti-Tank Guided Missiles should also be incorporated as part of the fire base when brigade/ division reserves are launched.

(b) Some medium and heavy anti-tank weapons are required in forward defended localities to counter enemy armour employed in fire support role and to engage enemy trawl tanks employed for the clearance of lanes.

**Advance**

2. In advance, Baktar Shikan weapon system can be used in the following echelons:-

a. ***Advance Guard***. If enemy is likely to use armour in strength against the advancing force Baktar Shikan can be kept well ahead in the advance guard. While moving in the advance care must be taken not to commit Baktar Shikan prematurely. So it would be better to use them along with the main guard. In this role Baktar Shikan can effectively cover the move forward of tanks. If gaps are found in enemy defences Baktar Shikan can cover the tanks which may be used to exploit them. Recoilless Rifles may not be used in this role due to lack of protection from small arms fire.

b. ***Main Body***. If tank threat is not too great Baktar Shikans may be kept in the main body. In this role Baktar Shikans and Recoilless Rifles can move by bounds and should any threat develop along the axis of advance, these are moved quickly and used to engage tanks at the maximum ranges. However it must be borne in mind that the Baktar Shikan cannot fire while on the move. It has to stop to be able to acquire and destroy its target as tracking requires complete stability of launching plate form.

**Attack**

3. In attack Baktar Shikan can be employed for the following:-

a. ***Before the Attack***. Baktar Shikan can be employed to cover the movement of tanks as well as attacking troops to their Forming Up Place. Due to its long-range and high hit probability Baktar Shikan can

be employed effectively to perform this task, as it can provide direct fire on identified targets. Recoilless Rifles may not be as suitable for this task because enemy tanks have greater range and accuracy than Recoilless Rifles.

b. ***Base of Fire***. Baktar Shikan and Recoilless Rifles cannot engage a target while on the move but they can be used effectively in fire support role. Baktar Shikan can support the attack by forming a base of fire along with other anti-tank/heavy weapons. If tanks are also available in the base of fire, Baktar Shikan supplements their fire. Long-range capability of Baktar Shikan must be exploited to the maximum. Proper coordination with tanks and other anti-tank weapons is imperative in case the base of fire has to be shifted to another location.

c. ***With Attack Echelons***. Some Baktar Shikans may follow the attack force. Baktar Shikan detachments move with the force and take up positions from where they can destroy enemy bunkers, neutralize his tanks and cover the move of own tanks to the objective. They can also isolate the objective by blocking withdrawal/reinforcement of tanks. Baktar Shikans are also readily available for any unforeseen eventualities and enemy counter-attack. Number of Baktar Shikans with attack echelon will depend on the tank threat and the nature of enemy defence construction.

d. ***Flank Protection***. To guard the exposed flanks of attack force Baktar Shikans are employed systematically on selected positions parallel to the direction of attack. Recoilless Rifles can also be employed for this task provided suitable positions are available. Maximum use should be made of cover and concealment.

e. ***In Reorganization Phase***. In this phase all the Baktar Shikan weapon systems, along with other anti-tank weapons will quickly move forward to the designated firing positions in order to meet enemy counter-attack by armour. Baktar Shikans are ideally suited for employment in this phase to beat back any enemy counter-attacks supported by tanks.

f. ***Exploitation Phase***. Baktar Shikans can be employed effectively in this phase of attack. Its mobility, flexibility and firepower helps in execution of this phase. Baktar Shikan can be used to destroy enemy tanks and at the same time afford protection to own armour in carrying out its exploitation missions.

**Withdrawal**

4. The Baktar Shikan weapon system can ideally exploit its long-range, accuracy, good communication and mobility during withdrawal. It can be used in the following echelons of a withdrawing force:-

a. ***Original Positions***. If the enemy is likely to use tanks in great numbers against the withdrawing force, some Baktar Shikan and Recoilless Rifles may be used in the original defensive positions to form part of the rear parties. In this role Baktar Shikan can help in showing an intact front to the enemy and also keep his tanks away at a great distance.

b. ***Rearguards***. As the rearguard actions involve occupation of defensive positions to cover the withdrawal, Baktar Shikan weapon system and Recoilless Rifles can be used effectively to form part of it. The number of Baktar Shikans to be used in this role will be dictated by the type of enemy pressure and the time required for the preparation of new main defences. As part of rearguards Baktar Shikan along with Recoilless Rifles cover the withdrawal of the main force by taking up successive defensive positions on the route of withdrawal.

c. ***Intermediate Positions***. Baktar Shikan with Recoilless Rifles can be very usefully employed in the intermediate positions. In this role Baktar Shikan and Recoilless Rifles occupy temporary defences to impose maximum delay on the advancing armour and also destroy enemy tanks making use of their accuracy and range. Efforts must be made to position Baktar Shikans in a manner so as to make full use of their destructive capability.

d. ***Counter-Attacks***. Counter-attacks by the withdrawing force with limited objectives can be supported by Baktar Shikans and Recoilless Rifles fire.

5. During withdrawal Baktar Shikan must be employed mounted on vehicles to ensure rapid movement to the firing positions or to promptly react to the unexpected enemy threat. When mounted the dust kicked by the back blast is also reduced. Baktar Shikan should only be dismounted in delaying role only when the terrain does not permit the use of vehicles.

6. Owing to the speed with which withdrawal takes place care must be exercised not to separate Baktar Shikans and Recoilless Rifles from the units/ subunits providing local protection. If for some reason Baktar Shikan/Recoilless Rifles vehicle breaks down, the weapon system should immediately be transferred to another vehicle. If in withdrawal capture of the weapon system is inevitable Baktar Shikan launcher, missiles and Recoilless Rifles with ammunition must be destroyed.

In doing so, priority must be given to the optical sight and the missile guidance set of the Baktar Shikan.

7. ***Methods of Movement***. Baktar Shikan detachments move by successive bounds or alternate bounds. In both these methods one detachment is moving while the other is covering it. The advantage is that continuous anti-armour coverage is provided against enemy threat. This method of movement is adopted when Baktar Shikans are employed with the leading elements.

8. ***Helicopter Mounted Baktar Shikan***. Attack helicopter adds a new dimension of anti-tank firepower and flexibility to the commander of an advancing force. Helicopter mounted Baktar Shikans are of immense value in advance.

## CHAPTER IX – SURFACE-TO-AIR MISSILES IN AN INFANTRY BATTALION

### SECTION 19 – CHARACTERISTICS

**General**

1. Air threat has assumed multiple dimensions and being very dynamic, is continuously changing with the rapid development in technology. Careful and continuous assessment of air threat, is therefore, essential to successfully combat it. It is an important guiding factor and should be thoroughly evaluated during the planning stage. The major threat to ground forces, operational assets and assets of strategic/national importance is generally from fighter bombers in Close Air Support (CAS) missions (both offensive and defensive operations) and multi role combat aircraft. An infantry battalion in defence does not provide any worthwhile target to air. In view of the role to be performed and enemy air threat, 4 x launchers have been authorised to standard infantry battalion of offensive formations only.

**Characteristics of Stinger Surface-to-Air Missile (SAM)**

2. Characteristics of Stinger Surface-to-Air Missile (SAM) are as follows:-

   a. Stinger SAM is of United State origin. The missiles is man portable and shoulder fired with Infra-red (heat seeking) guidance system. It requires no control of the operator after firing, therefore it is regarded as "fire and forget" type of weapon. It carries High Explosive (HE) warhead and uses solid propellant motor for propulsion.

   b. Stinger is designed to counter low level, high speed ground attack aircraft and provides short-range air defence for manoeuvring units. It is a lethal weapon against helicopters and transport aircraft. It has an Identification Friend or Foe (IFF) sub system which aids the missile operation. Operation at night or in adverse weather conditions is some what restricted due to missile operators' ability to see and identify the target.

3. ***Salient Features***

a. ***Physical Data***

| | | | |
|---|---|---|---|
| (1) | Length of missile | – | 147.32 centimetres. |
| (2) | Length of complete weapon | – | 152.4 centimetres. |
| (3) | Diameter of missile | – | 6.98 centimetres. |
| (4) | Weight of the missile | – | 12.65 kilograms. |
| (5) | Weight of complete weapon | – | 15.86 kilograms. |

b. ***Technical Data***

| | | | |
|---|---|---|---|
| (1) | Range | – | Excess of 4 kilometres (may extend upto 5.9 kilometres). |
| (2) | Effective altitude | – | 3000 metres. |
| (3) | Speed | – | 600 metres per second. |
| (4) | Guidance | – | IR having modified proportional navigation. |
| (5) | Warhead | – | High Explosive. |
| (6) | Kill probability | – | 70%. |
| (7) | Elevation limits | – | 10 to 65 degree. |
| (8) | Operation temperature | – | 4.4 to 52.2 centigrade. |
| (9) | Shelf life | – | 10 years. |
| (10) | Reaction time | – | 6 seconds. |
| (11) | Self-destruction time | – | 17 to 19 seconds. |

**Characteristics of Anza Surface-to-Air Missile (SAM)**

4. Characteristics of Anza Surface-to-Air Missile (SAM) are as follows:-

a. Anza Mark-1 SAM system is based on HN-5 SAM system (Chinese origin) with no structural changes. The system is capable of making tail on-engagement against Jet aircraft or head on-engagement against propeller driven aircraft and helicopters under visual aiming conditions.

b. Anza Mark-1, is man portable short-range shoulder fired SAM weapon system for low level air defence. Anza Mark-1 missile utilizes the heat energy i.e. Infra-red (IR) radiations, emitted from the exhaust/hot body of the aerial targets, as a source of guidance for the missiles.

Due to its light weight the missile is highly mobile and can be rapidly deployed in all sorts of terrain, making best use of the terrain features like hills, mounds, sand dunes, buildings etcetera.

c. ***Salient Features***

| | | | |
|---|---|---|---|
| (1) | Maximum altitude of target | – | 2300 metres. |
| (2) | Minimum altitude of target | – | 50 metres. |
| (3) | Maximum lethal slant distance (in tail-on attack) | – | 4200 metres. |
| (4) | Maximum firing slant distance (in head-on attack) | – | 2800 metres. |
| (5) | Maximum velocity of target 950 kilometres (in tail-on attack) | – | (260 metres per second). |
| (6) | Maximum velocity of target 550 kilometres (in head-on attack) | – | (150 metres per second). |
| (7) | Weight of the weapon system under combat conditions | – | 15 kilograms. |
| (8) | Time for transition from move | – | Less than 10 seconds. |
| (9) | Time for preparation | – | Less than 5 seconds. |
| (10) | Range of operational temperature degrees | – | 40 to +50. |

## SECTION 20 – PRINCIPLES OF EMPLOYMENT AND SITING OF WEAPONS IN AIR DEFENCE ROLE

**Principles of Employment**

1. Four principles which are almost universally accepted provide the doctrinal basis for the designing of air defence. A balanced application of these principles can enhance the effectiveness of air defence elements. These principles are:-

a. ***Mass***. Air defence weapons must be used in heavy concentration to effectively defend a critical point or area against air attacks. 12.7 millimetres and 7.62 millimetres Machine-gun in anti-aircraft role must be coordinated with Surface-to-Air Missiles.

b. ***Mixture***. It is with the combined employment of anti-aircraft guns, missiles and passive air defensive measures that the limitations of one system can be offset with the capabilities of the other. It forces the enemy aircraft to change its tactics besides effectively reducing its capability to neutralize all types of weapon systems on ground.

c. ***Mobility***. Surface-to-Air Missiles must be able to assume new task rapidly as dictated by the operational situation. Mobility is also required to move weapons use in air defence role to alternate positions within the air defensive layout of the same asset.

d. ***Integration***. It is the cohesiveness of various air defence elements to achieve maximum effectiveness and minimum interference with each other. It requires good Command, Control, Communication and Intelligence ($C^3I$) system. Within the framework of infantry battalion, it is by earmarking separate net for air warning within the battalion, air sentries and by keeping all weapons in air defence role ready to fire. Surface-to-Air Missile detachment will be employed on likely hostile aircraft approaches. Due to high speed of target, short time lag between first receipt of warning about intruder aircraft till fire orders reach the detachment owing to limited communication, a clear policy must be spelled out as executive order to fire and destroy the hostile aircraft.

**Principles of Siting of Weapons in Air Defence Role**

2. The principles for deployment of weapons in air defence role will generally be applicable regardless of the type of operation. The main principles are:-

a. ***All-Round Defence***. In mountainous terrain the likely directions of enemy air attack can be identified due to terrain restrictions, however, in plains an infantry battalion will be attacked by air from all the directions. The requirement of all-round defence is therefore of special significance for air defence.

b. ***Defence in Depth***. As the engagement time is generally too short, the weapons must be sited in depth to retain the ability to keep the attacking aircraft under fire, for maximum possible time irrespective of direction of attack. This calls for weapons ability to fire in 360 degree.

c. ***Mutual Support***. The spacing of weapons should be such that they can provide mutual support and at least two weapons should be able to engage an aircraft and be able to cover the dead zone of the adjacent weapon.

d. ***Concentration***. This principle implies siting of sufficient weapons, to carry-out the task effectively as opposed to dispersing Anti-Aircraft Machine-guns/missiles on several tasks, none of which may be covered properly. The aim is to achieve adequate concentration of firepower all around area.

e. ***Flexibility***. The pattern of air defence deployment should keep in step with the likely changes in the pattern of employment of the supporting unit/asset and threat parameters.

f. ***Control and Coordination of Fire***. Modern aircraft fly very fast and are difficult to recognize. For the safety of own aircraft and to cause maximum damage to those of the enemy, proper control and coordination of fire is necessary. Forward line of troops and bomb line must be worked out with precision. Information about safe corridors and timing of own aircrafts passage must be obtained from formation staff daily. Forward air controllers must be employed and all troops in area informed about the likely action by Pakistan Air Force. Safety of own aircrafts is supreme; battalion commander must personally supervise the training of Surface-to-Air Missile detachments. Theses detachments should always be employed under direct control of an officer may be rifle/combat support company commander. The officer should be given the authority to order fire without seeking permission when hostile aircraft are in his effective range.

## CHAPTER X – ARMOUR INFANTRY COMBINED OPERATIONS

### SECTION 21 – PRINCIPLES AND TASKS

**General**

1. In this chapter, the principles of armour infantry combined operations have been enumerated. Details of techniques have been covered in the various operations of war chapters. (This section should be read in conjunction with GSP-1854 – Tanks/Infantry Operations).

2. In modern war, success in combat can only be achieved by combined actions and complete cooperation of all arms and services. This is especially true of armour and infantry. To achieve this, a common technique is necessary, which must be practiced in training so that intimate cooperation in combat is achieved automatically and without any loss of time.

**Principles**

3. Employment of armour is governed by the fundamental principles given in subsequent paragraphs. Observance of these principles will ensure effective and successful use of armour in all operations.

4. ***Information***. Before armour is committed to battle all possible information must be obtained about the area of operation, enemy dispositions and likely intentions. This will entail:-

a. A careful and thorough terrain analysis through patrol reports, air reconnaissance, photographs and tank going maps, etcetera.

b. Collection and study of information regarding natural and artificial tank obstacles.

c. Visual terrain reconnaissance by armour commander.

d. Visual/air reconnaissance by flying over the proposed battle area, if possible.

e. Early visit by unit/subunit commanders to forward areas.

f. Joint reconnaissance by armour/infantry patrols.

5. ***Aggressiveness***. Tanks are aggressive weapons and as such should be used boldly. Offensive spirit must be maintained, at all times, in every operation of war, by all commanders. The momentum of attack must not be allowed to slow down and once the tanks have been launched, greater risks are justifiable, to maintain momentum.

6. ***Concentration***. Tanks must be used concentrated under centralised control, as far as possible, depending on the ground, enemy and own situations. It implies concentration of armour firepower at critical time and places. Concentration must be in depth. More the number of tanks used, lesser will be loser and greater the chances of success. Concentration combined with surprise and speed, produces the shock effect that paralyses the enemy. The tanks should be used to achieve decisive results.

7. ***Flexibility***. Every effort should be made to concentrate tanks, at the critical place, at the right time. Till then they may be in action elsewhere or dispersed in reserve. Good communications allow them, to be regrouped and switched, with great ease, from subunit to subunit, from area to area and from role to role as needed.

8. ***Surprise***. Surprise is very often the key factor in war. Although number of tanks employed, speed of their movement, bold use of ground, cover of darkness or bad visibility and deception, are all aids to surprise, shock action by armour is the principal means by which it is achieved.

9. ***Mobility***. The ability to move cross-country and armour protection provides battlefield mobility to the tanks. The tank develops its destructive firepower, because of its inherent mobility and firepower. This combination of firepower and mobility is one of the most important fundamental in the employment of armour. In armour operations, all plans must make maximum use of its mobility and firepower to achieve surprise and shock action.

10. ***Cooperation***. Cooperation between other arms, services and armour is essential for success in battle. This can only be achieved by intimate knowledge of each other's capabilities and limitations, integrated training and mutual confidence.

The personal touch must be fostered between commanders, staff officers and troops both in peace and in war.

11. ***Controlled Dispersion***. Paradoxically, the physical massing of tanks under enemy air superiority may lead to disaster. Even a squadron may become a lucrative target. The aim must be, to disperse to survive; and to concentrate briefly at the decisive point to fight before dispersing again. This concentration may be achieved by fire, by physical presence or a combination of both. The time which can be spent in concentration depends on:-

a. How closely engaged the forces are.

b. Enemy Target acquisition capability.

c. Enemy response time.

d. Air situation (this will include enemy capabilities with helicopter borne guided weapons).

**Tasks**

12. ***Tasks of Armour***

a. Destroy enemy armour supporting his infantry.

b. Support own infantry attacks by fire, by neutralizing or destroying enemy weapons likely to hold up its advance.

c. Physically assault the objective and reinforce infantry by forming an anti-tank screen in the reorganization phase.

d. Break-up/destroy enemy attacks by counter-attack and counter penetration actions.

e. Cover withdrawal of own infantry and help achieve a clean break.

f. Carry-out anti-helicopter borne operations.

13. ***Tasks of Infantry***

a. Seize and hold ground won by an integrated attack.

b. Destroy or neutralise enemy anti-tank weapons.

c. Protect own tanks from enemy tank hunting parties, while passing through close country or BUA, such as towns, villages, forests, jungles, narrow defiles or in a night leaguer.

d. Provide protection to the tanks from enemy short-range anti-tank weapons.

e. Overcome obstacles and establish a bridgehead.

f. Attack enemy's prepared defensive positions and field fortifications.

g. Mopping up of enemy on the objective.

h. Hold ground against attack by enemy armour and infantry; to provide a firm base for aggressive mobile action by armour.

i. Carry-out anti-helicopter borne/airborne operations.

**Methods of Tank and Infantry Attack**

14. Methods of tank and infantry attack are as follows:-

a. The manoeuvre force composed of tanks and infantry accomplishes its mission by closing with and destroying the enemy. There are three general methods of attack employed by the manoeuvre force:-

(1) Tanks and infantry attack on the same axis.

(2) Tanks and infantry attack on the converging axes.

(3) Tanks support the infantry attack by fire only.

b. ***Attack on the Same Parallel Axis***. Using this method, the entire manoeuvre force closes with the enemy on the same axis. It has following advantages:-

(1) ***Coordination***. As the force attacks in the same direction, control of all elements is simpler.

(2) ***Mutual Support***. Since all elements of the force are together throughout the attack, mutual support within the assaulting echelons is possible.

c. ***Attack on Converging Axes***. Using this method, the manoeuvre force closes with the enemy from more than one direction. This method entails following:-

(1) ***Surprise***. Since the attack converges on the enemy from multiple directions, it forces the enemy to fight in more than one direction and split his resources. The technique, therefore, results in achieving surprise.

(2) ***Coordination***. When elements of the manoeuvre force are attacking over difficult terrain with varying enemy resistance, the coordination, control and simultaneous assault may be difficult to achieve. Measures, therefore, must be taken to overcome this difficulty.

(3) ***Mutual Support***. Mutual support between elements of the force is more difficult especially in the earlier stages of attack. Later, when the force is closer to the objective, mutual support would be possible.

d. ***Tanks Support by Fire Only***. This method may have to be resorted, when difficult terrain and other obstacles prevent the tanks from moving in close support of the infantry. The method, therefore, does not fully utilize the mobility and shock effect of the tanks.

**Command Responsibility**

15. The command of combined tank and infantry operations cannot be governed by any hard and fast rules, but, to ensure that there is never any doubt in the battlefield, it is necessary that a commander for the combined operations must be appointed by the higher commander. Following guiding principles should be followed in deciding as to who will be the commander:-

a. As tank units cooperating with infantry battalion will usually be smaller in size, command will normally devolve on the infantry commander.

b. When exactly should the cooperating tank units or subunits be placed under command or in support of infantry will depend on the operation. This, in turn, will depend upon the factors of ground and strength and type of enemy opposition.

c. Tactical command should be exercised by the commander of the predominant arm. Thus, in a deliberate attack, as the infantry is normally the predominant arm, the infantry commander should therefore be in tactical command. In quick moving operation where armour plays the major role, the armour commander will naturally be in command.

16. It may sometimes be necessary for the command to pass from one commander to the other in the light of changing situations in the battle. This may sound unworkable or undesirable in theory but in actual operation, the changeover is normally smooth. For example, during the pursuit, when armour is predominant, it may become necessary for the deployment of a company or more of infantry to capture a position which has halted the advance of armour. The infantry commander either company or battalion, will assume command of the particular attack, supported by the tanks. As soon as the attack succeeds and the pursuit is resumed, the armour commander will assume command.

## SECTION 22 – ARMOUR – INFANTRY COMMUNICATION

**General**

1. Reliable and adequate communication between tanks and infantry is imperative for intimate cooperation and coordination in combined operations. It is also essential that the communication facilities of each are thoroughly known to the

other. This section deals with the communication facilities of armour and infantry and methods of intercommunication between the two.

2. ***Tank Wireless Set***. The principal means of communication between tanks and infantry is wireless. The supplementary means are tank telephone (or infantry head phone in T–series tanks) and visual signals, which have very limited utility. They are explained in the succeeding paragraphs. At present armour regiments are equipped with a variety of wireless sets for communication with infantry. The tank wireless set consists of the following three parts, each of which has separate transmitter and receiver:-

a. ***Set 1***. This is the main part which is meant to provide communications between tanks over long-ranges. It is used within a tank squadron or a regiment.

b. ***Set 2***. This set provides communication between tanks over a very short-range and is used within a tank troop or to communicate with infantry.

c. ***Intercommunication Set***. This is meant to provide facility for the tank crew to talk to each other with in the tank. The T–series tanks do not have set 2, rather they have RVC-8 for communicate with infantry. Tactical Communication between infantry battalion and squadron is shown in Figure 9.

d. Set 2 has frequency ranges which normally overlap with frequency ranges of infantry wireless sets thus making it possible for the two to establish direct communication with each other.

3. All tanks (less T–series, see paragraph 7 below) are equipped with an external telephone. This is fitted on the rear of the tank and can be used by the infantry to communicate with the individual tank commander in his tank. The telephone is connected to the internal communication circuit and all that is necessary to hold the telephone; one just needs to press a buzzer and speak. The telephone should later be replaced carefully at the end of the conversation as it is liable to become displaced or damaged, due to its exposed position.

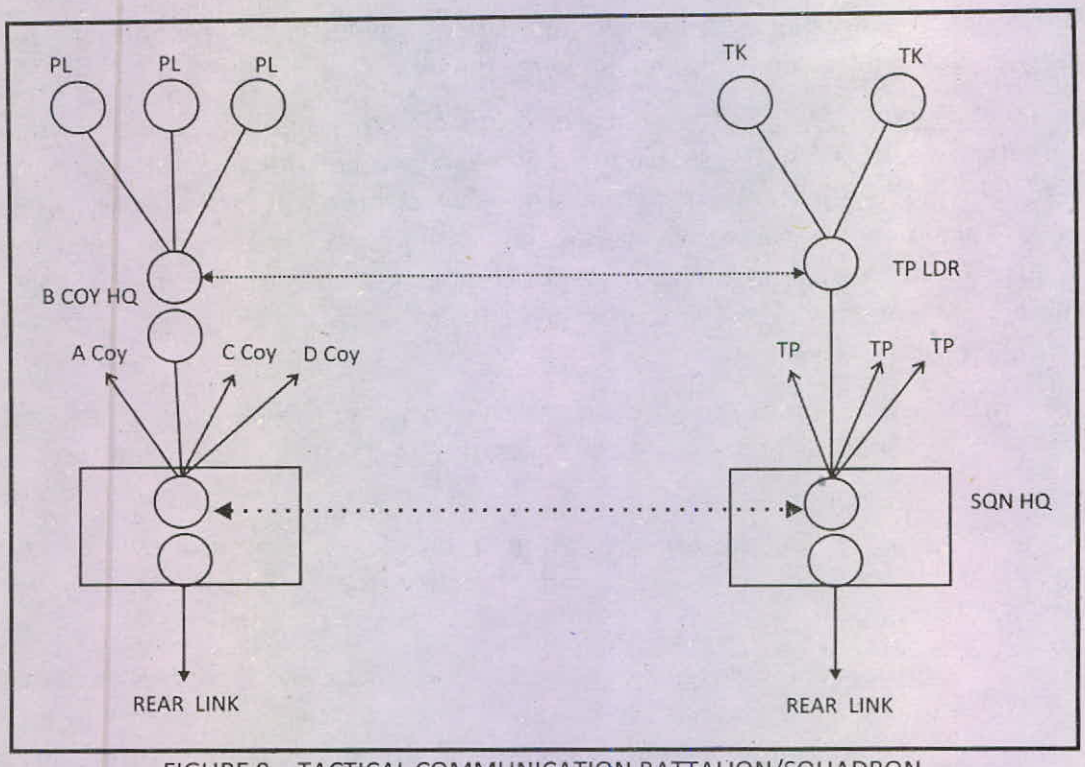


FIGURE 9 – TACTICAL COMMUNICATION BATTALION/SQUADRON.

**Infantry Headset**

4. T–series tanks do not have tank telephones, instead there is a headset placed at the back of the turret, which is connected to the intercommunication system of the tank. It is necessary to climb the tank, put on the headset and then speak to the tank commander.

**Visual Signals**

5. Very light pistol can be used upto a range of 200 metres to convey information from the infantry to the tanks in accordance with a prearranged code.

**Parachute Flares**

6. Fired by the 2 inch mortars of infantry to illuminate and is also used to indicate the general position of own troops in close country when contact with the supporting tanks has been temporarily lost. It is not possible to pinpoint a position with parachute flares as it is not very accurate.

**Tracer Ammunition**

7. This can be used by the automatics of the infantry and tank machine-guns to indicate targets to each other. In the battlefield when a lot of weapons will be firing, it may become difficult to differentiate fire, however, it helps in indicating targets when fire is explained using other means of communication.

**Flag/Weapons Signals**

8. These can be used in certain conditions. It is effective only if prior understanding and mutual observance has been achieved. It requires considerable training.

## SECTION 23 – TARGET INDICATION

**General**

1. When tanks and infantry are operating together, enemy targets are observed, located and dealt with automatically by tanks and infantry. The situation may, however, arise when one or the other is incapable of dealing with certain enemy targets, that the assistance of the other may be required. In such cases, there will be a requirement to clearly indicate the target. The method to indicate the target in such situations is termed as tank infantry target indication.

2. In most situations, infantry will need the assistance of armour to destroy enemy tanks or machine-guns; while the tanks will need the assistance of infantry for self-protection and for destruction of anti-tank weapons, which could be in positions where the fire of tanks is ineffective. It must be remembered that in most cases the tanks can destroyed whatever they can observe on ground.

**Means of Target Indication**

3. These are:-

   a. Radio.
   b. Tank telephone.
   c. Personal contact.
   d. Flag/hand/weapons signals.

**Sequence**

4. It is as under:-

   a. Attract the ATTENTION of the tank/infantry commander.
   b. Make him look in the desired DIRECTION.
   c. Give him the RANGE.
   d. Describe the details of the TARGET.
   e. Give him the EXECUTIVE ORDER.
   f. Observe and correct the fire when target is not correctly observed/ identified/engaged.

**OR**

(WHEN REFERENCE POINT IS PREARRANGED)

   a. As per para 4 a above (attract the Attention).

b. As per para 4 b above (Direction).

c. Indicate the Reference Point and indicate the target from the Reference Point, (Give Direction and Range from Reference Point).

d. As para 4 d above (Details of Target)

e. As per para 4 e above (Executive Order)

f. As per para 4 f above (Observe and Correct).

**OR**

(WHEN REFERENCE POINT IS NOT PREARRANGED)

a. As per para 4 a above (attract the attention).

b. As per para 4 b above (Direction).

c. As per para 4 c above (Give Range).

d. Give the description of reference point.

e. Give direction and range of target from the reference point and also describe the detail of the target.

f. As per para 4 e above (Executive Order).

g. As per para 4 f above (Observe and Correct).

**OR**

(CREATING AN ARTIFICIAL REFERENCE POINT)

a. As per para 4 a above (Attract the Attention).

b. As per para 4 b above (Right Direction).

c. As per para 4 c above (Give Range).

d. Explain the method of creating the artificial reference point.

e. Indicate target (explaining details of target) from artificial reference point (give direction and range of target from artificial reference point).

f. As per para 4 e above (Executive Order).

g. As per para 4 f above (Observe and Correct).

**Procedure – Using Radio**

5. ***Attracting the Attention***

a. The infantry radio will be used:-

(1) To offer the target using the tank prefix 99. This may be followed by the tank turret number or designation Alpha/Bravo etcetera when only that particular tank is visible to the infantry commander. Examples are as follows:-

(a) Hullo 2 for 99 target over.

(b) Hullo 2 for 99 target for turret 6 over

(c) Hullo 2 for 99 target for 2 Alpha over.

(d) Hullo 21 for 1 target over.

(e) Hullo 21 for 1 target for two Alpha over.

(f) Hullo 21 for 99 target for turret 6 over.

(g) Hullo 21 for 2 Alpha 99 target over.

(h) Hullo 21 for turret 6 target over.

***Note***: All these examples have been given to fulfil the requirements which are likely to occur due to numerous situations and variety of wireless sets or types of radio nets.

(2) To inform a tank commander that an infantry commander is about to move across to contact him either in person or through tank telephone.

b. The use of light or smoke signal is not recommended to attract the attention unless radio/telephone/personal contact is not possible and light or smoke signals are so well prearranged that no misunderstanding is possible.

6. ***Making the Commander Look in the Right Direction***. The infantryman has to imagine himself in the position of the tank and all indications are from that position:-

a. It is simple and quickest by prearranged reference points.

b. Using direction of tank's gun barrel/hull or axis of advance. For example:-

(1) Gun barrel – right – 3 O'clock; left – 7 O'clock etcetera (clearest method).

(2) Gun barrel quarter/half/three quarter/full right.

(3) Hull – left – 10 O'clock, right – 5 O'clock, etcetera (also very clear method but in hull down position it may not be possible to see the hull).

(4) Axis quarter/half/three quarter right/left etcetera (not very clear method; should be used when the other methods are not possible).

c. Firing WP smoke (best); composition; coloured not available with tanks but available with artillery; machine-gun burst (poor method – infantry may have to adopt it), mortar flares and high explosive bombs (infantry may have to adopt it, better than machine-gun) and very light fired in a direction (infantry may have to adopt it, poor method) for example:-

(1) 2 for 99 firing white phosphorus smoke for indication.

(2) 3 for 99 firing red very light for indication.

7. ***Range***. The range to the Reference Point being indicated or to the Target should be as accurate as possible. For example:-

a. 2 for 99, your gun axis; 2 O'clock; 500, AKELA DARAKHT; left 10 O'clock, 1000 CHAR JHARHIAN; sab se right wali JHARI, Recoilless Rifle on jeep, destroy over (The acknowledgement of the message has not been shown in the above example).

8. ***Description of Target***. Make full use of physical land marks; describe target in full detail; if target is obvious keep description minimum.

9. ***Indicating from Prearranged Reference Point***

a. Once the tank has been directed to look in the right direction, it is necessary to direct him to the Reference Point and from the Reference Point to the Target. This can be done either by linking the Reference Point to the Target along with the range by some physical land mark on the ground.

**OR**

b. Where above is not possible, by giving the distance and direction to the target from the Reference Point, in terms of yards LEFT, or RIGHT, ADD or DROP. In either case the possible description of the Target and its location should be given and again the infantryman must endeavour to imagine himself in the position of the tank commander.

10. ***Indicating From a New Reference Point***. On occasions, it may be necessary to indicate a new Reference Point when one has not been prearranged. In this case, first the direction will be given, followed by the approximate range to the Reference

Point and lastly a description of the Reference Point. The Reference Point will be 'named'. Once the Reference Point has been identified, then the same procedure as given in paragraph 9 above will be followed in directing the tank on to the target.

11. ***Creating an Artificial Reference Point***. Sometimes no suitable Reference Point can be found in the target area owning to the nature of the ground. In such cases, it will be necessary to create an artificial Reference Point, either by the infantry or by the tanks. The infantry will do this by firing a mortar bomb or ask main gun, giving the general direction and range. The point of burst of the shell then becomes the artificial Reference Point and the target is then indicated in the normal way.

12. ***Correction of Tank Fire when the Target is Not Identified***

a. If the tank commander fails to see the target or fires at the wrong one, the infantry will:-

(1) Give another description of the target.

(2) Use first shot for reference.

b. Correction will be given as:-

(1) Direction by clock rays and range method using first shot as reference.

**OR**

(2) "Left" or "Right", "Add" or "Drop" in hundreds of yards as the infantry observer sees it on the ground. The tank commander allows for variations in ground when applying gun corrections.

13. ***Executive Order or Disposal of the Target***. Once the target has been identified, he should be given the disposal instructions that is destroy, neutralize for so many minutes etcetera. The action being taken by the individual who is asking the fire should also be explained for example "I am moving around left flank" or "I am remaining behind cover". The decision about the weapon/ammunition will be that of tank/infantry commander, who is going to engage the target.

**Wireless Procedure**

14. The details have been given in the preceding paragraphs for various eventualities. The message below is an example only:-

a. Infantry – Hullo 2 for 99 target over.

b. Tank – 99 target over.

c. Infantry – 2 Axis your gun barrel right 1 O'clock – one thousand – TALL TREE – machine-gun destroy over.

d. Tank – 99 wilco out.

15. ***Method of Tank Infantry Designation of Reference Points***. As given in the preceding paragraphs.

**Procedure Using Weapons/Hand Signals**

16. These must be prearranged and practised. One should resort to these only when other means are not possible. For other signals see Armour Data and Reference Book 1972.

17. ***Attracting Attention of the Tanks***. This is done by the infantryman holding his rifle above his head at full extension of the arm. The rifle should be parallel with the ground, muzzle pointing in the direction of the enemy. The rifle is moved up and down till such time that tank acknowledges by moving its main gun up and down.

18. ***Distance to Target***. After the attention of the tank has been attracted, the infantryman gives distance to the target. This is done by stretching one arm parallel to the ground in the direction visible to tank and then carrying on the closing/ opening motion at the elbow. Every close and stretch motion indicates 25 metres.

19. ***Nature of Target***. The nature of the target is indicated with the help of thumb, rifle and steel helmet. Fist closed and thumb pointing down or muzzle of the rifle pointing to the ground indicates enemy anti-tank weapon. The thumb/muzzle of rifle pointing up means enemy machine-gun or any other automatic weapon. Removing the steel helmet from head and showing its top side indicates enemy armour.

## SECTION 24 – ORGANIZATION OF INFANTRY/TANK COMBINED TRAINING

**General**

1. Combined infantry/tank training could be organized as under:-

   a. To be conducted, during the individual training period, under arrangements of the battalion and armoured regiment.

   b. Combined exercises with troops during the collective training/battle inoculation period.

**Suggested Syllabus**

2. A suggested syllabus is given below:-

| Serial No. | Lecture/Demonstration | To be Attended by |
|---|---|---|
| a. | Organization and display of equipment of an infantry battalion. – 2 periods | All ranks of the armoured unit. |

| Serial No. | Lecture/Demonstration | To be Attended by |
|---|---|---|
| | Organization and display of equipment of an armoured regiment/squadron. – 2 periods | All ranks of the infantry battalion. |
| b. | ***Demonstration*** | |
| | Characteristics and capabilities of all weapons held in the infantry battalion and an armoured regiment/squadron. – 2 periods | All ranks of the infantry battalion and armoured regiment/squadron. |
| | Characteristics, capabilities and investigation of a tank. – 2 periods | All ranks of the infantry battalion. |
| c. | ***Lecture/Practice on the Sub Calibre Range*** | |
| | Target Indication. – 2 periods | Officers, Junior Commissioned Officers and Non-Commissioned Officers of the infantry battalion. |
| | Practice. – 6 periods | |
| d. | ***Lectures/Film*** | |
| | Recognition of armoured fighting vehicles. – 2 periods | Officers, Junior Commissioned Officers and Non-Commissioned Officers of the infantry battalion. |
| e. | ***Practice*** | |
| | Mounting and dismounting drills – 2 periods | Officers, Junior Commissioned Officers and Non-Commissioned Officers of the infantry battalion. |
| f. | ***Film/Demonstration*** | |
| | Troop tactics and drill film – 3 periods | Officers, Junior Commissioned Officers and Non-Commissioned Officers of the infantry battalion. |
| | Demonstration – 2 periods | |
| g. | ***Practice*** | |
| | Marrying up procedure – 2 periods | All ranks of both arms. |
| h. | ***Practice*** | |
| | Recognition of targets from inside a tank – 2 periods | Crew of 106 millimetres RRs only. |

| Serial No. | Lecture/Demonstration | | To be Attended by |
|---|---|---|---|
| i. | ***Lecture/Demonstration*** | | |
| | Tank hunting lecture | – 2 periods | All ranks of both arms. |
| | Demonstration | – 3 periods | |
| j. | ***Model Discussion/Tactical Exercise Without Troops*** | | |
| | Combined operations in advance and encounter Battle of both arms. Details as under:- | – 14 periods | Officers, Junior Commissioned Officers and Non-Commissioned Officers. |
| | (1) | Importance of teamwork. | |
| | (2) | Correct target indication. | |
| | (3) | Maximum utilization of tanks' capabilities. | |
| | (4) | Action of tanks on contacting opposition. | |
| | (5) | Action of infantry and tanks to clear a strong point. | |
| | (6) | Techniques of infantry/tank assault. | |
| k. | ***Model discussion/Tactical Exercise Without Troops*** | | |
| | Combined operations in deliberate attack. Details as under:- | – 14 periods | Officers, Junior Commissioned Officers and Non-Commissioned Officers of both arms. |
| | (1) | Importance of joint reconnaissance and planning. | |
| | (2) | Battle procedure and detailed marrying up in concentration and assembly areas and Forming Up Place. | |
| | (3) | Target indication and wireless procedure. | |
| | (4) | Assaulting the objective, when moving on same axis. | |
| | (5) | Reorganization. | |
| l. | ***Model Discussion/*Tactical Exercise Without Troops*** | | |
| | Night advance and attack. Details as under:- | – 14 periods | Officers, Junior Commissioned Officers and Non-Commissioned Officers of both arms. |
| | (1) | Capabilities/limitations of tanks at night. | |
| | (2) | Methods/techniques of guiding tanks at night. | |
| | (3) | Target indication methods at night. | |
| | (4) | Technique. | |
| | *To be conducted at night. | | |

| Serial No. | Lecture/Demonstration | To be Attended by |
|---|---|---|
| m. | ***Model Discussion/Tactical Exercise Without Troops*** | |
| | Defence. To study the employment of tanks in the counter penetration and counter-attack. – 14 periods | Officers, Junior Commissioned Officers and Non-Commissioned Officers of both arms. |

## SECTION 25 – TRANSPORTATION OF INFANTRY ON TANKS

**General**

1. During mobile operations and approach marches fatigue of infantry can be reduced and its movement speeded up by transporting it on tanks when no other means of transport are available. Besides this, it keeps the two arms together and conserves time for deployment. Also, the infantry may be mounted wherever wheeled vehicles are not able to operate.

2. This method has, however, the following limitations:-

   a. ***Vulnerability***. Infantry on tanks present a massed target particularly to High Explosive air bursts and mortars. They also prevent own tanks from traversing their guns. If infantry is carried, they must never be on the tanks of the leading troops, particularly when contact is likely.

   b. ***Fatigue***. Heat, dust and fumes can tire and daze infantry.

   c. ***Safety***. In close country, hazards of injury from trees or accidental falls are possible.

**Techniques of Transporting Infantry on Tanks**

3. ***Allotment of Infantry to Tanks***. Depending on the availability and type of tanks and tactical situation the allotment of infantry to tanks is as given in subsequent paragraphs. This is flexible and should be made according to the type of tanks and the modifications carried out to carry the infantry such as handle bars or ropes, or any other modification. Other considerations will be the enemy situation, terrain, the distance to be travelled and strength of infantry to be carried.

4. ***Carrying Capacity of Armoured Unit/Subunits***

   a. ***Normal***. One Infantry Company to one squadron of tanks.

   b. ***For Short Moves***. An armoured squadron may carry upto two companies mounted for short distance.

   c. During tactical moves companies should not be broken up to avoid loss of control.

5. ***Suggested Distribution of Company on a Squadron***

a. With a full strength of 14 tanks the suggested distribution is:-

(1) No infantry on the leading troop.

(2) Squadron headquarters carry company headquarters.

(3) 3 troops carry a platoon each. (One section per tank with platoon headquarters on the troop leaders tank).

b. With a reduced strength of 11 tanks the distribution should be:-

(1) No infantry on the leading troop.

(2) Squadron headquarters carry company headquarters.

(3) A platoon mounted on two tanks. One section and platoon headquarters on one tank and the remaining two sections on the second tank.

c. If the infantry commander feels the need, he could keep his 'O' group on the Forward Observation Officer's tank to reduce the time for issuance of further orders. During follow in zone operations, infantry commander may have to resort to such measures. However, the risk of losing the entire 'O' group through artillery fire or air strike should be considered.

6. ***Mounting Principles***

a. No infantry on leading tanks/troops.

b. Infantry to be dismounted on orders of tanks commander.

c. Infantry dismounted on mutual consent of tank and infantry commanders.

d. Safety of troops must be ensured.

7. ***Control***. Control over the mounted infantry is maintained by their commanders through:-

a. Infantry or squadron/troop radio communication, if radio silence is not imposed. It may not function satisfactorily when the tanks are moving.

b. ***Visual Signals***. Infantry must detail at least one person per tank to maintain visual direct personal contact with their respective commanders. Mounted infantry must not touch the tank radio antenna sections.

8. Tanks/Infantry Combined Operations in Various Operations of War. Refer Chapters III to VI of GSP-1595 – Tanks/Infantry Combined Operations.

## SECTION 26 – IDENTIFICATION OF TANKS

### General

1. Identification of own tanks is an important aspect of tanks and infantry operations. It is also important that troops should be able to distinguish between friendly and hostile tanks. Failure to do so, usually, results in engagements between friendly troops. Such engagements, besides causing casualties and damage to precious equipment, have an adverse effect on morale of troops. Nothing could be more distressing and demoralizing than to cause casualties by own action, even if by mistake. Correct identification of tank units in battle, therefore, is a subject of great importance.

### Methods of Identification

2. ***Recognition Training***. Peacetime training must include training of troops in recognizing own and enemy tanks. Following methods are suggested:-

a. Use of Armoured Forces Vehicles recognition charts.

b. Familiarization of infantry and other arms with our current tanks.

c. Familiarization with captured enemy tanks.

d. Use of miniature tank models.

e. Use of mock-up wooden/Hessian cloth models according to actual specifications and their location at various distances.

3. ***Identification Markings***

a. ***Arabic Numerals***. Tanks in an armoured regiment are serially numbered on the turret in bold Arabic numerals in white. The numbers commence from the commanding officers tank in the regimental headquarters and continue through A, B and C squadrons as under:-

| | | | |
|---|---|---|---|
| (1) | Regimental Headquarters | – | 1-2 |
| (2) | A Squadron | – | 3-16 |
| (3) | B Squadron | – | 17-30 |
| (4) | C Squadron | – | 31-44 |
| (5) | Dozers | – | 45-47 |
| (6) | Armoured Recovery Vehicles | – | 48-50 |

b. These numbers not only facilitate identification of own tanks, but, also indicate their subunit. For example, a tank bearing number 19 will belong to Number 1 troop of B Squadron.

c. B vehicles of an armoured regiment are painted on both sides with a squadron code and also serially numbered. The codes are:-

(1) A Squadron

(2) B Squadron

(3) C Squadron

(4) Regimental Headquarters

4. ***Recognition Signals***. Prearranged visual signals should be used by tanks and infantry for mutual recognition. One form of visual signals is very light flares in a prearranged combination to be used by each side just as a password or counter password. For security reasons the combination should be changed, periodically.

5. ***Gun Direction***. Main gun of the tanks must always point towards the enemy, irrespective of the direction of movement of the tanks. This practice provides a certain measure of recognition.

6. ***Fire Discipline***. This implies opening of fire after positive recognition of the target. The crews of tanks and anti-tank guns should be trained to hold their fire until the target is well within range not only for positive recognition but also for destruction.

7. ***Battle Procedure***. Following practices must always be adhered to in battle for mutual identification:-

a. Laying down of inter unit/formation boundaries.

b. ***Coordination and Liaison with Flanking Units***. The responsibilities for coordination in principle are mentioned below, however these do not absolve others from their responsibility:-

(1) From higher to lower headquarters and from left to right.

(2) The unit/formation manoeuvring from the flanks or passing through will be responsible for coordination with the formations/units/subunits being passed through or around the flanks.

(3) The relieving formations/units are responsible for coordination with those being relieved.

(4) Formation/units effecting any changes in their operations/ direction of movement are responsible to coordinate with all concerned.

# CHAPTER XI – COOPERATION OF ARTILLERY AND INFANTRY

## SECTION 27 – ALLOTMENT AND TYPES OF ARTILLERY SUPPORT

**Methods of Allotment**

1. Artillery units or subunits may be allotted to an infantry battalion during a particular operation. Artillery so allotted may be placed:-

a. Under command.

b. In direct support.

c. In support.

d. At priority call.

e. To reinforce.

2. ***Under Command***. When artillery is placed under command of an infantry battalion, the battalion commander is responsible not only for the operational activities like movement/deployment etcetera of the artillery but also for its full maintenance, which will include the provision of all types of ammunition. The field battery is the artillery subunit most likely to be placed under command of a battalion. It must be emphasized, however, that the placing of artillery under command of a battalion is not a normal procedure. It can be rarely done only during operations, which are generally so fluid that it is not possible for commanders to effectively control the battle as a whole.

3. ***In Direct Support***. An artillery unit, placed in direct support of a unit, has its fire guaranteed to that unit. It will be normal for infantry battalions to have a battery of field regiment in direct support, which provides liaison, communication and observation for the utilization of its fire. The direct support field battery commander is the main adviser to the battalion commander on the employment of all available fire support. Authority to order the guns to move into or out of action remains with the higher artillery commander. Artillery in direct support of one unit cannot be at priority call to another unit at the same time.

4. ***In Support***. Artillery is said to be in support of a battalion, when its fire support to that battalion, is subject to the availability conditions at any particular time and hence cannot be guaranteed. Artillery allotted in support of a unit may be in direct support or at 'priority call' to some other unit.

5. ***At Priority Call***. This term indicates a priority or reservation of fire support allotted to more than one formation or unit e.g. an artillery unit of one formation allotted to reinforce another formation, artillery may remain at priority call to its parent formation for, say defensive fire. Guns at priority call cannot be in direct

support to another unit/formation at the same time. Artillery units at priority call do not normally provide liaison and observation.

6. ***To Reinforce***. An artillery unit allotted to reinforce another artillery unit/formation will augment, subject to any restrictions or limitations imposed by the higher commander, the fire of the reinforced artillery unit/formation. It will establish liaison and communication with the reinforced artillery unit/formation and will deploy its observation facilities according to the order of the commander of reinforced artillery.

**Types of Artillery Tasks**

7. ***Concentrations***. Concentrations, in which the fire of a number of guns is put down on a particular target, are the most common form of artillery support. These are of two types:-

a. ***Simple Concentrations***. This is ordered as a grid reference or a set of coordinates. In reasonably good conditions, a simple concentration by a regiment should cover an area of about 275 x 275 metres.

b. ***Linear Concentration***

(1) This type of concentration is used when the fire of the guns is required to be spread out over a given line with no depth. It is indicated by registering both ends of the target as it extends on the ground. The length of the linear target could be upto 475 metres. The minimum number of guns required to engage a given length is as under:-

| Serial No. | Length | Field/Medium/ Heavy | 120 Millimetres Mortar |
|---|---|---|---|
| (a) | 183 metres | Battery | Troop |
| (b) | 475 metres | Two batteries | Two troops |

(2) This does not mean that a target of 183 metres frontage should be engaged only by a field battery etcetera. The officer planning the fire support may use the entire divisional artillery on that target. Concentrations are most economical form of artillery support, because all ammunition is fired at the known or suspected enemy position. For concentrations to be effective, accurate information regarding the enemy position must be available.

8. ***Standard Barrage***. This takes the form of a curtain of fire, which is made to move in front of the advancing troops. It is employed only in covering fire.

**Types of Artillery Support**

9. ***In Attack***

   a. Preparatory bombardment.

   b. Covering fire.

   c. Defensive fire, to cover the period of reorganization on the objective.

10. ***In Defence***

   a. Defensive Fire (DF).

   b. Support of counter-attacks (covering fire for counter-attacks).

11. ***In Attack or Defence***

   a. Counter-Bombardment (CB).

   b. Harassing Fire (HF).

   c. Smoke screens.

   d. Target indication.

   e. Battlefield illumination.

12. ***Preparatory Bombardment***

   a. This may have one of the three aims:-

      (1) To weaken the resistance of the enemy by inflicting casualties damaging his equipment and by destroying his fieldworks.

      (2) To demoralize the enemy so that he will offer little or no resistance to an assault.

      (3) Disrupt enemy command and control at critical stage of battle by destroying his communications.

   b. If resources for achieving demoralization are insufficient, long preparatory bombardment is probably not justified owing to:-

      (1) Heavy ammunition expenditure.

      (2) Loss of surprise.

      (3) Only temporary neutralization being achieved, although a certain degree of disorganization is still inflicted.

   c. A brief but intense preparatory bombardment on confirmed and suspected enemy weapons and headquarters immediately before an attack is very valuable. Disorganization and confusion may be inflicted on the enemy and his defensive plans may be interrupted. Such bombardment should not cause any appreciable loss of surprise.

13. ***Covering Fire***

a. The object of covering fire is to prevent enemy from operating his weapons, which can engage the assaulting troops, in an attack or counter-attack.

b. Covering fire for an attack should always be planned in advance and may be:-

(1) ***A Timed Programme***. May take the form of a concentration. This method is essential for the large scale set piece attacks. Disadvantages are, its rigidity, its effectiveness being dependent on the attacking troops maintaining their expected rate of advance, the time lag and element of risk involved in modifying it. But none of these snags are difficult to overcome.

(2) ***Concentrations on Call***. Useful for the support of subsequent phases of attack after a timed opening phase. It is a flexible form of support as the observer can control the fire to conform to the requirements of the attacker. This method depends on good observation, good visibility and first class communications, any of which may fail at the critical moment. This can be used in conjunction with a timed programme to provide prearranged tasks on targets likely to interfere.

(3) ***Impromptu Covering Fire***. Such fire may be required for dealing with opposition which suddenly discloses itself during the attack or which has been ineffectively dealt with by the fire support plan. It can also be used to extend the programme of attack beyond the period of planned support. The task of providing this fire devolves primarily on the Forward Observation Officer (FOO) accompanying the attack.

c. Usually a judicious mixture of the three foregoing methods will form the covering fire schedule for any attack.

d. ***Heavy Mortars***

(1) 120 millimetres Mortar must be included in the fire support plan. Due to their beaten zone and safety factor, they should not normally be used against targets closer to own troops.

(2) 120 millimetres Mortar will also be given the task of providing impromptu covering fire, after completion of timed programmes.

(3) The main task of heavy mortars, however, is counter mortar.

14. ***Defensive Fire (DF)***. The types of defensive fire are:-

a. ***Defensive Fire Tasks in Depth***. To disorganise the preparations of the enemy for attack, or counter-attack and to inflict casualties on his reserves during the assault. Targets include headquarters, communications and assemblies/concentrations.

b. ***Close Defensive Fire Task***. To break-up the concentration of attacking troops while forming up, crossing the Start Line or during the actual assault. The field regiment in direct support of each brigade can provide immediate response to calls for these tasks and the additional artillery, through the normal fire-control channels.

c. ***Defensive Fire (SOS)***. The close defensive fire tasks, which cover the most dangerous approaches, are nominated by battalion commander as defensive fire (SOS) tasks. There can be only one defensive fire (SOS) task for each artillery battery or mortar troop in direct support as mortars/guns in direct support will be laid on them, when actually not engaging other targets. They can be changed, at any time, at the request of the battalion commander.

15. The object of recording defensive tasks is to obtain quick and accurate fire. Following practical difficulties must also be appreciated:-

a. The work and time involved in recording a large number of targets.

b. Other commitments of artillery, for example counter-bombardment, which may limit their ability to fire defensive fire.

c. Limitations of ammunition.

d. It is not indented to limit the number of defensive fire tasks, however following points must be borne in mind in planning and calling for defensive fire tasks:-

(1) Artillery should not be expected to ring battalions with a series of regimental linear concentrations.

(2) Close defensive fire tasks must be reserved for the most likely lines of approach which the infantry cannot appropriately cover with their own weapons.

(3) Whenever possible a simple concentration should be used in preference to a linear concentration. This not only reduces the technical work of artillery, but is also easier to move, than a regimental or battery linear concentration. Linear concentrations should only be demanded when it is really inescapable.

(4) The weight of fire on a defensive fire task should not be greater than what the circumstances demand.

16. In attack, defensive fire tasks during reorganization on the objective must be laid down in the plan and selected off the map. Close defensive fire tasks may have to be appropriately adjusted after the objective has been captured.

17. ***Difference between Preparatory and Covering Fires***

a. ***Preparatory Fire***

(1) Aims at inflicting casualties and destroying equipment before the main battle.

(2) Not necessarily related to the movement of troops.

(3) Desirable and possible to concentrate firepower on one target at a time.

(4) Targets are generally engaged in no definite pattern.

b. ***Covering Fire***

(1) Related to the movement of troops.

(2) Firepower distributed and timed to prevent enemy from operating his weapons, that may hinder movement of own troops.

(3) Targets generally engaged from enemy's front to his rear.

18. ***Harassing Fire (HF)***

a. Objectives of harassing fire are:-

(1) To reduce the morale of enemy by interfering with the movement of his troops and supplies and by allowing him no rest.

(2) To delay the approach of enemy.

b. Harassing fire involves all long-range weapons and must be carefully coordinated along with any air effort if allotted and with own deep patrolling activity.

c. Harassing fire tasks are coordinated at the highest convenient level, usually at the corps level.

19. ***Smoke***

a. Smoke screens are used:-

(1) To blind enemy observation and to prevent observed fire by his artillery and aimed fire by his small arms and anti-tank weapons.

(2) To screen the movement of own troops.

(3) For deception.

b. Smoke may be fired:-

(1) To indicate bomb lines and targets to the Pakistan Air Force, particularly with coloured smoke.

(2) To indicate the beginning or end of a particular phase of fire support plan.

(3) In a concentration to impair the vision of and to confuse the enemy.

c. Care must be taken not to blind or impede own troops. Meteorological conditions considerably influence the effectiveness of smoke screen. Wind direction must always be kept in view when planning a smoke screen.

## SECTION 28 – ARTILLERY TARGET INDICATION

1. This procedure is used between artillery and other arms for indication and, if necessary, engagement of targets.

2. In this section the artillery observation post officer is referred to as 'The Observer'. The officer of the supported arm who is asking for fire is referred to as 'The Supported Arm Representative'.

**Communications**

3. Artillery observers, forward observation officers and battery commanders are provided with wireless sets of the type used by the unit that they are supporting. These sets are normally netted on the battalion net in case of infantry. In certain circumstances it may even be necessary for an observer or forward observation officer to be on a company net.

4. The artillery sets use the following standard call signs:-

| | | | |
|---|---|---|---|
| a. | Battery commander of field battery | – | 96 |
| b. | Senior observer of field battery | – | 96A |
| c. | Junior observer of field battery | – | 96B |
| d. | Junior observer of field battery | – | 96C |
| e. | Troop commander of artillery mortar battery | – | 96D |

5. ***Procedure***. This is based on normal voice procedure but is modified to comply with certain aspects of artillery fire orders, as follows:-

a. All transmissions are answered back in full.

b. Figures are not preceded by the pro-word 'Figure'.

c. Orders that are sent or answered back wrongly are corrected by the use of the word 'Wrong' followed by the correct order or part of the order. The reply is always prefaced by the word 'Wrong'. Illustrations of this procedure are shown in example (b).

**Indication and Engagement of Targets**

6. To warn all stations on the net, the first call from the supported arm representative to the observer will be 'target', 'target', 'target' over.

7. The observer should then be told all the needs to know about the target as briefly as possible, for example:-

a. Where it is?

b. Observer post to target bearing.

c. What it is?

d. What is to be done to it?

e. When and for how long?

8. Further explanation is as follows:-

a. ***Where it is***? The normal method of indicating location of the target is by a grid reference which is always given in clear e.g. 'grid reference 321502'. If for some reasons the infantry commander is unable to indicate the target by a grid reference then one of the following alternative methods may be used:-

(1) By a Target Grid Correction from a Reference Point. The reference point should be obvious and clearly visible, or previously agreed, that is:-

'Reference bridge square 5972. Oscar Tango 110 degrees. Go right 200 add 100'.

The reference point could also be previously recorded target known to both the infantry commander and the observer that is:-

'Reference target Mike 2468. Oscar Tango 45 degrees. Go left 200 add 100'.

Where the supported arms representative can identify the target he should also give the grids reference of the target to the observer along with the observer post to target bearing.

(2) By a target grid correction from a battery salvo. In badly mapped, featureless country the observer may be asked to provide a reference point in the shape of a rough indication as to where the salvo is required, for example, by reference to the axis of advance, a map square, etcetera e.g.:-

`Fire salvo for reference point in square 4250'.

The supported arm representative will then give a target grid correction from this salvo to the target (see Example C).

(3) By the firing of tracer or colour smoke at the target by the supported arm.

b. ***Observer Post to Target Bearing***. This is the grid bearing (unless specified as magnetic) of the line from the supported arm representative to the target. This bearing must be reported every time target is engaged, even if it has been engaged previously, since it is not recorded at the gun position. The bearing is sent to the nearest degree or 20 mils. If the bearing is sent as:-

'Oscar Tango six zero degrees' or 'Oscar Tango six zero degree magnetic or Oscar Tango 4840 mils'.

It will be assumed by the observer that it is a grid bearing. If the supporting arms representative is unable for any reason to convert the bearing to grid, he must state that it is a magnetic bearing, e.g.

'Oscar Tango six zero degrees magnetic'.

It is emphasized that the Observer post to target bearing is always the bearing from the supported arm representative to the target and not to any reference point that may be used for indicating the target.

c. ***What it is***? This information should include:-

(1) A tactical description of the target to enable the observer and possibly any higher artillery commander to assess the weight of fire merited by the target, for example:-

'Four machine-guns' or 'Five hundred infantry forming up for attack'.

(2) A physical description of the target to enable the observer to identify the target, for example 'Green bushes in brown field'; or twin trees on the knoll etcetera.

d. ***What is to be Done to it***? The type of fire support required, for example neutralize or smoke.

e. ***For How Long and When***? This refers to the timings of fire for effect. The most usual forms are:-

(1) ***'Neutralize Now'***. This implies that the fire is required immediately. The observer will bring down fire for effect as soon as possible.

(2) Neutralize At..... Hours, For ..... Minutes. Fire for effect is not required immediately but at the time stated for the duration stated.

(3) Neutralize For ..... Minutes. Report when ready to engage. The supported arm representative has no definite time in view. The observer will report `Ready' to the supported arm representative as soon as ranging is completed and he is ready to engage the target with fire for effect.

9. With this information the observer will normally be able to take on the engagement of the target. If, however, for any reason, he is unable to do so, then the supported arm representative will have to carry-out the shoot himself, passing his orders to the observer who will edit them and pass them back to the guns on artillery net using normal artillery procedure and communication discipline.

**Target Grid Correction** (see Figure 10)

10. The observer corrects fire on to the target with reference to the line Observer post to target:-

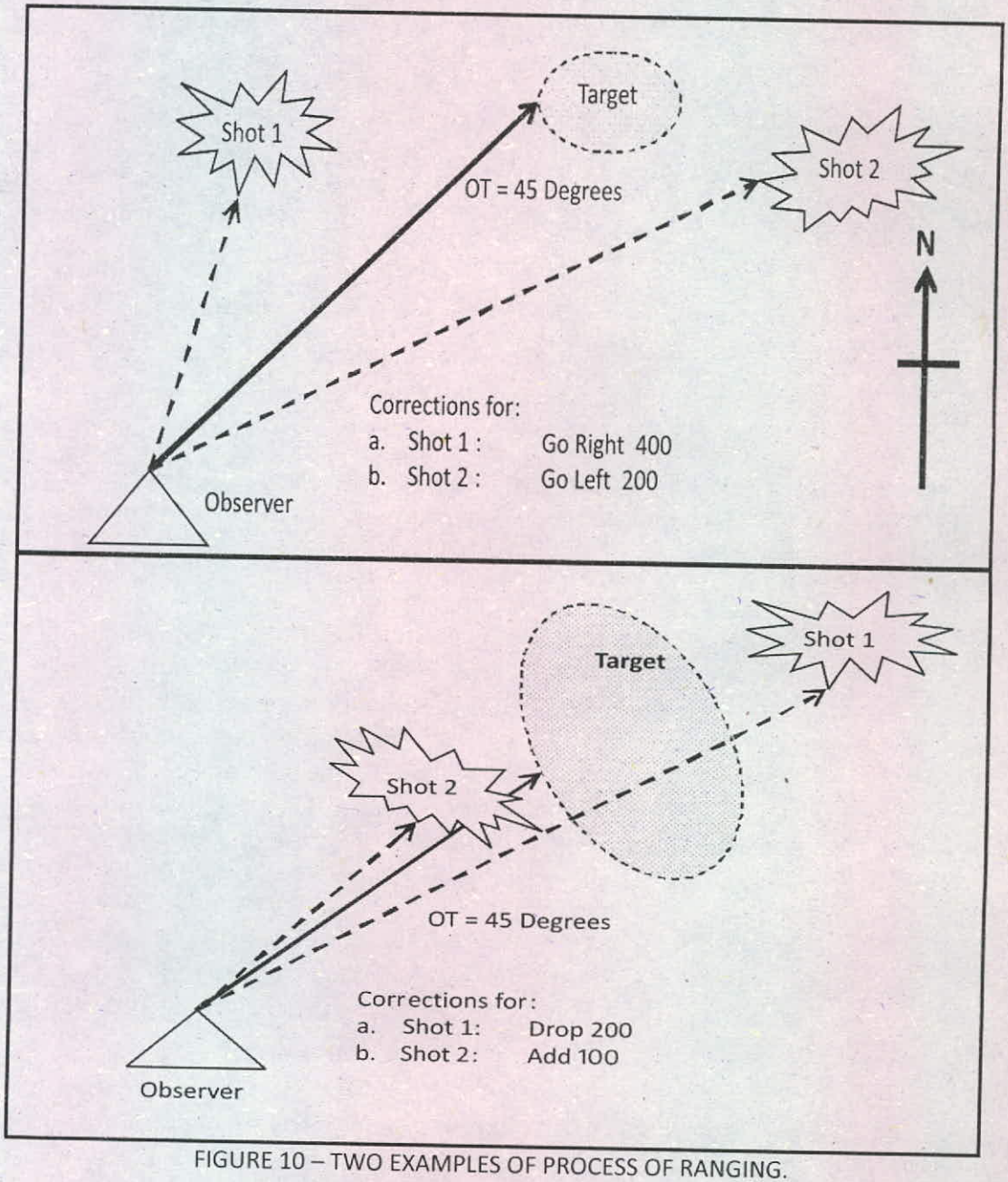


FIGURE 10 – TWO EXAMPLES OF PROCESS OF RANGING.

a. Corrections are given as follows:-

| Position of round in relation to the line observer post to target or target | Correction |
|---|---|
| Left of line observer post to target | – Go right 400 |
| Right of line observer post to target | – Go left 200 |
| Beyond the target | – Drop 200 |
| Short of the target | – Add 100 |
| On target | – Target |

b. ***Ranging***. The process of ranging consists of:-

(1) Correcting the fire on to the line observer post to target to the nearest 25 metres.

(2) Bracketing along the line observer post to target until a 'Short bracket' of 100 metres is obtained.

(3) When the supported arm representative is experienced and has good command, he may order a double correction, for example 'Go right 200, Add 400'.

c. ***Deflection Correction***. To judge how big a correction to order, measure with the graticules of a binocular, or with the hand, by how many degrees the round missed the line observer post to target, judge, or measure from the map, the distance observer post to target and apply the rule 'At 914 metres (1000 yards) one degree subtends 16 metres' and order a correction to the nearest 25 metres. For example, if the distance Observer post to target is 2000 metres and the round falls six degrees to the right of the line Observer Post to target, order 'Go left two hundred' (6 x 2 x 17 = 204).

d. ***Bracketing***

(1) Once rounds are falling on or near the line observer post to target, it is easy to see whether they are beyond the target 'over', or short of the target 'short'.

(2) It is difficult to judge how far 'over' or 'short' a round falls. If the first round is 'short' correct boldly enough to make sure that the next is 'over' and vice versa; at this stage always correct by 400 or 800 (or 200 or 1600) metres apart, which is also known as a long bracket.

(3) If the order that produced the long bracket was 'add 800', the next order should be 'drop 400'. If the next round falls 'short', the order should be 'add 200' and so on, progressively halving the bracket until one round falls 'over' and one 'short' with only 100 metres between them (a short bracket). Then the last order should be 'add or drop 50'. On target.

(4) If, after an initial 'drop or add ..... 800' correction, the next round still falls 'over' or 'short' order should be 'drop or add 400 or 800 or exceptionally even 1200 or 1,600' making sure that this correction is big enough to suitably bracket the target.

(5) If a round hits the target during ranging, order 'On target', at once.

**Procedure during the Shoot**

11. The observer will check back the orders and finish his transmission with 'wait out'. He will then see if he can identify the target. If he can, he will report 'target identified' and carry on with the ranging. If he cannot identify the target he will report to the supported arms representative 'target not identified, send corrections'. When this occurs, or when the supported arm representative has called for a salvo to be fired as a reference point, the procedure is as follows:-

a. The artillery will normally fire a battery salvo with the guns concentrated at the point indicated. However, if limited ammunition is available, only one gun may be used for ranging.

b. When the guns fire, the observer duly reports to the supported arm representative.

c. The supported arm representative then corrects the mean point of impact of the salvo on to the target as described in paragraph 10.

d. The observer will observe each salvo, if it is visible to him and on receipt of the supported arms representative's correction, will try to identify the target. If he does not identify it another salvo will be fired and the supported arm representative will again correct it. If at any time during the ranging the observer does identify the target he will immediately report `target identified'. He then takes full charge of the shoot and no further traffic passes on the net of the supported arm.

e. Any unobserved salvo may be reported by the supported arm representative as `unobserved', if he feels unable to order a correction and bring the salvo into his view. The observer will then decide whether to fire another salvo so as to give the supported arm

representative a better chance of observing it, or whether to fire smoke. The supported arm representative may also request smoke if he thinks it would be difficult to observe High Explosive.

**Fire for Effect**

12. The observer will bring down fire for effect on to the target at the time it is required by the supported arm representative who must inform whether or not the fire was effective and if necessary, should order a correction. If he also wishes the target to be recorded for future use, he may ask for it. He will then be given the target number by the observer.

13. ***Repeat***. During ranging or at fire for effect if the same number of round or rounds are required to be fired again the order 'repeat' is used.

**Examples**

14. In the following examples note the use of the particular rules for this procedure:-

a. All transmissions answered back in full.

b. Pro-word 'Figure' not used.

c. Use of pro-word 'Wrong' to correct an order or part of an order.

15. Where necessary transmission may be broken down into shorter and more convenient lengths, for example the transmission containing the initial orders. When this is done the link must still be maintained by the use of `Over', until all the initial orders are through.

16. ***Example (a)***. The simplest case where the observer immediately identifies the target:-

| Serial No. | B Company Commander | Observer | Notes |
|---|---|---|---|
| a. | Hello two for nine six Bravo. Target, target, target. Over. | – | Warning order. |
| b. | – | Nine Six Bravo, Target, target, target. Over. | – |
| c. | Two. Grid reference two four nine five two three. Oscar Tango five zero degree.<br>Two machine-guns in ditch at cross-roads. Neutralize for three minutes now. Over. | – | What it is observer post to target.<br>Bearing.<br>What it is?<br>What to do?<br>When and how long? |

| Serial No. | B Company Commander | Observer | Notes |
|---|---|---|---|
| d. | – | Nine six Bravo Gird reference two four nine five two three. Oscar Tango five zero degrees.<br>Two machine-guns in ditch at cross-roads. Neutralize for three minutes now. Wait. Out. | – |
| e. | – | Hello nine six Bravo for two, Target identified. Over. | – |
| f. | Two. Target identified. Out. | – | – |

The observer now carries out the shoot on his own artillery net and engages the target as soon as he is ready.

17. ***Example (b)***. The target is not identified by the observer and the supported arm representative completes the shoot. Example of use of word `wrong' in the procedures area given:-

| Serial No. | D Company Commander | Observer | Notes |
|---|---|---|---|
| a. | Hello four for nine six Charlie. Target, target, target. Over. | – | – |
| b. | – | Nine six Charlie, Target, target, target. Out. | – |
| c. | Four. Reference target Mike, Nine three two one. Oscar Tango one one degrees. Go left two hundred. Four mortars in depression. Neutralize for three minutes at one four one five hours. Over. | – | – |
| d. | – | Nine six Charlie. Reference target Mike Nine three two one. Oscar Tango two one degrees. Go left two hundred. Four mortars in depression neutralize for three minutes at one four one five hours. Wait. Out. | The observer answers back the Oscar Tango bearing correctly in. |

| Serial No. | D Company Commander | Observer | Notes |
|---|---|---|---|
| e. | Hello four for nine six Charlie. Wrong Oscar Tango one degrees. Over. | – | The supported arm representative sends that part of the order which was answered back incorrectly. The corrected word or words would be stressed. |
| f. | – | Nine six Charlie Wrong Oscar Tango one degrees. Out. | The observer fails to identify the target. |
| g. | – | Hello nine six Charlie for four Targets not identified send corrections. Over. | – |
| h. | Four Targets not identified. Send corrections. Out. | – | – |
| i. | – | Hello nine six Charlie for four Shot. Over. | – |
| j. | Four Shot. Out. | – | – |
| k. | Hello four for nine six Charlie Go left one hundred. Over. | – | – |
| l. | – | Nine six Charlie Go left one hundred. Out. | – |
| m. | – | Hello nine six Charlie for four shot. Over. | – |
| n. | Four Shot. Out. | – | – |
| o. | Hello four for nine six Charlie Drop two hundred. Over. | – | – |
| p. | – | Nine six Charlie Drop four hundred. Out. | The observer answers back incorrectly. |
| q. | Hello four for nine six Charlie. Wrong Drop two hundred. Over. | – | The supported arm representative sends the order again, stressing the part corrected. |

| Serial No. | D Company Commander | Observer | Notes |
|---|---|---|---|
| r. | – | Nine six Charlie Wrong Drop two hundred. Out. | – |
| s. | – | Hello nine six Charlie for four Shot. Over. | – |
| t. | Four Shot Out. | – | The salvo lands on the target. |
| u. | Hello four for nine six Charlie on target. Over. | – | – |
| v. | – | Nine six Charlie on target. Out. | The observer orders fire for effect at 1415 hours on artillery net. |
| w. | – | Hello nine six for four Shot. Over. | – |
| x. | Four Shot. Out. | – | – |
| y. | Hello four for nine six Charlie target neutralized – Record as target. Over. | – | – |
| z. | – | Nine six Charlie target neutralized, record as target. Out. | – |

18. ***Example (c)***. Using a battery salvo as a reference point.

| Serial No. | A Company Commander | Observer | Notes |
|---|---|---|---|
| a. | Hello one for nine six alpha. Target, target, target. Over. | – | – |
| b. | – | Nine six Alpha, Target, target, target. Out. | – |
| c. | One, fire salvo for reference point in square four two five six Oscar Tango five five degrees. Two machine-guns in bushes on hill side near track. Neutralize for three minutes from one four hundred hours. Over. | – | – |

| Serial No. | A Company Commander | Observer | Notes |
|---|---|---|---|
| d. | – | Nine six Alpha Fire salvo for reference point in square four two five six Oscar Tango five degrees Two machine-guns in light bushes on hill side near track. Neutralize for three minutes from one four hundred hours. Out. | – |
| e. | – | Hello nine six Alpha for one Shot. Over. | – |
| f. | One shot out. | – | A battery salvo is fired in square our two five six. |

## SECTION 29 – USEFUL DATA

1. ***Ranges and Ammunition***

| Serial No. | Equipment | Maximum Range (Metres) | Ammunition (Rounds per Gun) | |
|---|---|---|---|---|
| | | | First Line | Second Line |
| a. | 120 millimetres Mortar (POF) | 5700 | 120 | 60 |
| b. | 120 millimetres Mortar (Brandt) | 6602/5669 | 120 (20 Smoke) | 60 |
| c. | 3.7 inch Howitzer | 6218 | 120 (20 Smoke) | 60 |
| d. | 25 Pounder | 12253 | 120 | 60 |
| e. | 105 millimetres Howitzer | 11274 | 120 | 60 |
| f. | 5.5 inch Gun | 16551/14813 | 70 | 35 |
| g. | 155 millimetres Howitzer | 14596 | 70 | 35 |
| h. | 155 millimetres Gun | 23514 | 90 | 45 |
| i. | 8 inch Howitzer | 16925 | 60 | 30 |
| j. | 122 millimetres Howitzer | 11799 | 120 | 60 |
| k. | 130 millimetres Gun | 27478 | 100 | 50 |
| l. | 122 millimetres Multi Barrel Rocket Launcher | 20579 | 60 | 30 |

2. ***Frontages Covered by Fire Units***

| Serial No. | Equipment | One Battery/ Troop Mortar (Metres) | Regiment/Mortar Battery |
|---|---|---|---|
| a. | 120 millimetres Mortar (Pakistan Ordnance Factory)/ 120 millimetres Mortar (Brandt) | 200 | 400 |
| b. | 3.7 inch Howitzer | 150 | 200 |
| c. | 25 Pounder | 200 | 300 |
| d. | 105 millimetres Howitzer | 200 | 300 |
| e. | 5.5 inch Gun | 300 | 400 |
| f. | 155 millimetres Howitzer | 300 | 400 |
| g. | 155 millimetres Gun | 200 | 400 |
| h. | 8 inch Howitzer | 300 | 400 |
| i. | 122 millimetres Howitzer | 200 | 300 |
| j. | 130 millimetres Gun | 300 | 400 |
| k. | 122 millimetres Multi Barrel Rocket Launcher | 400 x 600 | Battery of 6 Multi Barrel Rocket Launchers |

3. ***Smoke (Guaranteed-in Ideal Meteorological Conditions)***

a. One battery 183 metres (200 yards) 5 points.

b. Two batteries 366 metres (400 yards) 10 points.

c. Regiment 548 metres (600 yards) 15 points.

4. ***Rate of Fire (Round per Gun per Minute)***

| Serial No. | Equipment | Intense | Rapid | Normal | Slow | Very Slow |
|---|---|---|---|---|---|---|
| a. | 25 Pounder | 5 | 3 | 2 | 1 | ½ |
| b. | 3.7 inches Howitzer | 5 | 3 | 2 | 1 | ½ |
| c. | 5.5 inch Gun | 2 | 1½ | 1 | 2/3 | 1/3 |
| d. | 105 millimetres Howitzer | 5 | 3 | 2 | 1 | ½ |

| Serial No. | Equipment | Intense | Rapid | Normal | Slow | Very Slow |
|---|---|---|---|---|---|---|
| e. | 155 millimetres Howitzer | 2 | 1½ | 1 | 2/3 | 1/3 |
| f. | 8 inch Howitzer | 1½ | 1 | ½ | 1/3 | 1/6 |
| g. | 120 millimetres Mortar | 5 | 3 | 2 | 1 | ½ |
| h. | 122 millimetres Howitzer | 5 | 3 | 2 | 1 | ½ |
| i. | 130 millimetres Gun | 2 | 1½ | 1 | 2/3 | 1/3 |
| j. | 155 millimetres Gun | 2 | 1½ | 1 | 2/3 | 1/3 |
| k. | 122 millimetres Multi Barrel Rocket Launcher | (Loaded Rocket Launcher can fire a salvo of 30 rockets in 15 seconds and the reload time is 10 minutes). | | | | |

5. ***Timings***. The following deployment timings are excluding the travelling time:-

a. ***Registration***. Generally, twenty minutes per successive registration plus twenty minutes for other preparations and reserve, may be used as basic guide.

b. ***Deployment***

| | | Field | Medium |
|---|---|---|---|
| (1) | **One Battery** | | |
| (a) | Day | 20 minutes | 30 minutes |
| (b) | Night | 40 minutes | 1 hour |
| (2) | **Regiment** | **Field** | **Medium** |
| (a) | Day | 45 minutes | 1 hour |
| (b) | Night | 90 minutes | 2 hours |

6. ***Frontage and Safe Distances***

| Serial No. | Equipment | Maximum Effective Frontage (in Metres) Covered by One HE Shell in Hard Ground Against the Troops in Open | Distance in Battle, Troops Can Approach Point of Burst in Reasonable Safety | | Maximum Distance in Peacetime Practice from the Point of Burst to the Spectators or the Range Boundary | |
|---|---|---|---|---|---|---|
| | | | Over-head (Metres) | From the Flank in (Metres) | Normal Safe Distance HE Shell | Reduced Safe Distance from Burst of HE Shell with Special Authorisation in Writing only |
| a. | 25 Pounder | 35 | 150 | 200 | 457 | 320 |
| b. | 105 mm Howitzer | 40 | 200 | 200 | 548 | 365 |
| c. | 5.5 inch Gun | 50 | 400 | 400 | 914 | 503 |
| d. | 155 mm Howitzer | 55 | 400 | 400 | 914 | 548 |
| e. | 155 mm Gun | 55 | 600 | 400 | 914 | 548 |
| f. | 8 inch Howitzer | 65 | 600 | 750 | 1829 | 914 |
| g. | 120 mm Mortar (Brandt) | 45 | 550 | 550 | 823 | No reduction allowed. |
| h. | 122 mm Howitzer | 66 | 300 | 300 | 688 | 457 |
| i. | 120 mm Mortar (POF) | 45 | 550 | 550 | 823 | No reduction allowed. |
| j. | 130 mm Gun | 45 | 600 | 400 | 950 | 550 |
| k. | 122 mm MBRL | 50 | 1000 | 1000 | 2012 | – |

***Note***: During normal practice troops must be away from the line of fire by the safe distance.

7. *Star Shell*

| Serial No. | Equipment | 25 Pounder | 105 Millimetres Howitzer | 155 Millimetres Howitzer & Gun |
|---|---|---|---|---|
| a. | Area illuminated (diameter) | 548 metres | 1097 metres | 1829 metres |
| b. | Correct height of burst | 1371 metres | 2194 metres | 1920 metres |
| c. | Rate of fire for continuous illumination | 4 rounds per minute | 2 rounds per minute | 2 rounds per minute |
| d. | Burning time | 43 seconds | 60 seconds | 75 seconds |

## CHAPTER XII – SECURITY

### SECTION 30 – SECURITY THREATS AND MEASURES

**Purpose of Security**

1. Purpose of security is to protect a command against espionage, sabotage, subversion and surprise by the enemy. Security is extremely essential for avoiding damage to troops as well as equipment and for gaining and own maintaining freedom of action and manoeuvre. The commander of every unit and subunit is responsible at all times for its protection.

**Security Threats**

2. Security can be threatened by any of the following:-

a. Ground action to include reconnaissance, fire, attack or interference by regular ground forces, fifth columnists' guerrillas or a combination of these.

b. Ground action similar to a above by air, or helicopter borne/ parachute dropped forces.

c. Air action to include reconnaissance, bombing, strafing etcetera.

d. Covert action to include espionage, sabotage and subversion.

e. Threat from atomic, biological and chemical warfare.

f. Attack on the minds, emotions and attitudes of troops through psychological warfare.

**General Security Measures**

3. The most essential step in security training is to appreciate as to what information enemy actually wants and how can he get it. Only then can the nature and purpose of security countermeasures be thoroughly understood. Certain security measures are designed to afford protection against a specific type of threat. These are discussed in the succeeding section. However, there are certain measures which are applicable to all or most of the threats. These are:-

a. ***Information***. Adequate and timely information is essential to detect threats to security. Agencies which are normally available to a battalion to gain information have been discussed in Chapter II Section 3.

b. ***Effective Communication***. Gathering of information can only be useful if it can reach all concerned in time. Whenever possible, multiple means should be provided to ensure reliable communication even if primary system fails because of malfunction or enemy interference.

c. ***Reliable Warning System***. Battalion's standing orders for war should lay down as to how warning about various types of threats is to be communicated to all concerned. Without this, all efforts on gaining information and providing good communication are liable to be wasted and the battalion will be exposed to avoidable threats.

d. ***Timely Reaction***. The battalion should be well prepared to react to various threats. The reaction may take the form of avoidance, neutralization or destruction of the threat. The basic principle is that the measures adopted should be the most economical in effort and should interfere the least with the primary mission of the battalion. Battalion standing orders should cover this aspect in detail for each types of threat.

## SECTION 31 – SPECIFIC SECURITY MEASURES

**Measures Against Ground Threat**

1. The first essential element in security against ground action is the proper use of terrain:-

a. Security from reconnaissance is afforded by counter-reconnaissance, concealment, camouflage, deception and movement. Security from fire is provided by counter-fire, movement, cover, fortifications and dispersion.

b. Security against ground attack is provided by air and ground reconnaissance forces, covering forces, flank, advance and rearguards and local security elements, according to the nature of the operation. Composition of these security forces is depends on the type of operation and the enemy forces. Strong in armour, security forces must be of equal or greater mobility and possess appropriate anti-tank means; natural and artificial terrain obstacles also assume great importance. In any case, security forces must be capable of effective operations under the same conditions as the enemy force that poses the threat.

c. Airborne forces, guerrillas and infiltrators, in addition to their physical capabilities, produce psychological shock. They are dependent upon local or isolated sources of supply and are not as heavily armed as conventional ground forces. Security measures against such forces are psychological indoctrination of personnel, constant patrolling, immediate reaction to contain the enemy or to prevent the capture of critical areas and rapid and continuing offensive operations to destroy the threat. In addition, it is important that local resources be denied to such forces. It should be ensured that undue dispersion does not permit the destruction of isolated friendly elements and that measures for the registration of the civilian population and right control over their movement be established and enforced.

**Security Against Helicopter Borne Attacks**

2. Essential elements of sound defence against helicopter borne threat are:-

a. Reconnaissance and mining of likely landing areas.

b. Sentries and reconnaissance patrols to be vigilant.

c. Sound system of communication.

d. Strong all-round defence.

e. Sound surveillance over vulnerable areas.

f. High standard of training.

g. Infantry personnel to be trained in air portability of equipment.

h. Retain special mobile reserves for the protection of the administrative installations and to keep open lines of communications.

i. All civilians in the area be trained to give early warning against such an attack.

j. Coordination with the civil defence authorities.

3. Use of protective detachments can also help in the security of troops in the defensive localities. These detachments are:-

a. Reconnaissance elements.

b. Covering troops/advance position.

c. Screens.

**Measures Against Enemy Air Threat**

4. An infantry battalion will be especially exposed to attack from enemy aircraft in the opening phase of a war when the enemy is likely to have air superiority. The speed of modern aircraft is such that the use of wide spread small arms anti-aircraft fire is uneconomical and can lead to men exposing themselves unnecessarily and disclosing their positions, unless properly planned (see Paragraph 6 below).

5. The measures which an infantry battalion can take to protect itself against air attack are as follows, however, their application will depend on circumstances:-

a. Warning of attack to be given as early as possible by such means as air sentries.

b. Digging-in of both men and material. Care must be taken to camouflage the excavated earth and shadow produced by dumped earth and trenches in open.

c. Concealment (including camouflage) and track discipline.

d. Dispersal, especially of vehicles.

e. Offensive action taken by automatic weapon teams as detailed.

f. All ranks knowing what action they must immediately take, if attacked.

g. Trenches dug in a non linear pattern to reduce the danger from strafing.

**Offensive Action Against Enemy Aircraft**

6. The use of small arms fire against aircraft will be limited as follows:-

a. Some of the machineguns in the battalion must be capable of firing, when required, against low flying aircraft carrying out attacks on infantry. These weapons will be sited in anti-aircraft role and be ever ready to take on the intruder. A portion of tracer rounds must be included in the ammunition of all weapons in anti-aircraft role to ensure accuracy of fire as the firer can track the tracer bullet on to the target.

b. All the area must engage a hostile aircraft so as to bring a very high volume of fire to bear against the aircraft, not only to distract the pilots aim but also to pose a serious threat to the aircraft itself.

c. In defence, concealment is of special importance. Fire from machine-guns will not be opened unless specific orders are given.

d. Surface-to-Air Missile detachments must be put under command company/platoon. Fire orders issued well in time. Detachment should only receive orders for not engaging a target in active operation.

**Measures Against Nuclear, Biological and Chemical (NBC) Warfare**

7. All units must be able to defend themselves against atomic, biological and chemical warfare. Since these subjects are dealt with detail in various other publications, only a brief summary of a unit's defence requirements is given below.

**Nuclear Warfare**

8. In the field, where special protection against the atomic bomb will not be provided, the standard fire trench, 600 metres from ground zero, will afford a large measure of protection against the immediate effects provided the individual does not expose any part of his body above ground level.

9. Residual effects depend on the height at which the bomb bursts and the weather conditions. For bombs bursting above 500 feet the effects are not likely to be serious after the first few minutes and even then only in a restricted area directly under the bomb. Where bombs burst below 500 feet, an area close to ground zero might not be safe to stay without considerable risk, though it might be possible to drive through in a vehicle under certain circumstances within 15 minutes of the bomb bursting. Personnel trained by the unit atomic, biological and chemical officer with the help of special instruments, will be able to:-

a. Plot areas of contamination.

b. Measure the intensity of radiation.

c. Assess the rate of decay of radiation.

**Biological Warfare**

10. Biological warfare may be defined as the mass dissemination of a specific bacteria, fungi, etcetera, including the poisonous product of living organisms, which would cause disease, injury or death to men, animals or plants. There will be an incubation period, according to the agent varying from a day to possibly two or three weeks, before disease breaks out. Agents can be persistent or non-persistent. The best action, which a unit can take to protect itself against biological warfare, is to ensure that a very high standard of unit hygiene is maintained at all times.

11. As soon as it is established that a unit is under biological attack, individuals must prevent the access of the agents to the body by wearing the respirators. The respirator may be removed as soon as personal decontamination has been carried out. Personal decontamination will probably consist of a change of clothing and a bath.

**Chemical Warfare**

12. Chemical warfare is the employment of chemical products to produce death or casualties to men, in order to create military advantages or to defend against such actions. A chemical compound when suitably disseminated produces incapacitating, lethal or damaging effects on the men, animals, plants or materials. The characteristics of these agents may be summarized as follows:-

a. ***Nerve Gases***. Nerve gas symptoms are extremely unpleasant and may seriously affect the morale of troops, who may not be themselves affected but who see their comrades exhibiting such symptoms. These gases as act with great rapidity the troops must be able to adjust their respirators within a matter of seconds. These are much more lethal than any other war gas.

b. ***Blister Gases***. These are primarily harassing and incapacitating agents. After a period of about 2 to 12 hours, blisters will be produced by the liquid or the vapours. Blister gas liquid contamination will not normally prevent determined troops from crossing ground, but the vapours may, if casualties are to be avoided, cause evacuation of an area for some time.

c. Apart from nerve and blister gases following are the killing agents:-

(1) Blood agents prevent body tissues from using the oxygen in the blood.

(2) Choking agents attack the breathing passages and lungs.

13. The actions necessary for battalion commander to ensure that the unit can protect itself against chemical warfare, may be summarized as follows:-

a. Teach all ranks about the chemical agents that may be encountered.

b. Ensure that all ranks' respirators fit correctly.

c. Accustom all ranks to wearing their respirators for long periods.

d. Ensure that all ranks have a high standard of respirator drill to adjust their respirators rapidly when required.

e. Organize a unit alarm system, to give rapid warning of chemical attack.

f. Ensure that all individuals know about the collective arrangements for unit defence.

**Measures Against Covert Threats**

14. Security against hostile threats of espionage, sabotage and subversion is best provided by effective counter-intelligence operations. The principal covert threats are espionage and sabotage, it is essential that the security measures to counter these be emphasized. These include training of all individuals in safeguarding of information, control of movement, security of communications, physical protection of sensitive areas, installations and individuals; and fostering of friendly relations with the civil population.

**Measures Against Psychological Warfare**

15. The best defence against psychological warfare is knowledge of our own aims and a free discussion of enemy efforts based on an understanding of his objectives and techniques. Completeness of information is the chief bulwark of propaganda defence system and is furnished primarily by the unit information officer.

## CHAPTER XIII – ADVANCE AND PURSUIT

### SECTION 32 – BASIC CONSIDERATIONS

**General**

1. ***Occasions when a Force Advances***

a. Attention is drawn to Chapter II "Advance" General Staff Publication-1991 "Operations of War (Volume-II) Offensive Operations – 2004".

b. In the operational environment of Pakistan, an advance is often likely to be through areas inhabited by a hostile population and where the reaction will be imminent. Under such conditions advance must be carried out in a manner that it provides optimum security against enemy reaction. Advance is a prelude to attack or defence. The purpose of advance must therefore be to facilitate future operations. It must aim at obtaining a position of tactical advantage for development of further operations.

c. ***Advance before Contact with the Enemy***. Advance before contact with enemy may take place when the enemy is still unbeaten in battle, has equal initiative and is operating according to a well considered and coordinated plan with his main forces uncommitted and ready for action. The leading elements are tasked to locate the enemy, secure ground that will assist future operations and to cover the deployment of the rest of the force. It will be very rare for

Pakistan Army to advance to establish contact with the enemy except in the desert/semi desert sectors. Further the pattern of enemy defensive preparations including construction of field fortifications, anti-tank ditches etcetera fairly close to the border indicates that we will have to launch an attack even to break into the enemy territory.

d. ***Advance to Follow-Up a Deliberate Enemy Withdrawal***. It is undertaken to follow-up a deliberate enemy withdrawal, while enemy forces may still be balanced, ready for battle and able to even counter-attack. In view of the preponderance of resources available with our potential enemy, it will be rare that he will conduct a deliberate withdrawal. However, there is a possibility of his staging a deliberate withdrawal as he possesses adequate geographical depth for a strategic manoeuvre. In this event, Pakistan Army may advance to regain contact with the enemy in order not to allow him to adopt an offensive or defensive posture on the ground and time of his own choosing. Such an advance operation should be a bold and rapid action, while maintaining balance.

e. ***Advance to Follow-Up an Enemy Withdrawing off Balance***. Advance under this situation will turn into a pursuit which is a separate phase of the offensive operations. There is every possibility of such an operation taking place in a sector where the Pakistan Army is on the offensive. For obvious reasons, its techniques must differ from those of conventional advance to contact. This calls for a bold use of armour, artillery and airborne assets with the ultimate purpose of annihilating the withdrawing enemy forces. Logistic support must be well planned. However, certain administrative risks can be taken so that tactical advantage can be pressed home ruthlessly.

**Basic Considerations**

2. The basic considerations in planning and execution of the advance are:-

a. ***Offensive and Bold Action***. Offensive action consists of rapid exploitation of opportunities, determination in execution, maintenance of momentum, retention of balance and timely employment of reserves. At the battalion level advance should be executed so boldly and aggressively that the enemy is thrown off balance from the very outset. This will be possible only if:-

(1) The details of the plan and the concept of operation are known to all concerned well before the commencement of advance.

(2) The commanders move well forward to take appropriate action in time.

(3) The deployment drills and battle procedures of a unit are perfect.

(4) Commanders at all levels should correctly visualize the likely enemy and own reactions and plans made well before hand to deal with such contingencies. If time and resources permit, out flanking manoeuvres around the enemy positions may also be considered. Such movements are likely to cause psychological imbalance to the enemy trying to hold up the advance, thereby compelling him to abandon the position as well as initiative.

b. ***Maintenance of Momentum***. By continuous and relentless offensive action, the enemy is prevented from carrying out an organized withdrawal, which is turned into a rout and his power to resist is broken. Speed is often the dominant factor. Minor oppositions unlikely to threaten the line of communications are generally bypassed. Momentum is maintained by good initial orders, balanced grouping and initiative to junior leaders.

c. ***Simplicity of Plans***. A simple plan, with the battalion commander's intentions clearly stated, enables junior leaders to use their initiative.

d. ***Balanced Grouping***. The battalion should be correctly balanced in order to be able to react to unforeseen eventualities. Grouping should be flexible so that the order of march can be changed and new situations appropriately dealt with, without loss of any time. Grouping will depend on:-

(1) Aim.

(2) Terrain and visibility conditions.

(3) Likely enemy opposition.

(4) Resources available.

(5) Any special requirement; river crossing or mine breaching etcetera.

e. ***Good Battle Procedure***. This will ensure that action is taken without delay. Commanders moving well forward with their groups and ready to receive orders will ensure that troops are not committed into battle piecemeal.

f. ***Good Communications***. In advance to contact reliable wireless communication is very essential for passing back timely information and for prompt receipt and issue of orders.

g. ***Detailed Reconnaissance and Forward Planning***. Detailed reconnaissance and acquisition of maximum information regarding terrain and enemy will enable the battalion commander to get a clear picture which will help him in forward planning and maintenance of initiative.

h. ***Concentration***. By rapid concentration of the main body intense artillery fire and air attacks enemy can be surprised, threatened and out manoeuvred.

i. ***Speed***. To obtain speed it is necessary to:-

(1) Advance on a wide front gain information and exploit quickly the enemy's weak points. At battalion level in open terrain clearance of 800 metres of area on either side of the axis is considered the minimum in a daylight advance.

(2) Have balanced grouping, adequate control and determination to push on.

(3) It is better not to restrict the advance to roads and tracks. If the going is good tanks and light anti-tank elements move cross-country to reach certain objectives ahead of the main guard.

j. ***Quick Reaction***. The overall rate of advance is governed more by quick reaction in dealing with the enemy when he is encountered than by undue haste between contacts. As a yard stick in encounter battle, a company may take maximum from 30-45 minutes and a battalion from 1-2 hours under normal conditions.

k. ***Use of Air***. A daylight advance in our environment presupposes that we have achieved local air superiority over the area of operation. Air, if allotted must be used for interdiction to cut off and disrupt the retreating enemy which will assist in maintaining the momentum. Close support and air reconnaissance will be of great assistance. Air control team must, therefore, be kept well forward to ensure its expedient use.

## SECTION 33 – PLANNING AND CONDUCT OF ADVANCE

**Planning**

1. The battalion commander formulates the plan based on the orders of the brigade commander. The brigade commander/brigade staff will provide maximum information regarding the enemy likely strength on the axis of advance, obstacles, demolitions, etcetera. Based on the information, with the help of the map and air photographs, the battalion commander visualizes the enemy's likely strength at each tactical feature on the axis of advance and plans to deal with each such opposition. Supporting arms commanders should be associated with the planning from the very start. The battalion commander should base his plan on the most likely course open to him thereby ensuring smooth execution of plans against an unexpected threat from the enemy. He should also prepare contingency plans and the entire operation should be planned as a whole.

2. The battalion commander should give out his plan to subordinate commanders in great detail for implementation. Detailed tactical guidance should also be provided to them so that correct actions are taken by them, without the loss of valuable time.

**Grouping**

3. Grouping of the force will depend on the type of terrain and likely enemy delaying positions. The vanguard company and the next company in order of march should be grouped alike so that in the event of next company assuming the role of vanguard, time is not wasted in grouping and regrouping. Minimum troops should be committed for advance thereby retaining the maximum in reserve for dealing with a strong position at the battalion level.

4. In case, sufficient elements of supporting arms are not available, regrouping of the next company in the order of march for assuming the duties of Vanguard Company becomes unavoidable, the battalions should work out their own drills during peacetime for the smooth change over. It is therefore, suggested that the second-in-command of the company should physically contact various elements of the supporting arms and keeps them concentrated and suitably placed on the axis of advance. An advancing force may be organised as under(see Figure 11):-

a. Covering troops.

b. Advance guard.

c. Main body.

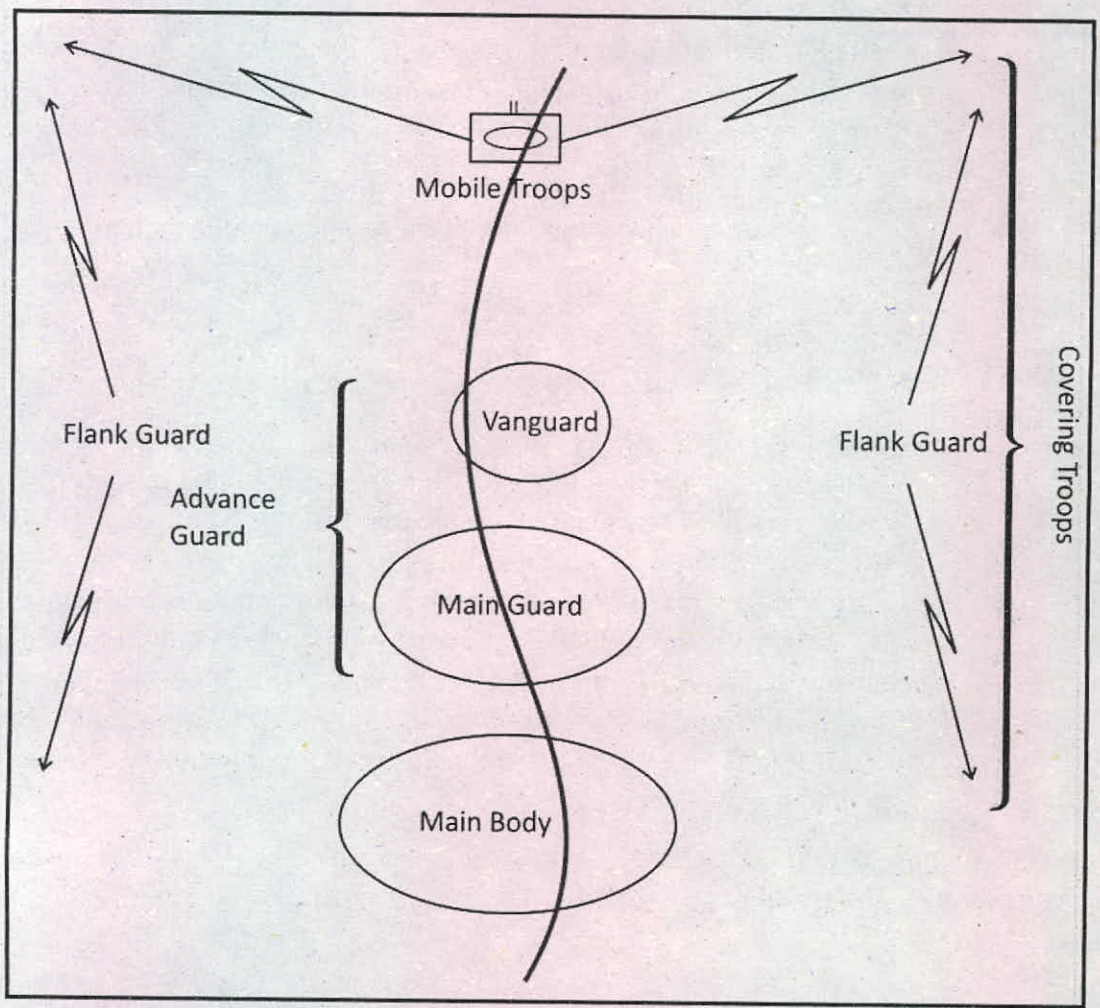


FIGURE 11 – GROUPING IN ADVANCE TO CONTACT.

5. The organization of groups and their mutual time and space relationship is for the following purpose:-

a. The forward echelons ensure speed, prevent rear echelons from being surprised and provide time for their deployment.

b. Enemy opposition is dealt with by the amount of force necessary for the purpose.

c. The main force is not unnecessarily engaged in dealing with minor opposition.

6. ***Covering Troops***. Consist of following:-

a. ***Mobile Troops***. These are the leading elements of force engaged in advance. In view of their role the mobile troops must be organised around mobile elements most suitable for the area of operations.

Elements of Reconnaissance/Armoured Regiments/Light Anti-Tank battalion and self-propelled artillery are ideally suited for the task. These however must be adequately grouped with mobile infantry for their intimate protection and to grant them requisite staying power. Tasks of mobile troops are:-

(1) Reconnaissance on a wide front and obtain information.

(2) Clear minor oppositions. They should not bypass resistance unless so authorized.

(3) Secure important tactical features such as bridges and defiles.

b. ***Flank Guards/Protection***. Their primary role is to watch and protect the exposed flanks, if any. They also give early warning of the enemy which can then be dealt with by the reserve forces. Flanks are protected either by moving on a route parallel to the main axis or by picketing along the lines of approach. They are organised more or less like Mobile Troops and their strength will depend on the envisaged enemy threat. However, it should be kept to the bare minimum as they take the troops away from the main force. They must, under all circumstances have very strong adequate communications and compatible mobility.

7. ***Advance Guard***. An advance guard is a group of all arms. It moves in front of the main body and provides security from the front. The advance guard is organised in two parts:-

a. Vanguard.

b. Main guard.

8. Normally the advance guard will comprise an armoured regiment and an infantry battalion group. The vanguard will consist of a squadron or a company group and will be deployed along/astride the main axis. The main guard (the remainder of the battalion group) will follow at a reasonable distance along the axis. The move will be tactical.

9. ***Main Body***. The main body comprises the bulk of the force's fighting power. Troops of the main body are employed after contact with the main enemy force. Main body should be moved from harbour to harbour to have a clear run. The time and space relationship between the main body and advance guard is primarily determined by the degree of enemy opposition likely to be met. In a brigade advance, troops less the advance guard will constitute the main body.

**Command and Control**

10. To avoid delay, commanders of the vanguard and advance guard must move well forward, followed closely by their 'O' groups, ready to receive orders. In certain cases, the company commanders may move with their own companies and subsequent orders for attack and deployment can be issued to them on wireless set. Although wireless will be the primary means of communication with forward and rear echelons, yet, cable should also be laid in advance by battalion and brigade Signal Company behind the vanguard and advance guard respectively. Line communications, comparatively, will be safe and free from enemy's interference.

11. Control is affected by laying down bounds, report lines and control points. Bounds and report lines should be given nicknames to ensure security. Good and duplicate communications are essential.

12. ***Phases of Advance***. There are two phases of advance:-

a. ***Preparatory***. It starts with the receipt of mission by the commander and ends when the advance guard crosses the start point. Depending on the enemy and own situation, covering troops may be dispatched to secure important tactical features immediately on receipt of mission.

b. ***Conduct***. Conduct phase commences from the time when advance guard crosses the start point and terminates when the main body deploys for attack or defence. Since the purpose of advance is to secure a position of tactical advantage for development of future operations, speed becomes the main feature of advance. At the same time, possible vulnerability of advancing force to enemy ambush makes security a very major concern. This conflicting requirement is resolved through exercise of command, which is best done by physical presence of the commander with the leading troops.

## SECTION 34 – ADVANCE GUARD

**General**

1. Advance guard will consist of two components:-

a. Vanguard.

b. Main guard.

**Tasks of Advance Guard**

2. Tasks are as under:-

a. To prevent the main body from running blindly into enemy opposition.

b. To provide a force strong enough to clear minor oppositions and to cover deployments of the main body.

c. To find flanks/gaps in the enemy's defences and exploit them, if possible and so directed.

d. To establish a firm base for under taking subsequent operations.

e. To pass back information regarding enemy dispositions, strength and obstacles both natural and artificial obstacles.

**Vanguard**

3. The vanguard is related to main guard in the same way as the advance guard is related to the main body. When a battalion is acting as advance guard, the vanguard should comprise a company with necessary supporting arms. A suggested composition of the vanguard company is as under:-

a. ***Allotment of Troops from the Battalion***

(1) One Rifle Company.

(2) One section battalion mortars.

(3) One section from assault pioneer platoon.

(4) Detachment from regimental police and intelligence section.

(5) Surface-to-Air Missiles (offensive formation authorized to infantry battalion).

(6) Anti-Tank Guided Missile platoon.

b. ***Troops from Outside Battalion Resources***

(1) Tanks, if allotted to advance guard.

(2) Elements of Light/Heavy Anti-Tank battalion, if allotted.

(3) One forward observation officer from direct support battery.

(4) Engineer's reconnaissance party.

(5) Air Control Team if allotted.

**Gaps**

4. The size of the gaps between various components of the advance guard in the order of march is calculated in term of time and not space. Gaps will vary with the speed of units, enemy resistance and with the type of country. The yard stick is that there should be no possibility of forces being allowed to drift into action, unaware or piecemeal.

5. Gaps will cease to exist after the contact has been established with the enemy. The advance to contact will terminate when enemy resistance necessitates deployment of the main body.

**Carriage of Infantry**

6. In an advance or in mobile operations depending upon availability of resources, type of terrain and enemy opposition infantry moves:-

a. On tanks (if allotted).

b. In Troops Carrying Vehicles (TCVs).

c. In Armoured Personnel Carriers (APCs).

d. By marching.

7. Infantry carried on tanks is vulnerable to enemy ground and air action and impedes tank actions. In no circumstances infantry should be carried on the leading tanks, who must be ready to engage the enemy, at all times.

8. When troops are conducting advance on wheels/tracks, following precautions will be taken:-

a. During day, gaps between the vehicles should be greater than in the normal move in order to avoid enemy air action. Under no circumstances these gaps will be reduced during advance.

b. Troops move from harbour to harbour.

c. Tarpaulins are rolled up and necessary anti-aircraft precautions are taken.

d. Move will be tactical.

9. When the troops are advancing on foot, following points will be taken into account:-

a. Vanguard company moves well dispersed on either side of the road/ track ensuring safety to the main guard from the enemy's small arms fire from the flanks.

b. Follow-up companies move in sections along the road, with a platoon in about 100-300 metres. Cross-country marching is quite tiring, therefore, the remaining troops should be kept as close to the road/ track as possible.

c. Vehicles, less carrying wireless sets, should move from harbour to harbour.

d. On establishing contact by the vanguard company, further move of the troops and the vehicles should be controlled. Regimental police detachment with the vanguard should establish a check post beyond which no vehicle will be allowed to pass.

e. Troops will continue marching until contact is made with the enemy.

## SECTION 35 – TACTICAL HANDLING OF BATTALION GROUP IN ADVANCE

**General**

1. Normally a battalion group will be acting as advance guard to a leading brigade. However, in certain cases, a battalion may also be assigned an independent role as part of an overall plan to carry-out a wide outflanking movement, with a view to facilitating the advance of the main force by establishing a block well in the rear of the enemy.

**Order of March**

2. The order of march of an advance guard will be decided by the following factors:-

a. Nature of the terrain i.e. plain country, hilly, jungle and desert etcetera.

b. Availability of supporting arms i.e. armour, artillery, engineers, air etcetera.

c. Day or night advance.

d. State of the enemy's preparedness and morale.

e. Means of transport-whether Armoured Personnel Carriers and Troops Carrying Vehicles are available.

f. Arrangements for flank protection.

g. Plans for dealing with enemy's likely positions en route.

3. The order of march for vanguard (infantry company with 2 troops armour) will be as follows:-

a. Two troops of tanks moving off the road.

b. Two rifle platoons made mobile.

c. Section assault pioneers.

d. 'R' group including forward observer officer/mobile fire controller.

e. 'O' group including engineer officer.

f. Detachment regimental police and intelligence.

g. Engineer reconnaissance party.

h. Company headquarters.

i. Number 3 Platoon.

j. Section battalion mortars, (they leap frog during the advance).

k. 'F' echelon.

4. The order of march for complete advance guard (infantry battalion with squadron armour) will be is follows:-

a. Vanguard company group (including engineer reconnaissance party).

b. Advance guard 'R' group.

c. Advance guard 'O' group.

d. Follow-up company, (grouping like vanguard).

e. Squadron tanks less two troops.

f. Engineer field platoon less reconnaissance party.

g. Rifle company.

h. Rifle company less platoon.

i. Advance guard headquarters.

j. Officer Commanding Combat Support Company less detachments with vanguard.

k. 'F' and 'A' echelons.

l. Rifle platoon.

**Formations in Advance**

5. ***General***. In an advance, the commander has to decide as to how the unit/ subunit is to advance to gain maximum advantage over the enemy. He has to make up his mind whether the unit should advance in a column or in line formation.

6. ***Methods***. Imminence of contact, knowledge of enemy strength/dispositions, required speed of advance and ground to be traversed will determine which of the following methods to be used:-

a. ***Column Formation***. This method may be used when enemy dispositions have not been ascertained, when greater security is required or when the commander desires to retain the bulk of his force in reserve. Commitment of the force into battle from this formation is more time consuming because of additional time required to deploy forces from single column. It also permits the

enemy to achieve maximum delay with minimum forces, thus slowing the momentum of advance. It adds depth to the advance and permits the force to be deployed to each flank quickly or pass through or around the leading elements to maintain the momentum of advance. It is characterized by relatively weaker firepower to the front but maximum firepower, flexibility and security to the flanks. The flexibility for foot mobile column is reduced because it requires greater time to bring up the rear echelons. The column formation is favoured:-

(1) For deep objectives, strongly held by the enemy.

(2) During periods of poor visibility, when maximum control is desired.

(3) When forced by restrictive terrain conditions.

(4) When enemy situation is vague and major strong points or troops locations are not known.

(5) When enemy resistance is light or disorganised and initial enemy resistance can be eliminated by one element.

(6) When there is a requirement for security against a counter-attack.

(7) When there is a requirement for strong flank security.

(8) When there is a need to concentrate supporting fire in one area.

(9) When our forces have a high degree of vehicular mobility.

b. ***Line Formation***. A line is a formation in which two or more subunits are abreast. This method may be used when greater speed of deployment is desired and manoeuvre on broad front is necessary. The formation presents several threats to the enemy which may not be able to counter it at one point. As the force is deployed on a wide area, the bulk of the force remains committed. The line formation permits maximum fire to the front or rear and minimum fire to the flanks. It is more difficult to control and does not provide as much depth as, the column. It should be employed by troops and tanks emerging from smoke, crests, woods/crops and while assaulting an objective, as it permits closing on to the objective in minimum time. It also permits flexibility in commitment of reserve elements. This formation is favoured by following factors:-

(1) Limited objectives.

(2) When major enemy strong points and troops locations are known.

(3) When objective is strongly held by the enemy and there is a requirement to place maximum fire on the objective.

(4) When situation requires maximum freedom of action by units/ subunits.

(5) When flanks are secured.

(6) When there are multiple approaches to the objective.

(7) When fire support is adequate to support all attacking elements.

(8) When strong reserves are not needed.

(9) When there is a need to close with the enemy rapidly with the bulk of the force.

(10) When the operation is to be conducted by infantry or dismounted troops.

7. ***Action on Meeting Opposition***

a. ***Advance without Armour (Infantry Company)***. As soon as position operator is met, the leading platoon should deploy and try to find detailed information about the terrain and enemy. The company commander with his 'R' group should start planning and preparation for action. The suggested technique is as follows:-

(1) The advance is led by one or two sections of the point platoon of the vanguard with the object of quickly finding maximum information. If the opposition expected is more and the battalion action may be required, the advance of the vanguard may be led by two platoons so that maximum information is obtained in minimum time for the battalion action.

(2) As the leading elements arrive closer to the tactical feature which had been appreciated to be held, or when the contact is imminent, or when the leading elements are fired upon, they quickly deploy and take up positions to carry-out speculative fire or creep forward fire and move. They try to observe from any vantage point including houses, trees, etcetera and bring maximum fire onto the enemy to obtain information which they continuously pass to the higher commander.

(3) If it is appreciated, as a result of information by the leading platoon(s) that the vanguard will have to launch a quick attack, it should preferably be done on the flanks of the enemy. In the mean time the advance guard commander must be kept continuously informed about the situation and the plan of action.

(4) The other situation could be where the vanguard or advance guard commanders would have appreciated and planned off the map that a certain tactical feature is such where there would be requirement of action by complete advance guard. In such a situation, the task of the vanguard will be to quickly deploy and obtain maximum information in minimum possible time to confirm the appreciation carried out off the map. The vanguard, therefore, should advance with two platoons up, if possible. While the vanguard is deploying to find out information including the extent of flanks of the enemy, the vanguard commander must keep in mind that eventually he has to form a base of fire to fix the enemy for the battalion attack. The purpose of the base of fire is to give fire support, fix the enemy so that he is unable to react and to provide a base against any enemy counter-attack.

(5) While the leading elements (point platoon(s) of vanguard) are deploying and obtaining information, the remaining elements of the vanguard/main guard should also be planning and completing preparations for the impending attack.

(6) The registration of targets by forward observation officer should start immediately on contact with the enemy.

(7) The vanguard commander will select suitable place from where he can brief the advance guard commander. Guides to be posted on the road.

(8) Regimental police to control the traffic on the main axis.

(9) Position will not be bypassed unless ordered otherwise by the higher commander.

b. ***Armour Infantry Combined Advance***. The tanks may be employed to advance with infantry in an open tank able country, or in the terrain which is restricted and/or in a very close country. Employment of two troops in the vanguard may not be able to overcome the opposition as in such types of terrain the enemy is also likely to employ his tanks. As a general rule, the squadron should not be distributed and should

be employed concentrated. The advance guard commander may impose the restriction that whenever more than two troops are required to be used for any action, prior permission must be obtained from him:-

(1) ***Advance in Open Tank able Country***

(a) The enemy is likely to employ tanks to delay/impede our advance. He is likely to deploy minimum of one tank troop. Although he will normally employ at least two troops, grouped with other arms in the delaying positions. Therefore, it is recommended that the vanguard should be organised on a complete squadron.

(b) The infantry company should be grouped with the squadron. The company should be mounted on Troops Carrying Vehicles (TCVs), which should follow the squadron at such a distance that they do not come under observed artillery fire or get involved immediately. In case Troops Carrying Vehicles are not available, the infantry company should be mounted on tanks less the leading troops.

(c) When enemy tanks are expected, the advance should be led by the tanks and not the infantry.

(d) Even if enemy tanks are not expected the advance will be much faster if it is led by the tanks.

(e) The commanding officer will always remain in picture about the actions of the vanguard; therefore, he can control any impending action by the squadron commander. He may even initially impose the restriction that whenever the squadron commander wants to use more than two troops, prior permission from the battalion commander must be obtained.

(2) ***Advance in a Terrain where the Use of Tanks is Restricted***. The terrain visualised is such where the country is close, the field of fire and observation are reduced to 400-500 metres or less, the enemy can make use of the terrain for employment of short-range anti-tank weapons but otherwise the area is tank able. In such cases:-

(a) The advance should be led by the infantry.

(b) The command of the vanguard should be with the company commander.

(c) The whole squadron should be placed under command of the vanguard commander on whom however, certain restrictions may be imposed.

(d) Alternatively, the squadron should remain under command of the battalion commander with an assurance to the vanguard commander that, the squadron will be placed under his command on requirement.

(e) The squadron should follow the vanguard company in order of march.

(3) ***Advance in a Very Close Country***. In this type of country, the field of fire and observation are limited to 100-200 metres and further less, due to BUA, jungle and mountainous country. However the going is otherwise good. The advance suggested is as follows:-

(a) The advance should be led by the infantry.

(b) Infantry company commander will be the vanguard commander.

(c) The vanguard company may be given a troop of tanks.

(d) The squadron less one troop should follow the follow-up company.

(e) In case of an action by the battalion, in certain situations, the other infantry companies may also be given a troop each, if considered necessary.

(4) ***Other Actions***. The other actions as per paragraph 5a (6) to (9) above are common to advance with armour in any type of terrain.

8. ***Action on Meeting Opposition***

a. Advance guard commander should immediately move up and personally contact the vanguard commander to get the latest information regarding the enemy, own troops and terrain etcetera. In fact the advance guard commander should have been mentally prepared for this eventuality and contingency plan already formulated.

b. Already prepared tactical plan to be quickly checked and reviewed if necessary. Otherwise he should go ahead with the earlier plan and clear the opposition as soon as possible.

c. In the event of an opposition being stronger that cannot be tackled by the advance guard, the commander should try to secure features/ ground of tactical importance in the area by deploying on a wider front thereby facilitating the launching of attack by the brigade .

d. Brigade headquarters must be provided necessary information throughout the operation.

e. Probing forward should continue and efforts should be made to gain maximum information about the enemy.

f. The methodology of clearing enemy opposition with the tanks is explained in GSP-1854 – Tanks/Infantry Combined Operations.

## SECTION 36 – ADVANCE BY NIGHT

**Purpose of the Night Advance**

1. A retreating enemy will normally try to take advantage of darkness to break contact, move back and organise his next position. In order to maintain pressure on the enemy it is often necessary to advance at night. Infantry, on its own or supported by tanks, can often advance successfully by night and thus may be able to achieve considerable surprise by the next day. For armoured troops, the advantages of advance by night should however be carefully weighed against the noise, fatigue to the crew and the requirements of replenishment. Nevertheless, opportunities must be fully exploited if the cover of darkness can help. The advantage of continuous advance by day and night may be great and may end in such prizes as the capture of bridges intact.

**Factors**

2. The most important factors to be considered in deciding whether to advance by night are:-

a. ***Pressure on Enemy***. The importance of maintaining pressure on the enemy.

b. ***Surprise***. The chances of achieving surprise.

c. ***Degree of Visibility***. Phase of the moon or availability of moon light.

d. ***Wear and Tear***. Tanks and vehicles need time for maintenance in long advances. Troops also need rest at sometime.

e. ***Passage of Defiles***. It may be safer at night when observation is denied to the enemy. However, extra vigilance must be exercised against likely enemy ambushes.

f. ***Ground***. Both tracked and wheeled vehicles require fairly good going for night movement.

g. ***Enemy Air Threat***. The degree of enemy air threat during day, enemy air is less effective at night.

h. ***Mission***. Mission given to the battalion by the brigade.

i. To exploit a tactical situation, which may not last till the next day.

j. Morale of own troops vis-à-vis that of the enemy.

3. Consideration should always be given to use of light during a night advance. Its use will reduce fatigue of drivers.

**Advance at Night**

4. The conduct of night advance will differ from the advance to contact during day in some respects. Points to be kept in view are summarised below:-

a. ***Planning***

(1) In night advance greater reliance will have to be placed on maps and air photographs for planning. Air photographs should be arranged at least 3 to 4 days in advance, when possible.

(2) Planning should be in greater detail than for day advance. All possible eventualities should be considered and catered for before hand.

(3) Appreciation of positions likely to be held by the enemy and assessment of the enemy strength will have to be made very incisively.

b. ***Control***

(1) Bounds and report lines should be fixed at short intervals as compared to the advance by day.

(2) Emphasis should be on steadiness in probing forward rather than on speed in advance at night.

(3) Commanders should be moving with their own subunits for efficient command and control and for orders to be passed effectively.

(4) Communication should be duplicated.

(5) Commanders should wear distinguishing signs.

c. ***Grouping***

(1) Unless mobile troops have infra-red devices they will not be able to move freely in darkness. Therefore, they will be seldom employed.

(2) Flank guards may be used, especially in moonlight. Light anti-tank battalion may be employed on this role. This task will be better performed by picketing the lines of approach rather than moving on routes parallel to the main axis at night.

(3) The advance on the main axis should be mainly by infantry. Tanks usually should be required to join the infantry by first light except as given in sub-paragraph f(2).

(4) Harbour, gun position officer and engineer reconnaissance parties will have to be well forward with the advance guard. An engineer platoon may also be placed under command of the advance guard commander.

(5) There should be a separate advance guard for each route of advance.

d. ***Formations***

(1) The point platoon should search both flanks of the axis of advance to the maximum visibility distance from the axis so that the enemy cannot ambush the advancing troops or bring down effective small arms fire on them.

(2) Assault pioneers should be required to move immediately behind the point platoon for clearance of any mines or booby-traps etcetera on the main axis.

(3) The follow-up platoons of the vanguard should also be made responsible for one flank each.

(4) The main guard should move at least 300-400 metres in the rear of vanguard to ensure that they do not come within effective range of enemy small arms fire immediately on contact with the enemy.

(5) The gap between the vanguard and main guard should be filled by connecting files provided by one of the companies in the main guard. The number will depend on visibility.

(6) Suggested formation of advance guard for night advance is given in Figure 12.

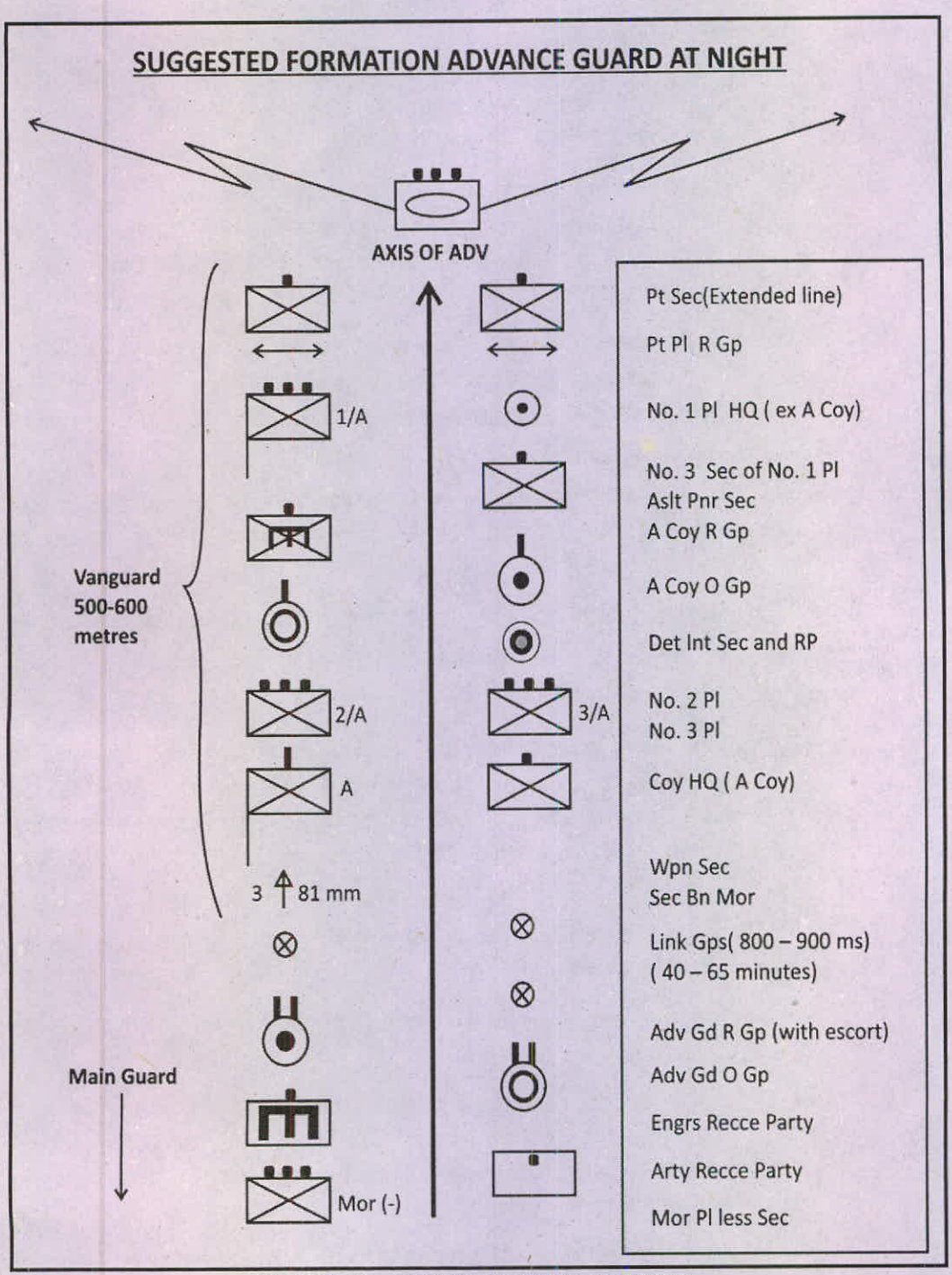


FIGURE 12 – ADVANCE GUARD AT NIGHT.

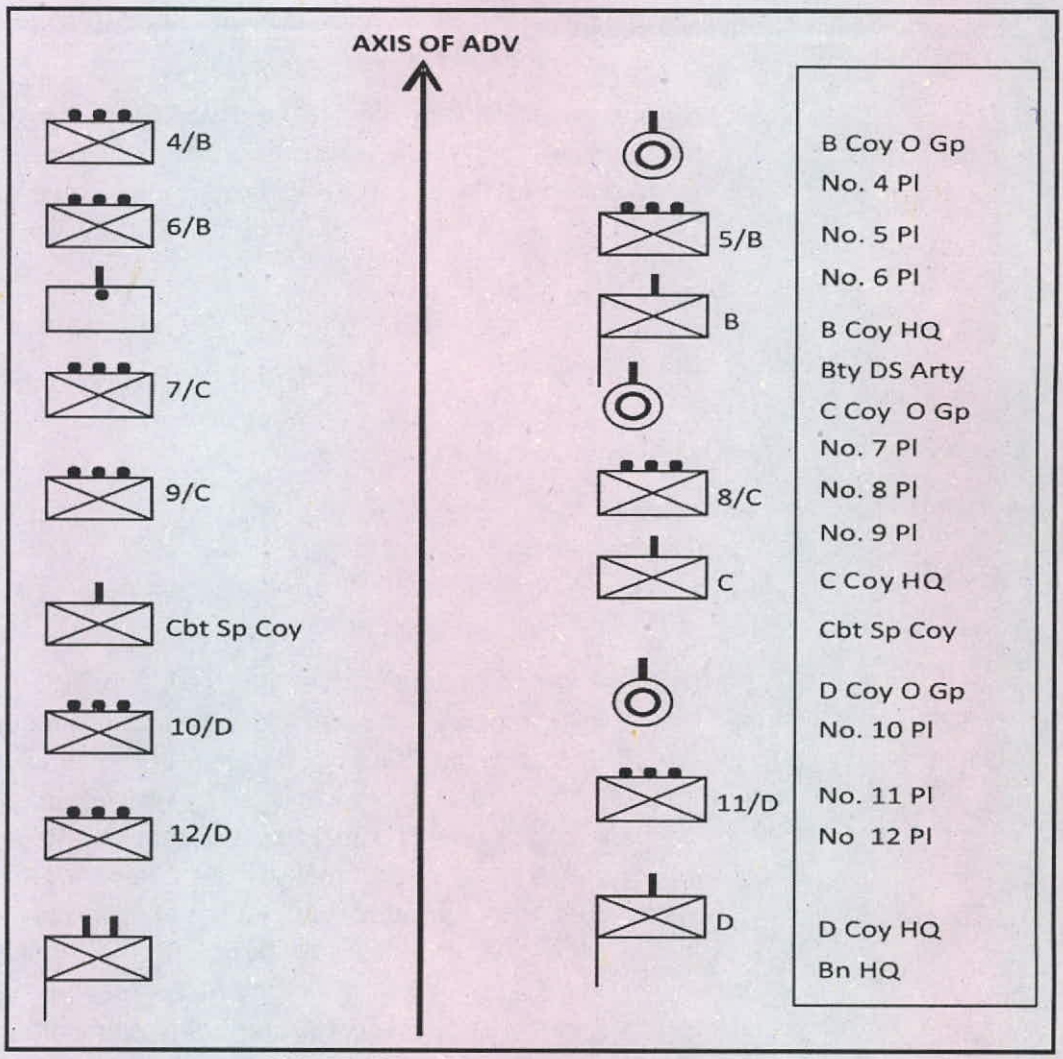


FIGURE 12A – ADVANCE GUARD AT NIGHT.

e. ***Conduct***

(1) The point platoon should lead the advance at a steady pace watching the flanks to ensure that the vanguard and main guard are not ambushed.

(2) On meeting enemy opposition the point platoon should try to draw enemy fire to determine his flanks.

(3) If the opposition is within the capability of the vanguard, it will clear it by attacking usually from a flank. The bearings may be worked out for possible attacks from both the flanks, while moving along a given axis. No plan survives the enemy comprehensiveness of contingency planning is hallmark of preparation.

(4) For limited manoeuvring of this kind at night, alternative aids can be:-

(a) Flank marking lights from the held up troops, to be switched on only when they are held up. However, great care must be exercised while using any type of light. It will be far better, if it is possible to do without lights.

(b) Tracer fire used by held up troops to indicate flanks of the objective.

(c) Physical liaison with held up troops and taking a guide from them.

(5) Infantry battalions would have flank marking lights for each platoon but platoons should use them only when held up while acting as point platoon. In night attack, these lights should normally be used at company level only.

(6) If the vanguard is not capable of clearing the opposition, it should probe further to obtain maximum information about the enemy and terrain and to find out flanks of the enemy. It can be done by keeping the enemy engaged by fire. Company heavy weapons and Baktar Shikans etcetera should also be moved up to prevent any clean break by the enemy and to provide base of fire for the attack by main guard. The main guard should then clear it preferably by launching a quick attack on his flank(s). For this the force should manoeuvre behind the vanguard company and the base of fire elements.

(7) If the main guard finds the opposition too strong, it should start probing the enemy's flanks. This situation will necessitate the employment of the main body for clearance of the positions, which normally will not be possible during the same night.

(8) In view of the strain on leading troops at night, bounds for vanguard should be kept relatively short.

(9) If the original advance guard battalion has succeeded in clearing enemy opposition, another battalion should pass through to resume the advance. This necessitates detailed coordination and must be pre-planned.

(10) Normally the leading troops should not bypass an opposition at night as they may be led into a trap.

(11) If the advance is after a deliberate night attack, the advance guard should be moved up from their concentration area in the mechanical transport, to start advance from the forward positions held by own troops. Troops carrying vehicles should move up without lights.

f. ***Armour***

(1) Unless there is moon light, tanks without Night Vision Devices and search lights will be a liability in advance.

(2) In moonlight, if fitted with night fighting equipment or when other artificial illumination arrangements exist, the tanks may be given to vanguard or they may follow the advance guard, which depends on the situation, particularly, the terrain. They will be able to produce shock effect in an attack even though their shooting may not be as accurate as during the day. They will usually be able to give fire support if close support to the infantry is not possible.

g. ***Light Anti-Tank Elements***

(1) The Armour Personnel Carrier borne light anti-tank elements can also be grouped with armour to lead the advance if terrain permits.

(2) In case any armour threat is visualized at night and it is considered necessary that light anti-tank elements should be grouped with the leading elements to provide close support to the advancing column, arrangements for artificial illumination, should be planned.

(3) It will, be best to let the light anti-tank elements advance by bounds behind the advancing column and be in a position to be available by first light to support the advance.

h. ***Heavy Anti-Tank Elements***

(1) Like light anti-tank elements, heavy anti-tank elements and armour personnel carriers have good cross-country mobility and night fighting capability. If it is visualized that they will be needed before first light they may be taken along with the advancing column and arrangements for artificial illuminations, if necessary, made to enable them to fight enemy armour.

(2) They should, therefore, advance by bounds at night and join the main body by first light so that enemy armour threat can be countered effectively.

i. ***Artillery***

(1) Artillery units should move as far forward along the axis as possible before last light and cover the advance from there as long as possible.

(2) Farther gun areas should be selected off the map/air photos and confirmed by the Gun Position Officers/reconnaissance parties moving with the advance guard.

(3) Deployment of guns at night and carrying forward of survey without prior daylight reconnaissance and preparations will take longer time. With preparations off the map, timings for deployment excluding travel time would be as follows:-

| | | | |
|---|---|---|---|
| (a) | Battery | – | 2 hours. |
| (b) | Regiment | – | 4-5 hours. |
| (c) | Divisional Artillery | – | 12-14 hours. |

(4) For deployment of Divisional Artillery, survey carried out at night must also be checked during daylight, which may require at least 3 hours. As such, quick support by divisional artillery from a new area at night is not easy. Support from a field regiment may be possible depending on the hours of darkness available for the move forward and deployment.

j. ***Engineers***

(1) Engineer's reconnaissance parties should be with the advance guard.

(2) Some engineers may also be given under command advance guard for clearance of mines and booby-traps etcetera.

(3) Engineers should put-up signs at night on roads/tracks cleared by them.

k. ***Air***

(1) Air strikes may be arranged on deep targets and enemy vehicular columns moving at night.

(2) Close Air Support and armed reconnaissance should be arranged soon after first light.

(3) Air reconnaissance should also be arranged just before last light on the day before the advance is due to start.

(4) In certain cases, air strikes on the objective are also possible. This requires detailed planning and coordination. At times, the air force may be asked to provide support for battlefield illumination.

(5) Air observations post (Air Operation) is extremely useful during advance for engagement of deeper targets and also for guidance of own mobile/infantry elements.

(6) Use of air for casualties evacuation can save precious lives and also lift the morale of troops.

l. ***Administration***

(1) Administration units should move up as far forward as possible on the night before advance is to start. There should be no move of administrative installations during the night in which advance is to be carried out. If necessary, only refilling points may be opened forward.

(2) Traffic control arrangements should be thoroughly coordinated. Provost may be reinforced by additional trained persons from infantry battalions.

(3) Electrical Mechanical Engineer recovery/repair teams should move immediately behind the advance guard.

m. ***Command and Signal***

(1) Headquarters should be close to the main axis and commanders immediately behind the leading elements of their respective commands.

(2) Communications should be duplicated. Although most of the intercommunication will be on wireless, lines to the leading units should be laid as soon as possible and completed before first light. Arrangements must also be made to provide line communication in all the harbour areas preferably before the units actually arrive there.

## SECTION 37 – EXPLOITATION AND PURSUIT

### Exploitation

1. Exploitation is a phase of an offensive operation designed to destroy enemy's ability to conduct in an orderly withdrawal or to reconstitute an organised defence. The battalion may exploit its own success, may be a part of the exploiting force of a higher echelon or may follow and support another exploiting force.

2. Since exploitation is normally characterised by rapid movement, mechanised battalions along with armour are the best suited for this type of operation. Mechanised battalions of armoured division frequently participate in exploitation type operations. However, regular infantry battalions may also exploit within their own capabilities. Helicopter borne forces may be profitably used during exploitation to seize key terrain and blocking enemy withdrawal routes.

### Planning for Exploitation

3. In attack, all units should be prepared to transit into the exploitation when required or when the opportunity presents itself. The main aim is to retain the initiative by preventing the enemy from reorganising his defences or conducting an organised withdrawal. Planning for exploitation is mostly conducted before the attack as part of the whole operation. Considerations for planning exploitation are essentially the same as outlined in planning of attack. However, particular emphasis is placed on organization for manoeuvre, administrative support and security. This is necessary since the exploitation normally entails seizure of deeper objectives. In such a situation calculated risks must be taken to retain momentum, even if at the expense of some security.

### Conduct of Exploitation

4. The exploitation is characterised by decentralised execution of orders and aggressive offensive action. Mission type orders may be used and minimum control measures (axis of advance, phase line, checkpoints, etcetera) are employed to facilitate a rapid advance. Forces in exploitation normally advance on a wide front, depending upon the mobility of the force, road network and other aspects of the terrain. Maximum companies of the battalion should be committed in exploitation, while essentially retaining balance.

5. In exploitation, enemy resistance will mainly consist of delaying actions by small units, defence of scattered strong points and a greater use of obstacles. The exploiting commander must, therefore, ensure that such a minor resistance does not deter him from his primary mission. Enemy forces that cannot seriously interfere with the accomplishment of unit mission are bypassed, remaining out of the ranges of Anti-Tank Guided Missiles, Recoilless Rifles and machine/Heavy Machine-guns. Use of smoke is often beneficial for bypassing action and presence of resistance

must be reported back for the following units to capture or destroy these. The entire action is characterised by boldness, maximum and prompt use of available firepower and a rapid and speedy employment of uncommitted units.

**Infantry Battalion – Follow in Zone Operation**

6. Follow in zone is an operation in which infantry units/formations follow the breaking out armoured formations within a tactical grouping and remain readily available to clear enemy opposition on axis of advance. When an infantry battalion is designated for this task, it is normally provided with sufficient transport to enable it to keep pace with the exploiting force. Infantry battalion will never be operating in isolation in this operation and will be part of a bigger force.

**Tasks**

7. During follow in zone operation an infantry battalion may be called upon to join a bigger force or to undertake any one of the following tasks:-

a. Clearance of minor oppositions bypassed by mechanized/armoured formations.

b. Assisting armoured elements in clearance of major oppositions to resume offensive.

c. Establishing lodgement when the armour comes across enemy's defence (may be second line), presenting a continuous front, that cannot be bypassed.

d. Seizing/holding of bottlenecks/restricted areas to facilitate passage of mechanized armoured units/formations particularly when such areas have strong anti-tank defence.

e. Establishment of firm base for development of armour operations.

f. Protection of flanks.

g. Maintenance/security of line of communication.

h. Relief of armoured/mechanized units by taking over the area captured by them.

**Grouping and Order of March**

8. The follow in zone force should have balanced grouping with combined arms teams/groups to enable them to undertake any type of task with speed. It will be based on analytical study of terrain, tasks and enemy reactions visualized during various phases of the operation. Grouping of engineers with infantry should be carried out for negotiating minor obstacles as well as provision of road expedients for the B vehicles. The correct order of march particularly move forward of the commanders and reconnaissance parties is also essential. Possible grouping of an

infantry brigade group, keeping in view the resources available at division and corps level is as under:-

a. Infantry Brigade.

b. Regiment Artillery.

c. Regiment/Squadron Armour.

d. Company Light Anti-Tank.

e. Company/Platoon Heavy Anti-Tank (if available).

f. Company Engineers.

g. Air Defence Detachments.

h. Elements of Electrical and Mechanical Engineers.

i. Advance Dressing Station.

j. Combat Aviation Squadron if available.

9. To ensure that the operation proceeds smoothly, the infantry and armoured commanders will be required to operate hands in glove. Duplicate communication arrangements must exist. In addition the contemporary commanders must move in a manner that they can establish physical contact when required. A suggested order of march for the leading battalion of the leading infantry brigade will be as under:-

a. Battalion 'R' group (would move with brigade 'O' group).

b. Harbour party.

c. Battalion 'O' group ('R' group of the companies).

d. ***Main Body***

(1) A and B Companies.

(2) C and D Companies.

e. Fire base group.

f. 'F' echelon group.

g. 'B' echelon (brigaded).

10. A suggested diagrammatical layout of a formation following in zone is explained in Figure 13. This is to highlight the spacing in the framework of time and to indicate their correct position with reference to each other on an axis. It would be realized that correct placing of commanders and reconnaissance parties can ensure speedy action by own force to deal with impromptu situations. This force will move from harbour area to harbour area (bound to bound) behind the breaking out armour.

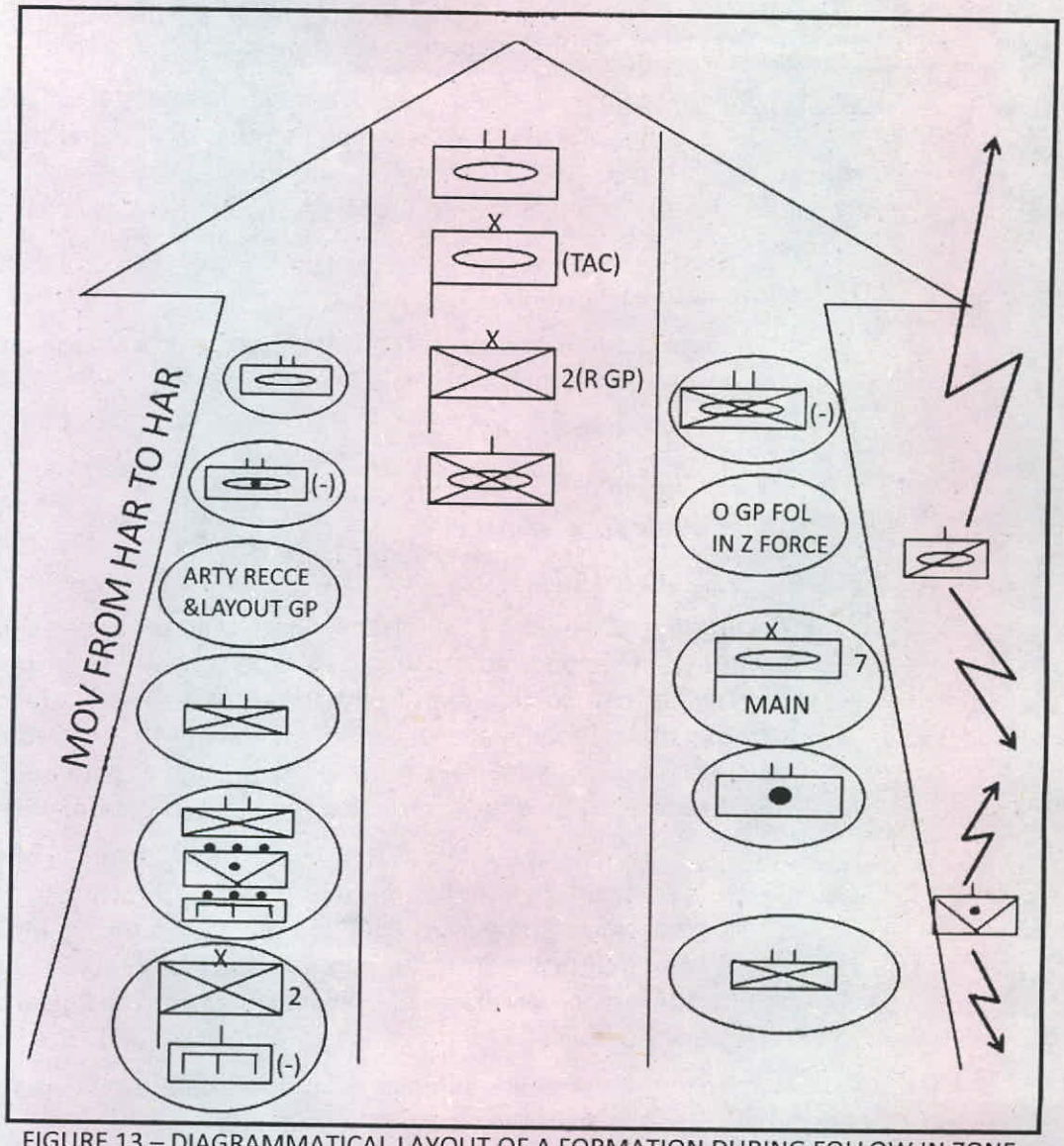


FIGURE 13 – DIAGRAMMATICAL LAYOUT OF A FORMATION DURING FOLLOW IN ZONE.

11. ***Harbour Areas***. The aim of harbour area is to dispose the units in the areas, where they can feed, rest and carry-out maintenance and replenishment, with the minimum delay and maximum security. A harbour is established when the troops are not in direct contact with the enemy and are not immediately needed for action. A harbour however should not be confused with the temporary defence occupied by the leading troops of an advancing column when halted for the night, even though contact with the enemy may not have been established. To save time and obviate the issuing of detailed orders, it is necessary to have a simple and workable Standing Operating Procedure (SOP) and method of occupying the harbour area for each

formation and unit. Units may often form part of a brigade harbour and an officer may be detailed from one of the battalions to act as the harbour party commander:-

a. ***Selection Considerations***. Harbour areas are basically selected off the map at the planning stage. However, commander of the harbour layout group can also decide and select the most suitable site for harbour at the spot. Selected harbour area should have following characteristics:-

(1) Close to the axis of advance.

(2) Should have some natural cover so that force can camouflage and conceal itself immediately with minimum effort.

(3) Easy entry and exit routes.

(4) Suitable for all-round defence.

(5) Spacious for dispersion.

b. ***Occupation/Vacation Drills***

(1) ***Occupation***. On receipt of orders, units will send harbour parties to the appointed rendezvous where they will meet the brigade harbour commander. Unit harbour parties will include guides from headquarters and subunits. A suggested composition of a battalion harbour party is given at paragraph 16a. Following should be ensured for smooth occupation:-

(a) Every man must know the action to be taken in the event of any air or ground threat. This must be rehearsed. Firing inside the harbour should be avoided. Any infiltration into the harbour must be repulsed by the use of the bayonet. Security measures inside the harbour will be enforced strictly and will entail:-

i. Immediate deployment of air defence weapons and other security arrangements.

ii. Electronic silence.

iii. No lights or fires.

iv. Minimum noise.

v. Track discipline.

(b) After evening 'stand to' no movement outside the harbour must be permitted except for patrols and listening posts. Only essential sentries should be posted to ensure that the maximum number of men in the battalion get complete rest and are ready for impending operation.

(c) For smooth occupation, battalion should train their personnel in quick embussing/debussing drills.

(d) Main body is received at the battalion checkpoint, where guides of each company/platoon are available in accordance with the order of march.

(e) Guides quickly move their respective elements to the allotted areas for immediate dispersal.

(f) Units/subunits are allotted space in the harbour areas according to the order of march for next move.

(g) Place of each vehicle is so selected that it facilitates move out in its tactical order of march.

(h) Three to four personnel are earmarked in each vehicle to help the driver in rapid camouflage/concealment of the vehicle.

(i) If the stay is less than one hour the personnel are made to rest close to the vehicles, but ensuring that they are safe from ground and aerial threat.

(j) After reaching harbour areas all companies coordinate local defence so that in case of alarm every individual knows his respective place.

(k) Slit trenches are dug in a non linear pattern for protection against aerial and ground attack.

(l) Anti-aircraft defence is coordinated.

(m) Sentries for aerial and ground observation are posted.

(n) Code word visual signals for occupation of defensive positions in case of ground/aerial attack be coordinated.

(o) Mortar platoon and artillery to select Direction Findings in likely direction of threat.

(p) Surface-to-Air Missiles are deployed to cover the area against air threat.

(2) ***Vacation***. Following actions will be taken:-

(a) On receipt of given code word preparation for vacation of harbour area commences.

(b) All weapons deployed on protection role are moved back ensuring anti-aircraft protection on the move.

(c) Companies embus and move in the given order of march.

(d) Adjutant ensures required interval between each group moving out of the harbour area. Vehicles remain dispersed in respective areas and move out in the given order of march.

(e) Search of harbour area carried out to ensure that nothing is left behind.

c. ***Security***. There are two types of threats in harbour area i.e. ground and aerial. The security of own troops is ensured against these by the following actions:-

(1) ***Ground***

(a) All-round perimeter defence.

(b) Earmarking suitable reserves with likely tasks.

(c) Posting of sentries and lookout posts.

(d) Deployment of automatic weapons.

(2) ***Air***

(a) All possible passive measures i.e. camouflage and concealment, dispersion, track discipline, etcetera.

(b) Coordination in deployment of Anza missiles, 12.7 millimetres Anti-Aircraft Machine-guns and other guns in anti-aircraft role.

(c) Appointment of air sentries to give early warning.

12. ***Coordination Points***. Infantry battalion will coordinate following points:-

a. ***Coordination with Higher Headquarters***

(1) ***Planning Stage***

(a) Units to obtain the following from higher headquarters:-

i. Operational orders.

ii. Information about regiment/squadron to which the unit is being affiliated (if peacetime affiliation cannot continue due to operational requirements).

iii. Details of the expected area of operation covering the following:-

aa. Route/axis on which own unit/ formation would follow in zone.

bb. State of roads and tracks.

cc. Water points.

dd. Cover in the shape of clumps, orchards, canal banks and standing crops etcetera.

ee. Dimensions of the distributaries, minors, nullahs, canals and their observation value. Raw material and plantation types of bunds in the area for use in making expedients.

ff. Built-up areas.

(b) Complete information about the enemy covering the following:-

i. Likely delaying positions with expected strength and composition.

ii. State of minefields and other natural/man made obstacles.

iii. Information on air activity.

iv. State of preparation and morale.

v. Details of actions on other fronts.

(c) Demand/availability of troops carrying vehicles allotted to the battalion and their collection point.

(d) Demand/allotment of artillery and engineers effort.

(e) List of special code words/nicknames and control points.

(f) Demand/collection of maps and air photos of the area.

(g) Coordination about move of elements attached to the unit like engineers/air control teams etcetera.

(h) Composition of own break-out forces, order of march etcetera.

(2) ***<u>During Operations</u>***

(a) Collection of information on enemy dispositions/ strength.

(b) State of own forces and location.

(c) Location of base of fire and strength.

(d) Employment of armour coming under command which was previously in the base of fire.

(e) Availability of artillery, air and combat aviation.

(f) Movement of armour elements for use in blocking position.

(g) Latest location/position of armour regiment/squadron.

b. ***<u>Coordination with Infantry Division Armour Regiment</u>***. Various aspects of coordination are as under:-

(1) Affiliation of battalion with squadron.

(2) Complete plan of armour regiment/formation and their likely action at each position/phase.

(3) Armour approach of the likely enemy actions and reactions.

(4) Their view of likely positions where infantry is required for clearance.

(5) Exchange of signal instructions and communication set-up, to include the frequency being used by own units/companies/ squadrons.

(6) List of code words, nicknames and control points.

(7) Arrangements for infantry tank target indication.

(8) Arrangements/coordination for communicating with armour regiment/squadron commander in contact with enemy.

(9) Location and allotment of tanks for mounting the infantry whenever required.

(10) Coordination for joint attack if required.

**Logistic Support**

13. The nature of the envisaged operation warrants that in addition to the logistic support at formation level, sound administrative arrangements are also required at individual and unit level. Following merit consideration:-

a. ***Individual Load***. Since the individual is expected to be invariably on the move during operation an effort should be made to make his load as light as possible. Nevertheless this should not compromise his ability to sustain himself during the battle. These items include light bedding, which should be carried personally by the individual in his rucksack. Some of the important items are:-

| | | | |
|---|---|---|---|
| (1) | Emergency pack rations | – | 1 day. |
| (2) | Pouch ammunition | – | Of the respective weapon. |
| (3) | Light bedding | – | A blanket with allied items. |
| (4) | Water Bottle | – | Should always be kept filled with drinking water taken from authorized source. |

b. ***Unit Administration***. The unit should join the battle self-contained particularly in the field of local administration. Loads should be prioritized and worked out on the basis of held number of vehicles rather than the authorized ones. This will ensure unhindered development of operation in the face of various constraints and bottlenecks as already discussed.

14. ***Composition of Various Groups***. Battalion 'O' group, 'R' group and fire base will be formed as per Standing Operating Procedures. Rest of the groups will be as follows:-

a. ***Unit Harbour Party***. Guidance is essential for the main body to occupy and vacate harbour areas smoothly in the minimum possible time without any chaos and confusion. This necessitates having harbour layout groups with proper composition and tasks. In fast moving operation, there may even be a requirement of duplicate layout groups. This is necessary to avoid miscoordination and confusion in fluid situation because, while one layout group is busy receiving and deploying the battalion, the other one has already moved forward to select the next harbour area. With some ingenuity and if the unit is well-trained, the second layout group can also be organized from within the first one. The suggested composition of a harbour layout party at the unit level is given below:-

(1) ***Commander***. Officer Commanding Combat Support Company.

(2) ***Manpower***

(a) Naib Subedar Adjutant.

(b) Regimental Police Havildar with 2 soldiers.

(c) Junior Commissioned Officer/Non-Commissioned Officer with 4 soldiers from each company.

(d) One guide for each specialist platoon.

(e) One Light-Machine gun per company.

(f) Wireless sets with operators.

(g) Line laying party (if stay is longer).

(h) Infantry platoon for protection.

(3) ***Transport***

(a) Two jeeps.

(b) One dodge.

(c) 2 x 2½ ton.

(d) Motorcycle.

15. ***Main Body***

a. ***Commander***. Second-in-Command.

b. ***Troops***. Battalion less 'R' group and 'O' group.

16. ***'F' Echelon Group***

a. ***Commander***. Battalion Havildar Major/Mechanical Transport Junior Commissioned Officer.

b. ***Transport***. All 'F' echelon vehicles of the companies/combat support company.

**Pursuit**

17. ***General***. Pursuit differs from exploitation in that its primary purpose is annihilation of the enemy force, rather than the seizure of terrain. In pursuit the battalion may mostly operate as part of the direct pressure force of the enveloping force. The actions in pursuit at battalion level are essentially the same as in exploitation.

18. As soon as the enemy is thrown off balance and loses the initiative, he should never be allowed to regain it. His organised withdrawal should be turned into a route. The morale of own troops will be at its highest at this time and great demands can be made on them. The greater the pressure on the enemy, the easier will the fighting become; the fewer our own casualties, the greater the prize.

19. The process of moving forward and confronting pockets of resistance may go on for some distance. The mobile troops may often be able to deal with these, though the time will come when they are checked and can make no further progress or when the vanguard meets a locality against which it can advance no further. The battalion commander may then:-

a. Order the leading company to take over from the mobile troops.

b. Order the vanguard company to remain in the fire base to give fire support to the battalion attack.

c. Use the vanguard company as part of the main battalion assault.

**Bypassing of Opposition**

20. Commanders may be required to outflank or bypass opposition. This call for initiative on the part of the advance guard commander who must, however, be given information about the opposition he is to bypass. He must pass this on to the vanguard commander. Information must be sent back about the enemy bypassed and arrangements made subsequently to deal with them.

**Air Support**

21. Skilful application of immediate support may greatly assist in maintaining the momentum of pursuit. Unit commanders must exercise great care in reporting their estimated future location at a given time, so that the bomb line can be appropriately established. Any changes must also be passed back immediately to ensure safety of own troops.

## SECTION 38 – FIRE SUPPORT PLAN

**Artillery Fire Support Plan**

1. While it is the responsibility of the direct support field battery commander to advise on the nature and scope of fire support for battalion operations, it is the battalion commander who must decide on the type of fire support that he would like and to state what he wishes the fire plan to achieve.

2. The fire support plan should cover movement and deal with the enemy threatening the advance during the three stages of attack as follows:-

a. ***Move from Assembly Area to Start Line***

(1) Blind enemy with smoke to cover own movement.

(2) Deception with HE or smoke on another part of the front.

(3) Preparatory bombardment of enemy defences to demoralise him and to cause casualties, or silent policy, if achievement of surprise is intended.

(4) Covering fire to support own physical assault against enemy opposition.

b. ***Assault, Fighting through the Objective and Exploitation***. Fire will first deal with the enemy on the near edge of the objective and will then be lifted to neutralise successive enemy positions in depth right through the objective in front of the advancing infantry.

c. ***Reorganization***. Possible/probable defensive fire tasks for artillery and mortars to break-up enemy counter-attack must be laid down before hand and corrected as soon as actual dispositions on the ground are known. Blinding of enemy observation posts with smoke must also be included in the plan.

**Sequence in Making the Fire Support Plan**

3. Fire support plan is an integral part of the tactical or manoeuvre plan and must be considered within the wider framework of the making of that plan. The sequence for considering the fire support plan is as follows:-

a. Brigade commander gives out his outline plan to include the fire support.

b. Battalion commander carries out reconnaissance accompanied by his direct support field battery commander and also decides about the weapons (Machine-guns, Infantry Mortars, Recoilless Rifles, Baktar Shikans, Tanks) that he can make available for the fire support plan.

c. Battalion commander gives out his manoeuvre (or attack) plan, to include the following:-

(1) Phases of attack - objectives and allotment of troops.

(2) Start Line.

(3) Timings (H hour etcetera).

(4) Rate of advance.

d. Battalion commander, in consultation with the battery commander, decides on a rough division of ammunition available to him for each phase and the proportion of the ammunition that will be kept as reserve for on call impromptu targets and defensive fires.

e. The fire support plan is made by phases, each phase being dealt with as follows:-

(1) Battalion commander states the enemy localities that are likely to interfere and as to when does he want them to be neutralized, relative to the manoeuvre plan. The targets may be divided into 'the essential' and 'the desirable' categories and the guns at call can, if necessary, deal with the latter.

(2) Direct support field battery commander allots target numbers and recommends the punishment for each target, i.e. number and type of guns, number and type of rounds and the rate of fire. The armour adviser is consulted with regard to the targets to be engaged by the tanks. The battalion commander approves or modifies these recommendations.

(3) If air support is available the battalion commander, advised by the battery commander, also decides on its employment.

(4) Direct support field battery commander keeps a running total of the ammunition expenditure. If resources are limited, the battalion commander decides on the priorities.

(5) Having decided on the fire support to fit the manoeuvre (or attack) plan, the battalion commander, in consultation with the battery commander, decides on the on call targets; targets on the flanks and prearranged defensive fires. The number of guns required to be at call to meet contingencies depend on circumstances. For the un-located targets that may appear during the battle, prearranged concentrations on likely places can be called.

f. When fire support for various phases has been planned, battalion commander, in consultation with the battery commander, will consider the following depending, of course, on the availability of ammunition:-

(1) Preparatory bombardment.

(2) Use of the artillery for deception.

(3) Smoke screens to cover move from Forming Up Place to the Start Line.

g. Depending on the time available, detailed timings for the engagement of targets are either worked out jointly by the battalion commander and battery commander, or by the latter alone, based on the

manoeuvre plan and the battalion commander's desire as when each target has to be neutralised relative to the plan.

h. Details of the fire support plan, including the tasks allotted to different support weapons (air, armour, artillery, infantry weapons) are entered on the task table (PAFB-6047) by the direct support field battery commander. He completes the task table, gets it approved by the battalion commander. This final approval of the completed task table by the battalion commander is the most essential. It will often be given only after the company commanders and their artillery observers have had a chance of examining and discussing the fire support plan and suggesting any modifications they think necessary.

i. The battalion fire support plan will then be submitted to the direct support field regiment commander by the battery commander. The former will coordinate the fire support plans of battalions, on behalf of the brigade commander (usually at a coordinating conference with the battery commanders). Approval of the fire support plan, with modifications, if necessary, is signified in the brigade commander's coordinating conference.

**Issue of Fire Support Plan**

4. The battalion fire support plan, as recorded on PAFB-6047, should be given out by the battalion commander during his verbal orders. It not only provides information to all concerned regarding the complete fire support plan, but also constitutes executive order for the supporting arms commanders, to carry-out the tasks allotted to them in it. It is not the responsibility of direct support field battery commander to issue orders for the implementation of the fire support plan to non-artillery units.

5. Infantry commanders must remember that preparation and issue of necessary artillery orders may take considerable time and may govern the timings of H hour. A simple fire support plan involving a battery or regiment may take only a few minutes to arrange, but a plan involving whole divisional artillery may even under the most favourable conditions, take more time.

**Registration**

6. Registration of targets before a fire plan is not always essential or acceptable. Provided necessary condition for accurate predicted shooting apply and the engagement of unregistered targets is satisfactory. When in the opinion of the artillery adviser concerned those conditions are not prevalent, battalion commander may be advised to allow registration by artillery even at the risk of losing surprise.

**Observation**

7. A field battery is organized to produce three observation posts/forward observation officers parties. It is only in very exceptional circumstances that a battery should be required to increase this number. When this is done the strain on the battery's resources is great and can only be borne for a limited time. Observation posts/forward observation officers deployed in support of an infantry battalion will normally be found by the affiliated battery, when this is not so they will come from the battery's parent regiment or other artillery resources.

8. In addition to field artillery observation posts, other artillery observation posts will often be deployed. The deployment of artillery observation posts in an infantry brigade area is controlled by the commander of the field regiment in direct support of that brigade, similarly observation posts deployed within a battalion area will be controlled by the field battery commander in direct support of the battalion. This prevents the battalion commander having to deal with a number of artillery officers. It also ensures that all observation posts can be kept in touch with the infantry intentions, patrols etcetera.

9. Field artillery observation posts are equipped with wireless sets which enable them to be out stationed on the battalion command net. Any rifle company commander can therefore call on an observation post for artillery support. If for any reason an Observation Post cannot see the target, infantry officer can control the artillery fire using target grid procedure which is described in Chapter IX, Section 41.

**Air Support**

10. When planning an attack, the battalion commander must consider whether air force can economically contribute to the success of his plan, at his level he will be mainly concerned with close air support and air reconnaissance information.

11. In a deliberate attack, air support will be largely pre-planned at a high level; battalion commander must pass his request for air support as soon as possible to give maximum time for detailed ground/air planning. Weather may interfere with air support in the battle. Decisions must, therefore, be made as to whether:-

a. The attack must proceed according to the set timing irrespective of the weather, or

b. The attack will be delayed till development of favourable weather to the air. In case of a above, two fire plans must be made, one with ground and air support full integrated while the other relying entirely on ground weapons.

# CHAPTER XIV – THE ATTACK

## SECTION 39 – GENERAL

**Introduction**

1. Attack is a dynamic action which defines the basis of war. Only a successful attack defines military success, therefore, extreme care should be taken to precisely define the military objective. This chapter should be read in conjunction with Sections 6 to 13 of the GSP-1991 – "Operations of War (Volume- II), Offensive Operations, 2004". Attack has one or a combination of the following as its aim:-

a. Destruction of enemy forces.

b. Capture of territory.

c. Destruction of enemy resources.

**Definitions**

2. Some of the attack definitions that must be understood are as follows:-

a. ***Deliberate Attack***. A carefully planned and executed offensive action, in which various elements of a command are employed in such a manner as to utilise their powers to the greatest advantage to the command, as a whole.

b. ***Counter-Attack***. Offensive action by a part or all of a defending force against an attacking force for the purpose of restoring original positions and trapping and destroying the hostile forces at a point particularly favourable to the counter-attack. It is carried out by a pre-designated part of the force.

c. ***Local Counter-Attack***. An attack organised by the local commander to destroy enemy forces which have penetrated or threaten to penetrate his defensive position, generally using a prearranged plan.

d. ***Diversionary Attack***. An attack, wherein a smaller force attacks, or threatens to attack, a target other than the main target, for the purpose of drawing enemy attention away from the main attack .

e. ***Feint Attack***. An attack which simulates intense activity of a large scale attack and keeps the enemy guessing about its weight and time.

f. ***Holding Attack***. Attack designed to hold the enemy in position, to deceive him as to where the main attack is being made, to prevent him from reinforcing the elements opposing the main attack and to cause him to commit his reserves prematurely and at an indecisive location. Also called containing attack.

g. ***Attack by Infiltration***. When infiltration is the main feature of an offensive operation, it will be described as an attack by infiltration.

h. ***Main Attack***. The principal attack or effort, into which the commander throws the full weight of the offensive power at his disposal, directed against the main objective of the campaign or battle.

i. ***Mopping Up***. The liquidation of remnants of enemy resistance in an area that has been surrounded or isolated, or from an area where other units have assembled/pass through without eliminating all active resistance.

j. ***Spoiling Attack***. A tactical manoeuvre employed to seriously impair or delay a hostile attack. It is normally launched while the enemy is in the process of assembling or forming up for an attack. Usually employed by the units in defence or in withdrawal on enemy assembly area or Forming Up Place.

k. ***Assembly Area***. Assembly area is the place where attacking units/ subunits organize for an attack. An ideal assembly area is free from enemy observation and harassing fire and has good entrances and exits. Here battalion, company, platoon and section commanders make contact with their counter parts of the supporting arms, if not already done. Communications are checked, ammunition issued, weapons examined for their serviceability and reorganization stores sorted out. If possible, troops are given hot tea or meals particularly in the case of night attack. Orders are rechecked.

l. ***Forming Up Place (FUP)***. From the assembly area assaulting troops move to the FUP and deploy in the battle formation in which they have to advance up to the objective. The FUP needs to be easily recognisable and well sign posted for a night attack, if necessary. It must not be under enemy observation. It is from here that the troops break cover at H hour to advance towards their objective. The battalion/company commander must ensure that he is in his proper place in relation to the supporting tanks.

m. ***H Hour***. H hour is the time when the leading elements of the assaulting troops cross the Start Line. All timings for an attack are given in relation to H hour.

n. ***Start Line***. Start Line is the line which the leading elements of the assaulting troops cross at H hour. It must be clearly recognisable. It may be a natural feature such as a road or a narrow water channel,

or even artificial marked with the help of tape niwar, cable, lime or lights etcetera. It should, if possible be at right angle to the line of advance and must be secure from the enemy interference. It must not be behind any obstacle. It should be as close to the enemy as possible but not under observed fire of his direct or indirect fire weapons. Usually, the forward edge of the Forming Up Place can be the Start Line, especially for a night attack.

o. ***Assault Line***. This is a line used to coordinate the final deployment of attacking troops in preparation for assault against an enemy position. It is located as close to enemy positions on the objective as attacking troops can move, before becoming dangerously exposed to own supporting fire. Artillery will cease to engage forward enemy positions when the attacking troops reach this line; arty fire is then shifted to engage targets in depth/on flanks. Troops carrying out the assault will continue their movement to the objective. There may, however, be occasions when the progress of attack is retarded or even temporarily halted by accurate enemy fire. In such situations, the assaulting troops must get as close to the enemy positions as safety conditions permit, taking maximum advantage of the supporting fire. The moment the fire is lifted they will assault.

**Stages of Attack**

3. At divisional and brigade level 'fixing and developing the manoeuvre' constitutes the first stage. The whole extent of the defensive position is frontally contained and continuous pressure is maintained against it. However, at a battalion level this stage will only be applicable when battalion is employed in an independent role. Under normal circumstances a battalion attack resolves itself into four stages (see Figure 14):-

a. Stage 1 – Preparatory.

b. Stage 2 – Assault.

c. Stage 3 – Destruction.

d. Stage 4 – Exploitation.

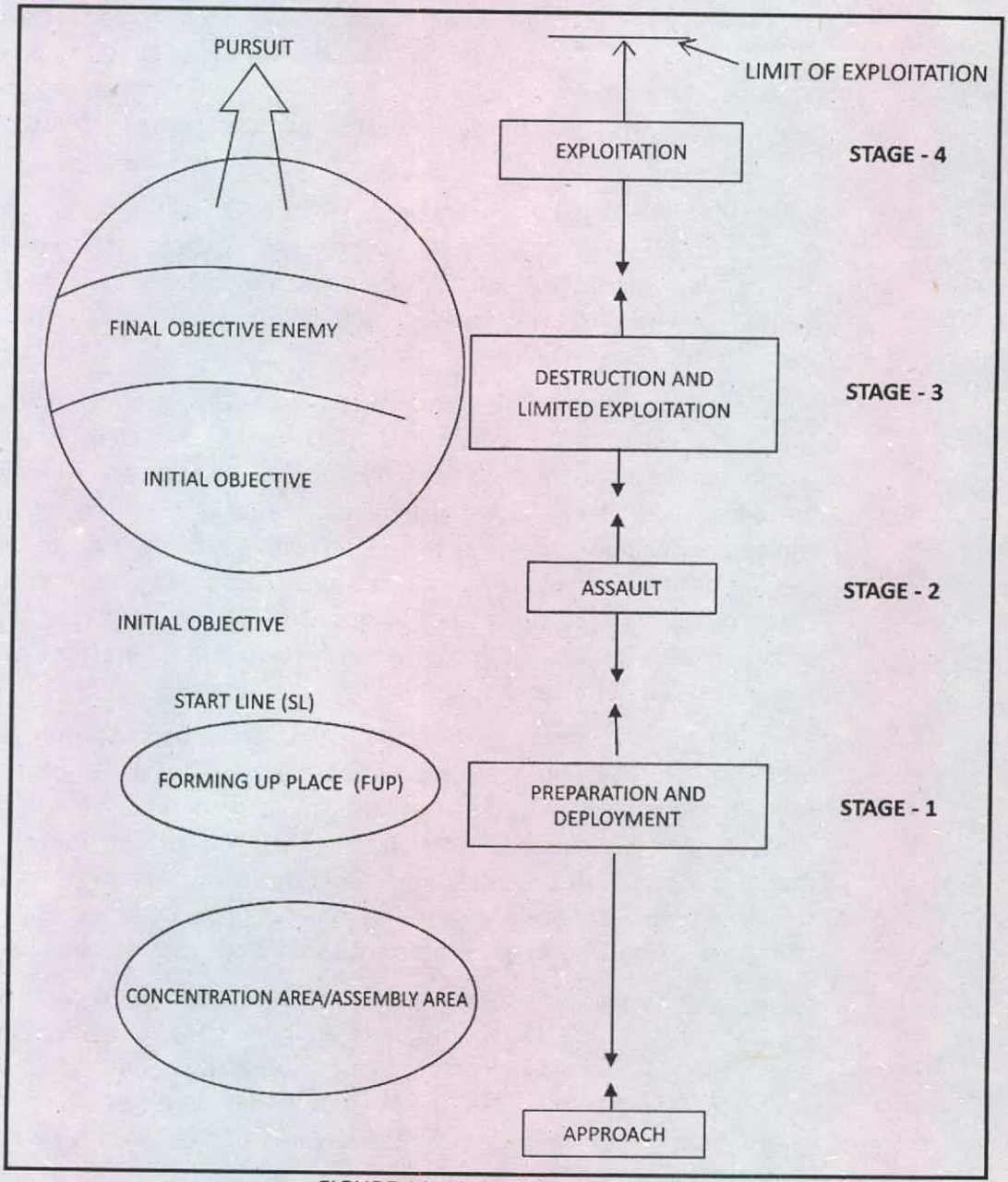


FIGURE 14 – STAGES OF ATTACK.

**Basic Considerations**

4. Whatever the type of attack following basic considerations must receive attention:-

a. ***Initiative***. It is vital to gain and retain own freedom of action while denying that to the enemy. The battalion/company/platoon commanders must maintain initiative to dominate the enemy by active ground reconnaissance. All efforts should be made to continue

pressure on the enemy and not to allow him any rest until the attack is finally launched. Reorganization to hold captured ground must be rapid. After reorganization No Man's Land to the front as well as flanks must once again be dominated through a strong network of patrols.

b. ***Selection of Objectives***. It should be ensured that the objectives selected are the positions that will hurt the enemy the most and their capture will neutralise him. The objectives should be easily identifiable and their capture must be possible within specified time and with the troops employed.

c. ***Impetus***. The impetus of the attack must not be allowed to die down. The attack must be organised in depth, so that sufficient troops remain readily available at all times for subsequent tasks and to press the advantages. This will entail immediate attack by the follow-up companies/platoons once the phase-I objective has been captured. Delay in attack or launching of subsequent phases will give reaction time to the enemy to either evict own troops through counter-attack or comfortably reorganise his defences on some next line/position. This may result in failure of attack.

d. ***Fire Support***. The battalion/company attacks must be supported by fire upto and beyond the objective. Supporting weapons of the battalion (usually deployed in fire base) must be moved forward as soon as possible according to the progress of attack. Tanks, if attacking along with the company should move as close as possible behind own artillery fire and provide intimate support. The assaulting infantry and tanks (when present) must use fire and move technique.

e. ***Intelligence***. The battalion/company commanders should at all times try to get real time information about enemy dispositions, weapons and obstacles. It would help in capturing of objectives. This should be done by use of patrols, Unmanned Aerial Vehicles and air photographs etcetera both before as well as during the attack and also getting whatever information is available with the higher headquarters.

f. ***Speed***. Effort should be made not to allow sufficient time to the enemy for preparation of his defences or to intervene with his reserves. The weaker the enemy defences, lesser the time needed for preparation of an attack, therefore the urgency to quickly get on with the battle procedure and deployment of troops to capture the objective must remain predominant for the battalion and company commanders.

## SECTION 40 – APPRECIATION

1. In a coordinated brigade attack, a battalion commander is not expected to go through the mechanics of a formal appreciation as it would have been done by the brigade commander. The brigade plan will include the following:-

a. Objective.

b. Phases of attack – allotment of troops.

c. Direction of attack.

d. Forming Up Place and Start Line.

e. Timings (H hour etcetera).

f. Limits/extent of exploitation.

2. In this case, the job of the battalion commander is very simple. Besides, designating companies for each phase of attack, he has to make sure that the battalion arrives in the Forming Up Place in time and the troops cross the Start Line on H hour. Very little planning is conducted by the battalion commander for attack but he has to ensure execution of the brigade commander's plan.

3. However, when the battalion is acting independently for launching an attack or is engaged in encounter battle, a battalion commander will be required to carry-out appreciation, which under the circumstances may be a mental one. For an appreciation and making an outline plan, the battalion commander will need following information:-

a. Mission/tasks

b. Ground and enemy dispositions.

c. Own and enemy resources.

d. Time available.

4. Following are the main sources of information, available to the battalion commander:-

a. Personal reconnaissance.

b. Maps and air photographs.

c. Units in contact with the enemy.

d. Higher headquarters.

e. Own intelligence staff.

f. Observation Posts – own and artillery.

g. Supporting arms.

h. Patrols.

i. Prisoners of War.

j. Locals.

5. As the battalion commander is formulating his attack plan, he should consult his supporting arms commanders and seek their advice on the employment of their particular arms. As soon as an outline plan has been conceived, a warning order should be issued to all concerned setting the entire machinery in motion. To save time and to ensure smooth execution of the plan, simultaneous actions on various aspects of the attack would be undertaken.

**Battalion Commanders' Decision Regarding Attack**

6. Battalion commander must decide the following while making his plan:-

a. Objectives and phases of the attack.

b. Organization of the battalion into assault and follow-up companies and attack formations in each phase.

c. Selection of assembly area, Forming Up Place and Start Line.

d. Method of crossing obstacles.

e. Boundaries during assault as well as reorganization.

f. H hour.

g. Support of light and heavy anti-tank elements, if allotted.

h. Movement plan for each phase of the attack.

i. Exploitation.

j. Reorganization, including the move forward of 'F' echelon transport.

k. Tank support, their grouping and tasks.

l. Fire plan and coordination of fire available from integral heavy weapons, artillery, tanks and air.

m. Administration.

n. Control of the battle.

## SECTION 41 – ORDERS AND BRIEFING

**Issue of Orders**

1. The battalion commanders' orders will usually be given verbally. The following points should be observed:-

   a. The 'O' group should also include supporting arms representatives.

   b. Those attending the 'O' group should preferably have previously carried out reconnaissance or, failing this, studied the ground on model or air photographs.

   c. Rendezvous (RV) for the orders should be properly sign posted. Arrangements should also be made for dispersion of vehicles in the vicinity.

   d. Movement to the view point should be controlled properly for maintaining secrecy.

   e. The setting should, if possible, be free from distraction and have an air of calmness. There should be no interruptions. The adjutant and intelligence officer are responsible for the setting (which must be protected) and for ensuring the presence of all concerned with the right maps.

   f. A short preliminary talk should be given to the 'O' group by the intelligence officer to explain the general situation and the maps. For this, it is preferable for the 'O' group to reach about fifteen minutes before the appointed time on the Rendezvous (RV).

2. Orders should cover clearly all the phases separately. For each phase, following details should be given:-

   a. The move to the Start Line.

   b. The assault and exploitation.

   c. Action of supporting arms.

   d. Reorganization.

   e. ***Fire Support Plan***. Detailed timings of fire support should be prepared before hand and issued in writing.

**Briefing**

3. Briefing of men will normally be done by company or platoon commanders, though on occasions a battalion commander may have the opportunity to brief the whole battalion himself. Men go into battle with greater confidence and higher morale if the nature of their future operation is well explained to them. Briefing is, however, no substitute for orders which must always be given in the correct form by the subunit commanders.

## SECTION 42 – STAGES OF ATTACK

**Stage 1 – The Preparation, Assembly and Move Forward to the Start Line**

1. ***Requirements of the Assembly Area***. The Assembly Area should:-

   a. Be secure, under cover and, if possible, free from shelling.

   b. Have a good entrance and exit for vehicles as it is here that supporting arms normally marry up with the battalion.

   c. Normally be located one to five miles from the enemy position/ objectives.

   d. Be carefully sign posted.

2. ***Actions in the Assembly Area***. The battalion with the necessary vehicles and support weapons will move into the assembly area during the reconnaissance/ orders period. Actions taken here will include:-

   a. Immediate arrangements for local protection.

   b. Checking of weapons and equipment.

   c. Administrative arrangements.

   d. Netting of wireless sets.

   e. Marrying up with the supporting arms.

3. ***Move from the Assembly Area to Forming Up Place***. The route (or routes) should be clearly marked and if necessary, guides provided. Tapes or glow-worm lights may be used. When tanks are cooperating with infantry the route marking and the marking of the Forming Up Place itself is the task of the infantry accompanied by the liaison officers or guides of the tank subunits taking part. It can only be done after a thorough reconnaissance.

4. ***Requirements of the Forming Up Place***. The selection of Forming Up Place requires considerable care, especially when tanks and infantry both, are to move. The Forming Up Place must be close to the Start Line, but clear of the likely enemy observed fire tasks. It should offer concealment, depending on the general air situation and the enemy observation points. It must have good exits forward and also be easy to approach. The latter, is especially important when the tanks have to approach from a flank as it entails changing of direction within the Forming Up Place area with the consequent danger of loss of direction particularly at night.

5. The enemy in defence will have patrols and listening posts suitably deployed to gain earliest possible information about own intentions and strength and the possible direction of the attack. The enemy will make efforts to interfere with the attack by artillery fire or by launching spoiling attack. It is, therefore, essential that

the No Man's Land is dominated, the routes to and the Forming Up Place itself are secured. If possible, no light should be used; otherwise, minimum number of lights for shortest possible duration should be used.

6. It is not necessary to have the same Forming Up Place for both infantry and tanks, even if they are attacking on the same axis, as one of the requirements is that it should not be so far away that by the time the infantry reaches the objective, it is too tired. On the other hand, as the tanks produce noise and also present a larger target, their Forming Up Place should be at a greater distance. Alternatively, if the forming up place has to be the same, the tanks should arrive there at the last possible moment. Even so, it should be out of the enemy anti-tank and tank guns effective range and/or it should have suitable cover.

7. Great care and coordination will be required for forming up during bad visibility as the tanks could over run own infantry in the Forming Up Place. The possible methods are:-

a. The infantry arrives first and forms up. The tanks arrive at the last possible moment and are led individually to their respective places.

b. The tanks arrive first and form up while the infantry arrives and forms up later.

c. The tanks arrive with infantry mounted and form up. The infantry dismounts and forms up and the attack is launched.

8. In almost all the cases, there will be a requirement for deception and drowning of the noises. The following deception measures may be adopted for a battalion attack:-

a. Show movement in areas other than the point of attack. This task may be assigned to the holding troops, if possible.

b. Carry-out shelling in the areas other than or adjacent to the point of attack.

c. Patrolling activity should be carried out over a wider frontage with the assistance of neighbouring units to conceal own intentions.

d. Carry-out feints or diversionary attacks.

e. Depending on the availability of additional wireless sets, operate a fake net to deceive the enemy regarding the direction and timings etcetera of attack.

f. Simulate tank noises with the help of loud speakers in areas other than the point of attack.

9. Plan anti spoiling attack measures which may be done by assigning the task of dealing with the enemy spoiling attacks to one of the companies. Additional artillery may also be asked to register flanks of the Forming Up Place as targets to beat off enemy spoiling attacks when required.

10. ***Actions in the Forming Up Place***. The assaulting troops and supporting arms deploy in battle formations and make final preparations for the attack. Time spent in the Forming Up Place should be as short as possible because attack may be seriously jeopardised if the troops come under heavy fire at this stage.

11. If the forming up is to be carried out during daylight necessary deployment must be carried out as rapidly as possible. Following is the normal sequence of events connected with the Forming Up Place when an armoured squadron is also under command of an infantry battalion:-

a. The routes to the Forming Up Place, the layout on arrival and the Start Line are marked by the infantry, usually by the battalion intelligence officer accompanied by the liaison officer of the armoured squadron.

b. Guides meet the troops on the routes from the assembly area to the Forming Up Place(s) and guide them to their respective areas. Except at night, the infantry should arrive at the Forming Up Place with the tanks or wait for them there.

c. Infantry and tank commanders meet and check all last minute arrangements.

d. Infantry and/or tanks move to cross the Start Line at H hour.

12. ***Requirements for the Start Line***. The Start Line should:-

a. Be square to the objective.

b. It should easily be identifiable. If possible, it should follow a natural feature such as a hedge or a track etcetera. Failing this, the intelligence officer should have it marked. There may a possibility that the tanks may miss an artificial Start Line and thus upset the timings of attack in the process.

c. It must be secure.

d. All tanks must be lined up on correct bearing for passing through the gaps in our own minefields where they exist. Where practicable the gaps should be made on the same bearing as for the attack and suitably marked. Tanks, however, should not be regarded as a framework on which the other units can adjust their positions on the Start Line.

13. If tanks are not in the lead for the initial phase of the attack then they must be timed to be in their initial firing positions before the infantry advance starts. If, on the other hand, tanks are to lead, they will generally have to start behind the infantry Start Line as the noise of their approach may give away the attack. In this case it may be necessary to have a tank Start Line separate from the infantry Start Line, or alternatively to start the tanks before the infantry. Timings in these circumstances will require careful calculation.

**Stage 2 – Assault**

14. ***General***. The assault is the stage when own artillery fire lifts to the depth objectives and the assaulting troops should expect a heavy volume of fire, of all types, from the enemy position, if the enemy is still holding on to his defences. It is at this stage that the attacking troops storm the forward defended localities of the enemy and do not allow them to retaliate by firing their weapons.

15. The assault stage is split in two parts i.e. "The Break-In" or "The Storming of the Forward Defended Localities" and "The Fighting on the Objective" or the "Dog Fight". In the "Break-In" the enemy defences are rushed/assaulted and in the "Fighting on the Objective", if the assaulting troops are held up and the assault is not making any progress the enemy defences are overcome by fire and movement.

16. Following points should be borne in mind during the assault stage:-

a. At H hour the leading groups break cover, cross the Start Line and advance towards the objective.

b. Organic machine-guns of assault companies and additional machine-guns, if allotted, move forward with the assaulting troops in accordance with the movement plan.

c. Depending upon the distance between the Start Line and the objective and the obstacles in between, cover available, the Recoilless Rifles and/or Baktar Shikan should either support the assault from a preselected position in the proximity of Start Line or move forward to suitable positions at a time decided by the battalion commander concerned.

d. During the move from the Forming Up Place to the objective enemy minefield is also likely to be encountered. In such an eventuality the Forming Up Place will be located on the home side (own side) of the enemy minefield. If the lanes through the enemy minefield have not been cleared before hand for the sake of surprise and secrecy,

infantry will have to assault through the minefield. There are the following two methods of doing so:-

(1) ***Assault through the Minefield in Extended Formation***. Advantages and disadvantages are:-

(a) ***Advantages***

i. If jumping mines have been combined with other mines by the enemy there will be lesser casualties.

ii. There will be lesser number of casualties from fire of enemy automatics and artillery.

iii. There will be no requirement for a fresh forming up on the enemy side of the minefield and thus there will be no waste of time and loss of control.

iv. Infantry approaches the objective in org battle group, more suitable to react to any unforeseen situation.

(b) ***Disadvantages***

i. There are likely to be more casualties by anti-personnel mines.

ii. Casualty evacuation will be extremely difficult and time consuming.

(2) ***Assault through the Minefield in Single File***. Advantages and disadvantages are:-

(a) ***Advantages***

i. There are likely to be less casualties due to anti-personnel mines.

ii. Better command and control.

iii. There will be lesser casualties due to enemy artillery and automatic fire.

(b) ***Disadvantages***

i. There will be a requirement for a fresh forming up on the enemy side of the minefield.

ii. Since fresh forming up is required which may be under enemy artillery and automatics, lot of valuable time will be lost.

iii. Morale will be adversely affected by casualties sustained under the very eyes of the troops.

iv. Strenuous for leadership.

e. Assaulting troops should expect heavy volume of fire of all types from the enemy position at the time when own artillery fire lifts to the depth positions. As a result, own troops are likely to suffer casualties. Momentum of the advance to be maintained by fire and movement and with the help of depth sections. Platoons should reduce the strong points one by one.

f. The assault troops may have to resort to fire and movement within the platoons and sections to cross undulating ground mounds or low ground that is likely to be dominated by the enemy troops.

g. An assaulting company is likely to come across one of the following situations in attack:-

(1) ***First***. Attack is successful and the objective is captured.

(2) ***Second***. One of the assaulting platoons is held up while the other one is maintaining a steady progress towards the objective. The 'held up' troops should under no circumstances, become passive and paralysed in that position. They should make good use of ground and fire and try to advance as far forward as possible. In case, no progress is being made due to enemy action, they should support the other platoon with their fire. Consequently, the company commander should commit his follow-up platoon in the assault (if possible from a flank) to capture the objective.

(3) ***Third***. Both the assaulting platoons are held up and attack is not making any progress. In this case, the 'held up' platoons should continue to creep forward and at the same time, support the third platoon by fire when it is finally committed by the company commander for accomplishing the assigned mission. Company commanders should drive his men hard and force them to close-up with the enemy positions. Determined troops cannot be stopped by the enemy. Commanders should be well forward to control the battle. However, if the company still fails to make any progress, it should nevertheless continue to exert pressure on the enemy by engaging him with fire. It should form a base of fire to facilitate the attack by one of the follow-up companies from a different direction. In this manner it can still contribute a lot towards success even while remaining pinned down.

h. ***Employment of Follow-Up Companies***

(1) Follow-up companies should be employed against the flank or rear of the enemy or against any other point of known or suspected weakness. Passing the follow-up troops through the subunits halted due to enemy action should be avoided whenever possible. The "follow-up" company should rather be committed from a new direction to avoid massing and to achieve surprise. They should actually be committed to exploit success and not to redeem failure.

(2) The "follow-up" companies should be located in a position from where they can be rapidly moved to the points of probable employment. As the attack progresses there are kept close enough to the attacking echelons to either press the momentum of the attack or to intervene if the enemy counter-attacks.

(3) The battalion commander commits only that portion of the follow-up company that is required to accomplish a specific task. When, in his judgement, the situation warrants employment of the remaining companies, he commits them without hesitation and accordingly informs the brigade commander.

i. Occasions may arise when all the companies will be committed in the attack. This information should be passed to brigade headquarters immediately so that appropriate action is taken by them.

17. ***The Axis of Advance when Infantry and Tanks are Cooperating***

a. Infantry and tanks may begin the assault by advancing on same axis, on converging axis, or on parallel axis, even the last two converge on the objective. The method adopted depends on the ground, the nature of the assault and also, to some extent, on the training of the units concerned. Whichever method is used, the principle is that the tanks and infantry should try to arrive together on the objective as far as possible.

b. In an assault on a strongly defended position, it is usual for the infantry and tanks to be on the same axis. It is so because it is more difficult to coordinate the arrival of the tanks and the infantry onto the objective when the axis of advance is not common to both.

18. ***Method of Attack when Infantry and Tanks are Cooperating***

a. ***The Leading Tank Troops***. The tanks and infantry may not have same Forming Up Place, even though the axis of attack may be the same. It is so because the tanks must remain outside the maximum effective range of enemy anti-tank weapons, while the infantry is required to be as close to the objective as possible, consistent with the other requirements of the Forming Up Place. At the start of attack, the leading echelons will generally consist of infantry and tanks. Preferably, the tanks may give fire support from their Forming Up Place and/or any other suitable firing positions, to shoot the infantry on to the objective. When infantry is close to its assault line (150/250 metres from the objective) the tanks of the assault echelon will rush forward, overtaking infantry and arriving on the objective just before or along with the infantry. The combined tanks-infantry advance/ assault will depend on terrain, enemy and own situation. They may cross the Start Line together, or leapfrog their way forward when forced by the enemy. The infantry will move at its best speed and by the best infantry route straight to the objective. The tanks in the leading echelon support the infantry on to the objective. To do this, they may not necessarily move with the infantry, but could also employ their own method of advance, keeping the infantry in view as far as possible. Tanks, whether leading or supporting, must not be tied down to the infantry speed. Except in open country, tanks will always require some infantry to accompany them if their axis of advance is different from the main infantry axis.

b. During the attack the leading tank troops have four clear tasks:-

(1) To be in direct support of the infantry and assist it in overcoming any unexpected opposition, between the Start Line and the objective.

(2) To be in a position to support the infantry by fire during the final stages, after the artillery fire has been lifted.

(3) To assist in fighting through the objective in close cooperation with the infantry subunits. They do this by initially over running the objective. They over run enemy trenches and bunkers and later run through the length of the objective, churning and milling around; shooting at enemy infantry trying to escape and destroying its bunkers, pillboxes and fortifications, until all the resistance ceases. Thereafter the tanks must go to the farther limits of the objective immediately.

(4) To protect the infantry from immediate enemy counter-attack during reorganization, in conjunction with the Anti-Tank Guided Missiles, if allotted.

c. ***The Rear Tank Troops***. The rear tank troops must make full use of the ground to enable them to cover the leading tank troops by observed fire. These tank troops will not be tied to the follow-up companies but will be available to advance in direct support of these subunits when required. In close country or if the ground is not suitable, the rear troops must be positioned on the flanks. In most situations, the troops deployed on the flank(s) may be able to provide better fire support.

d. ***Information***. During the assault, information is of the greatest importance to commanders and a steady stream of battle information should flow from the front to the rear. By the use of code words, nicknames and reference points this information can be passed back quickly and accurately over the wireless and all arms, tanks, artillery and infantry, must contribute to this.

e. ***Flank Protection***. In the initial stages of the assault, flank protection is usually provided by the units in line, through which the battalion is attacking. As the attack reaches greater depth, the battalion commander should make his own arrangements for flank protection by the use of artillery, 81 millimetres Mortars, Machine-guns, Baktar Shikans and tanks. On occasions it may be necessary to establish Machine-gun and Recoilless Rifle posts on the exposed flanks of an attack.

**Stage 3 – The Destruction**

19. ***Fighting through the Objective***. When the enemy forward defended localities have been cleared, the task of fighting through the objective begins. Close support from artillery and mortars may not be possible now and companies must advance using their own weapons. Forward companies, however, should be able to call for fire against unexpected targets through their Forward Observation Officers (FOOs) and Mobile Fire Controllers (MFCs). At this stage of the battle, the follow-up companies can be used to exploit success or to reinforce the assault; failure must not be reinforced. Proper fire support must be arranged for any task given to the follow-up troops and time allowed for the subunit commanders to give orders. The tanks can be of great assistance for fighting through the objective.

20. ***Mopping Up***. Unless special troops are detailed for mopping up, this is the responsibility of each assaulting company on its objective. To achieve more rapid penetration, the battalion commander may lay down that mopping up is to be the

task of "follow-up" companies or the brigade commander may assign this task to the reserve battalion. It is essential that the troops carryout their task systematically.

21. ***Success Signals***. When the objective has been secured company and battalion commanders must report to their higher headquarters by success signal and on wireless by the given code word.

22. ***Measures to be Taken***. The following measures should be taken with all possible speed:-

a. Supporting weapons, especially Recoilless Rifles, Baktar Shikans, Mortars and Machine-guns, must move up to pre-designated places very quickly as the enemy counter-attack is likely to be launched when the troops have just arrived on the objective or have as yet not gone into proper positions. The accompanying tanks should go immediately to the pre-planned positions to beat back any counter-attack.

b. Company commanders must coordinate the defensive layout of their platoons.

c. All commanders must ensure that digging proceeds immediately.

d. Patrols and observation posts must at once be sent out for early warning and local protection.

e. Ammunition must be checked and replenished.

f. Casualties must be immediately checked and evacuated as necessary.

g. As soon as the position appears secure, 'F' echelon should be moved up.

h. Tanks must be relieved as soon as anti-tank weapons are in position on the objective and a temporary defensive position has been effectively organised.

i. Line communication to be established.

j. Defensive fire adjusted and new defensive fire selected and passed to the observers for registration.

k. Mines must be laid and wire obstacles constructed. Even a hurriedly laid minefield of a few anti-personnel mines on the likely enemy approaches may prove very useful.

23. ***Reorganization***. As each objective is captured, it must immediately be organised for defence against a possible counter-attack and as a firm base for further operations. Initially, the framework of defence must be organised around infantry and the tanks that have carried out the assault. Reorganization stores required to be

carried, would vary from situation to situation depending upon commander's plan, ground and enemy situation, however they will include following:-

a. Machine-gun mounts.

b. Remaining first line ammunition.

c. Night Vision Devices of weapons.

d. Anti-tank/anti-personnel mines.

e. Spare batteries/telephone sets/exchanges/wire.

f. Ground surveillance radar.

g. Battery chargers for Baktar Shikan weapons system.

h. Explosive.

i. Aiming discs etcetera.

j. Digging tools.

k. Command post accessories.

l. Camouflage nets.

m. Night sign posts.

n. Foxhole digging aid GBP-131.

o. Assault pioneer stores.

p. Line bedding.

24. ***Battalion Commander***. It is essential that the battalion commander (with his 'R' group) arrives on the objective as soon as possible after the assaulting companies report themselves firmly established. He must carry-out a rapid reconnaissance and make certain decisions (also adjusting his former plans, when necessary) about:-

a. Anti-tank and machine-gun fire plans. Also to release any tanks those were allotted for the assault as soon as possible.

b. Artillery and mortar defensive fire and defensive fire (SOS) tasks.

c. Layout of localities (to be coordinated as soon as possible so that digging can quickly begin).

d. Location of the battalion headquarters so that a command post can be prepared and line is laid.

e. Visit to all the platoons.

f. Plan for dealing with the enemy counter-attack on the position.

25. ***Action of the Tanks***. The immediate tasks of the tanks during the period of reorganization will be to neutralise those enemy position, beyond the objective, that be interfering with the infantry and to help break-up any enemy counter-attack by affording anti-tank protection until the Baktar Shikans and Recoilless Rifles take over. To effect this, the tanks may have to regroup and reposition themselves, but should not be brought forward of the objective unless a previous plan has been made for a subsidiary operation of this nature. The factors which affect handling of the tanks at this stage (many of them conflicting), are as follows:

a. The choice of the best position for neutralising the enemy by fire.

b. Concealment of tanks from view and fire.

c. Covering likely enemy tank approaches.

d. To be part of any counter-attack force which the battalion commander may have designated to defeat the enemy's counter-attack after the objective has been captured by own battalion group.

e. The flank protection of infantry during mopping up.

f. The morale effect of tanks being close to the infantry as against the risk of attracting enemy fire.

26. If after these factors have been considered, it is felt appropriate to release the tanks, it must be done with speed. Tanks should preferably not be employed in a static role. The physical effect on the crew of fatigue suffered during assault may considerably reduce their subsequent efficiency. Except in an open country, however, it will normally be advantageous to retain a few tanks in well concealed positions on the objective, rather than releasing them all to the leaguer. The infantry commander must, therefore, give early consideration, even during the initial planning stage of the attack as to when and in what numbers he can release the supporting tanks after the attack.

**Stage 4 – Exploitation**

27. Full benefit from hard fighting is reaped and casualties saved, by exploitation. Every effort must be made to advance beyond the given objective, within the specified limit of exploration given by higher commander, thereby upsetting enemy's counter-attack plans and gaining more time for reorganization. In the event of great success, the battalion commander must use his initiative with regard to seizing tactical features ahead of his original task which, if left, might, assist the enemy to reoccupy the lost positions. But if secured by us, might further, assist subsequent operations. The principle of exploitation must, however, be firmly established in men mind. Exploitation will form part of the plan of attack and should be worked out in detail so that all are clear about it and minimum time is taken for the commencement of this phase.

## SECTION 43 – BASE OF FIRE

**General**

1. Base of fire encompasses the elements that provide intimate supporting fire to the assaulting troops, especially once the artillery fire is lifted. The success of an attack largely depends on the correct establishment and coordination between the base of fire and the assaulting troops.

2. ***Tasks***

   a. To cover by fire, the movement of assaulting troops on to the objective by providing intimate fire support.

   b. The base of fire must basically be given specific targets for neutralisation/destruction. However, it should also cater for impromptu targets and can provide suppressive fire.

   c. It should pin the enemy down and disallow any reinforcement/ readjustment of his defensive positions, thereby ensuring greater freedom of action to the assaulting force.

   d. In certain cases, be able to support assaulting troops in reorganization during first phase of the attack by engaging enemy weapons in depth.

   e. Be able to help in extricating own troops, if the attack somehow fails.

3. ***Composition***. It consists of minimum essential troops but maximum firepower. At battalion level, its composition shall vary with the requirement and availability of various types of weapons, type of enemy defences and terrain. Generally, it will consist of the following:-

   a. Anti-Tank Guided Missiles.

   b. Recoilless Rifles/SPG-9.

   c. Mortars (81 millimetres).

   d. Grenade Launcher Heavy-Machine gun.

   e. Machine-guns (12.7 and 7.62 millimetres).

   f. Tanks (if available).

   g. Artillery guns (if available), in direct firing role.

4. ***Characteristics***. Base of fire should preferably have following characteristics:-

   a. ***Dominating Position***. The base of fire elements are best positioned on a ground which is higher and to the flanks of the line of assault so that its fire is not masked by the assaulting troops till the last moment. It should give good observation of the objective area.

b. ***Cover and Concealment***. In the existing environment of Night Vision Devices, the covered and concealed approaches and routes leading to the deployment area and weapon positions should also be adopted.

c. ***Defensible Position***. A base of fire should be located on a ground that could be converted/developed into a suitable defensive position to guard against enemy counter action.

d. ***Command and Control***. The site must facilitate effective command and control over the elements of base of fire.

e. ***Security***. If the location of fire base is not within the positions of holding troops then necessary security measures must be ensured. Troops for the security should cover all possible approaches.

f. ***Communication***. The position must facilitate visual/wireless communication with the assaulting troops.

g. ***Flexibility***. The base of fire should be able to respond to fluid situations by carrying out minor changes within its position with a view to support maximum phases of attack and if possible, the reorganization.

**Type of Base of Fire**

5. Types, categorised in accordance with the employment of troops in a base of fire are:-

a. ***Integral Base of Fire***. It consists of integral troops/weapons of an infantry battalion. The Light-Machine guns may be employed to provide local protection if required. Preferably, an officer (Anti-Tank Guided Missile platoon commander) should be the commander of base of fire.

b. ***Integrated Base of Fire***. In this type tanks, elements of heavy anti-tank battalion, light anti-tank battalion, artillery guns firing in assault fire role, integral weapons of the infantry battalion and holding troops can be employed. Officer of the dominating arm/element should preferably be the overall commander of the base of fire.

**Establishment of Base of Fire**

6. A base of fire is normally established in three stages as under:-

a. Preparatory stage.

b. Movement stage.

c. Execution stage.

**Preparatory Stage**

7. This includes the preparation and will encompass the following:-

   a. Collection of information for planning.

   b. Orders and briefing.

   c. Coordination with attacking troops.

   d. Coordination with the holding troops/formation.

8. ***Collection of Information for Planning***. Information essentially required to help in the planning and conduct include:-

   a. Plan of attack including the artillery fire support plan.

   b. Length and width of the objective and type of defence.

   c. Location of Forming Up Place and direction of attack.

   d. Rate of advance.

   e. Location of friendly forces at forward defended localities/holding troops and their tasks during attack.

   f. Arrangements for protection of base of fire if it is located ahead of own forward defended localities/holding troops.

9. ***Orders and Briefing***

   a. ***Issue of Orders***. After receiving the orders from the battalion headquarters the base of fire commander will issue his orders. The orders should preferably be given from a vantage point from where both the objective and the proposed location of the base of fire are visible. The following points need attention:-

      (1) Supporting arms commanders are present in the 'O' group.

      (2) All commanders/representatives comprising 'O' group have preferably reconnoitred the area.

      (3) Orders should cover the following aspects:-

         (a) Layout with extent.

         (b) Move to the location with order of march.

         (c) Dispositions of various weapons.

         (d) Allocation of targets.

         (e) Actions during each phase of attack and after the attack.

(f) Command, control and coordination measures, with the battalion headquarters, fire base commander and within the fire base elements/detachments.

(g) Detailed timings of fire support (should be issued in writing).

(h) The scale of ammunition to be carried.

(i) Timings (H hour, A hour and move etcetera) of attack.

(j) Instructions regarding use of wireless sets and line communication.

(k) Dumping of ammunition (if any).

(l) Disposal of casualties.

(m) Code words/nicknames.

10. ***Coordination with Assaulting Troops***. Detailed coordination between the assaulting troops and the elements in the base of fire is of utmost importance. Lack of coordination between them may cause serious casualties to own troops. Following are some of the important aspects which must be coordinated:-

a. Exact and alternative location and composition of the base of fire.

b. Routes and timings of movement of base of fire troops from present position to the site of base of fire.

c. Time by which base of fire troops will be in position.

d. Fire program (this can form part of the fire support plan).

e. Details of specific targets to be destroyed/neutralised or extents of the area required to be suppressed.

f. Types/characteristics of weapons and ammunition being used i.e. smoke, incendiary, high explosive etcetera.

g. Communication arrangements.

h. Coordination of signals/field signals/codes/nicknames.

i. Arrangement for indication of line of assaulting troops during night, particularly in mountainous/restricted terrain. For example, white cloth or dim lights on the back of the troops on the flanks.

j. Timings to open/ceasefire.

11. ***Coordination with Holding Troops***

a. The location of base of fire.

b. Route to the position.

c. The fire support available from the holding troops/units.

d. Command and control measures.

e. Protection arrangements for base of fire.

f. Any other assistance required.

**Movement Stage**

12. It is the most important stage which must be executed stealthily and with complete secrecy. It involves following actions:-

a. Move from concentration area to assembly area.

b. Actions in assembly area.

c. Move from assembly area to RV.

d. Actions at the RV.

e. Occupation of base of fire.

13. ***Actions in Assembly Area***. On arrival, the company/platoon commander must:-

a. Post sentries.

b. Site and rehearse alarm posts and supervise digging if this has been ordered.

c. Carry-out a final inspection of the company/platoon to make sure that every man is fully prepared for battle.

d. Supervise or order cleaning of weapons, supervise changing of magazines, distribute ammunition.

e. Test wireless sets.

f. Rehearse formations and order of march for the move to the Forming Up Place.

g. Coordinate with supporting arms.

h. Order equipment to be taken off if there is time for the men to rest.

i. Supervise feeding, if time allows.

14. ***Move from Assembly Area to Rendezvous (RV)***

a. At the given time, Light-Machine gun groups (protection group) will move out from assembly area. They will go to RV and secure it against any enemy interference.

b. Other elements of base of fire will start at the assigned time on receiving signal from the protection group. It is advisable to move the weapons mounted on vehicles, such as Anti-Tank Guided Missiles, Recoilless Rifles and tanks at the last moment and position them away from the enemy in order to maintain secrecy and surprise.

c. Guidance arrangements from assembly area to the location of base of fire should be ensured. Move forward should be without lights if the base of fire is to be established at night.

15. ***Action at Rendezvous (RV)***. RV should be as close to the deployment positions as tactically feasible. Following actions are taken at the RV:-

a. Vehicles may or may not halt at the RV. Guides sent earlier or the detachment commanders should guide the vehicles to their respective locations. In case the vehicles can not go to their respective locations then weapons will be dismounted and carried by the crew. The vehicles in this case should be in camouflaged in the vicinity of the RV.

b. Maximum use of the Night Vision Devices should be made for move forward from the RV to ensure complete secrecy.

c. Last minute coordination, if any, will be done.

**Execution Stage**

16. Once all the elements are ready, a report to this effect will be given to the commander:-

a. Weapons are sited and laid on the specific targets required to be destroyed/neutralised. The weapons should preferably be able to shoot through the loopholes when fire base is established within location of holding troops using. Bearings of alternative targets should also be taken/confirmed.

b. The weapons will fire on their assigned objectives/targets at specified time/on orders.

c. If enemy weapons can not be located accurately, then fire should be placed on the likely enemy dispositions.

d. Commander of the base of fire must monitor the progress of the attack and accordingly give orders for adjustment of fire.

e. Fire should be shifted to other targets or ceased in case assaulting troops come within the danger zone of own weapons.

f. On completion of fire support, the base of fire elements should revert to the battalion headquarters unless specified otherwise.

**Siting of Weapons**

17. The correct siting of weapons plays an important role for bringing effective fire onto the objective, however, it will depend upon the ingenuity of the commander. The weapons should not hinder fire of each other and the advantage of their longer range should be exploited to the maximum. If the deployment has to be carried out at night, due care must be taken to precisely mark their positions on ground during daylight/dusk reconnaissance. Besides this, the bearings to the targets and, if possible, even the desired elevations for own weapons must also be recorded to facilitate laying of the weapons at night. Following two deployment patterns along with merits/demerits are given as a guideline:-

a. ***Layered Pattern***. (Figure 15). In this option the weapons are deployed in a layered pattern keeping in view their ranges from the objective:-

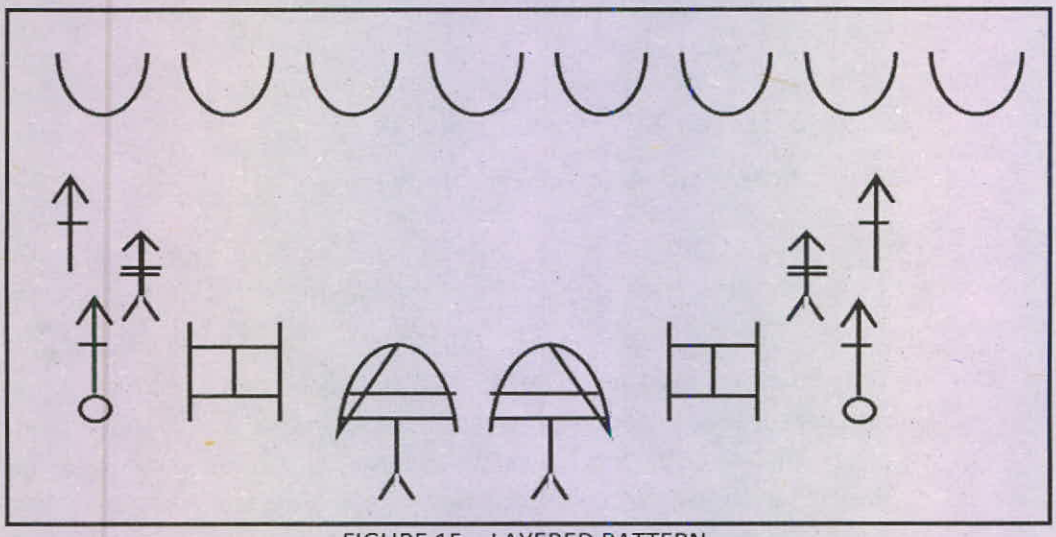

FIGURE 15 – LAYERED PATTERN.

(1) ***Merits***

(a) Less deployment area in width is required.

(b) More suited for mountainous terrain.

(c) Concentration of fire can be achieved.

(d) Weapons can be kept at stand-off distances from the enemy weapons.

(e) More chances to maintain secrecy while occupying the sites.

(2) ***Demerits***

(a) Overhead fire may create safety problems.

(b) Adjustment of position will be difficult.

(c) Wider objectives would be difficult to engage.

b. ***Linear Pattern***. In this option all the weapons are deployed in linear pattern irrespective of their ranges (see Figure 16):-

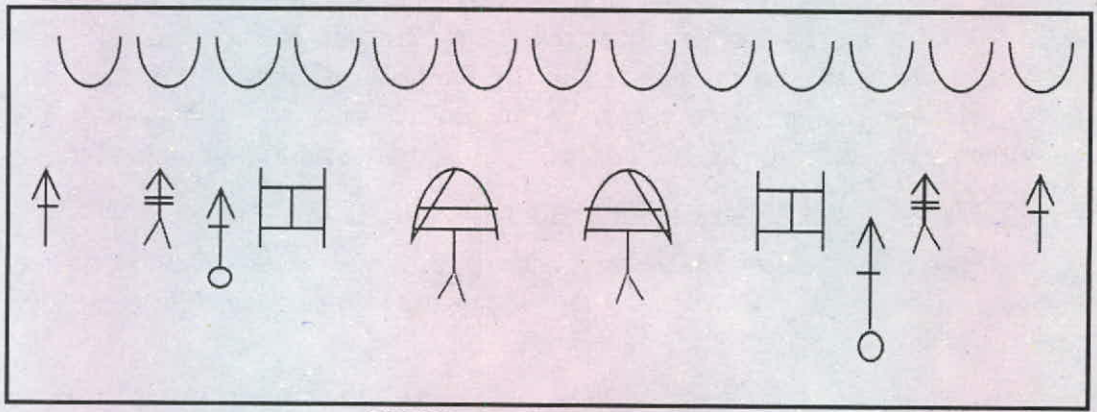

FIGURE 16 – LINEAR PATTERN.

(1) ***Merits***

(a) More suited for plains.

(b) Wider objectives can be engaged.

(2) ***Demerits***

(a) More deployment area in width is required. In most cases, so much space in operational area will not be available.

(b) Effective concentration of fire would be difficult to achieve.

(c) Difficult to maintain secrecy and security because of proximity to enemy Forward Defended Localities.

18. Very often, it is not possible to adequately engage the desired width of the target area due to limitations of terrain and the consequent angle of deployment of the base of fire weapons. Generally the angle is so narrow that the assaulting troops immediately begin to mask the fire of the weapons. Under the circumstances and regardless of the pattern adopted, it becomes necessary to deploy the weapons of base of fire on both the flanks. Sometimes a small group may also have to be deployed in the middle of the assault wave. This calls for much greater requirement of command and control within the base of fire groups and an intimate coordination

of these with the attacking troops. As a general rule, it is always better to first engage the targets in the middle of the objective area and then gradually shift the fire to the flanks as the attack progresses.

## SECTION 44 – NIGHT ATTACK

**General**

1. The advent of Night Vision Devices has enhanced the night fighting capability manifold. Since it is easier to use these devices when stationary, these are more advantageous to a defender than an attacker. The induction of Night Vision Devices has not yet necessitated a complete change of the existing doctrine of night attack. However, there is a need to recognize the impact of these devices on night operations and suitably modify our drills and procedures to neutralise their effect.

**Impact of Night Vision Devices on Night Attack**

2. Night Vision Devices enable the enemy to observe, recognise and effectively engage targets, at night at relatively longer ranges. Some of the effects of Night Vision Devices on night attack will be as under:-

a. Enemy can determine the location, size and nature of our activities at longer ranges, enabling him to take effective countermeasures.

b. Enemy will be able to carry-out coordination as well as exercise better command and control at night.

c. Enemy will be able to bring all types of observed defensive fire on the attacker, upto about 1000 metres.

d. Artillery observers, equipped with Night Observation Devices will be able to effectively engage targets upto a range of 1200 metres. Movement from the assembly area to the Forming Up Place and the process of forming up will be vulnerable to the observed enemy fire.

e. Availability of Night Vision Devices on tanks/Anti-Tank Guided Missiles and Recoilless Rifles will facilitate the defender to neutralise/ destroy targets at comparatively longer ranges.

f. Possibility of achieving surprise has been reduced considerably as the enemy has acquired ground surveillance capability of accurate detection of movement and concentration of troops/formations upto a depth of 40 to 60 kilometres.

g. Night vision capability will enable armour to operate in close support role, facilitating more rapid and potent counter-attacks.

h. Accurate fire support will be available from the base of fire due to the Night Vision Devices fitted on the weapons.

i. Night Vision Devices will assist commanders at all levels to observe and control movement of troops during all stages of attack.

3. Following guiding principles govern the planning and conduct of night operations:-

a. Best possible reconnaissance and use of all available sources of information.

b. A simple plan.

c. Detailed preparation.

4. ***Advantages of Night Vision Devices in Attack***

a. Maintenance of command and control and direction keeping is easy.

b. Greater frontages and depth can be maintained.

c. Weight/pressure of enemy counter-attacks can be sustained for a longer time.

d. Clearance of minefield and launching of bridges will be speedy.

e. Speed of operations will be faster. An advancing force can also follow-up its daylight success at night.

5. Following assume added importance for night attack:-

a. Time for detailed planning and preparation.

b. Detailed information of terrain, enemy and expected weather conditions.

c. Secrecy and surprise.

d. Detailed orders and a simple manoeuvre plan.

e. Easily identifiable objectives.

f. Well-trained units and subunits.

g. Use of camouflage, concealment and cover to neutralise the effects of enemy Night Vision Devices.

h. To be out of the detection range of Night Vision Devices, Forming Up Places should not be less than 1000-1200 metres from the objective.

i. Adverse weather conditions like rain, dust, mist and haze/fog reduce the effectiveness of Night Vision Devices.

j. Phases of the moon would also be an important consideration in determining the time of attack.

k. Time required for logistic support would be an important consideration.

**Types of Night Attack**

6. Night attack falls in two main categories:-

a. ***Noisy Attack***. Noisy attack uses all available fire support to cover the assault as in daylight attacks.

b. ***Silent Attack***. A silent attack is launched to obtain advantage of surprise and is generally resorted to when relative strength and terrain conditions do not favour a noisy attack. Fire support plan for a silent attack is made as for the noisy attack but the fire is opened only when surprise is lost.

7. In both types of attack, the aim is to maintain secrecy as long as possible and reduce the reaction time available to enemy. The degree to which secrecy can be maintained is usually the factor that determines whether the attack would be noisy or silent. For a noisy attack, deception measures must be planned to conceal the strength, direction and timings of attack. The surveillance devices like Night Vision Devices, radar etcetera have reduced the chances of achieving surprise.

8. Factors affecting the choice of objectives at night are:-

a. Impaired visual contact and difficulties in keeping control as it is difficult to use the Night Vision Devices on the move.

b. Difficulties in the effective application of unplanned fire support at night.

c. Coordination of fire and movement between sections and platoons during the attack because everyone will not be using Night Vision Devices.

d. Effectiveness of fire support from the fire base, although Night Vision Devices have reduced this difficulty to quite an extent.

**Stages of Night Attack**

9. Following are the stages of a night attack:-

a. Preparatory stage including assembly and forming up.

b. Assault.

c. Destruction

d. Reorganization.

10. ***Preparatory Stage***. During preparatory stage, following actions will be undertaken:-

a. Collection of maximum information about terrain, enemy positions, weapons, fixed lines, defensive fire tasks, natural and artificial obstacles, location of enemy reserves and location and capabilities of his Night Vision Devices.

b. Detailed reconnaissance of terrain and enemy positions by all commanders. In order to achieve surprise and deceive the enemy about the time and place of attack, a portion of the daylight reconnaissance can be relegated to dark hours.

c. Briefings and orders.

d. Rehearsals of plans, if possible.

e. Checking up of equipment, arms, ammunition, etcetera.

f. Marrying up with the supporting arms.

g. Arrangements for move to the Forming Up Places.

h. Preliminary operations, if necessary.

11. Following measures will be taken in the assembly area:-

a. Digging will depend on the period of stay.

b. Anti-aircraft defence, posting of air sentries, siting of automatic weapons and controlled dispersion of men, vehicles and equipment, if it is being occupied during daytime.

c. Security detachments around assembly area to ensure defence against enemy raid etcetera.

d. Assembly area and approach to it must be covered from direct enemy observation.

e. Parking and camouflage of the unit transport.

12. ***Forming Up Places***

a. ***Choice of Forming Up Places and Start Line for Night Attack***. For attacks mounted within the Forward Defended Localities (FDLs), normal tactical considerations like ground, radar coverage, enemy artillery, mortars and patrolling habits etcetera affect the choice of Forming Up Places and Start Line. For attacks mounted outside the FDLs, these factors together with the enemy Night Vision Devices capability, affect the distance between Forming Up Places and objective. Following points must be kept in mind:-

(1) Forming Up Places/Start Line be selected out of effective range of enemy direct firing weapons. Enemy direct firing weapons will be effective at longer ranges due to Night Vision Devices, therefore, the distance of Forming Up Places will be virtually dictated by the same considerations as are operative during day attack. Suggested distance of Forming Up Places in plains is 1000 to 1200 metres. In case of broken ground and BUAs, this distance may be reduced.

(2) Covered place should be selected for forming up if available after ensuring that it is not occupied by enemy.

(3) Route to the Forming Up Places should be prominent but concealed from the enemy observation.

(4) Marking signs should not be displayed so prominently that they give away the location.

(5) An alternative Forming Up Places must always be selected and be known to all ranks. Move to alternative Forming Up Places should be rehearsed.

(6) Domination of No Man's Land around Forming Up Places must be ensured.

(7) Depending upon the tactical situation, follow-up companies may be kept at a distance of 500-750 metres behind the forward companies when the latter are in the Forming Up Places with minimum marking. However, when the assaulting companies leave the Forming Up Places and move towards the objective, the follow-up companies move forward to the Forming Up Places and stay there as ordered or till the time leading companies have completed their phase.

b. ***Protection of Forming Up Places***. The ability of the enemy to observe at night will not allow elaborate arrangements for the protection of Forming Up Places. Therefore, the process of forming up should be completed with minimum possible time to reduce detection and reaction time by the enemy. Depending upon the degree of protection required, the protection party for the Forming Up Places can be from a platoon plus to a company.

c. ***Marking of Assembly Area, Forming Up Places and Start Line within the Forward Defended Localities***. This presents no particular difficulty. Drills and procedures should be included in unit standing orders and these should be well understood and well-practised by all ranks. The example given in Figure 17 illustrates layout of an Forming Up Places marked for four companies and tactical headquarters. Minimum use of passive night signs and other marking material is made for marking of Forming Up Places.

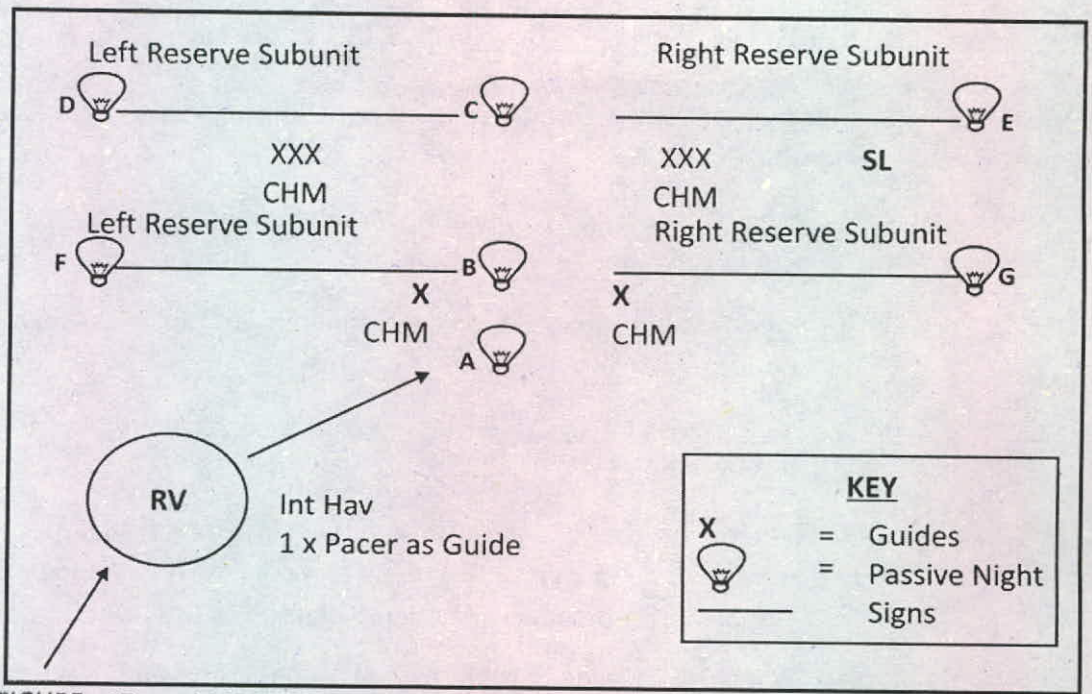


FIGURE 17 – MARKING OF FORMING UP PLACES AND START LINE WITHIN THE FORWARD DEFENDED LOCALITIES.

d. ***Explanation of Figure 17 – Marking of Forming Up Places and Start Line within the Forward Defended Localities***

(1) ***Rendezvous (RV)***. Intelligence Havildar and a pacer who must know bearing and distance from RV to point A.

(2) ***Point A***

(a) Intelligence Officer who must know bearing of centre line and distance in paces between points AB and BC.

(b) Centre line night sign.

(3) ***Point B***

(a) Company Havildar Major and a guide each for left and right reserve subunits, who must know bearings and distances between points BF and BG respectively.

(b) Centre line night sign, with arrows indicating right and left turn for respective subunit.

(4) ***Point C***

(a) Company Havildar Major and three guides each for left and right assault subunits, who must know bearings and distances between points CD and CE respectively.

(b) Centre line night sign, with arrows indicating right and left turn for respective subunit.

(5) ***Point D***. Night sign indicating extreme left limit of left assault subunit.

(6) ***Point E***. Night sign indicating extreme right limit of right assault subunit.

(7) ***Point F***. Night sign indicating extreme left limit of left reserve subunit.

(8) ***Point G***. Night sign indicating extreme right limit of right reserve subunit.

(9) Two guides of each assault subunit will lead forward platoons from point C to points D and E respectively. Third guide of each assault subunit will lead depth platoon to their location.

(10) Company Havildar Majors will lead company tactical headquarters to their locations.

(11) ***Battalion Tactical Headquarters***. Placed as required on the centre line by Intelligence Officer.

(12) Intelligence Havildar besides guiding the battalion from RV to point A, will also be responsible to guide the battalion from point A to B and C.

e. ***Marking of Forming Up Places and Start Line outside the Forward Defended Localities***. Directional help within Forming Up Places is provided by the centre line and Start Line itself. Security considerations rule out elaborate use of even passive night signs and tape. It will be necessary to plan an easily recognisable RV, preferably a natural feature, near the selected Forming Up Places and Start Line. In the absence of a natural or artificial feature, minimum help of marking material should be taken. From this rendezvous, subunits will be guided by the Forming Up Places marking party upto the centre line of forming up places and onward to their respective places. An example of this kind of Forming Up Places marking is shown in Figure 18. Move in and occupation of Forming Up Places will require a very high standard of training.

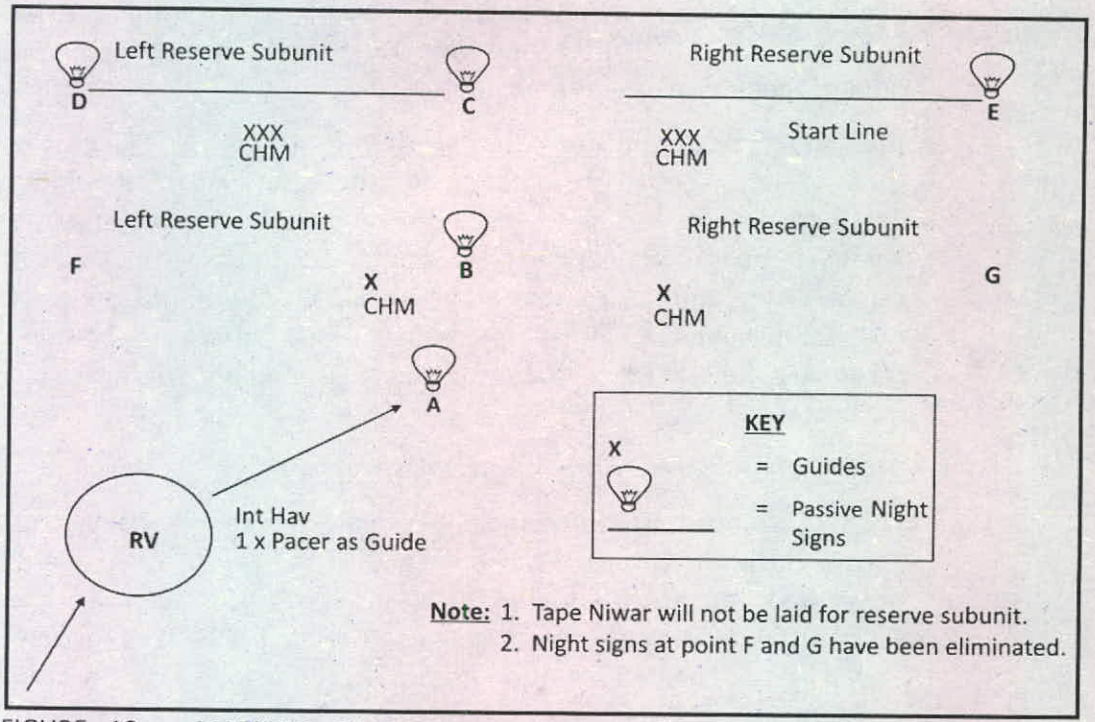


FIGURE 18 – MARKING OF FORMING UP PLACE AND START LINE OUTSIDE THE FORWARD DEFENDED LOCALITIES.

f. ***Attacks Mounted within Own Forward Defended Localities***. Assembly area, Forming Up Places and Start Line marking is done by the Forming Up Places marking party.

g. ***Attacks Mounted outside the Forward Defended Localities***. If Forming Up Places marking party is to be used, it must be constituted before the approach march and moved with the force detailed to protect Forming Up Places and Start Line. During the approach march, marking party must be organised as a fighting element ready to take part in any action on its way to Forming Up Places. All rifle companies should be capable of laying out a battalion Forming Up Places and Start Line. During the approach march, this force would move one navigational bound ahead of the main body and should be ready in Forming Up Places to receive the main body, when it arrives. Minimum strength be used for protection and marking of the Forming Up Places.

h. ***Marking of Routes***. Marking of unit routes within own Forward Defended Localities may be somewhat elaborate. Outside own Forward Defended Localities, marking should only be resorted to where absence of natural or artificial landmarks makes navigation difficult, or where the going is particularly bad. Because of its

significance, such marking should be backed up with static guide parties to ensure correct interpretation and to prevent their displacement by accident or enemy action.

i. ***Guiding Arrangement from Assembly Area to Forming Up Places***. The route(s) should be clearly marked and if necessary guides should be provided or passive night signs used. In case of armour infantry combined operation, marking of the routes and Forming Up Places is the task of infantry, accompanied by the liaison officer and guides of the tank subunits. It can only be done after a thorough reconnaissance. Marking should be done just before the arrival of troops.

j. ***Move from Assembly Area to Forming Up Places***

(1) For move from assembly area to Forming Up Places, the assaulting companies will move in single file on either side of the track followed by the battalion tactical headquarters and the remaining two companies. The unit and the subunits should be led by the respective commanders or in their absence by the second-in-commands.

(2) Company guides may either lead the companies from the assembly area to the Forming Up Places or meet them at the base of the Forming Up Places centre line. From here they can guide them to their respective places.

(3) In case tanks are taking part in assault, their move from assembly area to the Forming Up Places should to be covered by the artillery fire and a deception plan. Maximum precautions must be taken to avoid tanks overrunning own infantry.

(4) The enemy with his long-range Night Vision Devices and battlefield surveillance radars would generally be able to detect our move from assembly area onwards. Troops should therefore move well dispersed using a covered approach. The move should be as late as possible so that minimum reaction time is available to the enemy.

(5) Fire base elements would be able to move into their positions during darkness; hence their movement could be undertaken as late as possible. Weapons in the fire base would be able to bring accurate and effective fire on enemy positions and provide intimate fire support to own troops with better safety with the help of Night Vision Devices.

(6) It is not essential for follow-up companies to be deployed in assaulting formation in Forming Up Places as they will be required to move to another Start Line for attack in the next phase. They should remain readily available for subsequent move. Depending upon the task, terrain, moonlight and enemy situation, follow-up companies have following alternatives:-

(a) Move behind forward companies in deployed formation.

(b) Move behind forward companies in concentrated from i.e. file, single file or any other suitable formation.

(c) Stay in Forming Up Places and move on orders.

k. ***Navigation/Forming Up Places Marking Party***. Navigation party will mark the route from the assembly area to the Forming Up Place and also be responsible for guiding the battalion onward to objective. The party is also responsible for protection of the main body in the Forming Up Places. Thereafter, the guides will join their respective subunits. The suggested composition of this party is as under:-

| | | | |
|---|---|---|---|
| (1) | Intelligence Officer with his signaller | – | 2 |
| (2) | Intelligence Havildar | – | 1 |
| (3) | Soldier for pacing | – | 1 |
| (4) | Marker | – | 1 |
| (5) | Three soldiers for laying cable | – | 3 |
| (6) | One Havildar Major per company | – | 4 |
| (7) | Company guides | – | 8@ |

13. ***Assault Stage***. Most of the enemy weapons will be on fixed lines along expected approaches. Observation and detection are difficult at night, yet the location of enemy weapons can be found from their flash and also with the help of Night Vision Devices. The use of Night Vision Devices will allow the commander to exercise better command and control and also assist him in identifying the objective. It will also facilitate direction keeping. Following variations and techniques for the assault/dogfight at night should normally be used:-

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@3 guides per forward company and one per depth company. The number could increase if the area is difficult or the distance between assembly area and Forming Up Place is larger. In addition, guides from supporting armour and a suitable protection party may also be necessary.

a. The assaulting troops should be well dispersed to avoid enemy's observation and fire.

b. Greater use of stealth and cover of darkness to get closer to the enemy weapons holding up the advance.

c. Greater reliance on bayonets and grenades.

d. Due to lack of visibility, dogfight will mostly be by rifle sections and even by rifle groups, section commanders and section second-in-commands should therefore be well-trained to control this type of fighting.

e. Although it may be possible to get reasonable close support from tanks, yet the usual close support will be provided by integral infantry weapons and very often only by the platoon's own weapons. In tank-infantry combined attacks, tanks in the assault echelon usually lead, with the infantry following behind closely. Since tanks are blind, particularly for the dead ground near them, there is every likelihood of own infantry suffering casualties if they have to move in front of the tanks. In all cases, detailed coordination and recognition methods are required especially when the infantry is leading. Stationary tanks may preferably provide fire support from the flank when the infantry is going into attack. In this case the tanks stop firing on receipt of a message by wireless and join up with the infantry on the objective. This will also necessitate a higher degree of coordination so that own infantry is not overrun by friendly tanks.

f. Tanks, guided missiles, recoilless rifles and rocket launchers etcetera should be employed for destroying enemy bunkers and pillboxes at night. Bangalore Torpedoes are used quite often to make small lanes in enemy minefields and for blowing up wire obstacles. Troops, therefore, should be well-trained in use of Night Vision Devices and handling and firing of these weapons in darkness.

g. There should be effective arrangements for coordinating supporting fire with the brief manoeuvre plan of the destruction parties. The platoon commander concerned will carefully estimate the time that his assault parties are likely to take in reaching the assault line. He will then ensure that all supporting fire from his platoon is stopped a few minutes before the assault. If the supporting fire stops earlier, the assaulting troops will work their way to the assault positions by stealth under cover of darkness. If the fire does not stop in time, the assaulting troops may lie low or use a prearranged signal like shouting a specific battle-cry (Naarah) followed by firing of very light signals after a short pause so that the supporting fire stops.

h. Whenever possible small parties or individuals should try to approach the enemy holding up the assaulting troops without any supporting fire. They should take the help of darkness and then deal with the enemy with grenades and bayonets, etcetera.

i. Sometimes the assault on company objectives may involve negotiating nullahs, ravines and hillocks. Adoption of extended line or any other assault formations in such situations may not be possible. In such circumstances, infantry platoons may have to move by sections in single file. After crossing such terrain, sections, will revert to the normal assault formation.

14. ***Reorganization Stage***

a. On capture of objectives at night, attacking unit commanders will send out fighting patrols in front and to the flanks. Patrols and listening posts with Night Vision Devices (if available) will be sent out to occupy specific features of tactical importance which may either be unoccupied by the enemy or which he may have vacated. Tank hunting parties should also be sent out with Night Vision Devices on the likely tank approaches to guard against any counter-attack. This is local exploitation by the unit commander as well as a measure of protection for his reorganized position. All this necessitates prior planning and assigning of be prepared tasks in the orders.

b. Recoilless rifles and 'Fighting' echelon stores must be brought forward without any loss of time. In this regard timely provision of vehicle safe lanes would facilitate move forward of the fighting echelons. A commander must be detailed to ensure proper move of these weapons/vehicles. Communications and proper reception must be tied up during the preparatory stage.

c. Defensive fire must be adjusted.

d. Defence of battalion should be immediately organised and coordinated.

e. Ammunition should be replenished.

f. Casualty evacuation should start immediately.

g. Administrative arrangements tied up.

h. Maximum use of Night Vision Devices must be made by the commanders while siting weapons and during readjustment of positions.

**Maintenance of Command and Control**

15. Use of Night Vision Devices during night attack will greatly assist commanders and major weapon numbers to overcome problems faced at night. Following measures should also be taken to ensure proper command and control during night operations:-

a. While planning all possible eventualities should be considered beforehand and arrangements made to meet them.

b. Every individual must be told what part he is to play in the plan and what he is to do if something goes wrong. Whenever possible, rehearsals should be carried out on ground similar to the one on which operations are to be undertaken.

c. Avoid use of difficult terrain if it is likely to result in reduced effectiveness of command and control. However, it should be remembered that all convenient approaches may very well be covered by the enemy.

d. Objectives should be limited to ensure better control. This may however, mean employing more troops to capture a given area.

e. In deliberate night attacks, alternative Forming Up Places should also be selected and secured. It should be possible to move the alternative Forming Up Place from assembly area if the original Forming Up Place has been compromised.

f. Formations at night should be such that no man is out of visibility on flank, front or rear.

g. Detailed arrangements including 'foolproof' signals should be made to ensure coordination in actions of various units/subunits participating in the night operations.

h. Rate of advance should be worked out accurately to ensure proper control. It should neither be too optimistic nor too timid. During advance each man should adjust his move with the other. Individuals should take dressing from the centre where the commanders really will be.

i. As far as possible, reserves should be kept at all levels at night.

j. Distinguishing signs e.g. white armbands, white bands around helmets, white patches on the backs etcetera should be worn by all commanders down to the section second-in-commands and the guides.

k. White marking may also be used on the rear of tanks for easy recognition from a distance.

l. Flank markers carrying shaded coloured lights on the backs, should be used with rifle companies, to watch and control company's progress. Two such lights will be carried by each platoon.

m. Arrangements should be made to have duplicate communications at all levels. This is indispensable at the subunit level.

n. Traffic control should be thorough.

o. Limits of local exploitation should be laid down by the higher commander.

p. Junior leaders should be well-trained.

**Supporting Arms**

16. Roles of different supporting arms are as follows:-

a. ***Armour***. Despite proven advantages of shock effect, employment of armour during the hours of darkness, has traditionally remained inhibited essentially on account of two factors i.e. lack of tank's physical night fighting capability and a psychological inhibition/mental reservation of employing the tanks and the mechanised forces at night. The present/planned induction of Night Vision Devices/Night Observation Devices in the Indian Army has further added to the problem by offsetting the advantages of protection against aimed/ observed enemy fire and achievement of surprise during the darkness of night. However, employment of Night Vision Devices/Night Observation Devices on own tanks has also enhanced the offensive capability of own armour at night. There are two methods of infantry-tank combined night attack i.e. employment of tanks in close support role and fire support role. In this context, there are important aspects i.e. distance of Forming Up Places from the objective and the option of employing armour in the fire support role. These are discussed as under:-

(1) ***Distance of Forming Up Places during Employment of Tanks in Close Support***. Tanks are vulnerable to enemy detection and fire by Anti-Tank Guided Missiles at a distance ranging between 1500-3500 metres. Assuming the likely deployment of enemy Anti-Tank Guided Missiles, approximately 500-1000 metres in depth from the line of forward defences and even making a reasonable allowance for the degradation of Anti-Tank Guided Missiles due to factors of terrain and quality/ sophistication of Anti-Tank Guided Missiles, an increase in the

distance of Forming Up Places to about 2500-3000 metres is necessitated. Important implication is that common Forming Up Places for tanks and infantry is no longer feasible due to:-

(a) Infantry assault from a distance of 2500-3000 metres would result in fatigue and an unacceptable delay in reaching the objective as against the assault from a distance of 1000-1200 metres.

(b) Forming Up Places of tanks/location from where tanks assault, at a distance of 1000-1200 metres from the objective (to conform to the distance of Start Line for infantry) is also not feasible for evident reasons.

(2) ***Employment of Tanks in Fire Support Role***. Employment of tanks in fire support role at a distance between 1000-1200 metres from the objective, has become exceedingly difficult on account of defenders' ability to direct observed and accurate fire from its long-range anti-tank weapons. Although provision of night firing capability on our tanks provides good opportunity to dispense with the procedure of registration of targets for night firing/engagement, yet the limited night firing range of 800 metres of tanks' main gun with Night Vision Devices and its effective range with registration method of 1600 metres, does not completely obviate the requirement of registration. Following options are available with regard to occupation of base of fire:-

(a) ***First Option – Occupation of Base of Fire by Tanks after Commencement of Attack/Assault***. Prior registration of targets is not carried out and tanks do not occupy the base of fire before the commencement of attack (as is the technique/practice in vogue). Tanks remain out of range of enemy's Night Vision Devices/ Night Observation Devices and his anti-tank weapons that is generally at a distance of approximately 2500-3000 metres from the objective. The base of fire distance is kept within the night firing range capability of own tanks (presently night firing range of T-69 (II) M tank is 800 metres). After commencement of attack/ assault, tanks move towards the base of fire, in such a manner that they reach the base of fire at a time when the assaulting infantry is approximately at the same distance from the objective as that of base of fire from

objective. Availability of Night Vision Devices on own tanks enables them to move at the last moment, avoiding the risk of detection/neutralization at an early stage. The main advantage of this option/technique is that the tanks are able to provide intimate support to infantry during the critical stages of battle.

(b) ***Second Option – Intermediately Distance from the Objective***. In open terrain environment and an extensive use of anti-tank weapons and Night Vision Devices/Night Observation Devices by the enemy, the tanks provide fire support remaining between 1500-3000 metres from the objective. In this eventuality, an additional base of fire may be planned around tanks and Anti-Tank Guided Missiles. Depending upon the progress of attack, tanks/other mobile elements of base of fire should move forward.

(3) ***Miscellaneous***

(a) A new factor i.e. the importance of use of ground during day, for example, use of hull down positions, adoption of covered route/approach for movement, speed, bound to bound movement and camouflage and concealment would have equal relevance and application at night.

(b) Noise of tanks and use of Night Vision Devices militates against surprise at night. Deception incorporating tank noises should start before the tanks start moving to Forming Up Places. If tanks have to move into forward assembly areas a night before, deception should be arranged for that as well.

(c) Target indication by infantry to armour is comparatively difficult by night than by day. Tanks and interspersed infantry should preferably advance in line. Methods in order of priority for target indication are infantryman climbs the tank and indicates the target by using tank telephone or radio. The hazards of shelling to the infantryman when he is protected behind turret are much less than when he is on ground.

(d) Drills for indicating targets at night must be perfected. Indication by firing tracers, in addition to the use of

wireless and physical contact with a member of the crew, should be adopted. It should however, be remembered that in most cases, it may not be possible to identify tracers as the enemy as well as own troops will be firing a lot of tracer ammunition. Following may also help in target indication:-

i. Prearranged targets – reconnoitred during day with azimuths and elevations noted.

ii. Prearranged reference points for target indication.

(e) Laying down of drills for stopping the tank fire is also essential. Prearranged very light signal is one of the methods for stopping fire. The best method is through personal radio contact as it is very difficult to distinguish any type of fire at night while enemy and own troops are firing.

(f) Command and control of tanks in darkness is difficult. Squadron/troop leaders should therefore, remain well forward.

(g) Armour is very vulnerable to enemy tank hunting parties at night. Tanks should therefore, operate within the areas protected by own infantry. Once employed in base of fire, armour should be provided adequate protection.

b. ***Artillery***. Artillery support during different stages of night attack will be provided as under:-

(1) ***Reconnaissance and Preparation***. Artillery observers and battery commanders, equipped with thermal imagers if available (range 3000-4000 metres), will be of great help to infantry commanders during reconnaissance for identification and location of automatics/major weapons of enemy. This is important because owing to enemy Night Vision Devices, own patrols may not be able to get closer, beyond a certain distance.

(2) ***Move to Forming Up Places***. Move of own troops from assembly area/RV to Forming Up Place is likely to be picked up by the enemy's surveillance devices. Artillery fire support, will be employed as under:-

(a) Preparatory fire will perhaps become a necessity during this phase of operation. High Explosive (HE)

ammunition mixed with smoke will help in neutralising/blinding enemy Night Vision Devices/Night Observation Devices.

(b) The duration of fire will have to be increased. Rate of fire will be decided keeping in view the effects desired and the ammunition available.

(c) Simultaneous engagement of the objective area (all targets at the objective) will reduce the effect of enemy Night Vision Devices/Night Observation Devices.

(d) Expenditure of smoke, High Explosive and illuminating ammunition, will substantially increase. It would necessitate increase in authorisation of the number of vehicles to a unit including its ammunition holding capacity etcetera.

(3) ***Move From Forming Up Place to Objective***

(a) Artificial illumination and use of Night Observation Devices/Night Vision Devices are not complimentary to each other on the battlefield as the use of former renders the latter temporarily ineffective. So the use of illumination at night has to be viewed vis-à-vis its blinding effect on Night Vision Devices/Night Observation Devices used by the friendly troops.

(b) Star shells will be fired to exploit the susceptibility of Night Vision Devices/Night Observation Devices (Image Intensifier type) to sudden bursts of light. It is recommended that illumination should be fired in the depth and its height adjusted so as to avoid direct illumination of the area where own troops are moving. Another advantage of this technique is that the objective will be suitably silhouetted and own troops/ artillery observer will be able to observe the objective. Adoption of this technique should be possible as neutralisation of Night Vision Devices/Night Observation Devices does not necessarily require a sudden burst of light in front of them.

(c) Use of quick smoke shoot (weather permitting) should be resorted to blind enemy Night Vision Devices/Night Observation Devices.

(d) All concentrations will be fired with HE and smoke mixed. Safety distances for HE and smoke shells must be kept in mind while employing this technique.

(e) Dust and haze kicked up by artillery fire and smoke adversely affect the performance of Night Vision Devices/Night Observation Devices.

(f) Engagement of opportunity targets by artillery is difficult at night. Artillery units should however practise and engage such targets with the help of previously registered targets and by starting the shoots with the opening round firing plus of the target. Artificial illumination and use of Night Observation Devices will assist forward observation officers in their tasks.

(g) Changeover of forward observation officers/artillery advisors is difficult at night due to problems of finding correct location. Therefore, arrangements should be made to provide additional observers to avoid changing of observers/artillery advisers.

(h) Rotary wing aircraft of army aviation should be used by the airborne forward observation officers to engage deeper targets and supplement existing counter-bombardment resources.

(i) Increase in the distance of Forming Up Place from the objective and the requirement of illuminating/smoke shells will enhance ammunition expenditure, which has to be duly catered for.

(4) ***Deployment***

(a) Guns should be moved forward/deployed as late as possible. Their movement must be undertaken during night in order to avoid detection and giving minimum reaction time to the enemy.

(b) The requirement of moving guns at night and edging forward with the progress of operation will involve blind occupation. Issue of Night Vision Devices to reconnaissance parties will greatly facilitate blind deployment.

(c) ***Light Anti-Tank/Heavy Anti-Tank Elements***. A battalion may be allocated some light anti-tank/heavy

anti-tank elements from the divisional/corps resources in addition to its integral Anti-Tank Guided Missiles. Those should be employed to strengthen the fire-base. However, they must join the assaulting troops as soon as possible during the reorganization, to beat back any counter-attack.

(d) Anchor observation post equipped with thermal imagers (AN-USA-11) and laser range-finder will play a significant role in controlling artillery fire.

17. ***Air Support***

a. Close air support is difficult at night. However, the possibility of hitting deeper targets and the enemy guns that have opened up, must be explored especially, in the moonlit nights and where a well-defined bomb line e.g. a canal/road etcetera, separating the two forces is available.

b. In major attacks of divisional level and above, where substantial armour and infantry are employed, artificial illumination may be arranged with flares dropped by own aircrafts, in the rear of enemy.

c. Own air may also be asked to bomb the objective area before the pre-H hour bombardment by artillery.

**Passage through Lines in Darkness**

18. Deeper objectives will often have to be attacked by either envelopment or passing the troops through the units/subunits that had captured a portion of the enemy positions in earlier phase(s). The latter course should be adopted only as a last resort because the area through which the troops have to pass would already have attracted enemy's attention and could be under heavy fire. Besides, the passage itself is a quite difficult operation at night even with Night Vision Devices. However, when such an operation is indispensable, following measures should be ensured:-

a. Passage of lines at night should be carried out immediately after the last phase in order to take advantage of the enemy confusion created by the earlier phase.

b. The troops on ground and those passing through should be thoroughly briefed about the operation and the special measures necessary for its success.

c. A tape should be laid in the earlier phase by the parties from the following subunits to mark the axis on which these subunits would have to subsequently move up.

d. Guides should follow the troops attacking in the earlier phase to guide their subunits on arrival in the Forming Up Place.

e. Forward edge of the forming up place should be immediately behind the forward elements of the troops on ground. In case the Forming Up Place has to be forward ahead or on a flank, the troops on ground should arrange protection and also guide the passing troops to it.

f. Counter-bombardment should be arranged while the passing troops are forming up.

g. All available information should be quickly passed to the commanders of the passing subunits on their arrival, by the subunits being passed through. Forward observation officers of the passing subunits should also liaise with the forward observation officers of the troops in contact with the enemy.

h. Effective arrangements should be made to stop the troops being passed through from firing just before the next phase is to start. Assaulting troops should start firing from hip position, if necessary, to keep the enemy heads down as soon as they cross the Start Line.

## SECTION 45 – ATTACK BY INFILTRATION

1. A conventional attack against any enemy, defences strengthened with deep minefields and wire obstacles, is likely to be very expensive. Under these circumstances, the attackers will be required to rely mainly on surprise, deception and superior tactics. Attack by infiltration does afford this opportunity.

2. Planning, conduct and training for an attack by infiltration usually follows the methods different from normal attack, certain aspects of the techniques of planning, conduct and training, having special relevance to attack by infiltration, is outlined below.

**Planning**

3. ***Information***. Detailed information about enemy strength and disposition is a prerequisite for the success of infiltration. As compared to a normal operation of similar size, the information in infiltration operations must cover a much wider spectrum of the enemy deployment and other activities. Ideally, information as under must be available at an early stage of planning:-

a. ***Enemy***

(1) Deployment along forward and depth defended localities.

(2) Detailed dispositions.

(3) Minefields and gaps.

(4) Obstacles.

(5) Security arrangements i.e. standing patrols.

(6) Enemy habits, pattern of patrolling, night alarm system.

(7) Defensive fires, both artillery and mortars.

(8) Dispositions of reserves, especially those which can be launched against infiltrating force.

(9) Gun areas.

(10) Headquarters and administrative installations.

b. ***Terrain***. Study of terrain to include information regarding terrain along the entire length of routes from assembly areas to the objectives. Following information will be required:-

(1) Type of terrain, desert, hilly, jungle, plain, etcetera.

(2) Cover available.

(3) Natural obstacles.

(4) Features of tactical importance.

(5) State of routes especially in hilly terrain.

(6) Distances.

(7) Villages, habitations.

(8) Population.

c. ***Meteorological Conditions***

(1) Hours of darkness.

(2) Seasonal climatic conditions at the time of operations, fog and mist etcetera.

4. The above information should be gathered through the study of maps, air photographs, patrols, trained agents and prisoners of war. Under certain conditions, it would be useful to employ observation posts from the infiltration force near points of infiltration, two or three days in advance.

**Selection of Objectives**

5. The objective for an infiltrating force must be such that its seizure paralyses the entire defence of the force intended to be beaten. It must be appreciated that in any defensive system there can be only a very few such targets. Correct priority for their capture will be dictated by the tactical relevance of these to the overall

mission. However, to ensure that an infiltrating force does contribute positively to the overall plan it must also be given an alternative objective. Whatever the objective, it must be selected after careful consideration of following points:-

a. Tactical importance of the objective.

b. The effect of its capture on future operations.

c. Physical security of the infiltrating forces itself till the link-up.

d. Distance from assembly area to the objective. It must be within one nights marching distance. In hilly terrain a distance of upto 15 Kilometres can be traversed by well-trained infantry by night.

e. Likely opposition.

f. Routes and approaches.

g. Fire support. Ideally the objective should be within the field artillery range. However, it should be at least within the range of own medium artillery.

6. ***Supporting Operations***. Depending on the availability of troops and gaps, supporting operations with limited objectives can be launched in different parts of the front. Such operations will mislead and force the enemy, particularly the defender who expects an attack by infiltration to carry-out faulty and premature reactions. Launching of supporting operations to facilitate infiltration, must therefore be seriously considered and launched, if the resources permit.

**Fire Support**

7. An infiltrating force works within the division/brigade plan and will normally be operating within divisional artillery range. It is most essential to provide maximum possible fire support to the infiltrating columns. Where the enemy deployment is confirmed, timed programmes can be planned. H hours for these can be fixed and conveyed from the assembly area, once the unit has reassembled after infiltration. Majority of the targets however, will be registered for on call fire because of the uncertainty of own progress and requirements of secrecy.

8. Deficiencies in firepower must be compensated if possible, by the carriage and use of maximum number of integral mortars and Baktar Shikans. For this purpose, it is recommended that an infantry battalion must take with it all its 81 millimetres Mortars/60 millimetres Mortars and 1 x Baktar Shikan in hilly terrain. All Baktar Shikans may be carried in the tank able areas. It will be more advisable to carry maximum mortar ammunition and may be relatively fewer mortars, because the carriage problem makes it difficult to carry all the mortars and ammunition.

**Communication**

9. Operations of an infiltrating force usually have a very important bearing on other operations of the headquarters ordering it. The commander is naturally interested in the progress of the force. At the same time, it is important, from security point of view to cut down the radio traffic emanating from the force. A suggested course of action, satisfactory from the command and control as well as security point of view is:-

a. Radio silence the assault phase.

b. Prearranged short signals to indicate progress of the force, that includes:-

(1) Tapping/knocking on the mike.

(2) Blowing into the mike.

(3) Pressing hand switch to produce sound of clicks.

(4) Bird and animal calls.

(5) Use of line bedding for strings within groups.

(6) Use of fictitious net as deception.

c. Use of small line patrols for important messages only.

d. Laying of the cable behind the infiltrating battalion headquarters.

**Administration**

10. It is recommended that link-up with the infiltrating force must be established within 24-36 hours of its launching. Essential administrative cover to sustain the force in battle for this period must go with it. The plans for air supply should also be prepared, if the operation is stretched beyond its specified limits due to unavoidable circumstances. An infiltrating force is always in danger of being cut off from its base. It is, therefore, vital to ensure a thorough administrative cover for it. This aspect must be given careful thought at the time of planning. Some aspects of administration are discussed below:-

a. ***Ammunition***. A proper ammunition carriage plan should be worked out and troops should not be overburdened. Only essential ammunition needs to be carried along. Improvised containers may be provided for facilitating comfortable carriage of ammunition.

b. ***Rations***. For operations lasting 24 hours or so normal emergency pack rations supplemented by a 'MITHI ROTI' per man are considered adequate. Gur and Channa are not considered good as they make the men thirsty, creating water problem.

c. ***Water***. Troops must carry full water bottles with them. Besides they should be trained to observe strict economy in the use of water. Some empty chagals and water sterilizing tablets must also be taken along to be used when necessary, because even in hilly country consumable water is not found everywhere.

d. ***Medical***

(1) ***Evacuation***. Evacuation of casualties forward to a firm base is recommended. Casualties can be evacuated by helicopters if available or once the link-up has been achieved.

(2) ***Scope of Medical/First Aid***. Since it may not be possible to provide specialist treatment to patients for the first about 12 hours, the regimental aid post must be very well stocked with the essential life saving drugs. Details of an enlarged medical cover need to be worked out before the start of the operations and once the contents of medical aid have been decided, special portable containers, must be prepared and kept with the headquarters of the field ambulance, to be issued to the units on the orders of the brigade headquarters.

e. ***Boots***. Physical Training shoes issued by the Army are not a satisfactory footwear for long marches. They cause blisters and affect efficiency. Therefore, same superior quality rubber soled boots should be used to avoid fatigue to troops and for keeping the noise down.

**Conduct**

11. Conduct of an infiltration operation falls into following four stages:-

a. Preparatory stage.

b. Infiltration and assembly.

c. Assault and consolidation.

d. Link-up.

12. ***Preparatory Stage***. This stage covers the activities and movement of troops in the assembly area and move forward upto the infiltration point:-

a. ***Forward Concentration Areas***. The infiltrating columns should preferably assemble in areas which are outside enemy field artillery range. Apart from detailed orders and briefing other important details to be attended to in this area are:-

(1) Rehearsals.

(2) Testing of weapons.

(3) Preparation of ammunition loads and organization of porters/ mules.

In these forward concentration areas the troops should not stay for more than 48 hours. Throughout this period the area must be fully protected against detection by enemy ground/air observation.

b. ***Assembly Area***. Movement forward to assembly area will be governed by the need for secrecy. If suitable assembly area is available close to infiltrating points the troops should move into them a night prior to the commencement of the operation. The effort should be made to move infiltrating columns as close to the infiltrating points as possible. The assembly area must be protected. Troops must lie low during the intervening day as the detection of any activity by the enemy is likely to seriously jeopardize the whole operation. Final briefing and orientation of commanders is carried out here.

13. ***Movement to the Rendezvous (RV)***. Having sent forward patrols to dominate No Man's Land and to prevent enemy interference, infiltrating units/subunits, move upto the enemy's Forward Defended Localities (FDLs) along assigned routes. On reaching close to the enemy forward defended localities the force adopts appropriate grouping and formations for infiltration. The principal considerations are:-

a. Ruthless elimination of non-combatant elements from columns.

b. Only minimum essential administrative elements should be taken along.

c. Grouping should facilitate command and control.

14. Grouping for movement through the lanes depends upon terrain, number and size of gaps, enemy dispositions and his security measures. The whole force may move through one gap or it may split itself in groups and move through two or more lanes. A suggested grouping for a battalion infiltration operation is as follows:-

a. ***Two Lanes***

(1) Two companies.

(2) Section 81 millimetres Mortar.

(3) Forward observation officers.

(4) Battalion headquarters less second-in-command.

(5) Two companies.

(6) Section 81 millimetres Mortar.

(7) Forward observation officer.

(8) Second-in-command.

b. ***One Lane***

(1) Leading company.

(2) Forward observation officer.

(3) Reconnaissance group.

(4) Rear link.

(5) Three rifle companies.

(6) Mortar fire controllers.

(7) Battalion headquarters.

(8) Mortar platoon.

(9) Pioneer platoon.

(10) Anti-Tank Guided Missile platoon.

15. When using more than one lane, it may be better to move in echelons. A part of the force is detailed in groups, or small subunits. Each group/subunit infiltrates through its allotted lane. As soon as safe lanes of suitable size are reported by these leading groups the main body follows using these lanes.

16. Various techniques may be adopted to reconnoitre gaps and move undetected. It is possible that friendly patrols may previously have been through the enemy's defensive positions. In such cases personnel from these patrols may be employed to guide the subunits or groups. In other cases small reconnaissance patrols, equipped with wireless sets, may be detailed for reconnoitring gaps, ensuring that these are wide enough for undetected movement of the group, subunits, or unit, that they lead. It would be ideal for the patrol to return and lead the main body through the gaps. But the time may seldom permit this. Therefore, the patrol may send a prearranged signal to the main body to follow.

17. The patrol must inform the main body of any deviations from the planned route due to, obstacles, trip wires or other obstructions and particularly the enemy localities, found by it. These must either be marked by the patrol on the ground or preferably the patrol should guide the main body how to cross these obstructions.

18. Units, subunits and groups should take all precautions to avoid detection by the enemy and if detected to avoid decisive engagement. Procedures are laid down in advance for action in case enemy localities are encountered. Troops must also know how to silently dispose of enemy sentries.

19. Fire support may be employed prior to the start of infiltration in order to neutralize enemy surveillance capability. During the movement, fire support is used to disrupt activities of enemy patrols, to drown noise of movement, particularly if vehicles are accompanying and for deception in areas away from the planned lanes of infiltration.

20. Often a lane may be found blocked by the enemy. In such an eventuality following alternatives are open to the subunit or group using the lane:-

a. It may further split in smaller groups and cross the area.

b. It may withdraw to follow a safer lane.

21. ***Selection of Rendezvous (RV) and Firm Base for Assembly of the Force***. Having infiltrated through enemy positions, the groups move to the RV for assembly. Care is exercised by all groups in approaching the RV. To ensure that the RV is not occupied by the enemy, patrols may be sent forward well in advance to carry-out reconnaissance. The main body may then move in if the patrol confirm that the RV is clear of the enemy. If the enemy is found occupying the rendezvous, the group moves to an alternative RV employing the same procedure. Requirements for selection of RV and firm base are:-

a. ***Rendezvous (RV)***

(1) Should be away from likely defensive fire.

(2) Should be protected by specifically detailed groups.

(3) Avoid places hemmed in by obstacles.

b. ***Firm Base***

(1) Should be away from likely defensive fire.

(2) Should be protected.

(3) Features of tactical importance dominating it should be held.

(4) Should have alternative routes to the Forming Up Place.

(5) Alternative firm base be selected.

22. ***Protection during Movement***. Following precautions to be undertaken:-

a. Specific troops should be detailed to protect the route.

b. Important features along the route should be occupied by the route protection troops.

c. In the event of contact with enemy patrols, only the troops in contact should fire back. The contact with such enemy parties should be maintained by the troops detailed for the lane/route protection.

Unless the lane is physically blocked other troops should continue to advance.

23. ***Sealing Off the Gaps by the Enemy***. Once an infiltration is detected by the enemy, his first reaction would be to seal off any further infiltration of the troops as well as administrative echelons. Therefore, it is imperative that the shoulders of the gaps are secured by the leading troops, before the infiltration starts. These troops will stay in the position till the last man has cleared off.

24. ***Crossing of the Minefields***. The possibility of extensive minefields ahead of and in between the enemy localities can not be ignored. The crossing of minefields is not only hazardous but also time consuming. The problem is twofold, i.e. finding the gaps and the passage of troops through the gaps.

25. Infiltrating troops should carry-out extensive reconnaissance of the routes for finding out any minefields ahead or in the gaps of the enemy localities. Before undertaking the operation, lanes have to be cleared through the minefields during the preceding nights. Clearance of minefield, may even take 2 to 3 nights. If the lanes are not cleared, the troops may have to go through the minefields which may result in the failure of operation because the surprise will be lost due to the exploding mines. However, efforts should also be made to locate the existing gaps/ lanes in the enemy positions, which should be used by the infiltrating troops.

26. ***Command and Control***. We have yet to find an ideal solution to this very difficult problem connected with infiltration, yet some suggestions for improvement are as under:-

a. ***Bifurcation of Headquarters***. Battalion and company headquarters should move in two parts, each to be commanded by the battalion/ company commander and battalion/company second-in-command respectively. Each of these should be capable of taking over the command in case of the loss of the other. It also brings out the importance of thorough briefing of at least three top persons in every subunit about the plans. A clear line of succession must be established.

b. ***Place of Commanders/Second-in-Command***. Commander's operation group should move immediately because during this stage the leading troops will need guidance and possibly prodding. Because of tension a commander can project his strong will in a more effective way if he is well forward. Second-in-command must remain at the back to guard against any delays and straggling. At company level, a company officer may be detailed to command guides, protect RV and brief patrols. Company commander may move behind the leading elements in any of the lanes ensuring that he is at RV when about half the company

has reached there. The second-in-command must follow the lane other than that of the company commander and take all the administrative elements along with him.

c. ***Use of Bounds and Report Lines***. Sure methods of reporting progress without detection must be devised. Short signals on wireless should be an ideal way of reporting progress.

d. ***Recognition and Challenge by Night***. It is important to further practise the existing drills of challenge and recognition by night. Troops must be thoroughly practised in this to avoid firing on each other. They should also guard against falling into enemy trap.

27. ***Assault and Reorganization***. The only difference between the assault by an infiltrating force and by a conventional force is that of the opportunities that the former offers for success. By the very act of achieving successful infiltration, this force puts the enemy at a disadvantage to a large extent. The attack from an unexpected direction causes disruption of the highest degree and usually hits a soft and relatively weaker defended spot. The shock effect of such an attack must therefore, be exploited most vigorously. Every junior leader and soldier must be trained to grasp the fact that such an attack must be driven home with all the speed and energy. There is no room for unnecessary halts and delays. The following sequence of action is suggested:-

a. Softening up of the objective by artillery fire.

b. Capture of the given objective.

c. Capture of be prepared objectives.

d. Reorganization and replenishment.

e. Establishment of a ring of patrols to prevent counter-attacks. Tank approaches to be covered with anti-tank weapons.

f. Despatch of raiding parties/task force to attack enemy guns, administration areas or headquarters, if located.

28. ***Linking-Up***. Final linking-up between the infiltrating force and the troops employing conventional methods of attack is the most critical operation and requires detailed and thorough planning. Certain drills and procedures will have to be adopted to affect a smooth link-up. Some of the points which need special consideration are:-

a. Communications between the link-up forces.

b. Working out of RV and link-up points between the forces.

c. Communications between the link-up patrols that are sent out by the forces after reaching the designated RV before affecting the final link-up.

d. ***Artillery Fire Support***. Limits must be fixed beyond which artillery fire support should not be called by the link-up forces.

e. The link-up should be established preferably by a mobile/armoured task force.

f. Plans must be prepared for an eventuality when link-up does not materialize and the infiltrating force has to exfiltrate to rejoin the main force.

**Training**

29. A normal well-trained soldier is considered quite capable of undertaking infiltration. Some aspects of training to achieve greater operational effectiveness are highlighted below.

30. ***Fieldcraft and Minor Tactics***. Marksmanship, fire-control, camouflage and a correct use of ground must be instilled into troops. Section and platoon leaders must be able to act and react correctly to tactical situations and the troops should be able to respond to orders. A high standard must, therefore, be achieved in:-

a. Tactical knowledge of junior leaders.

b. Handling of weapons.

c. Fire-control.

d. Camouflage and concealment.

e. Close quarter battle.

f. Map-reading and navigation.

31. ***Battle Leadership***. On account of the peculiar challenges of command and control encountered in leading the infiltrating troops, success of such operations to a great degree depends on the skill and courage of junior officers. In these operations troops are exposed to a hostile environment for long periods. Not only that, but, the success of the entire operation often depends upon the vigour and determination with which attacks are carried out. Only properly motivated and highly trained officers can lead the troops to success. The training of junior officers must, therefore, be so organized that they are not only capable of making correct decisions but also implement orders with vigour and devotion. Such leaders have to be trained over years and through proper schooling in the units and brigades.

32. ***Physical Fitness***. Infiltration demands a very high standard of physical fitness and involves extra ordinary stress and strain. A well conditioned body capable of

marching 24.15 kilometres (15 miles) and assaulting an objective thereafter is a basic condition for undertaking infiltration. The standard set forth here, therefore, must be achieved by organizing physical training, games and exercises.

33. ***Night Training***. More often than not, infiltration will be undertaken during night, which makes it a difficult operation requiring a high degree of proficiency. Direction keeping, command and control, handling of weapons and equipment and stealthy movements are some of the major elements connected with night operations. We must, therefore, train to solve these problems.

## SECTION 46 – ASSAULT ACROSS WATER OBSTACLE

**General**

1. This section should be read in conjunction with GSP-1900 – 'Strike Corps Operations, Part I, Break-in Battle (Provisional), 2000'.

2. The crossing of a water obstacle, if strongly opposed, will be a highly difficult and complicated operation, it will seldom be carried out by a battalion on its own. The subject, therefore, is only dealt with in outline in this section.

3. ***Problems Facing the Attacker and Important Points***. The problems are as follows:-

a. ***Clearance and Securing of Home Bank***. This necessitates following actions:-

(1) Reconnaissance.

(2) Planning.

(3) Clearance and consolidation.

b. ***Reconnaissance for Assault Across***. The requirement would be the reconnaissance for:-

(1) Approaches to the obstacle suitable for the vehicles and foot infantry, including availability of cover.

(2) Various areas required for multifarious activities like forward assembly, boot off loading, vehicle marshalling and forming up etcetera.

(3) Objective(s), terrain, enemy deployment, weapons and strength.

(4) Suitable crossing and bridging sites.

(5) Approaches coming in/going out of the objective area.

c. ***Security***. The maintenance of secrecy will be a big problem as an assault across water obstacle is a major deliberate operation requiring detailed reconnaissance, planning and preparation. The security of the operation will be achieved by secrecy and deception, which should be carefully planned, in detail.

d. ***Assault Across and Consolidation***. This requires very meticulous planning and preparation, which are dealt with in detail in the subsequent paragraphs.

e. ***Direction Keeping***. It is a serious problem while crossing a water obstacle at night, particularly in poor visibility due to darkness or foliage, when the other bank cannot be seen. It is also difficult when there are no prominent landmarks, water current is strong and crossing has to be diagonal or when it has to be silent crossing and assistance of fire cannot be utilised to maintain direction. In such cases, it is suggested that initially a group of path finders should proceed the first wave, who should place dim coloured lights facing the Start Line, that can be switched on and off according to the H hour. This will be only possible if the other side is not held by the enemy. If the other side is held, the path finders should accompany the first wave for markings.

f. The most important points that will facilitate success of an assault across water obstacles are:-

(1) Secrecy.

(2) Deception.

(3) Speed.

**Basic Requirements**

4. Basic principles of the attack are:-

a. Clear orders and briefing based on sound planning and intelligence.

b. Simplicity.

c. Surprise.

d. Speed and determination in execution.

e. Rapid exploitation of success.

**Stages of an Assault Crossing**

5. The operation will normally be divided into nine stages, some of which will overlap each other:-

a. Reconnaissance.

b. Appreciation and plan.

c. Detailed planning and orders.

d. Assembly and preparation of troops and the crossing equipment.

e. Advance to the waterway.

f. The assault.

g. Establishment of bridgehead.

h. Construction of bridges and crossing of the main force (reserve battalions or brigades).

**Reconnaissance and Planning**

6. Information is needed on:-

a. Enemy defences, including mines on both sides of the water obstacle.

b. Details about the obstacle:-

(1) Width, depth, current and type of bed.

(2) Banks, height and slope.

(3) Islands, fords, weirs and banks.

c. Subsidiary obstacles, ditches or fences.

d. Approaches to and exits from the water for wheels and tracks.

e. Suitable crossing places for boats and rafts.

7. The sources of information in this operation are no different from any other; e.g. patrols, observation posts, ground surveillance radars, maps, air photographs, prisoners of war etcetera. The engineers can normally supply fairly detailed information about the obstacles, which can be checked or further enlarged by the patrols. Joint patrols, infantry and tank or infantry and engineer, will frequently be necessary. Care must be taken not to disclose the intended crossing places. Reconnaissance of the obstacle may be controlled by the brigade or division and patrolling activity kept along the whole front as part of a deception plan.

**Advance to the Waterway**

8. Areas to be used by assault troops are given in Figure 19.

a. ***Concentration Area***. If possible it should be out of enemy artillery range, where the troops can rehearse the coming operations.

b. ***Brigade Assembly Area***. One assembly area per brigade is usually required. Here the battalion vehicles, other than those crossing under

the brigade arrangements are assembled after the initial bridgehead has been secured.

c. ***Battalion Assembly Area***. Will be needed for each assault battalion. It should not be far from the obstacle. Here the battalion makes final preparations before moving to the Forming Up Place(s).

d. ***Boat Off Loading Point***. If the assembly area is some distance away from the obstacle and it is possible to take the transport forward of it before the assault without prejudicing security, a boat off-loading point is selected. Vehicles carrying the assault boats are driven to it before being unloaded.

e. ***Checkpoint***. This will ensure that companies move forward in the correct order at the right time.

f. ***Forming Up Place***. Area where the assault troops get into their assault formations. In this case, the formations in which they enter the water. When the home bank of the obstacle is secure, the waterline is the Start Line.

g. ***Crossing Places***. Where the troops put their boats into the water.

**Vehicles**

9. Vehicles of the assaulting battalions are collected in the vehicle marshalling area near battalion or brigade assembly areas and called forward in their order of priority to the battalion or brigade rafts or bridges, according to their classification. The organization includes:-

a. ***Vehicle Waiting Areas***. To ensure an even flow of vehicles over the bridges and ferries, it may be necessary to organize vehicle waiting areas between the assembly areas and the water obstacle. These areas should be reasonably close to the obstacle and should not contain more than a small group of vehicles at any one time lest they should present good targets for the enemy air and artillery fire.

b. ***Release Points***. These will be organized on the far bank and each assaulting battalion will normally require one. Here the vehicles will be met by the company guides and taken to their respective reorganization areas as in the normal attack.

c. ***Brigade Forward Release Areas***. Areas at which the vehicles crossing under the brigade arrangements can be met by the unit guides. Normally one of these areas is required per brigade.

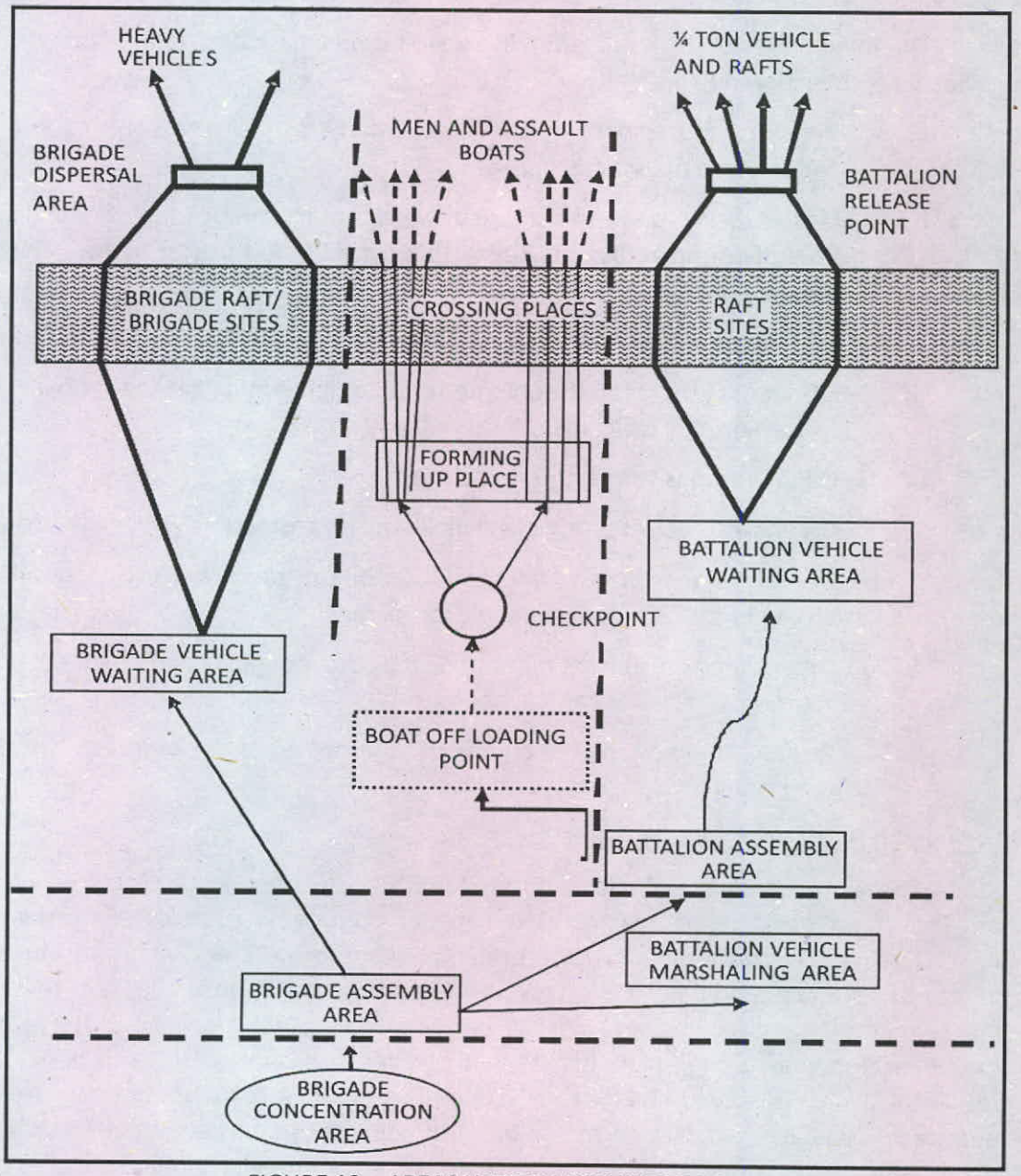



FIGURE 19 – AREAS USED BY ASSAULT TROOPS.

## Duties of the Bank Group

10. The smooth and orderly progress of the crossing depends largely upon efficient direction and control of the troops and vehicles, between the assembly areas and the release points on the far bank. These duties are undertaken either

entirely or in part by a bank group, specially formed for the operations. Usually a bank group is formed by a separate unit. However, an assaulting battalion may also have to supply some of the men to help organized a bank group and may sometimes have to form its own bank group.

11. The main duties of the bank group in an operation where a battalion does not provide its own bank group are:-

a. Directing and controlling movement of personnel and vehicles into and forward of assembly areas.

b. Close supervision of the crossing, which entails the upkeep of an even flow of traffic across the obstacle by making full use of all available bridges, rafts and crossing places.

c. Protection of routes to crossing places.

d. Off-loading the assault equipment and helping in its carriage forward to the Forming Up Place.

e. Guiding troops to the crossing places.

f. Organizing each crossing place including its protection.

g. Operating the boats, rafts and outboard motors. Ferrymen are provided at least for the assault companies.

h. Organizing and staffing medical posts, recovery posts and prisoners of war cages needed near the obstacle.

i. Keeping the next, higher commander informed of the progress of the operation.

**Bridgehead Battle – Important Aspects**

12. The aim of bridgehead battle is to establish a superior combat force on the far side of an obstacle with a view to conduct operations towards a strategic objective. The preparatory stage of the bridgehead battle therefore, assumes much greater importance than a normal attack. It has been generally observed, that there is a tendency to mark the extents of a projected bridgehead on the map, in a symmetrical manner during planning stage. However, the complicated and difficult operation of a bridgehead will generally not progress in a manner that the area indicated by a well drawn line on the map is ultimately secured as a bridgehead. The initial bridgehead plan must be based on securing a defensible terrain feature across the obstacle, in keeping with the requirements of a secure base, having sufficient depth for its defence as well as sufficient space for induction of forces.

13. The extents of the bridgehead would be designated by the headquarters having overall control of the operation. Major factors are as follows:-

a. It should be based on a defensible terrain.

b. It should cover terrain that facilitates exploitation, in continuation of the overall mission.

c. It should provide sufficient space to accommodate the troops, equipment and the support elements of the crossing units and formations without congestion.

14. The above considerations have to be kept in mind while planning the operation. Due to the complexity of the bridgehead battle, part of the plan may, if necessary, be based on assumptions. The plans will, therefore, have to be flexible enough to incorporate necessary changes with regards to the extent of the bridgehead as well as various timings.

**Requirement of Space in the Bridgehead**

15. The bridgehead battle is characterized by congestion arising from the lack of space. It will thus require a very meticulous plan for the build up of troops for breaking out from the bridgehead. These problems will further be compounded by an ongoing fighting either for the expansion of the bridgehead, or for dealing with enemy's counter-attack.

16. While designating space in the bridgehead, following aspects should be catered for:-

a. Space required for the deployment of the troops who established the bridgehead.

b. Various headquarters of the assault formations.

c. Sufficient space for induction of the break-out forces.

17. Generally, an area of about 48 square kilometres (8 x 6 kilometres) is required for accommodating an armoured division group in the bridgehead. As a guideline, 10 to 12 square kilometres will be used up by the break-in infantry division (mostly on the periphery), while 18 to 20 square kilometres will be required for the elements of the armoured division. The remaining 10 square kilometres may be distributed between the elements of follow in zone infantry division and other assets like air defence and engineers. Of course, it is always more prudent to undertake graduated induction in the bridgehead in order mitigate the problems of space.

**Break-in Phase**

18. Night assault is preferable, as it allows own forces to close-up, including move forward of the base of fire elements, artillery and engineer support, without being subjected to observed enemy fire.

19. Generally a division plus (four brigades) force is needed to secure a bridgehead for armoured division and the follow-in-zone infantry. However, at least a minimum of two to three brigades are needed in this phase, which will assault on a narrow front. They will be provided maximum possible fire support. The actual assault will be preceded by a massive fire on the enemy defences in order to cause meaningful destruction and a considerable reduction in his defensive capability. The aim is to reduce his ability to such an extent that success of own assaulting infantry is ensured, keeping the attrition to own troops to the bare minimum. This will also ensure speedy negotiation of the obstacles with a relative ease.

20. The brigades will undertake the break-in in two phases. The first phase is likely to terminate about 300 to 600 metres ahead of the bund with a width of about 1800 to 2000 metres for each brigade. The second phase will see the expansion of the base and the depth of the bridgehead to 3500 to 4000 metres and 1500 to 1800 metres respectively in each brigade area (see Figure 20).

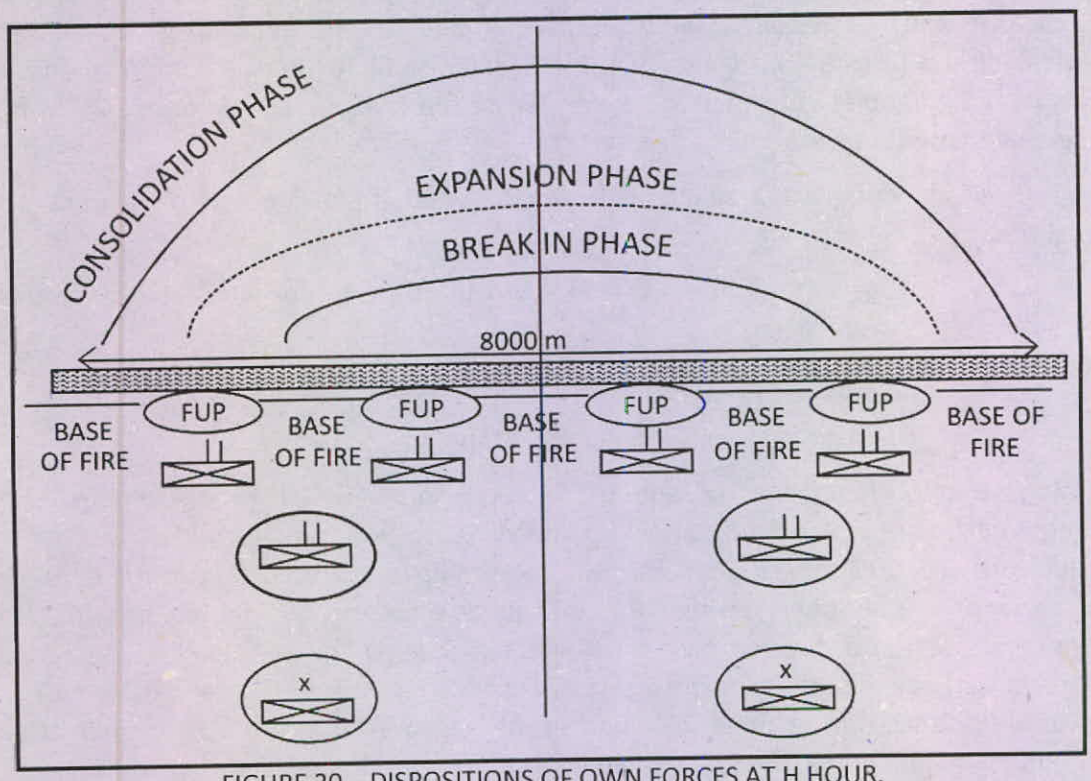



FIGURE 20 – DISPOSITIONS OF OWN FORCES AT H HOUR.

21. Two brigades are expected to secure an area of 7000 to 8000 metres in width and 1500 to 1800 metres in depth on completion of the break-in phase. Each brigade is thus responsible for an area of 3500 to 4000 metres in width and 1500 to 1800 metres in depth.

22. A brigade will normally assault with two battalions up, keeping the third one in reserve. The leading battalions should, initially attack on a narrow front of 500 to 800 metres keeping a gap of 500 to 600 metres in between. They should subsequently expanding sideways and on completion of the break-in phase, secure an area of about 1500 to 1800 metres in width and 600 metres in depth. The third battalion should pass through the area already captured and expand the bridgehead by securing the area up to about 1500 metres in depth (see Figure 21).

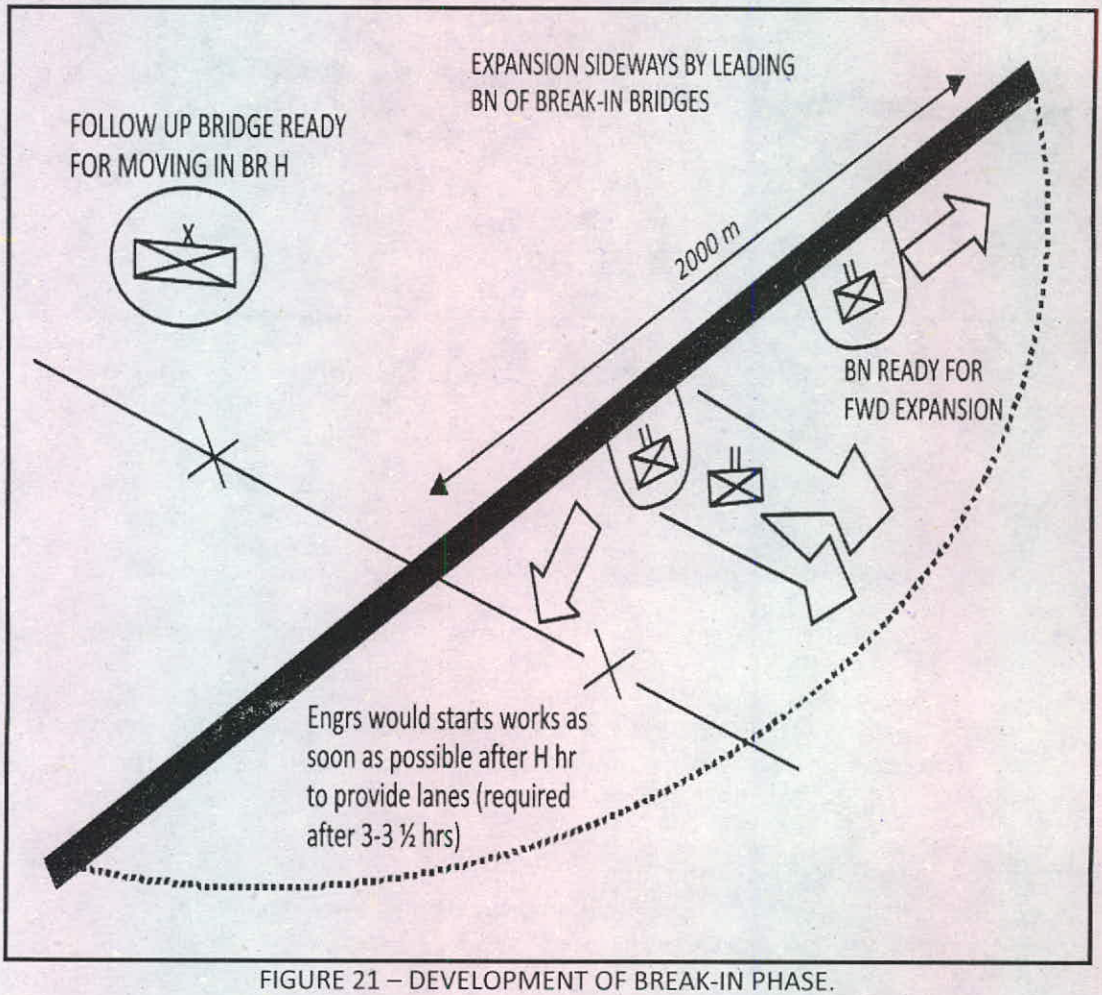



FIGURE 21 – DEVELOPMENT OF BREAK-IN PHASE.

23. It will be preferable to have four battalions in the break-in brigades. The fourth battalion will give the brigade commander reasonable flexibility and leverage to attain his objective. In fact, the brigade phase could then be extended even up to a depth of about 2500 metres (see Figure 22).

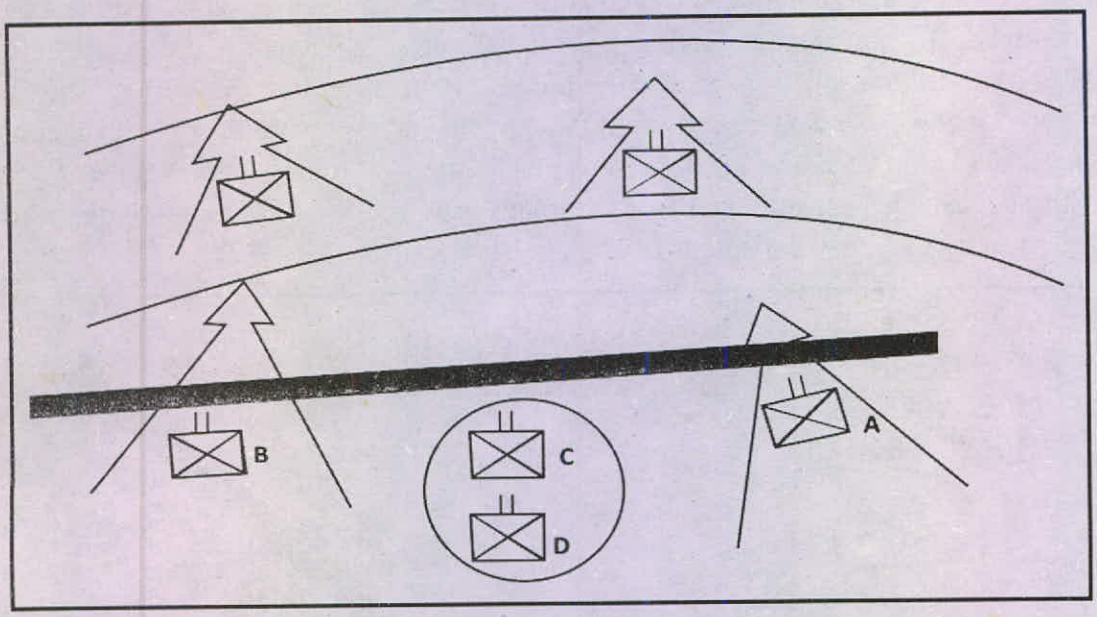



FIGURE 22 – FOUR BATTALION BRIGADE.

**Objectives**

24. Following considerations must be kept in mind while selecting the objectives:-

a. Must be limited, particularly at night.

b. Primary objectives are those positions from which the enemy can put aimed small arms fire onto the crossing places.

c. Secondary objectives are those positions which overlook the obstacle and from which the enemy can direct artillery fire onto the crossing places.

**H Hour**

25. The time at which the first wave of assault boats enter the water.

**Fire Support**

26. Following points are emphasized:-

a. If a silent crossing is attempted, fire support must be on call.

b. Flat trajectory weapons are useful for:-

(1) Flank protection.

(2) Shooting in assaulting troops on to the objective by firing from a salient to the flank.

(3) For close protection of the crossing places.

(4) Neutralizing enemy defences, particularly close to the far bank.

c. Tracer fire from the machine-gun is a useful aid for direction keeping.

d. Full use must be made of the firepower of troops already holding the home bank. In particular, their recoilless rifles can provide flank protection to the initial landings where required.

e. The fire from the tank gun and machine-guns can be utilized to provide fire support. However, this necessitates reconnaissance, selection of fire positions and move of tanks to the fire positions keeping in consideration the requirement of secrecy and silent attack.

**Mopping Up**

27. This must be thoroughly carried out by the forward troops and not left to the follow-up companies, because the bypassed enemy will continue to fight and may seriously impede both follow-up companies and the rear of the troops who have gone through them. The delay imposed by mopping up in front must be accepted.

**Anti-Tank Defence**

28. It will take some time before the recoilless rifles or tanks can reach the forward companies in the bridgehead. Anti-tank defence calls for guns to be put across the obstacle as early as possible. Platoon and section anti-tank weapons are, therefore, needed. Arrangements may also be made to carry some anti-tank mines, which should be laid immediately on the arrival of the assaulting troops for the protection of the bridgehead.

**'F' Echelon Transport**

29. Priority for crossing for most operations:-

a. Anti-Tank Guided Missiles.

b. Ammunition.

c. Forward Observation Officer's vehicle as applicable.

d. Command, signal and medical vehicles.

**Administration**

30. After a battalion has gained its foothold, it may be temporarily cut off by enemy counter-attack and shelling on the bridges, therefore:-

a. An adequate reserve of all important stores particularly ammunition, must be put across the obstacle with the least possible delay.

b. Arrangements must be made to hold the wounded and the prisoners of war for some time, until they can be evacuated.

c. Men must carry enough food and water for the next about 24 hours.

## SECTION 47 – ASSAULT ACROSS A WATER OBSTACLE WITH HIGH FAR BANK

### General

1. In any offensive operation across the international borders, following general pattern of enemy defences is likely to be encountered by own forces:-

a. ***Initial Line of Defence***. This consists of border out posts and the border villages converted into strong points.

b. ***First Line of Defence***. Where major water obstacles are some distances away from the borders, the enemy has constructed smaller water channels, to base his first line of resistance.

c. ***Second/Major Line of Defence***. Organized on the major water obstacle.

2. This section discusses various aspects/problems involved in assault across a major enemy water obstacle with high far bank and the measures to deal with these at battalion level.

### General Concept of Operations in Assault Across Water Obstacle

3. Notwithstanding minor operations of a local nature, such assaults would mainly be undertaken as part of the offensive operations of sizeable proportions. Therefore, the planning would usually be done at the division and higher levels since it involves employment and coordination of considerable armour, artillery, engineer and air efforts. Broadly, an operation of this nature involves:-

a. Clearance of the enemy security zone.

b. Establishment of bridgehead across water obstacle by the assaulting force and its consolidation and expansion.

c. Break-out from the bridgehead by the follow-up elements for missions deeper in the enemy area.

4. An infantry battalion may be employed in any one or more of the above mentioned stages of this operation. The scope of this section is limited to the task performed by the leading assault battalion employed in bridgehead operation.

### Characteristics of Enemy Defence on a Major Water Obstacle

5. ***Characteristics of the Water Obstacle***. A major enemy defence oriented water obstacle is generally over 80 feet wide so as to prevent bridging of the water gap by our existing bridge layer tanks in a rush. The depth of water varies between

6 to 8 feet. The banks are near vertical and rise 4 to 6 feet above the water surface. A 15 to 18 feet high steep bund is built by the enemy on the home bank to provide good observation and fields of fire over the far bank. The far bank is flushed with ground. The water may either be stationary or may have a current of upto 2-3 knots. The water level can be varied with the help of regulators. The water obstacle may vary from the standard width as given above to a smaller ditch; a single or twin canal. Canals may have brick lining. Each variation poses different problems of drills and technique. For a summary of different types of water obstacles see Annex P.

6. ***Enemy Defence/Frontages and Dispositions of Bunkers***. A water obstacle is further reinforced by various types of wire obstacles and 900-1000 metres deep mixed minefields on the far bank. Further ahead, the enemy deploys usual security elements like screens and standing patrols, leaving necessary lanes in the minefield for their speedy withdrawal when required.

7. Due to the nature of the obstacle, the enemy usually occupies a linear defence with all the three sections up, each section occupying a group of three bunkers, spaced approximately 15 metres apart (centre bunker being of RCC). With each section spread over 32-36 metres and with a gap of 77 metres between the sections, a platoon occupies a front of about 250 metres. An enemy company occupies a front of about 600-950 metres depending on whether it is deployed with 2 or 3 platoons up. A diagrammatic layout of enemy defence on a typical water obstacle is given at Figure 23.

**Problems of the Attacker**

8. The combination of a strong minefield and major water obstacle presents formidable problems. From a study of the enemy concept of defence, it is evident, that an assaulting infantry battalion would be confronted with four consecutive obstacles before it finally captures the far bank:-

a. Enemy minefield.

b. Wire obstacles.

c. The water gap.

d. Scaling of the high far bank and clearance of the enemy bunkers.

9. The implications arising from these obstacles and the remedial measures satisfying these implications are covered in the succeeding paragraphs:-

a. The assault troops have to go through and overcome these obstacles, giving the defender much longer time to bring the maximum weight of his fire on them from the elevated far bank. The attacking force would thus suffer a high rate of attrition, adversely affecting the required combat ratio during the final assault on the enemy bunkers.

b. The time lag between the capture of the far bank and the arrival of the reorganization stores and heavy weapons may be quite enough for the defender to launch an effective counter-attack. The leading assaulting groups, therefore, need to be immediately replenished, after the capture of the far bank.

c. The combination imposes serious limitations on the ability of the attacker to complete the lanes as well as bridge the water gap during one night, making him vulnerable to on observed daylight air and artillery punishment. With the attack not being able to gain an early momentum the defender has plenty of time to move his reserves etcetera for reaction.

d. Effective drills to maintain command and control and accurate navigation to the objective would have to be evolved especially at the section/platoon and company levels.

e. The combination also make it very difficult for the attacker to carry-out effective reconnaissance of the water obstacle, the defences on it and the layout of defences behind the water obstacle.

f. The high far bank is linear and lacks depth. Bunkers on it can be effectively engaged and neutralized.

g. The scaling of the high far bank will require special equipment and training by the assault force. Methods and techniques for smooth passage through the mines and wire obstacles will also have to be worked out and rehearsed.

h. Once the high far bank is gained by the attacker it could be used effectively to dominate the area behind it. Attack on the depth positions may be relatively simpler after this phase.

**Planning**

10. Some important aspects of planning for a clear understanding of the techniques and drills to be followed in such operations are discussed in the succeeding paragraphs.

11. ***Development of the Bridgehead***

a. ***Size***. This would depend on:-

(1) Type of terrain including availability of cover.

(2) Quantum of the troops, stores and vehicles which would form/move through the bridgehead.

(3) Own resources to establish the bridgehead within a reasonable time.

(4) Dispersion required in the bridgehead in view of the enemy threat. In an operation where an infantry division is employed in conjunction with an armoured brigade, a bridgehead should be at the minimum, 500-4500 metres wide and approximately 500 metres deep.

b. ***Assault on a Narrow or Broad Front***. This decision would be taken by the higher commander keeping in view the available resources, the terrain, enemy situation and the number of lanes or bridges/rafts required to be made. If an assault is initially made on a narrow front, it can be extended by working sideways after reaching the far bank.

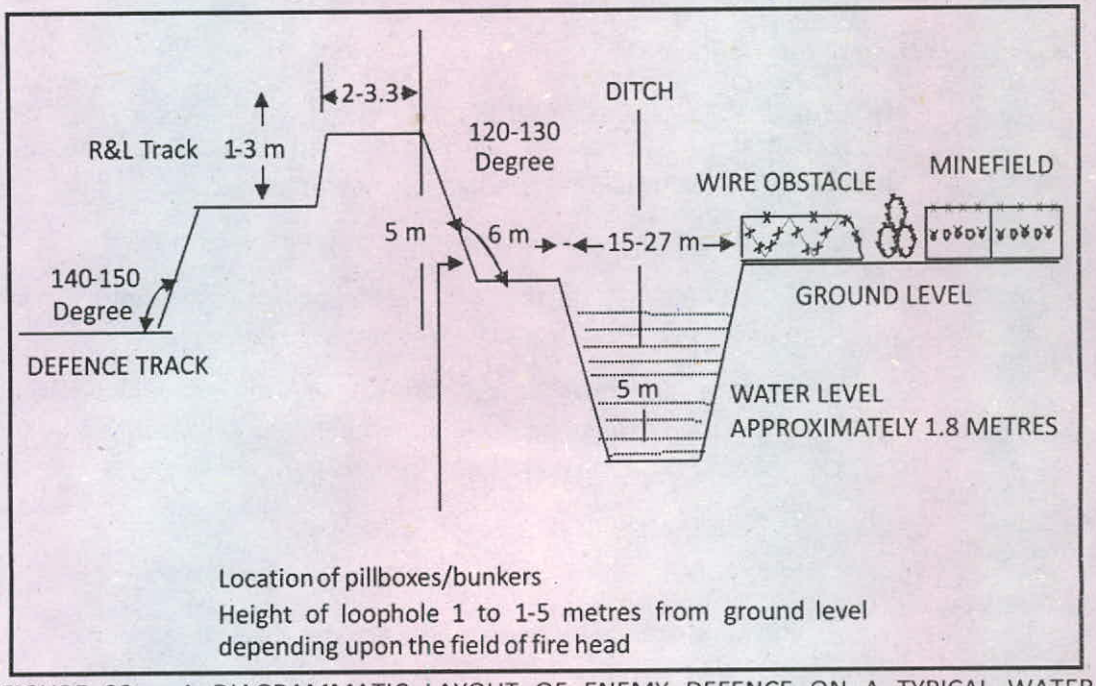



FIGURE 23 – A DIAGRAMMATIC LAYOUT OF ENEMY DEFENCE ON A TYPICAL WATER OBSTACLE.

12. ***Stages***. At the infantry battalion level, this operation would be undertaken in following general stages, which are discussed in detail in the subsequent paragraphs, assuming that the enemy security zone has already been cleared:-

a. Reconnaissance and gathering of maximum possible information.

b. Planning, preparation and rehearsals.

c. Move to the assembly/forward assembly area and the Forming Up Place.

d. The assault and capture of the far bank.

e. Replenishment of initial assault companies with essential reorganization, stores, weapons and equipment. For details refer to paragraph 28 below.

f. Capture of enemy depth positions close behind the bund.

g. Replenishment and carriage across of reorganization stores and weapons and equipment for the follow-up companies.

h. Carriage across of heavy weapons e.g. 106 millimetres Recoilless Rifles, 81 millimetres Mortars etcetera and other stores.

i. Readjustment, consolidation and completion of reorganization.

j. Staying ready for subsequent tasks.

13. ***Gathering of Information***

a. ***Organization of Reconnaissance***. Since it is vital that the actual point of assault should not be known to the enemy till the last possible moment, it is essential that the battalion commander:-

(1) Plans, organizes and coordinates the reconnaissance.

(2) Coordinates with the holding troops for the information gained by them while in contact with the enemy.

(3) For the purposes of deception, feigned reconnaissance of areas other than the point of assault may be arranged at the brigade/division level.

b. ***Sources of Information***

(1) Holding troops.

(2) Reconnaissance by commanders of the assaulting troops.

(3) Observation posts placed as far forward as possible.

(4) Patrols.

(5) Air photos.

(6) Maps.

(7) Intelligence reports/terrain analysis.

(8) Local civilians.

(9) Ground Surveillance Radars.

14. ***Determination of Time for the Assault***. Keeping in view own limitations of resources and overall superiority of the enemy air, it is essential that all critical phases of the bridgehead operation be conducted during dark hours. But at present, even if the clearance of the enemy security zone and the ultimate passage of the follow-up troops through the bridgehead is not taken into consideration, the bridgehead operation itself extends beyond one night as it involves:-

a. Move of troops to the Forming Up Places.

b. Security of the far bank and depth positions in its vicinity.

c. Mine breaching.

d. Construction of bridges and rafts.

e. Expansion of the bridgehead.

15. Since some daylight hours have to be per force utilized under all circumstances for the completion of the bridgehead, two workable alternatives emerge; namely, the adjustment of daylight hours at the beginning or towards the end of the operation. Unless the following are strongly catered for, the second alternative is more sound and appropriate:-

a. Local air superiority.

b. Anti-aircraft resources for the protection of bridges and raft sites.

c. Effective and large-scale counter-bombardment of enemy gun positions.

d. Units taking adequate anti-aircraft protection measures for vehicles, etcetera.

16. ***Advantages of Operations during Hours of Darkness***

a. Hazardous task of assaulting across the minefield and the water obstacle will be undertaken during the night, when the enemy's ability, to read the battle, bring down accurate fire and react quickly will be considerably reduced. The attacker will also have a psychological advantage.

b. The serious disadvantage of the troops during daylight, inviting punishment from enemy air and observed artillery fire is eliminated.

c. Surprise, so vital for the success of the operation, particularly in the opening stages of the battle, is easy to achieve during night than during day.

d. Duration of vulnerability will be reduced, because even though the complete build up runs into the day, sufficient heavy weapons and tanks would have already crossed before the first light, to give requisite staying power to the attacking infantry at first light.

e. Success of infantry attack against formidable enemy defensive system is surer at night than during day.

17. ***Selection of Objective***

a. From the study of enemy defences, frontages and general disposition of bunkers in the high far bank (paragraph 7 above refers), it can be assumed that keeping the usual 3:1 superiority, each section is capable of effectively assaulting one enemy bunker and a platoon is able to clear a group of three bunkers comprising one enemy section position. Similarly, a company is capable of capturing an enemy platoon spread over 250 metres of front. Coincidentally this corresponds to our conventional frontage for a company assault at night. Each battalion assaulting with two companies up would attack a frontage of 450-650 metres.

b. The actual place of attack should be the one where the enemy least expects the attack. The weakest spot in the enemy defences must be determined for the assault. Assault may even be made on comparatively difficult part of the far bank if the enemy opposition is weaker. Since layout of enemy defences on such water obstacles seldom offers flanks, a frontal attack would invariably have to be launched. However, once a gap has been created, it must be speedily expanded towards the flanks.

c. ***Bridges/Bridge Sites***. Main approaches and bridges over water obstacles are likely to be well defended, but at higher levels clearance of the main axis and capture of suitable bridges or bridge sites will be essential for tactical as well as administrative requirements. Following measures are suggested to overcome the problem:-

(1) By tackling these sites from the flanks/rear.

(2) Use of helicopter borne/airborne troops from the rear/flank.

(3) Problem of timings must be coordinated.

18. ***Fire Support***

a. Provision of accurate, large scale and effective fire support is a major factor for success in such operations. This would be planned at the divisional/formation level where all means of fire support can be well coordinated.

b. Following weapons/means will be incorporated in the overall fire support plan:-

(1) Aircraft.

(2) ***Artillery***

(a) Of holding formation.

(b) Of the assaulting formations. Both in the assault fire as well as conventional role.

(c) Additionally allotted from the Corps.

(3) Tanks and recoilless rifles of the holding/assaulting formations.

(4) 81 millimetres Mortars of infantry battalions.

c. ***Fire Support Requirements***. Following overall fire support requirements must be catered for in such operations:-

(1) Maximum destruction of enemy defences on the far bank.

(2) Neutralization of enemy fire support weapons i.e. counter-bombardment.

(3) Covering the move of own assaulting troops and neutralization of enemy weapons on the far bank.

(4) Providing fire support to own troops against enemy counter-attack.

(5) Providing anti-aircraft protection at the bridges/crossing sites and to the break-out force during their move through the bridgehead.

d. ***Fire Support for an Assaulting Infantry Battalion***. The aspects of fire support with which the assaulting infantry battalion would be concerned are:-

(1) ***Destruction of Enemy Defences on the Far Bank***

(a) ***Planning***. This would be planned at the division/ brigade/battalion level utilizing all available weapons.

(b) ***Weapons***. Use will be made of aircraft, tanks, Anti-Tank Guided Missiles, Grenade Launcher Heavy- Machine guns, Recoilless Rifles and heavy and medium artillery guns even in direct/assault fire role, if required.

(c) ***Timing***. Generally the destruction of enemy bunkers can be undertaken well before the H hour using daylight to maximum effect. However, elements of surprise must be given its due importance. In that case the fire will be brought to bear on the objectives at a suitable time prior to the H hour.

(d) ***Location of Fire Base***. For a brigade assault, the fire base for tanks, Anti-Tank Guided Missiles, Grenade

Launcher Heavy-Machine guns, Recoilless Rifles and the artillery pieces firing in the assault fire role, are located at suitable places on the flanks so as to ensure intimate fire support till the last possible moment, without endangering their safety. In addition to the fire base, some tanks, Anti-Tank Guided Missiles and recoilless rifles could also engage enemy bunkers from the flanks of the Forming Up Place upto the H hour. The fire base must be kept well forward.

(e) For the purpose of maintaining surprise, such weapons can stealthily move one/two days prior to the actual day of assault, to their firing positions to record all firing data of their targets, mark their positions and return to their localities. They can thus on the day of the assault quickly get into their firing positions and fire accurately without any wastage of time.

(2) ***Pre H Hour Bombardment***. This remains as in any conventional attack. However, it would require heavier calibre guns firing for a much longer duration.

(3) ***Covering Fire***. This will be provided by artillery in the usual manner. Commanders may consider reducing safety distances to provide additional covering fire at a stage when the troops are most vulnerable to enemy fire from the far bank. In addition, the weapons deployed in the fire base will also be used till the last possible moment as explained above. Signals between the assaulting troops and the weapons in the fire base must be very well coordinated so that fire from the fire base can be instantly stopped when the movement of own troops is endangered.

(4) ***Integral Fire Support during the Swim of Assaulting Troops Across the Water Gap***. This is discussed in detail in paragraph 23c below.

(5) ***Fire Support during Reorganization***. In this phase targets in depth of the enemy position are engaged by artillery and battalion mortars to give respite to the assaulting battalions from enemy interference during reorganization and to engage enemy counter-attack forces.

e. ***Some Useful Points in Fire Support***

(1) Forward observation officers must move with the assaulting troops for effective fire support and meeting any unforeseen situations. They must be practised in peacetime in swimming across the water gap with essential equipment.

(2) If enemy uses illumination devices, own artillery must also illuminate the area behind the high far bank to silhouette the enemy.

(3) Mortars must be used to engage the depth positions of the enemy.

(4) Enemy bunkers/pillboxes must be pinpointed during the reconnaissance and planning stage. Efforts must also be made during peacetime to acquire as much accurate information as possible through intelligence sources including aerial photography.

(5) At higher levels effective counter-bombardment must be organized to silence the enemy guns that are engaging own assaulting troops. This will ensure safety of the assaulting troops to a great extent, during crucial stages of the battle.

(6) Due to the linear disposition of targets, linear concentrations with the mean point of impact placed over the waterline would be more effective, especially for air bursts.

(7) Use of smoke can also be made to blind the enemy during the crucial stages of assault.

(8) Medium and heavy guns e.g. 8 inches Howitzer and 130 millimetres Guns are very effective in the assault fire role.

19. ***Deception***. Achievement of surprise in such operations is of prime importance. The actual place of the crossing must not be known to the enemy till the last possible moment. For this purpose following means can be employed:-

a. The troops clearing the enemy security zone must do so on a broad front.

b. Deploying all calibres of artillery in two or more possible directions of attack.

c. Assaulting troops to move to the Forming Up Place at night. As an alternative they could move some days before the attack so that enemy does not get to know their presence.

d. Diversionary attack-simultaneous, before or after the main attack, depending on the enemy situation.

**Techniques and Drills for the Assault**

20. ***Move to Forward Assembly Area***. The necessity of executing critical phases during the hours of darkness imposes the restriction to keep the H hour as early as possible after the last light. Consequently, the assault troops need to be moved as close to their Forming Up Places as tactically possible to maintain surprise. Following measures can be adopted in this regard:-

a. The brigades for bridgehead operation or the leading assaulting battalions must have forward assembly areas, where they should move the night previous to the assault, if possible.

b. The forward assembly areas should be kept well forward.

c. While the forward assembly areas are kept as far forward as possible the leading companies of the leading battalions should be moved before the last light, if possible. They move to the Forming Up Places stealthily and part of their move is covered by the pre H hour artillery/air tasks.

21. ***Technique of Passage through the Minefield and Negotiation of Water Obstacles***. Enemy minefield and wire obstacles pose serious problems of command and control for the attacker when the battalion is under direct fire of enemy weapons from the far bank. Various measures for ensuring smooth passage, lesser casualties, better command and control and better chances of success are discussed below:-

a. Near end of the minefield will be contacted by the holding troops all along the front/along major approaches. Thereafter an assaulting formation has three options:-

(1) ***Option A***. Cross the minefield on the first night, secure home bank and wait for the lanes to be ready in the minefield. Assault across the water obstacle, while F echelon and rafting/ bridging equipment move forward. Assaulting companies may swim across or use assault boats, brought forward through the gaps made in the minefield.

(2) ***Option B***. Treat the whole system as one obstacle and conduct one assault from near end of the minefield to the reduction of the far bank. In this case it would be difficult to carry the assault boats through the minefield. Therefore, the assault troops must carry the swimming aids with them to assault across the water obstacle.

(3) ***Option C***. Cross the minefield on first night, securing the home bank and assault across the waterline at or just after the

last light on the second night. During the first night the minefield would have been cleared, heavy equipment such as 85 mm assault guns, tanks and rafting equipment concentrated behind the home bank and fire base shifted on the home bank itself.

b. As already discussed the leading assaulting battalions/brigades should generally assault in two up formation. Since the main crux of the problem the capture of the far bank is the sub paragraphs below mainly pertain to the actions of the leading assaulting companies and battalions so that they do not have to make an opposed crossing and their battle is more or less conventional in nature.

(1) ***Extended Line Versus Single File/File Formation***. In order to minimise causalities due to mines it is recommended that the assaulting troops move in a single file or a file formation in preference to the conventional extended line formations for the following reasons:-

(a) ***Lesser Causalities due to Mines***. If men were to follow each other in a single file, only the leading man would be exposed to a mine, once on each strip, whereas when moving in an extended line each individual exposes himself more than once.

(b) ***Lesser Casualties due to Small Arms Fire***. Enfilade fire of the enemy weapons at night will not be as effective against the assaulting troops moving in a single file as it would be in case of an extended line.

(c) ***Command and Control***. The proximity of the troops to their commander and lesser number of casualties will help maintain effective command and control. There would be no confusion when passing through the gaps in the wire obstacles, swimming the water gap and when finally tackling the enemy bunkers.

(2) ***Three Up Versus Two Up Formation***. Considering the basic need of providing depth in attack and reaction capability to the commanders at all levels, it is suggested that subunits/ units and brigades should assault in two up formation, unless the situation demands otherwise. A diagrammatic layout of the suggested formation at the battalion level is given in Figure 24. However, at the platoon level, in view of the enemy

dispositions on the far bank, the three up formation is suggested as it offers following advantages:-

(a) ***Assault on a Broad Front***. All the three sections in a platoon, attack a group of three enemy bunkers, allowing six sections of the assaulting company to tackle a major portion of the objective simultaneously. The enemy is, thus, forced to divide his attention.

(b) ***Command and Control***. The move within the sections and platoons being compact, better command and control can be exercised.

(c) ***Ease of Passage through Existing Lanes/Gaps***. The entire platoon passes through the lanes in the minefield made by the leading elements of the platoon and the gaps in the wire obstacle blown by them thus saving time and effort.

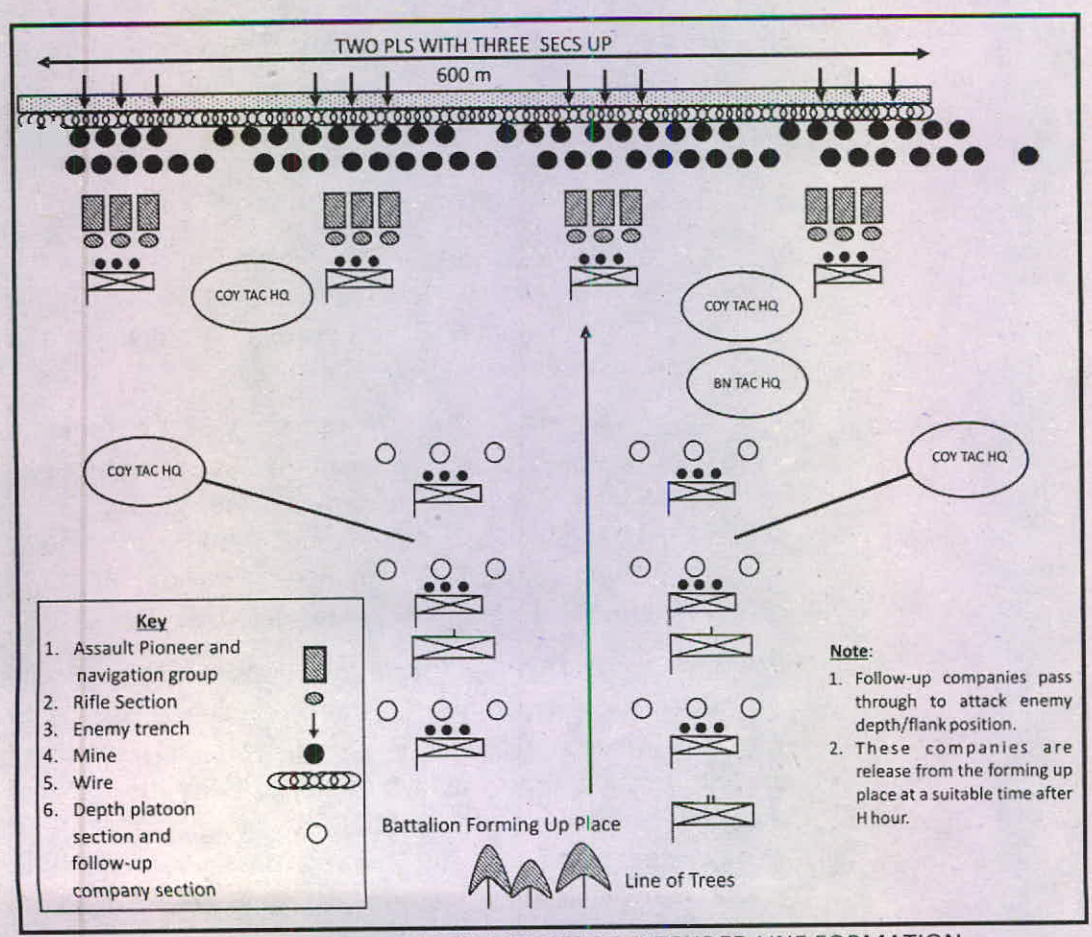



FIGURE 24 – DIAGRAMMATIC LAYOUT OF EXTENDED LINE FORMATION.

(d) ***Flexibility***. In the case of a failure of any one of the leading assaulting platoons, the platoons following in depth could be employed. The move of each section in file formation allows fire and move within the section and helps to overcome minor opposition en route.

(e) ***Speed***. Ensures enhanced speed and momentum in the assault.

(f) ***Exposure to Enemy Artillery and Small Arms Fire***. The assaulting troops are exposed to the artillery and enfilade small arms fire for a much shorter duration.

c. ***Forming Up for the Assault***. The Forming Up Places which ought to be directly square to the objectives, must be kept as far forward as is tactically possible. The assaulting companies and follow-up companies form up for the assault according to the formations suggested above. The marking and organization of the Forming Up Place would have to be suitably modified to conform to these formations. The time for move of troops in the various stages of the assault must be well regulated by the battalion commander to avoid congestion on the crossing site or premature involvement of the follow-up elements in the battle.

d. ***Employment of Assault Pioneer and Navigation Group***

(1) ***Necessity***. If the assaulting troops were to bear the brunt of casualties due to mines, cater for navigation, carry explosives for blasting gaps in the wire obstacles, also carry cumbersome expedients for crossing the water obstacle, they are bound to be tired and reduced in number. To ensure the desired combat ratio during the crucial stages of the assault, the employment of additional troops to perform the above tasks becomes inescapable.

(2) ***Provision of Personnel***. For this purpose, the assault pioneer platoon and persons from the field platoon engineers (under command for the operation), can be formed into an assault pioneer and navigation group consisting of four men who would move approximately 50 metres ahead of each assaulting section. In the absence of Field Platoon Engineers or in the event of engagement of the Assault Pioneer Platoon on some other task, these men would have to be drawn from the reserve battalion or one of the follow-up companies of the battalion. Also refer to paragraph 28 of Section 63.

(3) ***Tasks***. This group will be given following tasks:-

(a) Navigation from the Forming Up Place to the objective.

(b) Marking a path through the minefield with the help of a tape/rope for the assaulting/follow-up troops.

(c) Blowing a gap for each section through the enemy wire obstacles.

(d) Ultimately deploying on the home bank to give intimate fire support to the assaulting troops during water crossing.

e. ***Passage through the Minefield***. The assaulting platoons of the two leading assault companies move off from the Forming Up Place with three sections up, under the fire support provided by artillery and tanks/recoilless rifles from the fire bases located on the flanks. Each section moves in a file formation capable of resorting to fire and move when forced. An Assault Pioneer and Navigation Group consisting of four men moving under the direct command of section commander lead each section. These groups provide navigation and mark a path with a white tape/rope for the section personnel following them. Both the Assault Pioneers and Navigation Group and the assaulting sections carry appropriate equipment/explosives/ expedients required for the assault. In the event of any one becoming a casualty, the person following, picks up the expedient and is ready to perform the same task.

f. ***Negotiating the Wire Obstacle***. The approach of the Assault Pioneer and Navigation Groups upto wire obstacle should be so regulated that they contact the wire at one time, thereby enabling them to blow a gap for each assaulting section in the wire obstacle simultaneously. While these gaps are being blown, it must be ensured that the personnel remain at a safe distance. In case any one of the Assault Pioneer and Navigation Groups cannot blow a gap, the assaulting sections negotiate the wire obstacle with the help of expedients i.e. a pole ladder etcetera carried for this purpose. After performing this task they dash forward to the near bank and using all available cover, deploy themselves to provide intimate fire support to the assaulting sections when they are swimming the water gap.

22. ***Swimming the Water Gap***

a. ***Water Obstacle Crossing Plan***. The crossing plan should provide for movement of all elements during the crossing of the obstacles. This

includes ordinary infantrymen who have to swim across, using improvised rafts and other expedients. The crossing sites are suitably marked and all elements, assaulting personnel, weapon section, stores and personnel etcetera are included in the move tables. Techniques of crossing a water gap include when, where and on whose orders are the tubes to be inflated or the jackets worn. Whether they are to be discarded, if so, where and who picks them up. Similarly who carries the jackets to make light rafts for the company stores and weapons and who uses and retrieves them. These minor points must be practised and laid down in the battalion drills.

b. ***Technique of Swimming the Water Gap***. From the edge of the minefield, the assaulting sections, rush through the gaps in the wire obstacle, already blown by the assault pioneers and navigation groups. At this stage speed ensures minimum possible exposure to enemy fire. In case of flowing water, entry must be made some distance upstream to cater for the drift. The drift is roughly twice the width of the water obstacle. Each section enters the water a few yards upstream of its objective bunker/trench. Personnel in each of the assaulting sections must swim with their respective Light- Machine gun groups under the effective control of their section commanders/seconds-in-command. Men should swim together and not stray off.

c. ***Expedients Used for Crossing the Water Gap***

(1) ***Leading Assaulting Platoons***. The two leading assaulting platoons of the leading assaulting companies carry the minimum essential and light expedients. The main distribution is as follows:-

(a) A swimming pillow per individual.

(b) Four additional pillows per section (one for each Light-Machine gun group and one for crossing the wire obstacles).

(c) Three bamboo pole ladders per section (one for each Light-Machine gun group and one for crossing the wire obstacles).

(2) ***Follow-Up Platoon***. Since the follow-up platoons would not be making an opposed crossing they can carry following expedients:-

(a) A swimming pillow per individual.

(b) Four additional reserve pillows per section as above.

(c) One rope bamboo ladder per section.

(d) One set of ten iron bar 'US' per platoon or one set of ten single iron bars per platoon.

(3) ***Fording***. In case of small ditches with less that 4 feet of water, it should be possible for an infantryman to wade or jump across. Vehicles will need a small bridge span or an expedient to get them across.

(4) ***Light Rafting***. Rifle company stores and the weapons of weapons section would have to cross either by means of improvised rafts prepared with the help of swimming pillows, tubes and jackets and infantry support rafts provided by the formation headquarters. Only minimum essential rafts should be constructed.

(5) ***Recycling of Expedients***. In case small water obstacles occur at short distances, it may be necessary for the follow-up subunits to collect rafting equipment and carry it forward for the next phase, when follow-up units change their role into assault units.

d. ***Integral Fire Support during Swimming Across***. The fire support, organized for such operation as for a conventional attack and is lifted when the assaulting troops are close to the enemy positions, in view of their safety. At this crucial stage the assaulting troops are the most vulnerable to flat trajectory weapons from the far bank. Hence, there is a need to organize intimate integral fire support from the home bank. For this purpose, in addition to the fire support being provided by the assault pioneers and navigation groups, two additional Light-Machine gun groups and one rocket launcher team should also be deployed on the home bank. From within the each assaulting platoon. These weapons provide fire support till the leading assaulting troops touch the far bank. Thereafter, these weapons teams swim across and join their respective sections for the final assault on enemy bunkers. The commencement and stopping of this fire support must be meticulously coordinated beforehand.

e. In certain situations it may be possible to establish a fire base as under:-

(1) ***Initial Location***. 200 to 500 metres short of the near end of the minefield between the gap of the assembly areas of the

assault brigades. More range for the Anti-Tank Guided Missiles.

(2) ***Composition***. Tanks, Artillery, Anti-Tank Guided Missiles, Grenade Launcher Heavy-Machine guns, Recoilless Rifles, Mortars and 85 millimetres Cannons in direct fire support role.

(3) ***Time***. Each weapon/weapon system in the fire base is brought in position a night earlier. Range cards, data in respect of each assigned target is worked out and applied during daytime. Fire programme is then integrated with the overall Fire Support Plan to support the assaulting companies upto the home bank.

(4) ***Subsequent Location***. After the lanes are ready in the minefield and before the assault is launched across the water channel, the fire base will be shifted to the home bank to engage enemy bunkers on the far bank. Maximum illumination will be arranged to aid the fire base.

23. ***Scaling the High Far Bank and Bunker Clearance***. Following drills are suggested:-

a. On reaching the far bank, each group of the assaulting sections can use any of the following expedients to negotiate the far bank:-

(1) Single bamboo pole ladder.

(2) Rope ladder.

(3) Iron bar U steps.

(4) Single iron bar steps.

(5) Grapnels.

b. ***Bunker Clearance Drill***

(1) ***Neutralization of Enemy Bunkers***. In each of the assaulting sections, the group climbing nearest to the objective bunker/ trench, rushes up the slope, firing at the loopholes of the enemy bunkers and lobs in hand/smoke grenades into these. This group then moves beyond the bund to shoot at any escaping enemy or to stop any enemy reaction.

(2) ***Clearing the Enemy Bunkers***. The second group also rushes up the slope and moving behind the bund clears the bunkers by approaching it from the rear. Rocket launchers and Energa grenades can be used to blow off the steel door of the bunker.

To avoid cross fire within sections all bunkers/trenches should be tackled from the same direction.

24. ***Reorganization on the Far Bank***. In a bid to recapture a lost portion on the bund, the enemy will invariably to launch a quick and violent counter-attack, supported by armour. In order to forestall such a counter-attack, the reorganization on the objective must be quick and effective. The drills for reorganization for both the assaulting and the follow-up companies will vary with the existing tactical situation, terrain conditions and dispositions of the enemy. Following are a few guidelines:-

a. The most expected direction of a counter-attack is either from the front or a flank. The commanders must, therefore, deploy their subunits so as to be able to beat back such enemy counter-attacks.

b. The captured portion of the bund must be physically occupied but only by an adequate number of troops, while the remaining troops should deploy on a dominating ground close to the bund, covering the likely approaches.

c. Subunits located on a flank of the captured bund must deploy some elements to cover the exposed flanks.

d. Rocket launchers and Energa grenades, which are the only anti-tank weapons taken across with the assaulting and follow-up companies, should be so located as to be effective against an immediate tank threat. Additional rocket launchers would have to be given to the forward assaulting troops to offset the absence of the recoilless rifles and Baktar Shikans, which should be carried across the water obstacle as soon as the lanes have been made through the minefield. Carriage of these during the initial assault is not possible.

e. At least some mines should be laid on all the roads/tracks leading into the bridgehead area from the enemy sides, before the first light. This must be pre-planned and the necessary mines etcetera should be taken across, after the lanes have been made through the minefield. While laying such mines the ultimate routes for the break-out must be kept in view. The mines would be laid by the assaulting infantry battalions.

25. ***Move of Follow-Up Companies***. Follow-up platoons of the leading and follow-up companies of the assaulting battalion move off from the Forming Up Place at the appointed time. They walk along the tapes already laid in the minefield, pass through the troops on the captured far bank and proceed to execute the tasks given to them, as generally done in a conventional attack.

26. ***Replenishment of Ammunition and Carriage Across of Heavy Weapons/ Stores***

a. To maintain a favourable combat ratio and to keep the assaulting troops them light and fresh during the final stages following aspects need due consideration:-

(1) ***Replenishment of Ammunition to the Leading Assault Companies***. Keeping in view the exhausting combat action involved in such assault, the assaulting troops carry only their pouch ammunition with a view to travelling light. Due to the high rate of ammunition expenditure, the leading assaulting troops need to be promptly replenished after capture of the far bank, so as to be able to face a usually quick and violent enemy counter-attack.

(2) ***Carriage of Heavy Weapons/Stores***. After the battalion assault, there would also be need to carry across heavy weapons like Anti-Tank Guided Missiles, 12.7 millimetres Anti-Aircraft guns, Grenade Launcher Heavy-Machine guns, Recoilless Rifles, 81 millimetres Mortars and other stores required to effectively reorganize on the far bank.

b. Keeping in view the above, there would be a definite requirement of additional troops to carry-out the above tasks. It is, therefore, suggested that one of the companies of an assaulting battalion should be employed to meet these requirements. This company could also provide additional personnel required for the Assault Pioneer and Navigation Groups as discussed above.

c. The employment of a rifle company for the above mentioned tasks, however, could temporarily reduce the size of the bridgehead area assigned to the battalion as an objective.

27. ***Carriage Across of Heavy Weapons/Stores and Subsequent Actions of the Battalion***

a. Once the leading assaulting battalions have succeeded in gaining a foothold, the follow-up battalion of the bridgehead brigade would pass through to expand and consolidate the bridgehead. Meanwhile the engineers must move up to make the required number of lanes in the minefield and to construct rafts and bridges with utmost speed, so that armour and other anti-tank/aircraft weapons could move into the bridgehead as quickly as possible.

b. The assaulting infantry battalions would have to wait for the construction of lanes, for carriage across of its heavy equipment and stores. These can be brought to the waterline in vehicles and ferried across with improvised expedients. Essential vehicles of the battalion will be brought across after the construction of rafts/bridges, before the first light, if possible.

c. Plans for the above will be coordinated at the brigade level.

28. ***Use of Trawl Tanks for Mine Breaching***. Trawl tanks have following characteristics:-

a. One trawl tank moving singly clears a 2 metres wide lane.

b. A pair moving 0.6 metres apart clears a of 4.3 metres wide lane.

c. The operating speed is 8 kilometres per hour. It would thus take a pair of trawl tanks about 15 minutes to clear a lane of 91 metres deep and 4.3 metres wide through a minefield.

29. The trawl tank would considerably shorten the time for breaching lanes through the minefield and thus hasten the build up in the bridgehead. Besides, if it is found practicable to make the lanes even before the leading assaulting battalion are launched it would be done after some of the techniques explained above have been suitably modified. Additional technical data on the trawl tank, bridging/rafting equipment and minefield breaching is given at Annex Q.

30. ***Use of Paratroops***. Para and helicopter borne troops can be usefully employed when available for following tasks:-

a. Capturing a part of the bund by dropping close behind the far bank. This gap could later be exploited by the bridgehead battalions.

b. Capturing bridges/bridge sites intact.

c. Attacking enemy gun positions at a time when own assault is being launched.

d. Raiding enemy armour concentrations in the vicinity of the crossing sites.

e. Raiding enemy headquarters and disrupting his communications.

31. ***Training of Assaulting Troops***. This operation can be undertaken only by seasoned troops who have been thoroughly trained. Every individual must know his task and the troops must be thoroughly rehearsed in the drills to be followed at the Forming Up Place, through the minefield and wire obstacles, during crossing of the water gap, in scaling the high far bank and while clearing bunkers etcetera. Physical fitness, mental robustness alertness and motivation of a very high standard would ensure success in such operations.

## SECTION 48 – FIGHTING IN BUILT-UP AREAS

**General**

1. Fighting in built-up areas imposes additional problems for reasons of marked deviation in the layout of terrain. Major characteristics of operations in the built-up areas are, difficult command and control, lack of fire support of major weapons, abundance of cover and concealment, limited fields of fire and heavy dependence on and extensive use of Close Quarter Battle weapons.

**Factors**

2. Following factors affect operations in the BUAs:-

a. ***Cover***. Although in old cities or towns there is no regular or identical pattern of the lanes and streets. The buildings and houses provide cover both from view and fire. In a rigid pattern, however, even the slightest movement is attracted.

b. ***Movement***

(1) Movement is very slow and restricted. Although buildings can be bypassed, yet troops have to fight successively from building to building.

(2) Because of regular pattern, it is easy to detect movement but difficult to locate fire.

c. ***Command and Control***. Demands greater initiative and control by junior leaders. In most cases platoons and sections have to operate independently and act on their own.

d. ***Fire Support***. Tanks and artillery fire may not be very effective because of closeness of area. However, the direct support provided by the tanks can be of immense assistance in wider streets. The tanks can also be used for covering of the killing area. While tanks are employed for fighting in built-up area, they may have to be distributed by troops. Hence greater reliance has to be placed on infantry battalion's integral weapons, particularly anti-tanks, machine-guns and close range weapons.

e. ***Manpower***. Comparatively requires much greater forces both for the offensive and defensive operations. In defence, each and every house may have to be guarded. In attack, once a lane or a house has been cleared, men have to be posted to avoid its reoccupation.

f. ***Darkness***. Fighting in built-up areas may preferably be carried out at night but unless there is adequate moonlight, it will further aggravate command and control problems.

g. ***Civilians***

(1) The problem of dealing with civilians will almost always arise whether the inhabitants are friendly, indifferent or openly hostile. In the first instance, full use must be made of the civilian cooperation and assistance and in the last, adequate control and supervision must be exercised.

(2) In defence, before total evacuation of civilians is ordered, it must be remembered that some important public services such as medical, food and other utilities considered useful for the defence operation will cease. Any civilians not evacuated have to be fed and accommodated.

(3) In the attack, non-essential civilians should be evacuated as soon as possible from the important areas that have been captured.

(4) In all cases, adequate measures must be taken for security against activities of spies, fifth columnists and saboteurs. The staff must provide adequate and qualified personnel for the supervision of local authorities and interrogation of the doubtful civilians. It may be necessary to issue new passes or identification papers to the civilians. If civilians are not evacuated, they should at least be registered/segregated.

## SECTION 49 – ATTACK IN BUILT-UP AREAS

**General**

1. The general pattern of attack in a big town or small village may be the same; yet, each may require some variation in the technique. The major reasons are:-

a. Clearing of a big town is more complex and slower.

b. While attacking a small village it will be easier to cut and block all escape routes.

c. In case of big cities outflanking may be accepted but it would demand clearance of the town subsequently.

**Planning of Attack**

2. ***Basic Consideration***

a. ***Detailed Information***. To mount successful attack, detailed information about the city/towns is essential. Following can be the sources of information:-

(1) Air photograph coverage.

(2) Maps.

(3) Town plans.

(4) Tourist guides.

(5) Civilians.

(6) Stay behind agents.

b. ***Simplicity***. The junior leaders should be given greater latitude and therefore, all plans must be simple to afford flexibility.

c. ***Objective***. Because of uncertainty of enemy and restriction on observation the objectives allotted to subunits should be limited. Once a sector is allotted to a battalion it should be further subdivided into subsectors. This division can be made by using lanes and streets.

d. ***Deception***. To divide and scatter enemy effort the initial bridgehead should be on wide front to mislead the enemy about the direction of actual assault.

e. ***Reserves***. Unpredictability of the operation and possibility of heavy casualties necessitate maintenance of adequate reserves at all levels.

3. Planning of operation must be based on accurate information and should cover all the details. Normal attack in built-up area is carried out in following three phases. An infantry battalion within the overall brigade or divisional plan, may be involved in any of the three phases:-

a. Gaining of foot hold in the built-up area.

b. Systematic progress within the area.

c. Reduction of individual centres of resistance.

4. ***Gaining of Foot Hold***. This is like an ordinary operation. Its execution may involve crossing of water obstacle or a minefield. Some considerations regarding this are as follows:-

a. Maximum fire support.

b. Unless otherwise inevitable, operation be done at night.

c. Operation to cover a wider frontage to deceive the enemy.

5. ***Progress within the Area***. Methods by which this phase can be carried out are as follows:-

a. Frontal attack in conjunction with development on the flanks. Troops advance on a continuous front clearing all buildings and blocking any possible escape routes from the area. This method requires a large

number of troops and is usually practical only in a small town, where opposition is light.

b. Penetration on two or more narrow fronts on covering axis. This method is usually quick, but risky against an aggressive defence. It should only be used when the defence is hastily organized.

c. Attack on narrow axis with a view to cutting off part of the defence, which is then cleared before any further penetration is made. This is a slow but systematic method and probably the best against a very strong defence.

d. Attack on a broad front systematically clearing only one small area at is a time. This is modification of 'a' above. It needs fewer troops, is easier to organize and control and is effective against a strong defence. It is, however, slow.

**Conduct of Attack**

6. BUA provides good observation and fields of fire. The strong points effectively dominate and cover the killing areas, especially to their front. The positions are strengthened with the help of minefields, barbed wire and ditches, which make day attack very difficult. The frontal assault proves to be costly, therefore, the attack should be launched from the flanks or rear and that too during the hours of darkness.

7. The troops will be assaulting through the minefield. It is therefore, preferable to make lanes secretly in the minefield the same night or a night earlier. The assaulting troops can pass through these lanes in small groups.

8. The depth of the minefield and the number of lanes required would determine as to how much time will be taken by the engineers to clear these lanes. The F echelon transport of the battalion will be able to join the companies on the objectives after the lanes have been made in the minefield. Clearing the streets and passing of 'F' echelon transport through them is time consuming. It is therefore, suggested that the attack should be launched soon after the dusk so that sufficient time is available for the above actions.

9. Minefields are kept under effective small arms and artillery fire, especially when good observation and fields of fire are available. The weapons sited on the outer edge of the BUA can interfere in the mine breaching operation unless they are neutralized in the initial stages. It is therefore, necessary that all the enemy weapons which are likely to interfere in the breaching operation are destroyed/neutralized before the engineers parties commence making the lanes in the minefields.

10. Heavy weapons are sited close to the minefields to neutralize enemy weapons and positions at the outer periphery of the built-up area, before the lanes are prepared.

11. In the beginning, enemy troops are normally deployed on the periphery of the BUA. Therefore, the plan of attack including direction and stage of assault, fire support etcetera is made keeping in view the type of the perimeter. However, it does not mean that the inner parts of the built-up area are ignored even if they are not initially occupied by the enemy. Once, the buildings on the periphery, especially the ones on the flanks, are captured, the enemy position as a whole becomes untenable.

12. Due to the dominating enemy positions, further strengthened by the ditches, wire obstacles and protective minefields etcetera, it will not be possible for the troops to move to their objective in a group or in a proper assault formation. Therefore, the attack should proceed in section and platoons groups, following fire and movement. Rocket launchers and Energa grenades are used to make holes, routes and to demolish buildings.

13. Fire support plan should be prepared with lot of deliberation incorporating the views of the company commanders and the artillery observers. Contrary to the normal circumstances more accurate fire is required in an attack on the built-up area. Due to longer distances from the forming up area to the objective, artillery fire support will be required for a relatively longer duration.

14. To deal with the enemy's defensive positions, especially in the last stages of the attack, following explosives and other demolition materials will be required:-

| Serial No. | Explosives/Demolitions | Purpose |
|---|---|---|
| a. | Pole charges | For the destruction of strong points, houses and pillboxes. |
| b. | Shaped charges | |
| c. | Limpet | For the destruction of tanks. |
| d. | Bangalore Torpedo | For destruction of buildings and for making passages in the wire obstacles. |

15. ***Level of Attacks***. All attacks in built-up areas evolve themselves into a series of independent platoon operations, each against one house (or part of house). For each attack, a platoon requires three groups, probably section for the following tasks:-

a. To clear the house (or part of it).

b. To form a firm base to support and protect the house clearing group.

c. To provide a reserve to meet the unexpected situations. A platoon can, if the opposition is light, deal with the clearing of a whole street. However, in order to avoid fighting through the street, it is normal for the platoons to fight their way forward through the blocks using rooftops, sewers and the back gardens and by 'mouse holing' from house to house. A platoon frontage should generally not exceed 100 metres and if possible should preferably be confined to the width of one house.

16. ***Streets***. Any enemy expelled by platoons from the houses should be destroyed by low trajectory weapons sited to fire down/through the streets. Thus, streets are used both as boundaries between platoons and sections and also as killing areas for our own weapons. Later, they can be opened up as an axis for unit supplies. Against a weak enemy they may be used as the actual thrust lines for attack.

17. ***Covering Fire***. If it becomes necessary to cross open spaces, adequate covering fire, must be arranged. No section should ever move into the open until other troops are in position to give covering fire. There must always be 'one leg on the ground'.

18. ***Thoroughness in House Clearing***. Houses and outbuildings must be thoroughly searched, including the cellars and roofs. Furniture likely to conceal 'mouse holes' must be removed.

19. ***Firm Base and Reorganization***. Every house must be consolidated after its capture and put immediately to a state of all-round defence with snipers duly posted. The house thus becomes a firm base. This is vital because:-

a. Fresh attacks in built-up areas will only succeed if they are launched from firm bases.

b. Unless reorganization is thorough and each house and bound is properly secured made firm as the attack goes forward, the attack will be badly balanced to meet a counter-attack and the enemy may infiltrate back into the areas earlier.

**Administration**

20. It will be difficult for the normal system of supply to function in operations of this nature. Hard scale rations should be issued and possibly two water bottles to each man, as it is unlikely that water vehicles will be able to get forward. The battalion 'F' echelon transport may have to carry forward stores which are normally carried in 'A' and 'B' echelons. The main problems are as follows:-

a. ***Ammunition***. Dumps must be formed behind the assaulting troops as they advance, ammunition being delivered by transport or tanks to

company areas and then carried forward by previously arranged parties. Sandbags are very useful for this purpose. Large quantities of grenades will normally be required. Ammunition must be replaced after each bound.

b. ***Equipment***. Small packs, if worn and used for carrying grenades, should be slung and kept in place by a belt. Leather jerkins and toggle ropes are useful. Smoke grenades, owing to their toxic effect, are very effective for driving enemy out of cellars.

c. ***Casualties***. Casualties should be left in the houses. Stretcher bearers must always follow-up assaulting troops and evacuate casualties to collecting posts. All troops should carry shell dressings as well as field dressings.

**Communications**

21. Telephone lines will be difficult to maintain owing to the danger of falling buildings. They must be laid along routes to which easy access is possible. In towns it may be impossible to use the civil telephone system which may be under ground. Wireless may be interfered with both by the proximity of buildings, the presence of steel in their construction and power lines.

## SECTION 50 – BATTALION WEAPONS USED IN FIGHTING IN BUILT-UP AREA

**Characteristics**

1. Characteristics of some of the battalion weapons when used in fighting in built-up areas summarized below:-

a. ***Rifle and Bayonet***. The rifle in competent hands is an immensely valuable weapon. Fighting in built-up areas offers unrivalled opportunities for sniping which, in an enclosed area, can be itself bewilder and almost parleys the enemy. As opposing sides will be literally at grips with each other, the bayonet is most effective.

b. ***Hand Grenade***. Large numbers of these are required. Though hand grenades are lethal against unprotected men in rooms, cellars, etcetera, the delayed fuse may enable them to take cover behind furniture.

c. ***Light-Machine Gun (LMG)***. These are the main supporting weapons in attack. They are invaluable for covering streets and countering enemy covering fire. In defence they will be used at low levels to cover the flanks of the buildings held and the streets.

d. ***Sub-Machine Gun (SMG)***. This is valuable for close quarter fighting and for dealing with fleeting targets. However, it has little power of penetration through the doors and can not breaks open the locks.

e. ***Baktar Shikan/Recoilless Rifle***. In attack they can be used as follows:-

(1) To break into the houses.

(2) To fire through the windows and doors before an attack, particularly to clear the upper storey.

(3) To fire at the corner houses behind which enemy machine-guns and recoilless rifles are sited.

(4) To breakdown heaps of rubble on the roadways, to destroy enemy machine-gun posts and to scatter/bury the mines covering those posts.

(5) In defence their task is to destroy enemy tanks. They should be sited in defiladed positions to cover streets, barricades and open spaces

f. ***Rocket Launchers and Energa Grenades***. When fired through a window, these weapons have a considerable lethal and demoralizing effect. If fired at the corner of a normal brick house, they will make a hole sufficiently large for a man to climb through. These can also be effectively used against enemy tanks and in anti-personnel role.

g. ***81 Millimetres/82 Millimetres Mortars***. They are best used against enemy reserves and likely Forming Up Places. However, they can only penetrate the roofs which are smashed or badly built.

h. ***60 Millimetres Mortar***. It has considerable lethal effect if fired at a low angle through the windows. It can be used against snipers on the roofs or fired at high angles against possible enemy Forming Up Place (s) or communications in neighbouring streets. It is very useful for smoking tanks.

i. ***Machine-Guns***. These are valuable for sealing off the sectors by shooting down the streets to prevent movement of enemy reserves in the battle area.

j. ***Man Pack Flame Throwers***. Flame Thrower is a portable weapon comprising three fuel tanks fitted with a harness and a firing gun. On firing, the weapon emits a flame of fire which has a maximum range of 50 metres (54.66 yards). The weapon can have a considerable demoralizing effect on the enemy. It can be effectively used through

the loopholes of bunkers and doors/windows of the buildings to cause casualties to the enemy troops and to set fire to the material inside.

k. ***Grenade Launcher Heavy-Machine Gun (GLHMG)***. These are very effective against men and soft protections. These can be used to block the streets and to lob the grenades inside the buildings. These are very useful against the targets behind the cover, at comparatively longer ranges.

## SECTION 51 – EMPLOYMENT OF SUPPORTING ARMS DURING ATTACK IN BUILT-UP AREA

**Armour**

1. Armour may be used in three principal ways:-

a. ***Under Command***. Even a limited number of tanks decentralized to the companies and working behind them can be invaluable. However, their limitation i.e. restricted mobility and the risk of their destruction by close range weapons, should constantly be borne in mind.

b. ***Fire Support***. Some tanks firing direct into the houses from the flanks can be invaluable in assisting troops to clear small towns and villages. If communications are good, they can engage houses immediately in front of the leading infantry.

c. ***Cutting Off Withdrawal***. This role is an ideal task for the tanks in cooperation with infantry.

**Artillery**

2. After the initial break-in, artillery support is limited by observation, the trajectory of the guns and proximity of our own troops to the enemy.

3. The heavier type of artillery may often be employed, sited singly or in pairs to destroy individual buildings or strong points.

4. Pack artillery may be normally managed to the upper storeys or into the courtyards and used in the strong points. Recoilless rifles may also be of help in the reduction of strong points. A comprehensive fire support plan will be prepared by taking into account battalion weapons as well.

**Engineers**

5. In addition to normal tasks, engineers will have considerable call for offensive demolition, work inside the houses, making internal communications and strengthening the cellars. There may also be a heavy demand for mine clearance to facilitate the forward movement of the vehicles.

6. Majority of engineers should be decentralized, on occasions, even down to the rifle companies.

**Signals**

7. Very High Frequency (VHF) wireless equipment is liable to interruption due to screening effect in the built-up areas. Care must therefore be taken while siting these sets with a view to obtaining quasi-optical paths between them, for example, on rooftop. Line communication is also liable to interruption due to falling and burning buildings causing damage to cables. Under the circumstances the most reliable means of communication among forward troops in the BUAs is the runner.

**Air**

8. ***Heavy Bombing***. Whether the heavy bombers, if available, should be used or not, will depend partly on the buildings in the built-up area. If these are solidly constructed and equipped with cellars, heavy bombing is unlikely to help the attacking force. Its effects would generally be to bring large quantities of rubble down on the cellars, which in turn would increase their overhead cover and defence potentialities. If on the other hand, the buildings are of a weaker structure without cellars, heavy bombing will help considerably, especially, if closely followed up by the assaulting troops. An incendiary raid might be the most effective way of clearing a weakly flimsily constructed area.

9. ***Rockets***. May be of considerable use, especially, if employed on such tasks as harassment and possible destruction of the enemy headquarters or the softening of strong points when identified from the air.

10. ***Provision of Air Photographs***. This is one of the chief ways in which the air can help in this type of fighting. Detailed maps and plans of built-up areas are rarely available and it is essential that these, or air photographs, should be provided in the form of large scale mosaics which should be issued down to company commanders and, if possible, even to section and tank commanders. During a protracted operation of this nature it will be of great value if further photo coverage can be provided to show the area still to be cleared. Low oblique photographs are may valuable for locating enemy positions.

11. ***Prevention of Enemy Reinforcements***. Air can also greatly assist by sealing off enemy approaches to the town and by cutting his reinforcements and supplies.

## SECTION 52 – QUICK VERBAL ORDERS ON WIRELESS FOR ATTACK BY AN INFANTRY BATTALION

**General**

1. During offensive operations there will be many occasions when, due to the fleeting nature of opportunities, it may not be possible to adhere to the normal

battle procedure, particularly, to have the 'O' Group handy at all times for the issue of orders. Under such circumstances, it may well be necessary for a commanding officer to issue orders on wireless for a quick attack, to exploit a fleeting situation. If necessary these orders can subsequently be confirmed through other means. These situations most likely arise under following conditions:-

a. In an encounter battle, especially in pursuit, when the attacking troops are mobile.

b. During a set piece attack, when a unit is required to continue beyond the objective already given and quick orders have to be issued for a fresh attack. This is particularly so when the subunit commanders are away with their respective subunits.

c. During counter-attack when the fluid nature of the battle may require quick employment and reemployment of subunits in various offensive tasks in quick succession, with few pauses in between.

2. It is imperative to lay down standard procedure and codes for the issue of verbal orders for attack. Specimen is as at Annex R.

## CHAPTER XV – DEFENCE

### SECTION 53 – GENERAL

**Introduction**

1. Victory in war can be won by attack only. The defence is a temporary measure to gain time for the build up of sufficient strength, to allow the offensive to be launched or resumed, or to hold the enemy at one point while attacking the other. Since it is an attack halted; an aggressive spirit must be maintained throughout this phase. This should be studied in conjunction with 'GSP-1985– Operations of War (Volume-I), Defence 2004'.

**Definitions**

2. Important terms used in defence are:-

a. ***Defended Area***. An area of operational responsibility containing a group of mutually supporting localities organised in depth. Any enemy, who penetrates into it or passes around the flank, will be engaged by fire of all weapons that can be brought to bear on him.

b. ***Defended Locality***. It is an area of ground organized for all-round defence. General line of the foremost localities will be known as the line of "Forward Defended Localities (FDLs)". Enemy penetrations will not be accepted into a locality. Enemy that does penetrate will be attacked by fire, local counter-attack or by a deliberate counter-attack, ordered by the higher commander.

c. ***Defended Post***. A position held by a small subunit weapons/group of weapon such as an infantry section or platoon. Posts are grouped together in mutual support to form a defended locality.

d. ***Counter Penetration***. The action taken by a defender to halt penetration of his defence by the enemy and to destroy the enemy forces that have penetrated between his defended localities by fire.

e. ***Counter-Attack***. Attack by a part or all of a defending force against an enemy attacking force, for such specific purpose as regaining lost ground or cutting off or destroying enemy advance units and with the general objective of denying to the enemy the attainment of his purpose in attacking. In sustained defensive operations, it is undertaken to restore the battle position and is directed at limited objectives.

f. ***Local Counter-Attack***. An attack organized by the local commander to destroy enemy forces which have penetrated or threaten to penetrate his defensive position, generally using a prearranged plan. This may be organized at battalion or even company and platoon level.

g. ***Area of Responsibility***. A defined area of land in which responsibility is specifically assigned to the commander of an area for the development and maintenance of installations, control of movement and the conduct of tactical operations involving troops under his control along with parallel authority to exercise these functions.

h. ***Listening Posts***. A listening post normally consists of two to four men and is mounted during night on likely enemy approaches to give early warning of the enemy movement. They should have cover, concealment and good routes of withdrawal. Due to strenuous nature of their task, listening posts should be relieved as often as possible. If frequent relief is not possible then double sentries are posted to ensure proper vigilance and prevent surprise.

i. ***Defensive Fire (DF)***. Prearranged fire which can be quickly brought down on actual or suspected enemy movement. It will include:-

(1) Tasks closer to the defended localities to break-up enemy attack.

(2) Tasks in depth to disorganise enemy preparations just before and during his attack.

j. ***Defensive Fire Tasks (SOS)***. Defensive fire (SOS) tasks cover the most vulnerable approaches, usually close to the position. The guns and

mortars will remain laid on their defensive fire (SOS) tasks when not otherwise engaged.

k. ***Harbour Party***. In defence this party is in charge of laying out an assembly area or a labour and guiding the main body into it. It is normally commanded by Officer Commanding (OC) Combat Support Company and comprises at least one or more guide per rifle platoon.

l. ***Positions***

(1) ***Alternative Position***. A reconnoitred and prepared position to which a unit/subunit or weapons may move to carry-out the same tasks as from the main position.

(2) ***Secondary Position***. A previously prepared position to which a unit/subunit or guns/mortars may have to more carry-out a different task.

(3) ***Temporary Position***. An area of ground or weapon emplacement, occupied by a unit, subunit or weapon for a period, to deceive the enemy as to its real location and/or enable a particular task of a temporary nature to be carried out.

**Aim of Defensive Operations**

3. The aim of defensive operations is to hold off enemy whilst preparing a counter-offensive by which to defeat him. Defence is, therefore, the prelude to offensive operations and must be regarded as a temporary phase with a view to:-

a. Develop favourable conditions for attack.

b. Economise forces in one area in order to apply decisive force elsewhere.

c. Destroy hostile force.

d. Reduce the enemy capacity for offensive action.

e. Deny the enemy entry into an area.

4. All defence must be aggressive to retain the initiative and harass the enemy. Aggressiveness is also necessary to maintain morale.

**Classification of Defence**

5. ***According to Technique***

a. ***Area Defence***. A type of defence where the bulk of the defending force is deployed in selected tactical localities with the task of preventing enemy penetration and destroying his forces, in front of

the position by fire, by absorbing the weight of attack within the position and ultimately destroying him through counter-attack.

b. ***Mobile Defence***. In this primary reliance is placed on the use of strong mobile elements and fire for offensive actions. In mobile defence minimum forces are employed for physical defence around noodle points (may be strong points/fortifications) in the forward areas and strong mobile reserves are held in the rear. The decisive battle in this case is fought within the defended area accepting a temporary loss of some of the ground. The enemy is so effectively contained within the defended area that his destruction is facilitated by heavy concentrated fire and launching of reserves.

6. ***According to the Degree of Preparation***

a. ***Deliberate Defence***. It is normally organized when out of contact with the enemy or when contact with the enemy is not imminent and sufficient time for development of defence is available.

b. ***Hasty Defence***. Normally organized while in contact with the enemy or when contact is imminent and time available for development of defence is limited.

**Fundamentals**

7. In defence the following fundamentals must be considered:-

a. ***Enemy***. The battalion commander must first assess the nature of the enemy threat, his strength, composition and disposition; his capabilities and intentions.

b. ***Terrain***. A sound terrain analysis is the next requirement in the concept of a defensive battle to arrive at the following:-

(1) Likely enemy approaches.

(2) Grounds of tactical importance.

(3) Observation and fields of fire.

(4) Cover and concealment.

(5) Obstacles natural/artificial.

(6) Killing ground.

c. ***Frontages***. The frontage must be related to the type of terrain, the extent and nature of enemy threat and resources available to the battalion commander. In open terrain these would tend to be more extended than in the mountainous and jungle type of terrain. In the plains of Punjab a battalion will normally be required to occupy a

frontage of 3000-5000 metres. In all situations emphasis must be laid on holding the ground by fire and not by men. This will necessitate gaps between the defended localities which must be covered by mines, fire, observation and patrolling. Company localities will, therefore, per force have to be compact with gaps between the company defended localities.

d. ***Mutual Support***. The ideal is that rifle companies within the battalion should be able to cover the front, flank or rear of neighbouring companies with small arms fire. Where distances or ground configuration make this impossible, an interlock of anti-tank and machine-gun fire must be aimed at. Any intervening gaps must be kept under observation so that artillery and mortar fire can be brought down when required.

e. ***All-Round Defence***. A battalion will layout its area to defeat attack from all likely enemy lines of approach. Sections, platoons and companies must be prepared to fight in any directions, although they will normally be concentrated on the most likely directions. Where all-round defence cannot be achieved by initial disposition of the troops it should be achieved by subsequent pre-planned repositioning.

f. ***Depth***. Normally a battalion will be required to have a depth of upto 2000-3000 metres. Besides positioning of troops in depth, mines and obstacles must also be planned in depth. Adequate depth is essential for following reasons:-

(1) To deny enemy observation and reconnaissance of the defended area.

(2) To allow time for the commander to assess enemy intentions and take countermeasures.

(3) To absorb the weight of attack and break it up by a process of attrition.

(4) To deter outflanking of the position.

g. ***Reserves***. The battalion commander must be prepared to react to any unexpected situation created by the enemy. It will seldom be possible for the battalion commander to maintain uncommitted reserve, however, he can assign depth companies holding a piece of ground certain be prepared tasks to counter-attack or to occupy a counter penetration position, to stop the enemy onslaught.

h. ***Aggressiveness***. In defence no effort will be spared to wrest initiative from the enemy and to destroy his forces. Aggressiveness in defence at battalion level is achieved by:-

(1) Domination of No Man's Land by aggressive patrolling, raids, ambushes, tank hunting parties, sniping, shelling and harassing.

(2) Spoiling attack(s) to delay or disrupt enemy attacks.

(3) Counter-attack.

i. ***Security***. All necessary steps must be taken by the battalion commander to ensure security of troops and equipment and prevent any surprise. In this connection employment of following measures should be carefully considered:-

(1) Screens.

(2) Standing patrols/listening posts.

(3) Good battle intelligence.

(4) Camouflage and concealment.

(5) Communication security.

(6) Movement and fire-control.

**Deception**

8. Deception should constitute an inalienable part of the defensive planning at all stages. At battalion level the enemy may be deceived by the following measures:-

a. Construction of dummy positions and dummy weapons.

b. Operating fake wireless nets.

c. Showing strength where weak.

d. Showing movement during daylight at one place while occupying actual positions somewhere else during the hours of darkness.

e. Depicting tank noises with the help of loud speakers.

f. Educating the enemy through artillery shelling.

g. Any other viable/workable measures.

**Concealment from the Air**

9. The overall plan for concealment from air observation will be laid down by the formation. At battalion level this will be interpreted to cover such aspects as track discipline, use of roads, movement by day etcetera.

10. Although concealment may not always be effective against modern photographic techniques, it does offer valuable protection against visual air observation and detection by ground attack aircraft. Against an enemy holding air superiority, little movement or digging will be possible by day except under cover.

**Concealment from Ground**

11. Concealment from ground observation offers both defensive and offensive advantages. Concealment should not be wholly sacrificed to fields of fire. The minimum field of fire for a subunits primary task is 200 metres plus area under low wire entanglements between this distance and the trenches.

**Counter-Attacks**

12. The object of counter-attack is to recapture lost ground and destroy enemy penetration. It can be done by physical attack supported by fire:-

a. At company level there is no need to lay down reserves. In a two up position, if two forward platoons are under attack, the depth platoon becomes reserve and may be employed to counter-attack to evict the enemy, if he has penetrated a forward position.

b. A local counter-attack (company level) in all probabilities must be mounted immediately, within half an hour.

c. The situation can be restored by employment of troop of tanks (if available) either by physical movement or by fire only. Bringing down quick artillery concentration or defensive fires on own localities that have been over run will also help in evicting the enemy.

13. Counter-attack below company strength will be exceptional. Before moving reserves from their defensive positions and committing them in a counter-attack role, certain factors must be considered by the commander at the time. Some of these factors are:-

a. ***Importance of the Area Penetrated***. It may often happen that the enemy gets possession of an insignificant area either deliberately to deceive the defender or just to occupy a territory because the opportunity existed. Battalion commander should be careful to use the depth companies only on those areas which are important to his mission and not fritter away his strength on frivolous objectives.

b. ***Size of Enemy Penetration***. The extent of the area occupied by the enemy and quantum of troops in that area is yet another consideration which dictate the feasibility or otherwise of launching a counter-attack. While strength of the enemy may not be a serious disadvantage, the physical extent of the occupied area, may be a

factor against launching a counter-attack. Too large an area is clearly not within the capability of a platoon.

c. ***Time***. If the enemy has had sufficient time in the captured area to reorganize and dig in, a counter-attack may be futile. Unless the counter-attack can be quickly launched before the enemy has been able to get replenishments, it will more often than not be doomed to failure.

d. ***Vulnerability of Troops***. Troops after leaving their prepared positions will be most vulnerable to enemy artillery and small arms fire particularly during daytime. Battalion commander should regard this aspect as a disadvantage while launching the counter-attack.

e. ***Weakening of the Defensive Positions***. Removal of troops from depth localities will weaken the defensive position and may lead to speedy disintegration of the defences particularly by day. Battalion commander should carefully weigh the need for counter-attack against the possible dislocation of troops and weakening of the defences, especially when the counter-attack is also not one of the planned and rehearsed actions.

**Domination of Ground between Opposing Forces**

14. It is axiomatic that a thoroughly aggressive control over No Man's Land will always be maintained both by day and night. Failure to achieve this is indicative of loss of morale and offensive spirits. Domination is established as follows:-

a. ***By Day***. The No Man's Land is primarily dominated by aggressive patrolling. As far as possible, raids, ambushes, tank hunting parties, fighting patrols, etcetera may also be provided with artillery/armour support. In addition, best possible observation posts must be established to cover the entire area of responsibility and bring down effective artillery fire.

b. ***By Night***. The ground on the battalions' front and flanks must be dominated more directly at night than by day to prevent the enemy from securing valuable information about our position. This can only be done by patrols.

**Advantages of Night Vision Devices in Defence**

15. The Night Vision Devices will have following advantages in defence:-

a. Listening posts/standing patrols will become more effective, thus increasing strength in defences/reserves.

b. Protective detachments like screens etcetera will become more effective as they would be able to detect enemy at longer ranges.

c. Effective surveillance of the area of operations can be carried out thus lesser chances of being surprised.

d. With proper training, target acquisition will be possible and enemy could be engaged at longer ranges, which will impose caution on him.

e. Counter-attacks will now be conducted boldly with greater speed.

f. Occupation of counter penetration positions would now be possible in lesser time.

## SECTION 54 – PLANNING AND LAYOUT OF THE DEFENCE

1. The battalion commander will have received orders from his brigade commander or direct from the divisional commander in case the battalion is operating under direct command of divisional headquarters. The orders will include:-

a. Mission.

b. Forward and rear boundaries of the area of responsibility.

c. Any other instruction/advice affecting fulfilment of mission.

2. Having received the above orders, the battalion commander may proceed in the following sequence to carry-out the planning of his defensive task:-

a. Detailed analysis of the mission to sift out the main aim and limitations in order of priority.

b. Gathers all available information regarding salient aspects of enemy and terrain to determine its effect on the accomplishment of the mission assigned.

c. Depending upon the availability of time, carries out detailed reconnaissance along with the supporting arm commander.

d. Carries out appreciation and formulates his plan, which must cover following major aspects:-

(1) Identification of important tactical ground, the approaches leading into the area of responsibility and how this ground is to be defended.

(2) Company localities along with the platoon positions.

(3) Location/composition of screen(s).

(4) Use of armour and Anti-Tank Guided Missiles if allotted.

(5) Location of depth localities, battalion headquarters and mortar platoon.

(6) Obstacle plan including minefields.

(7) How to kill enemy tanks and where (anti-tank bias)?

(8) Early warning system to include location of standing patrols listening posts etcetera.

e. Issue of verbal orders.

f. Reconnaissance and planning by company and supporting weapons commanders to site platoon and section positions.

g. Coordination by the battalion commander on ground to carry-out readjustments to ensure the overall viability of defences to achieve the mission.

h. Issue of final orders.

3. ***Defence Layout***. The corner-stone of the defensive plan, therefore, will be where and how to kill the enemy which will largely dictate the dispositions of weapons so that they can be used to the greatest effect. In this, a study of the approaches, the use that can be made of obstacles and the need to achieve the maximum degree of observation at all times while denying it to the enemy will be major considerations. By this process, a commander will hope to construct his defensive position so that battle will be invoked and fought in accordance with his plans. However, a commander cannot ignore the probability that the enemy may not act or react as anticipated. The commanders' plan for battle must, therefore, be flexible so that by the judicious use of his reserves, he can counter the unexpected situation and force the battle back within the framework of his plan. An important point to remember is that wherever the enemy is likely to carry-out his operation with armour, the defensive layout should be based primarily on tank defeating defences. Therefore, the priority will go to the siting of anti-tank weapons and anti-tank obstacles. The remaining weapons and effort will be deployed and utilised in support of own anti-tank effort.

**Echelons of Defence**

4. ***Protective Detachments***. (Figure 25). All possible steps should be taken to reduce the initiative and superiority enjoyed by the enemy. Such actions should be aimed at gaining early warning of the enemy's intentions, disrupting his preparations, delaying his advance by forcing him to frequently deploy before he reaches the forward defended localities and inflicting maximum possible losses on

him. To do so protective detachments are deployed ahead of the forward defended localities. These detachments normally consist of the following:-

a. ***Reconnaissance Elements***. The main task of the reconnaissance elements is to keep a constant watch on the enemy's movement/ actions, pass back information and to harass the enemy. These troops will normally be provided by the Corps Reconnaissance Regiment. Reconnaissance elements will be employed in conjunction with the covering troops and or the advance position. When finally pushed back, they fall on the flanks or form part of the reserves to provide additional security to the main position, unless otherwise directed.

b. ***Covering Troops/Advance Position***. The tasks assigned to troops in these positions may include some or all of the following:-

(1) Imposing maximum delay on enemy by forcing him to frequently deploy, so as to gain time for development of the main position.

(2) Gaining maximum information about enemy's strength and intentions etcetera.

(3) Harassing and inflicting casualties on the enemy.

(4) Denying information about own main position.

(5) These will operate under the overall commander controlling the battle of the main position along that independent axis. Commanders of these troops receive mission type orders which include tasks, boundaries and timings etcetera and are allowed maximum possible latitude within that framework. Continuous flow back of information is essential. Enemy must not be allowed to bypass. Troops must not seriously involve themselves in any major action unless capable of extricating themselves quickly.

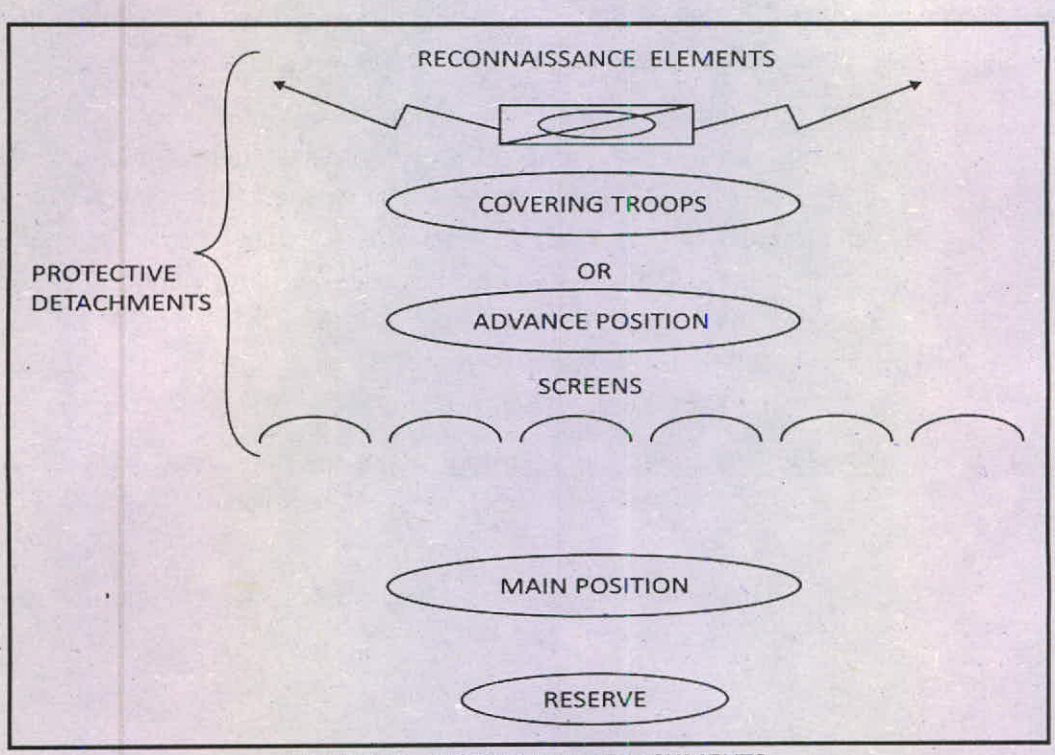



FIGURE 25 – PROTECTIVE DETACHMENTS.

(6) It is the type of terrain, degree of delay to be imposed and the effort which can be spared, that dictate the employment of one element or the other or both. Whenever more time is to be gained and if large numbers of mobile troops with sufficient supporting arms are available, covering troops may be deployed well ahead of the main position. Thus, they are capable of fighting rearguard action, effectively imposing necessary delay on the enemy. On the contrary advance position has comparatively smaller forces, mainly infantry, through reinforced with light anti-tank detachments some armour and few guns deployed in the forward temporary positions, both to gain maximum range and to conceal the main battle positions. Owing to limited mobility, advance position is reasonably close to the main position and is capable of delaying the enemy from one or two positions before finally falling back.

c. ***Screens***

(1) Screens are company/platoon/section size outposts established laterally with intervals in front and/or on the exposed flanks of the main defensive position. Screens should be

deployed only as far forward from the main positions as is necessary to deny the enemy observation and reconnaissance of the main positions and its obstacles. Screens should be deployed on the likely enemy approaches on a defensible ground that provides good observation and covered routes of withdrawal. Screens may also be employed by division/ brigade on seemingly unlikely approaches away from the main defences for surveillance and early warning.

(2) Situation permitting a brigade/battalion commander may deploy a company/platoon size screen to exploit advantages of a piece of ground which may otherwise not be falling within the defensive position. For example an isolated knoll some distance from the forward defended localities of company/ battalion can be occupied by a reinforced platoon strengthened by minefields etcetera, so that enemy is forced to attack the screen position before developing an attack on the main positions.

(3) ***Tasks***. The tasks of screens irrespective of the size are as follows:-

(a) Deny close observation of the main defensive position and its obstacles.

(b) Provide information about the enemy.

(c) Inflict casualties upon the enemy.

(4) ***Composition***. Troops for screens will be drawn from the depth companies/platoons holding the main positions. As such their number and size is kept as small as possible so that maximum numbers of troops are available for preparing and manning the defences. The considerations for deciding the strength of the screen are:-

(a) Frontage of the defence required to be screened.

(b) Period for which deployed.

(c) Direction and quantum of threat.

(d) Defensibility of the ground where deployed.

(5) Owing to the demanding nature of the tasks company/platoon commanders should be very selective in choosing the personnel for screens. Depending on the size, it will often be desirable to detail company/platoons/sections as a whole for

screens, ensuring however that the platoon/section commanders so selected are brave, imaginative and reliable. A suggested composition of a screen is as follows:-

(a) Company with artillery/mortar observer and necessary communications.

(b) Platoon with all or some of the following additional weapons/personnel:-

i. Extra tripod Machine-gun 1A3.

ii. 106/75 millimetres Recoilless Rifle.

iii. 60 millimetres Mortar or artillery/mortar observer.

iv. Baktar Shikan.

(c) Section with all or some of the following additional weapons and men:-

i. Tripod Machine-gun 1A3.

ii. Rocket Launcher.

iii. 60 millimetres Mortar detachment or mortar observer.

(6) Mines may also be laid in front and on the flanks of the screens if not already deployed in the minefield. Availability of mines and time/resources for laying will have to be considered.

(7) ***Control and Deployment***. Screens are controlled by the authority who orders it. Screens commander should therefore be clearly told the headquarters that will order its withdrawal. In most cases, it will be the battalion headquarters. However, when screens are deployed away from the battalion on the orders of the brigade/divisional headquarters, command channels should be clearly indicated. General area of the screen will be given by the commander ordering the establishment of the screen, yet the screen commander will have to select the most appropriate position for his troops/ weapons. Major criteria for a good screen position are, clear observation and field of fire and availability of covered routes for withdrawal. Effort should be made to present a large and continuous front to the enemy so as to deceive him about the actual line of own forward defended localities and precipitate

his attack on the screens; in front of the main defensive position. In order that larger areas are covered, platoons may be split in section posts, but the sections may preferably not be split or deployed over extended frontages. Capability of quick disengagement should always be retained while siting a screen. A screen should have a number of alternative positions for weapons to quickly switch the fire from one direction to another.

(8) ***Conduct of Screen Battle***. The battle of screens can be divided into three stages i.e. preparatory, contact and withdrawal. In all stages however, continuous passage of information about the enemy is very essential. Various actions of the screen during different stages will be as under:-

(a) ***Preparatory Stage***. This is the stage when attacker is assembling his forces and making preparations for the assault. During this stage the screen should:-

i. Gain information about enemy strength and concentration area by careful observation.

ii. Gain information about the areas of enemy interest by watching his patrolling and reconnaissance activities.

iii. Keep enemy patrols away from the main positions by a use of the minimum essential weapons.

iv. Plan, prepare and rehearse withdrawal of the screen. Withdrawal plan should be thoroughly coordinated with the concerned subunit in the main position.

(b) ***Contact Stage***. This is the stage when the attacker sets off to contact and assault the main positions. During this stage screens should:-

i. Engage the enemy with all available weapons at long-ranges.

ii. Call for artillery/mortar fire.

iii. Force the enemy to deploy.

iv. Make preparations for withdrawal.

v. Ensure that it does not get inextricably involved with the enemy.

(c) ***Withdrawal Stage***. This is the stage when the screen, having done its job, is ready to fall back in the main position. During this stage the screen should:-

i. Seek permission for withdrawal.

ii. Pass back all available information about the enemy.

iii. Continue exerting pressure on the enemy with all available weapons.

iv. Withdraw to the main position at the very last moment.

(9) It should be noted that withdrawal of a screen should be along covered routes. It cannot be carried out without the permission of the commander who had ordered establishment of that screen. In case the communications break, the screen commander may withdraw at his own, provided he has suffered 50% casualties or the time upto which he was required to stay has elapsed.

d. ***Patrols/Listening Posts***. Patrols and listening posts provide a measure of protection and early warning. Therefore, all the care should be taken to dominate those areas by observation and fire which cannot be covered from the main defensive position by sending out patrols/ outposts.

5. There will not be enough manpower to spare; hence, this weakness should be offset through the extra firepower of the automatics and anti-tank weapons.

6. Screen will normally be provided by the battalions the holding forward defended localities and their activities will be coordinated by the commander of the main position. Employment of forward observer officers with these troops should also be considered.

7. ***Pattern of Deployment***. A typical diagrammatic layout of a battalion and three variations are given in Figures 26 to 28. The frontages and depths as shown in the figures are only as a guide; these will vary in different tactical situations/terrain conditions. These are meant as guides and have to be interpreted intelligently.

8. ***Salient Points – Battalion Group Variation 1*** ( Figure 26)

a. Battalion occupies and holds a locality with equal defence emphasis to the front and flanks.

b. An ideal layout when flanks of a battalion are open and there are no friendly troops in the area.

c. Both depth companies will have to be given dual responsibility of defence and local counter-attacks depending on the tactical situation.

d. This layout tends to divide the firepower of a battalion to the front and flanks, therefore, a commander must pay particular attention to assessment of tasks and allotment of troops to area/localities.

e. It appears to be simple layout from the command and control point of view.

f. All avenues of approaches are guarded and there are lesser chances of enemy achieving surprise.

g. As all troops are committed to their primary tasks the plan seems to be less flexible, particularly to conduct subsequent phases of defensive battle and to meet unforeseen tactical situations.

h. This layout is advantageous to meet enemy attack by infiltration.

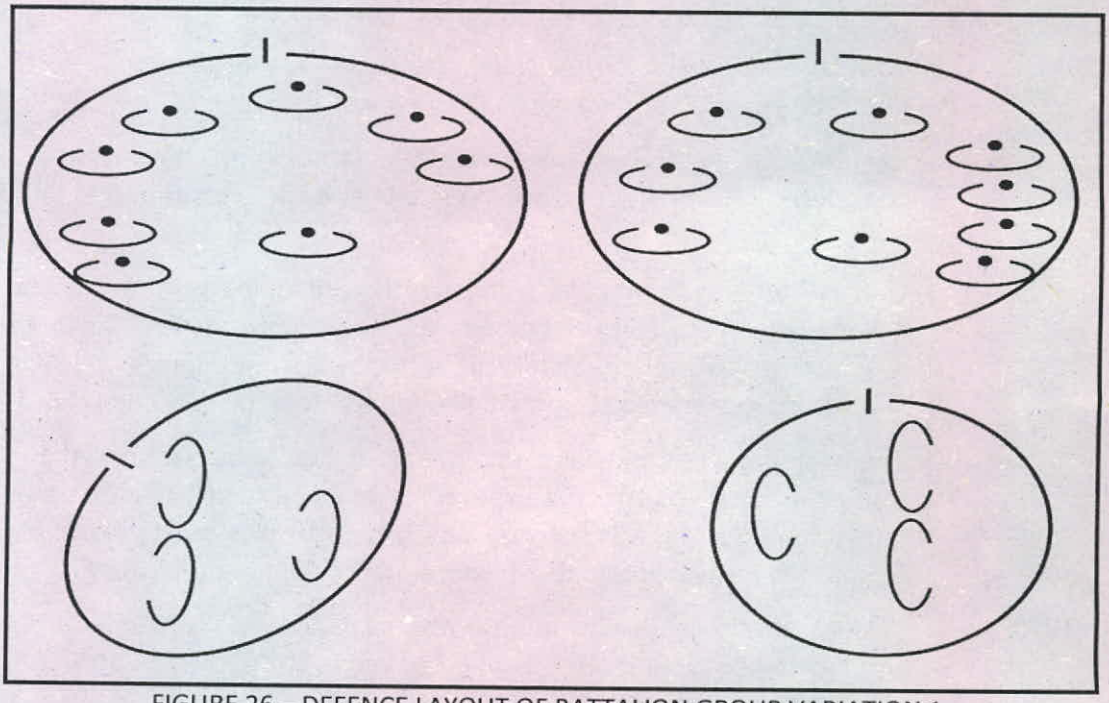


FIGURE 26 – DEFENCE LAYOUT OF BATTALION GROUP VARIATION 1.

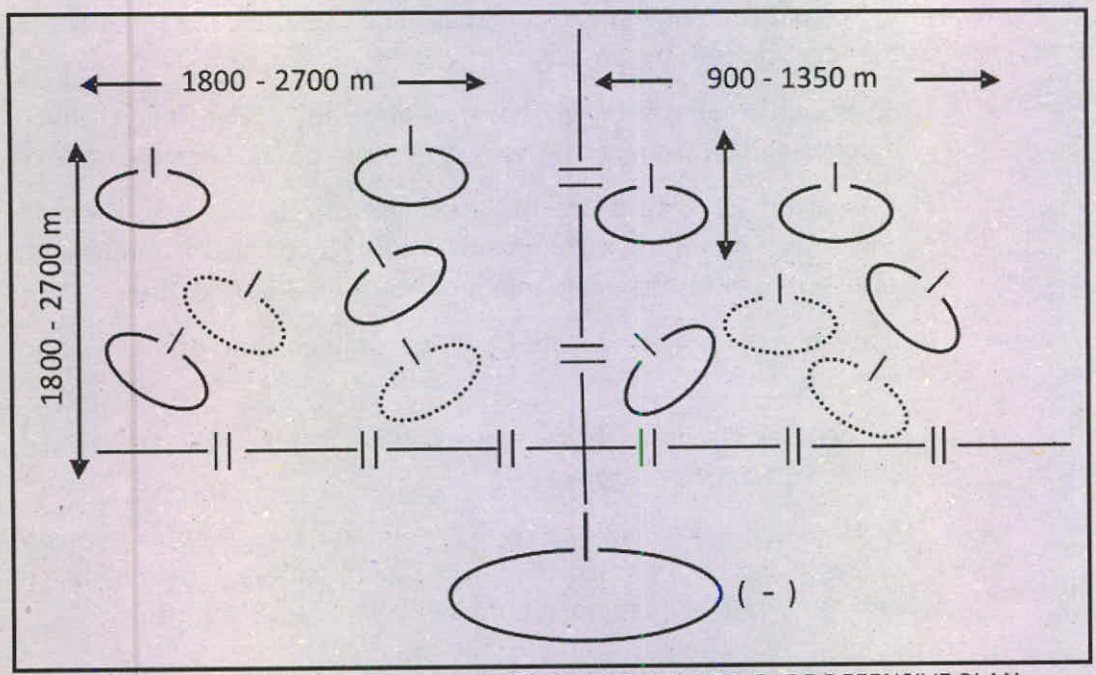



FIGURE 26A – INFANTRY BATTALION IN DEFENCE IN A BRIGADE DEFENSIVE PLAN.

9. ***Salient Points – Battalion Group Variation 2*** (Figure 27)

a. This layout is biased to a flank to cover a threatened flank to neutralize/destroy enemy outflanking movement.

b. Existence of obstacles and location of troops on the flank are considered to establish the degree and type of defence required on a particular flank.

c. Like the other variation both the depth companies will have to be given dual responsibility to hold ground and counter-attack. As there is less threat from one flank or less troops requirement, a small reserve may be created to provide added advantage in defence plan.

d. Enemy approaches and allied enemy tactics should be considered and due attention given to more exposed/threatened flanks. On the other hand, full advantage should be obtained from the existence of an obstacle and close proximity of troops on the flanks and to the front.

e. Taking advantage of the natural/artificial obstacles and creating a small reserve, gives more flexibility to conduct defensive battle and meet changing situation.

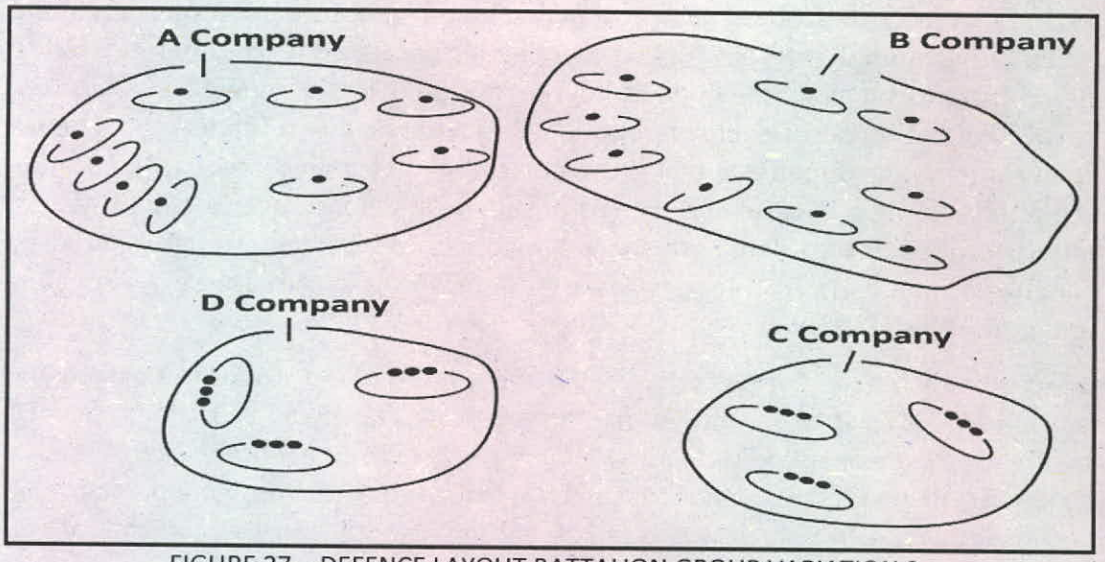



FIGURE 27 – DEFENCE LAYOUT BATTALION GROUP VARIATION 2.

10. ***Salient Points – Layout Battalion Group Variation 3*** (Figure 28)

a. All-round defence has been catered for.

b. Proper depth has been provided.

c. Rear most company acts as committed reserve in the hands of the battalion commander. The committal of the company outside the battalion locality in no way affects the layout of the battalion.

d. Battalion commander retains maximum flexibility.

e. The defence is compact, close and the localities support each other.

f. Command and control is easy.

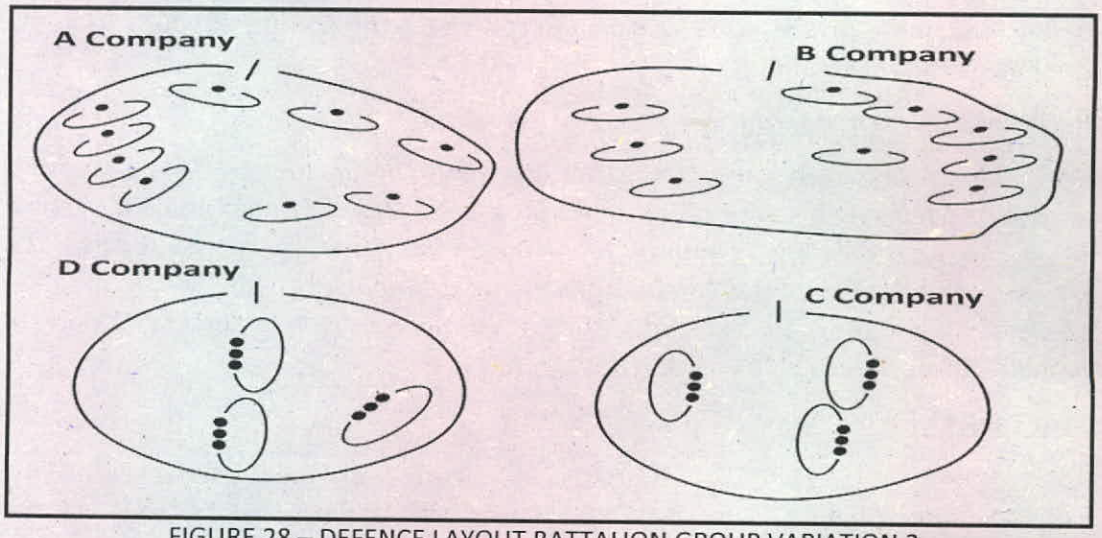



FIGURE 28 – DEFENCE LAYOUT BATTALION GROUP VARIATION 3.

11. Tendency to split sections and platoons into groups of two men each and string them across the front at a considerable distance from each other in a bid to cover the ground with the men is to be resisted. This is generally undertaken because of the threat of enemy infiltration. Such a deployment renders the defence rather insecure and ineffective and the enemy will not only succeed in infiltration but also in taking a few isolated men as prisoners. In defence over broad frontages, intensive patrolling to dominate the No Man's Land and gaps, supplemented by suitably located early warning devices are the only means of successfully detecting and preventing infiltration.

12. It would be a more effective deployment if the rifle platoons can occupy the best possible defensive grounds in their area of responsibility and cover the rest of it by fire. Deployment of the sections would be governed, among other factors, by the requirement of effective command and control by the section commanders. The sections could be grouped into two teams, each team occupying two trenches of two to three men each. The distances between these trenches may be between 5 to 20 metres depending on the nature of ground and observation.

13. There is also a tendency to seek depth in defence at section and platoon level. Weapons sited in echelons within a section are normally masked by the presence of friendly troops immediately in front and thus cannot be effectively employed. Fire effect of the subunit at the critical point of attack is thus reduced. An effective deployment could be to site the weapons of a section in an irregular lie so that they cover the front, curving on the flanks to cover gaps. To achieve all-round defence capability, trenches for one of the groups of the section could be prepared a short distance in depth facing the rear to be occupied only in case of need. Even at platoon level in defence, where ground, frontage and other conditions demand, we may consider the feasibility of deployment with three sections up. The depth in defence could be organised at company level where the line of weapons in depth can have a clear field of fire.

**Reserves at Battalion Level**

14. It will seldom be possible for the battalion commander to have a specially constituted reserve, because of the limited resources. However, he should task some of his subunits already committed, to remain prepared to carry-out the local counter-attacks to evict the enemy from the neighbouring localities; if the situation so demands. He may, however, bear the considerations given in Section 70 para 14 in mind before launching a counter-attack.

**Anti-Tank Layout**

15. The anti-tank layout will form the framework of the main defensive position. The commander having considered the likely armour threat will select localities and allocate anti-tank resources accordingly to counter such threat.

16. ***Siting of Anti-Tank Weapons***

a. At times the Energa grenades and rocket launchers can be sited together ahead, with 106 millimetres Recoilless Rifles covering them from the rear, both for provision of depth and should the situation so demand, to be in position to extricate them. This will produce following results:-

(1) The long-range of 106/75 millimetres Recoilless Rifles will not be fully utilised.

(2) The tanks will be given a free run upto the effective ranges of rocket launcher and Energa grenade.

(3) The tanks will have a very short run, say 200/300 metres, from the point the fire of all anti-tank weapons becomes effective and the position on the ground of the short-range anti-tank weapons. This distance of 200/300 metres, even at 40 kilometres per hour, the tanks will cover in approximately 16/24, seconds, which will encourage the tanks to "charge" and overrun the short-range anti-tank weapons positions. It will reduce very much the number of rounds, which can be fired at the tank before it overruns the position.

(4) Once the short-range anti-tank weapons, which are in maximum number in a battalion, are overrun (and these are the positions where the remaining infantry will also be positioned), the long-range anti-tank weapons will be divorced from the infantry and the short-range anti-tank weapons. Therefore, the long-range anti-tank weapons will be more easily made ineffective later on by the tanks/infantry combined team of the attacking enemy.

(5) In such type of siting, the depth of fire is reduced very much.

b. The following techniques therefore are suggested:-

(1) ***Technique 1***. The weapons are sited so as to be able to engage enemy at the maximum effective ranges. The short-range anti-tank weapons to occupy temporary positions at a distance of 500/600 metres ahead of the main position. Some of the long-range anti-tank weapons may also be positioned along with short-range weapons in the temporary positions. If the enemy tanks advance, they have to first pass through a fire curtain of the long-range and short-range weapons. This way, the tanks will be under continuous fire for a longer distance.

(2) ***Technique 2***. Only few long-range weapons are deployed ahead in the temporary positions that withdraw to the main defences at a suitable time after having engaged the enemy. All weapons as in technique 1, are resited in a manner that short-range weapons provide depth of fire to the fire of long-range weapons. The engagement of tanks starts at the maximum effective weapons. The net effect is that as the tank is getting closer to the main defences, it is getting in the thicker and thicker anti-tank fire.

c. At time it is quite likely that certain approaches are absolutely impossible for the tanks, while certain other are quite good. In such situations, all anti-tank weapons, including short-range weapons, may be concentrated in the threatened area. Alternative and secondary positions should be prepared for all anti-tank weapons. This improves their effectiveness.

d. Judicious, intelligent and clever use of mines and other natural and artificial obstacles will further assist in destruction of the enemy tanks.

17. However, the difference in the ranges of the Energa grenades and the rocket launchers is negligible where hundreds of metres are involved, when a comparison is drawn vis-à-vis the tank guns, with the result that to site them as explained above, would not be sound. Therefore, the correct deployment would be to site the Energas and the rocket launchers together, with 106 Recoilless Rifles of Baktar Shikan covering them from the rear-both for provision of depth and to be in a position to extricate them should the situation so demand.

18. The number of weapons so employed, will depend on the ground, the threat and the availability. However, the guiding principle should be that the number of anti-tank weapons should be more than the expected number of tanks, without denuding the main defences altogether.

19. When the advancing enemy tanks hit the protective minefields or are under an effective fire from the Recoilless Rifles/Baktar Shikan from the main defensive position, they would instinctively manoeuvre to get into a nearby hull down position. In this general area preselected sites are manned by tank hunting parties duly equipped with Rocket Launchers/Energa grenades/RPG-7. They take on the tanks from the flanks and destroy them at short-ranges.

20. ***Anti-Tank Guided Missiles***. The induction of Anti-Tank Guided Missiles (TOW) and Baktar Shikan has bestowed upon the defender a great advantage due to its greater range and accuracy. More often than not, the employment and siting of the Anti-Tank Guided Missiles will be planned and affected at brigade and above level.

However, in certain situations, Anti-Tank Guided Missiles may be allotted to the battalion to improve its anti-tank resources. These weapons, if skilfully combined with the organic battalion anti-tank weapons and boldly sited, would immensely add to the overall effectiveness of the battalion defence. A suggested method of combining/siting the Anti-Tank Guided Missiles with the battalion anti-tank weapons in the light of techniques given above is as under:-

a. The Anti-Tank Guided Missiles (TOW) have a long-range of 3000-400 metres. These should, therefore, be sited 500-600 metres behind the 106/75 Recoilless Rifles, in the depth localities. When so sited, these weapons will provide depth to the battalion short and medium range anti-tank weapons and will themselves be protected against observed artillery and automatic fire of the enemy.

b. The Anti-Tank Guided Missiles will engage the enemy armour at the optimum ranges and would thus be able to thicken the anti-tank fire of the short and medium range weapons deployed ahead, as well as within the main defences. The controlled withdrawal of the weapons deployed ahead would be further facilitated due to continuous engagement of enemy armour by the long-range fire of the Anti-Tank Guided Missiles from the depth localities.

c. The above siting, therefore, enables continuous engagement of the enemy tanks with the fire of all anti- tank weapons opening at the maximum ranges.

21. ***Development of Defences***. Refer to Chapters III and IV of 'GSP-1985 – Operations of War (Volume-I), Defence – 2004'.

**Occupation of Defences**

22. Normally the defences will be occupied during the hours of darkness, though the reconnaissance will be carried out during day. The drill for the occupation of defensive position during night will be as under:-

a. Commanders down to section level would have carried out daylight reconnaissance.

b. A section commander will take with him maximum of 2 men to assist him in marking the positions and guiding the troops to their respective positions at night.

c. Marking of weapon pits may be done as under:-

(1) Mark the trenches by scrapping the earth with pick and shovel.

(2) Put stones on the corners to mark the trench.

(3) Drive wooden pegs in the ground marking the corners of a trench.

(4) Put white tape to show the outline of a trench. This tape may be fastened with the pegs on the corners or putting stones on it. The tape should be spread at the later hours of the day lest it should be detected by the enemy ground/air observation. In the absence of tapes, the section commander, even if he has one man with him, marks the trenches with a pick and digs 4 to 6 inches of each trench. When the troops arrive during darkness, they can easily identify the markings and carry-out the digging.

d. Arrangements for fix line for the automatics.

e. Select listening post about 300 metres ahead of the position. One of the two men brought by the section commander with him, will man the listening post along with other two or three men as detailed by the platoon commander on the first night.

f. Select positions for his section for the alarm scheme.

g. Select a place where the loose earth will be dumped and concealed.

h. Method of guiding men to their positions:-

(1) Section commander will collect his men from the platoon dispersal points to the section positions.

(2) Follow easily recognisable route from the dispersal point to the section position.

(3) Lay white tape to facilitate move in the night from platoon dispersal point to his section position if possible.

(4) Take compass bearing of his position from the dispersal point during daytime and march on this at night.

(5) Having brought his men up, the section commander will collect them at a central place and show them their trenches in turn, starting first with the rifle groups. He will also explain them their arcs of fire.

(6) Having shown them their arcs he will show them the rest area and the place where they will deposit the lose earth.

(7) The digging will then start and supervised by the commanders. The digging must be completed during the night and by the first light, the position should be fully ready and camouflaged.

(8) The range card will be made as soon as possible but before darkness.

(9) On occasions, it may not be possible to complete digging of the whole battalion position in one night. Depending on the air situation, digging may continue at a reduced scale, during daytime as well. However, if the situation does permit, the position so far prepared will be camouflaged, men pulled back to rest areas and digging restarted after last light.

i. The platoon and company commanders are responsible for the guidance of their subunits beyond the company and battalion dispersal points respectively.

## SECTION 55 – THE FIRE SUPPORT IN DEFENCE

**General**

1. The aim is to destroy and stop the enemy in front of the Forward Defended Localities (FDLs). It is the responsibility of the battalion commander to provide the maximum weight of fire, both anti-personnel and anti-tank, on the approaches to his locality. This is planned with the advice of the commander of all supporting arms. Superimposed on the fire plan is the counter-bombardment organization.

**Platoon Weapons**

2. Platoon positions should, if possible, be so sited as to have a field of fire upto 400 metres. Riflemen should be trained to shoot with confidence upto 400 to 500 metres ranges. Though their most effective killing ranges are 300 metres and under, it may well be necessary for them to start destroying an enemy advancing in large numbers upto 500 metres ranges.

**Artillery and Mortar Defensive Fire**

3. The aim of defensive fire is to break an enemy attack before it reaches the assault line. When possible, this will be done by the Forward Observation Officers (FOOs) controlling fire by observation:-

a. ***Defensive Fire Tasks***. To be able to engage an enemy attack when observation is impossible e.g. in close country, bad visibility, or by night. The commander must prearrange defensive fire tasks to cover every possible enemy of approach and also catch the enemy in possible assembly areas, Forming Up Places and defiles. Close defensive fire tasks should be as near our own forward positions as safety of own troops permits.

b. ***Defensive Fire (SOS) Tasks***. To get the quickest possible response to a call for the defensive fire, one defensive fire (SOS) tasks can be selected for each troop/battery or regiment, or in the case of 81 millimetres Mortars, for each platoon. The task chosen as defensive fire (SOS) will normally be the defensive fire task on the most vulnerable approach. Guns allotted to these tasks remain laid on them when not otherwise engaged.

**Drill for Artillery Defensive Fire Planning**

4. The battalion commander, advised by his direct support battery commander and in consultation with his company commanders will select close defensive fires.

5. The battery commander will enter the battalion defensive fire plan on the Task Table (PAFZ-6047), have it approved by the battalion commander and forward a copy each to his regimental commander, adjutant and the battery command post.

6. The direct support regimental commander, on behalf of the brigade commander, will coordinate defensive fire near inter battalion boundaries. The brigade commander in consultation with the regimental commander will select defensive fires in depth as necessary and initiate or suggest any grouping of the targets. He may also recommend the desirable density of fire on selected defensive fires, particularly in the case of grouped targets defensive fires. The regimental commander will prepare the task table (PAFZ-6047) containing the coordinated brigade defensive fire support plan and forward one copy to the divisional artillery headquarters and one to his regimental headquarters.

**Armour**

7. The tank is designed primarily for aggressive operations and it should not be employed in a purely static defensive role. All defence is, temporary and fore-runner of the impending offensive operations. It must therefore, take part in the counter-attacks for which, the tanks should generally be centralised and not split up as local support for the units and subunits of infantry. However, when the enemy has temporary superiority some of the available tanks may be split up, allotted to battalions and used defensively. However, it is essential that the number so employed should remain minimum.

8. Penetration of any defensive position under modern conditions is generally inevitable. The defensive position must, therefore, be organised in depth with suitable areas selected for launching of a counter-attack. The majority of the tanks will thus be held in reserve, as part of a counter-attack force.

9. Following factors must be considered when allotting armour in support of infantry units and subunits:-

a. The enemy.

b. Ground, type of country, vital ground.

c. Infantry layout.

d. Tanks available and their layout.

e. Quality of the tanks compared with those of the enemy.

f. Number and quality of own anti-tank guns and mines.

**Light Anti-Tank Battalion**

10. Light anti-tank elements, if allotted, may be employed on following tasks:-

a. They may be given an independent locality to defend for a limited period when other troops are not available.

b. They may be employed to reinforce or thicken up the battalion anti-tank defence. However, it must be borne in mind not to dish them out to platoons and sections in penny packets.

c. Depending on the level of defensive planning, they may be employed in the reconnaissance elements or as part of the advanced positions/ screens.

d. They may be employed to cover gaps in the defended localities.

e. May be assigned the task of dealing with any enemy helicopter borne threat.

f. May be given under command the counter-attack force. In this role, they may support the counter-attack from a base of fire, may physically accompany the assaulting infantry or may be employed on flank protection of the counter-attacking force at a higher level.

**Air Support**

11. In defence, battalion commander is less likely to receive direct air support than in attack. The air effort would normally be earmarked for to indirect support planned at a higher level. However, it should certainly be considered when planning or rehearsing counter-attacks. The battalion commander should assist the divisional counter-bombardment officer with reports of enemy shelling and with flash and sound bearings. This information may lead to reconnaissance aircraft locating hostile battery positions which may be attacked by own guns directed from the air or by the ground attack aircraft. For this purpose, at least one set of cloth panels must be held by each battalion for indicating targets to the aircraft.

## SECTION 56 – THE CONDUCT OF DEFENSIVE BATTLE

### General

1. The conduct of the defence will depend upon the battalion commanders' concept of the battle but can best be considered in three phases as follows:-

a. ***The Preparatory Phase***. This is when the attacker is assembling his forces and making preparations for the assault, while the defender is preparing his defensive positions.

b. ***The Assault Phase***. This is when the attacker is attempting to break into and through the main position and the defender seeks to seal off any penetration and to stabilize the situation.

c. ***The Counter-Attack Phase***. This is when the enemy attack has lost its impetus and the moment has come to restore the position or to resume the offensive.

### The Preparatory Phase

2. This has been largely covered in planning, occupation and development of the defence. It is a period of intense activity for the defender. He will be striving to complete his position, to delay the enemy's advance so that the work can go on unhindered and unobserved and to discover all he can about the attackers' plans so that he can disrupt them at every opportunity.

3. ***Settling Down***. However, detailed planning, final adjustment of positions and coordination of fire can be made only when the troops are on ground. Battalion and company commanders will personally inspect every position and ensure the siting of every support weapon. Artillery and mortar tasks must be checked and whenever possible, registered without revealing the plan to the enemy. Obstacles must be inspected and the fire tasks covering them perfected.

4. Every commander and man must know the plan and his own job. He must have such confidence in himself and in the troops and weapons in support of his defence that he welcomes the prospect of attack as a means of inflicting a crippling blow to the enemy. Commanders at every level must be satisfied that they cannot be surprised by day or night and that the enemy reconnaissance patrols cannot approach or observe their positions. When the situation permits, it will be useful for the commanders to go out in front of their positions to see them and their weaknesses from the enemy's point of view. It will sometimes disclose a defect in their plan which had hitherto been overlooked.

5. All means of intercommunication must be checked. Arrangements for maintenance of line must be practised. As far as possible telephone wires should be

buried deep enough. The system of administration must be adjusted as necessary. It must also cover the detachments from the supporting arms, in the locality.

6. ***Domination of No Man's Land***. While the battalion is settling down, the second stage of battle will have started; No Man's Land must be dominated by patrols and snipers. Enemy standing patrols or advance post cannot be allowed to remain near our foremost defended positions. The operation may start with reconnaissance patrols and ambushes. It will progress until the enemy can never be certain which areas of No Man's Land are in fact occupied.

7. ***Aggressive Operations***. The whole enemy position must be subjected to aggression. Local attacks and deep penetration patrols can occasionally be launched, but they must be carefully planned to have utmost chance of success. The simplest method of reaching the enemy is by fire.

8. ***Harassing Fire***. The objects of harassing fire are:-

a. To cause enemy casualties.

b. To lower his morale.

c. To disturb his rest.

d. To interfere with his administrative routine.

e. To interfere with the concentration of men and material.

9. Harassing fire will include the fire of all weapons. Requests for day and night programmes are made by battalions and coordinated not below brigade level. Stereotyped methods be avoided that allow the enemy to predict as to when and where he will be hit next.

10. During the preparatory phase the battle is fought entirely ahead of the main position by a combination of the following:-

a. Protective detachments.

b. Demolitions, nuisance mines and route denial.

c. Harassing fire.

d. Deceptive measures.

e. Air reconnaissance and interdiction.

**The Assault Phase**

11. The assault phase begins when the enemy's advance has been halted by the main position. At this stage all the protective detachments except a few patrols will have fallen back and most of the lanes through the minefields and other obstacles closed. After contacting the main position enemy's patrolling and reconnaissance activities will increase. He may also launch a series of probing attacks to gain

information and feel the strength of the defences. All these activities should be firmly and decisively frustrated by an application of minimum firepower preferable from the positions other than the original. It will be difficult to differentiate between probing and the main attack. Company/platoon commanders should, therefore, exercise strictest fire-control so that they do not give away their positions or run out of ammunition required urgently at a later stage, to repel the main assault.

12. When the main assault is launched, the defender will bring down full weight of his defensive fire in an attempt to break-up the attack when the enemy troops are well concentrated in the assembly areas, Forming Up Places or are on the Start Line or crossing the obstacles.

13. Despite these efforts and depending on how carefully the preparatory phase had been conducted, determined enemy will press home his assault and some forward positions will probably be penetrated and others overrun. The localities under direct attack will be subjected to heavy fire and the troops there may have difficulty in using their weapons. This is the time when the full destructive fire of the defended area must be fully developed. Every weapon capable of doing so must engage the enemy and cause maximum possible destruction. For a short period, commanders on either side will not be absolutely sure of the situation. It is vital that, no matter how hard pressed they may be, the subunits send back all available and correct information on which the battalion commander can control the development of battle. At this critical stage, when determination can tip the balance in the defenders' favour, every soldier must hold on firmly to his position even if it is temporarily isolated. Neighbouring localities must also hold firm to their positions and continue to destroy the enemy even when they have been bypassed. After the initial onslaught has passed, these localities will provide the bases from which local counter-attacks can be launched or around which the reserves can operate.

14. The next step is to halt and contain the penetration. The battalion commander should be prepared to launch a local counter-attack if it seems likely that the situation can be restored by striking the enemy before he has had time to reorganise. The infantry for this purpose must be taken from whichever part of defensive position or reserves is relatively less exposed to the enemy at the time.

15. It may not be often possible to mount a counter-attack at the battalion level, because the removal of even one platoon from an intact locality might seriously weaken the whole position. It is to be remembered that an immediate counter-attack led determinedly by a small body of men can often restore the situation. Even a few tanks against unsupported infantry may well be successful. However, it is important that, when a counter-attack is not possible, previously prepared alternative/counter penetration positions may be occupied by adjusting troops quickly, with a view to halting and containing the enemy so as to stabilise the situation.

16. The contact of defensive battle will follow the sequence as mentioned below:-

a. During the preparatory phase of the defensive battle, the attacker has revealed his designs.

b. The warning system of the defender has provided timely information of the enemy's movements from assembly area to the Forming Up Place.

c. The defender will call artillery and mortar fire to cause maximum damage to the enemy while he is concentrated and exposed.

d. The defender will organise spoiling attacks by sending strong fighting patrols/raiding parties to prevent enemy from forming up for the attack. Such attacks during the night would cause considerable confusion in enemy ranks.

e. Once the attacker leaves the Forming Up Place and launches his assault, the defender will quickly move troops from the depth localities to pre-prepared positions on flanks. These troops will open fire from a flank to sweep the attackers from end to end. This will produce shock effect on the enemy as these weapons were not located by him earlier.

f. By the time the attacker closes with the main position, he should have suffered much attrition. Now the most accurate defensive fire (SOS) is brought on to the assaulting troops who have also been denuded of their own supporting fire. Hand grenades thrown by the defenders would take a heavy toll of the left over enemy. In case the enemy has closed up with the bunkers and trenches, he may lob grenades into the bunkers/trenches to evict the defenders. It will, therefore, be advisable for the defenders to have some open trenches connected with their bunkers for use under such eventuality.

g. It may happen, that some of the determined enemy soldiers may reach their objectives. But they will be physically exhausted and exposed to many dangers, as they have yet not been able to dig down.

h. The defender, who has been conducting the defensive battle with an open mind, must have moved some troops into the already prepared counter penetration positions, to halt the assault. All the troops in defence will not be directly involved in the battle, so, it would not be difficult for the defender to throw fresh troops into the battle at the decisive time and place.

i. Localities temporarily bypassed or isolated will close the gaps and go on inflicting casualties on the follow-up echelons of the enemy.

j. Once the situation has been stabilised, the battalion commander will launch his speedy local counter-attack to evict the enemy who has caused penetration into the defensive position.

k. While the local counter-attack is dealing with enemy penetration, artillery and mortar fire are causing great damage to the enemy follow-up echelons, who are exposed and much concentrated, to present a lucrative target to the gunners. Air can also play important interdiction role.

l. The whole art of defensive battle is to cripple the enemy with a well coordinated defensive fire and obstacle plan and to present a superior number of troops at the decisive time and place to off-set enemy balance.

17. Fire on artillery defensive fire tasks in depth will not be normally authorised below the brigade level. Fire on artillery and mortar close defensive fire tasks will be called for by the company commanders and in some cases by platoon commanders. Fire on defensive fire (SOS) tasks of all weapons may be called for by a company commander, or by a forward platoon commander through his company commander.

18. Fire of recoilless rifles Baktar Shikans and machine-guns, except when harassing, must be with held until a kill is certain. Orders for firing will be laid down by the battalion commanders and fire-control must be strictly observed.

19. Troops in defended posts must know and believe that they are in the best possible firing positions and that their task is to stand fast and fight it out where they are. There must not be any rear ward movement from the original positions, unless ordered by the higher commander.

**Counter-Attack Phase**

20. The counter-attack plan is conceived and integrated in the overall defence plan of the battalion commander, right from the start. The battalion commander must keep a flexible mind and carefully decide the correct moment for launching his counter-attack force, to ensure decisive results.

21. The success of local counter-attack at the battalion level depends on the speed and surprise. The emphasis therefore, should be on speed rather than the quantum of force. The form of local counter-attack most likely to succeed is spontaneous attack with bayonets against local and small scale penetration of the forward localities. Such an attack depends on the circumstances at the time and initiative of the local commander.

22. Troops, however, are required to be instantly available for immediate local counter-attack against enemy penetration. Where possible, certain platoons should be earmarked in the company plan for counter-attack roles, but platoons so earmarked will usually have the primary role of holding a part of the battalion locality. Once they become involved in the fire fight to defend their localities, they are not likely to be able to lunch a counter-attack. The smaller the force to be a lunched, the quicker the counter-attack will be conducted. An inspired rush by a few courageous men is likely to meet with much greater success than in a carefully prepared but delayed attack.

23. If a commanding officer appreciates that a threat is purely local, there are occasions when a depth company not yet committed to the battle, can be used effectively to counter-attack and restore the situation. However, it should not be a matter of course for a battalion commander to counter-attack with a depth company. The depth companies are normally committed to the defence of important ground and to also commit them to counter-attack in the face of serious enemy threat will probably achieve little, except to weaken the defence by abandoning ground of tactical importance.

24. It is neither possible nor prudent to lay down the unit commander's exact capability of local counter-attack against various dimensions of the enemy penetration. Since speed and surprise are the essence of a local counter-attack, a vigorous and bold battalion commander may be able to restore even an exceedingly adverse situation. Certain guidelines regarding the reaction ability of the company/ battalion commander under some situations are given below:-

a. Loss of a section or two in a forward company locality should be restored by the company commander by launching a quick counter-attack by the depth platoon. Since the enemy perpetration is being checked by the depth section of the platoon under attack, the company commander has reasonable chances of success against such penetration. Counter-attack against the loss of an entire platoon locality by the company commander would be a difficult proposition. He must, however, take measures to check the penetration by adjusting the defence/shifting into alternative positions to facilitate the counter-attack by the battalion commander.

b. As above, the battalion commander may launch a counter-attack against the capture/penetration of upto one or two platoons in any one of his forward companies. Since the loss of an entire forward company would entail induction of fresh troops by the enemy, the chances of success of a local counter-attack against such a situation (being late) would not be compatible. The battalion commander, therefore, may decide to occupy a counter penetration position to

check further build-up/exploitation by the enemy. Availability of some tanks, even if a troops, however, considerably adds to the counter-attack capability of the unit commander and can be effectively employed to eliminate such penetration.

c. Once the platoon/company commander has stabilised the situation after carrying out readjustment of defences and the enemy's penetration has been contained, the company/battalion commander must decide to immediately launch his counter-attack before the enemy follow-up troops are inducted. The company local counter-attack in all probabilities must be mounted within 30 minutes. Similarly, the battalion counter-attack must be launched within an hour, if it has to produce positive results.

25. In the event of simultaneous capture of some localities in the battalion defence, the battalion commander must exercise his option to counter-attack against the most important locality, while checking the enemy penetration in the other by adjusting his defence.

26. Maximum fire support from the troops, containing the enemy penetration and that of artillery must be integrated and brought to bear on the counter-attack objective. Anti-Tank Guided Missiles, if available, may also be grouped with the fire base to support the counter-attacking troops.

## SECTION 57 – COUNTER PENETRATION

**General**

1. Counter penetration is one of the very important actions taken in defence to forestall enemy's penetration for his ultimate eviction/destruction. It will seldom be possible for commanders to hold the entire area of responsibility in strength. Invariably a defence will be taken to meet the most likely/dangerous enemy option. To cater for various contingencies, cover relatively unimportant areas and to check enemy penetrations between defended localities and around the flanks, a number of counter penetration positions will have to be prepared. Companies/battalions in depth/reserve or occupying relatively less threatened areas will be tasked to be prepared to occupy previously prepared counter penetration positions. These positions will be prepared from the beginning as part of the defence plan and occupied when the degree of enemy threat/penetration so warrants.

2. To cater for all other options a number of counter penetration positions can be prepared. Generally their number will depend on following factors:-

a. Extent of area of responsibility.

b. Availability of time and resources.

c. The magnitude and direction of threat as visualized by the commander.

**Definition**

3. Counter penetration is defined as physical action taken by a defender to halt further penetration into his defence by the enemy and to cause maximum casualties/attrition to his forces that have penetrated the defended localities.

**Concept of Counter Penetration**

4. Counter penetration will generally precede a counter-attack by the next higher echelon. Careful planning and timely occupation of counter penetration positions will not only limit enemy penetration but also contribute towards the success of counter-attack.

5. A counter penetration position is a deliberately prepared position, sited in sufficient depth. It should not get simultaneously involved with the battle of forward defended localities. On the contrary, it should be adequately in depth to induce an additional phase in enemy's operation, thereby depleting his strength and also disrupting his time and space bondage. This, will allow the most needed time for own counter actions to be launched against him. The main philosophy and advantages that accrue from counter penetration battle are:-

a. To disturb and finally defeat the enemy attack plan.

b. Restrict space for further enemy operations and subsequent phases of attack.

c. To provide opportunity to the next higher commander to launch his counter-attack.

6. ***Readjustment vis-à-vis Counter Penetration***. It is pertinent to clarify that readjustment should not be mixed up with counter penetration. Readjustment is within the defences for which alternative positions are dug. Readjustments are normally carried out before the assault or any penetration. On the other hand, counter penetration positions are occupied only when the enemy has overrun a certain portion of forward positions or has achieved penetration within the defences. Occupation of counter penetration positions by the local commander will facilitate launching of counter-attack by the next higher commander.

7. ***Counter-Attack vis-à-vis Counter Penetration***. Due to obvious advantages, the foremost choice of any battalion commander should be to launch a counter-attack. However, if the opportunity for launching the counter-attack is lost then he

can resort to counter penetration. Some of the reasons which may force a commander to go for counter penetration include:-

a. The enemy has achieved a sizeable penetration which is beyond the capability of a counter-attack.

b. Lifting of troops for local counter-attack may seriously jeopardize/ weaken the defences.

c. Time and space dimensions do not permit launching of a counter-attack.

8. ***Fall Back of Forward Elements***. It is difficult to specify before hand as to where the remnants of forward elements will fall when they are overrun. However, the best option is to gather them and readjust within the existing locations so as to contain and limit the penetration. In case that is not possible, then the remnants may be ordered to fall back to the depth positions which have been vacated by the troops, moved out to occupy the counter penetration position. If both the options do not materialize, then for those who fail to fall back even to the depth localities, straggler's posts may be established at brigade/division level along the main axis/ route for this purpose.

**Planning and Tactical Considerations**

9. ***Planning Considerations***

a. The size of counter penetration position will depend upon the threat visualized by the commander. Usually at the battalion level it would be a company minus size positions. At brigade and above, infantry battalions constituting reserves could be tasked for the job.

b. These positions are sited to block likely penetration between or through the localities or on the flanks likely to be exploited by the manoeuvring elements.

c. These positions are to be extensively prepared and rehearsed by the troops who are earmarked to occupy them or by reserve.

d. Mines and wire obstacles may be laid in front to provide them strength and to check the enemy penetration.

e. To avoid confusion in execution, it is important to carry-out rehearsals during day/night for occupation, by the troops designated to occupy them.

f. Battle from counter penetration position is forced upon a defender because of enemy action. It implies that defender has to make last moment adjustments (though rehearsed) to contest/contain enemy ingress. Hence, before committing troops in this role a commander

must be clear of enemy intentions and the threat posed to the defences.

g. Artillery fire support plan should be carefully planned to effectively support battle from the counter penetration position.

h. Heavy anti-tank weapons if available, will so located so as to exploit their maximum ranges.

i. For efficient conduct of defensive battle, coordination of counter penetration positions should be carried out at the brigade/divisional level to ensure effectively/timely reaction for the eviction/destruction of enemy, penetrating the defences.

j. To ensure better command and control during occupation of counter penetration position and conduct of battle, line communication up to the prepared positions should be provided before its occupation.

10. ***Tactical Considerations***. Following tactical aspects must be considered before planning a counter penetration position:-

a. It should be sited against expected enemy axis/direction of advance.

b. Should deter outflanking of defences by the enemy. Should have sufficient frontage so that enemy can not out flank it without getting far away from his axis/direction of advance. Enemy should be forced to clear it before proceeding forward.

c. Should preferably be based on natural obstacle or built-up area.

d. Should be sufficiently in depth not to get involved in the battle of forward defended localities, thereby forcing a separate deliberate operation by the enemy for its clearance.

e. It may not always be possible to ensure "depth" in the counter penetrations position in view of the extended frontages required to be occupied vis-à-vis force available. Counter penetration positions be planned in such a manner that these are provided depth by the existing localities in the defences.

f. Considering time and availability of resources a number of positions should be selected to cater for all contingencies. However, as a guideline minimum of two to three positions at the battalion and brigade level should be planned.

g. The frontages of the counter penetrations position will invariably be more than the normal defensive position, depending upon the visualized penetration and availability of troops. Thus, there is a need for careful selection of counter penetration position based on

defensible feature, deriving strength from ground. Siting of automatics and anti-tank weapons also requires greater deliberation in order to cover complete frontage as well as to avoid outflanking.

**Time/Occasions for Occupation of a Counter Penetration Position**

11. ***Time***. Decision to occupy counter penetration position will primarily depend upon factors like own situation, progress of enemy attack and actual situation on ground. While analyzing own situation it must be remembered that if forward defended localities are completely overrun, the decision to occupy a counter penetration position will be late and will not bring the desired result. Such a decision has to be at a time when forward defended localities are not completely overrun. A commander should take this decision at such a time that even after spending time in the battle procedures and movement, the troops can occupy the counter penetration position in time. A commander will have to keep a close watch on the progress of enemy operation. While assessing the enemy situation/operation, a commander will have to analyze the size of enemy penetration and the quantum of enemy troops therein. Although, it would not be possible to lay down any specific time frame, yet one could consider certain rough timings planned for such operations, only as a guideline. Time when the enemy attack commences would be approximately known which will point to the approximate time of completion of enemy's battalion/brigade/division phases. This would facilitate decision making regarding the time of occupation of counter penetration position. However, it should not lead to any mental fixation and must be related to the situation on ground in terms of the progress of enemy attack. One could consider following timeframe as a general guideline:-

a. ***Within the Framework of a Battalion***. During the attack when own forward platoons (of forward companies) are about to be completely overrun i.e. enemy's forward battalion's phase 1 is yet not completed. Generally, one could consider this timeframe as not later than one and a half to two hours after the H hour. However, it may not always be possible to precisely determine the enemy's H hour. In this case, consideration of above mentioned factors would help in taking a decision for occupation of counter penetration position.

b. ***Within the Framework of a Brigade***. During the enemy attack before own forward companies have been completely overrun or enemy's brigades, phase I is yet not over. Generally, one could consider this time not later than three hours after the H hour. Besides, actual situation on ground which would actually influence any decision regarding occupation of counter penetration position.

12. ***Occasions***

a. Enemy has achieved penetration which is beyond the capability of local commander to eliminate.

b. To stop enemy's move/flow for destruction by the counter-attack force.

c. The time and space dimensions do not permit launching of counter-attack by the reserves.

d. As part of the overall design of the battle to commit enemy infantry and separate his armour for destruction piecemeal.

13. There may be occasions when appreciation of a commander goes wrong and enemy achieves complete surprise, for which a commander is not prepared. In such an eventuality, troops will have to be moved to an unprepared counter penetration position. For occupation of unprepared counter penetration position, commander along with his subordinate commanders will have to move earlier than the main body. They will carry-out a quick reconnaissance and the tasks/areas will be assigned to the subordinate subunits there and then. In such a case, company guides will accompany the order group to guide their companies to their locations on arrival. Central guidance arrangements will have to be made by the battalion. Time permitting, the troops must dig down, otherwise they will have to take position in the open to contain enemy penetration. If tanks are made available for such a task, they could immediately be deployed ahead of the counter penetration position to gain maximum time for preparation. Moreover, by this time composition and intentions of the attacking enemy will also become clearer and the tanks could then be deployed to guard against the likely armour threat.

**Conduct**

14. Counter penetration positions will be pre-planned and rehearsed by the troops earmarked to occupy them or by the reserves. At brigade or above level, battalion size counter penetration positions are prepared which are usually occupied by uncommitted reserves when the tactical situation so warrants. However, within the framework of an infantry battalion, counter penetration positions are planned to be occupied by the depth companies or by the troops from the areas likely to be the least committed in the battle. When asked to perform a counter penetration role would these companies continue to perform their original task with minimum of one third of the force in situation.

15. Once the enemy attack materializes and the forward locations have been overrun, the battalion commander will be in a position to correctly assess and visualize the size and the magnitude of threat. He can judge and decide whether the depth platoon(s) of the forward companies will be able to halt the attack for

sometime or not. If in his opinion the depth elements will be able to check the penetration, then he must put his counter-attack plan in execution, otherwise, he orders the occupation of appropriate counter penetration position(s), to facilitate the brigade counter-attack. Although this is a difficult decision but still the battalion commander due to his foresight and experience should be able to take a correct and bold decision.

16. To crystallize the rationale of occupation of counter penetration positions in relation to the conduct of defensive battle at the battalion level, a classic brigade attack plan against an orthodox battalion defensive locality is portrayed in Figure 29.

17. ***Brigade Attack Plan***. Normally in our scenario the enemy will plan his brigade attack at least in two phases to ensure adequate destruction and to attain requisite depth varying from 2000 to 3000 metres. For this purpose, he will have to allow optimum time between phase 1 and 2. This interval is vital for the defender to execute his actions of counter-attack or counter penetration. Within the brigade attack plan, probable tasks for the attacking battalions in various phases could be broadly as follows (see Figure 29):-

a. ***X and Y Battalions***. Capture locality A and B in phase I. Be prepared to capture depth locations in phase 2 on orders.

b. ***Z Battalion***. Capture locality C (or D) in phase II. Be prepared to assist X and Y battalions in phase I.

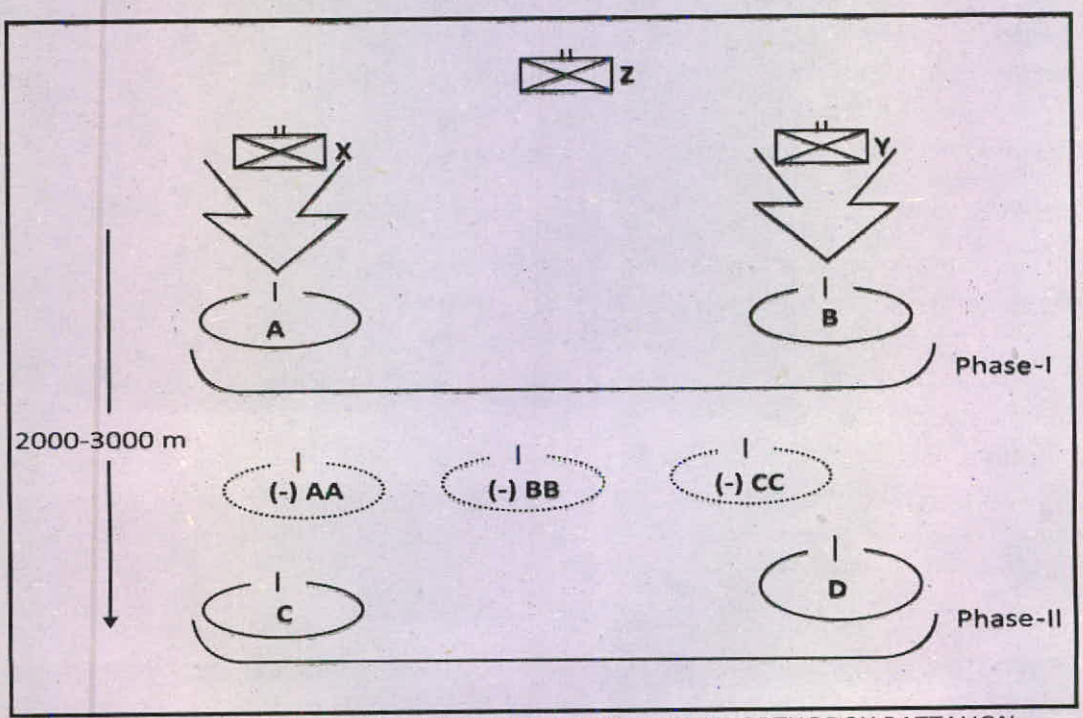



FIGURE 29 – ENEMY BRIGADE ATTACK PLAN AGAINST AN ORTHODOX BATTALION.

**Situation 1**

18. Once enemy attacks, there could be many options as far as the direction and weight of attack are concerned. The configuration of ground and the layout and depth of the defensive localities would determine the phases of attack. In this discussion, a hypothetical situation will be taken up to decide on the fundamentals and doctrinal aspects of the counter penetration battle that could apply equally to other situations. Situation 1 (see Figure 30) shows enemy penetration in B company locality, whereas attack against A company has been halted.

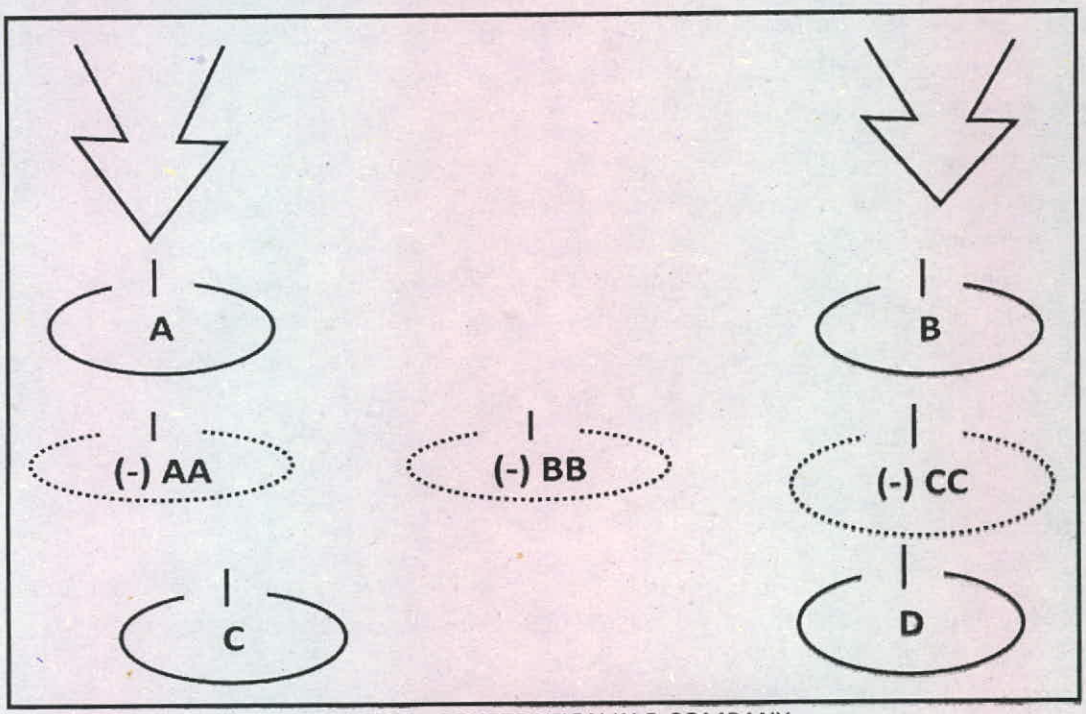



FIGURE 30 – PENETRATION IN B COMPANY.

19. We presume that in this case the battalion commander due to late information has missed the opportunity for launching a counter-attack against locality B. Now he has two options i.e., either dispatch C Company to occupy counter penetration position CC or order D Company minus to accomplish this task. This is a vital issue, where officers have difference of opinion and are usually vague. It will be more expeditious to trigger occupation of counter penetration position by oblique depth company rather than occupying the position with vertical depth company. In this situation C Company minus should be ordered to occupy counter penetration position CC. It will render following advantages:-

a. Imbalance will not be created in the defences because still 1/3 company is left in the defences.

b. Enemy attack will fall short of troops for clearance of two company positions.

c. More attrition can be caused on enemy.

d. Enemy penetration is effectively checked.

20. The onus of decision regarding move of troops from a particular depth company lies with the commander, who analyses the prevailing situation before taking such a decision. Advantages listed for initiation of oblique depth company for occupation of counter penetration positions should at best be taken as a guideline. Actual situation on ground would dictate the right course of action. Considerations like the threat and the fact as to which troops are likely to be less committed would also have a bearing on such a decision.

21. Depth companies in the event of enemy frontal attack as shown in Figure 30, are occupying well prepared defensive positions and may be considered suitable to contain enemy penetration from existing localities. This contention is not correct as the enemy has to be stopped/contained in further expansion and penetration. This space between the forward defence line and the depth companies must be denied by occupying counter penetration positions as far forward as possible. This action on the part of defender will ensure denial of the space well in time and thus deny opportunity to the enemy to launch phase II of his attack for final consolidation of his gains.

**Situation 2**

22. Figure 31 shows enemy main attack against B Company, which has been successfully contained. However, enemy has achieved a sizable penetration between the forward defended localities. Battalion commander must be clear about the location and size of enemy penetration before taking any counter action. Penetration within the gap of two companies or from the platoon area of inner platoon is exceptional, however, for understanding the concept, in this situation any one of the depth companies can be ordered to occupy locality BB with a platoon to check the enemy penetration. Preference should be given to that depth company which is behind the company still intact. If a portion of B company was also captured by enemy i.e. left forward platoon along with the penetration depicted, then only C company minus should be moved into counter penetration position BB, with bias towards right.

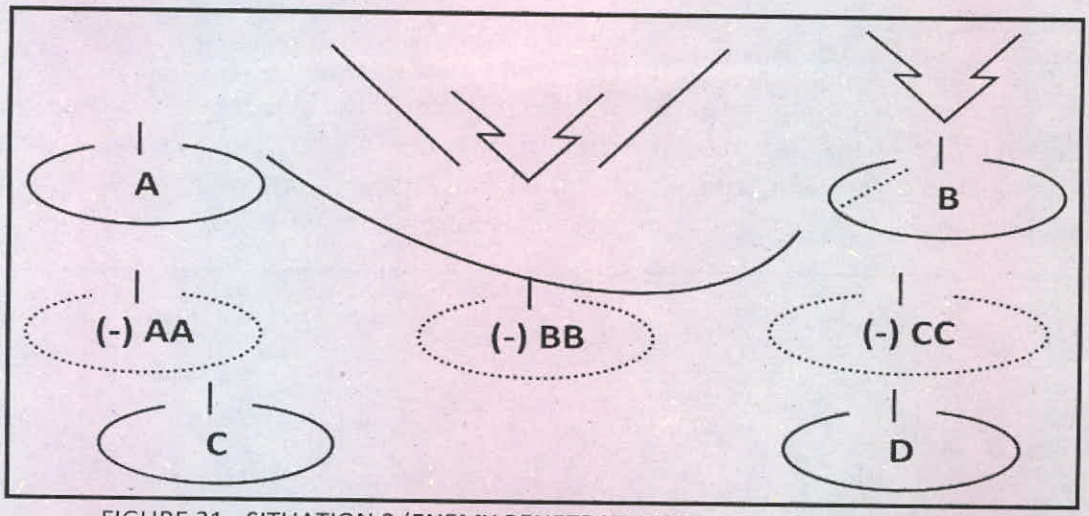



FIGURE 31 - SITUATION 2 (ENEMY PENETRATION IN INTER COMAPNY GAP).

**Situation 3**

23. Figure 32 depicts equal penetration of A and B company localities, therefore, the depth companies may be ordered to occupy counter penetration positions in front of their respective areas. In this situation although both the forward companies have been overrun the battalion commander must check further enemy penetration to gain time and prevent his further build-up, so that brigade counter-attack if planned can be launched under favourable situation. In this situation, since enemy's major threat/penetration has materialized in both the forward companies simultaneously, it is advocated that both depth localities should occupy counter penetration positions with a platoon each. This will guard against possible weakening/dislocation of defence and will also enable the depth companies to check the penetration as well as fight their major battle from well prepared defences.

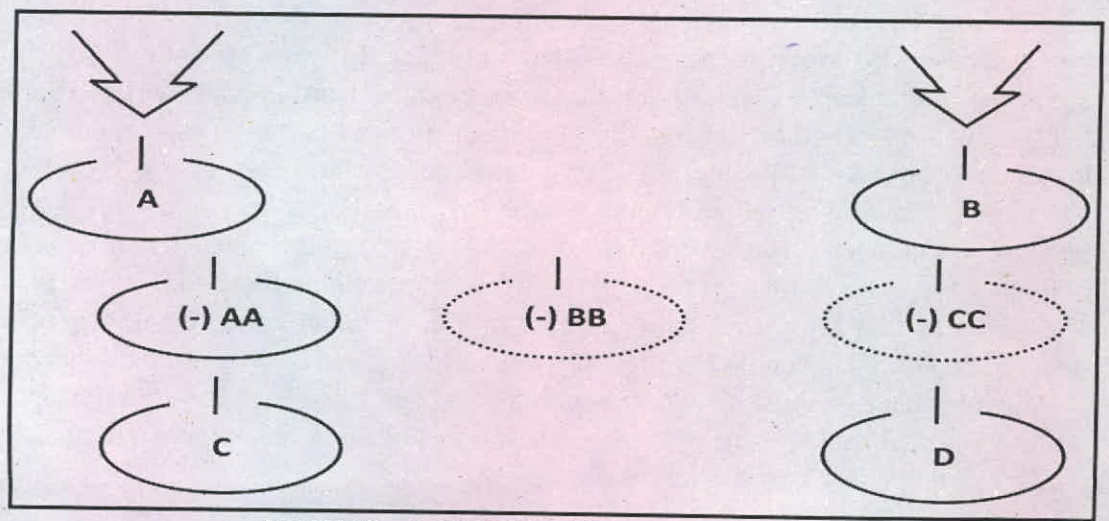



FIGURE 32 – SITUATION 3 (EQUAL PENETRATION).

**Situation 4**

24. Figure 33 shows enemy flanking attack. In this situation, B and D companies will readjust on confirmation of direction of attack. Readjustment is carried out within the company area to present front to enemy instead of flank. However, if the enemy presses home his attack and achieves penetration from the given flank, then occupation of locality CC by C company minus will be ordered.

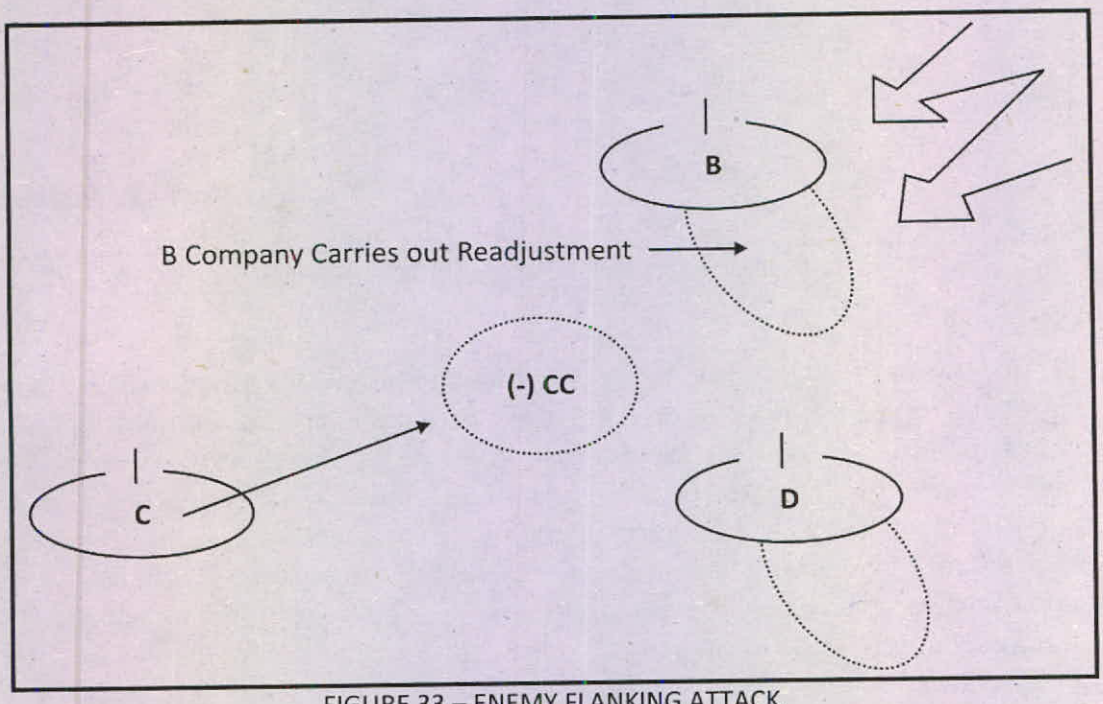



FIGURE 33 – ENEMY FLANKING ATTACK.

**Defence on an Obstacle (Plains of Punjab)**

25. Considering the fact that, our defences are mostly obstacle based and forward defended localities have extended frontages, forward assaulting brigades are likely to attack with two battalions up over a narrow front initially, with fixation/ feint elsewhere, intending to exploit intercompany or inter battalion gaps and break-in/gain a foothold. In the process he would acquire a fair idea of the layout of defences. Follow-up echelons would exploit the gaps/flanks to go for the depth localities. Frontal push wherever possible will not be adopted to avoid a coordinated defensive fire. Therefore, counter penetration positions would be sited to cover gaps and flanks as far forward as space permits to induct follow-up echelons to go for deeper objectives. A possible defence scenario against a hypothetical attack possibility which is most likely to occur across our obstacles based defence is depicted in Figure 34. Here counter penetration positions are sited to deter out flanking and to check penetration through gaps effectively.

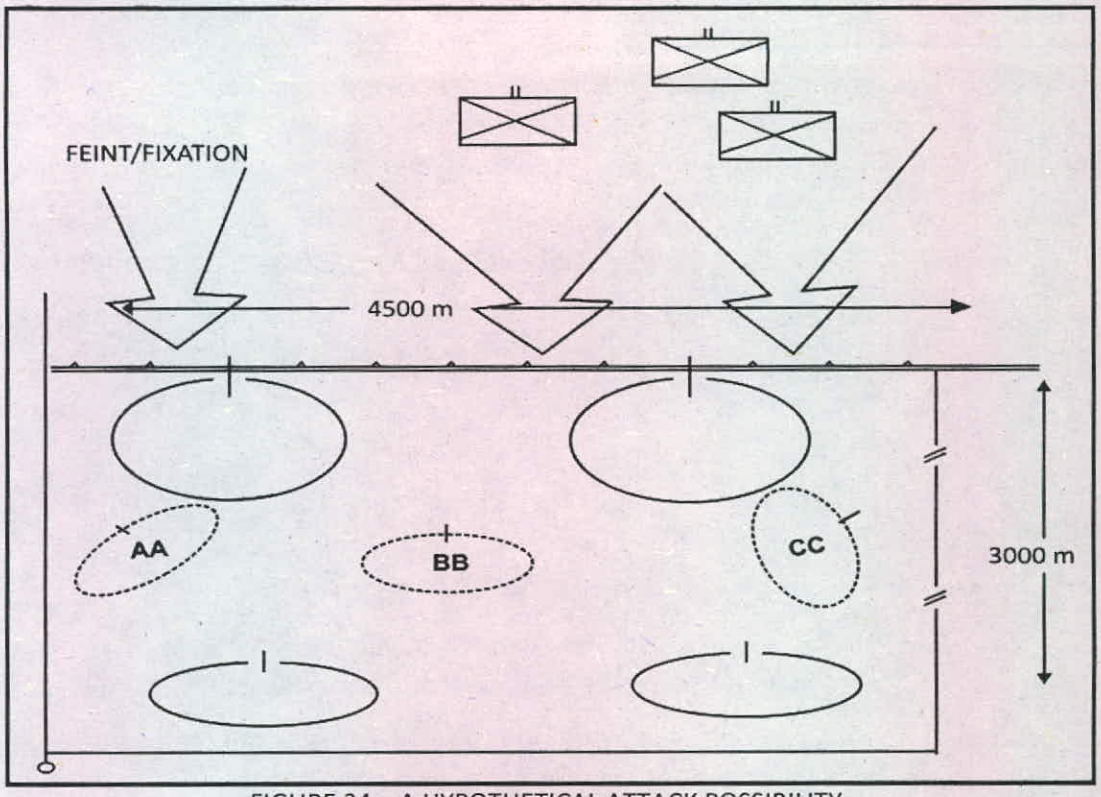



FIGURE 34 – A HYPOTHETICAL ATTACK POSSIBILITY.

a. ***Penetration through One of the Forward Company Position***. Counter penetration position BB should be occupied by oblique depth company. It will offer following advantages:-

(1) Immediate depth will not be depleted and enemy can be stopped by coordinated fire of position BB and Depth Company effectively. With the induction of Night Vision Devices, the space in between can be effectively dominated by battalion weapons and artillery defensive fire (own location).

(2) Position BB will sweep the flank of enemy pushing for depth location. Thus enemy will be forced to clear this position entailing delay in his operations.

(3) Position BB will sweep the flank of enemy pushing for the depth location. Thus enemy will be forced to clear this position entailing delay in his operation.

(4) This position will also effectively check penetration through gap between the forward companies.

(5) Could be occupied by either company as required.

b. ***Exploitation of Inter Battalion Gaps***. Positions AA and CC should be occupied obliquely to deter outflanking of defence. In case battalion commander is delayed he can order vertical occupation by immediate depth company. Though in this case he would be forced to order lateral move of minimum platoon strength from the depth. This will however, demand high training standards.

26. ***Battle Procedures/Drills for Occupation of Counter Penetration Position***

a. Battle procedures and drills for occupation of a counter penetration position within the framework of a battalion are given at Annex S.

b. Battle procedures and drills for occupation of a counter penetration position within the framework of a brigade/division are given at Annex T.

**Employment of Supporting Arms**

27. ***Armour***

a. If tanks are available, they can be moved ahead of infantry elements to immediately occupy counter penetration position.

b. Tanks can hold further penetration of enemy until infantry occupies counter penetration position.

c. Tanks can be kept in location with the depth companies to be employed against out flanking move by enemy.

d. Subsequently tanks can be assigned any of the following tasks:-

(1) Remain in the same position and hold the enemy along with infantry, until counter-attack is launched.

(2) Join counter-attack force to evict enemy penetration.

28. ***Light Anti-Tank Battalion***

a. Light anti-tank elements can be employed in the same manner as explained in armour tasks (paragraphs 26a and b).

b. These can be employed to counter the outflanking enemy.

c. Can provide base of fire to the counter-attacking force.

29. ***Anti-Tank Guided Missiles***

a. Engage enemy tanks from defensive position.

b. Readjust position to check enemy armour trying to outflank the position.

30. ***Engineers***

a. Make and improve tracks.

b. Lay minefields and obstacles in front of counter penetration positions.

c. Help in preparation of counter penetration positions.

31. ***Artillery***. Provide fire support to contain enemy penetration.

32. In the above paragraphs, concept of counter penetration battle and the drills and procedures involved in planning, preparation and execution of counter penetration tasks along with different situations have been discussed. There will be a number of variations to the above situations during actual operations, but judicious application of the above guidelines will help a great deal in meeting each contingency. The guidance provided for different contingencies should not cause any mental fixation. In no case the troops should be moved for counter penetration tasks haphazardly and without prior preparation/rehearsals.

## SECTION 58 – DEFENCE IN BUILT-UP AREAS

**General**

1. Subject to availability of manpower defence in BUA can be organized in two forms:-

a. ***Perimeter Defence***. An ideal method, when defence is organized in a continuous concentric rings on the basis of strong points, each mutually supporting, wherever possible. A number of reserves are placed by sector for counter-attacks. This indeed, is only possible if adequate manpower is available.

b. ***Defence Based on Number of Localities***. As problems of defence are accentuated by restricted fields of fire, that mitigate effective mutual support, the disadvantages can be offset by organizing defence in a number of defended localities, between which infiltration may be accepted with a view to isolating and destroying the attack. Due to restricted fields of fire, the localities must be of limited size and will normally be held by subunits. A reserve for counter-attack at all levels is essential and depending on the availability of troops should be as strong as possible.

**Basic Considerations**

2. Important considerations to be borne in mind while planning defence in built-up areas are:-

a. ***Control***. Lack of communication would cause isolation of the subunits who would have to fight independently.

b. ***Observation***. There may be a few high buildings which could be used for long-range observation. But generally speaking, the visibility of the immediate front may be restricted by buildings/houses, for which engineer effort may be required.

c. ***Field of Fire***. Mutual support between the posts is difficult and often impossible. Fighting is largely at the close quarters, requiring greater and effective use of the short-range weapons. Also the locality should be of limited size.

d. ***Difficulty of Locating Fire***. It is increased by the echoing effect of the fire amongst buildings.

e. ***Movement***. Movement within the built-up areas is either from the rooftops or through the walls, by making holes in the inter-connecting walls. Necessary reconnaissance must be done to make maximum use of both and denying their use to the enemy.

f. ***Cover and Concealment***. It is possible and proper that tank hides are organized to mislead and surprise the enemy by occupying mud houses, whose walls can be demolished for action or a hole can be made for the tank gun.

**Planning**

3. Planning of defence in the BUAs requires a detailed preparation and judicious use of all available resources on the basis of priorities. In general, the planning would involve the following:-

a. ***Reconnaissance***. A thorough and detailed knowledge of the area is essential and can only be gained by detailed reconnaissance and a study of air photographs, maps and town plans. All short cuts, concealed entrances to buildings, possible detours and observation posts should be known to all ranks. Routes for use by reserves must be reconnoitred and movement down these routes practised In addition, the reconnaissance should include discovering of necessary material which can be collected for development of defences.

b. ***Liaison with Civil Authorities***. Whether friendly or otherwise the civil efforts must be mobilized and exploited to the fullest.

c. ***Deception***. There should be screens of small mobile detachments in front of the forward defending lines, possibly located on the outer perimeter of the town or village. Their task will be to give the false impression about certain areas being occupied as the main defensive positions, when in fact they are not and to contest the enemy's initial bridgehead efforts. Their strength will depend on the size of the town

and the force provided for its defence. Withdrawal routes of these parties and the intermediate positions that can be taken up before the final withdrawal into the defended localities must be reconnoitred. Orders must state clearly the degree of resistance to be shown and who is to control the withdrawal.

d. ***Provision and Storage of Ammunition, Supplies and Water***. Necessary arrangements must be made to cater for all possible needs.

e. Efficient traffic control and marking of the routes and buildings.

f. ***Defended Localities***. Section and platoon posts will be grouped to form the defended localities, which in turn may be organized as components of a larger defended locality or area. In villages and small towns, perimeter defence is often possible as distances between the forward defending lines and reserves are not too great. In larger towns, manpower will probably be insufficient to defend the whole perimeter. It will therefore be necessary to concentrate on the defence of selected sectors such as main routes into the area and certain other key points, that must be denied to the enemy which can also be easily adopted for defence. Following must be taken into consideration for actual siting of the posts:-

(1) ***Fields of Fire***. To cover the main killing areas in streets and open spaces.

(2) ***Protection***. Bearing in mind the ease with which the enemy tanks may be able to demolish all but the strongest houses.

(3) ***Concealment***. This will lead to a variety of positions being selected. Sometimes the posts will be sited in open spaces between the houses. At other times, better positions may be available by occupying the ground floors of certain selected buildings, with sniper positions on the roofs above. Infiltration between the localities held may have to be accepted. Each locality must therefore be self-supporting and well stocked for defence. The defence, however, must always be aggressive and plans must be made to use reserves against the enemy who may infiltrate.

g. ***Strong Points***. Strongly constructed large buildings in each defended locality will be converted into strong points and held as the hard core of the defence. From these, the fire will be brought to bear on the main approaches to the locality. The main streets and open spaces will be blocked by barricades, wire obstacles and mines and these will be covered by fire from the strong points. Whenever possible strong

points should be mutually supporting and sited in depth, otherwise the enemy will have little difficulty in isolating and dealing with the each in detail.

h. ***Reserves***. It is essential that the defence is flexible as well as resolute. A mobile reserve must be maintained at all levels. It may have to ultimately hold a locality or a post, in order to meet a changing situation, but its primary role will be to undertake the following tasks:-

(1) Cutting off and destroying enemy penetration by moving in to an area from which the enemy movement can be observed and engaged in the open.

(2) Restoring a post or a locality, that has been over run by own troops after destroying the enemy in it. Routes for movement of the reserves can be prepared by mouse-holing through the building blocks.

**Defence of Buildings**

4. A defended house, including its interior and the roof, must be capable of all-round defence. It should, if possible, be inconspicuous. It must also be capable of withstanding bombardment. Priority of durability is:-

a. Steel or reinforced concrete frames.

b. Stone.

c. Brick.

d. ***Wood***. It should be avoided whenever possible.

5. ***Reconnaissance of a Building***. Points to look for are:-

a. Observation Posts.

b. Fields of fire with particular reference to the main task.

c. Requirement of loopholes.

d. Strength of building against small arms fire and artillery/air bombardment.

e. ***Entrances and Exits***. Too many of these are a source of weakness.

f. Communication with the adjacent houses.

g. Alternative positions.

h. Availability of water for drinking and fire-fighting.

i. Material available in the building and a rough estimate of the work required to prepare it for defence.

j. Dressing station.

k. Cookhouse.

l. Latrines.

m. Fire trenches in the gardens or outhouses, if any.

6. ***Local Protection***. One or two positions may have to be occupied by day and even by night, outside the building/premolar, to prevent the enemy from getting close enough to neutralize the defences inside the buildings.

7. If possible, sentries should be posted outside the building, especially at night. Fire trenches should be dug for the protection of sentries against bombing and if possible, as alternative positions. By night, small patrols may be used to provide additional protection. Dummy positions may mislead the enemy in his selection of the lines of advance.

8. Protection should be afforded inside the building by:-

a. Placing sandbags, mattresses, cupboards etcetera, against the walls particularly those with loopholes and on the floor which is not bulletproof.

b. All entrances not required for the use should be barricaded. Those not barricaded must be prepared for blocking at short notice. Staircases, and sometimes passages, may also require blocking. It may be possible to remove some of the stairs out of a staircase.

c. Communications between the rooms and to the roofs and cellars must be considered.

d. If sandbags or other protective materials are laid on floors, it may be necessary to shore up the ceiling of rooms or cellars below.

e. Water and sand must be available for dealing with any incidence of fire. All bath basins and buckets should be kept completely full to guard against bursting of mains. Gas and electricity should be turned off at the mains and the inflammable material not required for defence should be removed.

f. It may be necessary to demolish certain parts of the buildings to eliminate danger from the falling masonry or fire; and to provide overhead cover from the rubble, once the building has been strengthened.

9. ***Siting of Weapons***. The weapons should be sited as follows:-

a. All weapons should be sited well back from the windows. Sandbag walls should also be erected in the occupied rooms, well back from the windows. The glass of all the windows should be removed from the defended houses and from the adjacent houses in the vicinity.

b. In general, the machine-guns should fire from the lower floors to avoid plunging fire. The snipers should find the positions on the upper floors and roofs.

c. Methods of concealment and deception include dummy loopholes, use of lace curtains over windows and dummy figures at the apertures not in use.

d. Wire netting or similar material should be put over the ground floor windows to afford protection against the hand grenades.

**Conduct of Defensive Battle**

10. Screens will perform the conventional roles of delaying, deceiving and disorganizing the enemy to the maximum, without getting deeply involved.

11. The defender can remain concealed till the enemy comes in the close range. The attacker can also exploit the cover afforded by the buildings for closing in on the defender.

12. The defender is practically blind and an attacker can succeed in bypassing a locality or penetrating in the battalion defended area if he is not vigilant. The defender must do his best to prevent this because, if the attacker is able to once establish himself, it will be very difficult to dislodge him. The defender, though outnumbered and completely cut off, must fight whole heartedly as this can have considerable effect on the progress of attack. A BUA has a tremendous natural defence potential. Reinforcing a defended locality, especially in the village type of built-up area will be far more beneficial vis-à-vis launching a counter-attack in this type of warfare.

13. Built-up areas restrict employment of armour. The reserves therefore will primarily consist of infantry which may be supported by tanks. Though it is desirable to have uncommitted reserves yet, it may not be possible at the battalion level and they may have to hold ground. Central mobile reserves will have following tasks:-

a. Reinforce a defended locality.

b. Seal off and destroy enemy penetration between the defended localities.

c. Launch a counter-attack to regain a defended locality.

14. Successful conduct of defence will depend upon gaining and retaining initiative from the attacker. This could be achieved by:-

a. Remaining aggressive and exploiting the advantage of detailed knowledge of the area.

b. Observing enemy's movement constantly and avoiding being surprised.

c. Active and aggressive patrolling at night, since darkness favours the attacker.

15. For effective command and control, good communication is indispensable. Since radio communication gets affected, reliance has to be placed on line communication. Additional telephone sets must be arranged for the subunits as the line communication may have to be provided down to the section level. Training of runners assumes greater importance in this type of area.

## SECTION 59 – DEFENCE ON AN EXTENDED FRONTAGE

**General**

1. A battalion may be required to deploy on extended frontages when the approaches to be held are widely apart or due to the resource constraints. This Section deals with the problems of defence on an extended frontage and lays down guidelines to meet such an eventuality. Suggested frontages are given at Annex U and its Appendices 5, 6 & 7.

**Problems of Extended Frontages**

2. Problems encountered while taking deference on an extended frontages are:-

a. ***Command and Control***. It is affected due to limited signal resources and larger distances.

b. ***Protective Detachments***. These will have to be greater in number as well as mobile. This would tax the resources of the commander and may affect the strength of the main defensive position.

c. ***Reserve***. Size, composition and location of the reserves for timely reaction along extended frontages will be a serious problem. Troops may also be required to take care of the enemy helicopter borne and airborne threat.

d. ***Insufficient Artillery Support***. Large frontages always create the problems of location, observation and concentration of artillery fire.

e. ***Increased Threat of Infiltration***. Large frontages and big gaps increase the possibilities of infiltration by the enemy.

f. ***Problems of Surveillance***. Absence of any electronic means with the ground forces and the limited support of army aviation poses serious problems of surveillance of gaps in defence.

g. ***Mutual Support***. Since frontages will be extended, the gaps between defended localities will increase, making mutual support difficult. Mutual support in such situations will be only in terms of the ability of the next higher commander to readjust positions, or to reinforce the threatened locality in time.

h. ***Administration***. There will be an increased requirement of administrative support.

**Fundamental Considerations**

3. Even with the extended frontages, the standard fundamentals of defence remain operative which must not be disregarded. However, while planning the defence on extended frontage, a definite choice has to be made between presenting a continuous front or holding only selected areas in strength, with gaps between them.

4. ***Enemy***

a. The commander through his intelligence staff/agencies and protective detachments must first assess the nature, quantum, capability and direction of enemy threat in his area of responsibility.

b. Protective detachments employed should be strong enough to inflict maximum delay and attrition on the enemy.

5. ***Terrain***. The key areas should be held in sufficient strength and the defensive position should enable the defender to canalize the enemy into predetermined killing areas where he can be destroyed by fire and employment of reserves. Terrain friction should be increased by using counter mobility measures.

6. ***Frontages***

a. The governing factors are terrain, command and control, resource available and the capability of defence at night.

b. Frontages of company, platoon and sections cannot be extended, due mainly to limitations of command and control. However, the frontages of an infantry battalion may be extended by increasing the inter section/platoon and company gaps.

c. Additional positions should be dug within the defence location at the platoon, company and battalion level. These could be occupied once the points of attack have been ascertained.

d. Larger gaps between the defended localities should be accepted at the level where adequate reserves are available to counter the enemy efforts to exploit the gap(s). Such reserves are generally available at the brigade level and above.

e. Light anti-tank elements within the battalion/brigade defensive locations can be given sizeable areas to guard.

7. ***Depth***

a. With wide frontages some penetration of the defensive position is inevitable. But to prevent the enemy from exploiting the penetration beyond own reaction capability, selected localities in depth should be held firmly.

b. Lightly held positions in the depth must have prepared positions for occupation at short notice from the neighbouring localities. Troops should be earmarked to prevent the enemy from consolidating and acquiring strength beyond own reaction capability.

c. Depth at the company and battalion level must always be retained. A battalion may, however, deploy even three companies up.

8. ***Mutual Support***. Defensive position consists of a number of mutually supporting defended localities and the mutual support at the unit/subunit level is defined as "the ability of adjacent localities to engage the enemy in the intervening gap by direct fire". Mutual support in this case can be defined as the ability of the next higher commander to readjust positions or to reinforce the threatened locality in time to deny the attacker his objective(s).

9. ***Reserves***. At battalion level, it will be difficult to have reserves that are uncommitted. Depth troops may be given the tasks of counter penetration, reinforcement or even launching of quick local counter-attack. Reinforcement at the company and platoon level will normally be by weapons.

**Planning of Defence**

10. The sequence of planning in case of extended frontages remains the same as for area defence. Planning and preparation for defence on extended frontage is a process that should start, and have a major portion completed, well before the outbreak of hostilities. A 'hasty defence' on the extended frontage should be avoided. When sufficient time is available additional positions at platoon, company and battalion level should be dug in which can provide immeasurable flexibility in the conduct of battle.

**Organization and Layout**

11. Depending upon the number of approaches into the battalion area, the main position will consist of compact company/platoon defended localities with large gaps in between them. Various combinations/options suggesting frontages for platoon, company and battalion are attached as annex W and its appendices. These may be applied judiciously according to the situation.

**Anti-Tank Defence Plan**

12. Keeping in view the increased frontages and threat, the Baktar Shikans and other anti-tank resources should be judiciously employed and integrated with the obstacles. The main defensive position will be built around the anti-tank defence plan. Bias of the obstacle should also be anti-tank.

**Communication**

13. Signal resources of the battalion signal platoon will have to be augmented for effective communication.

**Surveillance**

14. A comprehensive surveillance plan, integrating all available resources, has to be planned and executed just like a fire support plan catering both for day and night surveillance. Field radars and platoon early warning systems should be incorporated.

**Fire Support Plan**

15. No change is required from the normal procedure except that the defensive fire plan will be made according to sectors. The battalion mortars may have to be split and made mobile for quick deployment. Artillery resources will also have to be augmented for effective fire support over the extended frontages.

## SECTION 60 – DEFENCE ALONG A WATER OBSTACLE – CANAL/ANTI-TANK DITCH

1. A series of irrigation canals/water channels exist parallel to our eastern borders which are effective obstacles to an attacker. Where these do not exist, anti-tank ditches have been dug for the same purpose. However, incorporation of these in the defensive position creates some peculiar problems of their own and force the defender to deploy the force in linear fashion.

2. This chapter deals with peculiar problems of defence along a canal/anti-tank ditch and lays down the guidelines for effective organization and conduct of defence along such obstacles.

**Types and Characteristics of Obstacles**

3. ***Canals***. There are three main types found in Punjab:-

   a. ***Link Canals***. They were constructed as a result of the Indus Basin Water Treaty for transfer of water from the three western rivers to the eastern rivers. They are generally not used for irrigation and are the "in-cut type", that is, below the ground level. Both the banks are raised 1-1.2 metres (3 to 4 feet) above the ground level. Their width varies from 300 to 400 feet and depth from 10 to 15 feet. Water level may be from 6 to 12 feet. Width of the banks may be about 3 metres (10 feet).

   b. ***Irrigation Canals***. They are the "in filled" types and raised above the ground level to facilitate flow of water into irrigation channels and minors. Width varies from 60-90 metres (200 to 300 feet). Banks are higher than the ground level to prevent flooding of the surrounding area. Width of the banks varies from 3 to 4.5 metres (11-15 feet).

   c. ***Combination of the First Two Types***. They are partially 'in-cut' and partially in-filled. These include irrigation canals and also some of the link canals used for irrigation. Width varies from 60-90 metres (200 to 300 feet). Depth varies from 3 to 5 metres (10 to 16 feet). Their banks are 1.8-3 metres (6 to 10 feet) higher than ground level and are as wide as 3 metres (10 feet) or more on both sides. (In case of any of the above canals that are located close to the international border and are to be used for the defence purposes, their home banks are further raised by 1-1.2 metres (3 to 4 feet), mainly astride the major enemy approaches, to give better observation and field of fire).

   d. There are some other characteristics of these canals which are common to all:-

      (1) ***Water Level***. 1.8-3.6 metres (6 to 12 feet). May be increased or decreased depending on the season and who controls the headwork and regulators.

      (2) ***Bed***. Mostly loamy. Can be forded by the tanks in most cases.

      (3) ***Brick Lining***. In some cases, brick lining of the canal banks has been done.

      (4) ***Gradient***. Mostly steep, particularly where the banks are brick lined.

      (5) ***Current***. 2 to 3 knots. In case of link canals, it may be 4 to 5 knots.

(6) ***Approaches/Exits***. Marshy at most places due to escaping water channels, particularly when flowing to capacity.

(7) ***Observation and Field of Fire***. High banks provide good observation and field of fire but these are poor at some places due to elephant grass and villages. In case both the banks are of equal height, the observation towards the front and the flank is considerably restricted. Although the field of fire may be clear, yet the effect beyond the far bank is reduced due to plunging fire. The problem is more acute if the far bank is slightly higher than the home bank.

(8) ***Concealment***. Limited cover from air is available due to lines of trees on both sides. Positions on the canals cannot be concealed from the enemy due to their prominent and regular layout.

(9) ***Bridges***. Class 9/12 on tracks connecting population centres and class 40/70 along the main roads or regulators and siphons. Also, there are some foot bridges that can take jeeps.

(10) ***Lateral Movement***. Good service tracks exist on both the sides for the traffic, limited only by low classification of the culverts. Cross-country movement in the areas close to canals is restricted due to numerous escape channels and marshy areas. However, the service tracks along most of the canals exist either on the home/far bank or between the bank and the waterline, restricting their use for lateral move during active hostilities.

(11) ***Tactical Features in the Vicinity***. Other than high banks of the canals, numerous villages on the both sides may be the only features of tactical importance that affect observation and fields of fire.

(12) ***Obstacle Value***. The amphibious, fording and bridging capabilities of our potential enemy have considerably reduced the obstacle value of canals.

4. Vertical banks and marshy approaches/exits would, however, require expedients and/or preparation. Brick lined canals with water flowing 1/1.2 metres (3-4 feet) below the banks would be a problem for the enemy tanks during entry and exit. While expedients like assault track way, PSP sheets and summer-field tracks are available with the enemy for negotiating marshy patches, dozing of canal banks, particularly of the 'in-filled' type, would flood the area of approaches and exits, creating further problems. However, our potential enemy has been experimenting

and developing canal crossing techniques. Banks as deep as 6-9 metres (20 to 30 feet) may have to be cut through for approaches and exits. The distinct possibility of vertical envelopment by our enemy in the future further reduces the obstacle value of the canals, for initial phases of an attack operation. The obstacle in that case becomes more relevant to enemy build up later.

5. ***Ditches***

a. ***Width***. 24 metres (80 feet). In some cases, it is only 4.5-6 metres (15-20 feet).

b. ***Depth***. 2-3 metres (6-10 feet).

c. ***Bed***. Dry or marshy when filled with stagnant water. May be full or half filled.

d. ***Banks***. Only home bank is high 2-2.5 metres (6-8 feet) which is known as `bund'. Width of the bank is 1-1.5 metres (3-5 feet).

e. ***Gradient***. Is steep, except where filled due to lack of maintenance.

f. ***Approaches/Exits***. Good cross-country or where tracks/roads are available.

g. ***Concealment***. Is difficult, except behind the 'bund' on the home side.

h. ***Observation and Field of Fire***. Good except where restricted by villages, high crops and in some cases, by elephant grass.

i. ***Bridges***. Nil except foot bridges. However, crossing places are deliberately left along tracks/roads.

j. ***Tactical Features***. Only the bund on the home side and the villages in the vicinity has tactical importance.

k. ***Lateral Movement***. Is possible along service tracks.

l. ***Obstacle Value***. Ditches of 24 metres (80 feet) width or more, are effective anti-tank obstacles, if well maintained. But the width must be increased from time to time with the increase in enemy's bridging capability.

## SECTION 61 – PLANNING AND LAYOUT OF THE DEFENCE

1. The sequence of planning defence along a canal/ditch is the same as for the area defence. This should be studied in conjunction with "GSP-1985 – Operations of War (Volume-I), Defence – 2004'. In the succeeding paragraphs, only the points of difference have been discussed.

**Protective Detachments**

2. In a situation where normally little space is available ahead of the main position, or where the space is available but restricted, due to obstacles and BUAs, use of strong mobile protective detachments like covering troops is precluded. However, it would be normal for the protective detachments to act as a yet another kind of defensive line/position, ahead of the obstacle (canal/ditch) on which the main defences have been based. At battalion level and below, mask positions for some major weapons should be occupied ahead of the obstacle to deceive the enemy about the location of main defences and to cause casualties.

3. ***Organization***. To execute their task, the protective detachments will include some or all of the following components:-

a. Armour and anti-tank resources including Anti-Tank Guided Missiles.

b. Most of the divisional artillery in support.

c. Substantial infantry element to grant staying power to the mobile elements.

d. Adequate engineer support to prepare the obstacles.

e. Aviation support for observation, surveillance and command and control.

**Main Position**

4. ***Organization and Layout***. Depending upon the number of approaches into the divisional area, the main position will consist of compact brigade/battalion defended areas/localities, with gaps in between them. Reserves will be employed to destroy the enemy forces who have been able to cross the canal and to restore the canal line by a counter-attack. In some cases where the situation gets out of control, they may be used for counter penetration role.

5. ***Battalion in Defence***. Siting of Forward Defended Localities:-

a. ***Ditches***. When the ditches are too close to the border, there is no choice. In these the bunds have been so sited that there is no advantage in going ahead of these for siting the Forward Defended Localities. However, some mask positions besides protective detachments must be sited forward of a ditch to achieve some measure of surprise on the enemy.

b. ***Canals***. A commander on the spot is the best judge to decide whether to site the forward defended localities forward, behind or along a canal. A commander is required to make use of the canal for a more effective defence, in view of its natural strength. He will, therefore, be occupying a lower ground level than the canal. He will also have

certain elements across the canal which might pose tactical as well as logistic problems, especially during the battle. If close to the borders, the defences should be sited on the obstacle itself. The bulk of troops should be located on the higher bank, or the far bank if it is of equal height. The forward elements should fall behind the obstacle after giving a battle, to provide further depth to defence.

6. ***Frontages***. Suggested frontages for battalion and below are given in the Annex V and its Appendices 8 & 9. There is no change except that in case of a battalion with three companies up, it may cover upto 4115 metres (4500 yards).

7. ***Depth***

a. Greater depth than in normal deployment is required to balance lateral extension of the defended area. This does not imply committing bulk of the troops to the defensive positions sited in depth along the axis of advance. There should be sufficient numbers of alternative positions prepared in depth to which the uncommitted troops, or those from the un-threatened sectors, can be moved rapidly.

b. In our environment, it may not be possible to achieve standard depth at all levels. Following measures should, therefore, be adopted:-

(1) Retaining depth at battalion level at all costs.

(2) Use of obstacles and lightly held positions in depth.

8. ***Mutual Support***. In defence of canals, mutual support within and between the forward positions will be easier to achieve. Also, for the depth positions movement of troops to reinforce the threatened locality in time with troops from other positions, will be easier. The routes for this purpose must however be improved and maintained.

9. ***Reserves***

a. At the level of battalion and below, it will not be possible to have uncommitted reserves. Depth troops may counter penetrate, reinforce or even launch local counter-attacks. Reinforcement at company and platoon levels will normally be by the company/platoon weapons.

b. Normally the reserves should not be employed forward of the obstacle, unless the problems of crossing, restricted area of manoeuvre and a high risk of its premature commitment, have been fully evaluated. Brigade and battalion reserves may be employed forward of the obstacle only when the bulk of the troops located are holding important tactical ground.

10. ***Fire Support Plan***. No change is recommended from the normal procedure.

11. ***Anti-Tank Defence Plan at Battalion Level***

a. When forward defended localities are ahead, or on the canal/ditch, some anti-tank weapons in the forward defended localities are needed to shoot enemy tanks.

b. When on the home side of the obstacle system bulk of these weapons would be sited in depth. Some of these should also be earmarked as reserve for the counter-attack and counter penetration tasks in positions, where enemy breakthrough is likely to take place.

c. In both cases, sufficient weapons should also be deployed ahead of the forward defended localities in protective detachments and the mask positions. Villages provide good areas for deployment of mask positions and laying tank ambushes.

d. When the far bank is also high, those few located in the forward defended localities should be sited on the far bank.

e. Emplacements/bunkers of recoilless rifles and Anti-Tank Guided Missiles must be of low silhouette (below ground level if possible) and well concealed with alternative weapon positions.

12. ***Obstacle Planning***

a. ***Minefields***

(1) Ideally, there should be minefields on both sides of the canal/ ditch, irrespective of which side of the obstacle the bulk of troops is located. This would, however, impose serious restrictions on the manoeuvre of own reserves. Therefore, the minefields on the home side of the obstacle should only be laid in the gaps. However, this observation does not apply to the protective minefields.

(2) Minefields must also be planned and laid in depth, particularly along the tank approaches/exits and in front of the counter penetration positions, leaving areas for the counter-attacks.

(3) Gaps and lanes left in the minefields for own troops must be plugged when no longer required, before the enemy can exploit them.

(4) Even on ditches, the minefields are better sited on the far side as well as some in depth.

b. ***Wire Obstacles***. Double concertina wire and low wire entanglement must always be used in conjunction with the minefields, preferably on both sides of the obstacles.

c. ***Under Water Obstacles***. Picketed wire entanglements or floating but anchored zebra wire may be used in canals as well as ditches approximately 6 inches below the water surface. Concertina wire on the home end of the waterline, about 6 inches above the water surface, will be very effective. Floating mines may also be used, if available. Planning must include creation of underwater obstacles when the far bank has been lost to the enemy.

d. ***Demolitions***

(1) All bridges forward of the main obstacle and some of these on the obstacle itself, may be designated as preliminary, while the others as reserve demolitions.

(2) Troops should rely on rafts/boats for routine crossing in the company areas.

(3) Foot bridges may be used and also prepared for demolition.

e. ***Inundation***. Inundation plans must be made/executed at the highest possible level i.e. Division/Corps.

f. ***Flame Throwers***. Flame throwers sited on the home bank will create a ball of fire, as the enemy assault comes in. Plans must also be made to float a sheet of burning oil from the upstream to the area where the enemy crossing is taking place.

**Field Defences**

13. While constructing field defences along the water obstacles following guidelines will be useful:-

a. Canals, specially "in filled" type, or the ones which are partially "in cut" and partially "in filled", are special structures and quite sensitive to any earth work disturbance. Excavation done at random may cause a breach in the canal and result in an unwanted inundation. Liaison should therefore be carried out with the Irrigation Department for technical advice before carrying out any construction or excavation on the canal. During war, it will be a good practice to have earth filled bags available down to platoon headquarters level, if the Irrigation Department does not approve of any excavations inside the bund.

b. Concrete bunkers with overhead and frontal protection should be made for major weapons only. The roofs and walls should be strong enough to take direct hits from the field guns. Steel doors and windows that can be locked from inside should be provided for the loopholes. This will guard against enemy attempts to throw hand grenades or use pole charges.

c. A bunker constructed at ground level inside the bund will be more stable than the one in the middle or upper half of the bund. But this must be seen in the light of observation and field of fire afforded by the bunker. Bunkers may preferably be made in the centre of the bund. Following advantages will accrue due to this positioning:-

(1) It will reduce plunging fire.

(2) It would be more stable than the one constructed in the upper half and more economical as compared to the one constructed at the base.

(3) Profile of the bunker will not break the skyline, hence it will be easy to conceal.

d. Weight of the bunker and the super imposed load of the bund may cause collapsing of the ditch. This would require leaving of adequate space between the base of the bund and the edge of the ditch. A ledge of 10-15 feet is recommended to guard against this eventuality. Ledge can also be used by drag lines for subsequent maintenance of ditches.

e. Approaches to canals, being generally marshy, may require use of road expedients for transportation of construction material to the site.

f. Route in, route out and communication trenches for defences on the home side of the canal ditch or bund should be made as in the other areas, because these would generally be higher than the far side. However, the defences sited in the bund on the far side, can either be excavated laterally inside the bund or made of the sandbags and duly camouflaged. Communication trenches should be made on the reverse slope, in case no space is provided between the obstacle and the bund. In case space is available, these could be made in that space.

g. Keeping in view the threat of vertical envelopment, loopholes should also be provided in the rear of the bunkers to cover the area effectively. There should also an overlap with the adjacent bunker.

h. Firing bays should be so designed that they enable the weapons to fire effectively without projecting the muzzle out in the open. This can be achieved by having platforms to support the weapons behind the loopholes and a projection ahead and over the loophole. Such projections will also help in concealment of the bunkers by filling earth on it.

i. Where a choice is available between linear and kidney shaped bunds, constructed in a zigzag pattern, the latter should preferred as they offer following advantages:-

(1) Better deployment positions.

(2) More difficult for the enemy to pinpoint and engage the bunkers.

(3) Easy camouflage and concealment.

j. Dummy bunkers or dummy firing bays be constructed for deception at different/irregular intervals. This will deceive the enemy about the location of actual ones and, even if located, will divide the enemy fire.

14. Home banks of canals, which are either level or lower than the far bank, should be raised for better observation and fields of fire. Vegetation on the far bank should be controlled during peacetime so that fire lanes can easily be cut through, during emergency. Service roads on the canals are on the canal banks which may not be used during hostilities. It is, therefore, necessary to construct tracks behind the home banks of the operationally important canals.

15. ***Bridges***. Bridges being the most suitable and likely areas for assault crossing should be defended as strong points including those in the depth.

16. ***Location of Command Post***. Battalion command post should be sited in the rear. However, the battalion commander must visit/supervise the battle from any of his forward company areas under following situations:-

a. When his personal presence is warranted to influence the battle.

b. When launching a counter-attack.

c. During the preparatory stage and lull periods.

17. Company and platoon command posts per force have to be located within their areas. If no suitable sites in the depth are available, these may have to be on the bank/bund itself. These commanders change locations only along with their forward troops. An alternative command post in the depth may be prepared for the purpose.

18. ***Surprise and Security at the Battalion Level***. Following measures will be useful:-

a. Protective detachments and mask positions must endeavour to deceive the enemy regarding location of the main position.

b. Field defences must be safe proof against enemy artillery, air and direct firing weapons as much as possible.

c. Communication trenches laterally and rearward should be a regular feature of such defences.

d. All movements on the banks/bunds should be carried out during darkness.

e. Major weapons should have alternative bunkers/positions.

f. Tunnelling and low silhouette of the weapon emplacements on the canal bank/bund.

g. Use of Night Vision Devices detection means and silent sentries should be made.

h. Intensive patrolling, listening posts at night and mask positions will enhance security and surprise.

19. ***Administration at Battalion Level***

a. Traffic control and recovery arrangements at each bridge site in the battalion area will be needed.

b. Regimental aid post should be located behind the obstacle, but company aid posts may have to be located forward if the bulk of troops are on the far side. In addition, an ambulance car with medical staff on both sides of each bridge and boats/rafts as alternative means of crossing are required at each site. In view of the change in role, casualty evacuation of protective detachments should be given special and separate attention.

c. For feeding and other arrangements for daily maintenance of troops, the canal and movement across may have to be restricted to the hours of darkness.

d. Administration within the platoon/sections for washing/drying of clothes, bathing, rest, weapons cleaning etcetera during daytime should be well-organized and carried out in concealed places and on reverse slopes.

## SECTION 62 – THE CONDUCT OF DEFENSIVE BATTLE

**General**

1. Defensive battle in this case should essentially be conducted in accordance with the existing doctrine laid down for the area defence.

2. A few particular aspects, however, need special consideration:-

   a. ***Preparatory Phase***. In the preparatory phase, maximum possible enemy, including armour must be destroyed forward of the obstacle by a combination of all available fire including air and a coordinated use of protective detachments and obstacle system. These must be suitably organised to achieve that end. Lanes and gaps in the minefields must be closed in time.

   b. ***Assault Phase***. During this stage:-

      (1) Reserve demolitions must be fired in time.

      (2) Gaps in the main defences are to be occupied by the withdrawing elements, if not already occupied by the task forces.

      (3) Troops from less threatened areas are to be shifted to the areas of likely enemy breakthrough to achieve concentration of force and flexibility.

   c. ***Counter-Attack Phase***

      (1) Brigade and battalion reserves may counter-attack forward of the obstacle in rare cases and only if the bulk of the troops is located on the far side, to hold important tactical features.

      (2) Otherwise, reserves at all levels must be used to quickly destroy all serious penetrations on the home side of the obstacle. Normally, commanders are likely to get sufficient reaction time to move their reserves if they have been reading the battle correctly and anticipatory moves have been undertaken.

      (3) Where counter-attack fails, or is forestalled, further penetration must be halted at all costs by counter penetration and all other means, including reserve anti-tank weapons, reserve engineer troops and mines, air and artillery fire, till the higher commanders reserves become effective.

## SECTION 63 – MASK POSITION

1. ***Definition***. A weapon position selected ahead of main defensive position from where the weapon can perform the same tasks as from the original position. A position which is prepared to deceive the enemy about the original position/layout of weapons in the main defence. After completion of the task, the weapon is moved back to its original/alternative position.

2. ***Situation Warranting Employment of Mask Positions***. A clever blend of mask positions with dummy positions will pay dividends. A weapon team can occupy mask positions in any of the following situations:-

   a. When it is desired to deny information to the enemy about original location of the weapons in the main defence.

   b. During day, weapons are placed approximately 300-500 metres ahead in the mask positions to gain additional range, in order to engage enemy at longer ranges thus gain time/cause attrition.

   c. To occupy a dominating ground ahead of the original position.

   d. These positions can also be meshed with depth position and/or towards the flanks when required.

3. ***Strength/Composition of Mask Positions at Company/Battalion Level***. This will be commensurate with the type of weapons being employed in the mask positions. Complete crew of the weapon will move to the mask position.

## SECTION 64 – ANTI-INFILTRATION MEASURES

1. Infiltration can be conducted by ground or air. For the latter type, air superiority is not essential. Infiltration operations may be launched to annihilate an isolated location or as part of a bigger plan. The presence of infiltrating forces in/around/behind a defensive location causes panic and induces wrong movements by the defenders, resulting in loss of balance. The vulnerabilities of infiltrators are many. A well-trained defender, who maintains his senses and balance, can annihilate the infiltrators and create a very favourable ratio of force for himself. The problem of anti-infiltration must, therefore, fall into following two categories:-

   a. Psychological training of commanders and men.

   b. Anti-infiltration tactics.

2. Commanders at all levels and troops must be prepared to accept the fact that the chances of infiltration can at best only be minimised but not completely eliminated. They must at the same time be made aware of the enormous difficulties that an infiltrating force faces. Its limited offensive capability fire support, administrative support and handicap of operating in an unknown area make it an

easy prey to encirclement and annihilation. The defenders must, of course, hold fast to their positions and act promptly but in accordance with the prepared orders and plans. Rumours and demoralization caused by enemy action, which may or may not have been intentional, should be curbed. First, in these moments the junior leaders must be in the forefront. Second, all commanders must ensure against any psychological breakdown of their troops by strict vigilance, tact and maintenance of effective control throughout. The training of our troops for anti-infiltration measures must be based on psychological hardening and taking necessary precautionary measures. The under mentioned drills are suggested to build upon:-

a. ***Alertness***. Once infiltration has been established the troops should be alerted and all unnecessary movements about the locations stopped. No movement is allowed without orders.

b. ***Establishment of Communication***. Communication all along the chain must be checked and established. All commanders must be in touch with their subordinates and superiors.

c. ***Passage of Information***. The latest and the most authentic information must be passed to troops immediately on occurrence, so that rumours are eliminated.

d. ***Rest***. The troops should be rested immediately after the operation/ alarm. Frequently, there will be false alarms regarding infiltration. We must harden the troops to bear the strain of such alarms etcetera.

**Anti-Infiltration Tactics**

3. Anti-infiltration tactics comprise all the important actions required to be taken during defence operation. Certain features requiring added emphasis are discussed in the succeeding paragraphs.

4. ***Terrain***. Analysis of terrain must include a study of the likely infiltration routes/helicopter landing areas, from where it is possible to attack by infiltration. It should be realized that even the best of infantry troops cannot undertake an infiltration march of more than 24 kilometres during a single night under operational conditions. However, against a brigade size objective, it will be seldom necessary for a force to infiltrate 24 kilometres, before putting in an attack. Terrain analysis must highlight aspect of critical importance for the defended area and the routes leading to them. This should lead to planning remedial measures against the infiltration threat.

5. ***Reserve***. Mobile and well-rehearsed reserve force must be created at brigade level. It should be so located that it initially holds important ground and can be launched against infiltrators at short notice. Infiltrating columns of a bigger force pose some tricky problems about launching of reserves. The defender has to decide to launch it against the infiltrating columns or hold it back till the main attack has

developed. The discovery of infiltration, its size, time when it can be attacked, coupled with the location of own reserve and its capability to regroup after attacking the infiltration force, are some of the factors which must be taken into account before the launching of reserve. It is essential to maintain integrity of the defence area. Therefore, the brigade reserve must be launched only after careful consideration of all the pertinent factors. The finer point of decision would be whether to commit the reserve too soon to fall into an easy trap, or to delay it for too long to be effective.

6. ***Defensive Battle***. Other elements of defensive battle as taught and practised remain applicable. Following measures assume added importance and must be further refined and practised:-

a. ***Security of Defence Area***. Patrolling of possible routes, assembly areas etcetera should be organized properly. Standing patrols, employment of listening posts and search operations of the area must be organized on a regular basis. These patrols must be carefully briefed/debriefed. Quick and efficient means of communication be established.

b. ***Maintenance of Control***. Once security troops contact an infiltrating column, they must maintain contact with it and try to establish its strength and direction by harassing fire and withdrawal to successive positions. Maximum artillery fire should also be brought down on the enemy troops.

c. ***Security of Headquarters and Installations***. Headquarters and administrative installations are lucrative targets for infiltrating columns. The headquarters must have well-organized defences. Everyone should know his place in the depth locations or should be so deployed that he is protected from direct assault. All administrative elements must be prepared to defend themselves. No chances should be taken by them merely because of their location in depth.

d. ***Challenge and Fire-Control***. Challenge system should be enforced. Troops should follow the laid down methods properly. Junior commanders must keep a watch on the actions of the troops. No one should fire without orders. A strict fire discipline is essential to prevent a general chaos in the depth areas.

e. ***Command and Control***. Section and platoon commanders must immediately on being alerted, apprise the troops of the situation and keep an effective check on their actions. No man should act without orders. Rumours etcetera must be suppressed. Orders of the appointed commanders only should be executed. All commanders must be available to each other on telephone.

f. ***Movement***. All unnecessary movement must be stopped. Unidentified troops loitering about the place may draw fire upon themselves due to tense nerves, thereby causing panic.

g. ***Isolating the Infiltrating Force***. Having located and established the direction of infiltration, the defenders must take measures to isolate it. This can be done by any one or a combination of following actions:-

(1) Occupation of counter penetration positions to check its progress.

(2) Harassment and occupation of tactical features on flanks.

(3) Subject it to artillery/mortar fire.

(4) Raid on administrative echelons of the force.

(5) Cutting it off from the base.

h. ***Counter-Attack***. The infiltrating force should be quickly eliminated by a combination of armour and infantry assaults.

## SECTION 65 – ANTI-AIRBORNE OPERATIONS

**General**

1. At brigade/battalion level, airborne threat to be posed by the enemy will be of a tactical nature. The aim of the enemy will be to prevent reinforcements/ reserves reaching at a critical stage/place for the assistance of the forward troops while they are subjected to a ground attack. The enemy may occupy a tactical feature by helicopter borne/airborne troops which lies between the brigade reserve and forward battalion or may seize a bridge or centre of communication behind the forward brigade to impose delay or cause disruption in launching of the divisional reserve. The former threat is likely to cause more worries and anxieties to the forward troops. Such air threats will be combined invariably with the ground attack.

2. ***Arrangements for Obtaining/Passage of Information***. The following arrangements should be made:-

a. Troops to be prepositioned on likely enemy objectives/launching/ dropping zones and to keep strict surveillance over enemy landings in the area for keeping higher headquarters informed.

b. Observation posts/listening posts to be located on vantage points ensuring all-round coverage.

c. Local police force to be integrated in the counter airborne operations.

d. Locals of the area to be educated and trained in this aspect and instructed for keeping a watch and informing the nearest troops.

e. Arrangements with locals/police for passage of information:-

(1) On horseback.

(2) Bicycles where possible.

(3) Very light pistols, if given.

(4) By lighting big fires in the vicinity of landing/dropping zones - a prearranged signal.

(5) By beating of the drums.

(6) By torches.

(7) By active patrolling in the area.

(8) On foot when nearby.

(9) By telephone, wireless, vehicles and fireworks crackers in different colours.

**Organization for Anti-Airborne Defence**

3. Based on the appreciation of the brigade/battalion commander, following arrangements for dealing with the enemy airborne troops are undertaken:-

a. ***Warning System***. Warning stations should be established with communications on the flanks and in the rear of the battalion on vantage points. However, the forward companies also keep strict vigilance over the area in front of them.

b. ***Local Defence Scheme***

(1) It will not be desirable to assign any local defence commitment to the forward troops. They should be kept free for dealing with the enemy ground threat. However, minimum of one platoon from one of the depth companies may be earmarked by the forward battalions for dealing with enemy helicopter borne troops, which are not expected to be more than a company.

(2) Patrols may be posted on the likely areas where helicopter borne troops may be dropped.

(3) We will not be in a position to spare adequate troops from the fighting formations/units for anti-airborne operations. Therefore, likely landing zones may be mined and appropriate obstacles erected.

c. ***Employment of Mobile Reserve***

(1) ***Organization of Move to Landing Zone***. The company earmarked as mobile reserve to deal with enemy air/ helicopter borne troops is deployed on the ground, in a way that it denies possible landing places to the enemy by initially positioning the platoons on such features. It has got to be compact, so that it can quickly move out to deal with the threat in the rear areas. To achieve the mission the company will make the following arrangements:-

(a) Position all the platoons to possible landing places to deny these to the enemy.

(b) As far as possible, keep the platoons compact.

(c) Send out the patrols and observer posts only from one platoon so that two platoons can move instantly in emergency.

(d) If one platoon cannot cope with patrolling and observation post assignments then ensure that at least one platoon (reserve platoon) is ready all the time and is close to the road/track.

(e) Keep transport ready for all the platoons to make them mobile.

(f) Order of march should be clearly spelled out to all concerned. The reserve platoon which has to move first must be able to move within 5 minutes of the receipt of the Signal.

(g) Signal should be prearranged to recall the patrols/ observers posts with minimum of delay.

(h) The drill should be known to all and rehearsed.

(2) ***Move Plan***

(a) On receipt of the move signal, the company commander will push out a light platoon at once. In this case, it will be the reserve platoon.

(b) The platoon which had sent out patrols will be the last to move.

(c) The suggested order of march is as under:-

i. One jeep
ii. One jeep } Machine-gun 1A3 fitted in both.

iii. One dodge carrying a section and two machine-guns. Platoon commander will travel in this vehicle.

iv. One 2½ ton carrying the remaining platoon.

v. Gap 1000 metres.

vi. One jeep for company reconnaissance group.

vii. Two 2½ ton lorries carrying the last platoon.

viii. Two 2½ ton lorries carrying next platoon.

ix. Gap 1000 metres.

x. Remaining personnel of company headquarters to move with second and third platoon.

xi. The distance between two vehicles in a group will be 200 metres.

(d) Each group of vehicles to be under an officer/Junior Commissioned Officer or Non-Commissioned Officer.

(e) Commander to position himself from where he can observe in all directions.

(f) Hoods to be removed for observation and for command and control. If need be, men should be able to fire from the vehicle, though ordinarily it is not done.

(g) No lights (head or tail).

(h) Brake lights to be covered as they light up when applied and thus become visible from a distance.

(3) ***Action on Arrival in Landing Zone***

(a) Deployment of two sections (keeping one section reserve) around the landing zone to ensure that it is contained by own troops.

(b) Maximum fire of all available weapons to be brought down on the enemy.

(c) Making use of very lights and flares to establish the assembly places and strength of the enemy troops.

(d) Maintain continuous contact with the enemy.

(e) Keep company headquarters informed of the latest situation.

(4) ***Tactical Plan***

(a) The platoon that moves first, reaches the landing zone and secures the firm base and immediately pushes out patrols to find out about the enemy.

(b) The second platoon manoeuvres from the flank and occupies some tactical feature.

(c) The third platoon manoeuvres from a secure flank and goes beyond the second platoon and captures another tactical feature. This platoon immediately pushes out one or two standing patrols to keep a watch on the enemy.

(d) The firm base platoon now up sticks and joins the second platoon. The two platoons advance forward together to attack the enemy and clear the site while the third platoon gives fire support.

(e) The key note of the plan is that the platoons continue advance by leap frog method till contact with the enemy is gained and he is finally destroyed.

(5) ***Light Anti-Tank Platoon***

(a) In case light anti-tank platoon is allotted to the company, they will move first as they are mobile. The platoon will keep on advancing in the direction of the enemy until fired upon by the enemy.

(b) Once contact is gained with the enemy, the platoon will make firm base and support the manoeuvre of the company.

(c) Infantry platoon will also leap frog in the same manner as explained above. The only difference being that the firm base in this case has been established by the light anti-tank platoon instead of a rifle platoon.

## SECTION 66 – RELIEF IN THE LINE

**General**

1. The execution of relief in the forward areas is one of the commonest operations of war. It may well be the first contact a unit will have with the enemy in a forward area. Since such operations are normally carried out by night, it is essential to have a simple procedure which is known to all ranks and in which they have been trained.

2. If this type of operation is successful, the morale of units concerned will be raised if its conduct lacks precision morale will be impaired.

**Sequence**

3. The problem must be considered, as under:-

   a. Planning and preparation for move.

   b. The procedure to be adopted in actual relief.

4. Following should be emphasised during the planning stage:-

   a. Necessity for completing the relief as quickly and smoothly as possible.

   b. Silence.

**Preparation for the Move for Relief**

5. ***Warning Order***. In an ideal relief, a battalion commander should receive a warning order at least 48 hours before his unit is required to take over part of the line.

6. ***Battalion 'O' Group***. The battalion commander will immediately pass on all available information to his 'O' group. He will then go forward to the unit to be relieved and meet its commanding officer, returning later to take up his advance party at the same time. This procedure is preferable as it saves time.

7. The relieving commanding officer will settle details with the outgoing commanding officer. They must decide on the order of relief. It will be more normal for the forward companies to be relieved before reserve companies. They must also agree as to when command is to change. This will normally be when two companies have reported relief completed. Command within companies will normally change when two platoons have reported relief completed.

8. ***Advance Party***. The advance party should normally consist of the following:-

   a. Battalion commander.

   b. Intelligence officer, who may be sent back to brief the 'O' group.

c. One officer from each company, usually the company commander with a runner.

d. One representative of each platoon (the platoon commander or platoon havildar) with a runner.

e. Mortar platoon commander.

f. Battalion headquarters' guides under an officer.

g. Guides for transport under an officer.

9. The battalion signal officer and the line and exchange parties will take over as early as possible to ensure that the battalion communications continue uninterrupted during the relief. Intelligence personnel should, if possible, take over 24 hours in advance, so that they can be correctly and adequately briefed by the outgoing unit. Patrol representatives should go out with the outgoing unit's patrols covering the relief, in order to find gaps and familiarize themselves with the ground in front of the positions.

10. ***Duties of Advance Parties***. The main task of each member of the advance party is to meet his opposite number in the outgoing unit and to collect all possible information to enable the relief to be carried out smoothly and quickly. Advance party representative must list all points and information required before going forward to the unit to be relieved. Following are the main points about which various officers should receive full detail:-

a. ***Commanding Officer***

(1) Enemy dispositions.

(2) Enemy habits, especially patrolling activities.

(3) Enemy reactions to movement and fire.

(4) Enemy minefields.

(5) Localities to be relieved.

(6) Disposition of units on flanks.

(7) Artillery and mortar defensive fire and defensive fire (SOS) tasks.

(8) Own observation posts.

(9) Patrol reports and patrol charts.

(10) Own minefields and gaps.

(11) Details of dead ground, particularly on approaches to the positions which permit movement by day.

(12) Own wire.

(13) Communications, especially to observation posts.

(14) Locations of transport lines.

(15) What vehicles, if any, are kept in the battalion area.

(16) Weapons, ammunition and trench stores, especially cable, to be taken over.

(17) Situation about civilians in the area.

(18) Cooking and feeding arrangements.

(19) Any special difficulties about evacuating casualties from forward posts.

(20) Location of regimental aid posts.

(21) Routes to brigade headquarters.

b. ***Intelligence Officer***

(1) A trace of enemy dispositions.

(2) A trace of own dispositions and the troops on flanks.

(3) Map showing enemy fixed lines.

(4) Patrol chart.

(5) Minefield charts – own and enemy's.

(6) Information about wire-own and enemy's.

(7) Location of own and suspected enemy operations.

(8) Trace of defensive fire and defensive fire (SOS) tasks.

(9) File of patrol reports.

(10) Intelligence section log.

(11) Shelling reports.

(12) Track discipline.

(13) Enemy habits.

c. ***Mortar Officer***

(1) Dispositions.

(2) Task and ranges.

(3) Alternative positions.

(4) Operations.

(5) Range cards and panoramas.

(6) Enemy dispositions and habits.

(7) Any orders, particularly affecting mortars.

(8) Communications.

(9) Positions of headquarters mortar platoon, companies and battalion.

(10) Administrative arrangements.

(11) Ammunition supply and dumps.

d. ***Company Commander***

(1) Battalion dispositions, maps and air photographs.

(2) Information about enemy habits, snipers, patrols, machine-guns and mortar positions. Mines, wire and shelling.

(3) Detailed company dispositions, including number of men in each section.

(4) Arrangements for artillery and mortar supports.

(5) Light signals.

(6) Intercommunication plan.

(7) Positions of units on flanks.

(8) Company and other local observation posts.

(9) Night dispositions including listening posts.

(10) Anti-tank weapons and their tasks.

(11) Minefields.

(12) Trench stores.

(13) Runners' routes.

(14) Dead ground.

(15) Gas alarm.

(16) Vehicles in the area.

(17) Feeding and sanitary arrangements.

11. ***Preparation at Platoon Level***. Assuming that the platoon havildar is going forward with the advance party, the platoon commander should do the following:-

a. Brief the platoon havildar, ensuring that he takes:-

(1) Strength of the platoon to enable him to estimate the number of fire trenches required.

(2) A written list of all information required.

(3) A compass.

b. ***Inspect the Men and the Stores***. This will entail:-

(1) Ensuring that all men have a thorough wash.

(2) Inspecting clothing, necessaries and equipment.

c. ***Brief the Platoon***. This will mean:-

(1) Explaining exactly what is going to happen.

(2) Emphasizing the need for maintaining contact with one another and for a slow space in front. Give information about darkness and the moon.

(3) Demanding absolute silent movement so as not to arouse enemy's suspicion.

(4) Giving details of actions to be taken should the enemy attack during relief. Orders will be based on fighting back.

(5) Warning the platoon to expect night battle noises and shelling, since this is normal enemy routine and is not directed at them in particular.

(6) Explaining the password and its use.

## SECTION 67 – RELIEF PROCEDURES

**Move from Rest Area to Debussing Point**

1. The battalion will usually move from the rest area to the debussing point in transport. It will debus at the debussing point.

**Marking of Debussing Point**

2. The outgoing unit will be responsible for marking the debussing point and the route forward.

3. The outgoing unit will mark the route forward using the incoming unit's things, which will be brought forward by the advance party.

4. The incoming subunit will meet the guides from the equivalent subunit of the outgoing unit. It will be led along its route by the guides to the battalion checkpoint. The route will be taped and lit, if necessary.

5. Companies will move to the battalion checkpoint in single file. This procedure will be laid down in the initial orders and will depend on the degree of darkness and enemy harassing fire.

**Battalion Checkpoint**

6. A convenient battalion checkpoint must be selected to collect reports and to report progress. It should normally be in the rear of the battalion area and must be at a point where all outgoing and relieving troops pass. It is marked by the outgoing unit but manned by one outgoing and one incoming unit officer. They will have two PRC-77/PRC-4620 wireless sets and line laid to the outgoing battalion headquarters.

7. These officers will check their battalions out and in respectively and report progress to their commanding officers who will remain at the battalion headquarters throughout the relief.

8. Normally one PRC-77/PRC-4620 set will be used, the second being kept in reserve.

9. The arrival of the unit at the battalion checkpoint is reported as follows:-

a. If possible, by line laid specially for the occasion.

b. By wireless, if line breaks down.

10. Each subunit is checked through the battalion checkpoint to the company checkpoint and the commanding officer is informed.

**Company Checkpoints**

11. The routes from the battalion checkpoint to the company checkpoints may be lit. They will certainly be taped in difficult places. Platoon representative with runners and a guide from each outgoing platoon will be at the company checkpoint. They will meet their respective platoons and lead them off independently to their positions.

12. If any of the platoons spread out during the march from the battalion checkpoint, they will reorganise here before going further. Company headquarters should meet its company commander or second-in-command at this point. Company headquarters should meet at its company point. They will also reform here before going further. Generally, the outgoing Company Havildar Major is in-charge of the company checkpoint.

13. The incoming company commander must immediately report his arrival at company headquarters to the battalion commander.

14. If the battalion checkpoint is suitable as a company checkpoint, it may be so used. Platoon representatives will then be found at the battalion checkpoint. Company checkpoints are normally provided for forward companies only.

15. Subunits must detail a responsible sepoy or Non-Commissioned Officer, to report when his subunit is clear of the checkpoints.

**Platoon Positions**

16. At platoon headquarters, guides from the outgoing sections will guide the incoming sections to their positions. The incoming platoon representative must brief the guide and give him:-

a. The names of the incoming section commanders.

b. Strength of the sections.

17. The outgoing sections will be "stand to", while the incoming sections on arrival in their positions will get down behind the fire trenches which they are to occupy.

18. The incoming platoon commander will visit all trenches with his opposite number and make a final check. When he is satisfied that everything is ready, he will at his own discretion or on orders from his company commander, begin the handing/taking over.

**Handing Over of Command**

19. The actual handing over should be supervised by the platoon havildar and the relief be conducted section by section. Men should be "stand to" until ordered to "stand down". They should also be given information about all the patrols out in front. When a particular section has been relieved, it will immediately withdraw to its platoon RV. After the whole platoon has reached the platoon RV, the platoon commander will take it to the company RV.

**Runners**

20. As soon as the platoon arrives in the position, the incoming platoon commander will send two runners to the company headquarters. One of them will report the platoon's arrival and will remain in the company headquarters, while the other will make himself familiar with the route and return to platoon headquarters.

21. The runner retained by the company commander may well be the advance party runner as he would have had the advantage of seeing the localities and the general area by daylight.

**Battalion Supporting Arms**

22. Relief of the supporting arms and of subunits of the headquarters' company is carried out concurrently with that of the infantry subunits.

**Protection during Relief**

23. Protection during the relief must be given by the covering parties. These are normally provided by the outgoing unit but, whenever possible, men of the incoming unit should be included in them to get first-hand knowledge of the ground. Patrolling can then be undertaken more effectively by the incoming unit after the relief. Patrols provided by the outgoing unit will only be withdrawn after the relief and the relieving unit must know where to direct these patrols on their withdrawal.

24. If the alarm is given during the operation, it is the responsibility of the outgoing company commander to coordinate actions to deal with the emergency.

**Speed in Relief**

25. It is essential that by first light the incoming unit should be completely settled and concealed in its new area. Unnecessary movement must be strictly controlled as it can draw fire. If a simple drill for relief is known to every man and if the subunit commanders have been properly briefed and given sufficient time for planning, before moving up, a successful relief will result.

**Secrecy**

26. It must be impressed on all ranks that the relief should be carried out by silent movement. Noise will make the enemy suspicious and it may lead to an attack during the relief. Normal harassing fire and patrolling should continue as usual. Complete silence and apparent inactivity can also make enemy equally suspicious. The normal communication should continue, but the incoming unit must have complete silence as new voices can disclose their presence in the area. If possible, only telephone should be used.

**Day Relief**

27. It is wrong to assume that relief will always be carried out by night only. In certain circumstances, they will be carried out by day. A relief by day has certain advantages. Circumstances that favour a daylight relief are:-

a. When the enemy has no direct observation over the area in which the relief is to take place.

b. When our air superiority over the battle front is such as to preclude the probability of enemy air action.

c. When the nature of the country is so difficult for infantry and motor transport movement that the hours of darkness are not considered sufficient to complete the operation.

d. Relief of reserve companies to speed up actual night relief.

28. After the outgoing unit has left, the incoming unit or subunit commander must check positions to confirm that they are the most suitable. The outgoing units positions must not be accepted unless a thorough study of the ground has been made.

## SECTION 68 – ORDERS FOR NIGHT RELIEF

1. Battalion orders for night relief are often embodied in a order. The following points, related directly to the relief, should be included:-

a. ***Advance Parties***

(1) Composition.

(2) Transport.

(3) RV.

(4) Whether to return to the battalion or not.

b. Details of units to be relieved.

c. Allotment of battalion headquarters and companies to the area and details of the companies to be relieved.

d. Allotment of support weapons to the company area.

e. Location of debussing point.

f. Time of arrival at debussing point.

g. What guidance will be arranged/marked?

h. How the route will be marked after the debussing point.

i. Order of march from the debussing point and gaps.

j. Whether transport is to move with the companies or to come up afterwards at the rear.

k. What transport is to remain in the battalion area.

l. Location of distinguishing signs at the checkpoint.

m. What guides will be at the checkpoint.

n. Stores and weapons, if any, to be taken over on the site.

o. ***Patrols***

(1) What covering patrols are out.

(2) When and where will they come in.

(3) Whether own unit is sending any patrols.

p. Actions if the enemy attacks during the relief.

q. When are the companies required to stand down after the relief.

r. ***Actions at the First Light***

(1) Time of stand to.

(2) Fires.

(3) Movement.

(4) Time by which observation posts are to be established.

s. Password.

t. Code words for artillery defensive fire and defensive fire/(SOS) tasks.

u. Code words denoting completion of relief.

v. Evacuation and feeding of casualties during and after the relief.

w. The need for silence.

x. The change of responsibility for communications must be clearly laid down.

## SECTION 69 – REVERSE SLOPE DEFENCE

**General**

1. ***Use***. A reverse slope defence is the one organized on the portion of terrain feature that is masked by a crest from the enemy's direct fire and ground observation. All or any part of the force on the forward defended localities may be on the reverse slope, depending upon the terrain in the area to be defended and the situation. A successful reverse slope defence depends on control of the crest by fire, or by physical occupation. Reverse slope defence may be advantageous when any of the following conditions exist:-

a. When the forward slope is untenable because of enemy fire.

b. When the forward slope has been lost, or not yet regained.

c. When the terrain on the reverse slope gives better fields of fire than the forward slope.

d. When necessary to avoid occupying disadvantageous terrain such as a dangerous salient or re-entrant.

e. When possession of the forward slope is not essential for observation/defence.

f. When necessary to assist in achieving deception and surprise.

g. When the defender can retain the ability to ensure that the crest and the forward slopes can be made untenable to the enemy. This is best done by fire from the defences on the flanking features.

2. ***Advantages and Disadvantages***

a. ***Advantages***. In a reverse slope defence, the forward elements of the battle area are protected from the enemy ground observation and his direct firing weapons. Enemy's indirect firing weapons will normally deliver unobserved fire, which reduces their effectiveness. Besides, the defender may effectively employ deceptive measures such as dummy positions etcetera on the forward slopes. These advantages permit greater freedom of movement, more detailed improvement of the position, ease of supply and rest, for the troops.

b. ***Disadvantages***. The major disadvantages are the difficulty of maintaining observation of the enemy and the restricted range for the direct firing weapons. Without observation, effectiveness of the friendly direct and indirect fire is limited. Obstacles and minefields on the forward slope cannot be covered by direct firing weapons. If the enemy seizes the crest, he can generally have the advantage of attacking and developing further operations moving downhill, while a counter-attack to eject him will involve moving uphill.

**Organization of the Reverse Slope Defence**

3. ***Disposition of Troops***. The reverse slope is generally organized according to the fundamentals of defence. Special considerations include the following:-

a. The lay of sand dunes in desert and tops in mountainous terrain, lends itself to occupation of reverse slope defence. Anticipated direction of enemy assault should be visualized while siting positions on the reverse slope. One must guard against, a false sense of security, where reverse slope positions may be under observation from a distant feature or from a flank. When a reverse slope position has been established, the forward slope may just be occupied as under:-

(1) ***By Day***. By observation posts and protective patrols. Forward observation officers and mobile fire controllers must be capable of bringing down accurate fire all along the approaches and onto the forward slopes.

(2) ***By Night***. The crest must be occupied in sufficient strength by infantry to prevent the enemy from capturing it.

b. The crest in front of the forward companies/platoons must be within the effective small arms range. The troops are deployed in order to

bring maximum fire on the crest, the approaches around the crest and the forward slope of the adjacent terrain features. Desirable location for the reserve company/platoon is the crest of the next high ground to the rear but within the supporting range of the forward elements.

4. ***Fire Support***

a. Machine-guns and other automatic weapons are placed where they can deliver the most effective surprise fire on the enemy as he crosses the crest. Their sectors of fire provide maximum coverage between the forward defended localities and the crest.

b. In the reverse slope defence, anti-tank weapons may preferably be positioned in the reserve platoon area. Firing positions on the next high ground behind the forward positions often give better fields of fire than the positions in the forward area.

c. The reserve company/platoon in a reverse slope defence can often provide more effective fire support for forward platoons.

d. Finally protective firing from indirect fire weapons are normally placed along the topographical crest or for crest of the hill to deny it to the enemy. When the forward defended localities are as much as 500 metres behind the crest, it may be more advisable to locate the artillery and mortar fire closer to the forward defended localities than to the crest. In this case, targets are planned on both sides of the crest to keep the enemy under fire as he advances into the area where artillery and mortar fire have been planned.

**Conduct**

5. The conduct of a reverse slope defence is generally on the lines of forward slope defence. The observation and security groups give an early warning of enemy's approach and delay and disorganize him with long-range fire. To give a better coverage of the front during periods of darkness or poor visibility, warning and illuminating devices such as trip flares, searchlights, infra-red equipment, Night Vision Devices and mines are used extensively. As the enemy forces the observations and security groups to withdraw, rearranged targets are fired at on the forward slope to disrupt the enemy attempts to mass for an assault. Automatic weapons with the security groups are withdrawn first so that they can occupy their primary firing positions before the enemy reaches the crest. Direct firing weapons within the battle area withhold their fire until a suitable target appears. As the enemy makes a penetration, the counter-attacking force destroys or ejects him and re-establishes the security group on the crest.

## SECTION 70 – PLANNING AND CONDUCT OF SPOILING ATTACK

1. A spoiling attack is a tactical manoeuvre employed to impair or delay a hostile attack. It is normally launched while the enemy is in the process of forming up or assembling for an attack. It is executed to:-

a. Destroy a portion of enemy force.

b. Throw the enemy off balance.

c. Deny the enemy surveillance of the forward defended localities.

d. Delay his attack and upset the timings.

e. Discourage and disorganize all further attempts to attack own defensive position.

f. Gain time and force the enemy to disperse his strength for providing protection to his troops and launching areas.

**Essential Requirements**

2. Following points must be kept in mind:-

a. The success of spoiling attack will depend upon the initiative and violent action by the junior leaders. No physical assault on the enemy is envisaged. Instead, only attack by fire is carried out.

b. The plan will have to be well rehearsed and fire support coordinated. At times the troops may have to be kept in the close proximity of the likely enemy Forming Up Places and Assembly Areas.

c. The decision to commit a force in spoiling attack must be carefully considered against the possible non availability of these troops and weapons in the main defensive position if the enemy is still able to launch an attack. At company level, only about one section strength may be adequate.

d. The plan for spoiling attack will normally not be initiated below company level.

e. Before launching spoiling attack, it is important to detect, identify and confirm enemy intentions as early as possible.

**Early Detection**

3. Deliberate effort must be made at all times to detect, identify, pinpoint enemy concentration for attack as early as possible. Some of the means are:-

a. Observation posts (Intelligence and Artillery).

b. Listening posts/standing patrols on likely avenues of enemy approaches.

c. Patrolling.

d. Radio interception.

e. Air observation posts.

f. Interrogation of prisoners of war or civilians of the area.

g. Ground surveillance radars.

h. Unmanned Armoured Vehicles/Recovery Personnel Vehicles.

**Conduct and Planning**

4. Deliberate pre-planning is essential for launching a successful spoiling attack. Therefore, it is essential to:-

a. Earmark troops and commanders for each contingency.

b. Rehearse movement and actions during day and night.

c. Integrate all available fire support for each action.

d. At battalion level upto a platoon with automatic weapons (Machine-gun/Grenade Launcher Heavy-Machine gun etcetera), 60 millimetres Mortar, RPG-7 etcetera may be employed. At times, standing patrols may be assigned to take on the spoiling attack.

e. Spoiling attacks differ from raids, in that these are invariably launched to spoil an imminent hostile attack by disrupting and delaying it. To this extent, these are in the nature of pre-emptive offensive actions by the defender.

## SECTION 71 – DEFENCE AGAINST NAPALM BOMB

**Aim**

1. The aim of this section is to provide information on the capabilities and limitations of napalm bomb and the defensive measures against its effects.

**Effects of Napalm Bomb**

2. Napalm bomb may contain 450-700 litres of gelatinized fuel, weighing 350-450 kilograms. Fighter or fighter-bomber aircraft usually carry two bombs, one under each wing.

3. On impact, the bomb produces a fireball and spreads thickened fuel over an elliptical area, approximately 100 metres long and 40 metres wide. The fireball will burn for 6-10 seconds with intense heat, thereafter the mixture will continue to burn for 6-10 minutes with reduced intensity. The temperature generated is as high as $1000^{0}$-$1200^{0}$ C. Some types of napalm thickened with polymeric substance will also emit pungent fumes that irritate the eyes and the respiratory system.

4. The general effect of napalm bomb on various types of targets is as under:-

a. ***Troops***. Personnel in the open are extremely vulnerable to napalm attack. Those in the bunkers or trenches with overhead cover are less susceptible to the fire hazards. However, they suffer from heat and the effects of hot gases. Psychologically, the fire has a great demoralizing effect, unless the troops are well-trained in defence against a napalm bomb.

b. ***Tanks, Vehicles and Other Weapons***. Napalm is very effective against armour, vehicles and weapons. The thickened fuel will stick to the surface and may get drawn into the vehicle through any open ports. If it remains on the surface, the damage may not be as serious. Once the napalm gets inside the tanks or vehicles, it may set fire to the ammunition, patrol etcetera causing serious damage. Besides this barrels of guns or other weapons may be damaged by the intense heat, if they are exposed to sufficient quantity of burning fuel.

c. ***Building and Other Structures***. Napalm is most effective against structures, containing inflammable materials such as petroleum products explosives, wood and straw. Against reinforced concrete buildings, the effects of Napalm are negligible. Brick and masonry structures, such as permanent fortified positions, factories, bridges, tunnels etcetera, cannot be easily penetrated and are thus less susceptible to harmful effect of napalm.

d. ***Machinery and Installations***. Non-combustible machinery and tools may be twisted out of shape by the heat. Radar and radio equipment may also be seriously damaged.

5. ***Targets***. Following targets are suitable for napalm attack:-

a. Concentrations of troops in the open.

b. Vehicles, including armour.

c. Gun positions.

d. Supply or equipment concentrations.

e. Command posts.

f. Airfields, aircraft and radar installations.

**Limitations of Napalm Attack**

6. Napalm Bomb Attack has following limitations which can help in defensive measures against such attacks:-

a. The attacking aircraft has to fly on a constant course upto the target.

b. The aircraft has to fly at a relatively slower speed i.e. 480-510 kilometres (300-350 metres per hour).

c. The attack on the target is delivered from a very low altitude i.e. about 50 feet.

d. The duration of effects of napalm attack is relatively short.

**Protective Action**

7. The greatest impact of napalm attack is its psychological effect. The demoralization and confusion caused may result in abandonment of defensive measures resulting in great damage. Well-trained and motivated troops, displaying calm and presence of mind, will minimize damage to personnel and equipment and continue to perform the assigned tasks. A thorough indoctrination of the individual soldier in the capabilities and limitations of napalm and efficient defensive measures is essential to prevent unnecessary damage.

8. When subjected to napalm attack, protection is provided by the following:-

a. Shelters and dugouts.

b. Trenches with overhead cover.

c. Natural cover, such as ravines, ditches and pits.

d. Stone or brick buildings.

e. Locally available non-combustible material such as green grass. Green branches or thick clumps of trees. These however, provide only partial/temporary protection because they are themselves susceptible to fire.

f. Tanks and Armoured Personnel Carriers when closed down, offer temporary protection.

g. Short time protection can be provided by overcoats, ground sheets and capes etcetera, wrapped over the body.

9. Following preparation/action by the troops in their respective fields of activity would serve to reduce the effects of napalm attack:-

a. Well dug in and camouflaged positions.

b. Overhead cover especially for ammunition pits and command posts.

c. Burying of telephone cable.

d. Coordinated siting and effective use of all weapons in AA role.

e. Knowledge of correct fire-fighting drills to minimise damage to personnel, weapons and equipment.

f. Use of fire extinguishers and other improvised means such as water, mud, earth, capes, canvas etcetera for putting out fire.

g. When engulfed by fire, the crew should keep the cupola closed, jettison external fuel tanks and quickly drive out of the fire area at an acute angle.

h. Areas where fires are easy to start, such as dry crops, grass etcetera should be avoided when occupying positions. If this is not possible, fire lanes should be cleared. Inflammable material and equipment, such as ammunition, petrol, vehicles etcetera should be isolated to prevent the spread of fire.

i. Good knowledge for providing prompt first aid to the victims of burns.

**Active Defence**

10. Aircraft making napalm bomb attack is highly vulnerable to ground fire due to the limitations given in paragraphs 9 above. Even if a 'hit' is not achieved, a barrage of automatic fire from all arms will make the task of accurate delivery of napalm bomb more difficult. A high standard of training in the use of machine-guns and small arms in anti-aircraft role, therefore, assumes a great significance. Good Anti-Aircraft discipline will make the job of the pilot hazardous.

11. ***Fire-Fighting***. When napalm attack has been delivered, the first action required to be taken is to extinguish or localize any fire which may affect personnel, weapons, equipment and ammunition etcetera. Use should be made of all available fire-fighting means, such as water, fire-extinguishers, earth, sand and mud or any other local material. An important element of fire-fighting is the administration of prompt first aid to personnel affected by burns.

12. ***First Aid***. Following first aid treatment is recommended irrespective of the degree of the burn:-

a. Extinguish flame by wrapping the individual in a blanket, ground sheet or by pouring water on him. Putting out fire by water will help in reducing the temperature of the burnt surface as well.

b. The individual should lie at rest.

c. Cover the burnt area with a clean sheet. Face should be left exposed.

d. Topical applications should be avoided, i.e. no ointment etcetera be applied to the burnt area.

e. Hot and sweet drinks containing salt should be administered.

f. Pain killing drugs available should be administered and record of the type and dose given be sent along with the casualty.

13. Minimum handling of the casualty, especially the burnt surface, is important to avoid further damage to the part and eventual complications.

**Training – Specific Measures**

14. Following measures should be specially stressed during training:-

a. Thorough indoctrination of troops in the enemy's napalm capabilities and limitations.

b. Use of natural cover, when attacked by napalm.

c. Knowledge and use of correct fire-fighting drills/equipment.

d. Use of improvised means for putting out fire.

e. Salvage of equipment.

f. Correct siting and effective use of weapons in the anti-aircraft role.

g. Providing prompt first aid to burn victims.

h. Treatment of victims for shock effect.

## CHAPTER XVI – WITHDRAWAL

### SECTION 72 – GENERAL

**Introduction**

1. This chapter should be read in conjunction with 'GSP-6-100-O – Operations of War (Volume-III), Withdrawal – 2006'.

2. Withdrawal is an operation of war in which a deployed force disengages from an enemy force and executes a planned move to the rear or away from it. It may be carried out either as part of a tactical manoeuvre within the framework of a tactical battle, or under one of the following conditions:-

a. In order to disengage from an unsuccessful battle or as a result of defeat.

b. To avoid accepting a battle under unfavourable conditions and thereby gaining time for creating overall favourable conditions to resume the offensive.

c. In order to coordinate defences with neighbouring units/formations on the flanks.

d. On accomplishment of mission in one area and/or change of role/ posture, when the unit/formation may be required in some other area/sector, for a similar or different task.

3. Troops engaged in withdrawal will be involved in one or more of the following:-

a. Defence of the original area of responsibility.

b. Imposing delay or gaining time for the preparation of the new defensive position. This is done normally through the following:-

(1) Covering troops.

(2) Intermediate position.

(3) Flank guards, rearguards.

(4) Counter-attacks and spoiling attacks.

c. Occupying the new defensive position.

4. Some of the frequently used terminologies in withdrawal are defined in following paragraphs:-

a. ***Covering Troops***. A mobile force that operates ahead, in the rear or on the flanks of the main force. In withdrawal, their tasks resemble those of the rearguard. The tasks of covering troops are:-

(1) Providing information about the enemy and the terrain.

(2) Delaying and harassing enemy forces.

(3) Preventing the enemy from gaining information and contact with the main force.

b. ***Delaying Position***. Position taken to slow down/delay the advance of the enemy without being decisively engaged.

c. ***Intermediate Position***. During withdrawal, a defensive position organised by an all arms combined force, but predominately by infantry, and designed to halt/delay the enemy advance to gain more time for preparation of the next main position in the rear. The intermediate position should be back enough so that the enemy is forced to redeploy his artillery, if he wishes to attack it.

d. ***Rearguards***. The rearmost element of an advancing or withdrawing force. It has following functions:-

(1) To protect the rear of a column from a hostile force.

(2) During withdrawal, to delay the enemy and to ensure safety of the main force from any chances of involvement in a running fight.

e. ***Rear Parties***. Troops, who remain in position in the forward defended localities in withdrawal until time for final abandonment has been reached, in order to prevent a quick follow-up by the enemy.

**Night Withdrawal**

5. Night withdrawal is usually preferred to daylight withdrawal in order to maintain secrecy, achieve surprise and reduce casualties. A night withdrawal normally starts at such a time as to ensure its completion before daylight. Hours of darkness are utilised to the maximum for the rearward movement of the main body.

**Daylight Withdrawal**

6. Daylight withdrawal should normally be avoided unless weather conditions favour it or ground conditions necessitate it. A daylight withdrawal is subject to enemy observation and its success depends on mobility, control, long-range firepower and the employment of security elements. However, surprise is very difficult to achieve. Withdrawing troops are exposed to enemy air attack, as there are more chances of the enemy enjoying air superiority during withdrawal. Hence, there is every likelihood of the enemy air being able to interfere so effectively with the progress of withdrawal that it gets turned it into a rout or unacceptable losses are inflicted on men and material.

**Day Versus Night**

7. It is, therefore, clear that the battalion can withdraw more easily by night than by day. In the latter case, unless surprise is achieved and there are good covered lines of withdrawal combined with an obstacle, it will be difficult to avoid a running fight and severe casualties.

## SECTION 73 – GENERAL CONSIDERATIONS

**General**

1. A force in defence may have to withdraw under one of the following situations:-

a. When the withdrawing force itself has suffered no major reverses and the withdrawal is due to enemy action elsewhere.

b. When a withdrawal is the result of a local tactical defeat suffered by the withdrawing force.

2. Whatever the situation:-

a. The withdrawing battalion may be moving back to a depth position behind another unit, which is already deployed on ground.

b. It may have to prepare another position of its own at a suitable place in the rear and at the same time carry-out a withdrawal from the present position.

c. A withdrawal is never carried out by a unit, except on the orders of the higher commander.

3. ***Implications***

a. If the withdrawal is the result of enemy action elsewhere, the withdrawing troops will have enough time and strength for the occupation of the new and intermediate positions. The withdrawal in such a case will therefore be carried out comparatively easily.

b. If on the other hand, the withdrawal is the result of a local tactical defeat, the problems of the withdrawing commander will vary according to the nature and scale of the setback.

c. However, as long as the commander is prepared for all eventualities, there should be no difficulty in conducting a withdrawal even under the most adverse circumstances. The key to success is that the commander must foresee and plan for all possible eventualities in advance. He must not, under any circumstance, allow events to surprise him.

**Aim of the Battalion Commander in Withdrawal**

4. Whatever the situation, the aim of the battalion commander in all cases must be to maintain an intact front throughout and to occupy the new defensive position with as little interference from the enemy as possible.

**Fundamentals of Withdrawal**

5. Fundamentals of withdrawal are:-

a. ***Control***

(1) Withdrawal at any level is a difficult operation to control. The highest commander must make a coordinated plan which will be controlled by a timed programme along the whole front. Orders must be sent down to the foremost troops in good time. Finally, if the commander is to contend with the unpredictable events that so often take place during withdrawal, the plan must be both simple and flexible. The following will assist in formulating a simple plan and will ensure a degree of flexibility:-

(a) Avoidance of tight timings.

(b) Early move back of unwanted vehicles and equipment.

(c) Maintenance of units and subunits intact during withdrawal.

(d) Reduction in the numbers of commanders to be sent back to reconnoitre the new positions.

(2) At platoon level, this is achieved by:-

(a) The platoon commander being the last man to leave the defensive position.

(b) Keeping the platoon' RV close behind the platoon positions.

(c) Ensuring that all section commanders know the routes back to the platoon and company checkpoints and RV.

(d) Use of single file formation, with the commander at the head and a Non-Commission Officer at the tail.

b. ***Clean Break***. To avoid a running fight and to achieve a successful withdrawal, a clean break is essential. A clean break may be achieved by one or a combination of the following actions:-

(1) Maintenance of intact front with a portion of the force and normal activity along the front till the very last minute.

(2) The establishment of one or more intermediate positions.

(3) The use of rear parties/covering troops/rearguards.

(4) Secrecy.

(5) Night withdrawal.

(6) A counter-attack or a spoiling attack.

(7) ***Security***. To achieve this, leap frog must be resorted to. While doing this, rearward hop should traverse so much space that the enemy is forced to redeploy at least his field guns and also reassemble/regroup his attacking forces. This is how adequate time is gained and enough attrition is caused to the enemy.

c. ***Maintenance of Morale***

(1) A withdrawal entails much greater strain on the morale of troops than any other operation of war. Low morale may turn a withdrawal into a rout. Therefore, it is important to ensure that everything is done to counter this. Following will assist in the maintenance of morale:-

(a) ***Information***. All ranks must be given immediate and reliable information. Lack of it will encourage rumours,

mostly of a depressing nature. A sound system of security is essential. The enemy must also be kept under observation by all means available, particularly against outflanking moves.

(b) ***Administration***. Good arrangements must be made throughout the withdrawal for the supply of food, ammunition and petrol.

(c) ***Discipline***. Discipline is vital. There must be no relaxing of the normal safeguarding and maintenance of equipment. It may be necessary to abandon certain stores and equipment, but this will only be done on the instructions of the highest commander.

(d) ***Leadership***. As always, leadership will take the troops through anything.

(2) At company/platoon level this is achieved by:-

(a) The setting of a personal example by the commanders.

(b) Counteracting false rumours and exaggerated reports by keeping the men informed.

(c) Making full efforts to prevent wounded men from being captured by the enemy.

d. ***Gaining Time for Occupation of the New Position***. Adequate time is required for preparation of new defensive position(s). This can only be achieved if all possible measures are adapted to trade minimum space for maximum time. A platoon may be asked to contribute to this effort by being made part of the rear party. Time can be gained by adopting following measures:-

(1) Having the new position well in the rear i.e. trading space for time. This will depend on the terrain.

(2) Establishment of intermediate position(s).

(3) The use of covering troops/rearguards/rear parties.

} This will however be applicable at higher levels.

(4) Counter-attack/spoiling attack.

(5) Making full use of obstacles and demolitions.

e. ***Secrecy***. All possible steps must be taken to keep the time of withdrawal secret from the enemy. This is important if a clean break is

to be achieved. This will not be easy because the tactical situation will probably make it obvious that a withdrawal is likely. The following will help in maintaining secrecy:-

(1) All abnormal movement to take place during the hours of darkness.

(2) Normal activity along the front to be maintained.

(3) Only those individuals who have to know about the withdrawal in order to start preliminary planning should be told early.

(4) Deception.

(5) Wireless security.

(6) Harassing fire to cover the noise of transport etcetera.

f. ***Security***. All possible measures must be taken by the higher commander to provide security to the withdrawing force. The company commanders will be informed of the employment of covering troops, rearguards and, if necessary, flank guards.

g. ***Balance***. Balance is essential if a withdrawing force is to avoid running fight or carry-out controlled movement to the rear. The withdrawing force must not be caught off balance on the move. At battalion/ company level, it is essential to have security to the front, flanks and rear at all times.

h. ***Intelligence***. Every effort must be made to get early information of the enemy movement. Only accurate and early intelligence will permit the adjustment of plans to minimise enemy interference with our withdrawal and to delay his advance through bold counter-strokes. A company/platoon may contribute to this effort by sending out patrols. However, for the most part, it will depend on higher headquarters for any intelligence that is relevant to its withdrawal. Intelligence, pertinent to subordinate/flanking commanders, should be passed on to them promptly and in sufficient detail.

**Rear Main Position**

1. ***General***. As already described, the withdrawing battalion may:-

   a. Be moving back to a position in rear behind another battalion or formation which is already deployed on ground. Such situations, though rare in practice, would be ideal from the point of view of the withdrawing battalion commander.

   b. Have to carry-out withdrawal from the present position and, at the same time, prepare the position on its own at a suitable place in the rear. This would be a more normal and also a more difficult situation and is therefore discussed here.

2. ***Selection of the Rear Main Position***. The rear main position to be occupied by a withdrawing battalion will normally be laid down by the higher commander, the main factors governing its selection will be as follows:-

   a. Future plans of the higher commander and situation on the flanks.

   b. Suitability of the ground for the new mission allotted to the battalion.

   c. ***Distance from the Old Position***. This will normally be governed by the time required for preparation of the new position and the amount of delay it is possible to impose on the enemy before he can contact the new main position.

3. ***Organization and Layout***. If the withdrawing battalion forms part of a division and has been given the task of taking up another defensive position, its organization and layout will be according to its location and will conform to the general layout of a normal defensive position.

**Intermediate Position**

4. ***Selection and Occupation***. Following are some of the considerations:-

   a. As discussed earlier, if the withdrawing battalion has also to prepare and occupy a new defensive position in the rear, it will be necessary to impose greater delay on the enemy than before. This will increase the number of intermediate positions on which delaying actions will have to be fought.

   b. The force responsible for an intermediate position will consist of specific units/subunits and a commander is appointed with the authority to make such regrouping within his force as may be necessary. However, this should be kept to the minimum. Normal organization should not be broken up and a complete unit or subunit

should, whenever possible, be given the task of occupying an intermediate position. A commander must invariably be nominated for the entire position.

c. Basically, Infantry with Anti-Tank Guided Missiles predominate these positions. However, availability of some armour and engineers elements will considerably add to the strength of the position and will also assist in the final extrication of troops. If a daylight withdrawal becomes necessary, maximum fire support is provided to these positions and the troops are made mobile to be able to move speedily after disengagement.

d. Because of the nature of its mission, the layout of an intermediate position requires that there is greater firepower in front to force the enemy to deploy, less depth to economise in strength and adequate frontage to make it difficult for the enemy to turn the position by a quick outflanking movement. Some reserves may always be needed for counter penetration, limited counter-attacks and for other unforeseen eventualities. Maximum use of obstacles should be made to impose delay and inflict heavy casualties on the enemy.

e. The intermediate positions must fulfil following requirements:-

(1) Be held in sufficient strength so that the enemy is compelled to deploy.

(2) Depending on the type of ground and the space available, be sufficiently apart to force the enemy to at least move his field artillery before he can bring it within range of the next position.

(3) Be far enough to permit a clean break and avoid a running fight.

5. ***Ground Factors***. In order to prepare the new main defensive position quickly, the force used for the intermediate position(s) will be kept to the minimum. These positions will lack the depth of a normal defensive position. It therefore follows that:-

a. Maximum use must be made of obstacles and demolitions.

b. There must be good observation over the battle area and good fields of fire.

c. There should be good covered lines of withdrawal to enable the force to break contact with the enemy, even in the daylight, if inevitable.

d. If flank(s) protection is provided by the obstacles, the position will be greatly strengthened.

**Counter-Attack**

6. When enemy has been able to effect penetration in an area which is likely to upset the overall plan of withdrawal and the scale of penetration is beyond the capabilities of the local troops to deal with, it may be necessary for the withdrawing commander to carry-out a local counter-attack to re-establish the intact front. In certain cases, the only way to relieve pressure of the enemy generally and to extricate the forward troops, might be to launch a counter-attack from a flank, in order to impose caution on and to divert the efforts of the advancing enemy. In all cases, the counter-attack must be launched with an adequate force and should have limited objectives.

**Spoiling Attack**

7. A situation may develop when the withdrawal is in progress and enemy may resort to deliberate attack to maintain contact and upset withdrawal plan. The battalion commander should be prepared for such an eventuality and think of putting in a spoiling attack beforehand. To accomplish this, the battalion commander should carry-out ground appreciation and think of all likely Forming Up Places. Spoiling attack is a difficult operation and requires bold action. For this, the troops should be earmarked in advance and all the actions movement should be rehearsed. Arrangements for timely information about enemy movement and forming up, are the most important for a successful spoiling attack.

**Grouping**

8. In the light of the earlier discussion it is apparent that grouping will fall under following headings:-

a. Troops holding the original position.

b. Troops required for the new position.

c. Troops required to man any intermediate position(s).

d. Reserves will be earmarked by the commanders of the original, new and intermediate positions. Besides, overall reserve may also be required and earmarked for such contingencies as unexpected breakthrough, outflanking move by the enemy, or extrication of the troops, that cannot be done by themselves. However, these reserves can normally be useful only if they are made mobile.

9. ***Rearguard***

a. A rearguard will normally be detailed by the formation/unit(s) carrying out withdrawal from the old positions and/or an intermediate position on the principle that every commander is responsible for his own protection.

b. The composition of the rearguard will depend on the mission, terrain, the extent of delay to be imposed and the expected scale of the enemy follow-up. Tanks, if available, must be allotted to the rearguard.

10. Covering troops delay the enemy forward of the new defensive position for a specified period. They achieve this by:-

a. Deceiving the enemy about the true location of the forward defended localities of the new defensive position and forcing him to deploy early.

b. Disorganising the attacking enemy as much as possible, by inflicting maximum casualties as he advances.

c. Providing early warning of his movements.

11. Their composition depends upon the mission, terrain, the enemy strength and the troops available. As the withdrawal is undertaken to preserve the integrity of the command under the pressure of the enemy's superior strength, it is imperative that the covering troops are sufficiently strong to execute the assigned mission.

12. Delaying positions for the covering troops may be designated by the higher commander or left to the discretion of the covering troops' commander. In executing the mission, the covering troops hold the enemy as far forward and for as long as possible, without becoming decisively involved in the battle. In certain types of close country, boldly handled covering troops can impose delay and inflict casualties out of all proportion to their size. However, it is unrealistic to expect smaller forces to hold on independently, against a superior enemy on a ground, where there is room for manoeuvre.

**Mobility**

13. Since the enemy would invariably be enjoying local numerical superiority, the withdrawing force must upset it either by its own superior mobility or by the denial of mobility to the enemy. It would be usual to detail mobile covering troops composed of armour, anti-tank elements and mobile infantry to cover the withdrawal. Similarly, the troops left behind at the old defensive position and the intermediate positions, must have armour and sufficient transport to allow speedy movement of the troops to the rear after final extrication. In addition, the withdrawal plan must ensure maximum denial of mobility to the pursuing enemy by means of interdiction and a coordinated obstacle and denial plan. By a combination of these measures, the commander tries to ensure greater mobility for the withdrawing force in the battle area, so that the enemy cannot bring to bear his superiority decisively, against any portion of the withdrawing force, at any stage of the withdrawal.

**Control and Coordination**

14. ***Control***. Withdrawal is a difficult operation to control and hence demands a simpler plan. Control is exercised through the following:-

a. A carefully phased and timed programme which must allow adequate timings for the completion of each part of the operation. Timings must allow forward troops to smoothly break contact with the enemy.

b. Subunits should be kept intact throughout the withdrawal.

c. Non-essential vehicles, equipment, stores and dumps should be moved out early.

d. The number of officers to reconnoitre and coordinate the new position in the rear should be kept to a minimum.

e. Clear demarcation of:-

(1) Inter-unit and intercompany boundaries.

(2) Phases, report lines, bounds and harbour areas.

(3) Routes of withdrawal, use of checkpoints, Rendezvous or assembly area, embussing and debussing points and start and release points.

f. Controlled rearward movement.

g. Clear battle procedures and efficient battle drills.

h. Efficient line and radio communications.

15. ***Coordination***. Coordination is particularly needed in:-

a. The withdrawal of troops from the original defensive position, especially of the rear parties.

b. Between the covering troops, main body, flank guards and rearguards, at the intermediate position and the new defensive position, especially while passing through the latter in the course of the withdrawal.

c. Artillery fire support, air effort and demolitions and obstacles.

d. Inter-unit and intercompany boundaries.

e. Counter-attacks, spoiling attacks and employment of reserves.

f. The movement of transport, equipment and stores from forward positions to the new defensive position, especially if the vehicles have to be brought forward to ferry back the stores and troops.

**Timings**

16. ***General***. Careful and accurate timings are essential if a withdrawal is to be successful. Certain standard timings will always be laid down, additional timings are often included. These are explained below.

17. ***Standard Timings***. The higher commander will be responsible for laying down the following timings:-

a. The time upto which a position will be denied to the enemy.

b. The time before which there will be no rearward movement. In order to arrive at this timing, a commander will usually have to balance the following factors:-

(1) The strength required on the present position in order to deny it upto the time laid down by the higher authorities.

(2) The amount of time required to withdraw troops from the present position.

(3) The degree of secrecy required to prevent the enemy any knowledge of the realizing that withdrawal.

c. The time by which all troops must be clear of a particular line. This is required to enable the supporting artillery and air to engage any enemy on the abandoned positions.

18. ***Additional Timings***. Following additional timings may also be laid down by the Brigade Commander:-

a. The time at which forward units may start thinning out.

b. The time by which the position will be finally abandoned.

**Sequence of Withdrawal**

19. Based on the timings given above, the normal sequence of withdrawal is as follows:-

a. All non-essential vehicles, equipment and personnel are withdrawn from the forward area.

b. Troops not actually in contact with the enemy withdraw.

c. Progressive thinning out by the forward units; so that, by the time the position had to be positively denied, the bulk of the troops withdraw, leaving behind only rear parties.

d. Rear parties leave the position at the time laid down for final abandonment of the position.

20. Having broken contact with the enemy, the last unit(s) formation(s) on the original defensive position will withdraw through the next position. The enemy will be delayed in his advance to the next position by the covering troops and/or the rearguard, if detailed. The process of leapfrog will be followed until the new main position is finally reached.

21. No position should be abandoned until the next position in the rear has been organized and at least partially manned.

## SECTION 75 – CONDUCT OF WITHDRAWAL

**Introduction**

1. The withdrawal of an infantry battalion by day and by night is two different types of operations. In either case, there is always a danger that the enemy may attack in strength, shortly before the last of the troops leave the position (particularly, when no reserves for immediate counter-attack are available either). In such an instance, a breakthrough may be achieved by the enemy, with the result that a running fight may ensue and the whole withdrawal operation may be jeopardized. The other danger is that the enemy may contain one position and make an outflanking move to cut off our position in the rear. The brigade commander will, therefore, decide on the minimum strength to be maintained in the position until the time it must be denied to the enemy.

**Withdrawal by Night**

2. By night the rearmost subunits leave first and those in contact with the enemy, at the end. If normal activity (e.g., patrols and harassing fire) continues on the front till the last moment, it will be difficult for the enemy to determine the exact time of withdrawal. This activity will be maintained by artillery and the rear parties of infantry battalion who will stay back until the rest of the battalion has withdrawn. These rear parties must be strong enough to give the impression that the position was still being held, but weak enough to get away unobtrusively. The move can take place cross-country and/or by vehicles, to break contact quickly.

3. Depth companies withdraw before the forward companies. Depth platoons of the forward companies provide the local counter-attack force, until they themselves withdraw, covered by the forward platoons, who provide the rear parties.

4. It will usually be possible to pull back majority of the supporting arms during early part of the night. However, if terrain is suitable and illuminating arrangements exist, the tanks can even be the last ones to withdraw along with the rear parties.

5. With the availability of improved night driving devices, there may be a threat of the enemy using armour by night. Recoilless rifles of the companies may therefore have to be left behind till the very end.

6. Withdrawal by night helps to minimize enemy air superiority. However, improved equipment may discount this to some extent.

**Withdrawal by Day**

7. Extrication of the forward troops is much more difficult by day. They will probably have to fight their way and also be subjected to enemy air attacks.

8. Their departure must be covered by the fire of all available weapons. Thus by day, the depth units and subunits do not leave first. It will be usual for the forward troops to withdraw through the depth subunits. This process is repeated till a breakaway is achieved.

9. During a daylight withdrawal the tanks will be left with the infantry till the very last, if there is a threat from the enemy armour. Tanks and machine-guns are particularly useful in extricating forward infantry by fire. If there is an enemy armour threat, majority of the Baktar Shikans must also remain in position till the end.

**Day Versus Night**

10. It is clear that the battalion can withdraw more easily by night, than by day. In the latter case, unless total surprise is achieved and good covered lines of withdrawal are available along with an obstacle to slow the enemy's advance, it will be difficult to avoid a running fight and consequent heavy casualties.

**Sequence of a Battalion Withdrawal by Night**

11. Following programme is given as a general guide the way a battalion supported by a troop of tanks might withdraw by night from a position in contact with the enemy. Figure 18 gives the layout of the battalion:-

a. Position to be denied to the enemy till 2300 hours.

b. Last light is at 2000 hours.

c. 2015 hours – 'A' and 'B' echelons to withdraw in small packets.

d. 2145 hours – Troops required to man battalion checkpoints, RV and embussing point etcetera take up their positions. One platoon from the depth companies takes up a position covering the battalion embussing point.

e. 2215 hours – Depth companies and battalion headquarters less command posts withdraw. Rear party starts taking over from the forward companies.

f. 2245 hours – Rear party in position.

g. 2300 hours – Battalion commander hands over the charge to the officer commanding rear party. Battalion mortars and command post withdraw. Forward companies along with troops of tanks begin to withdraw. Platoon from depth companies withdraws after rest of the battalion has embussed. All the troops clear the line specified in the orders by the given time.

12. ***Notes on the above Programme***

a. Command Post must remain functional till the time of denial. It would normally be established at the battalion checkpoint.

b. Some machine-guns from the depth companies may be given to forward companies.

c. Fire support should be so organised that the withdrawal is constantly covered by artillery or mortars.

d. Tanks should normally be the last to withdraw.

e. If possible, representatives from the companies and platoons should reconnoitre the routes of withdrawal during daylight and also at night.

f. If the situation permits, the entire battalion may not be required to assemble at the battalion RV/embussing point. Instead, the companies may be allowed to continue their move-back as and when they arrive. All the movements should, however, be well controlled.

13. Working on the above battalion programme, the sequence for the withdrawal of one of the forward companies might be:-

| | | |
|---|---|---|
| Timings | – | Surplus personnel of the company headquarters withdraw. |
| Depend on circumstances | – | Depth platoon withdraws. |
| 2300 hours | – | Forward platoons begin to withdraw by sections. Company headquarters including company commander, forward observation officer, mobile fire controller and one or two other (ranks) withdraw. |

14. ***Notes on the above Programme***

a. The company will withdraw from its positions to a company rendezvous (probably 200 to 500 yards in rear). Here the company can be formed up, ready for the move back to the battalion rendezvous.

b. If the route to the company rendezvous is long or difficult, a company checkpoint may be included. It would be where the platoon routes converge.

c. When necessary, representative from platoons and sections should reconnoitre routes of withdrawal during daylight and at night.

15. In a withdrawal by night, a battalion in the front line, may not only have to prepare and occupy a forward locality in the new main position, but also delay the enemy on an intermediate position. The commander should withdraw only when the major portion of his command has withdrawn. The sequence of events for such an operation might be:-

a. The reconnaissance of the intermediate position in the rear and the reconnaissance of the new main position. At the same time the route back should be reconnoitred and prepared.

b. The intermediate position will be occupied, if possible, by a company group.

c. The withdrawal from the main position of the depth companies.

d. Thinning out of the forward companies and supporting arms in these company localities.

e. Abandonment of the main position.

f. Occupation of the new main position in the rear.

g. Abandonment of the intermediate position.

**The Battalion Reconnaissance Party**

16. A battalion reconnaissance party for a position in the rear might consist of the following:-

a. Battalion Second-in-Command.

b. Seconds-in-Command of the rifle companies and the specialist platoons.

c. One Non-Commissioned Officer and two soldiers from each rifle platoon.

d. Signal officer, two signallers and two despatch riders.

e. Intelligence havildar with one soldier.

f. Provost havildar and one Regimental Policeman.

g. One representative from each supporting arm under common command.

17. ***Organization and Conduct of Rear Parties***. Following courses are open to the battalion commander:-

a. Nominate one of the companies which should take over the entire front from the forward companies. Advantages and disadvantages are as under:-

(1) Command and control will be facilitated.

(2) Responsibility delegated to one subunit.

(3) Coordination for withdrawal of the rear parties will be easier.

(4) Positions continue to show strength.

(5) Unnecessary movement for taking over from other subunits may jeopardize security and secrecy of the plan.

b. Designate one platoon each from the forward companies. Merits and demerits are as under:-

(1) Command and control will be complicated.

(2) Minimum movement involved, thereby ensuring greater secrecy of the plan.

(3) Troops will be more familiar with the area and will fight better from their respective positions.

(4) Positions are likely to be weakened.

(5) Coordination in withdrawal is likely to be difficult due to the fact that the platoons will be spread over a large area.

(6) Wide gaps between the platoon will be inevitable.

c. Detail one section each from the forward platoons under an officer. Salient points of this course are as under:-

(1) Less movements are involved ensuring greater secrecy and security.

(2) Fairly continuous and compact front is maintained.

(3) Command and control is more difficult.

(4) Coordination of withdrawal of the sections is likely to pose problems.

**Control Points**

18. ***Checkpoints***. They should be established on the routes back to the rendezvous at an easily recognisable place. There should be no halting at the checkpoints.

19. ***Rendezvous***

a. For each unit and subunit, a rendezvous must be selected by its commander, in order to regain control of his command. The RV should be reconnoitred and known to all concerned.

b. If due to the distance involved, it is not possible to have the battalion rendezvous behind the next covering troops position, it should be placed behind a reasonable obstacle where all the subunits can conveniently converge. Adequate arrangements must be made for its protection.

20. ***Embussing Points***

a. If parallel routes are available for transport, it will be the best to have company embussing points about 1500-2000 metres behind the old position on the company routes of withdrawal. This will save marching time and also facilitate a clean break. In such cases,, strict orders must be issued and enforced against the use of vehicle at night.

b. If due to non-availability of suitable tracks or other tactical reasons, it is not possible to have company embussing points, then the battalion embussing point should be established at behind the battalion rendezvous. In that case, the latter should be located well forward in order to save marching time.

**Morale**

21. A withdrawal always creates the impression of reverse and imposes much greater strain on the morale of troops than any other operation of war. Everything possible must be done to maintain a high standard of morale. Once the withdrawal has begun, all ranks must be promptly given reliable information about what is happening. This will help counter the spreading of depressing rumours and the consequent lowering of morale. Good administrative arrangements must be made throughout for feeding. An adequate supply of ammunition and petroleum, oils and lubricants must always be available. Efficient arrangements must be made for traffic control.

## SECTION 76 – SUPPORTING ARMS IN WITHDRAWAL

**Armour**

1. An armoured regiment or a part of it may be employed in any one or more of the following roles:-

a. To assist the withdrawing force in breaking contact.

b. To assist the withdrawing force in its final extrication from the old position.

c. To act as part of covering troops.

d. To act as rearguard or a part of it. Tanks provide the most effective rearguards, particularly during day and in open country. However, in our operational environment where enemy enjoys preponderance in armour, it will be necessary to supplement our tank resources with the Anti-Tank Guided Missiles. Daylight employment of Anti-Tank Guided Missiles should be carefully planned and the problem of their concealment, routes of withdrawal and resupply, carefully considered.

e. To be the commander's reserve, for employment in counter penetration role against enemy gun areas and for launching of spoiling/counter-attacks.

f. Protection of flank(s), vital points and demolitions.

2. Due to larger frontages and the difficulties of achieving a clean break-in a day withdrawal, it would be advisable to also allot some armour to the troops holding forward positions and/or acting as rear parties.

**Artillery**

3. ***Principles***. The following are the main principles for the handling of artillery in the withdrawal:-

a. The range of artillery will be used to force an early deployment of the enemy and to delay his approach.

b. Continuous artillery support must be available.

c. Early reconnaissance of gun positions and observation posts is essential.

4. ***Support of the Covering Troops***. It will normally be necessary to allot artillery to the covering troops. Self-propelled artillery is particularly suitable in this role. The range of medium guns should also be used to support the covering troops. However, there will revert to command of the commander artillery on arrival at the new main position.

5. ***Repositioning***. Before the main or intermediate positions are abandoned, part of the artillery must be withdrawn and repositioned some 5000-6000 metres back so as to cover the withdrawal of the last troops.

6. ***Support of Intermediate Positions***. In order to provide continuous support, some artillery will have to be deployed behind intermediate positions. The number of guns so deployed will depend on the strength of the intermediate position and the time for which it is to be held. The artillery remaining in the original gun area will stay until the last possible moment, increasing its activity to conceal the reduction of artillery on the front.

7. ***Covering Demolitions***. One of the most effective ways of increasing the delaying power of demolitions is to ensure that they are covered by artillery fire. As long as a demolition can be engaged by observed fire it is virtually impossible for the enemy engineers to fill or bridge the gap without suffering heavy casualties.

8. ***Reconnaissance and Occupation of Position***. In order that guns can come into action immediately on arrival at each position, early reconnaissance and survey are essential. Observation parties must accompany the infantry to each position so that observation posts are dug in and ready by the time the position is to be occupied.

9. ***Anti-Aircraft Artillery***. Hostile air attack is likely to be particularly aggressive. Throughout the withdrawal, any element of light anti-aircraft artillery allotted to the division will normally remain under centralized control and will be used for the protection of defiles and gun areas.

10. ***Air Observation***. If the air situation permits, artillery reconnaissance and air observation posts will be of great value in making use of the long-range of artillery, particularly the heavier types.

**Engineers**

11. ***Tasks***. Engineer tasks in the withdrawal will include:-

a. A first priority task is to keep rearward communications open. This is a task which will chiefly involve keeping bottlenecks open. Dozers must be positioned at such places to fill in craters and clear the rubble. Temporary bridges or ferries may have to be constructed as a safeguard in case a vital bridge is destroyed by the enemy action.

b. Demolitions and mining, with the aim of inflicting maximum possible delay and casualties on the enemy.

c. Defence works in the new main position. Intermediate positions may also require engineer efforts.

d. Destruction of the abandoned equipment. Special demolition detachments will have to be left behind.

e. Destruction of railways. These are the best destroyed at the bridges, culverts, junctions and signal boxes.

f. Destruction of civilian administrative services and food stocks etcetera (scorched earth). This will include workshops, power and transformer stations, reservoirs, telephone exchanges and stocks of all commodities.

g. Booby-trapping the abandoned positions.

12. ***Control and Deployment of Engineers***. Opportunities for the engineer work in withdrawal are almost limitless. There will never be enough engineers to carry-out all the tasks required. Deployment of engineers will need careful balancing between centralized and decentralized control. Close integration with the tactical plan is vital. Adequate time must be allowed for a successful demolition scheme.

**Signals**

13. In withdrawal, distances between the headquarters tend to be stretched. Headquarters are too often split to allow for control on one position and the reconnaissance and coordination on another. Battalion signal resources are likely to be taxed to the limit, since they do not cater for the splitting of headquarters for prolonged periods. In addition, wireless detachments will be required for traffic control and administrative purposes.

14. Security considerations may preclude the use of wireless, particularly in the initial stages of withdrawal. This may necessitate laying of extra lines at a time when this is also the most difficult and the most undesirable.

15. On occasions, the existing line network may be used. In order to do it, the headquarters will have to withdraw rather slowly, but exactly according to the a predetermined plan, which may not often be possible. Every use must however, be made of these line networks to preserve the resources and to allow for work.

**Light Anti-Tank (LAT) Battalion Elements**

16. The Anti-Tank Guided Missiles held by the Light Anti-Tank battalion can fully exploit their long-range, accuracy, efficient communication and mobility in the withdrawal operation. Keeping in mind their capabilities, the Anti-Tank Guided Missiles can be used in the following echelons of a withdrawing force:-

a. ***Rear Parties***. If the enemy is likely to use tanks in great number against a withdrawing force some Anti-Tank Guided Missiles may be used in the original defensive position to form part of the rear parties. In this role, the Anti-Tank Guided Missiles can keep the enemy tanks

away at a greater distance and, at the same time, help in showing an intact front to the enemy.

b. ***Rearguards/Covering Troops***. As part of rearguards/covering troops, Anti-Tank Guided Missiles cover the withdrawal of the main force by taking up successive defensive positions or forming part of the combined arms teams on the route of withdrawal. These delaying actions involve occupation of successive positions to cover the withdrawal and the Anti-Tank Guided Missiles can be used effectively to form part of these. The number of Anti-Tank Guided Missiles to be used in this role will be dictated by the type of enemy pressure and the time required for the preparation of new main defences.

c. ***Intermediate Positions***. Anti-Tank Guided Missiles can be very usefully employed in the intermediate positions. In this role, the Anti-Tank Guided Missiles occupy temporary defences to impose maximum delay on the advancing armour and also to destroy enemy tanks making use of their accuracy and range. Efforts must be made to position the Anti-Tank Guided Missiles in such a manner that full use is made of their destructive capability.

17. In all eventualities, the Anti-Tank Guided Missiles must be employed mounted on the armoured personnel carriers during withdrawal to ensure rapid movement to the firing positions or to react to any unexpected enemy threats. Grenade Launcher Heavy-Machine guns, Machine-gun and Light-Machine guns etcetera can however be deployed dismounted.

18. Owing to the speed with which withdrawal takes place, care must be exercised not to separate Anti-Tank Guided Missiles from the support platoons and the units/subunits providing local protection. If for some reasons, the Anti-Tank Guided Missiles carrier breaks down, the weapon system should immediately be transferred to another carrier. If in withdrawal, capture of the weapon system is obvious, the Anti-Tank Guided Missiles launcher and the missiles must be destroyed. In doing so, priority must be given to the optical sight and the missile guidance set.

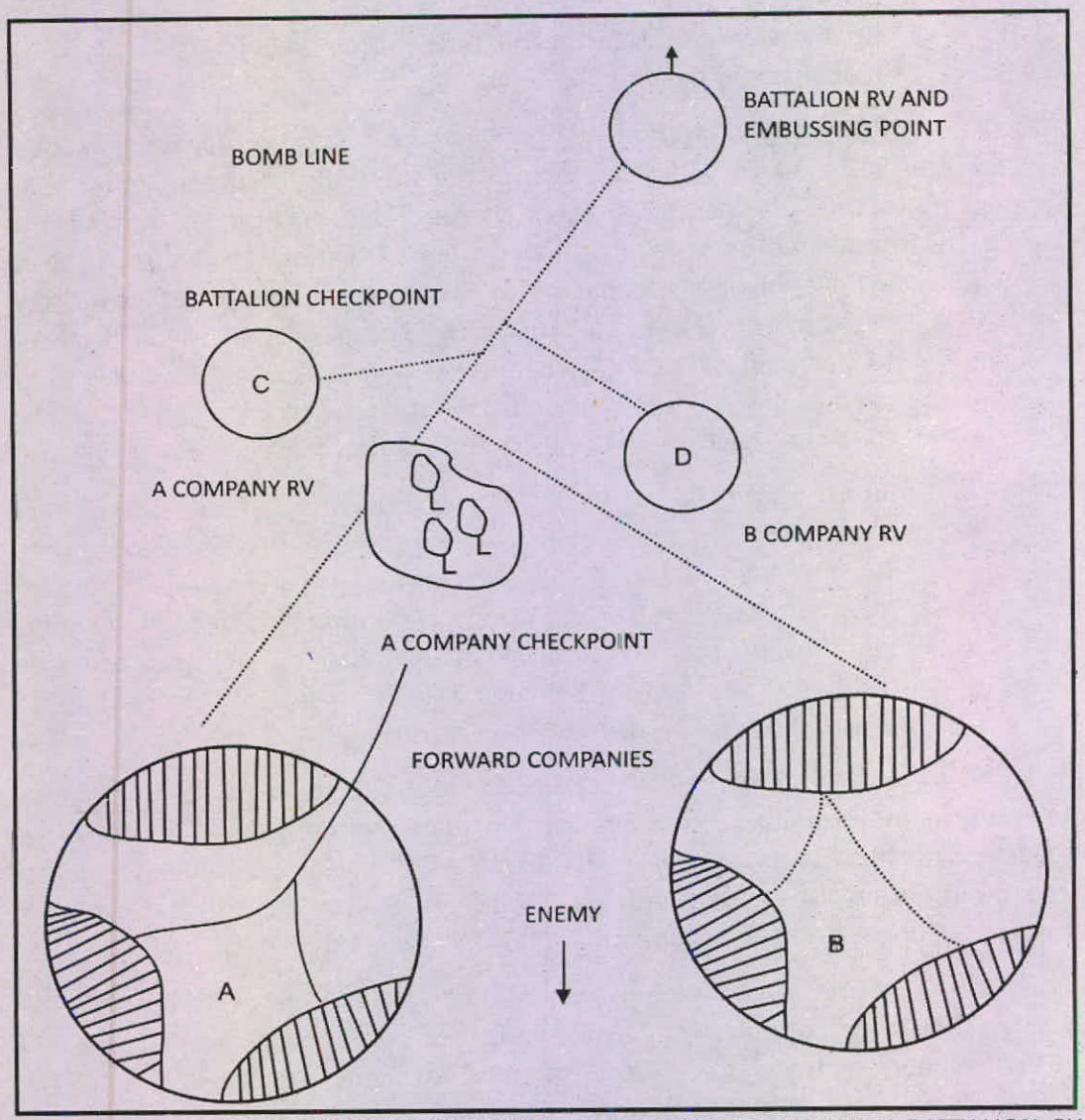



FIGURE 35 – DIAGRAMMATIC WITHDRAWAL OF A FORWARD INFANTRY BATTALION BY NIGHT.

**Annex A**
**To Section 1**

# DETAIL OF PERSONNEL

| Serial No. | Details | Bn HQ | Four Rifle Coys ( Each) | | | | | | | | | | | Cbt Sp Coy | | | | | | | | | | | | | Adm Pl | | | Total Bn | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Three Pls (Each) | | | | Wpn Sec | | | | | | | ATGM Pl | | | Mor Pl | | | | Aslt Pnr Pl | | | | | | | | | |
| | | | Coy HQ | Pl HQ | 3 Sec Each | Total One Pl | Total Three Pl | 60 MM Mor Sec | GLHMG Det | 106 MM RR Sec | Total Wpn Sec | Total One Rifle Coy | Total Four Coy | Coy HQ | Pl HQ | 2 Sec Each | Total Pl | Pl HQ | 2 Sec Each | Total Mor Pl | Sig Pl | Pl HQ | 3 Sec Each | Total Pl | Total Cbt Sp Coy | Adm Sec | Tpt Sec | Total Adm Pl | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
| **Officers** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 1. | Bn Comd (Lt Col) | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| 2. | 2IC (Maj) | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| 3. | Coy Comd (Maj) | | 1 | – | – | – | – | – | – | – | – | 1 | 4 | 1 | – | – | – | – | – | – | – | – | – | – | 1 | – | – | – | 5 | – |
| 4. | Adjt (Capt) | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | Act as Sig Offr |
| 5. | QM (Capt) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | 1 | Act as MTO |
| 6. | Pl Comd (Capt/ Subaltern) | – | – | 1 | – | 1 | 3 | – | – | – | – | 3 | 12 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 12 | – |

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 7. | ATGM Pl Comd (Capt) | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | – | – | – | – | – | – | – | 1 | – | – | – | 1 | – |
| 8. | Mor Pl Comd (Capt) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | – | – | – | – | 1 | – | – | – | 1 | – |
| 9. | IO (Subaltern) | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| | **Total Offrs** | **4** | **1** | **1** | **–** | **1** | **3** | **–** | **–** | **–** | **–** | **4** | **16** | **1** | **1** | **–** | **1** | **1** | **–** | **1** | **–** | **–** | **–** | **–** | **3** | **1** | **–** | **1** | **24** | **–** |
| **Junior Commissioned Officers (JCOs)** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 10. | Sub Maj | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| 11. | Sub (Coy Senior JCO) | – | 1 | – | – | – | – | – | – | – | – | 1 | 4 | 1 | – | – | – | – | – | – | – | – | – | – | 1 | – | – | – | 5 | – |
| 12. | Sub (Pl Comd) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | 1 | – | – | – | 1 | – |
| 13. | Sub (MT) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | 1 | 1 | – |
| 14. | N/Sub (Adjt) | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| 15. | N/Sub (QM) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | 1 | – |
| 16. | N/Sub (Pl 2IC) | – | – | 1 | | 1 | 3 | – | – | – | – | 3 | 12 | – | – | – | – | 1 | – | 1 | 1 | – | – | – | 2 | – | – | – | 14 | – |
| | **Total JCOs** | **2** | **1** | **1** | | **1** | **3** | **–** | **–** | **–** | **–** | **4** | **16** | **1** | **–** | **–** | **–** | **1** | | **1** | **1** | **1** | **–** | **1** | **4** | **1** | **1** | **2** | **24** | **–** |

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Havildars (Havs)** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 17. | BHM | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| 18. | BQMH | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | 1 | – |
| 19. | CHM | – | 1 | – | – | – | – | – | – | – | – | 1 | 4 | 1 | – | – | – | – | – | – | – | – | – | – | 1 | – | – | – | 5 | – |
| 20. | CQMH | – | 1 | – | – | – | – | – | – | – | – | 1 | 4 | 1 | – | – | – | – | – | – | – | – | – | – | 1 | – | – | – | 5 | – |
| 21. | Hav (Prov) | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| 22. | Hav (Int/ Radar Op) | 2 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – |
| 23. | Hav (Mess) | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| 24. | Hav (Sec/ Det Comd) | – | – | – | 1 | 3 | 9 | 1 | 1 | 1 | 3 | 12 | 48 | – | – | 1 | 2 | – | – | – | – | – | – | – | 2 | – | – | – | 50 | – |
| 25. | Hav (Sig) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | – | – | 1 | – | – | – | 1 | – |
| 26. | Hav (MPO) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – | 2 | – | – | – | – | 2 | – | – | – | 2 | – |
| 27. | Hav (OP) (MFC) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – | 2 | – | – | – | – | 2 | – | – | – | 2 | – |
| 28. | Hav (MT) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | 1 | 1 | – |
| | **Total Havs** | **5** | **2** | **–** | **1** | **3** | **9** | **1** | **1** | **1** | **3** | **14** | **56** | **2** | **–** | **1** | **2** | **4** | **–** | **4** | **1** | **–** | **–** | **–** | **9** | **1** | **1** | **2** | **72** | **–** |
| **Naiks (Nks)** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 29. | Nk (Int) | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| 30. | Nk (MT/ POL) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | 1 | 1 | – |
| 31. | Nk (Sig/ Storeman) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | – | – | 1 | – | – | – | 1 | – |

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 32. | Nk (Radio and Wire) | 2 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – |
| 33. | Nk (Wire Line Party) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – | – | – | 2 | – | – | – | 2 | – |
| 34. | Nk Pay/ Kote) | – | 1 | – | – | – | – | – | – | – | – | 1 | 4 | 1 | – | – | – | – | – | – | – | – | – | – | 1 | – | – | – | 5 | – |
| 35. | Nk (Sec/ Det Comd) | – | – | – | 1 | 3 | 9 | 1 | | 1 | 2 | 11 | 44 | – | – | 1 | 2 | – | 1 | 2 | – | – | 1 | 2 | 6 | – | – | – | 50 | – |
| 36. | Nk (Op) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – | 2 | – | – | – | – | 2 | – | – | – | 2 | – |
| 37. | Nk (Ammo) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | – | – | – | – | 1 | 1 | – | 1 | 2 | – |
| 38. | Nk (Op Key Board) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – | – | – | 2 | – | – | – | 2 | – |
| | **Total Nks** | **3** | **1** | – | **1** | **3** | **9** | **1** | – | **1** | **2** | **12** | **48** | **1** | – | **1** | **2** | **3** | **1** | **5** | **5** | – | **1** | **2** | **15** | **1** | **1** | **2** | **68** | – |
| **Lance Naiks/Sepoys (L/Nks/Seps)** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 39. | L/Nk (Int/ Radar Op) | 2 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – |
| 40. | Sep (RP) | 4 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 4 | – |
| 41. | Sep GD | 5 | 1 | 1 | 4 | 13 | 39 | – | – | – | – | 40 | 160 | 1 | 1 | | 1 | 2 | | 2 | 2 | – | – | – | 6 | 1 | 1 | 2 | 173 | – |
| 42. | Sep (MT Dvr) | 6 | 2 | – | – | – | – | 1 | 1 | 3 | 5 | 7 | 28 | 2 | 1 | 4 | 9 | 4 | | 4 | 3 | 2 | – | 2 | 20 | 3 | 2 | 5 | 59 | – |
| 43. | Sep (DR) | 2 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – |
| 44. | Sep (Op Radio & Wire) | 2 | 3 | 1 | – | 1 | 3 | – | – | – | – | 6 | 24 | 1 | 1 | – | 1 | 2 | – | 2 | 8 | – | – | – | 12 | 2 | – | 2 | 40 | – |

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 45. | Sep (Op Radio & Key Board) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – | – | – | 2 | – | – | – | 2 | – |
| 46. | Sep (106 mm RR) | – | – | – | – | – | – | – | – | 4 | 4 | 4 | 16 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 16 | – |
| 47. | Sep (GLHMG No) | – | – | – | – | – | – | – | 1 | | 1 | 1 | 4 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 4 | – |
| 48. | Sep (ATGM No) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | 4 | – | – | – | – | – | – | – | 4 | – | – | – | 4 | – |
| 49. | Sep (60 mm No) | – | – | – | – | – | – | 2 | – | – | 2 | 2 | 8 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 8 | – |
| 50. | Sep (Pnr) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 6 | 18 | 18 | – | – | – | 18 | – |
| 51. | Sep (Mor No) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 9 | 18 | – | – | – | – | 18 | – | – | – | 18 | – |
| 52. | Sep (RL No) | – | – | 2 | | 2 | 6 | – | – | – | – | 6 | 24 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 24 | – |
| 53. | Sep (LMG/ MG1A3) | – | – | – | 4 | 12 | 36 | – | – | – | – | 36 | 144 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 144 | – |
| 54. | Sep (Storeman) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | 1 | – |
| 55. | Cook (Mess) | 2 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – |
| 56. | Cook (Unit) | 2 | 4 | – | – | – | – | – | – | – | – | 4 | 16 | 3 | – | – | – | – | – | – | – | – | – | – | 3 | – | – | – | 21 | – |
| | **Total Lnks/ Seps** | **25** | **10** | **4** | **8** | **28** | **84** | **3** | **2** | **7** | **12** | **106** | **424** | **7** | **3** | **6** | **15** | **8** | **9** | **26** | **15** | **2** | **6** | **20** | **83** | **7** | **3** | **10** | **542** | – |

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 57. | Clks | 6 | 1 | – | – | – | – | – | – | – | – | 1 | 4 | 1 | – | – | – | – | – | – | – | – | – | – | 1 | 2 | 1 | 3 | 14 | – |
| | **Total Clks** | **6** | **1** | – | – | – | – | – | – | – | – | **1** | **4** | **1** | – | – | – | – | – | – | – | – | – | – | **1** | **2** | **1** | **3** | **14** | – |
| | **Total Sldrs** | **41** | **15** | **5** | **10** | **35** | **105** | **5** | **3** | **9** | **17** | **137** | **548** | **12** | **3** | **8** | **19** | **16** | **10** | **36** | **22** | **3** | **7** | **23** | **112** | **12** | **7** | **19** | **720** | – |
| | **Tota' All Ranks** | **45** | **16** | **6** | **10** | **36** | **108** | **5** | **3** | **9** | **17** | **141** | **564** | **13** | **4** | **8** | **20** | **17** | **10** | **37** | **22** | **3** | **7** | **23** | **115** | **13** | **7** | **20** | **744** | – |
| 58. | Sweepers | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 16 | – | 16 | 16 | – |
| | **Total NCs(E)** | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | **16** | – | **16** | **16** | – |
| **Civilians (Civs)** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 59. | Masalchi | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| 60. | Waiters | 3 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 3 | – |
| | **Total Civs** | 4 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 4 | – |
| **Attach Engineers** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 61. | Hav (IC) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | 1 | – | – | – | 1 | – |
| 62. | Nk Sec Comd | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | 1 | 1 | – | – | – | 1 | – |
| **Corps of Electrical and Mechanical Engineers (EME)** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 63. | Hav (Armourer) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | 1 | – |
| 64. | Cfn (Armourer) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – | 2 | 2 | – |
| 65. | Nk (VM) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | | 1 | 1 | 1 | – |

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 66. | Cfn (VM) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | 2 | 2 | – |
| 67. | Cfn (Elec) | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | 2 | 2 | – |
| **Army Medical Corps (AMC)** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 68. | Hav (Nursing Asst) | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |
| | **Total Attached** | **1** | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | **1** | **1** | **2** | **2** | **3** | **5** | **8** | **11** | – |
| | **Grand Total (Incl NCs(E), Civs and Attached** | **50** | **16** | **6** | **10** | **36** | **108** | **5** | **3** | **9** | **17** | **141** | **564** | **13** | **4** | **8** | **20** | **17** | **10** | **37** | **22** | **4** | **7** | **25** | **117** | **32** | **12** | **44** | **774** | – |

**Annex B**
**To Section 1**

# SUMMARY OF TRANSPORT

| Serial No. | Details | Bn HQ | Four Rifle Coys (Each) | | | | | | | | | | | Cbt Sp Coy | | | | | | | | | | | | | Adm Pl | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Three Pls (Each) | | | | Weapon Sec | | | | | | | ATGM Pl | | | Mor Pl | | | | Aslt Pnr Pl | | | | | | | | | |
| | | | Coy HQ | Pl HQ | 3 Sec Each | Total One Pl | Total Three Pls | 60 mm Mor Sec | GLHMG Det | 106 mm RR Sec | Total Wpn Sec | Total One Rifle Coy | Total Four Coy | Coy HQ | Pl HQ | 2 Sec Each | Total Pl | Pl HQ | 2 Sec Each | Total Pl | Sig Pl | Pl HQ | 3 Sec Each | Total Pl | Total Cbt Sp Coy | Adm Sec | Tpt Sec | Total Adm Pl | Total Bn | Remarks |
| 1. | Trk ¼ ton 4 x 4 GS | 2 | 1 | – | – | – | – | – | – | – | – | 1 | 4 | 1 | 1 | – | 1 | 1 | – | 1 | 1 | – | – | – | 4 | 1 | – | 1 | 11 | – |
| 2. | Trk Utility ¼ ton 4 x 4 106 mm RR Modified | – | – | – | – | – | – | – | – | 2 | 2 | 2 | 8 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 8 | – |
| 3. | Trk Utility ¼ ton 4 x 4 106 mm RR Ammo | – | – | – | – | – | – | – | – | 1 | 1 | 1 | 4 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 4 | – |
| 4. | Trk ¼ ton Jeep ATGM Launcher | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | 4 | – | – | – | – | – | – | – | 4 | – | – | – | 4 | – |
| 5. | Trk ¼ ton Jeep ATGM Ammo | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | 4 | – | – | – | – | – | – | – | 4 | – | – | – | 4 | – |
| 6. | Trk Amb ¼ ton 4 x 4 | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – |

| Serial No. | Details | Bn HQ | Four Rifle Coys (Each) | | | | | | | | | | | Cbt Sp Coy | | | | | | | | | | | | | Adm Pl | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Three Pls (Each) | | | | Weapon Sec | | | | | | | ATGM Pl | | | Mor Pl | | | | Aslt Pnr Pl | | | | | | | | | |
| | | | Coy HQ | Pl HQ | 3 Sec Each | Total One Pl | Total Three Pls | 60 mm Mor Sec | GLHMG Det | 106 mm RR Sec | Total Wpn Sec | Total One Rifle Coy | Total Four Coy | Coy HQ | Pl HQ | 2 Sec Each | Total Pl | Pl HQ | 2 Sec Each | Total Pl | Sig Pl | Pl HQ | 3 Sec Each | Total Pl | Total Cbt Sp Coy | Adm Sec | Tpt Sec | Total Adm Pl | Total Bn | Remarks |
| 7. | Trk ¼ ton 60 mm Mor Ammo | – | – | – | – | – | – | 1 | – | – | 1 | 1 | 4 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 4 | – |
| 8. | Trk ¼ ton Jeep Radar | 2 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – |
| 9. | Trk ¼ ton 4 x 4 Jeep for HMG/ GLHMG | – | – | – | – | – | – | – | 1 | – | 1 | 1 | 4 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 4 | – |
| 10. | Trk ¾ ton 4 x 4 15 cwt/ 1.5 ton 4 x 4 Dodge | 1 | 1 | – | – | – | – | – | – | – | – | 1 | 4 | – | – | – | – | 3 | – | 3 | 2 | 2 | – | 2 | 7 | – | 1 | 1 | 13 | – |
| 11. | Trk 2-1/2 ton 6 x 6 Lorry 3 ton | 1 | 1 | – | – | – | – | – | – | – | – | 1 | 4 | 1 | – | – | – | – | – | – | – | – | – | – | 1 | 2 | – | 2 | 8 | – |
| 12. | Tlr Cargo ¼ ton | 2 | – | – | – | – | – | 1 | - | 1 | 2 | 2 | 8 | 1 | 1 | – | 1 | 1 | – | 1 | 1 | – | – | – | 4 | 1 | 1 | 2 | 16 | – |
| 13. | Tlr Cargo ¾ ton | 1 | 1 | – | – | – | – | – | – | – | – | 1 | 4 | – | – | – | – | 3 | – | 3 | - | 2 | – | 2 | 5 | – | 1 | 1 | 11 | – |
| 14. | Tlr Cargo 1½ ton | – | 1 | – | – | – | – | – | – | – | – | 1 | 4 | 1 | – | – | – | – | – | – | – | – | – | – | 1 | 1 | – | 1 | 6 | – |

| Serial No. | Details | Bn HQ | Four Rifle Coys (Each) | | | | | | | | | | | Cbt Sp Coy | | | | | | | | | | | | Adm Pl | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Three Pls (Each) | | | | Weapon Sec | | | | | | | ATGM Pl | | | Mor Pl | | | | Aslt Pnr Pl | | | | | | | | |
| | | | Coy HQ | Pl HQ | 3 Sec Each | Total One Pl | Total Three Pls | 60 mm Mor Sec | GLHMG Det | 106 mm RR Sec | Total Wpn Sec | Total One Rifle Coy | Total Four Coy | Coy HQ | Pl HQ | 2 Sec Each | Total Pl | Pl HQ | 2 Sec Each | Total Pl | Sig Pl | Pl HQ | 3 Sec Each | Total Pl | Total Cbt Sp Coy | Adm Sec | Tpt Sec | Total Adm Pl | Total Bn | Remarks |
| 15. | Mobile Fd Kitchen (MFK) 1½ ton | – | 1 | – | – | – | – | – | – | – | – | 1 | 4 | 1 | - | - | - | - | - | - | - | - | - | - | 1s | - | - | - | 5 | – |
| 16. | Water Tlr 1-½ ton 400/450 Gals | 1 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | 2 | – |
| 17. | Motorcycle | 2 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 2 | – |
| 18. | Bicycles | 2 | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | 1 | – | 1 | 3 | – |
| 19. | **Total Vehs (Less Motor-cycles, Bicycle and Tlrs** | **7** | **3** | – | – | – | – | **1** | **1** | **3** | **5** | **8** | **32** | **2** | **1** | **4** | **9** | **4** | – | **4** | **3** | **2** | – | **2** | **20** | **3** | **1** | **4** | **63** | – |

**Appendix 1**
**To Annex B**

## SUGGESTED DISTRIBUTION OF BATTALION TRANSPORT INTO ECHELONS

| Echelon | Details | Motor-cycles | ¼ ton | ¾ ton | 2 ½ ton | Trailor | Remarks |
|---|---|---|---|---|---|---|---|
| 'F' | ***Battalion Headquarters*** | | | | | | |
| | Commanding Officer | – | 1 | – | – | 1 | – |
| | Second-in-Command | – | 1 | – | – | 1 | – |
| | Despatch Rider | 2 | - | – | – | – | – |
| | ***Combat Support Company*** | – | 1 | – | – | 1 | – |
| | Signal platoon | – | 1 | 1 | – | 1 | Signal JCO and line laying party. |
| | Mortar platoon | – | 1 | 2 | – | 3 | Platoon commander and mortar detachments. |
| | Anti-Tank Guided Missile platoon | – | 8 | – | – | – | Four for ammunition. |
| | Heavy Machine-gun/ Grenade Launcher Heavy- Machine gun detachment | – | 4 | – | – | – | – |
| | Ground Surveillance Radar | – | 2 | – | – | – | – |
| | Rifle companies reserve ammunition and reorganisation stores | – | – | 4 | – | 4 | – |
| | 60 millimetres Mortar section | – | 4 | – | – | 4 | Four for ammunition. |
| | Recoilless Rifles | – | 12 | – | – | 4 | – |
| | **Total:** | **2** | **35** | **7** | – | **19** | **Combat Support Company Commander may command 'F' echelon.** |

| Echelon | Details | Motor-cycles | ¼ ton | ¾ ton | 2 ½ ton | Trailor | Remarks |
|---|---|---|---|---|---|---|---|
| 'A' | ***Battalion Headquarters*** | | | | | | |
| | Adjutant | – | – | 1 | – | 1 | A Office. |
| | Regimental Aid Post | – | 1 | – | – | – | Ambulance. |
| | ***Combat Support Company*** | | | | | | |
| | Battalion's reserve ammunition | – | – | – | 1 | 1 | – |
| | Signal stores | – | – | 1 | – | 1 | – |
| | Mortar reserve ammunition | – | – | 1 | – | 1 | – |
| | Mechanical Transport Officer and Unit Repair Organization | – | – | 1 | – | 1 | – |
| | Assault pioneer stores | – | – | 2 | – | – | – |
| | **Total:** | **–** | **1** | **6** | **1** | **5** | **–** |
| 'B' | ***Battalion Headquarters*** | | | | | | |
| | Officers mess | – | – | – | 1 | 1 | Water Trailor. |
| | ***Combat Support Company*** | | | | | | |
| | Quartermaster | – | 1 | – | 1 | 1 | Commander 'B' Echelon. |
| | Cooking and bedding | – | – | – | 1 | 1 | Water Trailor. |
| | **Total:** | **–** | **1** | **–** | **3** | **3** | |
| | ***Rifle Companies*** | | | | | | |
| | Cooking and bedding | – | – | – | 4 | 4 | – |
| | **Grand Total:** | **2** | **37** | **13** | **8** | **31** | **–** |

**Annex C**
**To Section 1**

## CHARACTERISTICS OF BATTALION WEAPONS

| Serial No. | Weapon | Calibre | Weight (lbs/kg) | Weight of Bomb/Shell (lbs/kg) | Weight of Magazine | | Rates of Fire (rpm) | | | Effective Range (Metres) | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Empty (lbs/kg) | Filled (lbs/kg) | Maximum Effective | Sustained | Cyclic | | |
| 1. | a. Revolver .38 | 0.38 | 1–11½ | – | – | – | 12 | 12 | – | 15 m | – |
| | b. Walther 9 mm | 9 mm | 1–11½ | – | – | – | 24 | 8 | – | 45 m | – |
| 2. | G-3 Rifle | 7.62 mm | 4.083 kgs | – | – | – | 40 | 10 | 550-650 | 300 m | – |
| 3. | RPG-7 | 40 mm | 6.3 kgs | 2.25 kgs | – | – | 4-6 | – | – | 500/300 m | – |
| 4. | MG1A3 | 7.62 mm | 11.045 kgs | – | – | – | 100 | 50 | 1100-1300 | 457 m | Maximum effective range in Machine-gun role 2194 m (approximately). |
| 5. | a. 106 mm RR | 106 mm | 204.12 kgs | 16. 783 kgs | – | – | – | 1 | – | 1188 m (1300 yards) | ***Penetration*** Armour 16-20 inches. Indirect fire Range 4389 m. |
| | b. 75 mm RR | 75 mm | 86. 864 kgs | HE: 21 kgs HEAT:14 kgs | – | – | – | 1 | – | 600 m | – |
| 6. | 60 mm Mortar | 60.75 mm | 12. 474 kgs | 2. 27 lbs (907.2 grams) | – | – | 15-20 | – | – | 1897 with max charge | – |

| Serial No. | Weapon | Calibre | Weight (lbs/kg) | Weight of Bomb/Shell (lbs/kg) | Weight of Magazine | | Rates of Fire (rpm) | | | Effective Range (Metres) | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Empty (lbs/kg) | Filled (lbs/kg) | Maximum Effective | Sustained | Cyclic | | |
| 7. | 81 mm Mortar | 81 mm | 61.508 kgs | 7 lbs (3.175 kgs) | – | – | Slow – 2<br>Normal– 4<br>Rapid – 6<br>Intense–8 | – | – | Max: 4160<br>Min : 96 | – |
| 8. | Grenade Launcher Heavy-Machine gun | 40 mm | – | Latter | | | | | | 1500-1800 m | – |
| 9. | Baktar Shikan | | 88 kgs (with one missile) | 11.2 kgs | – | – | – | – | – | Min: 65 m<br>Max: 3000 m | a. Crew: 4 Men<br>b. ***Firing Platoon Forms***<br>(1) Ground.<br>(2) Jeep.<br>c. ***APC***. Scale of missile:-<br>(1) Ground and jeep 8 on weapon.<br>(2) 4 Reserve. Guidance Method – Comd to line sight. |

**Annex D**
**To Section 1**

## AUTHORISATION OF NIGHT VISION DEVICES/NIGHT OBSERVATION DEVICES IN AN INFANTRY BATTALION

| Serial No. | Night Vision Devices/ Night Observation Devices | Authorized | Remarks |
|---|---|---|---|
| 1. | AN/PVS-4 | 74 | Used with Light-Machine gun/Machine-gun/Heavy-Machine gun – one per weapon. |
| 2. | AN/TVS-5A | 8 | Used with 106 millimetres Recoilless Rifle – one per weapon. |
| 3. | AN/UAS-12A | 4 | Used with TOW/Green Arrow – one per weapon. |
| 4. | AN/UAS-11 (Night Vision Sight) | 7 | a. ***Distribution***<br>(1) One per Mobile Fire Controller – 2<br>(2) One each Rifle Company – 4<br>(3) Battalion Headquarters – 1<br>b. It has AN/TAS-6 for observation |
| 5. | AN/PVS-5A (Goggles) | 54 | a. 4 per Rifle Company – 16<br>b. 1 per fighting and administrative echelon vehicle – 38 |
| 6. | Night Vision Bino-4 (Binocular) | 33 | a. One per Recoilless Rifle Detachment – 8<br>b. 81 millimetres Mortar platoon – 1<br>c. Grenade Launcher Heavy-Machine gun Detachment – 4<br>d. Anti-Tank Guided Missile platoon – 1 |

| Serial No. | Night Vision Devices/ Night Observation Devices | Authorized | Remarks | | | |
|---|---|---|---|---|---|---|
| | | | e. | One per Company Headquarters | – | 5 |
| | | | f. | One per Rifle Platoon | – | 12 |
| | | | g. | Battalion Headquarters | – | 2 |

**Appendix 2**
**To Annex D**

## NIGHT VISION DEVICES/NIGHT OBSERVATION DEVICES HELD IN AN INFANTRY BATTALION

| Serial No. | Type of System | Equipment | Used With | Range | Capabilities/Limitations | Remarks |
|---|---|---|---|---|---|---|
| 1. | 25 mm Second Generation, (Image Intensifier) | AN/PVS-4 | MG1A3 | Moonlight – 450 m<br>Starlight – 300 m | a. Weight 1.7 kg<br>b. Magnification 3.5 times<br>c. Battery life at 20 degree centigrade 15 hours | Can be used with Rifle G-3 and RPG-7. |
| 2. | -do- | AN/TVS-5A | 106 mm Recoilless Rifle | Moonlight – 1000 m<br>Starlight – 600 m | a. Weight 3 kg<br>b. Magnification 5.6 times<br>c. Battery life at 20 degree centigrade. 15 hours | Can be used with certain wire guided missiles launchers also. |
| 3. | AN/TAS-4A (Thermal Imaging) | AN/UAS-12A | TOW Green Arrow | 3750 m | a. Weight 10.9 kg<br>b. ***Magnification***<br>(1) Wide Field of View (WFOV) 4 times<br>(2) Narrow Field of View (NFOV) 12 times | Bracket Interface is required when mounted on Baktar Shikan (Green Arrow). |
| 4. | AN/TAS-6 (Thermal Imaging) | AN/UAS-11 (Night Vision Sight). | – | 3000 m | a. Weight 11 kg<br>b. Tripod mounted | Can be used during reconnaissance, patrolling and for surveillance. |

| Serial No. | Type of System | Equipment | Used With | Range | Capabilities/Limitations | Remarks |
|---|---|---|---|---|---|---|
| | | | | | c. ***Magnification***<br>(1) Wide Field of View (WFOV) 3 times<br>(2) Narrow Field of View (NFOV) 9 times | |
| 5. | 18 mm, Micro Channel Wafer Type, (Image Intensifier) | AN/PVS-5A (Night Vision Goggles) | – | 25 cm to 250 m | a. Battery life at 20 degree centigrade 15 hours | Used for night driving, sentry duty, detection of low level helicopter and map-reading. |
| 6. | 18 mm, Micro Channel Wafer Type, (Image Intensifier) | GP/NVB-4A (Binocular) | – | – | a. Magnification 4 times<br>b. Focus adjustment range 10 m to infinity | Can be used for long range night observation by tank commanders. |

**Appendix 3**
**To Annex D**

## DATA/CAPABILITIES OF NIGHT VISION DEVICES OF OWN TANKS

| Serial No. | Type of Tanks | Type of Night Vision Devices | Driver (Metres) | Ranges Gunner (Metres) | Commander (Metres) |
|---|---|---|---|---|---|
| 1. | T-59 M Type-II | Infra-red | 200 | 800 | 800 |
| 2. | T-69 II M | Image Intensifiers | 75 | 800 | 800 |
| 3. | T-69 II MP | Image Intensifiers | 75 | 800 | 800 |
| 4. | T-85 II AP | Image Intensifiers | 75 | 800 | 800 |
| 5. | T-80 UD | Image Intensifiers | 150-300 | 600-800 | Moonlight – 1200-1400<br>Starlight – 800-1000 |

**Appendix 4**
**To Annex D**

## NIGHT VISION DEVICES – INDIAN ARMY

9. Following sights are under development/already in use by Indian Army:-

a. ***Tank Commander/Gunner Sight***. A Thermal Imager (TI) sight is being developed for Tank Commander/Gunner with a detection range of 2500 metres and an engagement range of 1800 metres. It is also planned to replace the existing Infra-red and image intensifier Night Vision Devices on T-72 tanks at the scale of one per tank, with the second generation image intensifier tubes.

b. ***Tank/Infantry Combat Vehicle Driver's Sight***. Image intensifier passive Night Vision Devices with a range of 200 metres has been developed for the tank/Infantry Combat Vehicle drivers and is being considered for induction at a scale of one Night Vision Device per tank (infantry combat vehicle).

c. ***BMP Commander/Gunner Sight***. Image intensifier sights with a detection range of 1500 metres and engagement range of 1000 metres are to be issued to BMP-I and BMP-II, at a scale of 2 per BMP and one per BRDM.

d. ***Anti-Tank Guided Missiles***. Missiles Infra-Red Attachment (MIRA) night sight has been developed for Milan Anti-Tank Guided Missile. It is also being procured for FAGOT (anti-tank missile). Technical data of MIRA night sight is as under:-

| | | | |
|---|---|---|---|
| (1) | Target detection range | – | 4000 metres. |
| (2) | Target recognition range | – | 2000 metres. |
| (3) | Target engagement range | – | 1500 metres. |
| (4) | Field of view | – | 6 x 3 degrees. |

e. ***Passive Night Observation Devices (PNOD)***. Thermal imaging based Night Vision Devices with a range of 3000/2500 metres are being proposed at a scale of 4 to 6 devices per regiment. Image Intensifier devices will be used as a substitute in the absence of Thermal Image devices. Night Vision Devices Mark-I and II (Image Intensifier devices) for artillery observers with ranges of 1200 and 1000 metres respectively is in use.

f. ***Medium-Machine Gun Sight***. Image intensifier machine-gun sight with a range of 1500/1000 metres is being planned for each infantry

battalion at a scale of one per weapon. A lightweight man portable/ weapon mounted sight with a range of 800 metres is in use as interim arrangements.

g. ***Recoilless Rifle Sight***. Mark-I and II sights with ranges of 600 and 1000 metres respectively will be issued to infantry battalions at a scale of one device per weapon.

h. ***Rocket Launcher, Light-Machine Gun (LMG) and Rifle Sights***. Night Vision Devices having engagement ranges upto 750 metres for Rocket Launchers and 500 metres for Light-Machine Guns/rifles are under development.

i. ***Night Vision Goggles (NVGs)***. Night Vision Goggles are image intensifier and are suitable for driving, engineer tasks (mine laying and breaching etcetera) and for all other tasks requiring short-range observation upto 100-200 metres. These can also be switched over to short distance mode for map-reading at night. Goggles can be hand-held or helmet mounted. It is planned to issue these goggles to division/brigade headquarters at a scale of 4 per headquarters and to artillery units at a scale of 15 goggles per regiment.

j. ***Binocular***. A general purpose Image Intensifier binocular with a range of 800-1000 metres has been developed. This will be issued to all the headquarters and units.

**Annex E**
**To Section 4**

## BATTLE PROCEDURE/COMPOSITION OF GROUPS AT BATTALION LEVEL

| Group | Battalion | Company | Platoon | Remarks |
|---|---|---|---|---|
| **RECEE OR 'R' GROUP** | Commander, Runner<br>Intelligence Officer or Adjutant *<br>Signal Officer<br>2 Soldiers – Intelligence Section<br>2 Soldiers – Signal Section (Wireless Sets)<br>Commanders in support %<br>Escort transport | Commander with Company Havildar Major/Runner, Wireless Set | Commander, Runner and Wireless Set | * Adjutant is normally at headquarters ready to operate rear link wireless set to brigade.<br>% Only those required by commanding officer to assist in making plan. |
| **ORDER OR 'O' GROUP** | Officer commanding all companies with one Runner each.<br>Mortar Platoon Commander @f<br>Pioneer Platoon Commander @f<br>Signal Officer, Medical Officer 'R' Groups of arms in support. | Platoon 'R' group | Section commanders | @ Some commanding officers like to issue orders for specialist platoon through officer commanding headquarters company.<br>f One orderly each. Second-in-command should also attend the orders if possible for keeping himself in picture so that subsequently take over of the battalion by him at any stage of the battle is facilitated. |
| **MAIN BODY** | Battalion less 'A' and 'B' echelon transport and personnel** | Company less 'A' and 'B' echelon transport and personnel** | Platoon headquarters and three sections ** | ** Normally commanded by second-in-command. |

| Group | Battalion | Company | Platoon | Remarks |
|---|---|---|---|---|
| **'A' ECHELON** | 'A' echelon personnel, and transport, e.g., consolidation stores, vehicles, weapons and personnel. | 'A' echelon personnel and transport, e.g., company truck with reserve ammunition wire etcetera. | Nil | – |
| **'B' ECHELON** | 'B' Echelon personnel, transport and stores normally includes, Quartermaster Havildar and transport platoon less 'A' echelon %% | Nil | Nil | %% Normally commanded in case of battalions by transport officers or Quartermaster. When 'B' echelon is brigaded, the whole is normally under Staff Captain under direction of Deputy Assistant Adjutant and Quartermaster General. |

**Note**: 1. The maximum number likely to be included in any one group is given. It does NOT necessarily follow that they are always present; it will depend on the situation.

2. No details of the transport required can be included. It will always be kept to the minimum and the type of vehicle will depend upon terrain and situation.

3. In a withdrawal, the reconnaissance party will normally be composed of seconds-in-command.

**Annex F**
**To Section 4**

## BATTLE PROCEDURE – ADVANCE

| Serial No. | Sequence of Events | Action by Commanding Officer | Action by Company Commander | Action by Platoon Commander | Remarks |
|---|---|---|---|---|---|
| 1. | **Receipt of Warning Order** | a. Studies map and carries out a quick appreciation.<br>b. Issues warning order.<br>c. Instructs second-in-command and adjutant before leaving for brigade commander's orders. | a. Receives warning order from battalion headquarters.<br>b. Studies map and carries out quick appreciation.<br>c. Issues warning order verbally.<br>d. Prepares to leave for battalion 'O' Group. | a. Receives verbal warning order from company commander.<br>b. Issues warning order to section commanders verbally.<br>c. Briefs platoon Junior Commissioned Officer.<br>d. Prepares to leave for company 'O' group. | |
| 2. | **Recce/ Orders** | a. Receives brigade commander's orders.<br>b. Carries out a study of the map/air photos.<br>c. Proceeds to rendezvous and issue orders. If possible will show whatever ground is visible to his 'O' group then issue orders on model/map depending on time available. | a. Receives battalion commander's orders.<br>b. Carries out his recce mostly off the map.<br>c. Prepares his plan in consultation with his 'R' group.<br>d. Proceeds to rendezvous and issues orders. If possible will show whatever ground is visible to his 'O' group and then issue orders on a model. | a. Receives order from his company commander.<br>b. Carries out his recce and issues orders to section commanders.<br>c. Prepares his plan. | |

| Serial No. | Sequence of Events | Action by Commanding Officer | Action by Company Commander | Action by Platoon Commander | Remarks |
|---|---|---|---|---|---|
| 3. | **Advance** | a. Ensures battalion crosses start point at correct time.<br>b. Keeps a constant watch and controls the progress of advance.<br>c. Tries to visualize possible enemy actions and issue orders accordingly.<br>d. Ensures that necessary momentum is maintained.<br>e. Keeps brigade headquarters informed about progress of advance. | a. Ensures company crosses start point at given time.<br>b. Controls the advance of his company.<br>c. Keeps battalion headquarters informed about progress of advance.<br>d. Keeps his subordinate commanders informed about the general progress of advance. | a. Controls the advance of his platoon.<br>b. Keeps company commander informed about progress of advance. | |

**Annex G**
**To Section 4**

## BATTLE PROCEDURE – ATTACK

| Serial No. | Sequence of Events | Action by Commanding Officer | Action by Company Commander | Action by Platoon Commander | Remarks |
|---|---|---|---|---|---|
| 1. | **Receipt of Warning Order from Brigade** | a. Studies the map and carries out time appreciation (recce plan).<br>b. Issues warning order.<br>c. Instructs second-in command and adjutant before leaving for brigade's orders/recce. | a. Receives warning order from battalion headquarters.<br>b. Studies map and carries out time appreciation.<br>c. Issues warning order verbally.<br>d. Prepares to leave for battalion 'O' group with his 'R' group. | a. Receives warning order verbally.<br>b. Issues warning order to section commanders verbally.<br>c. Briefs platoon Junior Commissioned Officer<br>d. Prepares to leave for company 'O' group. | |
| 2. | **Recce/ Orders** | a. If time permits, carries out preliminary recce.<br>b. Receives brigade commander's orders.<br>c. Carries out recce with battalion 'R' group and prepares his plan. | a. Carries out preliminary recce if time permits.<br>b. Reports at rendezvous and receives orders.<br>c. Carries out detailed recce. | a. Reports at rendezvous and receive orders.<br>b. Carries out recce.<br>c. Issues orders. | |

| Serial No. | Sequence of Events | Action by Commanding Officer | Action by Company Commander | Action by Platoon Commander | Remarks |
|---|---|---|---|---|---|
| | | d. Proceeds to his rendezvous and issue orders to his 'O' group.<br>e. Joins the battalion in the Forming Up Place/ assembly area/concentration area. | d. Issue orders to his 'O' group<br>e. Joins his company in the Forming Up Place/ assembly area/concentration area. | d. Joins his platoon in the Forming Up Place/ assembly area/concentration area. | |
| 3. | **Attack on Objective** | a. Conducts/controls the battle. | a. Moves with the company and controls progress of attack. | a. Moves with and controls the platoon. | |

**Annex H**
**To Section 4**

## BATTLE PROCEDURE – DEFENCE

| Serial No. | Sequence of Events | Action by Commanding Officer | Action by Company Commander | Action by Platoon Commander | Remarks |
|---|---|---|---|---|---|
| 1. | **Receipt of Warning Order** | a. Studies the map and carries out time appreciation.<br>b. Issues warning order.<br>c. Instructs second-in-command before leaving for brigade's orders/recce. | a. Receives warning order.<br>b. Studies the map and carries out time appreciation.<br>c. Issue warning order.<br>d. Briefs company second-in-command before leaving for battalion commander's orders. | a. Receives warning order.<br>b. Makes arrangements for move.<br>c. Briefs platoon havildar.<br>d. Prepares to leave for quick recce and company commander's orders. | |
| 2. | **Quick Recce and Outline Plan** | a. Carries out a quick recce.<br>b. Gives out his outline plan to 'O' group.<br>c. Proceeds for brigade commander's orders/ recce. | a. Receives outline plan.<br>b. Carries out a quick recce.<br>c. Issues outline plan to platoon commanders. | a. Receives outline plan from company commander.<br>b. Carries out a quick recce.<br>c. Issues outline orders. | |

| Serial No. | Sequence of Events | Action by Commanding Officer | Action by Company Commander | Action by Platoon Commander | Remarks |
|---|---|---|---|---|---|
| 3. | **Detailed Recce and Orders** | a. Receives orders from brigade commander.<br>b. Carries out detailed recce.<br>c. Coordinates intercompany details.<br>d. Attends brigade commander's coordination conference.<br>e. Issues detailed orders. | a. Carries out detailed recce.<br>b. Coordinates inter platoon detail.<br>c. Reports to commanding officer for coordination at given time and rendezvous.<br>d. Receives commanding officer's final orders.<br>e. Intimates change from previous orders, if any to platoon. | a. Carries out further recce.<br>b. Coordinates section and platoon weapons.<br>c. Carries out coordination with his company commander.<br>d. Issues changes if any. | |
| 4. | **Battle Procedure if time is very short** | a. Receives brigade outline plan in brigade warning order.<br>b. Issues warning order including outline plan after a quick ground recce if time permits otherwise off the map. | a. Receives outline plan in warning order.<br>b. Carries out quick recce and issues warning order and outline plan. | a. Receives warning order and outline plan.<br>b. Deploys platoon on the ground. | |

| Serial No. | Sequence of Events | Action by Commanding Officer | Action by Company Commander | Action by Platoon Commander | Remarks |
|---|---|---|---|---|---|
| | | c. Carries out coordination on the ground with company commanders and brigade commander. Decisions on the spot. | c. Coordinates on the ground with platoon commanders.<br><br>d. Receives on the spot decisions from battalion commander and similarly issues on the spot decisions to platoon commander. | c. Coordinates section and platoon weapons.<br><br>d. Receives on the spot decisions from company commander and acts accordingly. | |

**Annex I**
**To Section 4**

## BATTLE PROCEDURE – WITHDRAWAL

| Serial No. | Sequence of Events | Action by Commanding Officer and Second-in-Command | Action by Company Commander and Company Second-in-Command | Action by Platoon Commander and Platoon Havildar | Remarks |
|---|---|---|---|---|---|
| 1. | **Receipt of Warning Order** | a. ***Commanding Officer***<br>(1) Carries out recce off the map and time appreciation.<br>(2) Briefs second-in-command.<br>(3) Issues warning order.<br>b. ***Second-in-Command***<br>(1) Gets briefing from commanding officer.<br>(2) Briefs recce parties and arranges their move. | a. ***Company Commander***<br>(1) Receives warning order.<br>(2) Carries out time appreciation.<br>(3) Issues warning order.<br>(4) Briefs second-in-command.<br>b. ***Second-in-Command***<br>(1) Gets briefing from company commander.<br>(2) Briefs company recce parties. | a. ***Platoon Commander***<br>(1) Receives warning order.<br>(2) Issue warning order.<br>b. ***Platoon Havildar***<br>(1) Reports to company second-in-command and gets briefing about move of platoon recce parties.<br>(2) Leaves for the rendezvous for recce parties under command. | |

| Serial No. | Sequence of Events | Action by Commanding Officer and Second-in-Command | Action by Company Commander and Company Second-in-Command | Action by Platoon Commander and Platoon Havildar | Remarks |
|---|---|---|---|---|---|
| | | | (3) Leaves for the given RV for battalion recce parties. | (3) Company second-in-command. | |
| 2. | **Action by Recce Parties** | a. ***Second-in-Command***<br>(1) Carries out recce of new defensive position (i.e. intermediate position/main position).<br>(2) Prepares and issues his orders. | a. ***Second-in-Command***<br>(1) Leaves for new defensive position under command battalion second-in-command.<br>(2) Receives orders and carries out recce.<br>(3) Issues orders to platoon recce parties. | a. ***Platoon Havildar***<br>(1) Leaves for new defensive position under command battalion second-in-command.<br>(2) Receives order from company second-in-command.<br>(3) Makes arrangements for deployment. | |
| 3. | **Recce and Orders** | a. ***Commanding Officer***<br>(1) Receives orders from the brigade commander.<br>(2) Carries out his recce.<br>(3) Issues orders. | a. ***Company Commander***<br>(1) Receives orders form the battalion commander.<br>(2) Carries out his recce.<br>(3) Issues orders. | a. ***Platoon Commander***<br>(1) Receives orders from company commander.<br>(2) Carries out recce.<br>(3) Issues orders. | |

| Serial No. | Sequence of Events | Action by Commanding Officer and Second-in-Command | Action by Company Commander and Company Second-in-Command | Action by Platoon Commander and Platoon Havildar | Remarks |
|---|---|---|---|---|---|
| 4. | **Withdrawal of Troops** | a. Supervises move.<br>b. Remains at command post till time of denial. | a. Supervises move.<br>b. Stays behind till time of denial. | a. Stays behind till time of denial. | |
| 5. | **Occupation of New Defensive Position** | ***Commanding Officer***<br>a. Moves to new position.<br>b. Takes over from second-in-command.<br>c. Supervises the deployment of battalion in the defensive position. | ***Company Commander***<br>a. Takes over from second-in-command on reaching new position.<br>b. Supervises deployment of company. | ***Platoon Commander***<br>a. Takes over from platoon havildar on reaching new position and supervises deployment of platoon. | |

**Annex J**
**To Section 5**

## BATTLE PROCEDURE – PATROLS

| Serial No. | Sequence of Events | Action by Commanding Officer and Patrol Master | Action by Company Commander and Company Second-in-Command | Action by Patrol Master | Remarks |
|---|---|---|---|---|---|
| 1. | **Warning Order/ Brigade Patrol Programme** | a. ***Commanding Officer***<br>(1) Receives brigade patrol (warning order in the case of any special patrol. Some time only general patrol programme is issued by brigade headquarters.<br>(2) Based on the above carries out necessary planning and briefs the patrol master giving out his policy. | a. Receives warning order.<br>b. Details selects the patrol leader.<br>c. Issues necessary instructions (Warning order) to patrol leader.<br>d. Issues necessary instructions for administrative arrangements. | a. Receives instructions (warning order) from company commander.<br>b. Carries out time appreciation.<br>c. Selects personnel for patrol and issues instruction (warning order).<br>d. Briefs patrol second-in-command. | |

| Serial No. | Sequence of Events | Action by Commanding Officer and Patrol Master | Action by Company Commander and Company Second-in-Command | Action by Patrol Master | Remarks |
|---|---|---|---|---|---|
| | | b. ***Patrol Master (Second-in-Command/Intelligence Officer)***<br>(1) Prepares patrol programme based on commanding officer's instructions/ brigade patrol programme.<br>(2) Issues warning order to respective companies followed by patrol programme.<br>(3) Prepares to the patrol.<br>(4) In case commanding officer wants to brief some special patrol, makes necessary arrangements. | | | |

| Serial No. | Sequence of Events | Action by Commanding Officer and Patrol Master | Action by Company Commander and Company Second-in-Command | Action by Patrol Master | Remarks |
|---|---|---|---|---|---|
| 2. | **Briefing** | In special cases, briefing may be done by the commanding officer otherwise it will be done by the patrol master. | Attends briefing of Patrol Master. | a. Receives briefing from commanding officer/patrol master.<br>b. Prepares his recce plan and moves for recce.<br>c. Briefs second-in-command regarding further details. | |
| 3. | **Recce and Orders** | | a. Assists patrol leader in planning.<br>b. Coordinates the move out of patrol from own position. | a. Carries out recce and planning with the assistance of company commander.<br>b. During recce liaises with units affecting his route in and out.<br>c. Prepares orders.<br>d. Receives patrol at given rendezvous and issues orders. | |
| 4. | **Debriefing** | a. ***Commanding Officer***. May debrief the special patrol or attend debriefing (in some cases debriefing may be done at brigade headquarters, where briefing was also done by them). | | | |

| Serial No. | Sequence of Events | Action by Commanding Officer and Patrol Master | Action by Company Commander and Company Second-in-Command | Action by Patrol Master | Remarks |
|---|---|---|---|---|---|
| | | b. ***Patrol Master***<br>(1) Receives and debriefs patrol leader.<br>(2) If required the whole patrol may be debriefed at a later stage.<br>(3) Debriefing is carried out by the person who ordered the patrol to be sent out before the detailed report is prepared.<br>(4) Patrol report is prepared and shown to commanding officer before its despatch to brigade headquarters. | | | |

**Annex K**
**To Section 4**

## BATTLE DRILLS – ADVANCE

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| 1. | **Receipt of Warning Order** | ***Commanding Officer***. Action as per battle procedure.<br>a. ***Second-in-Command***<br>(1) Gets briefing from commanding officer.<br>(2) Ensures that necessary preparations for advance are made in the concentration area.<br>b. ***Adjutant***<br>(1) Issues instruction and makes necessary arrangements for move of battalion headquarters.<br>(2) Makes arrangements for manning of start point if the same is not being done under Brigade arrangements. | ***Company Commander***. Actions as per battle procedure.<br>a. ***Company Second-in-Command***<br>(1) Gets briefing from company commander.<br>(2) Checks administrative arrangements as per warning order.<br>b. ***Company Havildar Major***. Checks ammunition. | ***Platoon Commander***. Actions as per battle procedure.<br>a. ***Platoon Havildar***<br>(1) Inspects weapons ammunition and equipment of platoon.<br>(2) Collects unnecessary stores and hands them over to Company Quartermaster Havildar.<br>(3) Ensures issue of meals, tea, water and ammunition. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (3) Allots duties to Regimental Police personnel and arranges sign posting etcetera.<br>(4) Mans battalion fwd and rear links.<br>c. ***Intelligence Officer***<br>(1) Collects maps/air photos and other necessary information of the operation.<br>(2) Marks the map and accompanies commanding officer for order.<br>(3) Arrange preparation of model for commanding officer's orders.<br>d. ***Quartermaster***<br>(1) Makes arrangements for the following:-<br>(a) Cooking. | c. ***Company Quartermaster Havildar***<br>(1) Collects unnecessary stores and hands them over to the Quartermaster.<br>(2) Receives necessary orders from Quartermaster for cooking and 'B' echelon. | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (b) Provision of water.<br>(c) Collection of un-necessary stores from companies.<br>(2) Coordination of administrative platoon activity.<br>(3) Coordination of 'B' echelon transport.<br>(4) Collection of un-necessary stores.<br>e. ***Signal Officer***<br>(1) Reallocation of wire-less equipment if required.<br>(2) Preparation of wire-less diagram and allotment of channels and frequencies.<br>(3) Preparation and issue of griddle code and slidex. | | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | f. ***Mechanical Transport Officer***<br>(1) Drawing and distribution of petroleum, oil and lubricants.<br>(2) Arrangements for recovery.<br>(3) Ensuring that the battalion transport is in a fit state for the operation. | | | |
| 2. | **Recce/Orders** | a. ***Second-in-Command***<br>(1) Coordinates battalion activities in absence of commanding officer.<br>(2) Supervise arrangements for the advance.<br>(3) Helps Quartermaster in tying up administrative details. | a. ***Company Second-in-Command***<br>(1) Ensure dissemination of all information.<br>(2) Supervises:-<br>(a) Administrative arrangements.<br>(b) Checking of weapons and ammunition.<br>(c) Checking of signal communication. | a. ***Platoon Havildar***<br>(1) Ensures all information is passed on to the men.<br>(2) Arrangements for meals and water and coordinated with Company Quartermaster Havildar. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | b. ***Adjutant***<br>(1) In case troops lifting transport made available, he sub allots the same in consultation with Mechanical Transport Officer.<br>(2) Ensures issue of control points as an annex if battalion commander's orders are in writing.<br>(3) Details harbour parties.<br>(4) Coordinates move of supporting arms allotted.<br>c. ***Intelligence Officer***<br>(1) Liaison with brigade headquarters staff.<br>(2) Ensures that model is ready for commanding officer's orders and orientates 'O' group on the model. | b. ***Company Havildar Major***. Distribution of troops lifting transport if made available.<br>c. ***Company Quartermaster Havildar***. Ensures that all administrative arrangements have been completed. | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | d. ***Signal Officer***. Ensures that wireless communication signals on battalion commander net are in perfect condition.<br>e. ***Mechanical Transport Officer***. Ensures that all vehicles have tanks full and necessary reserve of petroleum, oil and lubricants.<br>f. ***Quartermaster***. Ties up arrangements for cooking and delivery of meals. | | ***Platoon Havildar***. Distribution of meals/tea and water. | |
| 3. | **Advance** | a. ***Second-in-Command***. Keeps himself in picture in order to take over if commanding officer becomes a casualty.<br>b. ***Adjutant***<br>(1) Ensures correct order of march. | a. ***Company Second-in-Command***<br>(1) Ensures that meals and water are delivered in time.<br>(2) Keeps himself in picture and takes over if company commander becomes causality.<br>b. ***Company Havildar Major***<br>(1) Intimates Mechanical Transport Officer in case of break of vehicles. | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (2) Ensures up to date information regarding advance is available all the time in command post vehicle.<br>c. ***Intelligence Officer***<br>(1) Keeps his map marked with up to date information regarding advance.<br>(2) Remains with the commanding officer and ensures protect-tion of 'R' group.<br>d. ***Quartermaster***. Ensures timely delivery of meal and water.<br>e. ***Signal Officer***. Ensures that all wireless communi-cation function properly.<br>f. ***Mechanical Transport Officer***<br>(1) Supply of petroleum, oil and lubricants. | (2) Supply of ammunition Company Quartermaster Havildar. Arrange timely delivery of meals/tea and water. | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (2) Prompt action for recovery of vehicles.<br>g. ***Officer Commanding Combat Support Company***. Controls move of 'B' and 'A' echelons. | | | |

**Annex L**
**To Section 4**

## BATTLE DRILLS – ATTACK

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| 1. | **Receipt of Warning Order** | ***Commanding Officer***. Action as per battle procedure.<br>a. ***Second-in-Command***<br>(1) Gets briefing from commanding officer.<br>(2) Takes following actions in assembly/concentration area:-<br>(a) Recce.<br>(b) Deployment. | ***Company Commander***. Actions as per battle procedure.<br>a. ***Company Second-in-Command***<br>(1) Gets briefing from company commander.<br>(2) Checks administrative arrangements as per warning order.<br>(3) Ensures departure of company 'O' group at proper time.<br>(4) Details guides for the company area. | ***Platoon Commander***. Actions as per battle procedure.<br>a. ***Platoon Havildar***<br>(1) Receives and issues ammunition/stores.<br>(2) Collects unnecessary bedding and stores for handing over to Company Quartermaster Havildar.<br>(3) Inspects equipment and weapons of the platoon and other connected arrangements.<br>(4) Ensures water bottles are full and other administrative arrangements are complied with. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (c) Reception arrangements.<br><br>(3) Ensures compliance with necessary instructions regarding transport and wireless communication.<br><br>b. ***Adjutant***<br><br>(1) If required details protection party for `R' group and at rendezvous for `O' group.<br><br>(2) Ensures that necessary transport is available for `R' group.<br><br>(3) Coordinates arrangements for assembly area guide, Regimental Police and sign posting etcetera. | (5) Receives and complies with instructions regarding transport and wireless communication.<br><br>b. ***Company Havildar Major***<br><br>(1) Checks ammunition and reorganisation stores.<br><br>(2) Organises the company for the task under instructions from second-in-command. | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (4) Considers and estimates ammunition requirement/supply.<br>c. ***Intelligence Officer***<br>(1) Marks the map and accompanies commanding officer.<br>(2) Briefs the intelligence section and allots duties to them.<br>d. ***Quartermaster***<br>(1) Preparation of issue of griddle code and slidex.<br>(2) Briefs signal Junior commissioned officer to look after arrange-ments and prepares to leave with the battalion `R' group. | c. ***Company Quartermaster Havildar***<br>(1) Draws and issues ration and stores.<br>(2) Collects unnecessary bedding and stores to be kept in 'A' and 'B' echelons.<br>(3) Receives and complies with any instruction from Quartermaster. | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | e. ***Mechanical Transport Officer***<br>(1) Drawing and distribution of petroleum, oils and lubricants if necessary.<br>(2) Arrangements for recovery.<br>(3) Provides maintenance assistance to companies. | | | |
| 2. | **Recce/Orders** | a. ***Second-in-Command***<br>(1) Coordination of battalion activities in absence of commanding officer.<br>(2) Briefs battalion personnel regarding battle situation.<br>(3) Coordinates the move of main body to assembly area.<br>(4) In assembly area supervises. | a. ***Second-in-Command***<br>(1) Takes the company to assembly area under battalion arrangements.<br>(2) Supervision of following:-<br>(a) Deployment of platoon.<br>(b) Protection against enemy ground and air action. | a. ***Platoon Havildar***<br>(1) Takes platoon to assembly area.<br>(2) The following:-<br>(a) Protection arrangements i.e. alarm post and anti-aircraft.<br>(b) Checking of weapons and ammunition. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (5) ***Marrying Up***<br><br>(a) Protection against enemy ground and air action.<br>(b) Deployment.<br>(c) Administrative arrangements.<br>(d) Marrying up.<br>b. ***Adjutant***. Assist second-in-command.<br><br>c. ***Intelligence Officer***<br>(1) Liaison with brigade staff and keeps the maps up to date.<br>(2) After the brigade commander's orders if the commanding officer has any instructions he would inform battalion head-quarters accordingly. | (c) Administrative arrangements.<br>(d) Rehearsals if required.<br><br>b. ***Company Havildar Major***. Remains with 'F' echelon transport under command headquarters company commander.<br><br>c. ***Company Quartermaster Havildar***. Makes arrangements for provision of meals/tea and supply of water. | (c) Distribution of meals/tea. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (3) Proceeds with commanding officer for recce and prepares himself for giving out land marks and situation before commanding officer's orders.<br>(4) Prepares sketch of the area of operation and gives out the limits of the Observation Post to help in company commanders making sketch.<br>(5) Ensures local protect-tion of 'O' group.<br>(6) Arranges navigation parties to guide troops to objective in case of night attack (if required).<br>d. ***Quartermaster***<br>(1) Makes arrangements for delivering of meals after the attack. | d. ***Platoon Havildar***. Accom-panies platoon headquarters. | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (2) Makes arrangements for bringing bedding and other stores after the attack.<br>e. ***Combat Support Company Commander***. Supervises and coordinates move of 'F' echelon. | e. ***Second-in-Command***. Accompanies the company and assists the company commander.<br>f. ***Company Havildar Major***. Stays with 'F' echelon transport with combat support company commander. | | |
| 3. | **Start Line to Objective** | a. ***Commanding Officer***. Ensures crossing of Start Line at H hr and controls the battle.<br>b. ***Intelligence Officer***. Remains with commanding officer in case of night attack with navigation party if required.<br>c. ***Signal Officer***. Arranges laying of line from Forming Up Place to objective if required.<br>d. ***Regimental Medical Officer***. Arranges treatment and evacuation of causalities. Takes over temporarily, till relieved by second-in-command in case the commanding officer becomes a casualty. | a. ***Company Commander***<br>(1) Ensures company crosses at the appointed H hr.<br>(2) Controls the progress of attack and brings down supporting fire as necessary. | a. ***Platoon Commander***. Moves with the platoon. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| 4. | Reorganisation | a. ***Commanding Officer***<br>(1) Supervises deployment/reorganisation on the objective.<br>(2) Readjustment of defensive fires in consultation with the artillery representative.<br>(3) If situation permits, allows armour under command to pull back to "Rally". | a. ***Company Commander***<br>(1) Success signal.<br>(2) Sends guides to given rendezvous for bringing `F' echelon transport or reorganisation stores to company locality.<br>(3) Deploy company as per reorganisation orders.<br>(4) Coordinates reorganisation of platoons.<br>(5) Sends location report.<br>(6) Reports casualties and ammunition expenditure.<br>(7) Arranges evacuation of wounded to Regimental Aid Post<br>(8) Details patrols and other parties.<br>(9) Adjusts defensive fires if necessary. | a. ***Platoon Havildar***<br>(1) Success signal.<br>(2) Sends report regarding casualties, ammunition expenditure and Previous Wastage Service.<br>(3) Deploys platoon and coordinates evacuation of casualties to Company Aid Post/ Regimental Aid Post. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | b. ***Second-in-Command***<br>(1) Supervises burials with the assistance of Subedar Major.<br>(2) Assists the commanding officer in reorganisation.<br>c. ***Adjutant***<br>(1) Informs brigade headquarters about situation.<br>(2) Brings up battalion headquarters and establishes it at the given place.<br>(3) Consolidates reports regarding the following:-<br>(a) Expenditure of ammunition.<br>(b) Casualty.<br>(c) Previous Wastage Service.<br>(d) Location report. | b. ***Company Havildar Major***<br>(1) Brings up 'F' echelon transport/reorganisation stores.<br>(2) Distributes reorganisation stores and ammunition.<br>c. ***Company Quartermaster Havildar***. Brings up meals/tea and bedding if required. | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | d. ***Intelligence Officer***<br>(1) Brings the situation map up to date.<br>(2) Carries out documentation of Previous Wastage Service.<br>(3) Coordinates detailing of patrols.<br>(4) Sifting of information.<br>e. ***Signal Officer***. Establishes line communications.<br>f. ***Regimental Medical Officer***. Arranges treatment and evacuation of causalities.<br>g. ***Quartermaster***. Brings up meals, bedding and required stores as per commanding officer's orders.<br>h. ***Officer Commanding Combat Support Company***. Brings up 'F' and 'A' echelons transport soon after success signal. | | | |

**Annex M**
**To Section 4**

## BATTLE DRILLS – DEFENCE

| Serial | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| 1. | **Receipt of Warning Order** | ***Commanding Officer***. Action as per battle procedure.<br>a. ***Second-in-Command***<br>(1) Gets briefing from commanding officer.<br>(2) If the battalion has to move to concentration area/ assembly area, he makes arrangements for move of main body or starts planning its move directly to defensive position. | ***Company Commander***. Action as per battle procedure.<br>a. ***Second-in-Command***<br>(1) Gets briefing from company commander.<br>(2) If main body has to move to assembly area etc he details necessary guides and moves the company.<br>(3) Supervises the issue of stores and equipment.<br>(4) Checks the state of weapons.<br>(5) Checks hygiene and sanitary arrangements.<br>(6) Distributes available transport to the company.<br>(7) Organises company if required. | ***Platoon Commander***. Action as per battle procedure.<br>a. ***Platoon Havildar***<br>(1) Receives and issues orders.<br>(2) Checks administrative arrangements.<br>(3) Receives orders, coordinates and works on behalf of platoon commander.<br>(4) Issues ammunition. | |

| Serial | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | b. ***Adjutant***. Ensures the following:-<br>(1) R' group is present at the due time.<br>(2) Required transport is available.<br>(3) Details protection party if required.<br>(4) Battalion 'O' group leaves for orders at the correct time.<br>c. ***Intelligence Officer***<br>(1) Keeps the map marked and accompanies the commanding officer.<br>(2) Briefs the intelligence section and allots duties to them. | b. ***Company Havildar Major***<br>(1) Assists second-in-command in carrying out company commander's orders and other arrangements.<br>(2) Draws and distributes ammunition.<br>c. ***Company Quartermaster Havildar***<br>(1) Draws ration and defensive stores from Quartermaster.<br>(2) Distributes stores to platoon.<br>(3) Arranges the checking and minor repairs of weapons with Quartermaster.<br>(4) Withdraws unessential stores and bedding.<br>(5) Receives instruction regarding `B' echelon and cooking etcetera. | (1) Draws rations and defensive stores from Quarter-master. | |

| Serial | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | d. ***Quartermaster***<br>(1) Issue of required stores.<br>(2) Collection of unnecessary stores and equipment.<br>(3) Provision of the following:-<br>(a) Rations.<br>(b) Sterilising tablets.<br>(c) Anti-malaria items.<br>(d) Issue of gas equipment if required.<br>(4) Assists company in the repair and replacement of their weapons.<br>(5) Planning sanitary arrangements.<br>(6) Coordinates administrative arrangements.<br>(7) If cooking arrangements are to be centralised, he plans and coordinates administrative details and move to `B' echelon area. | | | |

| Serial | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | e. ***Signal Officer***<br>(1) Checking, reallocation of wireless equipment if required.<br>(2) Preparation of wireless diagram, allotment of channels and frequencies.<br>(3) Issue of codes and slidex etcetera.<br>(4) Planning of line communications.<br>(5) Briefs Signal Junior Commissioned Officer and prepares to leave with 'R' group.<br>f. ***Mechanical Transport Officer***<br>(1) Drawing and distribution of petroleum, oil and lubricants.<br>(2) Arrangements for recovery of vehicles.<br>(3) Provides maintenance and assistance to companies. | | | |

| Serial | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| 2. | **Quick Recce and Outline Plan** | a. ***Intelligence Officer***<br>(1) Receives the `O' group at the recovery vehicle.<br>(2) Acquaints them with the lands marks and situation. Hands over the sketch if made. | a. ***Second-in-Command***<br>(1) If the `O' group has not already accompanied the company commander, he ensures that it leaves at correct time.<br>(2) Moves with the main body at the given time. | | |
| 3. | **Detailed Recce and Orders** | a. ***Second-in-Command***<br>(1) Sends the `O' group forward if not gone with the commanding officer. Leads the main body to the given area at the given time.<br>(2) Controls the main body and arranges its protection during move.<br>b. ***Adjutant***<br>(1) Assists second-in-command in the move of the main body by arranging the following:-<br>(a) Distribution of trans-port. | a. ***Second-in-Command***<br>(1) Controls move of the main body.<br>(2) Remains in contact with company commander by wireless if possible.<br>b. ***Company Havildar Major***. Assists second-in command during move. | a. ***Platoon Havildar***<br>(1) Sends forward `O' group.<br>(2) Brings the platoon into company area under company second-in-command.<br>(3) Directs sections to their respective areas. | |

| Serial | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (b) Traffic control.<br>(c) Guides etcetera trafic control.<br>(2) Keeps in touch with the commanding officer and brigade headquarters.<br>c. ***Quartermaster***<br>(1) Gets briefing about the location of `B' echelon area.<br>(2) Proceeds to the allotted area with `B' echelon transport and personnel.<br>(3) Maintains contact with battalion headquarters.<br>d. ***Intelligence Officer***<br>(1) Liaises with brigade and brings map up to date.<br>(2) Assists commanding officer by:-<br>(a) Providing information about the area. | c. ***Company Quartermaster Havildar***<br>(1) Reports to Quartermaster for move to `B' echelon if cooking is centralised.<br>(2) Moves under command Quartermaster to `B' echelon area. | | |

| Serial | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (b) Prepares sketch map of the battalion defence for brigade coordination conference.<br>(c) Accompanies commanding officer to brigade coordination conference.<br>(d) Helps company commanders in marking maps.<br>(e) Carries out briefing on behalf of the commanding officer if required.<br>(f) Receives battalion 'O' group and briefs them about land marks and situation. | | | |
| 4. | **Organisation of Defence/ Deployment** | a. ***Second-in- Command***<br>(1) Carries out arrangements for security with the assistance of Subedar Major and Regimental Police section. | a. ***Second-in-Command***<br>(1) Keeps himself in picture and assists company commander. | a. ***Platoon Havildar***<br>(1) Arrangements regarding siting and protection of platoon headquarters. | |

| Serial | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (2) Looks after the postal arrangements with the assistance of Subedar Major.<br>(3) Responsible for burials.<br>(4) Acts as battalion patrol master.<br>b. ***Adjutant***<br>(1) Sites command post as per commanding officer's orders.<br>(2) Arranges protection and functioning of battalion Command Post.<br>(3) Sends reports and returns as per standing orders/ instruction.<br>(4) Arranges replenishment of ammunition as required.<br>(5) Ensures that runners from respective companies are present. | (2) Remains in company headquarters.<br>(3) Supervises cooking arrangements.<br>b. ***Company Quartermaster Havildar***<br>(1) Establishes cooking arrangements under Quartermaster's orders.<br>(2) Moves forward with cooked meals at specified timings.<br>(3) Makes cooking arrange-ments in company area if they are not centralised. | (2) Assists platoon commander in his work. | |

| Serial | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (6) Ensures that from each company two runners come to battalion headquarters, one of them stays and the other returns to company headquarters after seeing location of battalion headquarters.<br><br>c. ***Intelligence Officer***<br><br>(1) Marks the map in the Command Post.<br><br>(2) Assists second-in-command in making patrol programme.<br><br>(3) Carries out briefing and debriefing of patrols.<br><br>(4) Handles Previous Wastage Service.<br><br>(5) Keeps himself ready to take over Adjutant's duties if required.<br><br>(6) Details intelligence section personnel for establishing Observation Post. | | | |

| Serial | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | d. ***Quartermaster***<br>(1) Deploys `B' echelon transport and personnel in the allotted area if centralised.<br>(2) Ensures:-<br>(a) Drawing of rations and other supplies at appointed time.<br>(b) Cooking of meals if arrangement are centralised.<br>(c) Sending meals to troops at specified timings.<br>(d) Protection and camou-flage/concealment of `B' echelon area.<br>e. ***Mechanical Transport Officer***<br>(1) Establishes unit recovery post.<br>(2) Replenishes petroleum, oil and lubricants.<br>(3) Supervises repair and maintenance of vehicles. | | | |

## BATTLE DRILLS – WITHDRAWAL

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| 1. | **Receipt of Warning Order** | ***Commanding Officer – Second-in-Command***. Also see action as per battle procedure.<br>a. ***Second-in-Command***. Arranges guides for companies to new positions.<br>b. ***Adjutant***. Details transport for recce parties.<br>c. ***Quartermaster***<br>(1) Receives instructions regarding move of non-essential stores.<br>(2) Arranges move of 'B' echelon. | ***Company Commander – Company Second-in-Command***. Also see actions as per battle procedure.<br>a. ***Company Commander***<br>(1) Briefs or details company recce party.<br>(2) Ensures that company recce party reports to the rendezvous at the correct time.<br>(3) Issues orders regarding collection of non-essential stores.<br>(4) Details guides for companies if required.<br>b. ***Company Havildar Major***<br>(1) Arranges move of company recce party to battalion rendezvous. | ***Platoon Commander-Platoon Havildar.*** Also see actions per battle procedure.<br>a. ***Platoon Commander***<br>(1) Details platoon recce party.<br>(2) Coordinates the collection of non-essential stores. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | d. ***Intelligence Officer***<br><br>(1) Accompanies the commanding officer for recce and orders.<br><br>(2) Keeps map up to date from brigade headquarters and marks new defensive positions.<br><br>e. ***Signal Officer***<br><br>(1) Reallocation of wireless sets to recce parties.<br><br>(2) Arrangements for line communication at new position.<br><br>(3) Any other instructions regarding wireless communications.<br><br>f. ***Mechanical Transport Officer***<br><br>(1) Allotment of transport in consultation with Adjutant if available. | (2) Details guides with the recce party as per company commander's orders.<br><br>c. ***Company Quartermaster Havildar***<br><br>(1) Arranges collection of non-essential stores.<br><br>(2) Receives instructions from Battalion Quartermaster regarding move of stores. | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (2) Arrangements for opening up embussing point if applicable. | | | |
| 2. | **Move of Recce Parties** | a. ***Second-in-Command***<br>(1) Receives the recce party at given time and rendezvous and check its composition.<br>(2) Arranges guides and dispersal at new positions<br>(3) Maintains contact with battalion headquarters and complies with Instructions if any. | a. ***Company Commander***. Ensures that the company recce party moves out in time.<br>b. ***Company Second-in-Command***. Commands company recce party and reports to battalion second-in-command at given rendezvous and time. | a. ***Platoon Havildar***. Acts as commander platoon recce party and reports to the company second-in-command. | |
| 3. | **Recce and Orders** | a. ***Adjutant***<br>(1) Supervises/coordinates execution of commanding officer's orders.<br>(2) Keeps brigade headquarters informed about progress. | a. ***Company Commander***. Coordinates administrative arrangements. | a. ***Platoon Commander***<br>(1) Details protection party and personnel for establishing platoon checkpoint and rendezvous. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | b. ***Intelligence Officer***<br>(1) Prepares model for commanding officer's orders if required.<br>(2) Receives `O' group and helps marking their maps.<br>(3) Briefs `O' group on land marks and situation.<br>(4) Allotment and allocation of duties to intelligence section. | b. ***Company Havildar Major***<br>(1) Carries out the arrangement for marking of company checkpoint and rendezvous with the assistance of Kote Naik.<br>(2) Details protection party for company rendezvous according to company commander's instructions.<br>c. ***Company Quartermaster Havildar***<br>(1) Collects non-essential stores of the company.<br>(2) Reports to Quartermaster with stores at given time and place. | (2) Collects non-essential stores and hands them over to Company Quartermaster Havildar. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| 4. | Withdrawal of Troops to New Positions | a. ***Adjutant***<br><br>(1) Briefs the following:-<br>(a) Protection parties for battalion check-point, battalion rendezvous and embussing point.<br>(b) Naib Subedar Adjutant for manning battalion checkpoint.<br>(c) Subedar Major for manning battalion rendezvous includeing the following:-<br>i. Protection.<br>ii. Guides.<br>iii. Allocation of company areas in battalion rendezvous. | a. ***Company Commander***. Informs Adjutant at battalion rendezvous about progress of withdrawal.<br>b. ***Company Havildar Major***. Establishes company rendezvous and arranges the following:-<br>(1) Guidance to company rendezvous.<br>(2) Allotment of platoon areas in company rendezvous.<br>(3) Protection of company rendezvous.<br>(4) Closing down the rendezvous.<br>c. ***Kote Naik***. He is responsible for establishing the company check-point. | a. ***Platoon Commander***<br><br>(1) Ensures that check-point and platoon rendezvous (if required) are established at the given time.<br>(2) Deploys his platoon in the allotted area in the company rendezvous.<br>(3) Reports to company commander about withdrawal of his platoon at the company rendezvous. | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | (2) Supervises move of battalion headquarters.<br>b. ***Intelligence Officer***<br>(1) Stays with commanding officer.<br>(2) Assists adjutant in briefing various (above mentioned parties).<br>c. ***Mechanical Transport Officer***<br>(1) Carries out marking/ direction of vehicles as detailed by adjutant.<br>(2) Establishes embussing point as per instructions.<br>(3) Accompanies transport to new defensive position. | | | |
| 5. | **Occupation of New Defensive Position** | a. ***Second-in-Command***. Hands over the position to the commanding officer and explains the layout. | a. ***Company Quartermaster Havildar***<br>(1) Re-issues defensive stores to personnel if collected earlier as non-essential stores. | | |

| Serial No. | Sequence of Events | Action at Battalion Headquarters | Action at Company Headquarters | Action at Platoon Headquarters | Remarks |
|---|---|---|---|---|---|
| | | | (2) Establishes cooking arrangements if the same are decentralized. | | |
| | | b. ***Adjutant***<br>(1) Reports to commanding officer regarding completion of withdrawal.<br>(2) Informs brigade about latest situation.<br>c. ***Quartermaster***<br>(1) Delivers required stores.<br>(2) Deploys `B' echelon in new allotted area. | | | |

**Annex O**
**To Section 4**

## BATTLE DRILLS – PATROLS

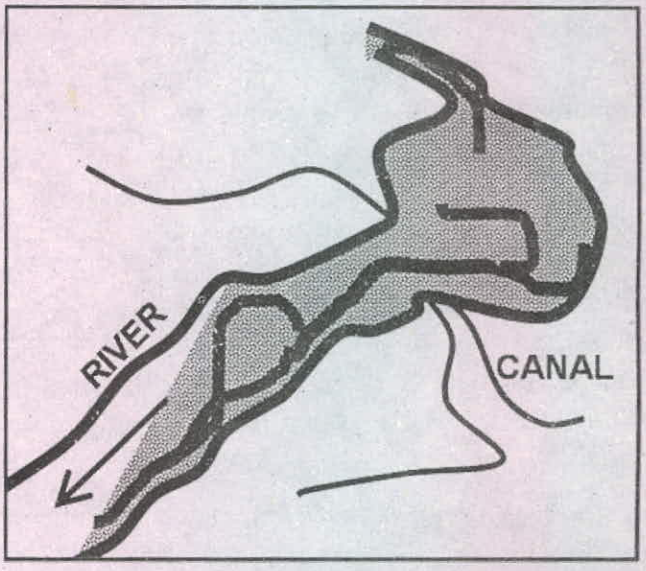
| Serial No. | Sequence of Events | Action by Commanding Officer/Patrol Master | Action by Company Commander | Action by Patrol Leader/Patrol Second-in-Command | Remarks |
|---|---|---|---|---|---|
| 1. | **Warning Order** | a. ***Commanding Officer Patrol Master***. See action as per battle procedure. | a. ***Company Commander***. Also see actions as per battle procedure. | a. ***Patrol Leader***. Also see action as per battle procedure.<br>b. ***Patrol Second-in-Command***<br>(1) Receives briefing from patrol leader.<br>(2) Checks and supervises administrative arrangements.<br>(3) Arranges for weapons/ ammunition as per orders. | |
| 2. | **Preparation** | | | a. ***Patrol Leader***<br>(1) Carries out rehearsals if necessary.<br>(2) Tests weapons of patrol.<br>(3) Makes arrangements for resting of the patrol. | |

| Serial No. | Sequence of Events | Action by Commanding Officer/Patrol Master | Action by Company Commander | Action by Patrol Leader/Patrol Second-in-Command | Remarks |
|---|---|---|---|---|---|
| | | | | (4) Checks administrative arrangements, arms/ ammunition as per warning orders. | |
| 3. | **Move of the Patrol** | – | a. Checks administrative arrangements.<br><br>b. Checks navigational arrangements.<br><br>c. Sends off the patrol. | a. ***Patrol Leader.***<br>(1) Carries out final inspection<br>(2) Explains the compass and march chart to company commander.<br>(3) Checks that weapons are loaded.<br>(4) Leaves with the patrol. | |

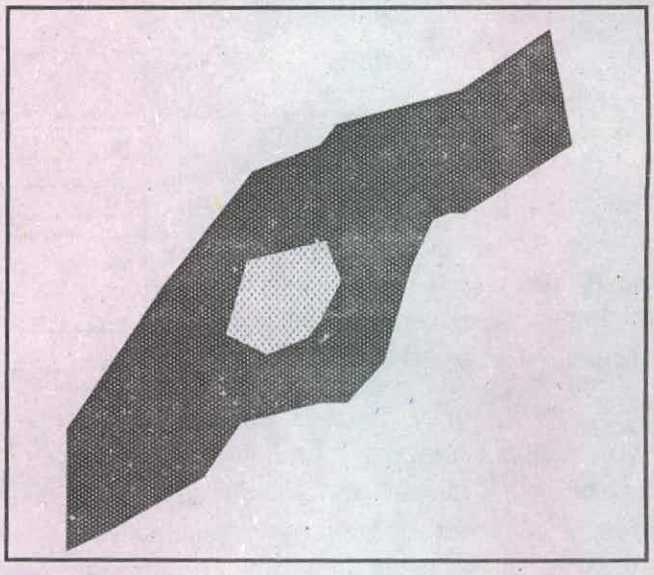
**Annex P**
**To Section 47**

## TYPE OF WATER OBSTACLE AND THEIR MAIN CHARACTERISTICS

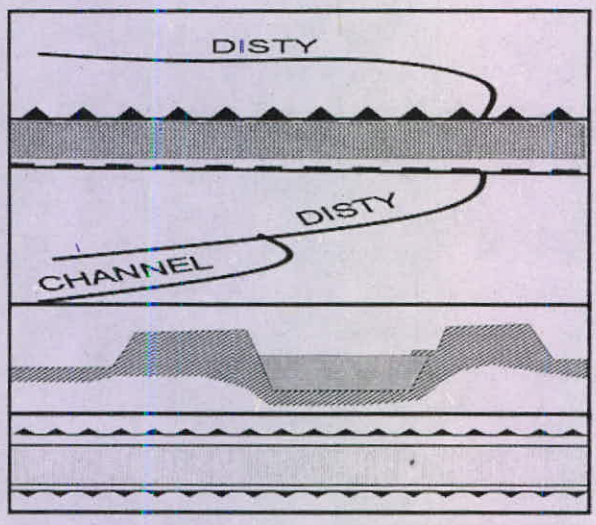
1. ***Dammed River***. Water flow in most of the rivers is controlled by dams in our operational area. In dry season very little water is released, hence such rivers become almost dry. In rainy season besides the rain water from nullahs joining the river, excess water is released from dams into the river turning it into a substantial obstacle. In dry season the width of water belt is negligible, depth however varies. Bridges and crossing sites exist. The river can be crossed at most places in dry season. Water can be regulated from dams to suit operational needs. Generally adjacent area of the river has been reclaimed which has normal crops and planned plantation. Old flood protection bunds exist, due to change in direction of river flow, the bunds may be away from main river now.



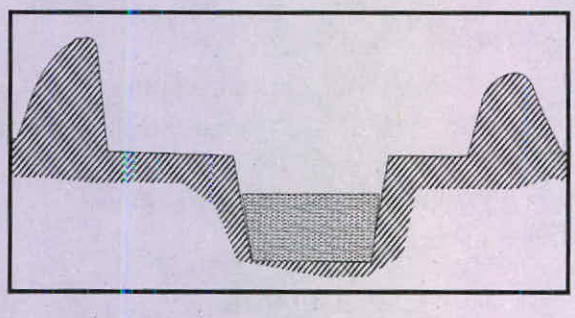

2. ***Undammed River***. It has sandy bed, marshy area astride and serves as an effective observations throughout. In dry season the wet span is over 500 feet, however has varying depth. At places flood protection bunds exist, they offer good observation. River may be crossed at existing bridges and ferry sites. Plenty of wild growth particularly elephant grass is common. The traffic ability is poor than dammed river.



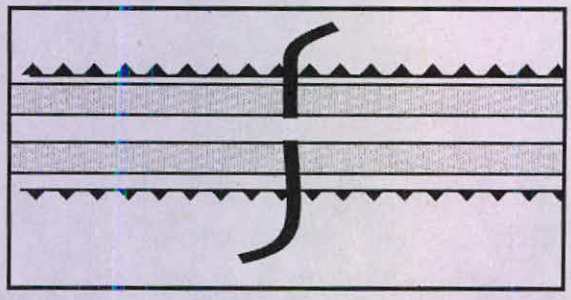
3. ***Irrigation Canal***. Irrigation canals have varying capacity, major 200–300 feet wide at head have capacity from 5000 to 10,000 cusecs. Both width and capacity gradually reduce towards the tail. At least one bank has a service track and embankment. Vehicle moving on canal road can be observed from a distance. Besides major crossing places like rail and road bridges, it has cattle crossing bridges and SWR (Steel Wire Rope) ferries. Tree plantation on banks offer cover besides protecting the bund from erosion. Water table in canal irrigated areas usually rises, marshy patches created along the canal due to seepage of water from canals.




4. ***Link Canal***. Capacity depends on the size of the link canal, may be 10,000 to 20,000 cusecs. Has road and rail bridges, and a few other bridges to link main towns? May have only wild growth of trees and shrubs. Rain water is drained into the canals. Less seepage of water from the canal.

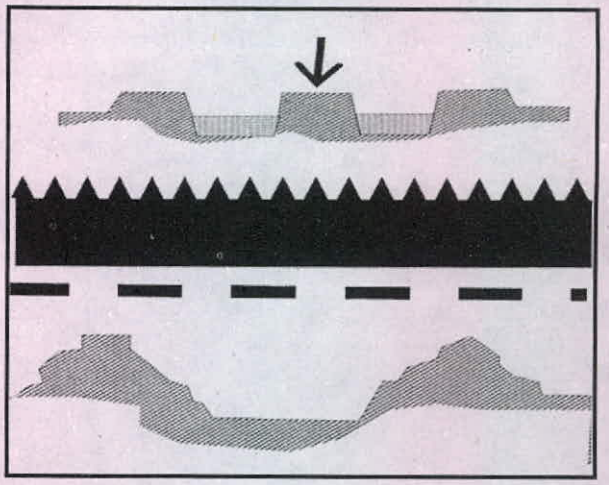


5. ***Feeder Canal***. Capacity depends on the size and purpose of the canal which remains constant till it starts feeding other canals. All known feeder canals have a capacity of over 2000 cusecs, hence width varies, has depth from 8 to 10 feet. May have the characteristics of either of the above two. It normally has wild growth of trees and shrubs.

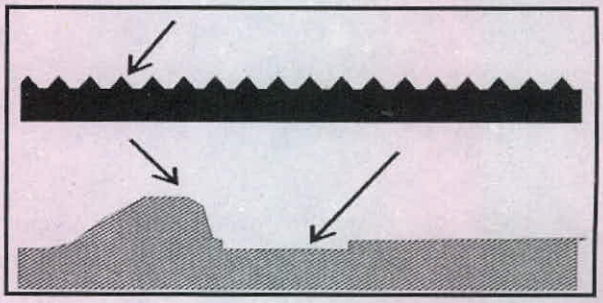


6. ***Twin Canal***. Twin canals is not a separate category of canals as such. The name is applicable to any two parallel canals running very close to each other. These may be both irrigation canals or feeder canals or one irrigation and one feeder canal. Bridges may exist for main roads/railway. Availability of other bridges will depend on the types of canals and the area through which the twin canal pass. Likely to have planned plantation. Factors affecting traffic ability are the same as for other canals.

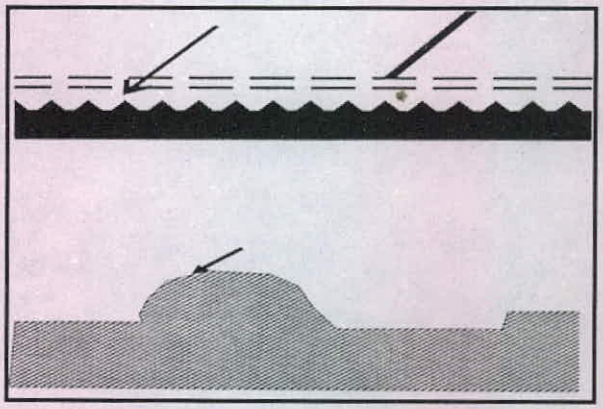
7. ***Drain***. Drains are of different sizes and do not normally have flowing water except in rainy season. They vary in size and have depth from 6 to 15 feet. Drains usually pass thorough well populated areas and in addition to road/rail bridges will have bridges linking main towns/villages. Not likely to have much vegetation. In dry season the traffic ability is good. In wet season and sometime after that, the traffic ability in low lying wet patches is poor.

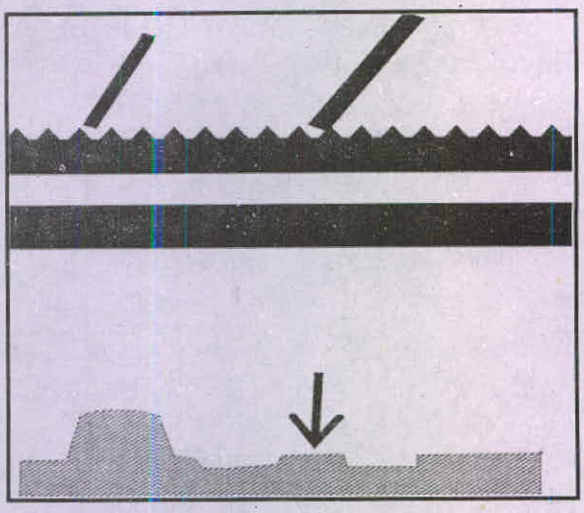


8. ***Anti-Tank Ditch***. It may be dry or stagnant water keeps the bed boggy. Generally it is not connected with any water source. Has width from 20 to 50 feet and depth varies from 5 to 6 feet. Far bank is usually at ground level. Earth dug is placed on the home bank to raise it, usually 5 to 10 feet.



9. ***Defence Drain***. Usually serve dual purpose i.e. drainage as well as defence obstacles, arrangements usually exist to feed water into defence drains in dry season. It is 80 to 100 feet wide, 8 to 10 feet deep. Far bank is usually at ground level. Home bank is raised to provide observation and for use as defence positions. May have defence works at important places. Has 11 to 15 feet high banks, with regular shape and constant height. The drain has tracks for vehicles behind the home bank. In addition to road/rail bridges it has limited number of bridges for operational use by the defender. If it is possible that the areas in front may be planned for inundation. It is fully developed as obstacles. Arrangements for filling it with water exist.





10. ***Double Ditch***. A double ditch may be a combination of an anti-tank ditch and a defence drain or two defence drains. Its characteristics are generally the same as a single defence drain. It may have stagnant water or water may be fed into it during emergency only. Both may be about 100 feet each or one may be 20 to 30 feet and the other about 100 feet. They are 8 to 10 feet deep. Far bank is at ground level. Middle bank may be 2 to 3 feet higher and home bank like a defence drain/ditch. Home bank will invariably have extensive defence work. Will have tracks for vehicles behind the home bank. In addition to road/rail it will be limited no of bridges for operational use by the defender. If it is possible, the area in front may be planned for inundation. It is fully developed as an obstacle. Arrangements for filling it with water exist.

**Annex Q**
**To Section 47**

# ENGINEER TECHNICAL DATA

1. ***Bridge and Assault Equipment***

| Equipment | Infantry Division | Armoured Division | Corps Engineers |
|---|---|---|---|
| M2 Assault Boats | 48 | 48 | 48 |
| Infantry Support Rafts | 6 | 6 | 6 |
| Class 9 (M2) Bridge | 84 metres | 84 metres | 84 metres |
| Class 50 (Hollow Plate Bridge) | – | 84 metres | 252 metres |
| Assault Track-way Class-30 | – | – | 891 metres |

2. ***Bridge Construction Timings***

| Type of Bridge | Span | Engineer Troops Required | Construction Timings (in hours) | | |
|---|---|---|---|---|---|
| | | | Day | Moonlight | Dark Night |
| M2 Bridge | 30 metres | 2 x Platoons | 1.3 | 1.7 | 2.00 |
| | 84 metres | 3 x Platoons | 2.25 | 3.0 | 3.75 |
| | 195 metres | 3 x Platoons | 4.00 | 5.00 | 6.5 |
| Hollow Plate Bridge | 30 metres | 3 x Platoons | 1.5 | 2.00 | 3.00 |
| | 84 metres | 3 x Platoons | 2.5 | 4.00 | 5.5 |
| | 168 metres | 3 x Platoons | 5.0 | 6.00 | 8.00 |

3. ***Boats and Raft***

| Type | Propulsion | | Typical Load | Construction Timings (in minutes) | | | Round Trip 30 Metres Gap (in minutes) |
|---|---|---|---|---|---|---|---|
| | Troops | Crew | | Day | Moon-light | Dark Night | |
| M2 Assault Boat | Paddles infantry | 3 | 12 armed men | – | – | – | 10 |
| 3 Pontoon OBM Raft | Infantry | 11 | ¾ ton | 20 | 25 | 30 | 10 |
| 5 Pontoon OBM Raft | Engineers | 11 | 2½ ton | 30 | 35 | 45 | 10 |
| 7 Pontoon OBM Raft | Engineers | 11 | 2½ ton with trailor | 40 | 50 | 60 | 10 |

| Type | Propulsion | | Typical Load | Construction Timings (in minutes) | | | Round Trip 30 Metres Gap (in minutes) |
|---|---|---|---|---|---|---|---|
| | Troops | Crew | | Day | Moon-light | Dark Night | |
| Class 50 OBM Raft | Engineers | 16 | Tanks | 60 | 90 | 120 | 15 |

4. ***Minefield Breaching***

a. To breach a 108 x 7.2 metres lane, an engineer platoon requires:-

| | | Days | Moonlight Hours | Dark Hours |
|---|---|---|---|---|
| (1) | No viper | 4 | 5 | 6 |
| (2) | Using baby viper | 3 | 4 | 4 |

b. Each 108 metres depth of minefield requires one engineer platoon.

c. Subsequent platoons can be committed as under:-

(1) No viper – After every 100 minutes.

(2) Using baby viper – After every 40 minutes.

d. Infantry soldiers' capability including that of the assault pioneers is 50 % of the above.

5. ***Trawl Tank***. An ordinary tank with two forward and one rear rollers for mine clearing is called a trawl tank. This equipment is under trials and is shortly likely to be introduced as standard equipment. A trawl tank can clear a lane of 2 metres width in one run. A second trawl tank moving .6 metres apart gives lane 4.2 metres wide. The operating speed of a trawl tank is 8 kilometres per hour and a 4.2 metre lane through 910 metres deep minefield can be made in about 15 minutes by a pair of trawl tanks. However, marking of the lanes and other drills to confirm absence of any live mines in the lane, take much more time. Following timings have been found to apply during trials in daylight:-

| Depth of Minefield | Troops Required | Time Taken |
|---|---|---|
| 270 metres | 1 x Field Platoon | 1 hours |
| 250 metres | 2 x Field Platoons | 2 hours |
| 120 metres | 2 x Field Platoons | 3 hours |
| 910 metres | 3 x Field Platoons | 3½ hours |

**Annex R**
**To Section 52**

## PROCEDURE FOR ISSUE OF VERBAL ORDERS ON WIRELESS

### Preliminary Requirements

1. To enable the procedure to be correctly and successfully applied, the following preliminary requirements must be fulfilled:-

   a. Clear drills must exist for quick deployment.

   b. Wireless discipline within the battalion must be of a high order.

   c. Orders must be issued in a set sequence.

   d. Code words must be allotted beforehand for deployment action.

### Deployment Drills

2. In the choice of a tactical plan, a battalion group has the following alternatives open to it for the employment of its subunits etcetera:-

   a. Attack with rifle companies either "One Up", "Two Up", or "Three Up", ("One Up", though NOT a normal formation for a battalion attack, may have to be adopted on occasions when a deep and narrow objective has to be attacked in phases).

   b. The attack will be led either by tanks or by infantry. The two elements will advance either on the same or on converging axes.

3. All troops should be ready in a suitable assembly area.

4. Forward Observation Officer, Mobile Fire Controllers, detachments of supporting arms including armour and engineers, and the elements of Support Company which are allotted to the assaulting companies, will report to their respective companies in the assembly area (depending on the time and space factor) before H hour and on timings as laid down by the commanding officer.

### Procedure for Issue of Orders

5. ***Call Signs***. These will be the standard call signs already in use in the infantry battalion. The collective call for orders for attack on wireless will be Charlie Charlie One Zero, on which only call signs 1, 2, 3 and 4 (rifle companies) will answer. Other stations will listen-in and record the transmissions. In certain exceptional cases, where other stations are also required to answer, the particular station (s) will be included in the call sign as follows:-

"Charlie Charlie One Zero and (call sign)

One to Alpha"

6. ***Warning Order***. A warning order will be given to all stations sufficiently at least 5 minutes before the commanding officer is due to issue his orders on the wireless. This warning order will consist of the following details:-

a. Designation of assaulting companies concerned.

b. Time for issue of the commanding officer's orders on wireless.

7. ***Orders by the Commanding Officer***

a. These will be issued NOT until at least 5 minutes after the issue of the warning order.

b. The orders will be given on the wireless in the standard sequence as per paragraph 9 below.

c. The orders will normally be addressed to rifle companies only. (See also paragraph 5 above). Subunits NOT mentioned in the orders will receive their orders by other means later.

8. ***Action after Orders***

a. The commanders of supporting arms will give detailed orders to their subunits, with regard to re-affiliation etc for the attack.

b. Orders issued to subunits on wireless may be confirmed by other means if possible and necessary.

**Sequence of Orders by the Commanding Officer**

9. Following will be the standard sequence for issue of orders on wireless by the commanding officer:-

a. ALPHA

(1) Whether the attack is to be led by infantry or tanks? (May be omitted if tanks are NOT participating).

(2) Formation of rifle companies in the attack (i.e. One Up, Two Up or Three Up).

(3) Objectives to companies. (The first company mentioned will be on the LEFT). The following is an example:-

b. BRAVO — Axis of attack, Normally two grid references will be given to describe the axis from the rear to the front. (To be given only when required, e.g. when tanks and infantry are moving on separate axes). The following is an example.

BRAVO – A Company axis GR 361429 to Control Point 162.

B Company axis GR 372418 to GR 373417.

Armour - - - - - - - - - - - - - - - - - - - - - - -
Control Point 637 to Control Point 492.

c. CHARLIE H Hour. (This is the time when it is intended that the troops will cross the Start Line).

d. DELTA Inter-Company boundary(ies). A boundary will normally be described giving two grid references from the rear to the front. In case of an attack with three companies up, the boundary between the LEFT and the centre companies will be given first and will be preceded by figure ONE, while the boundary between the centre and the RIGHT companies will be given next and will be preceded by figure TWO, e.g.:-

DELTA ONE GR 402331 to GR 402354.

TWO GR 413332 to GR 413335

e. ECHO Assembly Area(s). Coded Grid References or Control Points may be used.

f. FOXTROT Artillery, possibly 81 millimetres Mortar, 106 millimetres Recoilless Rifle and Machine-gun fire plan for the attack. The machine-gun used in the fire plan will be those which are NOT accompanying the assaulting companies.

g. GOLF Location of battalion headquarters.

h. HOTEL Success signals (visual as well as voice).

10. An example of the sequence and form of the verbal orders in clear using the appointment codes only is given below. These forms of orders are used in an attack when the time available is so short that it does not permit the use of code and when the success of the operation is not likely to be jeopardized by the orders being passed in clear. The specimen is based on the assumption that the battalion is attacking with two companies up, with infantry leading and with the support of tanks using the same axis:-

a. Control "Hello All Stations – Warning Order Attack – Call signs 2 and 4 – Stand by for orders 1455 hours. Over".

b. All Stations (in turn) "Roger – Out" (1, 2, 3, 4, 5, 7, 10 and 11) (After a minimum of 5 minutes pause and at the appointed time.)

c. Control "Hello Charlie Charlie One Zero".

ALFA – Infantry leading – Two Up – Left Call Sign 2 – Objective GR-375248 – Right Call Sign 4 – Objective SHERPUR.

BRAVO – Call sign 2 axis GR 363249 to RAWAL – Call sign 4 axis GR 400135 to GR 398075 – Roger so far – Over".

(Bravo may be omitted if axis of attack is NOT required to be given).

d. Call Signs (in turn) 1,2,3, and 4 – "Roger – Over"

e. Control "Charlie H Hour – 1530 Hours. DELTA GR 402339 to GR 434118. ECHO FOREST RESERVE – Roger so far – Over".

f. Call Signs (in turn) 1, 2, 3 and 4 – "Roger – Over".

g. Control "Charlie Charlie One Zero FOXTROT GHARBEEN from 1532 hours to 1600 hours concentrations GR 425311 – Mortar from 1535 hours to 1600 hours GR 434312 – Machine-guns LOW HILL – Roger so far – Over".

h. Call Signs (in turn) 1, 2, 3 and 4 – "Roger – Over".

i. Control "Charlie Charlie One Zero GOLF after Call Sign 4

HOTEL Red over green over green – Wireless GULDUM – Over".

j. Call Sigs (in turn) 1, 2, 3 and 4 – "Wilco – Out".

**Codes**

11. The following codes will be used in all orders issued on wireless and will be laid down by the next superior headquarters every 24 hours (except for the Griddle

Code which will be issued by the battalion):-

a. ***Code Words for Deployment Drill***

| | Detail | No of Code Words |
|---|---|---|
| (1) | Formation of attack One Up "Two Up" or "Three Up" | "One Up" Three |
| (2) | Attack to be led by tanks or by infantry. | Two |

b. Time letter Code for H Hour and Fire Plan Timings. This will be in the form of a letter code , e.g.:-

K O J V B U N D G H

0 1 2 3 4 5 6 7 8 9 M (Repeater)

C X Y N E F A T R

c. ***Control Points***. Normal issue.

d. ***Griddle Code***. Normal issue (under battalion arrangements).

e. ***Success Signals Code***. A list of nine sets of success signals will be issued. The serial numbers will be used as codes to indicate, the success signals (visual as well as voice) fixed for a particular attack. Any serial may be chosen at random.

12. A specimen of the codes required by commanding officer to give verbal orders on wireless is as under:-

a. ***Deployment Code Words***

| | | |
|---|---|---|
| SHERDIL | – | Warning order for attack. |
| JANGJOO | – | Attack will be with one company up. |
| SHARBAT | – | Attack will be with two companies up. |
| KHAKI | – | Attack will be with three companies up. |
| KARWAN | – | Infantry to lead the assault. |
| TEZGAM | – | Tanks to lead the assault. |
| SHAMSHIR | – | Stand by for orders. |

b. ***Time Letter Code***

K O J V B U W D G H

01 2 3 4 5 6 7 8 9 M (Repeater)

C X Y L N E F A T R

c. ***Control Points (Normal Issue)***

| | | | |
|---|---|---|---|
| (1) | SHERPUR | – | 635 |
| (2) | RAWAL | – | 129 |
| (3) | FOREST RESERVE | – | 784 |
| (4) | LOW HILL | – | 211 |

d. ***Griddle***. Normal Issue.

e. ***Appointment Titles*** (As per GSP-1620 – Staff Duties in the Field). In addition, standard titles for the following will also be used for giving out the fire plan to the assaulting companies:-

| | | | |
|---|---|---|---|
| (1) | Mortars | – | ABABEEL |
| (2) | Machine-guns | – | SURKHAB |
| (3) | Recoilless Rifles | – | HUDHUD |
| (4) | Anti-Tank Guided Missile | – | ZARAR |
| (5) | Grenade Launcher Heavy-Machine gun | – | NAG |

f. ***Success Signals Codes***. A list of nine success signals will be issued as per following specimen:-

| Serial No. | Visual | Wireless |
|---|---|---|
| (1) | Red over green over red | SURKHAB |
| (2) | Red over red over red | DAFTAR |
| (3) | Red over green over green | GULDUM |

13. An example of a coded version of verbal orders on wireless using the clear text and codes is given below. This form will be used in a situation where sufficient time is available and the success of the operation is likely to be jeopardized if the orders are passed in clear in the form indicated in paragraph 10 above.

a. Control – Hello All Stations

SHERDIL

Call Signs 2 and 4 –

SHAMSHIR. 1 spell – OBUK Over.

**Annex S**
**To Section 57**

## BATTLE PROCEDURES/DRILLS – OCCUPATION OF COUNTER PENETRATION POSITION WITHIN THE FRAMEWORK OF A BATTALION

1. ***Phase 1 – Planning/Preparation***. Planning and preparation of counter penetration positions is a concurrent activity. This phase starts with the planning/ preparation of main defences. During the planning stage of main defence counter penetration positions are planned, selected, reconnoitred, prepared and rehearsed to confirm routes and methods of their occupation.

2. ***Phase 2 – Execution***. During this phase, depending upon the tactical situation, counter penetration positions may be occupied on orders, for continuing the battle as visualized by the commander. This phase can be further sub-divided in two stages:-

a. ***Stage 1 – Move from Concentration/Original Area to the RV/ Debussing Point***. Following sequence of actions will expedite the operation:-

(1) Adjutant/Internal Officer (Intelligence Officer) informs brigade headquarters immediately on telephone/wireless or through dispatch rider regarding move.

(2) Commanding officer gives executive order to the company commander and gives latest information about enemy and time upto which the position has to be occupied.

(3) Adjutant instructs regimental police section to post guides at critical points along the route. Route would have been reconnoitred/rehearsed before move. This will not only ensure guidance at critical moments but also help to maintain control of operation.

(4) Immediately on receipt of orders, company heavy weapons including Anti-Tank Guided Missile and guides move with the company order group to the counter penetration position under the company commander.

(5) Company to march under second-in-command to the rendezvous.

b. ***Stage 2 – Occupation of Position***

(1) At rendezvous, guides receive the company and take them to their respective locations without halting.

(2) Company commander on reaching the location gives arrival of the company and ensures smooth occupation of position and deployment of heavy weapons.

(3) Company commander will site anti-tank weapons on the flanks to deter out-flanking of positions if visualized.

(4) Where possible anti-tank weapons and automatics may be sited in the minefield and surveillance of flanks will be ensured.

(5) Shoulders of the position should be held strongly.

Annex T
To Section 57

## BATTLE PROCEDURES/DRILLS – OCCUPATION OF COUNTER PENETRATION POSITION WITHIN THE FRAMEWORK OF BRIGADE/DIVISION

1. ***Planning/Execution***. An infantry battalion or battalion group will conduct this operation in two stages as under:-

a. ***Stage 1 – Move from Concentration Area upto Rendezvous/ Debussing Point***

(1) On receipt of orders, commanding officer along with his order group, guides, anti-tank weapons and mortars moves to the counter penetration position.

(2) Commanding officer gives out last moment briefing and communicates any changes required from already rehearsed task in view of the enemy situation.

(3) Without wasting time heavy weapons will be deployed by the respective commanders to face enemy, and the guides will fall back to the rendezvous/debussing point to receive the main body.

(4) Battalion second-in-command will bring up the main body to the rendezvous/debussing point (if Troops Carrying Vehicle borne). Guides receive respective companies and lead them immediately to their locations. Troops need not halt at this location.

(5) In the meantime, Adjutant will ensure that the regimental police are in position at critical points before the start of the move of main body.

(6) In case the battalion is Troops Carrying Vehicle borne, then sufficient troops may be taken with the order group, to ensure security of the force during occupation stage.

(7) Adjutant/Intelligence Officer keeps the brigade headquarters informed about the progress of operation.

b. ***Stage 2 – Occupation of Counter Penetration Position***

(1) Guides will receive the main body at the Rendezvous and lead them to respective defences.

(2) Battalion will preferably deploy with three companies and three platoons up to cover maximum frontage and to ensure delivery of maximum volume of fire to the front.

(3) Heavy anti-tank elements if allotted should be deployed on the flanks to strengthen the shoulders and to deter outflanking of the position by the enemy armour.

(4) Tanks if allotted should be kept with the depth company to react against out-flanking enemy.

(5) Effective surveillance of gaps and front will be ensured to give early warning.

(6) Raids/ambushes to be carried out to harass the enemy.

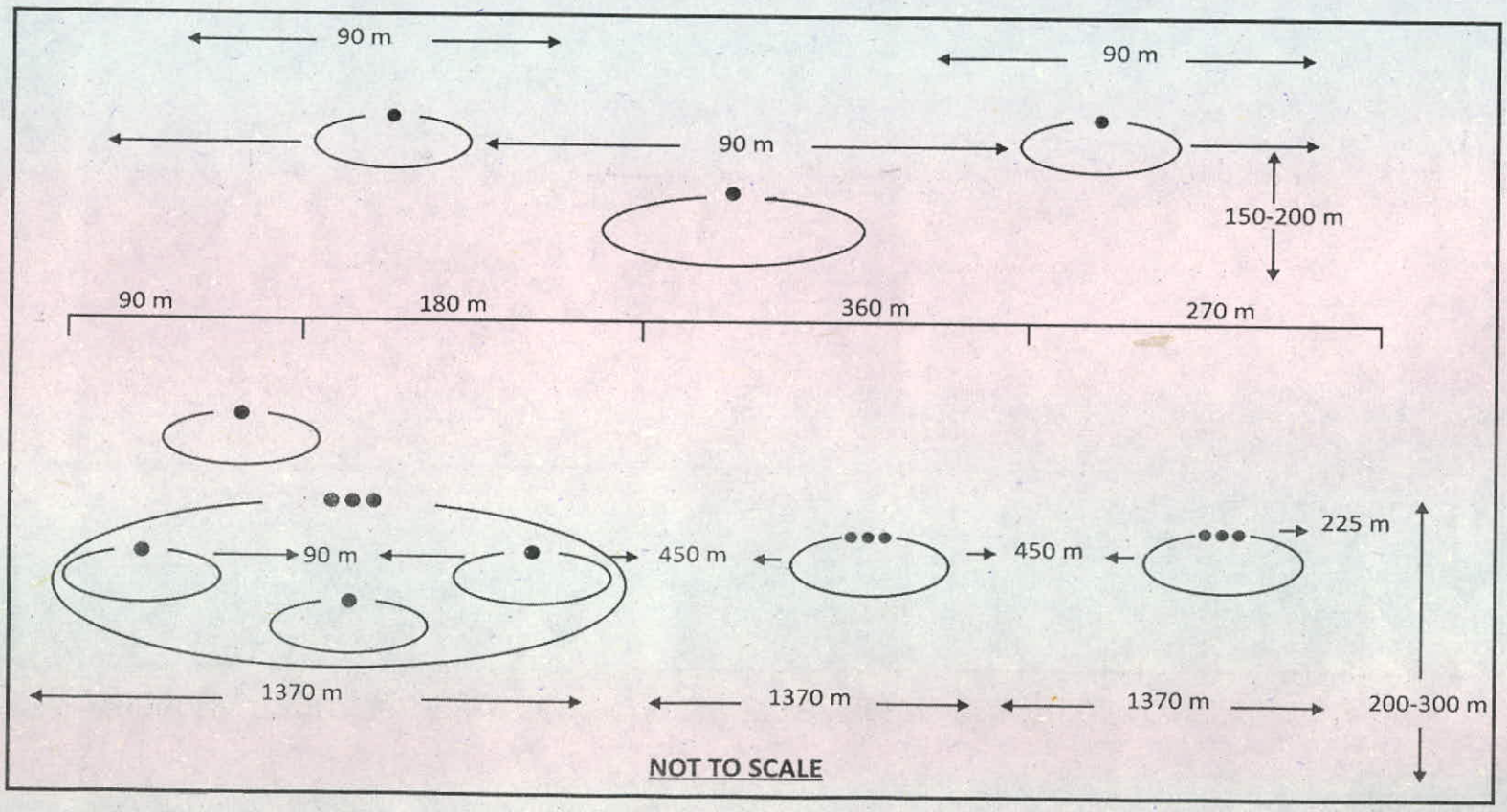
(7) Intelligent use of anti-tank and automatic weapon positions in the minefield will cause delay/attrition to the enemy.

(8) Battalion mortars and direct support battery to be deployed to effectively cover the front and flanks of the position.

(9) Battlefield illumination should be worked out in detail with the help of battery commander to ensure effective utilization of all weapons at critical stages of battle at night.

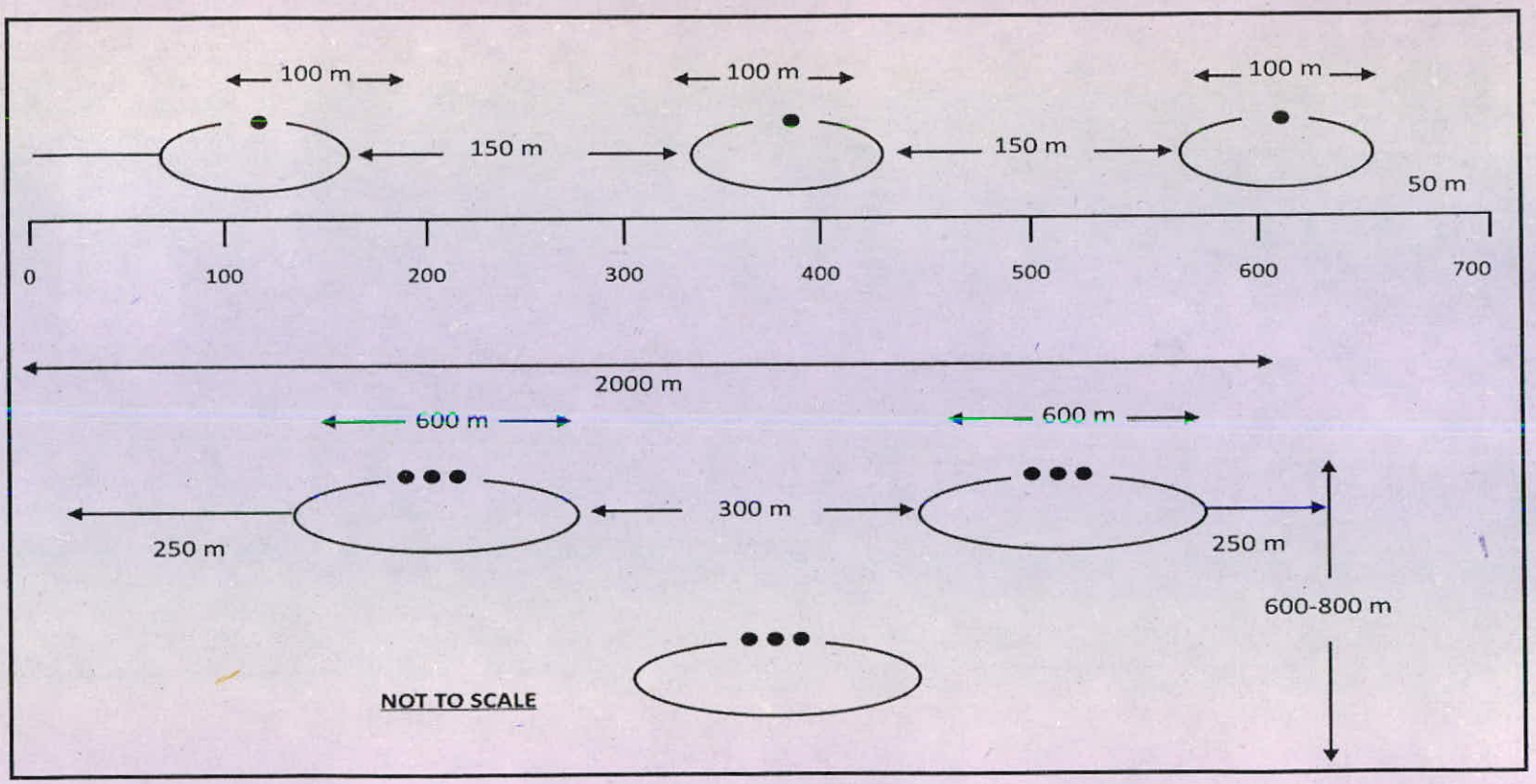
**Annex U**
**To Section 59**

## SUGGESTED FRONTAGES FOR SECTION/PLATOON IN DEFENCE ON EXTENDED FRONTAGES



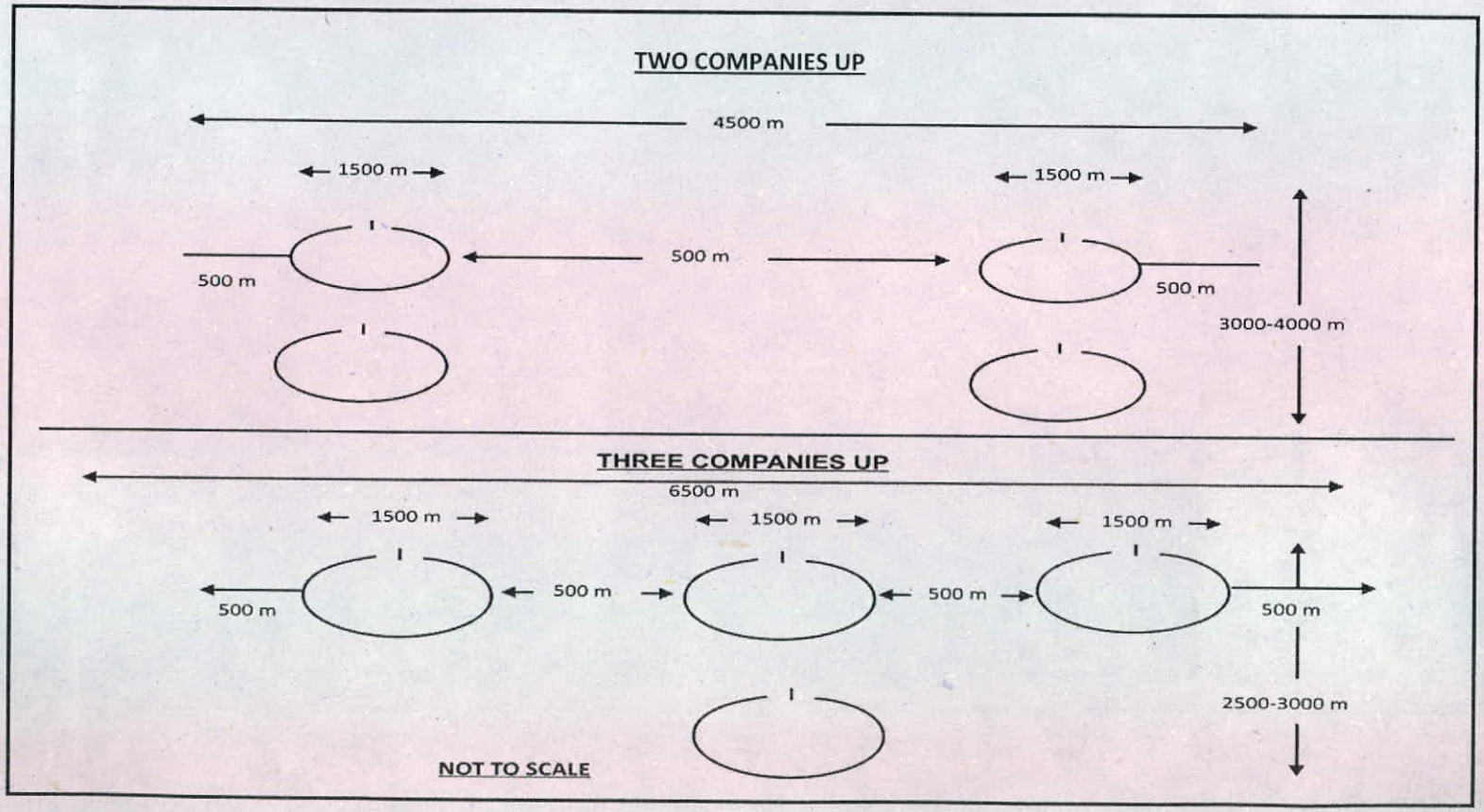
Appendix 5
To Annex U

## SUGGESTED FRONTAGES FOR SECTION/PLATOON/COMPANY IN DEFENCE ON EXTENDED FRONTAGES

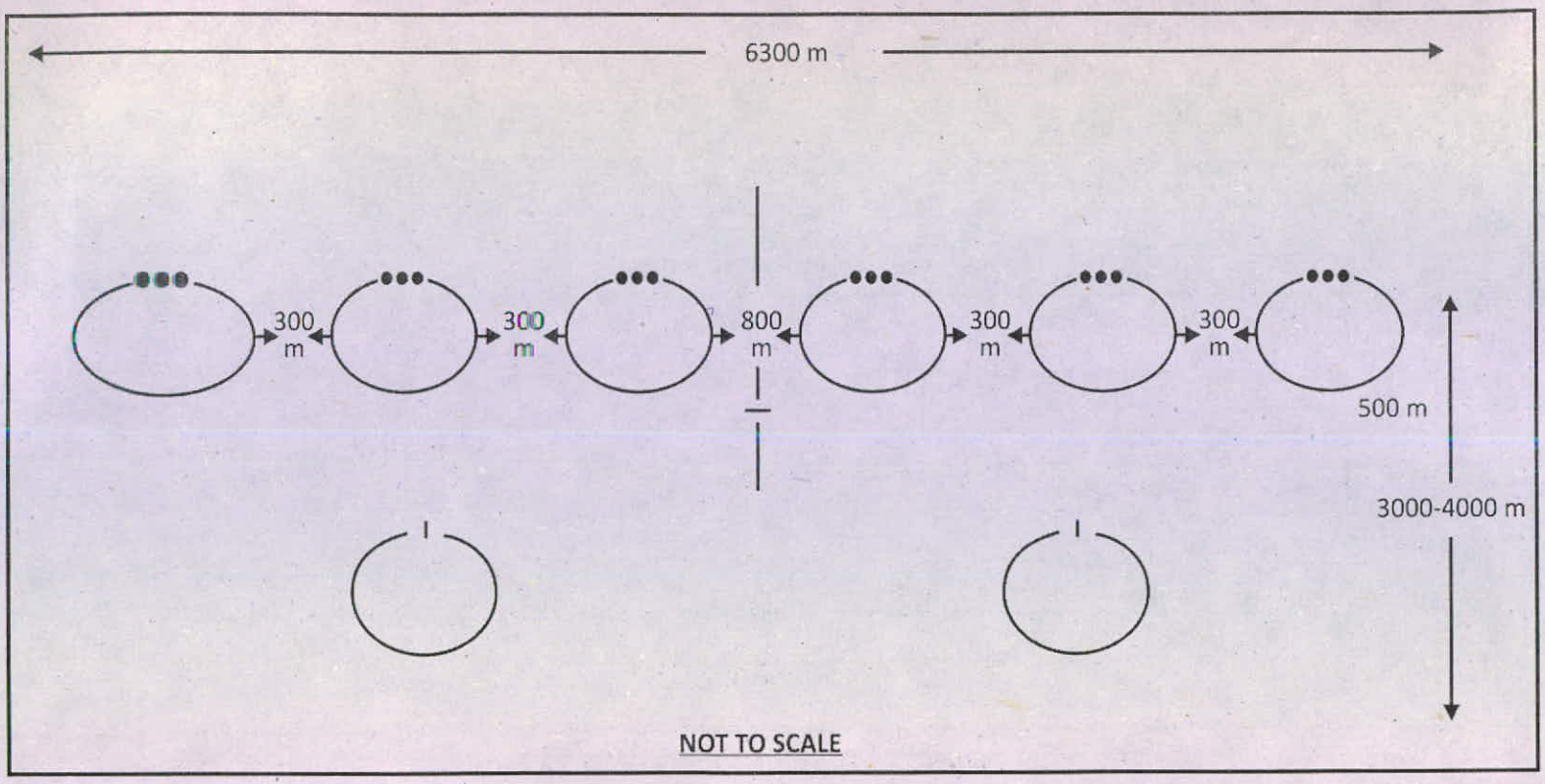


**Appendix 6**
**To Annex U**

## BATTALION FRONTAGES – TWO/THREE COMPANIES UP

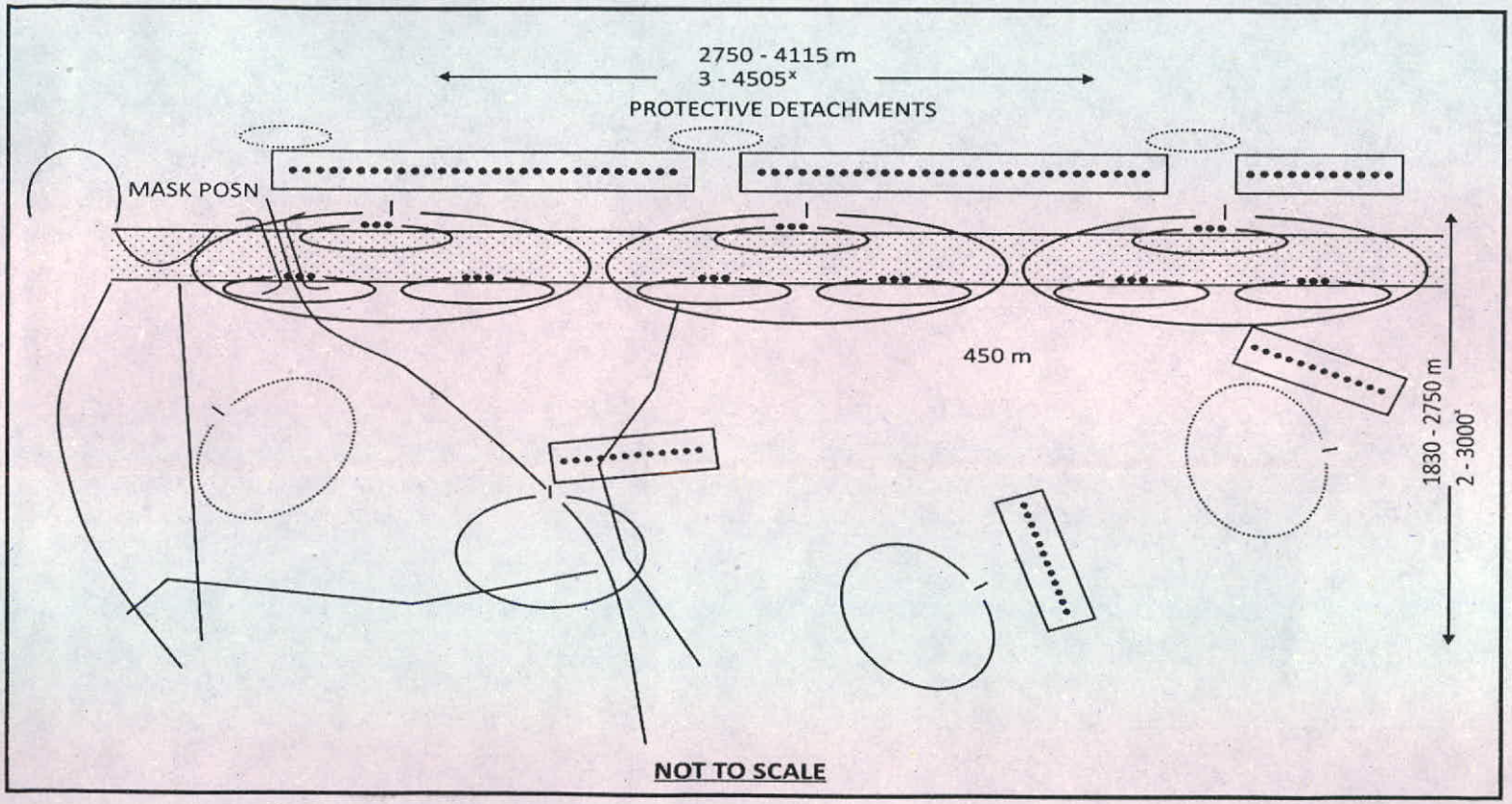


## BATTALION ON EXTENDED FRONTAGES WITH TWO COMPANIES AND THREE PLATOONS UP

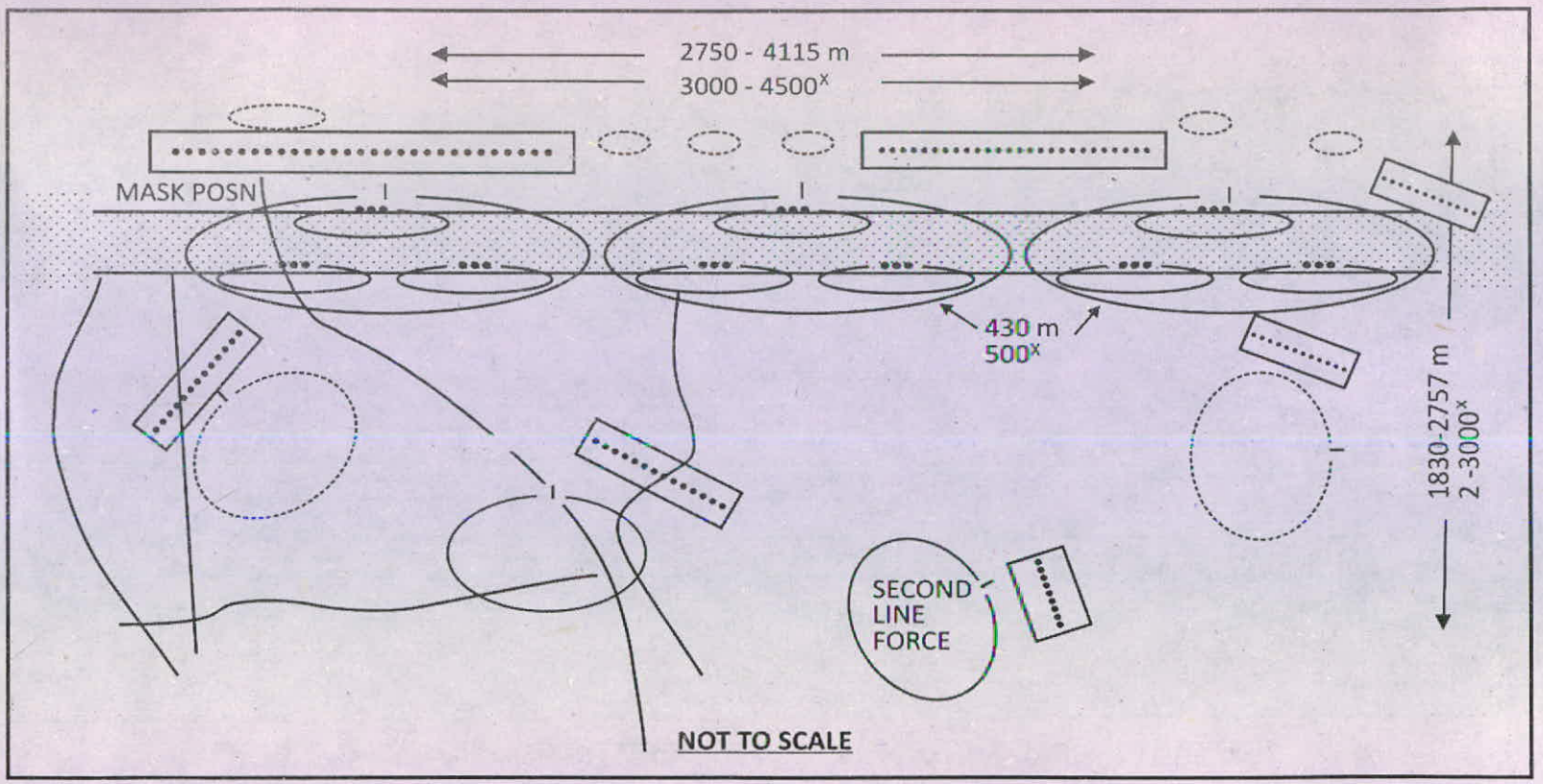


**Annex V**
**To Section 61**

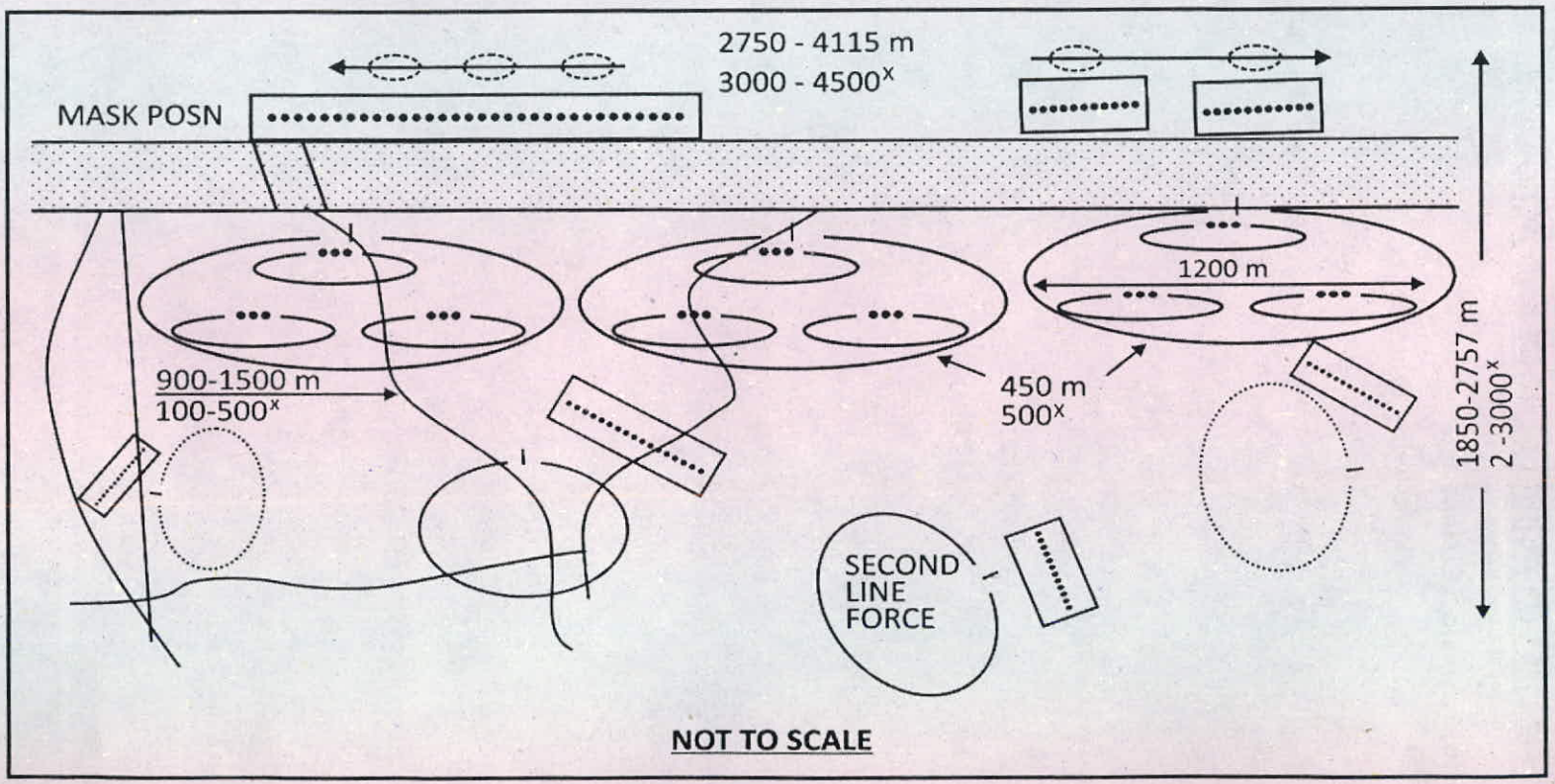
## INFANTRY BATTALION IN DEFENCE ALONG A CANAL WITH HIGHER HOME BANK



## INFANTRY BATTALION IN DEFENCE ALONG A CANAL WITH HIGHER FAR BANK OR BANKS OF EQUAL HEIGHT



## INFANTRY BATTALION IN DEFENCE ALONG AN ANTI-TANK DITCH




NOT TO SCALE

# LIST OF ABBREVIATIONS AND ACRONYMS

| Abbreviation/Acronym | Explanation |
|---|---|
| Ack | Acknowledge, (ed), (ment) |
| Adm | Administration, Administrative |
| Adv | Advance, Advantage, Advantageous, Advancing |
| Add | Addition |
| Adjt | Adjutant |
| AMC | Army Medical Corps |
| Ammo | Ammunition |
| Arty | Artillery |
| Aslt | Assault, (ed) |
| ATGM | Anti-Tank Guided Missile(s) |
| Attk | Attack |
| Bde | Brigade |
| BE | Base Ejection |
| BHM | Battalion Havildar Major, Battery Havildar Major |
| Bn | Battalion |
| BQMH | Battalion Quartermaster Havildar, Battery Quartermaster Havildar |
| Bty | Battery |
| BUAs | Built-Up Areas |
| Capt | Captain |
| Cbt | Combat |
| Cfn | Craftsman; Craftsmen |
| CHM | Company Havildar Major |
| $C^3I$ | Command, Control, Communication and Intelligence |
| Ck | Cook |
| Comd | Commander |
| CO | Commanding Officer |
| Coy | Company |
| CP | Command Post, Check Post, Contact Point |
| CQMH | Company Quartermaster Havildar |

| Abbreviation/Acronym | Explanation |
|---|---|
| Det | Detach, (ed), (ment) |
| DR | Despatch Rider |
| Dvr | Driver |
| Edn | Education |
| Eff | Effect, (ed), (ive), (iveness) |
| EME | Electrical and Mechanical Engineers |
| Engrs | Engineers |
| Estb | Establish, (ed), (ment) |
| Exch | Exchange |
| Fd | Filed |
| Fmn | Formation |
| Fol | Follow, (ed), (ing) |
| FUP | Forming Up Place |
| Fwd | Forward, (ed) |
| GD | General Duties |
| GHQ | General Headquarters |
| GLHMG | Grenade Launcher Heavy-Machine Gun |
| Gp | Group |
| GPS | Global Positioning System |
| GS | General Staff; Guidance System |
| Hav | Havildar |
| HE | High Explosive |
| HQ | Headquarters |
| Hr(s) | Hour(s) |
| Int | Intelligent, Intelligence |
| JCO | Junior Commissioned Offer |
| kg | Kilogram |
| km | Kilometre |
| kmph | Kilometres per hour |
| LAT | Light Anti-Tank |
| Ldr | Leader |

| Abbreviation/Acronym | Explanation |
|---|---|
| LMG | Light-Machine Gun |
| Lnk | Lance Naik |
| Lt Col | Lieutenant Colonel |
| Maj | Major |
| MBRL | Multi Barrel Rocket Launcher |
| MFC | Mobile Fire Controller |
| MG | Machine-Gun |
| Min | Minimum |
| Misc | Miscellaneous |
| mm | Millimetre |
| Mor | Mortar |
| MT | Military Transport |
| NCO | Non-Commissioned Officer |
| Nk | Naik |
| NOD | Night Observation Device(s) |
| NVD | Night Vision Device(s) |
| NVG | Night Vision Goggle(s) |
| PEWS | Platoon Early Warning System |
| Pl | Platoon |
| Pnr | Pioneer |
| POF | Pakistan Ordnance Factories |
| Pt | Point |
| QM | Quartermaster |
| RAP | Regimental Aid Post |
| Rd | Road, Round |
| Res | Reserve, (s) |
| Recce | Reconnaissance; Reconnoitre, (d) |
| RHQ | Regiment Headquarters |
| RL | Rocket Launcher |
| R&L | Reconnaissance and Layout |
| RP | Refilling Point, Regimental Police |

| Abbreviation/Acronym | Explanation |
|---|---|
| RR | Radio Relay, Recoilless Rifle, (s) |
| RV | Receipt Voucher, Rendezvous |
| 2IC | Second-in-Command |
| SAM | Surface-to-Air Missile |
| Sec | Section |
| SP | Start Point |
| Sep | Sepoy |
| SI | Seriously Ill |
| Sig | Signal |
| SOS | Save our Souls; Struck of Strength |
| Sqn | Squadron |
| Tac | Tactical; Tactics |
| TCV | Troop Carrying Vehicle |
| Tech | Technician, Technique, Technology, Technical, (ly) |
| Tk | Tank |
| TP | Traffic Post |
| Tpt | Transport, (ed) |
| Trg | Training |
| VHF | Very High Frequency |
| VIP | Very Important Person |
| WP | White Phosphorus |
| Wpn | Weapon |
| WL | Wagon Line |
| UHF | Ultra-High Frequency |

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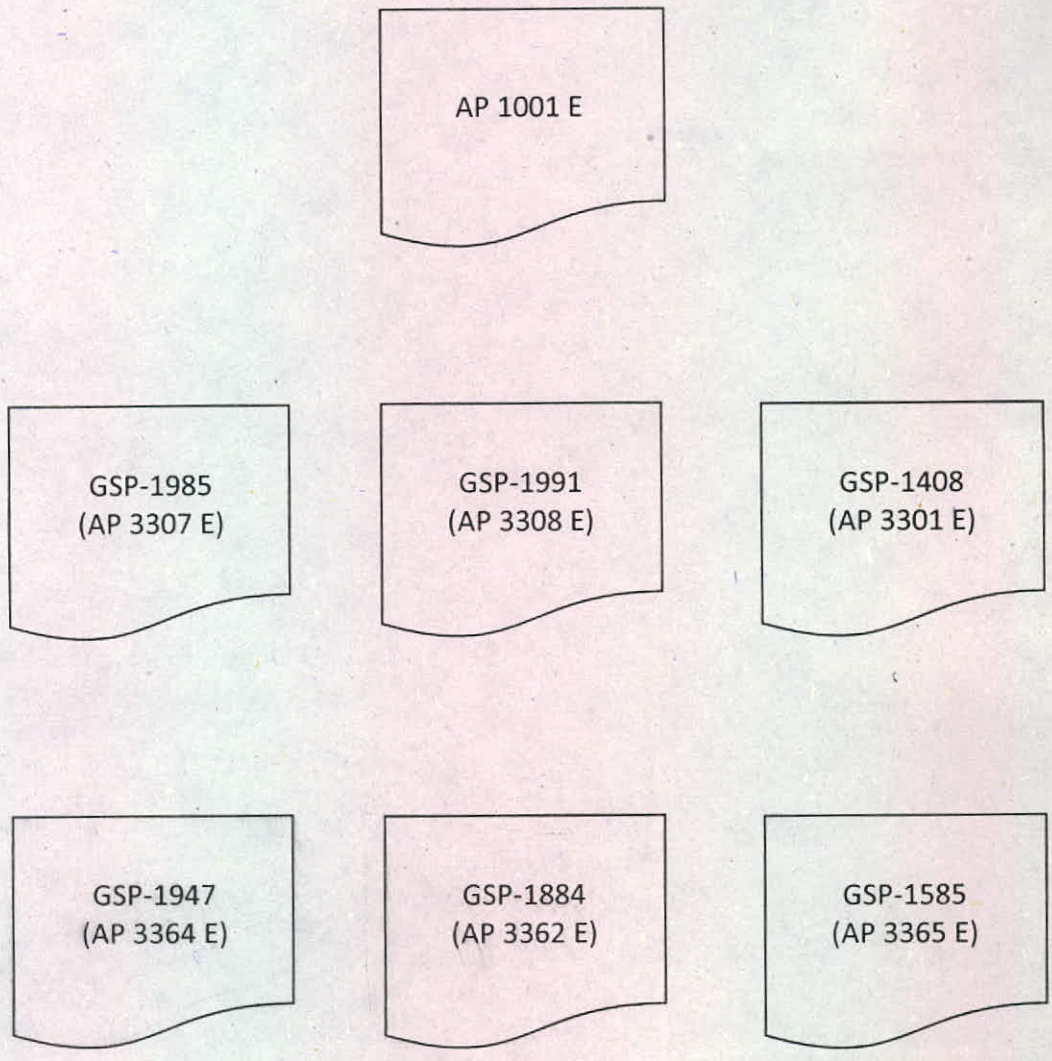
# RELATIONAL MAP

**AP 3367**



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